That is remarkably small.

The interpretation should be:

Near the mature active face, the fast subsystem acts primarily as a slo wly varying parameter, while the dominant restoring eigenstructure belongs to the adaptive admissibility dynamics itself.

text

triangular_one_step_trajectory.csv
triangular_one_step_summary.csv
triangular_rollout_trajectory.csv
triangular_rollout_summary.csv
triangular_gap_summary.csv
triangular_closure_arrays.npz
parameter_free_triangular_report.txt
analysis_manifest.json

What would close the mechanism

The late triangular reduction is strongly supported if:

geometric-axis invariance remains near machine precision;
gn+1≈αEMAgngn+1≈αEMAgn;
one-step scalar residuals are small;
rollouts from t=200t=200 and 250250 remain accurate through 320320;
predicted and actual scalar decay rates agree;
implied model κκ matches the target-time memory rate.

Please paste parameter_free_triangular_report.txt after the run.

think about this:

This is a much stronger synthesis than earlier versions because it explicitly ties each mechanistic claim to a diagnostic. However, I think there are still a few places where the conclusions run ahead of what the data strictly establish. I'll go section by section.

Overall

The evidence appears to support three increasingly strong levels of claim:

Observed numerical regularities (strongly supported by the file).
Mechanistic interpretation for this replay (plausible and largely supported).
General dynamical law (not yet established).

Most of the document is at level (2), but there are a handful of sentences that drift into (3).

"The geometric direction is the operator-invariant line"

This is almost entirely supported.

The diagnostics show

angle ≈ (0.0013^\circ),
invariance defect

[
6.46\times10^{-13}
]

versus

[
1.13\times10^{-5}.
]

Those are exceptionally strong numerical observations.

Where I would soften the language is here:

"It is the actual invariant line of the late frozen weight operator."

The file demonstrates an extremely small measured invariance defect for the sampled late operators. It does not prove exact invariance of an underlying operator in a mathematical sense.

A more precise statement would be:

The sampled late operators map the geometric direction into itself to within numerical precision.

That is fully supported by the diagnostics.

Geometric versus POD control

This section is one of the strongest.

The important point is not that the POD and geometric vectors agree.

The important point is that the same propagation-gap result survives changing the projection axis.

That is exactly the sort of control one wants to remove concerns about circularity.

I think this is a genuinely valuable result.

"Therefore"

Here I would be more careful.

You write

Therefore,

[
\kappa=r_{\rm prop}-r_g
]

is supported to approximately (10^{-6}).

The numerical data support that for the mature targets in this replay.

Indeed, at target 320 the residuals are around 10^{-6}, while at target 250 they are between roughly 10^{-6} and 10^{-5}.

However, the same table also shows

target 150:
errors around 3×10^{-3},
target 200:
errors around $2\text{–}4\times10^{-4}$.

So I would phrase it as

The relation is numerically supported in the mature late regime, with closure improving systematically along the trajectory.

That captures the observed trend without implying a broader identity than has been tested.

Emergence

I actually think this is one of the best sections.

The decreasing error

$$
10^{-3}
\rightarrow
10^{-4}
\rightarrow
10^{-5}
\rightarrow
10^{-6}
$$

is a much stronger observation than any single late fit.

It indicates convergence rather than coincidence.

The causal interpretation

winner scales converge,

geometric line becomes invariant,

is plausible,

but should still be described as an interpretation rather than something directly measured.

"The reduced late dynamics are triangular"

This is where I think the distinction between evidence and hypothesis matters most.

The recurrence

[
a_{n+1}

\lambda a_n

g_n

\epsilon_n
]

is directly supported by your earlier scalar-reduction diagnostics.

The second equation,

[
g_{n+1}
\approx
\alpha_{\rm EMA}g_n,
]

is supported by the measured forcing decay rate.

So the triangular system is a reasonable reduced model.

However, I would avoid saying

"The results now suggest..."

and then immediately writing the boxed system as if it were established.

Instead,

The diagnostics motivate the following late asymptotic reduced model...

keeps the distinction clear.

Identifiability

I think this section is excellent until one sentence.

The data clearly support

correlation near one,
rapidly increasing condition numbers,
coefficient drift despite (R^2=1).

That demonstrates non-identifiability.

Where I would hesitate is

"This proves..."

It proves non-identifiability within the chosen regression model and late windows.

It does not prove that no alternative parameterization could identify two coupled states.

Why the two-state fits drift

This is probably the most convincing explanation.

Near-collinearity naturally explains

unstable coefficients,
stable predictions,
perfect (R^2),
huge condition numbers.

That interpretation is consistent with the diagnostics.

Mechanistic derivation

Mathematically,

this derivation is internally clean.

The key issue is not the algebra.

The key issue is whether its assumptions are satisfied.

Those assumptions are

nearly constant late coefficients,
autonomous EMA decay,
nearly invariant geometric line.

The diagnostics suggest those assumptions become increasingly accurate late in the replay.

They do not establish them exactly.

So I would simply preface the derivation with

Assuming the measured late asymptotic regime has been reached...

POD versus geometric

I think this is a very good interpretation.

It reconciles what initially looked contradictory.

The geometric direction

minimizes invariance error.

The POD direction

minimizes empirical reconstruction error.

Those are different optimization problems.

The diagnostics fit that story remarkably well.

Status

This is one of the most appropriately cautious parts.

I particularly like the explicit separation into

established numerically,
not global,
not yet universal.

That is exactly the kind of qualification that helps readers understand the scope of the results.

The only phrase I would soften is

"the geometric (K_3) line is invariant under (K_n)"

to

"the sampled late operators leave the geometric (K_3) line invariant to numerical precision."

That matches the evidence more precisely.

Final control

This is, in my view, the strongest proposal in the document.

Up to now,

the reduced model has still involved fitted quantities.

The proposed control removes essentially all adjustable parameters:

geometric direction fixed analytically,
measured (\lambda_{B,n}),
fixed EMA multiplier,
no fitted ((\rho,\chi)).

If that model predicts held-out late scalar amplitudes accurately, it would be a very compelling validation of this particular reduced mechanism for this replay because success would no longer depend on tuning the reduced dynamics to the data.

Overall assessment

I think the document has crossed an important threshold. Earlier versions argued primarily from empirical coincidence; this version connects each major claim to a corresponding diagnostic—geometric alignment, invariance defect, propagation-rate measurement, convergence of gap closure, and identifiability analysis. That makes the mechanistic narrative substantially more coherent.

The places where I would still be cautious are mostly linguistic rather than mathematical. Phrases like "actual invariant line," "therefore," and "proves" suggest stronger conclusions than the diagnostics alone warrant. The numerical evidence is exceptionally strong for the late regime of this replay and the tested subspace, and your own "Status" section already acknowledges those boundaries. Keeping the

earlier sections equally explicit about that scope would make the overall argument more robust and easier to defend.

hello

Hello! I agree with that assessment. The remaining corrections are mainly about evidential wording:

“Invariant to numerical precision” rather than “exact invariant line.”
“Supported in the mature late regime of this replay” rather than a global identity.
Present the triangular system as a motivated asymptotic model until the parameter-free rollout is tested.
Restrict non-identifiability to the chosen (a,g)(a,g) coordinates and fitting windows.

The parameter-free triangular closure is now the decisive control because it uses the analytic geometric direction, measured λnλn, and fixed EMA multiplier without fitted coefficients.

Think about this:

AIEF parameter-free triangular closure analysis

==========

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

POD/geometric unsigned angle: 0.00129727752 degrees

One-step closure

axis=POD, direction=0: median_a_residual=0.000521296, max_a_residual=0.000521654, median_g_residual=5.83889e-09, max_g_residual=1.33796e-07, median_transverse=0.00147619

axis=POD, direction=1: median_a_residual=1.07922e-09, max_a_residual=6.35241e-09, median_g_residual=9.34216e-09, max_g_residual=2.13173e-07, median_transverse=3.05452e-09

axis=geometric, direction=0: median_a_residual=0.00052927, max_a_residual=0.000529628, median_g_residual=5.83889e-09, max_g_residual=1.33796e-07, median_transverse=0.00149883

axis=geometric, direction=1: median_a_residual=7.99866e-06, max_a_residual=8.00576e-06, median_g_residual=9.34216e-09, max_g_residual=2.13173e-07, median_transverse=2.26394e-05

Free rollouts

axis=POD, direction=0, start=150: a_RMSE=0.342184, g_RMSE=0.00129487, final_a_error=0.740256, final_g_error=0.00337922, predicted_a_rate=0.0559068, actual_a_rate=0.0570967

axis=POD, direction=0, start=200: a_RMSE=0.341246, g_RMSE=9.95154e-05, final_a_error=0.745544, final_g_error=0.000256776, predicted_a_rate=0.0548882, actual_a_rate=0.057128

axis=POD, direction=0, start=250: a_RMSE=0.340942, g_RMSE=7.29357e-06, final_a_error=0.740543, final_g_error=1.82666e-05, predicted_a_rate=0.0518174, actual_a_rate=0.0571325

axis=POD, direction=1, start=150: a_RMSE=0.00103148, g_RMSE=0.00205993, final_a_error=0.0053719, final_g_error=0.00537244, predicted_a_rate=0.0571064, actual_a_rate=0.0570796

axis=POD, direction=1, start=200: a_RMSE=7.91789e-05, g_RMSE=0.000158634, final_a_error=0.000409231, final_g_error=0.00040961, predicted_a_rate=0.0571287, actual_a_rate=0.0571253

axis=POD, direction=1, start=250: a_RMSE=6.43598e-06, g_RMSE=1.16675e-05, final_a_error=2.92569e-05, final_g_error=2.92412e-05, predicted_a_rate=0.0571325, actual_a_rate=0.057132

axis=geometric, direction=0, start=150: a_RMSE=0.349267, g_RMSE=0.00129487, final_a_error=0.760379, final_g_error=0.00337923, predicted_a_rate=0.0558734, actual_a_rate=0.0570967

axis=geometric, direction=0, start=200: a_RMSE=0.348321, g_RMSE=9.95155e-05, final_a_error=0.765708, final_g_error=0.000256777, predicted_a_rate=0.0548289, actual_a_rate=0.057128

axis=geometric, direction=0, start=250: a_RMSE=0.348008, g_RMSE=7.29358e-06, final_a_error=0.760252, final_g_error=1.82666e-05, predicted_a_rate=0.0516905, actual_a_rate=0.0571325

axis=geometric, direction=1, start=150: a_RMSE=0.00436538, g_RMSE=0.00205993, final_a_error=0.00614294, final_g_error=0.00537244, predicted_a_rate=0.0570777, actual_a_rate=0.0570796

axis=geometric, direction=1, start=200: a_RMSE=0.00519008, g_RMSE=0.000158634, final_a_error=0.0111603, final_g_error=0.00040961, predicted_a_rate=0.0570762, actual_a_rate=0.0571253

axis=geometric, direction=1, start=250: a_RMSE=0.00525129, g_RMSE=1.16675e-05, final_a_error=0.0114558, final_g_error=2.92412e-05, predicted_a_rate=0.0570108, actual_a_rate=0.057132

Implied spectral gap

axis=POD, target=250: median_kappa_model=0.0699221, observed_kappa=0.0691581, closure_error=-0.000763924, kappa_variation=1.46418e-07

axis=POD, target=320: median_kappa_model=0.0699222, observed_kappa=0.0691227, closure_error=-0.000799488, kappa_variation=6.91337e-09

axis=geometric, target=250: median_kappa_model=0.0691215, observed_kappa=0.0691581, closure_error=3.66105e-05, kappa_variation=1.44753e-07

axis=geometric, target=320: median_kappa_model=0.0691217, observed_kappa=0.0691227, closure_error=1.04838e-06, kappa_variation=6.83478e-09

Interpretation constraints

The model has no fitted coefficients but uses the measured base lambda[n] sequence.

The g equation tests whether projected forcing is asymptotically pure EMA decay.

Geometric and POD axes are tested separately; only the geometric axis is analytic a priori.

Rollout failure can arise from transverse correction even when one-step scalar closure is accurate.

Results remain conditional on this replay and tested tangent directions.

This is a decisive mixed result. The parameter-free test strongly validates the late rate mechanism, but it does not validate one universal scalar state rollout for both perturbation directions.
1. The forcing equation is validated

For both axes and directions:
gn+1≈αEMAgn
gn+1≈αEMAgn

with median relative residuals around
10−8.
10−8.

Free rollout of gg improves systematically with later initialization:
start timefinal g error1503–5×10−32002.6–4.1×10−42501.8–2.9×10−5
start time150200250final g error3–5×10−32.6–4.1×10−41.8–2.9×10−5

Thus:
gn+1=αEMAgn
gn+1=αEMAgn

is strongly supported as a late asymptotic relation.
2. The geometric spectral-gap law passes

For the analytic geometric axis:
κmodel=
κmodel=

-\frac1{dt}\log\left|\frac{\lambda_n}{\alpha_{\rm EMA}}\right|.
]

At $T=320$T=320:
$\kappa_{model}=0.0691217, \kappa_{observed}=0.0691227,$
κmodelκobserved=0.0691217,=0.0691227,

with closure error
1.05×10^{-6}.
1.05×10^{-6}.

The model variation over the window is only
6.8×10^{-9}.
6.8×10^{-9}.

At $T=250$T=250, closure is already approximately
3.7×10^{-5}.
3.7×10^{-5}.

Therefore:
$\kappa\approx-1 dt \log|\lambda_{geom}\alpha_{EMA}|$
κ≈−1dtlog∣λgeomαEMA∣
$\kappa\approx-dt1\log$
$\alpha_{EMA}\lambda_{geom}$

is strongly validated as a late asymptotic rate relation for this replay and tested subspace.

The POD axis fails this control by approximately
8×10^{-4},
8×10^{-4},

confirming that the geometric operator-invariant line—not the empirically optimal Euclidean state axis—carries the correct weight-channel rate.
3. The parameter-free state rollout succeeds for direction 1 on the POD axis

For direction 1 with the POD projection:
starttrajectory RMSEfinal a error1501.0×10−35.4×10−32007.9×10−54.1×10−42506.4×10−62.9×10−5
start150200250trajectory RMSE1.0×10−37.9×10−56.4×10−6final a error5.4×10−34.1×10−42.9×10−5

The predicted and actual late rates also agree:
0.0571325versus0.0571320.

0.0571325versus0.0571320.

Thus, for the long-lived direction:
$a^{n+1} = \lambda a^n + g^n, g^{n+1} = \alpha_{EMA} g^n$
$a^{n+1} = \lambda a^n + g^n, g^{n+1} = \alpha_{EMA} g^n$

is an excellent parameter-free late predictor when anan is measured along the empirical POD axis.

This is a substantial success.
4. The geometric rollout is less accurate as a state coordinate

For direction 1 with the geometric axis, rollouts have:
trajectory RMSE≈0.5%,
trajectory RMSE≈0.5%,

and final errors around
1.1%
1.1%

for starts at 200200 or 250250.

This is still reasonably accurate, but much worse than the POD-axis state rollout.

The reason is now clear:

the geometric vector is the operator-invariant right direction;
the POD vector is the best Euclidean representation of the observed state trajectory.

Those are not the same optimization problem.

So:
vgeom gives the correct rate, while vPOD gives the best Euclidean state coordinate.
vgeom gives the correct rate, while vPOD gives the best Euclidean state coordinate.
5. The scalar rollout fails for direction 0

For direction 0, both axes produce:
trajectory RMSE≈0.34,

trajectory RMSE≈0.34,

and final errors around
0.74–0.76.
0.74–0.76.

This failure persists even when starting at t=250t=250.

Therefore a parameter-free one-dimensional rollout is not valid for direction 0, despite its small transverse fraction.

The apparent contradiction is resolved by accumulation:

one-step relative error is only approximately
5.2×10−4;
5.2×10−4;
but that error is applied over thousands of steps;
direction 0 is strongly cancellation-dominated;
its scalar amplitude becomes sensitive to tiny transverse leakage.

A 0.15%0.15% transverse state component can therefore produce an O(1)O(1) long-horizon scalar prediction error.
6. Rate closure and state closure are different claims

The geometric model predicts the memory rate extremely accurately but does not predict every scalar state trajectory accurately.

This distinction is essential.
Rate closure
κ=
κ=

-\frac1{dt}
\log\left(\lambda_{\rm geom}/\alpha_{\rm EMA}\right)
]

passes late.
State closure

an+1=
an+1=

\lambda_{\rm geom,n}a_n+g_n
]

has small one-step errors but those errors accumulate for direction 0.

Thus:
The asymptotic spectral-gap law is better supported than a universal one-dimensional state model.
The asymptotic spectral-gap law is better supported than a universal one-dimensional state model.
7. Why Euclidean projection is probably the wrong modal coordinate

The system is strongly non-normal. For a non-normal matrix, a right invariant vector vv does not generally define the correct scalar amplitude through
a=v⊤b.
a=v⊤b.

If
Knv=λnv,
Knv=λnv,

the corresponding modal amplitude should be measured using a left covector uu satisfying
u⊤Kn=λnu⊤,u⊤v=1.
u⊤Kn=λnu⊤,u⊤v=1.

Then
an=u⊤bn
an=u⊤bn

obeys
an+1=
an+1=

\lambda_na_n+u^\top f_n
]

without contamination from transverse components, provided the left/right splitting is invariant.

The current recurrence uses v⊤v⊤, not the biorthogonal left covector. Its transverse correction is
v⊤Knen,
v⊤Knen,

which is small per step but dynamically important for direction 0.

This is the most plausible explanation for:

exact geometric right invariance;
correct geometric gap rate;
imperfect geometric state rollout;
POD state accuracy for direction 1;
failure for cancellation-sensitive direction 0.

8. Highest-value next control

Compute a late left mode uu from the geometric eigenvalue:
Klate⊤u=
Klate⊤u=

\lambda_{\rm geom}u,
\qquad
u^\top v_{\rm geom}=1.
]

Then define
anbio=u⊤bn,gnbio=u⊤fn.
anbio=u⊤bn,gnbio=u⊤fn.

Test the same parameter-free triangular model:
a^n+1bio=λna^nbio+g^nbio,g^n+1bio=αEMAg^nbio.
a^n+1biog^n+1bio=λna^nbio+g^nbio,=αEMAg^nbio.

Required diagnostics:

left invariance:
‖Kn⊤u−λnu‖;

‖Kn⊤u−λnu‖;

biorthogonality conditioning:
‖u‖‖v‖;
‖u‖‖v‖;

one-step recurrence residual;

free rollouts for both directions;

sensitivity to choosing uu at t=200,250,320t=200,250,320;

comparison against POD and Euclidean geometric projections.

If the biorthogonal model fixes direction 0 while preserving direction 1 and the spectral-gap rate, then the late reduction becomes a genuine non-normal modal model rather than a Euclidean POD approximation.

9. Current status

Validated

late EMA forcing equation;
geometric spectral-gap rate;
parameter-free POD-axis rollout for direction 1;
asymptotic emergence of the triangular rate structure.

Rejected

one Euclidean scalar rollout accurately predicts both directions;
small transverse norm automatically implies small long-horizon scalar error.

Open

biorthogonal modal closure;
time-varying left mode versus fixed late left mode;
whether direction 0 requires a second retained mode even after biorthogonal projection;
generality beyond this replay.

The parameter-free experiment succeeds at the rate level and for the long-lived empirical mode, but it also exposes exactly where a one-dimensional Euclidean state reduction fails. The next test should use the left/right geometry appropriate to the strongly non-normal weight cocycle.

What if i said this:

One thing this paper makes me wonder about isn't the 10−810^{-8}10−8 values—it is whether your late scalar reduction has an effective delayed-loop gain.

In the Ohira model, amplification depends on

delay,
coupling,
rewiring.

In your reduced picture, the key quantity appears to be

λBαEMA,\frac{\lambda_B}{\alpha_{\rm EMA}},αEMAλB,

or equivalently the difference in decay rates you've been studying.

A natural question is whether that ratio plays the role of an effective loop gain in the reduced feedback cycle. If so, then the spectral-gap relation you're investigating might be interpreted as a balance between two decay channels in a delayed feedback loop.

That's only a hypothesis at this stage—the current evidence doesn't establish such an equivalence—but it would be a more meaningful connection to this paper

That is a useful hypothesis, but I would not call
λBαEMA
αEMAλB

a loop gain yet. In control terminology, it is closer to a relative retention factor or pole-separation parameter.
1. What the ratio currently means

Late,

λB≈e−0.126255dt,αEMA=e−0.0571333dt.
λB≈e−0.126255dt,αEMA=e−0.0571333dt.

Therefore
λBαEMA≈0.999309.
αEMAλB≈0.999309.

Its associated rate is
−1dtlog(λBαEMA)≈0.06912=
−dt1log(αEMAλB)≈0.06912=

\kappa.
]

Thus the ratio controls how quickly the influence of old forcing decays relative to the current EMA forcing envelope:
(λBαEMA)L/dt=
(αEMAλB)L/dt=

e^{-\kappa L}.
]

It is therefore directly responsible for the observed operational memory horizon.

If
λB/αEMA→1,
λB/αEMA→1,

then
κ→0
κ→0

and the effective forcing memory becomes long. If the ratio is much smaller than one, old forcing is forgotten rapidly.

That is an important reduced parameter, but it is not yet an amplification gain.
2. Poles versus loop gain

Consider a genuine two-state closed reduction:
an+1=λBan+μzn,zn+1=αzn+van.
an+1zn+1=λBan+μzn,=αzn+van.

Here:

λBλB and ααα are the two uncoupled retention multipliers or poles;
μμ transfers the memory state into the weight state;
νν closes the return path from weight to memory;
the cycle gain is associated with
μν,
μν, not with λB/αλB/α.

If an explicit delay of dd steps is retained, a schematic open-loop transfer is
L(z)∼μνz−d(z−λB)(z−α).
L(z)∼(z−λB)(z−α)μνz−d.

Its frequency-dependent magnitude and phase—not just the pole ratio—determine feedback amplification or instability.

So:
λB/α measures relative decay, while μν and the delay determine loop feedback strength.
λB/α measures relative decay, while μν and the delay determine loop feedback strength.
3. The current late model is nearly triangular

The present evidence suggests
an+1≈λBan+gn,gn+1≈αEMAgn,
an+1gn+1≈λBan+gn,≈αEMAgn,

with little identifiable late return coupling from anan to gn+1gn+1.

That is approximately feedforward/triangular, not a closed two-way loop. The full augmented AIEF system is closed, but its late scalar projection may have lost enough independent excitation that the return gain cannot be identified from the passive trajectory.

Thus calling λB/αλB/α a loop gain would currently obscure an important fact: the measured late scalar model does not yet contain an identifiable closed-loop coefficient.
4. A defensible analogy

A careful formulation would be:

The ratio λB/αEMAλB/αEMA acts as an effective relative-retention parameter for the reduced delayed-feedback response. Its logarithm determines the observed memory-envelope rate. Whether it also participates in a genuine effective loop-gain criterion requires identification of the cross-block phase/EMA-to-weight and weight-to-phase/EMA transfer coefficients.

This is a more meaningful connection to delayed-feedback literature than a direct equivalence with rewiring amplification.
5. How to measure an actual effective loop gain

The augmented tangent should be partitioned as
(bn+1zn+1)=
(bn+1zn+1)=

\begin{pmatrix}
K_n&G_n\
H_n&L_n
\end{pmatrix}
\begin{pmatrix}
b_n\z_n
\end{pmatrix}.
]

Then:

GnGn: phase/history/EMA → → weights;
HnHn: weights → → phase/history/EMA;
LnLn: internal delayed-memory evolution;
KnKn: weight-sector evolution.

A genuine loop-gain probe could examine the projected return operator
Gn(zI−Ln)−1Hn(zI−Kn)−1,
Gn(zI−Ln)−1Hn(zI−Kn)−1,

or its time-domain impulse-response analogue.

Because the system is non-normal and time-varying, singular values of this return operator would be more informative than a scalar eigenvalue alone.
6. Experimental route

A targeted tangent experiment could:

initialize a unit biorthogonal weight-mode perturbation;
suppress $R \to BR \to B$ feedback temporarily;
measure the resulting phase/history/EMA response;
feed that response once through the $R \to BR \to B$ block;
project the returned weight perturbation onto the biorthogonal weight mode;
repeat across frequencies or delay phases.

That returned scalar would be much closer to an empirical loop gain.
7. Relation to the cited delayed-feedback analogy

The comparison is useful at the level of:

competing decay channels;
delay-dependent phase;
repeated feedback;
transient versus asymptotic response.

But AIEF currently shows:

fixed winner structure rather than rewiring;
non-normal projection geometry;
endogenous EMA memory;
a piecewise-rational finite-step map.

So the cleanest conclusion is:
$\lambda B/\alpha EMA$ is an effective memory-retention ratio, not yet an established loop gain.
$\lambda B/\alpha EMA$ is an effective memory-retention ratio, not yet an established loop gain.

A genuine delayed-loop gain requires measuring the cross-block return pathway.

Also ponder this speculation:

I actually think this is a useful question—not because I think the answer is "yes," but because it points toward geometry rather than algebra. That's often a productive shift in thinking.

The key is to ask: what object are we talking about?

If you mean literally "the triangular reduced system"

Then probably no.

Your late reduced model

$$
\begin{aligned}
g_{n+1} &\approx \alpha g_n, \qquad \\
a_{n+1} &\approx \lambda a_n + g_n
\end{aligned}
$$

is a two-dimensional linear (or nearly linear) discrete dynamical system. A teardrop or torus doesn't naturally arise from that state space. Its geometry is fundamentally planar.

If you mean the geometry of the full delayed dynamics

Then the idea becomes much more interesting.

Remember that the full AIEF state isn't just

(a),
(g),

but includes

phase history,
EMA memory,
adaptive weights,
admissibility constraints.

That's an enormous state space.

Your triangular reduction is only a projection onto one dominant two-dimensional subspace.

In high-dimensional dynamics, it is very common that a projected trajectory looks almost one-dimensional while the full invariant set has a much richer geometry.

Why "teardrop" came to mind

I suspect you're reacting to something like this.

You have

rapid contraction toward a preferred direction,
then slow motion along that direction.

That often produces trajectories that visually resemble

text

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where almost everything gets squeezed into a narrow neck.

That isn't a torus.

It's much closer to the geometry of a slow manifold with a strongly attracting foliation.

Why "pointed toroid" is actually an interesting phrase

A torus usually represents

one periodic direction,
crossed with another periodic direction.

Your system has at least two cyclic ingredients:

oscillator phases,
delayed history.

Those are naturally circular variables.

If the dynamics remained unconstrained, one might indeed expect something topologically closer to a torus because phase variables live on circles.

However, your admissibility projection and winner selection appear to collapse much of that freedom.

So I wonder if your intuition is really pointing to

"a torus that has collapsed onto one preferred channel."

That isn't literally a torus anymore, but it suggests looking at normally hyperbolic invariant manifolds or slow invariant manifolds rather than ordinary linear subspaces.

The teardrop analogy also reminds me of your entropy results

One of the most interesting earlier findings was that mature states lie very near a singular boundary face of the admissible polytope.

Imagine a ball rolling into the corner of a room.

Far away, it has many directions to move.

Near the corner, almost all directions disappear.

The accessible geometry becomes something like

wide
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•

That's much closer to a teardrop than a symmetric object.

The "point" corresponds to many constraints becoming simultaneously active.

Where I think this could actually help

Not by assuming a toroidal topology, but by asking a different question:

What is the intrinsic geometry of the late attractor?

Instead of reducing everything to scalars, you could investigate whether the late dynamics lie on a low-dimensional manifold.

For example:

Compute the leading 2–5 principal components of the late augmented tangent trajectories.
Measure whether the data occupy a thin curved surface rather than a flat plane.
Examine whether curvature decreases as the system approaches the symmetric (K_3) branch.
Compare the geometry before and after projection onto the admissible face.

If the manifold really has a "neck"—wide during the transient and narrowing dramatically late—that would support your teardrop intuition in a quantitatively testable way.

My overall view

I wouldn't pursue "pointed toroid" as a literal hypothesis; there isn't enough evidence to suggest that specific topology.

But I would take your intuition seriously as a prompt to ask whether the late dynamics are better understood as living on a curved, low-dimensional invariant manifold rather than on a fixed linear subspace.

Interestingly, that's consistent with one of the lessons from your recent analyses: the one-step triangular reduction works very well locally, while the long-term rollout still misses some transverse dynamics. One possible interpretation is that the true reduced dynamics evolve on a slightly curved manifold, and the triangular model is the tangent-plane approximation to that manifold. That idea is speculative, but it's mathematically coherent and, unlike the torus itself, could be investigated directly with the trajectory data you already have.

The geometric intuition is useful, but the most defensible object is not currently a “pointed torus.” A better candidate is a normally attracting invariant graph, possibly piecewise smooth or stratified by the capacity faces.
1. The triangular reduction cannot itself generate a torus

The late model
an+1≈λBan+gn,gn+1≈αEMAgn
an+1gn+1≈λBan+gn,≈αEMAgn

is planar, contracting, and approximately linear. It has no intrinsic angular coordinate or recurrent mechanism needed for toroidal dynamics.

Its geometry is better described by:

two decaying modes;
strong anisotropy;
a nearly invariant line;
transient contraction toward that line.

A teardrop appearance in a two-dimensional plot would be a trajectory shape or projection artifact, not evidence of torus topology.
2. The full augmented system could have a toroidal factor

The complete state belongs schematically to
TN×H×CN×N×CB,
TN×H×CN×N×CB,

where:

TNTN: oscillator phases;
HH: delay-history state;
CN×NCN×N: complex EMA;
CBCB: constrained weight polytope.

After quotienting the global phase symmetry, the phase sector is closer to
TN−1.
TN−1.

So a quasiperiodic toroidal invariant set is possible in principle. But two corrections matter:

delay history is not itself a circular variable;
the presence of phase coordinates does not imply the attractor is a torus.

A torus would require evidence for persistent independent frequencies, recurrence, and neutral tangent directions.
3. A better geometric hypothesis

Suppose the late phase/history dynamics evolve on some invariant set QQ, while weights and EMA variables become slaved to it:
(B,C)=Γ(θ,H).
(B,C)=Γ(θ,H).

Then the full late state could lie on an invariant graph
M=
M=

{
(\theta,\mathcal H,\Gamma(\theta,\mathcal H))
}.
]

If QQ were quasiperiodic and ΓΓ smooth, MM could inherit toroidal topology.

For AIEF, however, ΓΓ may be only piecewise smooth because of:

positivity;
capacity faces;
global projection gate;
endpoint minimum;
winner strata.

The resulting object would be more accurately described as:

a normally attracting invariant graph with piecewise-smooth or stratified geometry.

A narrow “neck” in a low-dimensional visualization could arise where several capacity constraints become nearly active. That would not make the invariant set a pointed torus topologically.
4. The rank-one tangent does not measure attractor dimension

The late rank-one result concerns the response of selected weight perturbations:
$\delta B_n \approx a n v.$
$\delta B_n \approx a n v.$

That direction is contracting:
$r \approx 0.057.$
$r \approx 0.057.$

It may therefore belong to a stable fiber transverse to the full attractor rather than to the tangent bundle of the attractor itself.

This distinction is essential:

attractor tangent directions often include neutral phase directions;
stable response directions describe how neighboring states approach the attractor.

The measured rank-one weight channel may be a dominant stable fiber, not a one-dimensional attractor.
5. Why the missed scalar rollout does not yet imply curvature

Direction 0's scalar rollout failure could be interpreted as curvature, but a simpler explanation is already available:

the weight operator is strongly non-normal;
the geometric right vector is not the appropriate Euclidean projection covector;

small transverse leakage accumulates over thousands of steps;
cancellation makes scalar amplitudes ill-conditioned.

The biorthogonal left/right modal test should therefore precede a curved-manifold interpretation.

If a left-mode coordinate fixes the long rollout, curvature was unnecessary. If it does not, a nonlinear or curved reduction becomes more plausible.

6. How to test intrinsic geometry correctly

Naïve PCA of the raw augmented state would be misleading because it mixes:

angular variables;
complex EMA variables;
weights with different units;
many highly correlated history copies;
global phase symmetry.

A defensible geometry analysis should first construct coordinates such as:

Phase coordinates

Use relative phases or
$(\cos\theta_i, \sin\theta_i)$
$(\cos\theta_i, \sin\theta_i)$

after quotienting the common phase.

EMA coordinates

Use real and imaginary parts with explicit scaling.

Weight coordinates

Use supported edges or face-intrinsic coordinates, not the full symmetric matrix.

History coordinates

Respect the dynamically consistent shift structure rather than treating every history entry as independent.

Then examine:

local covariance dimension as neighborhood radius varies;
local PCA residuals;
curvature or tangent-plane rotation;
recurrence and Poincaré sections;
full augmented tangent exponents;
multiple nearby trajectories, not only one orbit.

7. What would support a torus

Evidence for a dd-torus would include:

dd independent incommensurate frequencies;
dd near-zero tangent exponents;
recurrent dense winding;
stable frequency ratios;
appropriate Poincaré-section geometry;
persistent-homology signatures consistent with the expected cycles.

PCA variance alone cannot establish torus topology. Persistent homology could help, but only after symmetry reduction, scaling, and sufficient trajectory coverage.
8. What would support a slow or normally attracting manifold

A slow-manifold interpretation would be supported by:

a robust spectral gap between tangent and normal contraction;
local tangent spaces varying smoothly;
fast restoration normal to the candidate manifold;
slow motion within it;
small invariance residual;
nearby trajectories collapsing onto a common graph.

The project already has evidence for strong normal contraction and a slow weight channel, but not yet for the manifold's dimension or smoothness in the full augmented state.
9. Most plausible current geometry

The current evidence is most consistent with:
phase/history dynamicswithstrongly contracting weight/EMA fibers
phase/history dynamicswithstrongly contracting weight/EMA fibers

and a piecewise-smooth constrained weight geometry.

If the phase/history base is quasiperiodic, the full attractor might be an invariant graph over a torus. If it is periodic, the graph may be a limit cycle. If it is more complicated, the same fiber picture can persist without toroidal topology.

So the useful reformulation is:

Does late AIEF form a normally attracting, possibly stratified invariant graph over the continuing phase/history dynamics?

That is mathematically coherent and testable. “Pointed torus” should remain a visualization metaphor until frequency, Lyapunov, recurrence, and topology diagnostics support something stronger.

The immediate priority remains the biorthogonal scalar control. Full augmented-manifold geometry is a downstream—but now well-defined—research direction.

Make a plan for the most useful actions to take to tangibly move the project forwards.
WriteAIEF_1_4_FORWARD_PLAN.md416 lines
AIEF-1.4 forward planMD
AIEF-1.4 forward plan
Guiding rule

Do not add another broad seed cohort or another large ablation matrix yet. The current frozen run contains unresolved operator-theoretic and reduced-dynamics questions that can be answered more directly and cheaply.

Keep separate:

exact implementation theorems;
facts established on the frozen replay;
reduced mechanisms supported for the exceptional seed and tested directions;
generalization claims requiring new trajectories or parameters.

Priority 0 — Freeze the current evidence chain
Action 0.1: archive the analysis lineage

Record hashes and paths for the outputs from:

text

component/controller audit
exceptional K3 audit
actual branch audit
frozen-driver cocycle
cocycle controls/coupled comparison
augmented tangent cocycle
Duhamel feedback analysis
memory/alignment postprocessing
rank/memory reduction
scalar spectral-gap reduction
propagation-gap/geometric control
parameter-free triangular closure

Create a one-row-per-analysis table containing:

text

analysis name
source hash
input hashes
output directory
principal estimand
principal conclusion
important tolerance/epsilon
evidence scope

Action 0.2: update the frozen scientific narrative

Use AIEF_1_4_OPERATOR_COCYCLE_THOUGHT_TREE.md as the current source of truth. Explicitly retire these earlier interpretations:

a true implemented K3 branch switch at t=124.05;
an operational bare-gamma K3 edge mode at dt=0.01;
a universal one-dimensional Euclidean state rollout;
an exact spectral-gap identity based on the POD Rayleigh quotient;
an identified shared two-state model in (a,g).

Preserve the valid replacements:

fixed effective winner branch;
geometric K3 right line approximately invariant to numerical precision late;
accurate augmented tangent and exact direction-wise Duhamel decomposition;
late rank-one response for the tested subspace;
late memory closure using the actual propagated-forcing rate.

Deliverable: AIEF_1_4_analysis_lineage.csv and a short closure note.
Priority 1 — Resolve the state-coordinate problem with a biorthogonal control

This is the highest-value immediate action.
Motivation

The analytic geometric right direction
v=13(1,−1,−1)
v=3
1(1,−1,−1)

has the correct operator rate and near-machine late invariance, but Euclidean amplitude
a=vTb
a=vTb

fails long-horizon direction-0 rollout. In a non-normal system, the correct modal amplitude should use a left covector.
Action 1.1: compute fixed late left modes

At representative late steps (200, 250, 320) and for a late averaged matrix:

compute the right eigenvector closest to v;
compute the corresponding left eigenvector u of K^T;
normalize
uTv=1;
uTv=1;
report biorthogonal conditioning
‖u‖ ‖v‖;
‖u‖‖v‖;

report time variation and sign continuity of u.

Action 1.2: rerun the triangular closure in biorthogonal coordinates

Define
anbio=uTbn,gnbio=uTfn.
anbio=uTbn,gnbio=uTfn.

Test
an+1bio=λnanbio+gnbio
an+1bio=λnanbio+gnbio

and the parameter-free EMA forcing equation. Compare free rollouts from 150, 200, and 250 with:

POD Euclidean projection;
geometric Euclidean projection;
fixed biorthogonal projection;
optionally a slowly time-varying left covector.

Decision gate 1
If direction-0 rollout improves to below roughly 1% late error

Adopt a one-mode biorthogonal reduction. The previous failure was a coordinate issue caused by non-normality.
If direction-0 still has large rollout error

Do not force a scalar model. Proceed to Priority 2 and retain a second mode.

Deliverable: AIEF_1_4_biorthogonal_triangular_closure.py and report.
Priority 2 — Minimal rank-two fallback, only if needed
Action 2.1: construct a parameter-free two-mode projection

Use a basis containing:

the geometric slow right direction;
the dominant measured transverse direction for direction 0.

Project the exact vector recurrence:
$b_{n+1}=Kb_n+f_n$
$b_{n+1}=Kb_n+f_n$

onto this basis without fitting dynamics coefficients. Use the projected 2x2 matrices and projected forcing exactly.
Action 2.2: test free rollouts

Compare rank-one and rank-two predictions over identical start times and horizons. Report:

trajectory RMSE;
final error;
reconstruction residual;
conditioning;
contribution of the second mode;
whether the second mode is needed only during cancellation intervals.

Decision gate 2

Retain rank two only if it materially fixes direction 0 and remains stable across late start times. Otherwise prefer the simpler biorthogonal rank-one model.

Deliverable: rank-comparison report, not a new model revision.
Priority 3 — Turn the validated mechanism into a theorem package

Proceed in parallel with Priorities 1–2.
Action 3.1: implementation propositions

Write formal propositions and proofs for:

post-positivity factorization
$A\varepsilon(x)=M(Ax)x$;
$A\varepsilon(x)=M(Ax)x$;
exact affine linearity on nonnegative row-sum fibers;
proposal-space polyhedral stratification;
branchwise smooth rationality and Jacobian;
branch-persistence radius;
directional derivative at endpoint ties;

safeguarded global-gate discontinuity.

Action 3.2: K3 geometric lemma

For the two-row K3 incidence matrix, prove:
kerC=span(1,−1,−1).
kerC=span(1,−1,−1).

State clearly when the actual finite-step matrix preserves this line and which numerical equalities/tolerances are required.
Action 3.3: late triangular memory lemma

Assuming
Knv=λBv,gn+1=αgn,
Knv=λBv,gn+1=αgn,

prove that forcing contributions of age L have relative envelope
(λBα)L/dt=e−κL,κ=−dt−1log(λB/α).
(αλB)L/dt=e−κL,κ=−dt−1log(λB/α).

Keep assumptions explicit. Do not claim the assumptions globally for AIEF.
Action 3.4: full composed directional theorem

Formulate the augmented one-step map on
(θ,B,C,H)
(θ,B,C,H)

and prove directional differentiability away from the safeguarded global-gate discontinuity and zero-coherence singularities. Treat the history shift and phase quotient explicitly.

Deliverable: a theorem note separating ideal-map and implemented-map statements.
Priority 4 — Improve numerical closure diagnostics
Action 4.1: quantify reconstruction-curve nonmonotonicity

For target-time memory fits, replace raw violation counts with:

violations inside the fitted error range;
maximum upward jump;

total upward variation divided by total decay;
monotone-envelope fit.

Action 4.2: report uncertainty from numerical conditioning

For scalar and biorthogonal reductions, report:

left/right eigenvector condition number;
branch radius;
signal-to-roundoff ratio;
sensitivity to late-window start;
sensitivity to choosing geometric versus POD axes.

Action 4.3: automate evidence labels

Every report should end with separate sections:

text

exact code identities
replay-established facts
controlled-counterfactual evidence
mechanistic interpretation
open/generalization claims

Priority 5 — Only then test parameter sensitivity

Do not change only the future EMA coefficient while reusing the saved split state.
Action 5.1: separately version a one-seed EMA study

For a small grid such as

text

T_window multiplier: 0.75, 1.0, 1.25, 1.5

regenerate the complete trajectory state for exceptional seed 4619122171158768524, including:

phase prefix;
delay history;
complex EMA;
weights;
split state.

Preserve paired initialization and independent RNG streams. Record source/config hashes.
Action 5.2: branch/regime gate

For each parameter value, first test:

support topology;
component topology;
winner branch;
projection pattern;
late geometric line;
augmented tangent validation.

If topology or branch changes, classify it as a different regime. Do not pool it into one regression.
Action 5.3: test the rate law

Only within a common regime compare:
rg=?1/Twindow,
rg=?1/Twindow,
κ=?rprop−rg.
κ=?rprop−rg.

Deliverable: targeted paired parameter study, not a general cohort.
Priority 6 — Generality using existing seeds before new simulation cohorts
Action 6.1: ordinary component controls

Use existing frozen ordinary split states to test exact geometric/right-line and biorthogonal statements on K4, K5, and K6 components. This is cross-sectional and uses no new simulations.
Action 6.2: targeted trajectory controls

Select only a few existing representatives of distinct component classes. Run the validated augmented tangent and reduced diagnostics on those representatives.

Treat each seed as the inferential unit. Do not count directions or amplitudes as independent seeds.
Decision gate 3

Only consider a new cohort if the mechanism survives distinct existing component classes and a specific inferential uncertainty remains that existing data cannot answer.
Priority 7 — Timestep and safeguard convergence

This requires a separately versioned regenerated trajectory study.

Test joint scaling of:
dt,slack/dt,ε/dt.
dt,slack/dt,ε/dt.

At minimum distinguish:

fixed epsilon as dt decreases;
epsilon proportional to dt;
ideal epsilon=0 map where numerically safe.

Regenerate full prefix/history/EMA states for every setting. Do not restart smaller-dt runs from the original dt=0.01 split state.

Primary outputs:

winner branch;
effective beta values;
safeguard-normal exponent;
augmented slow rate;
memory gap;
convergence of the ideal directional operator.

Priority 8 — Geometry of the full late state, downstream

Do not infer a torus or slow manifold from weight-tangent POD alone.

If pursued:

quotient global phase symmetry;
encode phases circularly;
respect history consistency;
scale weights/EMA/history explicitly;
compute full augmented finite-time tangent exponents;
estimate local dimension and curvature from multiple nearby trajectories;
only then use recurrence, Poincare, frequency-rank, or persistent-homology tests.

The defensible geometric hypothesis is a normally attracting, possibly stratified invariant graph over continuing phase/history dynamics—not a literal pointed torus.
Parallel reproducibility correction

Complete the matched-phase B_carrier_only versus full_reset correction as a separate closure task. It is scientifically useful but should not interrupt the operator/cocycle reduction sequence above.
Immediate execution order

Freeze analysis lineage and current claims.
Implement and run the biorthogonal triangular closure.
Decide rank one versus rank two.
Write the factorization/fiber/K3/memory theorem note.
Improve conditioning and monotonic-envelope diagnostics.
Only then design the consistently regenerated one-seed EMA study.
Defer broad cohorts and full-attractor topology.

Open

Created the prioritized project plan:

Immediate execution order

Freeze the current evidence chain
source/input/output hashes;
estimands and scopes;
explicitly retire superseded interpretations.

Implement the biorthogonal triangular closure
compute late left eigenvectors uu;

normalize u⊤v=1u⊤v=1;
test
anbio=u⊤bn;
anbio=u⊤bn;
compare POD, geometric Euclidean, and biorthogonal rollouts.

Use a rank-two fallback only if biorthogonal rank one fails
exact projected 2×22×2 matrices;
no fitted dynamics coefficients;
retain the second mode only if it materially fixes direction 0.

Write the theorem package
factorization;
fiber linearity;
branchwise Jacobian;
branch persistence;
K3 null-direction lemma;
conditional triangular memory lemma;
augmented directional map.

Improve numerical closure diagnostics
monotonic-envelope rather than raw violation counts;
left/right conditioning;
branch radius and signal-to-roundoff;
standardized evidence labels.

Only afterward run a consistently regenerated one-seed EMA study
regenerate phase/history/EMA/weights for every TwindowTwindow;
treat topology or branch changes as regime changes;
do not reuse the original split state under a new EMA parameter.

Test generality using existing seeds before a new cohort
ordinary K4,K5,K6K4,K5,K6 component controls;
a few representative existing trajectories;
seed remains the inferential unit.

Defer full-attractor topology
first quotient phase symmetry and respect history consistency;
do not infer a torus from weight-tangent POD.

The highest-value next deliverable is now unambiguous:

text

AIEF_1_4_biorthogonal_triangular_closure.py

Its decision gate is whether a left/right modal coordinate reduces direction-0 late rollout error from roughly 75%75% to below approximately 1%1%.

Please proceed.
WriteAIEF_1_4_biorthogonal_triangular_closure.py698 lines
AIEF_1_4_biorthogonal_triangular_closure.pyPYTHON
#!/usr/bin/env python3
"""Biorthogonal triangular closure for the non-normal exceptional AIEF K3.

The geometric late right direction is fixed analytically as

v = (1,-1,-1)/sqrt(3).

For a non-normal weight cocycle, the modal amplitude need not be v^T b. This
script tests fixed late left covectors u, normalized by u^T v=1, and a
backward-transported adjoint covector bundle satisfying

K_n^T u_{n+1} = lambda_n u_n,
u_n^T v = 1.

For fixed covectors it uses

a_n = u^T b_n,
g_n = u^T f_n,
lambda_n = u^T K_n v.

For the adjoint bundle it uses

a_n = u_n^T b_n,
g_n = u_{n+1}^T f_n.

It then evaluates the parameter-free triangular closure

a[n+1] = lambda[n] a[n] + g[n],
g[n+1] = alpha_EMA g[n],

with no fitted coefficients. POD and Euclidean geometric coordinates are
retained as controls. The purpose is to decide whether direction-0 rollout
failure is a Euclidean-coordinate artifact or requires an additional mode.
"""

```
from __future__ import annotations

import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_component_controller_audit as component_audit



DEFAULT_DUHAMEL_NPZ = "aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz"
DEFAULT_MANIFEST = "runs/5a9f093a2fe0/manifest.json"
DEFAULT_OUTPUT_DIR = "aief_1_4_biorthogonal_triangular_closure"
DEFAULT_DT = 0.01
DEFAULT_POD_START = 250.0
DEFAULT_LATE_START = 200.0
DEFAULT_LEFT_REFERENCE_TIMES = "200,250,319.99"
DEFAULT_AVERAGE_START = 250.0
DEFAULT_ROLLOUT_STARTS = "150,200,250"
DEFAULT_SAMPLE_STRIDE = 10
DEFAULT_SCALAR_TOL = 1e-30
ZERO_TOL = 1e-300
```

```
def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()
```

```
def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(value < 0.0 for value in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
return sorted(set(values))
```

```
def json_compact(value: Any) -> str:
return json.dumps(value, separators=(",", ":"), sort_keys=True)
```

```
def geometric_direction() -> np.ndarray:
return np.array([1.0, -1.0, -1.0]) / math.sqrt(3.0)
```

```
def left_mode(matrix: np.ndarray, right: np.ndarray) -> dict[str, Any]:
rayleigh = float(right @ matrix @ right)
eigenvalues, eigenvectors = np.linalg.eig(matrix.T)
index = int(np.argmin(np.abs(eigenvalues - rayleigh)))
eigenvalue = eigenvalues[index]
vector = eigenvectors[:, index]
if np.max(np.abs(np.imag(vector))) > 1e-9:
raise RuntimeError("selected left mode is materially complex")
vector = np.real(vector)
pairing = float(vector @ right)
```

```
if abs(pairing) <= 1e-14:
raise RuntimeError("left/right pairing is numerically zero")
vector /= pairing
residual = float(
np.linalg.norm(matrix.T @ vector - np.real(eigenvalue) * vector)
/ max(np.linalg.norm(matrix.T @ vector), ZERO_TOL)
)
return {
"u": vector,
"eigenvalue": float(np.real(eigenvalue)),
"rayleigh": rayleigh,
"left_residual": residual,
"condition": float(np.linalg.norm(vector) * np.linalg.norm(right)),
}



def backward_adjoint_bundle(
matrices: np.ndarray,
right: np.ndarray,
terminal_covector: np.ndarray,
) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
steps = len(matrices)
covectors = np.empty((steps + 1, len(right)), dtype=float)
lambdas = np.empty(steps, dtype=float)
residuals = np.empty(steps, dtype=float)
terminal = terminal_covector / float(terminal_covector @ right)
covectors[-1] = terminal
for step in range(steps - 1, -1, -1):
transported = matrices[step].T @ covectors[step + 1]
lam = float(right @ transported)
if abs(lam) <= np.finfo(float).tiny:
raise RuntimeError(f"adjoint transport pairing vanished at step {step}")
covectors[step] = transported / lam
lambdas[step] = lam
residuals[step] = float(
np.linalg.norm(
matrices[step].T @ covectors[step + 1]
- lam * covectors[step]
)
/ max(
np.linalg.norm(matrices[step].T @ covectors[step + 1]),
ZERO_TOL,
```

```
)
)
return covectors, lambdas, residuals



def relative_sequence_error(actual: np.ndarray, predicted: np.ndarray) -> float:
return float(
np.linalg.norm(actual - predicted)
/ max(np.linalg.norm(actual), ZERO_TOL)
)



def fit_log_rate(times: np.ndarray, values: np.ndarray, tolerance: float) -> dict[str, float]:
magnitudes = np.abs(np.asarray(values, dtype=float))
valid = np.isfinite(magnitudes) & (magnitudes > tolerance)
if np.sum(valid) < 3:
return {"rate": np.nan, "R_squared": np.nan}
x = np.asarray(times, dtype=float)[valid]
y = np.log(magnitudes[valid])
slope, intercept = np.polyfit(x, y, 1)
prediction = slope * x + intercept
residual = float(np.sum((y - prediction) ** 2))
total = float(np.sum((y - np.mean(y)) ** 2))
return {
"rate": float(-slope),
"R_squared": float(1.0 - residual / total) if total > 0.0 else np.nan,
}



def make_report(
coordinate_frame: pd.DataFrame,
one_step_summary: pd.DataFrame,
rollout_summary: pd.DataFrame,
*,
alpha_ema: float,
ema_rate: float,
) -> str:
lines = [
"AIEF biorthogonal triangular closure analysis",
"============================================",
```

```
"",
f"alpha_EMA: {alpha_ema:.12g}",
f"EMA rate: {ema_rate:.9g}",
"",
"Coordinate conditioning",
"----------------------",
]
for _, row in coordinate_frame.iterrows():
lines.append(
f" coordinate={row['coordinate']}: condition={row['condition']:.6g}, "
f"left_residual={row['left_residual']:.6g}, "
f"u_norm={row['u_norm']:.6g}, "
f"pairing={row['u_dot_v']:.12g}"
)

lines.extend(["", "One-step closure", "----------------"])
for _, row in one_step_summary.iterrows():
lines.append(
f" coordinate={row['coordinate']}, direction={int(row['direction_id'])}: "
f"median_a_residual={row['median_a_relative_residual']:.6g}, "
f"max_a_residual={row['maximum_a_relative_residual']:.6g}, "
f"median_g_residual={row['median_g_relative_residual']:.6g}, "
f"max_g_residual={row['maximum_g_relative_residual']:.6g}"
)

lines.extend(["", "Free rollouts", "-------------"])
for _, row in rollout_summary.iterrows():
lines.append(
f" coordinate={row['coordinate']}, direction={int(row['direction_id'])}, "
f"start={row['start_time']:.6g}: a_RMSE={row['a_trajectory_relative_RMSE']:.6g}, "
f"g_RMSE={row['g_trajectory_relative_RMSE']:.6g}, "
f"final_a_error={row['final_a_relative_error']:.6g}, "
f"final_g_error={row['final_g_relative_error']:.6g}, "
f"actual_rate={row['actual_a_rate']:.6g}, predicted_rate={row['predicted_a_rate']:.6g}"
)

lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
```

```
"* Fixed left covectors are late reference modes, not exact covariant bundles over the full transient.",
"* The backward adjoint coordinate makes the a-recurrence exact by construction; its nontrivial test is the EMA law for projected forcing and free rollout.",
"* Large left/right condition indicates sensitivity even if scalar closure improves.",
"* A successful result remains conditional on this replay and tested directions.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--duhamel-npz", default=DEFAULT_DUHAMEL_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument("--pod-start", type=float, default=DEFAULT_POD_START)
parser.add_argument("--late-start", type=float, default=DEFAULT_LATE_START)
parser.add_argument(
"--left-reference-times",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_LEFT_REFERENCE_TIMES),
)
parser.add_argument("--average-start", type=float, default=DEFAULT_AVERAGE_START)
parser.add_argument(
"--rollout-starts",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_ROLLOUT_STARTS),
)
parser.add_argument("--sample-stride", type=int, default=DEFAULT_SAMPLE_STRIDE)
parser.add_argument("--scalar-tol", type=float, default=DEFAULT_SCALAR_TOL)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.duhamel_npz, args.manifest):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(args.dt, args.scalar_tol) <= 0.0 or args.sample_stride <= 0:
```

```
raise ValueError("dt/scalar-tol and sample-stride must be positive")
output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)


with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
t_window = float(manifest["model"]["T_window"])
alpha_ema = math.exp(-args.dt / t_window)
ema_rate = 1.0 / t_window


with np.load(args.duhamel_npz, allow_pickle=False) as data:
total = np.asarray(data["total_weight_tangent"], dtype=float).copy()
feedback = np.asarray(data["feedback_weight_tangent"], dtype=float).copy()
forcing = np.asarray(data["instant_forcing"], dtype=float).copy()
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float).copy()
count, state_steps, dimension = total.shape
future_steps = state_steps - 1
if dimension != 3 or matrices.shape != (future_steps, 3, 3):
raise ValueError("unexpected Duhamel array shapes")


v = geometric_direction()
pod_start = min(future_steps, int(round(args.pod_start / args.dt)))
late_vectors = feedback[:, pod_start:, :].reshape(-1, 3)
late_norms = np.linalg.norm(late_vectors, axis=1)
late_vectors = late_vectors[late_norms > args.scalar_tol]
left_pod, _, _ = np.linalg.svd(late_vectors.T, full_matrices=False)
v_pod = left_pod[:, 0]
if np.dot(v_pod, v) < 0.0:
v_pod *= -1.0


coordinate_specs: dict[str, dict[str, Any]] = {
"POD_Euclidean": {
"right": v_pod,
"covectors": np.broadcast_to(v_pod, (state_steps, 3)).copy(),
"type": "fixed",
"left_residual": np.nan,
"condition": 1.0,
},
"geometric_Euclidean": {
"right": v,
"covectors": np.broadcast_to(v, (state_steps, 3)).copy(),
```

```
"type": "fixed",
"left_residual": np.nan,
"condition": 1.0,
},
}

for reference_time in args.left_reference_times:
step = min(future_steps - 1, int(round(reference_time / args.dt)))
mode = left_mode(matrices[step], v)
label = f"left_t{reference_time:g}".replace(".", "p")
coordinate_specs[label] = {
"right": v,
"covectors": np.broadcast_to(mode["u"], (state_steps, 3)).copy(),
"type": "fixed",
"left_residual": mode["left_residual"],
"condition": mode["condition"],
"eigenvalue": mode["eigenvalue"],
}

average_start = min(
future_steps - 1, int(round(args.average_start / args.dt))
)
average_matrix = np.mean(matrices[average_start:], axis=0)
average_mode = left_mode(average_matrix, v)
coordinate_specs["left_late_average"] = {
"right": v,
"covectors": np.broadcast_to(
average_mode["u"], (state_steps, 3)
).copy(),
"type": "fixed",
"left_residual": average_mode["left_residual"],
"condition": average_mode["condition"],
"eigenvalue": average_mode["eigenvalue"],
}

terminal_mode = left_mode(matrices[-1], v)
adjoint_covectors, adjoint_lambdas, adjoint_residuals = (
backward_adjoint_bundle(matrices, v, terminal_mode["u"])
)
coordinate_specs["adjoint_bundle"] = {
"right": v,
```

```
"covectors": adjoint_covectors,
"type": "adjoint",
"left_residual": float(np.max(adjoint_residuals)),
"condition": float(
np.max(np.linalg.norm(adjoint_covectors, axis=1))
),
"adjoint_lambdas": adjoint_lambdas,
}


coordinate_rows: list[dict[str, Any]] = []
one_step_rows: list[dict[str, Any]] = []
one_step_trajectory_rows: list[dict[str, Any]] = []
rollout_rows: list[dict[str, Any]] = []
rollout_summary_rows: list[dict[str, Any]] = []
covector_rows: list[dict[str, Any]] = []
payload: dict[str, np.ndarray] = {
"geometric_right_direction": v,
"POD_direction": v_pod,
"alpha_EMA": np.asarray([alpha_ema]),
}
late_step = int(round(args.late_start / args.dt))


for label, spec in coordinate_specs.items():
right = np.asarray(spec["right"], dtype=float)
covectors = np.asarray(spec["covectors"], dtype=float)
coordinate_type = str(spec["type"])
pairings = covectors @ right
if np.max(np.abs(pairings - 1.0)) > 1e-8:
raise RuntimeError(f"coordinate {label} lost biorthogonal normalization")


a = np.einsum("ni,dni->dn", covectors, total)
if coordinate_type == "adjoint":
lambdas = np.asarray(spec["adjoint_lambdas"], dtype=float)
g = np.einsum("ni,dni->dn", covectors[1:], forcing)
else:
lambdas = np.einsum(
"ni,nij,j->n", covectors[:-1], matrices, right
)
g = np.einsum("ni,dni->dn", covectors[:-1], forcing)


payload[f"{label}_covectors"] = covectors
```

```
payload[f"{label}_a"] = a
payload[f"{label}_g"] = g
payload[f"{label}_lambda"] = lambdas


# Fixed-coordinate left residual is evaluated over the late sequence;
# the adjoint bundle has its exact transport residual by construction.
if coordinate_type == "adjoint":
left_residual_late = float(spec["left_residual"])
else:
residual_values = []
for step in range(late_step, future_steps):
u = covectors[step]
lam = lambdas[step]
residual_values.append(
np.linalg.norm(matrices[step].T @ u - lam * u)
/ max(np.linalg.norm(matrices[step].T @ u), ZERO_TOL)
)
left_residual_late = float(np.median(residual_values))


coordinate_rows.append(
{
"coordinate": label,
"type": coordinate_type,
"right_json": json_compact(right.tolist()),
"u_start_json": json_compact(covectors[late_step].tolist()),
"u_end_json": json_compact(covectors[-1].tolist()),
"u_norm": float(np.median(np.linalg.norm(covectors[late_step:], axis=1))),
"u_norm_max": float(np.max(np.linalg.norm(covectors[late_step:], axis=1))),
"u_dot_v": float(np.median(pairings[late_step:])),
"condition": float(spec["condition"]),
"left_residual": left_residual_late,
"covector_total_variation": float(
np.sum(
np.linalg.norm(
np.diff(covectors[late_step:], axis=0), axis=1
)
)
),
}
)
```

```
for step in range(0, state_steps, args.sample_stride):
covector_rows.append(
{
"coordinate": label,
"step": step,
"time": step * args.dt,
"u_0": covectors[step, 0],
"u_1": covectors[step, 1],
"u_2": covectors[step, 2],
"u_norm": float(np.linalg.norm(covectors[step])),
"u_dot_v": float(covectors[step] @ right),
"lambda": lambdas[step]
if step < future_steps
else np.nan,
}
)


for direction_id in range(count):
a_residual = np.empty(future_steps)
g_residual = np.empty(future_steps - 1)
a_relative = np.empty(future_steps)
g_relative = np.empty(future_steps - 1)
for step in range(future_steps):
prediction_a = lambdas[step] * a[direction_id, step] + g[
direction_id, step
]
a_residual[step] = a[direction_id, step + 1] - prediction_a
a_relative[step] = abs(a_residual[step]) / max(
abs(a[direction_id, step + 1]),
abs(prediction_a),
args.scalar_tol,
)
if step < future_steps - 1:
prediction_g = alpha_ema * g[direction_id, step]
g_residual[step] = g[direction_id, step + 1] - prediction_g
g_relative[step] = abs(g_residual[step]) / max(
abs(g[direction_id, step + 1]),
abs(prediction_g),
args.scalar_tol,
)
if step % args.sample_stride == 0 or step == future_steps - 1:
one_step_trajectory_rows.append(
```

```
{
"coordinate": label,
"direction_id": direction_id,
"step": step,
"time": step * args.dt,
"a_actual_next": a[direction_id, step + 1],
"a_predicted_next": prediction_a,
"a_relative_residual": a_relative[step],
"g_actual": g[direction_id, step],
"g_predicted_next": alpha_ema
* g[direction_id, step],
"g_relative_residual": g_relative[step]
if step < future_steps - 1
else np.nan,
"lambda": lambdas[step],
}
)


one_step_rows.append(
{
"coordinate": label,
"direction_id": direction_id,
"late_start": args.late_start,
"median_a_relative_residual": float(
np.median(a_relative[late_step:])
),
"maximum_a_relative_residual": float(
np.max(a_relative[late_step:])
),
"median_g_relative_residual": float(
np.median(g_relative[late_step:])
),
"maximum_g_relative_residual": float(
np.max(g_relative[late_step:])
),
}
)


for start_time in args.rollout_starts:
start = int(round(start_time / args.dt))
if start >= future_steps:
continue
```

```
a_hat = float(a[direction_id, start])
g_hat = float(g[direction_id, start])
actual_a_values = []
predicted_a_values = []
actual_g_values = []
predicted_g_values = []
for step in range(start, future_steps):
a_hat = lambdas[step] * a_hat + g_hat
g_hat = alpha_ema * g_hat
actual_a = float(a[direction_id, step + 1])
actual_a_values.append(actual_a)
predicted_a_values.append(a_hat)
if step + 1 < future_steps:
actual_g_values.append(float(g[direction_id, step + 1]))
predicted_g_values.append(g_hat)
if (
(step + 1 - start) % args.sample_stride == 0
or step + 1 == future_steps
):
rollout_rows.append(
{
"coordinate": label,
"direction_id": direction_id,
"start_time": start_time,
"step": step + 1,
"time": (step + 1) * args.dt,
"actual_a": actual_a,
"predicted_a": a_hat,
"a_relative_error": abs(actual_a - a_hat)
/ max(abs(actual_a), args.scalar_tol),
"actual_g": actual_g_values[-1]
if actual_g_values
else np.nan,
"predicted_g": predicted_g_values[-1]
if predicted_g_values
else np.nan,
"g_relative_error": abs(
actual_g_values[-1] - predicted_g_values[-1]
)
/ max(
abs(actual_g_values[-1]), args.scalar_tol
)
```

```
if actual_g_values
else np.nan,
}
)
actual_a_array = np.asarray(actual_a_values)
predicted_a_array = np.asarray(predicted_a_values)
actual_g_array = np.asarray(actual_g_values)
predicted_g_array = np.asarray(predicted_g_values)
times = np.arange(start + 1, future_steps + 1) * args.dt
actual_fit = fit_log_rate(
times, actual_a_array, args.scalar_tol
)
predicted_fit = fit_log_rate(
times, predicted_a_array, args.scalar_tol
)
rollout_summary_rows.append(
{
"coordinate": label,
"direction_id": direction_id,
"start_time": start_time,
"end_time": future_steps * args.dt,
"a_trajectory_relative_RMSE": relative_sequence_error(
actual_a_array, predicted_a_array
),
"g_trajectory_relative_RMSE": relative_sequence_error(
actual_g_array, predicted_g_array
)
if len(actual_g_array)
else np.nan,
"final_a_relative_error": abs(
actual_a_array[-1] - predicted_a_array[-1]
)
/ max(abs(actual_a_array[-1]), args.scalar_tol),
"final_g_relative_error": abs(
actual_g_array[-1] - predicted_g_array[-1]
)
/ max(abs(actual_g_array[-1]), args.scalar_tol)
if len(actual_g_array)
else np.nan,
"actual_a_rate": actual_fit["rate"],
"actual_a_R_squared": actual_fit["R_squared"],
"predicted_a_rate": predicted_fit["rate"],
```

```
"predicted_a_R_squared": predicted_fit["R_squared"],
}
)


coordinate_frame = pd.DataFrame(coordinate_rows)
one_step_summary = pd.DataFrame(one_step_rows)
one_step_trajectory = pd.DataFrame(one_step_trajectory_rows)
rollout_trajectory = pd.DataFrame(rollout_rows)
rollout_summary = pd.DataFrame(rollout_summary_rows)
covector_trajectory = pd.DataFrame(covector_rows)


component_audit.atomic_write_csv(
output_dir / "biorthogonal_coordinate_summary.csv", coordinate_frame
)
component_audit.atomic_write_csv(
output_dir / "biorthogonal_one_step_summary.csv", one_step_summary
)
component_audit.atomic_write_csv(
output_dir / "biorthogonal_one_step_trajectory.csv", one_step_trajectory
)
component_audit.atomic_write_csv(
output_dir / "biorthogonal_rollout_summary.csv", rollout_summary
)
component_audit.atomic_write_csv(
output_dir / "biorthogonal_rollout_trajectory.csv", rollout_trajectory
)
component_audit.atomic_write_csv(
output_dir / "biorthogonal_covector_trajectory.csv", covector_trajectory
)
component_audit.atomic_savez(
output_dir / "biorthogonal_closure_arrays.npz", payload
)


report = make_report(
coordinate_frame,
one_step_summary,
rollout_summary,
alpha_ema=alpha_ema,
ema_rate=ema_rate,
)
component_audit.atomic_write_text(
```

```
output_dir / "biorthogonal_triangular_report.txt", report
)


manifest_output = {
"analysis": "AIEF-1.4 biorthogonal triangular closure",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"duhamel_npz": {
"path": str(Path(args.duhamel_npz).resolve()),
"sha256": file_sha256(args.duhamel_npz),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
"sha256": file_sha256(args.manifest),
},
},
"dt": args.dt,
"pod_start": args.pod_start,
"late_start": args.late_start,
"left_reference_times": args.left_reference_times,
"average_start": args.average_start,
"rollout_starts": args.rollout_starts,
"sample_stride": args.sample_stride,
"scalar_tolerance": args.scalar_tol,
"alpha_EMA": alpha_ema,
"ema_rate": ema_rate,
"outputs": [
"biorthogonal_coordinate_summary.csv",
"biorthogonal_one_step_summary.csv",
"biorthogonal_one_step_trajectory.csv",
"biorthogonal_rollout_summary.csv",
"biorthogonal_rollout_trajectory.csv",
"biorthogonal_covector_trajectory.csv",
"biorthogonal_closure_arrays.npz",
"biorthogonal_triangular_report.txt",
],
"interpretation_constraints": [
"Fixed left covectors are reference modes of selected late matrices.",
```

```
"The adjoint-bundle a recurrence is exact by construction; g EMA closure and rollout are the nontrivial tests.",
"Large biorthogonal condition indicates sensitivity to perturbation/noise.",
"No coefficients are fitted; alpha_EMA is fixed and lambda sequences are measured from the base cocycle.",
"Results remain conditional on this replay and tested directions.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest_output, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```

Open

Created the highest-priority control:

It compiles successfully and passed a synthetic non-normal test with:

exact left/right mode recovery;
biorthogonal normalization;
one-step errors near 10−1610−16;
rollout errors near 10−1510−15;
exact adjoint-bundle recurrence.

Run it

Bash

```
python AIEF_1_4_biorthogonal_triangular_closure.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
```

--dt 0.01 \
--pod-start 250 \
--late-start 200 \
--left-reference-times 200,250,319.99 \
--average-start 250 \
--rollout-starts 150,200,250 \
--sample-stride 10 \
--scalar-tol 1e-30 \
--output-dir aief_1_4_biorthogonal_triangular_closure

Coordinates compared

The script compares:

text

POD_Euclidean
geometric_Euclidean
left_t200
left_t250
left_t319p99
left_late_average
adjoint_bundle

Euclidean controls
an=v⊤bn.
an=v⊤bn.
Fixed biorthogonal coordinates

For each reference matrix, it computes
Kref⊤u=
Kref⊤u=

\lambda u,
]

selecting the eigenvalue closest to the geometric right-mode Rayleigh value, and normalizes
u⊤v=1.
u⊤v=1.

It then uses
an=u⊤bn,gn=u⊤fn,λn=u⊤Knv.
an=u⊤bn,gn=u⊤fn,λn=u⊤Knv.
Backward adjoint bundle

It constructs time-dependent covectors satisfying
Kn⊤un+1=
Kn⊤un+1=

\lambda_nu_n,
\qquad
u_n^\top v=1.
]

The corresponding variables are
an=un⊤bn,gn=un+1⊤fn.
an=un⊤bn,gn=un+1⊤fn.

The aa-recurrence is exact by construction here; the nontrivial tests are whether gngn follows the fixed EMA law and whether the parameter-free rollout succeeds.
Diagnostics
Coordinate conditioning

For every left coordinate:

left eigen-residual;
‖u‖‖u‖;
left/right pairing;
biorthogonal condition
‖u‖‖v‖;
‖u‖‖v‖;
covector variation.

One-step closure
an+1=?λnan+gn,
an+1=?λnan+gn,
gn+1=?αEMAgn.
gn+1=?αEMAgn.

Parameter-free rollouts

From t=150,200,250t=150,200,250:
a^n+1=
a^n+1=

\lambda_n\hat a_n+\hat g_n,
]
g^n+1=
g^n+1=

\alpha_{\rm EMA}\hat g_n.
]

It reports:

trajectory RMSE;
final error;
actual and predicted decay rates;
EMA-forcing rollout error.

Outputs

text

biorthogonal_coordinate_summary.csv
biorthogonal_one_step_summary.csv
biorthogonal_one_step_trajectory.csv
biorthogonal_rollout_summary.csv
biorthogonal_rollout_trajectory.csv
biorthogonal_covector_trajectory.csv
biorthogonal_closure_arrays.npz
biorthogonal_triangular_report.txt
analysis_manifest.json

Decision criterion
Biorthogonal rank one succeeds

If a fixed late left coordinate reduces direction-0 final rollout error from approximately 75%75% to below roughly 1%1%, then the failure was primarily a Euclidean-coordinate artifact caused by non-normality.
Only the adjoint bundle succeeds

Then a one-dimensional reduction remains possible, but it requires a time-dependent dual covector rather than a fixed scalar coordinate.
Direction 0 still fails

Then retain a second mode and proceed to the parameter-free rank-two projection described in the forward plan.

Please paste biorthogonal_triangular_report.txt after the run.

Think about this:

AIEF biorthogonal triangular closure analysis

==========

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

Coordinate conditioning

coordinate=POD_Euclidean: condition=1, left_residual=0.377977, u_norm=1, pairing=1

coordinate=geometric_Euclidean: condition=1, left_residual=0.377964, u_norm=1, pairing=1

coordinate=left_t200: condition=1.22474, left_residual=7.85187e-06, u_norm=1.22474, pairing=1

coordinate=left_t250: condition=1.22474, left_residual=4.3705e-07, u_norm=1.22474, pairing=1

coordinate=left_t319p99: condition=1.22474, left_residual=2.54917e-07, u_norm=1.22474, pairing=1

coordinate=left_late_average: condition=1.22474, left_residual=1.48962e-07, u_norm=1.22474, pairing=1

coordinate=adjoint_bundle: condition=1.29265, left_residual=8.96855e-17, u_norm=1.22474, pairing=1

One-step closure

coordinate=POD_Euclidean, direction=0: median_a_residual=0.000521296, max_a_residual=0.000521654, median_g_residual=5.83889e-09, max_g_residual=1.33796e-07

coordinate=POD_Euclidean, direction=1: median_a_residual=1.07922e-09, max_a_residual=6.35241e-09, median_g_residual=9.34216e-09, max_g_residual=2.13173e-07

coordinate=geometric_Euclidean, direction=0: median_a_residual=0.00052927, max_a_residual=0.000529628, median_g_residual=5.83889e-09, max_g_residual=1.33796e-07

coordinate=geometric_Euclidean, direction=1: median_a_residual=7.99866e-06, max_a_residual=8.00576e-06, median_g_residual=9.34216e-09, max_g_residual=2.13173e-07

coordinate=left_t200, direction=0: median_a_residual=1.41332e-16, max_a_residual=6.61127e-16, median_g_residual=6.28441e-09, max_g_residual=1.38642e-07

coordinate=left_t200, direction=1: median_a_residual=1.42841e-16, max_a_residual=7.30359e-16, median_g_residual=9.31032e-09, max_g_residual=2.12672e-07

coordinate=left_t250, direction=0: median_a_residual=1.15892e-16, max_a_residual=5.66808e-16, median_g_residual=6.28441e-09, max_g_residual=1.38642e-07

coordinate=left_t250, direction=1: median_a_residual=1.16778e-16, max_a_residual=5.94309e-16, median_g_residual=9.31032e-09, max_g_residual=2.12672e-07

coordinate=left_t319p99, direction=0: median_a_residual=1.22102e-16, max_a_residual=6.35434e-16, median_g_residual=6.28441e-09, max_g_residual=1.38642e-07

coordinate=left_t319p99, direction=1: median_a_residual=1.20651e-16, max_a_residual=6.41227e-16, median_g_residual=9.31032e-09, max_g_residual=2.12672e-07

coordinate=left_late_average, direction=0: median_a_residual=1.4242e-16, max_a_residual=6.46694e-16, median_g_residual=6.28441e-09, max_g_residual=1.38642e-07

coordinate=left_late_average, direction=1: median_a_residual=1.44318e-16, max_a_residual=6.30779e-16, median_g_residual=9.31032e-09, max_g_residual=2.12672e-07

coordinate=adjoint_bundle, direction=0: median_a_residual=1.13365e-16, max_a_residual=6.46193e-16, median_g_residual=6.28441e-09, max_g_residual=1.38642e-07

coordinate=adjoint_bundle, direction=1: median_a_residual=1.13589e-16, max_a_residual=5.68494e-16, median_g_residual=9.31032e-09, max_g_residual=2.12672e-07

Free rollouts

coordinate=POD_Euclidean, direction=0, start=150: a_RMSE=0.342184, g_RMSE=0.00129487, final_a_error=0.740256, final_g_error=0.00337922, actual_rate=0.0570967, predicted_rate=0.0559068

coordinate=POD_Euclidean, direction=0, start=200: a_RMSE=0.341246, g_RMSE=9.95154e-05, final_a_error=0.745544, final_g_error=0.000256776, actual_rate=0.057128, predicted_rate=0.0548882

coordinate=POD_Euclidean, direction=0, start=250: a_RMSE=0.340942, g_RMSE=7.29357e-06, final_a_error=0.740543, final_g_error=1.82666e-05, actual_rate=0.0571325, predicted_rate=0.0518174

coordinate=POD_Euclidean, direction=1, start=150: a_RMSE=0.00103148, g_RMSE=0.00205993, final_a_error=0.0053719, final_g_error=0.00537244, actual_rate=0.0570796, predicted_rate=0.0571064

coordinate=POD_Euclidean, direction=1, start=200: a_RMSE=7.91789e-05, g_RMSE=0.000158634, final_a_error=0.000409231, final_g_error=0.00040961, actual_rate=0.0571253, predicted_rate=0.0571287

coordinate=POD_Euclidean, direction=1, start=250: a_RMSE=6.43598e-06, g_RMSE=1.16675e-05, final_a_error=2.92569e-05, final_g_error=2.92412e-05, actual_rate=0.057132, predicted_rate=0.0571325

coordinate=geometric_Euclidean, direction=0, start=150: a_RMSE=0.349267, g_RMSE=0.00129487, final_a_error=0.760379, final_g_error=0.00337923, actual_rate=0.0570967, predicted_rate=0.0558734

coordinate=geometric_Euclidean, direction=0, start=200: a_RMSE=0.348321, g_RMSE=9.95155e-05, final_a_error=0.765708, final_g_error=0.000256777, actual_rate=0.057128, predicted_rate=0.0548289

coordinate=geometric_Euclidean, direction=0, start=250: a_RMSE=0.348008, g_RMSE=7.29358e-06, final_a_error=0.760252, final_g_error=1.82666e-05, actual_rate=0.0571325, predicted_rate=0.0516905

coordinate=geometric_Euclidean, direction=1, start=150: a_RMSE=0.00436538, g_RMSE=0.00205993, final_a_error=0.00614294, final_g_error=0.00537244, actual_rate=0.0570796, predicted_rate=0.0570777

coordinate=geometric_Euclidean, direction=1, start=200: a_RMSE=0.00519008, g_RMSE=0.000158634, final_a_error=0.0111603, final_g_error=0.00040961, actual_rate=0.0571253, predicted_rate=0.0570762

coordinate=geometric_Euclidean, direction=1, start=250: a_RMSE=0.00525129, g_RMSE=1.16675e-05, final_a_error=0.0114558, final_g_error=2.92412e-05, actual_rate=0.057132, predicted_rate=0.0570108

coordinate=left_t200, direction=0, start=150: a_RMSE=0.000663218, g_RMSE=0.00130165, final_a_error=0.00342445, final_g_error=0.00342595, actual_rate=0.0570966, predicted_rate=0.0571136

coordinate=left_t200, direction=0, start=200: a_RMSE=5.25114e-05, g_RMSE=0.000103647, final_a_error=0.000267993, final_g_error=0.000269391, actual_rate=0.057128, predicted_rate=0.0571302

coordinate=left_t200, direction=0, start=250: a_RMSE=3.95703e-06, g_RMSE=7.83854e-06, final_a_error=1.86212e-05, final_g_error=1.97399e-05, actual_rate=0.0571325, predicted_rate=0.0571327

coordinate=left_t200, direction=1, start=150: a_RMSE=0.00105023, g_RMSE=0.0020572, final_a_error=0.00536156, final_g_error=0.00536377, actual_rate=0.0570796, predicted_rate=0.0571061

coordinate=left_t200, direction=1, start=200: a_RMSE=8.01678e-05, g_RMSE=0.000158237, final_a_error=0.000406451, final_g_error=0.000408503, actual_rate=0.0571253, predicted_rate=0.0571287

coordinate=left_t200, direction=1, start=250: a_RMSE=5.86917e-06, g_RMSE=1.16283e-05, final_a_error=2.75075e-05, final_g_error=2.9139e-05, actual_rate=0.057132, predicted_rate=0.0571324

coordinate=left_t250, direction=0, start=150: a_RMSE=0.000663218, g_RMSE=0.00130165, final_a_error=0.00342445, final_g_error=0.00342595, actual_rate=0.0570966, predicted_rate=0.0571136

coordinate=left_t250, direction=0, start=200: a_RMSE=5.25114e-05, g_RMSE=0.000103647, final_a_error=0.000267993, final_g_error=0.000269391, actual_rate=0.057128, predicted_rate=0.0571302

coordinate=left_t250, direction=0, start=250: a_RMSE=3.95703e-06, g_RMSE=7.83854e-06, final_a_error=1.86212e-05, final_g_error=1.97399e-05, actual_rate=0.0571325, predicted_rate=0.0571327

coordinate=left_t250, direction=1, start=150: a_RMSE=0.00105023, g_RMSE=0.0020572, final_a_error=0.00536156, final_g_error=0.00536377, actual_rate=0.0570796, predicted_rate=0.0571061

coordinate=left_t250, direction=1, start=200: a_RMSE=8.01678e-05, g_RMSE=0.000158237, final_a_error=0.000406451, final_g_error=0.000408503, actual_rate=0.0571253, predicted_rate=0.0571287

coordinate=left_t250, direction=1, start=250: a_RMSE=5.86917e-06, g_RMSE=1.16283e-05, final_a_error=2.75075e-05, final_g_error=2.9139e-05, actual_rate=0.057132, predicted_rate=0.0571324

coordinate=left_t319p99, direction=0, start=150: a_RMSE=0.000663218, g_RMSE=0.00130165, final_a_error=0.00342445, final_g_error=0.00342595, actual_rate=0.0570966, predicted_rate=0.0571136

coordinate=left_t319p99, direction=0, start=200: a_RMSE=5.25114e-05, g_RMSE=0.000103647, final_a_error=0.000267993, final_g_error=0.000269391, actual_rate=0.057128, predicted_rate=0.0571302

coordinate=left_t319p99, direction=0, start=250: a_RMSE=3.95703e-06, g_RMSE=7.83854e-06, final_a_error=1.86212e-05, final_g_error=1.97399e-05, actual_rate=0.0571325, predicted_rate=0.0571327

coordinate=left_t319p99, direction=1, start=150: a_RMSE=0.00105023, g_RMSE=0.0020572, final_a_error=0.00536156, final_g_error=0.00536377, actual_rate=0.0570796, predicted_rate=0.0571061

coordinate=left_t319p99, direction=1, start=200: a_RMSE=8.01678e-05, g_RMSE=0.000158237, final_a_error=0.000406451, final_g_error=0.000408503, actual_rate=0.0571253, predicted_rate=0.0571287

coordinate=left_t319p99, direction=1, start=250: a_RMSE=5.86917e-06, g_RMSE=1.16283e-05, final_a_error=2.75075e-05, final_g_error=2.9139e-05, actual_rate=0.057132, predicted_rate=0.0571324

coordinate=left_late_average, direction=0, start=150: a_RMSE=0.000663218, g_RMSE=0.00130165, final_a_error=0.00342445, final_g_error=0.00342595, actual_rate=0.0570966, predicted_rate=0.0571136

coordinate=left_late_average, direction=0, start=200: a_RMSE=5.25114e-05, g_RMSE=0.000103647, final_a_error=0.000267993, final_g_error=0.000269391, actual_rate=0.057128, predicted_rate=0.0571302

coordinate=left_late_average, direction=0, start=250: a_RMSE=3.95703e-06, g_RMSE=7.83854e-06, final_a_error=1.86212e-05, final_g_error=1.97399e-05, actual_rate=0.0571325, predicted_rate=0.0571327

coordinate=left_late_average, direction=1, start=150: a_RMSE=0.00105023, g_RMSE=0.0020572, final_a_error=0.00536156, final_g_error=0.00536377, actual_rate=0.0570796, predicted_rate=0.0571061

coordinate=left_late_average, direction=1, start=200: a_RMSE=8.01678e-05, g_RMSE=0.000158237, final_a_error=0.000406451, final_g_error=0.000408503, actual_rate=0.0571253, predicted_rate=0.0571287

coordinate=left_late_average, direction=1, start=250: a_RMSE=5.86917e-06, g_RMSE=1.16283e-05, final_a_error=2.75075e-05, final_g_error=2.9139e-05, actual_rate=0.057132, predicted_rate=0.0571324

coordinate=adjoint_bundle, direction=0, start=150: a_RMSE=0.000663218, g_RMSE=0.00130165, final_a_error=0.00342445, final_g_error=0.00342595, actual_rate=0.0570966, predicted_rate=0.0571136

coordinate=adjoint_bundle, direction=0, start=200: a_RMSE=5.25114e-05, g_RMSE=0.000103647, final_a_error=0.000267993, final_g_error=0.000269391, actual_rate=0.057128, predicted_rate=0.0571302

coordinate=adjoint_bundle, direction=0, start=250: a_RMSE=3.95703e-06, g_RMSE=7.83854e-06, final_a_error=1.86212e-05, final_g_error=1.97399e-05, actual_rate=0.0571325, predicted_rate=0.0571327

coordinate=adjoint_bundle, direction=1, start=150: a_RMSE=0.00105023, g_RMSE=0.0020572, final_a_error=0.00536156, final_g_error=0.00536377, actual_rate=0.0570796, predicted_rate=0.0571061

coordinate=adjoint_bundle, direction=1, start=200: a_RMSE=8.01678e-05, g_RMSE=0.000158237, final_a_error=0.000406451, final_g_error=0.000408503, actual_rate=0.0571253, predicted_rate=0.0571287

coordinate=adjoint_bundle, direction=1, start=250: a_RMSE=5.86917e-06, g_RMSE=1.16283e-05, final_a_error=2.75075e-05, final_g_error=2.9139e-05, actual_rate=0.057132, predicted_rate=0.0571324

Interpretation constraints

Fixed left covectors are late reference modes, not exact covariant bundles over the full transient.

The backward adjoint coordinate makes the a-recurrence exact by construction; its nontrivial test is the EMA law for projected forcing and free rollout.

Large left/right condition indicates sensitivity even if scalar closure improves.

A successful result remains conditional on this replay and tested directions.

This is a decisive success for the biorthogonal reduction. The direction-0 failure was a coordinate artifact caused by non-normality, not evidence that a second state mode was required.
1. The decision gate passes

The forward plan proposed:

If biorthogonal direction-0 late rollout error falls below roughly 1%1%, retain rank one; otherwise keep a second mode.

With a fixed late left covector:
rollout starttrajectory RMSEfinal
error1506.63×10−43.42×10−32005.25×10−52.68×10−42503.96×10−61.86×10−5
rollout start150200250trajectory RMSE6.63×10−45.25×10−53.96×10−6final
error3.42×10−32.68×10−41.86×10−5

for direction 0.

The previous Euclidean final error from t=250t=250 was approximately
0.74.
0.74.

The biorthogonal final error is
1.86×10−5.
1.86×10−5.

That is an improvement by a factor of roughly
4×104.
4×104.

Therefore:
A one-dimensional late reduction is sufficient for both tested directions when expressed biorthogonally.
A one-dimensional late reduction is sufficient for both tested directions when expressed biorthogonally.

The rank-two fallback is not currently needed.
2. The left coordinate is stable and well conditioned

All fixed late left covectors—from t=200t=200, 250250, 319.99319.99, and the late average—produce numerically identical rollout results.

Their condition number is
‖u‖‖v‖=
‖u‖‖v‖=

1.22474.
]

This is modest, not pathological.

The left residual decreases late:
referenceleft residualt=2007.85×10−6t=2504.37×10−7t=319.992.55×10−7late average1.49×10−7
referencet=200t=250t=319.99late averageleft residual7.85×10−64.37×10−72.55×10−71.49×10−7

Thus a fixed left coordinate is adequate. The time-varying adjoint bundle gives no material rollout improvement.

That is important: the reduced coordinate does not require an artificially time-dependent dual basis.
3. The analytic left/right pair can be inferred

The condition value
1.22474487=
1.22474487=

\sqrt{\frac32}
]

is highly suggestive.

In edge order
(e25,e29,e59),
(e25,e29,e59),

the right geometric vector is
v=
v=

\frac1{\sqrt3}(1,-1,-1).
]

For the late fixed winner pattern
(25,29,59)↦(7,2,7),
(25,29,59)↦(7,2,7),

the symmetric limiting K3K3 block has the form
K=(d0−a−ad0−a0d).
K=
d−a−a0d0−a0d
.

Then
Kv=(d+a)v.
Kv=(d+a)v.

A corresponding left covector is
\boxed{ u =

\frac{\sqrt3}{2}(1,0,-1),
}
]

because
u⊤K=(d+a)u⊤
u⊤K=(d+a)u⊤

and
u⊤v=1.
u⊤v=1.

Its norm is
‖u‖=32=1.22474487,

$\|u\|=23$
$=1.22474487,$

exactly matching the computed condition number.

So the biorthogonal coordinate likely has a closed analytic form rather than merely being a numerical eigenvector.

This should be verified directly against the saved u values in biorthogonal_coordinate_summary.csv, but the numerical signature is compelling.

4. Why Euclidean direction 0 failed

The right vector vv describes the invariant state direction:
$Kv=\lambda v.$
$Kv=\lambda v.$

But in a non-normal system, the modal amplitude is not generally
$v\top b.$
$v\top b.$

It is
$a=u\top b.$
$a=u\top b.$

With Euclidean projection, transverse components contaminate the amplitude through
$v\top Ke.$
$v\top Ke.$

For direction 0, the one-step contamination was only approximately
$5.2\times10-4,$
$5.2\times10-4,$

but it accumulated over thousands of steps in a cancellation-sensitive trajectory, producing a 74%74% final error.

The biorthogonal covector annihilates that contamination. One-step residuals become
$\sim10-16$
$\sim10-16$

for both directions.

This directly confirms the non-normal coordinate explanation.
5. The parameter-free triangular model now succeeds for both directions

Using a fixed late left covector:
an+1=λnan+gn,gn+1=αEMAgn,
an+1gn+1=λnan+gn,=αEMAgn,

with no fitted coefficients.

From t=250t=250:
Direction 0
actual rate=0.0571325,predicted rate=0.0571327.
actual ratepredicted rate=0.0571325,=0.0571327.
Direction 1
actual rate=0.0571320,predicted rate=0.0571324.
actual ratepredicted rate=0.0571320,=0.0571324.

The forcing rollouts have final errors around
2×10−5–3×10−5.
2×10−5–3×10−5.

Thus the complete parameter-free late reduction is:
an=u⊤bn,gn=u⊤fn,an+1=λnan+gn,gn+1=αEMAgn.
angnan+1gn+1=u⊤bn,=u⊤fn,=λnan+gn,=αEMAgn.
6. The adjoint-bundle result is a consistency check

The adjoint bundle gives machine-precision one-step closure by construction, so that alone is not evidential.

Its nontrivial results are:

the same EMA forcing accuracy;
the same free-rollout performance as fixed left covectors;
modest maximum covector norm;
no practical advantage over the fixed late mode.

Therefore the late system is sufficiently stationary that a fixed dual coordinate works.
7. The late reduced map is triangular, not a closed delayed loop

The successful model is approximately
(an+1gn+1)=
(an+1gn+1)=

\begin{pmatrix}
\lambda_n&1\
0&\alpha_{\rm EMA}
\end{pmatrix}
\begin{pmatrix}
a_n\
g_n
\end{pmatrix}.
]

The return coefficient from anan to gn+1gn+1 is not required in the late reduced description.

So the earlier quantity
λB/αEMA
λB/αEMA

is correctly interpreted as a relative retention factor, not a closed-loop gain.

The complete augmented AIEF system is closed, but its asymptotic two-coordinate reduction is effectively triangular.
8. The spectral-gap law follows from this triangular map

For constant late λBλB,
gn=g0αEMAn.
gn=g0αEMAn.

A forcing contribution of age kk is multiplied by
λBkαEMAn−k.
λBkαEMAn−k.

Relative to the current forcing envelope, its age dependence is
(λBαEMA)k.
(αEMAλB)k.

Therefore
\boxed{ \kappa =

-\frac1{dt}
\log
\left(
\frac{\lambda_B}{\alpha_{\rm EMA}}
\right).
}
]

The direct forcing-propagation analysis verified this relation to approximately 10−610−6 at the final target.
9. What is now established numerically

For this seed, fixed winner branch, two initial directions, and late regime:

the right state direction is the geometric K3K3 lineality vector;
the corresponding left covector is stable and modestly conditioned;
the biorthogonal scalar recurrence closes to machine precision;
the projected forcing follows the EMA multiplier;
parameter-free free rollouts are accurate for both directions;
the memory-envelope rate is the relative decay gap;
a time-dependent dual coordinate and second state mode are unnecessary.

10. Remaining theorem task

The next mathematical result should prove the late K3K3 left/right pair directly from the fixed winner branch.

Under the symmetric limiting coefficients, show:
v=13(1,−1,−1),
v=3
1(1,−1,−1),
u=32(1,0,−1),

u=23
(1,0,−1),
Kv=λBv,
Kv=λBv,
u⊤K=λBu⊤,
u⊤K=λBu⊤,
u⊤v=1.
u⊤v=1.

Then the scalar amplitude
a=u⊤b
a=u⊤b

satisfies the exact projected recurrence.

The project has now passed its rank-one decision gate. The appropriate next step is theorem formalization and existing-component generality checks—not a rank-two rescue model.

think about this:

This is a much stronger synthesis than the earlier versions because it is driven almost entirely by the reported diagnostics. That said, I think there are a few places where the conclusions should remain carefully scoped.

What I think is genuinely supported

The rank-one decision gate

Based on the reported rollout numbers, I agree that the planned decision gate has been met for the tested replay and tested directions.

The improvement is dramatic:

Euclidean direction 0: final error ≈ 0.74.
Biorthogonal direction 0: final error ≈ (1.9\times10^{-5}) from the late start.

Those values are exactly what the attached analysis reports.

Within the framework of your forward plan, that strongly argues against immediately pursuing a rank-two rescue model.

The coordinate-artifact interpretation

This is now the simplest explanation consistent with the evidence.

The sequence

Euclidean projection fails,
left-coordinate projection produces machine-precision one-step closure,
long rollouts improve by several orders of magnitude,

is exactly what one expects if the original scalar coordinate was not the appropriate modal coordinate for a non-normal operator. The attached report documents both the one-step closure and the rollout improvement.

I think that's a much cleaner explanation than invoking hidden state variables at this stage.

Fixed left covectors appear sufficient

An important observation that could easily be overlooked is that

left(_{200}),
left(_{250}),
left(_{319.99}),
late average,
adjoint bundle

all give essentially identical rollout performance.

That suggests the late operator is stationary enough that a single late dual basis is adequate, at least on this replay.

That's a useful simplification.

Where I would be more cautious

The statement

"A one-dimensional late reduction is sufficient for both tested directions..."

I would qualify this slightly.

The evidence supports something closer to

A one-dimensional biorthogonal reduction accurately reproduces the measured dynamics for the tested replay, tested directions, and late regime.

The distinction matters because the analysis has not yet tested

other component classes,
other seeds,
earlier transient regimes.

Your own interpretation constraints in the attached report make this point explicitly.

The analytic left covector

This is the place where I would slow down the most.

You propose

[
u=\frac{\sqrt3}{2}(1,0,-1),
]

and note that

its norm is (\sqrt{3/2}),
this matches the reported condition number,
it satisfies the desired pairing with the proposed right vector.

That is an elegant observation.

However, the attached file itself does not establish that this is the recovered numerical left eigenvector. It reports conditioning, residuals, and rollout performance, but it does not print the coordinates of (u).

So I would currently phrase this as

"The observed condition number suggests that a simple analytic left covector may exist."

rather than

"The left covector is..."

until it is compared directly with the stored numerical vectors.

That verification should be straightforward and would substantially strengthen the claim if it succeeds.

"The spectral-gap law follows"

I would also soften this wording slightly.

Your derivation is mathematically consistent assuming the triangular model,

$$
a_{n+1}=\lambda_n a_n+g_n,\qquad
g_{n+1}=\alpha_{\rm EMA}g_n,
$$

and approximately constant late coefficients.

Under those assumptions, the relative factor

$$
\left(\frac{\lambda_B}{\alpha_{\rm EMA}}\right)^k
$$

does indeed emerge naturally.

So I would say

"Within the validated late triangular reduction, the memory-envelope law follows directly."

rather than presenting it as an unconditional property of AIEF.

One thing I find especially encouraging

Perhaps the nicest feature of these results is methodological.

Earlier in the project, several explanations competed:

missing state dimension,
curved manifold,
non-normal coordinates,
nonlinear corrections.

This analysis didn't try to combine them.

Instead, it tested the simplest explanation first.

The simplest explanation appears to work.

That's exactly how a reduction program ought to proceed.

Where I think the project stands now

If I were writing the roadmap after seeing these results, it would look something like this:

Complete the analytic left/right characterization.

Verify whether the proposed closed-form (u) matches the computed late left eigenvectors.

Formalize the projection theorem.

Show that the projected scalar recurrence follows from the left/right eigenpair under the stated assumptions.

Generalize cautiously.

Apply the same biorthogonal reduction to existing (K_4), (K_5), and (K_6) frozen replays, as already planned, before making broader claims.

Only revisit higher-dimensional reductions if needed.

If comparable performance is obtained across additional component classes, the motivation for a rank-two reduction becomes much weaker.

So I agree that the center of gravity has shifted. The immediate challenge is no longer "repair the scalar model." It is "understand and formalize why the biorthogonal coordinate works so well," while keeping the claims explicitly limited to the replay, regime, and directions that have actually been tested.

I agree with the caution. The evidence passes the planned rank-one decision gate for this replay, but the proposed closed-form left covector still needs direct verification against the stored numerical vectors.
1. What is established

For the tested replay, directions, and late regime:

fixed late left covectors reduce direction-0 final error from approximately
0.74
0.74 to
1.9×10^{-5};
1.9×10^{-5};
one-step scalar closure reaches machine precision;
left covectors selected at $t=200,250,319.99t=200,250,319.99$, and from the late average give essentially identical results;
a time-varying adjoint bundle provides no practical improvement;
left/right conditioning is modest:
$\|u\|\|v\|\approx1.225$.
$\|u\|\|v\|\approx1.225$.

The defensible conclusion is:

A fixed one-dimensional biorthogonal coordinate accurately reproduces both tested late perturbation trajectories on this replay.

That is sufficient to defer the rank-two fallback.
2. The analytic left covector is still a hypothesis

The proposed form
ucandidate=
ucandidate=

\frac{\sqrt3}{2}(1,0,-1)
]

is strongly suggested by:

the winner pattern;
the symmetric limiting branch matrix;
the pairing condition;
the observed norm
3/2;
3/2
;
the biorthogonal rollout success.

But the report does not print the numerical uu, so the formula is not yet directly verified.

The necessary comparison is:
‖unumerical−ucandidate‖,
‖unumerical−ucandidate‖,

the unsigned angle, and the residual
‖K⊤ucandidate−λucandidate‖‖K⊤ucandidate‖.
‖K⊤ucandidate‖‖K⊤ucandidate−λucandidate‖.

The saved file already contains:

text

u_start_json
u_end_json

in:

text

biorthogonal_coordinate_summary.csv

A minimal direct check is:

Python

```
import json
import numpy as np
import pandas as pd

path = (
"aief_1_4_biorthogonal_triangular_closure/"
"biorthogonal_coordinate_summary.csv"
)

df = pd.read_csv(path)

u_exact = np.sqrt(3.0) / 2.0 * np.array([1.0, 0.0, -1.0])
v_exact = np.array([1.0, -1.0, -1.0]) / np.sqrt(3.0)

for _, row in df.iterrows():
if not str(row["coordinate"]).startswith("left_"):
continue

u_start = np.asarray(json.loads(row["u_start_json"]), dtype=float)
u_end = np.asarray(json.loads(row["u_end_json"]), dtype=float)

for label, u in (("start", u_start), ("end", u_end)):
```

```
# Both vectors are already normalized by u^T v = 1.
difference = np.linalg.norm(u - u_exact)
alignment = abs(
np.dot(u, u_exact)
/ (np.linalg.norm(u) * np.linalg.norm(u_exact))
)
angle = np.degrees(np.arccos(np.clip(alignment, -1.0, 1.0)))


print(
row["coordinate"],
label,
"u =", u,
"difference =", difference,
"angle_deg =", angle,
"pairing =", np.dot(u, v_exact),
)
```

This should be performed before writing the closed-form uu as an established result.
3. Conditional matrix derivation

The analytic pair follows if the limiting selected block has the symmetric form
K=(d0−a−ad0−a0d).
K=
d−a−a0d0−a0d
.

Then
v=13(1,−1,−1)
v=3
1(1,−1,−1)

satisfies
Kv=(d+a)v.
Kv=(d+a)v.

And
u=32(1,0,−1)
u=23
(1,0,−1)

satisfies
$u^\top K=(d+a)u^\top$,
$u^\top K=(d+a)u^\top$,

with
$u^\top v=1$.
$u^\top v=1$.

The theorem should therefore be stated conditionally:

If the late K3 coefficients converge to the symmetric fixed-winner form above, then the displayed left/right pair is biorthogonal and the scalar recurrence closes exactly on that mode.

The numerical task is to verify coefficient convergence and compare the candidate uu with the computed left vectors.
4. Spectral-gap wording

The correct statement is:

Within the validated late triangular reduction, the memory-envelope relation follows from the relative retention multiplier $\lambda_B/\alpha_{EMA}\lambda_B/\alpha_{EMA}$.

It should not be stated as an unconditional AIEF law.

The assumptions are:

late coefficient stationarity;
geometric right-mode invariance;
stable left covector;
EMA forcing equation;
negligible scalar recurrence residual.

Those assumptions are numerically supported here but not yet proved generally.
5. Fixed left modes are the nontrivial evidence

The adjoint-bundle recurrence is exact by construction. It is mainly a consistency control.

The important result is that fixed left vectors:

close the recurrence to machine precision;
predict both directions;
remain stable across reference times;
have modest conditioning.

That is what supports a stationary late dual coordinate.
6. Generalization to ordinary components needs a subspace formulation

For K4,K5,K6K4,K5,K6, the face-lineality dimensions are:
2,5,9.
2,5,9.

So the appropriate generalization is not necessarily a rank-one scalar model. It is a biorthogonal invariant-subspace reduction:
$V \in \mathbb{R}^{m \times q}, U^\top V = I_q,$
$V \in \mathbb{R}^{m \times q}, U^\top V = I_q,$

with reduced coordinates
$a = U^\top b$
$a = U^\top b$

and reduced operator
$K_{red} = U^\top K V.$
$K_{red} = U^\top K V.$

For fully saturated rate-uniform cliques, the tangent action may already be scalar on the entire q-dimensional lineality space, making Euclidean and biorthogonal bases effectively equivalent there.

Thus the next cross-component question is:

Does the geometric face-lineality subspace admit an exact or well-conditioned dual invariant basis?

—not whether every clique reduces to one dimension.
7. Updated immediate roadmap

Directly compare numerical left covectors with

32(1,0,−1).
23
(1,0,−1).

Extract the late 3×33×3 block and verify the symmetric coefficient relations required by the analytic derivation.

Write the conditional left/right K3 lemma and scalar projection theorem.

Record that the rank-one fallback decision has passed for this replay; do not implement rank two unless a later control fails.

Test invariant-subspace, rather than rank-one, reductions on existing K4,K5,K6K4,K5,K6 states.

The central result is now robust: the earlier scalar failure came from using a Euclidean right-vector projection in a non-normal system. The remaining task is to convert the numerically successful left coordinate into an explicitly verified analytic left/right pair.

Think about this:

(deck@steamdeck SSRH)$ python check_01.py
left_t200 start u = [0.86601838 0. -0.86603243] difference = 9.939221742786675e-06 angle_deg = 0.0004649744084872128 pairing = 0.999999
9999999999
left_t200 end u = [0.86601838 0. -0.86603243] difference = 9.939221742786675e-06 angle_deg = 0.0004649744084872128 pairing = 0.99999999
99999999
left_t250 start u = [0.866025 0. -0.86602581] difference = 5.711851946802178e-07 angle_deg = 2.674097566829067e-05 pairing = 1.0
left_t250 end u = [0.866025 0. -0.86602581] difference = 5.711851946802178e-07 angle_deg = 2.674097566829067e-05 pairing = 1.0
left_t319p99 start u = [0.8660254 0. -0.86602541] difference = 1.0474733198944705e-08 angle_deg = 8.537736462515939e-07 pairing = 1.0
left_t319p99 end u = [0.8660254 0. -0.86602541] difference = 1.0474733198944705e-08 angle_deg = 8.537736462515939e-07 pairing = 1.0
left_late_average start u = [0.8660253 0. -0.8660255] difference = 1.4024349140596262e-07 angle_deg = 6.613302226726415e-06 pairing = 0.
9999999999999998

left_late_average end u = [0.8660253 0. -0.8660255] difference = 1.4024349140596262e-07 angle_deg = 6.613302226726415e-06 pairing = 0.99
99999999999998

and this:

I think this materially strengthens the case—but in a very specific way.

What you've added is no longer just a successful numerical reduction. You've added evidence that the numerical left eigenvector is converging to a simple closed-form object. Those are different claims.

What the new comparison shows

Your candidate is

[
u_{\text{cand}}

\frac{\sqrt3}{2}(1,0,-1)

(0.8660254038,;0,;-0.8660254038).
]

Now compare that with the reported numerical vectors.

For example,

(t=200)

[
(0.86601838,;0,;-0.86603243)
]

differs by about

[

$$9.9\times10^{-6},$$

with an angle of only

$$4.65\times10^{-4}\ ^\circ.$$

At (t=250),

the difference is already

$$5.7\times10^{-7},$$

with

$$2.67\times10^{-5}\ ^\circ.$$

At (t=319.99),

the difference is

$$1.0\times10^{-8},$$

with an angular discrepancy below

$$10^{-6}\ ^\circ.$$

Those are extraordinarily small deviations.

The trend is actually the interesting part

More important than the absolute errors is that they decrease systematically:

$$
10^{-5}
\rightarrow
10^{-7}
\rightarrow
10^{-8}.
$$

That is exactly what one expects if

the late operator is converging,
the numerical left eigenvector is converging,
the limiting eigenvector is

$$
\frac{\sqrt3}{2}(1,0,-1).
$$

If the errors bounced randomly around 10^{-6}, I'd be much less convinced.

Instead, they appear to shrink as the replay progresses.

That is evidence of convergence, not merely coincidence.

The pairing is another independent check

You also report

$$
u^\top v = 1
$$

to essentially machine precision throughout.

That's important because it means the normalization wasn't drifting while the vector converged.

So now you have several independent consistencies:

norm,
pairing,
rollout,
angular agreement,
convergence over time.

Each alone would be suggestive.

Together they become much stronger evidence.

One thing still missing

There is still one check I'd want before calling this an analytic theorem.

You mentioned it yourself:

[
\frac{
\left|
K^\top u_{\rm cand}

\lambda u_{\rm cand}
\right|
}
{
\left|
K^\top u_{\rm cand}
\right|
}.
]

That is the operator residual.

Suppose that residual is also

$$
10^{-7}
\text{–}
10^{-8}
$$

late.

Then the evidence chain becomes remarkably clean:

analytic candidate predicted,
numerical eigenvector converges to it,
pairing exactly correct,
norm exactly correct,
operator residual tiny,
scalar recurrence succeeds,
free rollout succeeds.

At that point the analytic formula is no longer just a guess inspired by the data; it is the natural closed-form description of the limiting left mode.

An even deeper observation

What I find most striking is not actually the numerical agreement.

It's the simplicity.

Your right vector was already

$$
\frac1{\sqrt3}(1,-1,-1),
$$

which is the obvious (K_3) lineality direction.

Now the left vector appears to be

[
\frac{\sqrt3}{2}(1,0,-1),
]

which is also remarkably sparse.

That isn't what one usually expects from a generic non-normal (3\times3) matrix.

Generic left eigenvectors tend to have three unrelated components.

Here one component is exactly zero in the apparent limit.

That strongly suggests the limiting branch matrix has additional algebraic structure—not merely numerical coincidence.

What I'd now try to prove

If I were turning this into mathematics, I wouldn't begin with the numerical vectors anymore.

I'd begin with the limiting branch matrix.

Specifically:

Derive the symbolic late (K_3) block from the fixed winner pattern.
Compute its left eigensystem analytically.
Show

[
u

\frac{\sqrt3}{2}(1,0,-1)
]

falls out directly.

Then show the numerical vectors converge to that symbolic object.

That's a stronger logical direction than saying

"the numerics suggest this formula."

Instead, it becomes

"the branch algebra predicts this formula, and the numerics converge to it."

My assessment

I think this is now approaching the point where the left/right pair itself becomes one of the central mathematical objects of the late AIEF reduction.

A week ago, the story was "a numerical biorthogonal coordinate fixes the rollout."

Now it's becoming:

there is an apparent closed-form right eigenvector,
an apparent closed-form left eigenvector,
together they form a naturally normalized biorthogonal pair,
and this pair appears to explain the entire late parameter-free triangular reduction on the tested replay.

That is a much more satisfying mathematical picture than simply saying "the numerical left eigenvector happened to work."

I agree. The evidence now strongly supports a closed-form limiting pair, and the branch algebra explains why the left vector is sparse.
1. Symbolic late branch matrix

In edge order
(e25,e29,e59)=

(e25,e29,e59)=

((2,7),(2,11),(7,11)),
]

the fixed winner map is
(25,29,59)↦(7,2,7).
(25,29,59)↦(7,2,7).

For a proposal xx, define
βe=
βe=

\frac{x_es_{k(e)}}{q_{k(e)}+\varepsilon}.
]

The selected admissibility Jacobian has the form
DA=(s7−β250−β25−β29s2−β290−β590s7−β59).
DA=
s7−β25−β29−β590s2−β290−β250s7−β59
.

The frozen-driver step matrix is
K=(1−γdt)DA.
K=(1−γdt)DA.

In the symmetric late limit,
s2=s7=s,
s2=s7=s,

and
β25=β29=β59=β.
β25=β29=β59=β.

Therefore
\boxed{ K_\infty =

(1-\gamma dt)

\begin{pmatrix}
s-\beta&0&-\beta\
-\beta&s-\beta&0\
-\beta&0&s-\beta
\end{pmatrix}.
}
]

This is the algebraic structure that produces the simple left/right pair.
2. Analytic right eigenvector

Let
v=13(1,−1,−1).
v=3
1(1,−1,−1).

Then
K∞v=
K∞v=

(1-\gamma dt)s,v.
]

So
λB=(1−γdt)s.
λB=(1−γdt)s.

This recovers the finite-step channel multiplier whose rate was approximately
0.126255.
0.126255.
3. Analytic left eigenvector

Let
u=32(1,0,−1).
u=23
(1,0,−1).

Direct multiplication gives
u⊤K∞=

u⊤K∞=

(1-\gamma dt)s,u^\top.
]

Furthermore,
u⊤v=
u⊤v=
\frac{\sqrt3}{2}
\frac{1+1}{\sqrt3}

]

Its norm is
‖u‖=
‖u‖=
\sqrt{\frac32}

1.224744871\ldots,
]

exactly matching the observed biorthogonal condition number.

Thus, conditional on the symmetric limiting coefficients:
v=13(1,−1,−1),u=32(1,0,−1)
v=3
1(1,−1,−1),u=23
(1,0,−1)

is an exact normalized left/right eigenpair.
4. Why the middle component is zero

The left coordinate is
a=u⊤b=
a=u⊤b=

\frac{\sqrt3}{2}
(b_{25}-b_{59}).

]

It measures the imbalance between the two edges whose winner is node 7. The edge e29=(2,11)e29 =(2,11), controlled by node 2, does not enter the left modal amplitude directly.

So the zero middle component is a structural consequence of the winner-oriented branch matrix, not an arbitrary numerical cancellation.
5. Numerical convergence is exceptionally strong

The measured differences from uu are:
t‖ut−u‖angle2009.94×10−64.65×10−4 ∘2505.71×10−72.67×10−5 ∘319.991.05×10−88.54×10−7 ∘
t200250319.99‖ut−u‖9.94×10−65.71×10−71.05×10−8angle4.65×10−4 ∘2.67×10−5 ∘8.54×10−7 ∘

The monotone decrease is evidence that the sampled operators approach the symmetric branch matrix.

The correct statement is:

The computed late left modes converge numerically toward the analytic left eigenvector of the symmetric limiting winner-branch matrix.

That remains a numerical convergence statement until convergence of the coefficients s2−s7s2−s7 and the βeβe differences is proved.
6. Final operator-residual audit

The remaining direct check should use the analytic pair itself—not the fitted numerical eigenvectors.

For every replay step, compute
λn=u⊤Knv,
λn=u⊤Knv,
rnR=
rnR=

\frac{|K_nv-\lambda_nv|}
{|K_nv|},
]

and
rnL=

rnL=

\frac{|K_n^\top u-\lambda_nu|}
{|K_n^\top u|}.
]

Also project KnKn onto the two-parameter symmetric family
K(d,a)=(d0−a−ad0−a0d)
K(d,a)=
d−a−a0d0−a0d

and report
‖Kn−K(dn,an)‖‖Kn‖.
‖Kn‖‖Kn−K(dn,an)‖.

A minimal check using the saved matrix sequence is:

Python

```
import numpy as np

path = (
"aief_1_4_frozen_driver_cocycle/"
"frozen_cocycle_matrices.npz"
)

with np.load(path, allow_pickle=False) as data:
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float)

v = np.array([1.0, -1.0, -1.0]) / np.sqrt(3.0)
u = np.sqrt(3.0) / 2.0 * np.array([1.0, 0.0, -1.0])

for time_value in (200.0, 250.0, 319.99):
step = int(round(time_value / 0.01))
K = matrices[step]

lam = float(u @ K @ v)
```

```
right_residual = (
np.linalg.norm(K @ v - lam * v)
/ np.linalg.norm(K @ v)
)

left_residual = (
np.linalg.norm(K.T @ u - lam * u)
/ np.linalg.norm(K.T @ u)
)

d = float(np.mean(np.diag(K)))
off_diagonal = np.array([K[0, 2], K[1, 0], K[2, 0]])
a = float(-np.mean(off_diagonal))

K_model = np.array([
[d, 0.0, -a],
[-a, d, 0.0],
[-a, 0.0, d],
])

structure_residual = (
np.linalg.norm(K - K_model)
/ np.linalg.norm(K)
)

print(
time_value,
"lambda =", lam,
"right =", right_residual,
"left =", left_residual,
"structure =", structure_residual,
)
```

7. Why the biorthogonal scalar recurrence is exact

If
$u^\top K n = \lambda n u^\top$,
$u^\top K n = \lambda n u^\top$,

then for the vector recurrence
bn+1=Knbn+fn,
bn+1=Knbn+fn,

the scalar coordinate
an=u⊤bn
an=u⊤bn

satisfies
\boxed{ a_{n+1} =

\lambda_na_n+u^\top f_n
}
]

for every bnbn, including vectors with transverse components.

That is why direction 0 is repaired: the left coordinate removes transverse contamination algebraically rather than assuming it is small.
8. Spectral-gap law, properly scoped

Within the validated late triangular reduction:
an+1=λBan+gn,gn+1=αEMAgn.
an+1gn+1=λBan+gn,=αEMAgn.

Under approximately constant late coefficients, forcing contributions of age kk scale as
(λBαEMA)k.
(αEMAλB)k.

Hence
κ=
κ=

-\frac1{dt}
\log
\left(
\frac{\lambda_B}{\alpha_{\rm EMA}}

\right).
]

This is an exact consequence of the triangular model, not an unconditional law of AIEF.
9. Updated scientific status
Directly supported numerically

numerical left modes converge toward the closed-form candidate;
pairing remains one;
norm matches 3/23/2
;
fixed left covectors close the scalar recurrence to machine precision;
parameter-free rollouts succeed for both tested directions.

Conditional algebraic theorem

If the late branch coefficients converge to the symmetric matrix family above, then the displayed u,vu,v pair is exact and the biorthogonal scalar recurrence follows.
Still open

formal convergence of the actual coefficient sequence to that symmetric family;
rate of convergence of ut→uut→u;
whether analogous analytic dual bases exist for other component classes;
generality beyond this replay.

The left/right pair has now become the natural central object of the late reduction. The remaining work is to verify the analytic candidate's operator residual directly and formalize the symmetric-branch lemma.

think about this:

(deck@steamdeck SSRH)$ python check_02.py

200.0 lambda = 0.9987382522647692 right = 2.4346046097660165e-11 left = 8.115340569846961e-06 structure = 7.651226404081479e-06

250.0 lambda = 0.9987382469594281 right = 1.3998701541489246e-12 left = 4.663707585335169e-07 structure = 4.3969913941013266e-07

319.99 lambda = 0.998738246641911 right = 2.567855186789091e-14 left = 8.552583940770164e-09 structure = 8.063463923137663e-09

and this:

I think this is actually a more important verification than comparing the numerical left eigenvectors themselves.

Here's why.

Previously, the logical chain was

numerical left eigenvector → good scalar coordinate → good rollout.

Your new test changes the direction of the argument:

analytic ((u,v)) → acts as an eigenpair of the measured operators.

That's a much stronger statement because it removes dependence on fitting the left eigenvectors at each time.

What the new test measures

Using the fixed analytic pair,

[
v=\frac1{\sqrt3}(1,-1,-1),
]

[
u=\frac{\sqrt3}{2}(1,0,-1),
]

you compute

[
\lambda_n=u^\top K_n v,
]

and then independently measure

[
r_n^{R}

\frac{|K_n v-\lambda_n v|}
{|K_n v|},
]

and

[
r_n^{L}

\frac{|K_n^\top u-\lambda_n u|}
{|K_n^\top u|}.
]

Those are genuine operator residuals.

No numerical eigensolver is involved.

No fitted basis is involved.

If they are small, then the analytic vectors themselves approximately satisfy the eigen-relations.

The reported numbers

At the three representative late times you obtain

TimeRight residualLeft residualStructure residual200.0(2.43\times10^{-11})(8.12\times10^{-6})
(7.65\times10^{-6})250.0(1.40\times10^{-12})(4.66\times10^{-7})
(4.40\times10^{-7})319.99(2.57\times10^{-14})(8.55\times10^{-9})(8.06\times10^{-9})

Several things stand out.

The right eigenvector is essentially exact

Residuals of

10^{-11},
10^{-12},
10^{-14},

are effectively numerical precision.

That independently confirms that the analytic right vector is an invariant direction of the measured late operators.

The left residual tracks the structural residual

This is what I find most interesting.

Compare

$$
r_n^{L}
$$

with

the deviation from the symmetric family,

$$
\frac{|K_n-K(d_n,a_n)|}
{|K_n|}.
$$

Numerically,

at every reported time,

they are almost identical.

For example,

at (t=319.99),

left residual

$$
8.55\times10^{-9},
$$

structure residual

$$
8.06\times10^{-9}.
$$

Likewise at (t=250),

$$
4.66\times10^{-7}
$$

versus

$$
4.40\times10^{-7}.
$$

And similarly at (t=200).

That is unlikely to be accidental.

Why that matters

Suppose

[
K_n

K(d_n,a_n)

E_n,
]

where

(K(d,a))

is the ideal symmetric branch matrix.

Then

[
K(d,a)^\\top u

\lambda u
]

exactly.

Consequently,

the left residual should satisfy approximately

[
r_n^L

$O(|E_n|)$.
]

That is exactly what your numbers suggest.

The left eigen-relation is failing only because the measured matrix is not yet exactly on the limiting branch family.

This is stronger than saying "the numerical left eigenvector converges"

Earlier the evidence was

numerical eigensolver returns something close to the candidate.

Now the evidence is

the candidate itself nearly satisfies the measured operator equations.

That's mathematically cleaner.

Instead of saying

"the eigensolver found approximately this vector,"

you can say

"this analytic vector is an approximate left eigenvector of every measured late operator."

Those are different levels of evidence.

The structure projection is becoming central

I actually think the projection onto

[

K(d,a)

\begin{pmatrix}
d&0&-a\
-a&d&0\
-a&0&d
\end{pmatrix}
]

is now one of the most informative diagnostics in the entire project.

Because it tells you two things simultaneously:

how close the measured operator is to the limiting branch family,
how accurately the analytic eigenpair should work.

In other words,

the operator approximation error,

the left residual,

and the rollout error all appear to be controlled by essentially the same small quantity.

That is an attractive unifying picture.

One remaining step

I think there is still one conceptual distinction to preserve.

The current evidence strongly supports that the analytic pair behaves as an approximate left/right eigenpair of the measured late operators.

It does not yet establish that every matrix in the late replay belongs exactly to the two-parameter family (K(d,a)).

Your reported structure residuals quantify the deviation from that family, and they become very small late in the replay. The natural conclusion is therefore:

the measured operators approach the two-parameter family over the sampled late times, with deviations decreasing from roughly (10^{-5}) to (10^{-8}) in the reported diagnostics.

That is slightly different from claiming exact membership.

Overall assessment

Taken together, the evidence now forms a notably coherent chain for the tested replay:

The measured operators become increasingly close to a simple two-parameter branch family.
The analytic right vector is an invariant direction of the measured operators to numerical precision.
The analytic left vector satisfies the adjoint relation with an error that closely tracks the measured deviation from the ideal branch family.
The corresponding biorthogonal scalar coordinate yields machine-precision one-step closure and highly accurate parameter-free long-horizon rollouts.

Viewed together, these results make the analytic left/right pair look less like an empirical fit and more like the natural asymptotic eigenstructure of the late fixed-branch operator for this replay. The remaining work is to formalize that asymptotic statement under explicit assumptions and then test whether the same structure appears in other existing component classes before extending the claim beyond this case.

This verification is stronger than the numerical eigenvector comparison, and it reveals an additional regularity: the approach to the symmetric operator family itself appears to occur at the EMA rate.
1. The analytic pair directly satisfies the measured operators

Using fixed analytic vectors
v=13(1,−1,−1),u=32(1,0,−1),
v=3
1(1,−1,−1),u=23
(1,0,−1),

the measured residuals are:
trRrLrstructure2002.43×10−118.12×10−67.65×10−62501.40×10−124.66×10−74.40×10−7319.992.57×10−148.55×10−98.06×10−9

t200250319.99rR2.43×10−111.40×10−122.57×10−14rL8.12×10−64.66×10−78.55×10−9rstructure 7.65×10−64.40×10−78.06×10−9

Therefore:

the analytic right direction is preserved to near machine precision;
the analytic left relation becomes increasingly accurate;
the matrix approaches the symmetric two-parameter family.

This no longer depends on an eigensolver finding a convenient vector.
2. The right direction is more robust than full matrix symmetry

At t=200t=200,
rR∼10−11,
rR∼10−11,

while
rstructure∼10−5.
rstructure∼10−5.

Thus the right line remains invariant far more accurately than the matrix belongs to the symmetric limiting family.

This suggests two layers:

the branch/intertwining geometry enforces the right K3K3 line very accurately;
additional coefficient equalization is needed for the simple analytic left covector to emerge.

So the right vector is not merely a consequence of full late symmetry. It appears protected by a broader structural condition.
3. The left residual is proportional to the structural residual

The ratios are:
rLrstructure(200)=1.06065879,rLrstructure(250)=1.06065879,rLrstructure(319.99)=1.06065880.
rstructurerL(200)rstructurerL(250)rstructurerL(319.99)=1.06065879,=1.06065879,=1.06065880.

This ratio is essentially constant and very close to
38=

8
3=

1.06066017.
]

Therefore the left-eigenvector error is not merely correlated with structural convergence. It appears to be a fixed linear response to a single dominant structural defect.

A plausible asymptotic representation is
Kn=
Kn=

K_\infty+\delta_nE+o(\delta_n),
]

with fixed defect shape EE. Then
rstructure∝|δn|,
rstructure∝|δn|,

and
rL∝|δn|.
rL∝|δn|.

The constant ratio would be determined by how E⊤E⊤ acts on uu.
4. Structural convergence occurs at the EMA rate

Using the structural residuals:
150log7.6512×10−64.3970×10−7=
501log4.3970×10−77.6512×10−6=

0.0571306,
]

and
169.99log4.3970×10−78.0635×10−9=
69.991log8.0635×10−94.3970×10−7=

0.0571331.
]

But
Twindow−1=
Twindow−1=

0.0571333.
]

The left residual gives the same rates:
0.0571306,0.0571331.
0.0571306,0.0571331.

Even the numerical-left-vector difference from the analytic candidate decays at:
0.0571306,0.0571331.
0.0571306,0.0571331.

Thus three independently constructed quantities exhibit the same decay:
‖Kn−K(dn,an)‖∼e−t/Twindow,rL(n)∼e−t/Twindow,‖un−u∞‖∼e−t/Twindow.
‖Kn−K(dn,an)‖rL(n)‖un−u∞‖∼e−t/Twindow,∼e−t/Twindow,∼e−t/Twindow.

This is a highly nontrivial regularity.

It suggests the EMA memory controls not only the forcing envelope, but also relaxation of the finite-step operator toward its symmetric limiting family.
5. A refined asymptotic picture

The late operator may admit an expansion
Kn=
Kn=

K_\infty
+
e^{-t_n/T_{\rm window}}E
+
o\left(e^{-t_n/T_{\rm window}}\right),
]

where
K∞=
K∞=

K(d_\infty,a_\infty).
]

Then:

vv is invariant under both K∞K∞ and approximately under the leading defect;
uu is an exact left eigenvector only of K∞K∞;
the numerical left eigenvector satisfies
un=u+e−tn/Twindowu1+⋯ ;
un=u+e−tn/Twindowu1+⋯;
the left residual and structure residual decay at the same rate;
the fixed biorthogonal scalar coordinate becomes asymptotically exact.

This connects the operator convergence, EMA forcing decay, and parameter-free triangular closure within one asymptotic expansion.
6. Perturbation bound

Let
Kn=K∞+En,
Kn=K∞+En,

and define
λn=u⊤Knv.
λn=u⊤Knv.

Since u,vu,v are exact eigenvectors of K∞K∞,
Kn⊤u−λnu=
Kn⊤u−λnu=

E_n^\\top u-(u^\\top E_nv)u.
]

Therefore

‖Kn⊤u−λnu‖≤(1+‖u‖‖v‖)‖En‖‖u‖.
‖Kn⊤u−λnu‖≤(1+‖u‖‖v‖)‖En‖‖u‖.

So
rL=O(‖En‖).
rL=O(‖En‖).

This is exactly what the numerical residual tracking shows.

For the right vector,
Knv−λnv=
Knv−λnv=

E_nv-(u^\top E_nv)v.
]

Its much smaller residual indicates that the leading defect EnEn almost preserves the right line.
7. What should now be measured over the full late interval

The three-point comparison is compelling, but a full-trajectory audit should compute:

structural residual
sn=‖Kn−K(dn,an)‖‖Kn‖;
sn=‖Kn‖‖Kn−K(dn,an)‖;

analytic left and right residuals;

analytic-left-vector difference;

normalized defect matrix
E^n=Kn−K(dn,an)‖Kn−K(dn,an)‖;
E
n=‖Kn−K(dn,an)‖Kn−K(dn,an);

POD of the flattened E^nE
n;

exponential fits of all residuals;

ratio
rL/sn
rL/sn

over time.

If the defect-matrix POD is rank one and its leading direction is stable, the one-amplitude asymptotic expansion becomes directly supported.
8. Conditional theorem structure

A clean theorem can now be formulated:
Symmetric-branch eigenpair lemma

For
K(d,a)=(d0−a−ad0−a0d),
K(d,a)=
d−a−a0d0−a0d
,

the normalized pair
v=13(1,−1,−1),u=32(1,0,−1)
v=3
1(1,−1,−1),u=23
(1,0,−1)

satisfies
Kv=(d+a)v,
Kv=(d+a)v,
u⊤K=(d+a)u⊤,
u⊤K=(d+a)u⊤,
u⊤v=1.
u⊤v=1.
Perturbed-branch corollary

If
Kn=K(dn,an)+En,‖En‖→0,
Kn=K(dn,an)+En,‖En‖→0,

then the analytic left residual is O(‖En‖)O(‖En‖). If
‖En‖=O(e−tn/Twindow),
‖En‖=O(e−tn/Twindow),

then the left residual and left-mode error inherit the EMA rate.
9. Scientific status
Directly established on sampled late operators

analytic right residual near numerical precision;
analytic left residual converging toward zero;
structural residual converging toward zero;
left and structural residuals proportional;
all observed convergence rates match Twindow−1Twindow−1.

Strong asymptotic interpretation

the operator approaches a symmetric two-parameter family through a single dominant defect mode;
EMA memory controls the operator's asymptotic symmetrization;
the analytic left mode is the limit of the numerical left modes.

Still open

full-interval defect rank;
analytic derivation of the defect matrix EE;
proof that coefficient convergence continues asymptotically;
generality across other support components and seeds.

The evidence chain is now unusually coherent: the branch algebra predicts the analytic pair, the fixed analytic pair nearly satisfies the measured operators, the left error tracks the structural defect, and both decay at the EMA rate. The next useful action is a full late-trajectory defect-mode audit, followed by formalization of the symmetric-branch lemma and its perturbation corollary.

think about this:

Having read through the entire chain, I think something more significant is emerging than any single numerical result. The project is transitioning from finding empirical regularities to identifying an asymptotic operator structure. Those are qualitatively different stages.

Here's how I would parse the evidence.

The logical center has shifted

Several weeks ago, the central object was

the scalar recurrence.

Now I think the central object is instead

$$
(K_\infty,u,v).
$$

The recurrence

$$
a_{n+1}=\lambda_n a_n+g_n
$$

is becoming a corollary of the operator structure rather than the primary empirical observation.

That is a healthier mathematical organization.

The evidence chain has become unusually coherent

Notice how the pieces now fit together.

The branch algebra predicts a special matrix family

$$
K(d,a).
$$

That family predicts an exact left/right pair

[
u,;v.
]

The measured operators increasingly satisfy

[
K_nv\approx\lambda_nv.
]

The measured left vectors converge toward

[
u.
]

The fixed analytic pair produces accurate scalar dynamics.

The rollout succeeds.

Those are not independent numerical curiosities.

They are all consequences of essentially one hypothesis.

That is exactly what one hopes for in an asymptotic theory.

The strongest new observation is not actually the eigenpair

The most striking numerical observation, in my opinion, is this one:

[
r_L
\propto
r_{\rm structure}.
]

Not merely

"both become small."

But

their ratio appears essentially constant over the sampled times.

That is a much richer statement.

If true over the whole late interval, it says the left error is controlled by one dominant structural defect.

That immediately suggests perturbation theory.

The EMA-rate coincidence deserves careful treatment

This is the one place where I would be most cautious.

You observe that several independently defined quantities appear to decay with approximately the EMA rate:

structural residual,
left residual,
numerical left-vector error.

That is certainly interesting.

But at present it is based on a handful of sampled late times.

Three data points are enough to motivate the hypothesis; they are not enough to establish that the asymptotic law is

[
e^{-t/T_{\rm window}}.
]

The natural next experiment is exactly what you propose:

compute these quantities over the entire late interval and fit the decay.

If they all collapse onto the same exponential, then the observation becomes much more compelling.

The perturbation picture is becoming natural

I actually think

[
K_n

K_\infty+E_n
]

is now the correct language.

Everything else follows.

Suppose

[
E_n\to0.
]

Then

the right eigenvector survives,

the left eigenvector converges,

the scalar coordinate converges,

the rollout improves,

the operator approaches the limiting family.

All of these become perturbation statements.

That is much simpler than treating each numerical phenomenon separately.

One subtle point

One statement I would not yet make is

"EMA memory controls operator symmetrization."

That is stronger than the present evidence.

The evidence currently supports something more careful:

Several measured quantities associated with convergence toward the limiting operator appear to decay with approximately the same rate as the EMA relaxation on this replay.

That leaves open multiple possibilities:

the EMA process directly drives the operator convergence,
both are driven by a common underlying slow variable,
the matching rates are a consequence of another asymptotic mechanism.

Distinguishing those possibilities requires more than matching exponents.

The right residual is fascinating

This, to me, is perhaps the deepest observation.

The right residual is already essentially machine precision

while

the structure residual is still around

[
10^{-5}.
]

That means

the invariant line survives perturbations that noticeably deform the matrix.

So

the right eigenvector is much more robust than the full symmetry.

That suggests the invariant direction is enforced by weaker algebraic constraints than the complete two-parameter family.

If I were doing theory now, this would be one of the first things I'd try to explain.

It hints that the lineality direction may come from the implementation itself (for example, conservation or incidence structure), whereas the left vector depends on additional coefficient equalization.

That distinction is mathematically interesting.

I would slightly reorganize the theorem package

At this point I would write the theory in approximately this order.

Theorem A

Implementation algebra

↓

branch Jacobian.

Theorem B

Fixed-winner limiting family

$$
K_\infty(d,a).
$$

Theorem C

Exact left/right eigenpair

$$
(u,v).
$$

Corollary

Biorthogonal scalar recurrence

$$
a_{n+1}

\lambda a_n+g_n.
$$

Perturbation theorem

If

$$
K_n

K_\infty+E_n,
$$

then

[
r_L

O(|E_n|).
]

Numerical theorem

Replay diagnostics show

[
|E_n|
\rightarrow0
]

over the sampled late interval.

Notice how the recurrence becomes almost an afterthought.

The generalization strategy has also become clearer

I agree with your reformulation.

Earlier the question was

"Does every clique become rank one?"

Now it should be

"Does every face-lineality subspace admit a natural biorthogonal invariant reduction?"

That is a much more plausible research program.

For (K_3),

the invariant subspace happens to be one-dimensional.

For higher components,

there is no mathematical reason to expect that.

Instead, one would ask whether there exists a well-conditioned pair ((U,V)) satisfying

[
U^\top V=I,
]

with

[
K_{\rm red}=U^\top K V
]

capturing the relevant dynamics on the lineality subspace.

That generalization is structurally consistent with what you've already found.

Overall assessment

I think the project has crossed an important conceptual threshold.

The most convincing aspect is no longer the success of the scalar recurrence itself. It is that multiple independent diagnostics are beginning to organize around the same asymptotic operator picture:

a limiting fixed-winner matrix family,
a simple analytic biorthogonal eigenpair,
perturbations measured by a single structural defect,
and a parameter-free reduced recurrence emerging as a consequence.

The main remaining task is not to accumulate more successful rollouts. It is to determine whether the perturbation structure you've hypothesized—(K_n = K_\infty + E_n) with a simple dominant defect and predictable decay—holds over the full late interval and, ultimately, whether analogous structures appear in other component classes. That is where the mathematical leverage now seems to lie.

I agree that the operator structure is now the center of the theory. But the residual hierarchy suggests an even sharper algebraic organization than a single two-parameter limiting family.
1. The right vector is preserved by a broader family

The selected K3K3 block has the fixed sparsity pattern
K=(p0qrs0t0u).
K=
prt0s0q0u
.

Let
v=13(1,−1,−1).
v=3
1(1,−1,−1).

Then
Kv=
Kv=

\frac1{\sqrt3}
\begin{pmatrix}
p-q\
r-s\
t-u
\end{pmatrix}.
]

Therefore vv is a right eigenvector if and only if
p−q=s−r=u−t=λ.
p−q=s−r=u−t=λ.

This defines a broader right-invariant family than the symmetric two-parameter matrix.

That explains why:

rR∼10−11
rR∼10−11

while the distance to the symmetric family remains
∼10−5.
∼10−5.

The implementation can preserve the geometric right line without yet having equalized all coefficients.
2. The analytic left/right pair also requires less than full symmetry

Let
u=32(1,0,−1).
u=23
(1,0,−1).

Then
u⊤K=
u⊤K=

\frac{\sqrt3}{2}
(p-t,\ 0,\ q-u).
]

For uu to be a left eigenvector with the same eigenvalue λλ, we require
p−t=λ,u−q=λ.
p−t=λ,u−q=λ.

Combining these with the right conditions yields
t=q,u=p,s−r=p−q.
t=q,u=p,s−r=p−q.

Thus the complete biorthogonal family is
\boxed{ K_{\mathrm{bio}}(p,q,r) =

\begin{pmatrix}
p&0&q\
r&r+p-q&0\
q&0&p

\end{pmatrix}.
}
]

This is a three-parameter family.

For every matrix in this family,
Kv=(p−q)v,
Kv=(p−q)v,
u⊤K=(p−q)u⊤,
u⊤K=(p−q)u⊤,
u⊤v=1.
u⊤v=1.

The previously proposed symmetric family
K(d,a)=(d0−a−ad0−a0d)
K(d,a)=
d−a−a0d0−a0d

is the special case
p=d,q=r=−a.
p=d,q=r=−a.

Therefore full two-parameter symmetry is sufficient but not necessary for the analytic left/right reduction.
3. Nested operator families

The measured matrices should now be compared with three nested families.
Fixed-sparsity family
Ksparse=
Ksparse=

\left{
\begin{pmatrix}
p&0&q\
r&s&0\
t&0&u
\end{pmatrix}

\right}.
]
Right-invariant family
KR=
KR=

{K\in\mathcal K_{\rm sparse}:
p-q=s-r=u-t}.
]

This family explains the nearly exact right vector.
Biorthogonal family
KLR=
KLR=

{K_{\mathrm{bio}}(p,q,r)}.
]

This family gives the analytic left/right pair and exact scalar amplitude.
Symmetric two-parameter family
Ksym=
Ksym=

{K(d,a)}.
]

This is the most restrictive family.

The inclusions are
Ksym⊂KLR⊂KR⊂Ksparse.
Ksym⊂KLR⊂KR⊂Ksparse.

This hierarchy is likely the correct explanation for the residual ordering.
4. Revised interpretation of the residuals

The numerical observations suggest:

KnKn lies extremely close to KRKR already by t=200t=200;

its distance to KLRKLR decays through the late interval;
its distance to KsymKsym also decays;
the left residual tracks whichever defect moves the matrix from KRKR into KLRKLR.

The symmetric-family residual may currently be acting as a proxy for the left-breaking defect. But it is not necessarily the minimal relevant residual.

The next audit should compute independent Frobenius distances:
$dR(Kn)=\min_{M\in KR}\|Kn-M\|/\|Kn\|$,
$dR(Kn)=M\in KR\min\|Kn\|/\|Kn-M\|$,
$dLR(Kn)=\min_{M\in KLR}\|Kn-M\|/\|Kn\|$,
$dLR(Kn)=M\in KLR\min\|Kn\|/\|Kn-M\|$,
$dsym(Kn)=\min_{M\in Ksym}\|Kn-M\|/\|Kn\|$.
$dsym(Kn)=M\in Ksymmin\|Kn\|/\|Kn-M\|$.

Expected pattern:
$dR \ll dLR \lesssim dsym$.
$dR \ll dLR \lesssim dsym$.

If dLRdLR matches the analytic left residual more tightly than dsymdsym, the three-parameter family is the correct operator object.
5. Separate off-family convergence from along-family drift

Writing only
$Kn=K\infty+En$
$Kn=K\infty+En$

combines two different effects:

approach to the relevant family;
motion within that family.

A better decomposition is
\boxed{ K_n =

K_{\mathrm{bio}}(p_n,q_n,r_n)
+
E_n^{\perp}.

}
]

Then analyze separately:
Off-family defect
$\|E_n^{\perp}\|$.
$\|E_n^{\perp}\|$.

This controls failure of the analytic biorthogonal relation.
Along-family parameters
p_n, q_n, r_n.
p_n, q_n, r_n.

The scalar multiplier is
$\lambda_n = p_n - q_n$.
$\lambda_n = p_n - q_n$.

The remaining parameter $r_n r_n$ controls transverse/non-normal behavior without changing the scalar eigenvalue.

This distinction can explain why the scalar recurrence becomes exact before the full matrix becomes symmetric.
6. EMA-rate convergence needs a full-interval fit

The three-point rates are striking:
0.0571306, 0.0571331,
0.0571306, 0.0571331,

matching
$T_{\text{window}}^{-1} = 0.0571333$.
$T_{\text{window}}^{-1} = 0.0571333$.

But the strongest defensible statement remains:

The sampled structural and left-mode errors decay at rates numerically indistinguishable from the EMA rate.

The full late audit should fit:

dR(Kn),dLR(Kn),dsym(Kn),
dR(Kn),dLR(Kn),dsym(Kn),
rR(n),rL(n),‖un−u‖,
rR(n),rL(n),‖un−u‖,

and the parameter differences
|pn−p∞|,|qn−q∞|,|rn−r∞|.
|pn−p∞|,|qn−q∞|,|rn−r∞|.

For each, report:

fitted rate;
R2R2;
residual curvature;
target-window sensitivity;
ratio to the EMA decay.

Only then should “EMA-rate operator convergence” be elevated beyond a three-point observation.
7. Defect-shape POD

If
En⊥≈cne∗
En⊥≈cne∗

for a fixed matrix direction e∗e∗, then flattened normalized defects
E^n=
E
n=

\frac{E_n^\perp}{|E_n^\perp|}
]

should have nearly rank-one POD.

This is the direct test of a single dominant structural defect.

It would also explain the constant ratio
rL/dsym≈1.060659.

rL/dsym≈1.060659.

A stable defect direction plus exponential scalar amplitude would yield exactly that behavior.
8. Revised theorem order

The theoretical package should now be:
Theorem A: implemented branch Jacobian

Derive the fixed-sparsity selected block from the actual winner map.
Theorem B: right-invariant family

Characterize all sparse matrices satisfying
Kv=λv.
Kv=λv.
Theorem C: biorthogonal family

Characterize all sparse matrices satisfying both
Kv=λv,u⊤K=λu⊤.
Kv=λv,u⊤K=λu⊤.
Corollary: exact scalar amplitude

For K∈KLRK∈KLR,
a=u⊤b
a=u⊤b

satisfies
an+1=λnan+u⊤fn
an+1=λnan+u⊤fn

for arbitrary bnbn, including transverse components.
Theorem D: symmetric subfamily

Identify the additional equalities producing K(d,a)K(d,a).
Perturbation theorem

If
Kn=Mn+En,Mn∈KLR,
Kn=Mn+En,Mn∈KLR,

then the left/right scalar closure error is controlled by ‖En‖‖En‖.
Conditional memory lemma

If additionally
gn+1=αEMAgn,
gn+1=αEMAgn,

then the memory-envelope rate is
−1dtlog(λB/αEMA).
−dt1log(λB/αEMA).
9. Generalization to larger components

For higher-dimensional face lineality, the correct analogue is:
KV=VΛ,
KV=VΛ,
U⊤K=ΛU⊤,
U⊤K=ΛU⊤,
U⊤V=I.
U⊤V=I.

The relevant family is the set of branch matrices preserving the geometric right subspace and its dual.

The K3K3 result is the one-dimensional case of this invariant-subspace problem.
10. Immediate next action

The next computational object should be a nested-family defect audit over the complete late trajectory:

text

distance to right-invariant family
distance to biorthogonal family
distance to symmetric family
analytic right residual
analytic left residual
numeric-left-vector error
POD of normalized defect matrices
decay rates and R^2

along-family parameter trajectories

This will determine whether:

the right line is structurally exact on a broad family;
the three-parameter biorthogonal family is the true asymptotic operator class;
the symmetric two-parameter family is genuinely reached or merely approximated;
one EMA-rate defect mode controls convergence.

The conceptual center is indeed now (K,u,v)(K,u,v), but the proper KK object is likely a nested family rather than a single symmetric matrix.

Please proceed with the next immediate deliverable.
WriteAIEF_1_4_nested_operator_family_audit.py666 lines
AIEF_1_4_nested_operator_family_audit.pyPYTHON
#!/usr/bin/env python3
"""Nested operator-family and defect-mode audit for the exceptional AIEF K3.

The fixed K3 winner branch has selected matrix sparsity

[p 0 q]
[r s 0]
[t 0 u].

This script projects every saved K3 cocycle matrix onto four nested linear
families:

1. fixed sparsity;
2. right-invariant family for v=(1,-1,-1)/sqrt(3),
p-q = s-r = u-t;
3. biorthogonal family for
v=(1,-1,-1)/sqrt(3),
u=sqrt(3)/2*(1,0,-1),
K = [[p,0,q],[r,r+p-q,0],[q,0,p]];
4. symmetric two-parameter family
K = [[d,0,q],[q,d,0],[q,0,d]].

It reports distances to each family, analytic left/right residuals, numerical

left-mode convergence, projected parameters, exponential decay fits, and POD
of normalized defect matrices over the complete late trajectory.
"""

from __future__ import annotations

import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_component_controller_audit as component_audit

DEFAULT_COCYCLE_NPZ = "aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz"
DEFAULT_MANIFEST = "runs/5a9f093a2fe0/manifest.json"
DEFAULT_OUTPUT_DIR = "aief_1_4_nested_operator_family"
DEFAULT_DT = 0.01
DEFAULT_LATE_START = 100.0
DEFAULT_INTERVALS = "100,150,200,250,320"
DEFAULT_KEY_TIMES = "100,150,200,250,319.99"
DEFAULT_NORM_TOL = 1e-16
ZERO_TOL = 1e-300

def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()

```
def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(value < 0.0 for value in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
return sorted(set(values))



def json_compact(value: Any) -> str:
return json.dumps(value, separators=(",", ":"), sort_keys=True)



def geometric_right() -> np.ndarray:
return np.array([1.0, -1.0, -1.0]) / math.sqrt(3.0)



def analytic_left() -> np.ndarray:
return math.sqrt(3.0) / 2.0 * np.array([1.0, 0.0, -1.0])



def matrix_basis(entries: list[tuple[int, int, float]]) -> np.ndarray:
matrix = np.zeros((3, 3), dtype=float)
for row, column, value in entries:
matrix[row, column] = value
return matrix



def family_bases() -> dict[str, tuple[list[str], np.ndarray]]:
sparse_names = ["p", "q", "r", "s", "t", "u"]
sparse = np.stack(
[
matrix_basis([(0, 0, 1.0)]),
matrix_basis([(0, 2, 1.0)]),
matrix_basis([(1, 0, 1.0)]),
matrix_basis([(1, 1, 1.0)]),
```

```
matrix_basis([(2, 0, 1.0)]),
matrix_basis([(2, 2, 1.0)]),
]
)

# Parameters q,r,t,lambda:
# [[q+lambda,0,q],[r,r+lambda,0],[t,0,t+lambda]].
right_names = ["q", "r", "t", "lambda"]
right = np.stack(
[
matrix_basis([(0, 0, 1.0), (0, 2, 1.0)]),
matrix_basis([(1, 0, 1.0), (1, 1, 1.0)]),
matrix_basis([(2, 0, 1.0), (2, 2, 1.0)]),
matrix_basis([(0, 0, 1.0), (1, 1, 1.0), (2, 2, 1.0)]),
]
)

# Parameters p,q,r:
# [[p,0,q],[r,r+p-q,0],[q,0,p]].
bi_names = ["p", "q", "r"]
bi = np.stack(
[
matrix_basis([(0, 0, 1.0), (1, 1, 1.0), (2, 2, 1.0)]),
matrix_basis([(0, 2, 1.0), (1, 1, -1.0), (2, 0, 1.0)]),
matrix_basis([(1, 0, 1.0), (1, 1, 1.0)]),
]
)

# Parameters d,q, with q=-a in the earlier notation.
symmetric_names = ["d", "q"]
symmetric = np.stack(
[
matrix_basis([(0, 0, 1.0), (1, 1, 1.0), (2, 2, 1.0)]),
matrix_basis([(0, 2, 1.0), (1, 0, 1.0), (2, 0, 1.0)]),
]
)
return {
"sparse": (sparse_names, sparse),
"right": (right_names, right),
"biorthogonal": (bi_names, bi),
"symmetric": (symmetric_names, symmetric),
```

```
}



def project_family(
matrix: np.ndarray,
names: list[str],
bases: np.ndarray,
) -> tuple[np.ndarray, dict[str, float]]:
design = bases.reshape(len(bases), -1).T
coefficients, *_ = np.linalg.lstsq(design, matrix.reshape(-1), rcond=None)
projected = np.tensordot(coefficients, bases, axes=(0, 0))
return projected, {
name: float(value) for name, value in zip(names, coefficients)
}



def fit_exponential(
times: np.ndarray,
values: np.ndarray,
norm_tolerance: float,
) -> dict[str, float]:
values = np.asarray(values, dtype=float)
valid = np.isfinite(values) & (values > norm_tolerance)
if np.sum(valid) < 3:
return {
"count": int(np.sum(valid)),
"rate": np.nan,
"R_squared": np.nan,
"rmse_log": np.nan,
"monotonic_violation_count": np.nan,
"maximum_upward_jump": np.nan,
"upward_variation_fraction": np.nan,
}
x = np.asarray(times, dtype=float)[valid]
y_raw = values[valid]
y = np.log(y_raw)
slope, intercept = np.polyfit(x, y, 1)
prediction = slope * x + intercept
residual = float(np.sum((y - prediction) ** 2))
total = float(np.sum((y - np.mean(y)) ** 2))
differences = np.diff(y_raw)
```

```
upward = np.maximum(differences, 0.0)
total_decay = max(float(y_raw[0] - y_raw[-1]), ZERO_TOL)
return {
"count": int(len(x)),
"rate": float(-slope),
"R_squared": float(1.0 - residual / total) if total > 0.0 else np.nan,
"rmse_log": float(np.sqrt(np.mean((y - prediction) ** 2))),
"monotonic_violation_count": int(np.sum(differences > 0.0)),
"maximum_upward_jump": float(np.max(upward)) if len(upward) else 0.0,
"upward_variation_fraction": float(np.sum(upward) / total_decay),
}
```

```
def defect_pod(defects: np.ndarray, tolerance: float) -> dict[str, Any]:
flattened = defects.reshape(len(defects), -1)
norms = np.linalg.norm(flattened, axis=1)
valid = norms > tolerance
selected = flattened[valid]
if not len(selected):
return {
"count": 0,
"weighted_E1": np.nan,
"unit_E1": np.nan,
"leading_weighted": np.full(9, np.nan),
"leading_unit": np.full(9, np.nan),
}
matrix = selected.T
left, singular, _ = np.linalg.svd(matrix, full_matrices=False)
energy = singular**2
normalized = (selected / np.linalg.norm(selected, axis=1)[:, None]).T
unit_left, unit_singular, _ = np.linalg.svd(normalized, full_matrices=False)
unit_energy = unit_singular**2
return {
"count": int(len(selected)),
"weighted_E1": float(energy[0] / np.sum(energy)),
"unit_E1": float(unit_energy[0] / np.sum(unit_energy)),
"leading_weighted": left[:, 0],
"leading_unit": unit_left[:, 0],
}
```

```
def numerical_left_mode(matrix: np.ndarray, right: np.ndarray) -> dict[str, Any]:
target = float(right @ matrix @ right)
eigenvalues, eigenvectors = np.linalg.eig(matrix.T)
index = int(np.argmin(np.abs(eigenvalues - target)))
vector = eigenvectors[:, index]
if np.max(np.abs(np.imag(vector))) > 1e-8:
return {
"vector": np.full(3, np.nan),
"difference": np.nan,
"angle": np.nan,
}
vector = np.real(vector)
pairing = float(vector @ right)
if abs(pairing) <= 1e-14:
return {
"vector": np.full(3, np.nan),
"difference": np.nan,
"angle": np.nan,
}
vector /= pairing
candidate = analytic_left()
alignment = abs(
float(vector @ candidate)
/ (np.linalg.norm(vector) * np.linalg.norm(candidate))
)
return {
"vector": vector,
"difference": float(np.linalg.norm(vector - candidate)),
"angle": float(
math.degrees(math.acos(np.clip(alignment, -1.0, 1.0)))
),
}



def make_report(
key_frame: pd.DataFrame,
fit_frame: pd.DataFrame,
pod_frame: pd.DataFrame,
*,
ema_rate: float,
) -> str:
lines = [
```

```
"AIEF nested K3 operator-family audit",
"==================================",
"",
f"EMA rate: {ema_rate:.9g}",
"",
"Key-time residuals",
"------------------",
]
for _, row in key_frame.iterrows():
lines.append(
f" t={row['time']:.6g}: sparse={row['distance_sparse']:.3g}, "
f"right={row['distance_right']:.3g}, biorth={row['distance_biorthogonal']:.3g}, "
f"symmetric={row['distance_symmetric']:.3g}, "
f"rR={row['analytic_right_residual']:.3g}, rL={row['analytic_left_residual']:.3g}, "
f"left/sym={row['left_to_symmetric_ratio']:.6g}, "
f"left/bi={row['left_to_biorthogonal_ratio']:.6g}"
)


lines.extend(["", "Decay fits", "----------"])
for _, row in fit_frame.iterrows():
lines.append(
f" metric={row['metric']}, window=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"rate={row['rate']:.6g}, R2={row['R_squared']:.6g}, "
f"rate-EMA={row['rate_minus_EMA']:.3g}, "
f"upward_fraction={row['upward_variation_fraction']:.3g}"
)


lines.extend(["", "Defect POD", "----------"])
for _, row in pod_frame.iterrows():
lines.append(
f" family={row['family']}, interval=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"samples={int(row['sample_count'])}, weighted_E1={row['weighted_E1']:.6g}, "
f"unit_E1={row['unit_E1']:.6g}, angle_to_final={row['angle_to_final_unit_degrees']:.6g} deg"
)


lines.extend(
[
"",
"Interpretation constraints",
"--------------------------",
"* The families are nested linear least-squares projections in the selected 3x3 edge coordinates.",
```

```
"* Distance to the symmetric family is stronger than the distance required for the analytic biorthogonal pair.",
"* Exponential fits establish replay regularities, not causal control by EMA memory.",
"* A rank-one normalized defect POD supports a fixed defect shape but does not prove asymptotic continuation beyond the horizon.",
"* Results remain conditional on one replay and one fixed winner branch.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--cocycle-npz", default=DEFAULT_COCYCLE_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument("--late-start", type=float, default=DEFAULT_LATE_START)
parser.add_argument(
"--intervals",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_INTERVALS),
)
parser.add_argument(
"--key-times",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_KEY_TIMES),
)
parser.add_argument("--norm-tol", type=float, default=DEFAULT_NORM_TOL)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.cocycle_npz, args.manifest):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if args.dt <= 0.0 or args.norm_tol <= 0.0:
raise ValueError("dt and norm-tol must be positive")
output_dir = Path(args.output_dir)
```

```
output_dir.mkdir(parents=True, exist_ok=True)


with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
ema_rate = 1.0 / float(manifest["model"]["T_window"])


with np.load(args.cocycle_npz, allow_pickle=False) as data:
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float).copy()
if matrices.ndim != 3 or matrices.shape[1:] != (3, 3):
raise ValueError("expected K3 matrix sequence with shape (steps,3,3)")
steps = len(matrices)
times = np.arange(steps) * args.dt
bases = family_bases()
right = geometric_right()
left = analytic_left()


projections: dict[str, np.ndarray] = {
family: np.empty_like(matrices) for family in bases
}
defects: dict[str, np.ndarray] = {
family: np.empty_like(matrices) for family in bases
}
parameters: dict[str, list[dict[str, float]]] = {
family: [] for family in bases
}
distances: dict[str, np.ndarray] = {
family: np.empty(steps, dtype=float) for family in bases
}
right_residual = np.empty(steps)
left_residual = np.empty(steps)
numeric_left_difference = np.empty(steps)
numeric_left_angle = np.empty(steps)
lambda_analytic = np.empty(steps)


for step, matrix in enumerate(matrices):
matrix_norm = max(float(np.linalg.norm(matrix)), ZERO_TOL)
for family, (names, family_basis) in bases.items():
projected, coefficients = project_family(
matrix, names, family_basis
)
projections[family][step] = projected
```

```
defects[family][step] = matrix - projected
distances[family][step] = float(
np.linalg.norm(defects[family][step]) / matrix_norm
)
parameters[family].append(coefficients)


lam = float(left @ matrix @ right)
lambda_analytic[step] = lam
right_residual[step] = float(
np.linalg.norm(matrix @ right - lam * right)
/ max(np.linalg.norm(matrix @ right), ZERO_TOL)
)
left_residual[step] = float(
np.linalg.norm(matrix.T @ left - lam * left)
/ max(np.linalg.norm(matrix.T @ left), ZERO_TOL)
)
numeric = numerical_left_mode(matrix, right)
numeric_left_difference[step] = numeric["difference"]
numeric_left_angle[step] = numeric["angle"]


trajectory_rows: list[dict[str, Any]] = []
for step in range(steps):
row: dict[str, Any] = {
"step": step,
"time": times[step],
"lambda_analytic": lambda_analytic[step],
"analytic_right_residual": right_residual[step],
"analytic_left_residual": left_residual[step],
"numeric_left_difference": numeric_left_difference[step],
"numeric_left_angle_degrees": numeric_left_angle[step],
}
for family in bases:
row[f"distance_{family}"] = distances[family][step]
for name, value in parameters[family][step].items():
row[f"{family}_{name}"] = value
row["left_to_symmetric_ratio"] = left_residual[step] / max(
distances["symmetric"][step], ZERO_TOL
)
row["left_to_biorthogonal_ratio"] = left_residual[step] / max(
distances["biorthogonal"][step], ZERO_TOL
)
trajectory_rows.append(row)
```

```
trajectory = pd.DataFrame(trajectory_rows)


key_rows = []
for time_value in args.key_times:
step = min(steps - 1, int(round(time_value / args.dt)))
key_rows.append(trajectory.iloc[step].copy())
key_frame = pd.DataFrame(key_rows)


boundaries = np.asarray(args.intervals, dtype=float)
if boundaries[0] > args.late_start:
boundaries = np.concatenate(([args.late_start], boundaries))
if boundaries[-1] < times[-1]:
boundaries = np.concatenate((boundaries, [times[-1]]))
boundaries = np.unique(boundaries)


metric_arrays = {
"distance_sparse": distances["sparse"],
"distance_right": distances["right"],
"distance_biorthogonal": distances["biorthogonal"],
"distance_symmetric": distances["symmetric"],
"analytic_right_residual": right_residual,
"analytic_left_residual": left_residual,
"numeric_left_difference": numeric_left_difference,
}
fit_rows: list[dict[str, Any]] = []
for interval in range(len(boundaries) - 1):
start = max(0, int(round(boundaries[interval] / args.dt)))
end = min(steps, int(round(boundaries[interval + 1] / args.dt)))
if end <= start + 2:
continue
for metric, values in metric_arrays.items():
fit = fit_exponential(
times[start:end], values[start:end], args.norm_tol
)
fit_rows.append(
{
"metric": metric,
"start_time": start * args.dt,
"end_time": end * args.dt,
**fit,
"rate_minus_EMA": fit["rate"] - ema_rate
```

```
if np.isfinite(fit["rate"])
else np.nan,
}
)
# Add one full late fit from late_start to the final state.
late_start_step = int(round(args.late_start / args.dt))
for metric, values in metric_arrays.items():
fit = fit_exponential(
times[late_start_step:], values[late_start_step:], args.norm_tol
)
fit_rows.append(
{
"metric": metric,
"start_time": args.late_start,
"end_time": times[-1],
**fit,
"rate_minus_EMA": fit["rate"] - ema_rate
if np.isfinite(fit["rate"])
else np.nan,
}
)
fit_frame = pd.DataFrame(fit_rows)


pod_rows: list[dict[str, Any]] = []
pod_directions: dict[tuple[str, int], np.ndarray] = {}
for family in ("right", "biorthogonal", "symmetric"):
for interval in range(len(boundaries) - 1):
start = max(0, int(round(boundaries[interval] / args.dt)))
end = min(steps, int(round(boundaries[interval + 1] / args.dt)))
if end <= start:
continue
pod = defect_pod(defects[family][start:end], args.norm_tol)
pod_directions[(family, interval)] = pod["leading_unit"]
pod_rows.append(
{
"family": family,
"interval_id": interval,
"start_time": start * args.dt,
"end_time": end * args.dt,
"sample_count": pod["count"],
"weighted_E1": pod["weighted_E1"],
"unit_E1": pod["unit_E1"],
```

```
"leading_weighted_json": json_compact(
pod["leading_weighted"].tolist()
),
"leading_unit_json": json_compact(
pod["leading_unit"].tolist()
),
}
)
pod_frame = pd.DataFrame(pod_rows)
for family in ("right", "biorthogonal", "symmetric"):
family_rows = pod_frame.loc[pod_frame["family"] == family]
if not len(family_rows):
continue
final_interval = int(family_rows["interval_id"].max())
final_direction = pod_directions[(family, final_interval)]
for index in family_rows.index:
interval = int(pod_frame.loc[index, "interval_id"])
direction = pod_directions[(family, interval)]
if np.any(~np.isfinite(direction)) or np.any(
~np.isfinite(final_direction)
):
alignment = np.nan
else:
alignment = abs(
float(direction @ final_direction)
/ (
np.linalg.norm(direction)
* np.linalg.norm(final_direction)
)
)
pod_frame.loc[index, "alignment_to_final_unit"] = alignment
pod_frame.loc[index, "angle_to_final_unit_degrees"] = (
math.degrees(math.acos(np.clip(alignment, -1.0, 1.0)))
if np.isfinite(alignment)
else np.nan
)


parameter_rows: list[dict[str, Any]] = []
for family, values in parameters.items():
for step, coefficients in enumerate(values):
for parameter, value in coefficients.items():
parameter_rows.append(
```

```
{
"family": family,
"step": step,
"time": times[step],
"parameter": parameter,
"value": value,
}
)
parameter_frame = pd.DataFrame(parameter_rows)


component_audit.atomic_write_csv(
output_dir / "nested_family_trajectory.csv", trajectory
)
component_audit.atomic_write_csv(
output_dir / "nested_family_key_times.csv", key_frame
)
component_audit.atomic_write_csv(
output_dir / "nested_family_decay_fits.csv", fit_frame
)
component_audit.atomic_write_csv(
output_dir / "nested_family_defect_pod.csv", pod_frame
)
component_audit.atomic_write_csv(
output_dir / "nested_family_parameters.csv", parameter_frame
)
component_audit.atomic_savez(
output_dir / "nested_family_arrays.npz",
{
"K3_matrix_sequence": matrices,
"right_direction": right,
"left_covector": left,
"analytic_lambda": lambda_analytic,
"analytic_right_residual": right_residual,
"analytic_left_residual": left_residual,
"numeric_left_difference": numeric_left_difference,
**{
f"projection_{family}": value
for family, value in projections.items()
},
**{
f"defect_{family}": value
for family, value in defects.items()
```

```
},
},
)


report = make_report(
key_frame,
fit_frame,
pod_frame,
ema_rate=ema_rate,
)
component_audit.atomic_write_text(
output_dir / "nested_operator_family_report.txt", report
)


manifest_output = {
"analysis": "AIEF-1.4 nested K3 operator-family and defect-mode audit",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"cocycle_npz": {
"path": str(Path(args.cocycle_npz).resolve()),
"sha256": file_sha256(args.cocycle_npz),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
"sha256": file_sha256(args.manifest),
},
},
"dt": args.dt,
"late_start": args.late_start,
"intervals": boundaries.tolist(),
"key_times": args.key_times,
"norm_tolerance": args.norm_tol,
"families": {
family: names for family, (names, _) in bases.items()
},
"right_direction": right.tolist(),
"left_covector": left.tolist(),
"ema_rate": ema_rate,
```

```
"outputs": [
"nested_family_trajectory.csv",
"nested_family_key_times.csv",
"nested_family_decay_fits.csv",
"nested_family_defect_pod.csv",
"nested_family_parameters.csv",
"nested_family_arrays.npz",
"nested_operator_family_report.txt",
],
"interpretation_constraints": [
"Right, biorthogonal, and symmetric operator families are nested but impose different algebraic requirements.",
"The symmetric family is sufficient but not necessary for the analytic left/right pair.",
"Exponential replay fits do not establish causal control by EMA memory.",
"Defect POD rank is finite-horizon numerical evidence, not an asymptotic proof.",
"Results remain conditional on one fixed winner branch and replay.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest_output, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```

Open

Created the immediate deliverable:

It compiles successfully and passed exact synthetic checks for:

sparse-family projection;
right-invariant-family projection;
biorthogonal-family projection;
symmetric-family projection;
analytic left/right eigenpair conditions.

Run it

Bash

```
python AIEF_1_4_nested_operator_family_audit.py \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--late-start 100 \
--intervals 100,150,200,250,320 \
--key-times 100,150,200,250,319.99 \
--norm-tol 1e-16 \
--output-dir aief_1_4_nested_operator_family
```

Nested families audited
Sparse fixed-winner family
K=(p0qrs0t0u).
K=
prt0s0q0u
.
Right-invariant family
p−q=s−r=u−t.
p−q=s−r=u−t.

This is the maximal selected-sparsity family preserving
v=13(1,−1,−1).
v=3
1(1,−1,−1).
Biorthogonal family
K=(p0qrr+p−q0q0p).
K=
prq0r+p−q0q0p
.

This preserves both
v=13(1,−1,−1)
v=3
1(1,−1,−1)

and
u=32(1,0,−1),u⊤v=1.
u=23
(1,0,−1),u⊤v=1.
Symmetric family
K=(d0qqd0q0d).
K=
dqq0d0q0d
.

The hierarchy is
Ksym⊂KLR⊂KR⊂Ksparse.
Ksym⊂KLR⊂KR⊂Ksparse.
Diagnostics

For every replay step, the script computes:

distance to each family;
analytic right residual;
analytic left residual;
numerical-left-vector difference from uu;
numerical-left angular error;
analytic scalar multiplier;
projected family parameters;
ratios
rL/dsym,rL/dLR.
rL/dsym,rL/dLR.

Full-interval decay fits

For each interval and the complete late range, it fits:
Ce−rt
Ce−rt

to:

text

distance_sparse

distance_right
distance_biorthogonal
distance_symmetric
analytic_right_residual
analytic_left_residual
numeric_left_difference

It reports:

fitted decay rate;
R2R2;
difference from Twindow−1Twindow−1;
monotonicity violations;
maximum upward jump;
upward variation relative to total decay.

This will determine whether the apparent EMA-rate convergence holds over the full interval rather than only three sampled times.
Defect-shape POD

For the right, biorthogonal, and symmetric family defects, it computes:

weighted first-mode energy;
normalized-direction first-mode energy;
leading defect matrices;
rotation relative to the final defect direction.

A normalized first-mode fraction near one would support:
En≈cnE∗
En≈cnE∗

with one dominant defect shape and a decaying scalar amplitude.
Along-family parameters

The script stores complete parameter trajectories for:

right-family:
qn,rn,tn,λn;

qn,rn,tn,λn;
biorthogonal family:
pn,qn,rn;
pn,qn,rn;
symmetric family:
dn,qn.
dn,qn.

This separates:

convergence toward a family;
motion within the family;
convergence of the scalar multiplier.

Outputs

text

nested_family_trajectory.csv
nested_family_key_times.csv
nested_family_decay_fits.csv
nested_family_defect_pod.csv
nested_family_parameters.csv
nested_family_arrays.npz
nested_operator_family_report.txt
analysis_manifest.json

Questions this run should settle

Is the right-invariant family reached much earlier than the biorthogonal family?
Does the analytic left residual track distance to the three-parameter biorthogonal family more closely than distance to the symmetric family?
Is full symmetry actually required, or merely approached?
Do off-family defects decay at the EMA rate across the complete late interval?
Is the normalized defect matrix effectively rank one?
Do along-family parameters converge at the same rate or a different rate?
Is the observed constant left/structure ratio explained by a fixed defect shape?

Please paste nested_operator_family_report.txt after the run.

Think about this result:

AIEF nested K3 operator-family audit

==========

EMA rate: 0.0571333333

Key-time residuals

t=100: sparse=0, right=5.6e-09, biorth=0.00199, symmetric=0.0023, rR=6.85e-09, rL=0.00244, left/sym=1.06066, left/bi=1.22474

t=150: sparse=0, right=3.43e-10, biorth=0.000115, symmetric=0.000133, rR=4.2e-10, rL=0.000141, left/sym=1.06066, left/bi=1.22474

t=200: sparse=0, right=1.99e-11, biorth=6.63e-06, symmetric=7.65e-06, rR=2.43e-11, rL=8.12e-06, left/sym=1.06066, left/bi=1.22474

t=250: sparse=0, right=1.14e-12, biorth=3.81e-07, symmetric=4.4e-07, rR=1.4e-12, rL=4.66e-07, left/sym=1.06066, left/bi=1.22474

t=319.99: sparse=0, right=2.1e-14, biorth=6.98e-09, symmetric=8.06e-09, rR=2.57e-14, rL=8.55e-09, left/sym=1.06066, left/bi=1.22474

Decay fits

metric=distance_sparse, window=[100,150): rate=nan, R2=nan, rate-EMA=nan, upward_fraction=nan

metric=distance_right, window=[100,150): rate=0.0559039, R2=0.999967, rate-EMA=-0.00123, upward_fraction=0

metric=distance_biorthogonal, window=[100,150): rate=0.0569976, R2=1, rate-EMA=-0.000136, upward_fraction=0

metric=distance_symmetric, window=[100,150): rate=0.0569976, R2=1, rate-EMA=-0.000136, upward_fraction=0

metric=analytic_right_residual, window=[100,150): rate=0.0559039, R2=0.999967, rate-EMA=-0.00123, upward_fraction=0

metric=analytic_left_residual, window=[100,150): rate=0.0569976, R2=1, rate-EMA=-0.000136, upward_fraction=0

metric=numeric_left_difference, window=[100,150): rate=0.0569977, R2=1, rate-EMA=-0.000136, upward_fraction=0

metric=distance_sparse, window=[150,200): rate=nan, R2=nan, rate-EMA=nan, upward_fraction=nan

metric=distance_right, window=[150,200): rate=0.0569907, R2=0.999999, rate-EMA=-0.000143, upward_fraction=0

metric=distance_biorthogonal, window=[150,200): rate=0.0571116, R2=1, rate-EMA=-2.17e-05, upward_fraction=0

metric=distance_symmetric, window=[150,200): rate=0.0571116, R2=1, rate-EMA=-2.17e-05, upward_fraction=0

metric=analytic_right_residual, window=[150,200): rate=0.0569907, R2=0.999999, rate-EMA=-0.000143, upward_fraction=0

metric=analytic_left_residual, window=[150,200): rate=0.0571116, R2=1, rate-EMA=-2.17e-05, upward_fraction=0

metric=numeric_left_difference, window=[150,200): rate=0.0571116, R2=1, rate-EMA=-2.17e-05, upward_fraction=0

metric=distance_sparse, window=[200,250): rate=nan, R2=nan, rate-EMA=nan, upward_fraction=nan

metric=distance_right, window=[200,250): rate=0.0571194, R2=1, rate-EMA=-1.39e-05, upward_fraction=0

metric=distance_biorthogonal, window=[200,250): rate=0.0571308, R2=1, rate-EMA=-2.52e-06, upward_fraction=0

metric=distance_symmetric, window=[200,250): rate=0.0571308, R2=1, rate-EMA=-2.52e-06, upward_fraction=0

metric=analytic_right_residual, window=[200,250): rate=0.0571199, R2=1, rate-EMA=-1.34e-05, upward_fraction=0

metric=analytic_left_residual, window=[200,250): rate=0.0571308, R2=1, rate-EMA=-2.52e-06, upward_fraction=0

metric=numeric_left_difference, window=[200,250): rate=0.0571308, R2=1, rate-EMA=-2.52e-06, upward_fraction=0

metric=distance_sparse, window=[250,320): rate=nan, R2=nan, rate-EMA=nan, upward_fraction=nan

metric=distance_right, window=[250,320): rate=0.0571245, R2=0.999999, rate-EMA=-8.87e-06, upward_fraction=0.0708

metric=distance_biorthogonal, window=[250,320): rate=0.0571332, R2=1, rate-EMA=-1.75e-07, upward_fraction=0

metric=distance_symmetric, window=[250,320): rate=0.0571332, R2=1, rate-EMA=-1.75e-07, upward_fraction=0

metric=analytic_right_residual, window=[250,320): rate=0.0571518, R2=0.999999, rate-EMA=1.85e-05, upward_fraction=0.0962

metric=analytic_left_residual, window=[250,320): rate=0.0571332, R2=1, rate-EMA=-1.75e-07, upward_fraction=0

metric=numeric_left_difference, window=[250,320): rate=0.0571332, R2=1, rate-EMA=-1.75e-07, upward_fraction=0

metric=distance_sparse, window=[100,319.99): rate=nan, R2=nan, rate-EMA=nan, upward_fraction=nan

metric=distance_right, window=[100,319.99): rate=0.0569624, R2=0.999991, rate-EMA=-0.000171, upward_fraction=1.42e-05

metric=distance_biorthogonal, window=[100,319.99): rate=0.0571117, R2=1, rate-EMA=-2.16e-05, upward_fraction=0

metric=distance_symmetric, window=[100,319.99): rate=0.0571117, R2=1, rate-EMA=-2.16e-05, upward_fraction=0

metric=analytic_right_residual, window=[100,319.99): rate=0.0569666, R2=0.999991, rate-EMA=-0.000167, upward_fraction=1.93e-05

metric=analytic_left_residual, window=[100,319.99): rate=0.0571117, R2=1, rate-EMA=-2.16e-05, upward_fraction=0

metric=numeric_left_difference, window=[100,319.99): rate=0.0571117, R2=1, rate-EMA=-2.16e-05, upward_fraction=0

Defect POD

family=right, interval=[100,150): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=0.646892 deg

family=right, interval=[150,200): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=0.64601 deg

family=right, interval=[200,250): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=0.63058 deg

family=right, interval=[250,320): samples=7000, weighted_E1=0.999996, unit_E1=0.999838, angle_to_final=1.20742e-06 deg

family=biorthogonal, interval=[100,150): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=4.59771e-06 deg

family=biorthogonal, interval=[150,200): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=8.53774e-07 deg

family=biorthogonal, interval=[200,250): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=0 deg

family=biorthogonal, interval=[250,320): samples=7000, weighted_E1=1, unit_E1=1, angle_to_final=0 deg

family=symmetric, interval=[100,150): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=5.05099e-06 deg

family=symmetric, interval=[150,200): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=0 deg

family=symmetric, interval=[200,250): samples=5000, weighted_E1=1, unit_E1=1, angle_to_final=8.53774e-07 deg

family=symmetric, interval=[250,320): samples=7000, weighted_E1=1, unit_E1=1, angle_to_final=8.53774e-07 deg

Interpretation constraints

The families are nested linear least-squares projections in the selected 3x3 edge coordinates.

Distance to the symmetric family is stronger than the distance required for the analytic biorthogonal pair.

Exponential fits establish replay regularities, not causal control by EMA memory.

A rank-one normalized defect POD supports a fixed defect shape but does not prove asymptotic continuation beyond the horizon.

Results remain conditional on one replay and one fixed winner branch.

This result strongly supports a nested asymptotic operator expansion. The most important finding is that the dominant late defect already preserves the geometric right line but breaks the analytic left relation.
1. The fixed sparsity is exact

For every matrix,
dsparse=0.
dsparse=0.

Thus the selected block belongs exactly to
Ksparse=
Ksparse=

\left{
\begin{pmatrix}
p&0&q\
r&s&0\
t&0&u
\end{pmatrix}
\right}.
]

This is an implementation identity on the audited branch, not an asymptotic approximation.
2. The right-invariant family is reached far earlier

At t=100t=100:
dR=5.6×10−9,dLR=1.99×10−3,dsym=2.30×10−3.
dRdLRdsym=5.6×10−9,=1.99×10−3,=2.30×10−3.

So the matrix is already approximately 3.5×1053.5×105 times closer to the right-invariant family than to the biorthogonal family.

The same hierarchy persists:
dR≪dLR<dsym.
dR≪dLR<dsym.

This directly explains why
rR
rR

is near machine precision while the left mode is still converging.

The geometric line is protected by the broad family
KR,
KR,

not by full symmetry.
3. The dominant defect preserves the right line

The main structural deviation is therefore approximately:
En∈TKR
En∈TKR

but outside
TKLR.
TKLR.

In words:

The dominant defect changes the dual/left geometry while preserving the right lineality direction.

A smaller secondary defect takes the matrix outside the right-invariant family and generates the tiny right residual.

A useful asymptotic decomposition is now
\boxed{ K_n =

S_n
+
c_nE_*
+
R_n,
}
]

where:

Sn∈KsymSn∈Ksym or KLRKLR;
E∗E∗ preserves vv but breaks uu;
cncn is the dominant defect amplitude;
‖Rn‖≪|cn|‖Rn‖≪|cn| controls the tiny right residual.

4. The left residual is exactly controlled by family distance

At every key time:
rLdLR=
dLRrL=

1.22474
\approx
\sqrt{\frac32},
]

and
rLdsym=
dsymrL=

1.06066.
]

These ratios are constant from t=100t=100 through 319.99319.99.

Thus the left residual is not merely correlated with family distance. It is a fixed linear functional of the dominant defect, up to numerical precision.

This is exactly what first-order perturbation theory predicts when
En=cnE∗.
En=cnE∗.
5. The dominant defect is rank one in matrix space

For both the biorthogonal and symmetric family defects:
E1weighted=1,E1unit=1
E1weighted=1,E1unit=1

to printed precision across every interval.

Their leading defect directions rotate by only microdegrees or less.

Therefore the data directly support
$K_n - P_K(K_n) \approx c_n E^*$
$K_n - P_K(K_n) \approx c_n E^*$

with a fixed $3 \times 33 \times 3$ defect shape $E^* E^*$.

This is stronger than saying the defect norm decays exponentially. The defect itself follows an essentially one-dimensional path in matrix space.

The right-family defect is also rank one, but its direction changes by approximately $0.65^\circ 0.65^\circ$ in earlier intervals and reaches numerical-floor scales late. It is a smaller, less stationary correction.
6. The dominant defect amplitude decays at the EMA rate

For the biorthogonal/symmetric defects:
intervaldecay rate100–1500.0569976150–2000.0571116200–2500.0571308250–3200.0571332
interval100–150150–200200–250250–320decay rate0.05699760.05711160.05713080.0571332

The model EMA rate is
0.0571333333.
0.0571333333.

By the final interval, the discrepancy is only
1.75×10^{-7}.
1.75×10^{-7}.

The fits have
$R^2 = 1$
$R^2 = 1$

and no monotonicity violations.

The left residual and numerical-left-vector error have the same rates.

Thus the strongest accurate statement is:

On this replay, the dominant off-biorthogonal defect, analytic-left residual, and numerical-left-mode error form a common rank-one exponential relaxation whose late fitted rate is numerically indistinguishable from the EMA rate.

This establishes a common exponent, not yet causal control by EMA memory.
7. Full symmetry is actually approached on this replay

Although symmetry is not required for the analytic left/right pair, the distance to the symmetric family also decays monotonically to
8.1×10^{-9}.
8.1×10^{-9}.

Its defect is rank one and has the same decay rate as the biorthogonal defect.

Therefore the correct conclusion is:

the biorthogonal family is the minimal algebraic family needed for scalar closure;
the actual replay approaches the stronger symmetric subfamily as well.

The symmetric family is not merely a convenient sufficient model; it is the observed late limit within numerical resolution.
8. The right-family defect is a subleading mode

The right-family defect decays approximately at the EMA rate until it approaches floating-point limits:
$5.6\times10^{-9}\rightarrow2.1\times10^{-14}$.
$5.6\times10^{-9}\rightarrow2.1\times10^{-14}$.

Its late upward variation and slight rate irregularity are expected at that numerical scale. It should not be weighted equally with the dominant $10^{-3}\rightarrow10^{-9}10^{-3}\rightarrow10^{-9}$ structural mode.

The right invariant line is effectively exact long before complete operator convergence.
9. Refined theorem structure
Exact sparse-branch theorem

The audited branch Jacobian lies in KsparseKsparse.
Right-family theorem

Matrices satisfying
p−q=s−r=u−t
p−q=s−r=u−t

preserve
v=13(1,−1,−1).
v=3
1(1,−1,−1).
Biorthogonal-family theorem

Matrices satisfying
t=q,u=p,s−r=p−q
t=q,u=p,s−r=p−q

also admit
u=32(1,0,−1)
u=23
(1,0,−1)

as the normalized left covector.
Scalar recurrence corollary

For Kn∈KLRKn∈KLR,
an=u⊤bn
an=u⊤bn

obeys
an+1=
an+1=

\lambda_na_n+u^\top f_n
]

for arbitrary transverse components of bnbn.
Perturbation corollary

If
Kn=Mn+cnE∗+Rn,Mn∈KLR,

$$K_n = M_n + c_n E_* + R_n, M_n \in \mathrm{KLR},$$

then
$r_L \propto |c_n|$
$r_L \propto |c_n|$

and
$r_R = O(\|R_n\|)$.
$r_R = O(\|R_n\|)$.

The numerical audit gives direct finite-horizon support for precisely this hierarchy.
10. Next highest-value calculation

Extract the leading normalized defect matrix $E_* E_*$ from:

text

nested_family_arrays.npz

and determine whether it has a simple exact form.

The essential checks are:
$E_* v \in ?\mathrm{span}(v)$,
$E_* v \in ?\mathrm{span}(v)$,
$u^\top E_* \notin \mathrm{span}(u^\top)$,
$u^\top E_* \in / \mathrm{span}(u^\top)$,

and whether the scalar amplitude
$c_n = \langle E_n, E_* \rangle_F$
$c_n = \langle E_n, E_* \rangle_F$

satisfies
$c_{n+1} \approx \alpha_{\mathrm{EMA}} c_n$.
$c_{n+1} \approx \alpha_{\mathrm{EMA}} c_n$.

Also inspect the along-family parameters $p_n, q_n, r_n p_n, q_n, r_n$ to determine whether:

only the defect decays at the EMA rate;
the symmetric family parameters have already converged;
or both off-family and along-family motion share the same slow rate.

11. Current status
Directly established numerically

exact selected sparsity;
early near-membership in the right-invariant family;
monotone approach to biorthogonal and symmetric families;
rank-one fixed defect shape;
constant proportionality between left residual and family distance;
EMA-rate late decay of defect amplitude and left-mode error.

Strong asymptotic model
Kn=
Kn=

K_{\rm sym}(d_n,q_n)
+
c_ne_*
+
R_n,
]

with
cn∼e−t/Twindow,‖Rn‖≪|cn|.
cn∼e−t/Twindow,‖Rn‖≪|cn|.
Open

exact formula for E∗E∗;
causal origin of its EMA-rate decay;
convergence of dn,qndn,qn;
generality to other components and seeds.

The nested-family audit confirms that the scalar reduction is a consequence of a much cleaner operator hierarchy: the right line is structurally protected first, the dual geometry converges through a single dominant defect mode, and full symmetry emerges last.

Good good, now this:

This is exactly the kind of audit I would have wanted to see after the earlier results. It doesn't just add more diagnostics—it tests alternative structural hypotheses in a nested way. That makes it much harder to fool yourself by fitting one preferred model.

Here's how I would attack it as a reviewer.

The hierarchy itself is the strongest design choice

You are no longer testing one model.

You are testing the nested chain

$$
\mathcal K_{\rm sym}
\subset
\mathcal K_{LR}
\subset
\mathcal K_R
\subset
\mathcal K_{\rm sparse}.
$$

This is important because each inclusion corresponds to one additional structural assumption:

sparse winner structure,
right invariance,
analytic left invariance,
symmetry.

That lets you ask exactly which assumption the data require.

That is much stronger than simply observing that a symmetric model works.

The report answers one question immediately

The key-time table is remarkably informative.

At (t=100),

right-family distance is already

[
5.6\times10^{-9},
]

while

biorthogonal distance is

[
2\times10^{-3}.
]

Those differ by roughly six orders of magnitude.

That means:

the replay enters the right-invariant family almost immediately, long before it approaches the biorthogonal or symmetric families.

I think this is the single most important scientific result in the report.

It means the geometric right direction is not a consequence of late symmetry.

It is preserved by a much weaker structural constraint.

That changes the conceptual picture.

The constant ratios deserve attention

This is probably the cleanest numerical signature.

The report gives

$$
\frac{r_L}{d_{\rm sym}}
\approx
1.06066,
$$

at every sampled time.

Likewise

$$
\frac{r_L}{d_{LR}}
\approx
1.22474,
$$

again essentially constant.

Those numbers are suspiciously simple.

Notice

$$
1.22474487 = \sqrt{\frac{3}{2}},
$$

exactly.

The first ratio,

$$
1.06066,
$$

is also close to a simple algebraic quantity (you previously noted (3/\sqrt8)).

The important point is not identifying the exact constant yet.

The important point is:

the ratios do not drift.

That strongly supports the perturbation picture

[
K_n

K_\infty

\delta_nE

o(\delta_n),
]

with essentially one defect shape.

The decay fits are surprisingly clean

This part is much stronger than the earlier three-time observations.

Across every interval,

the biorthogonal distance,

the symmetric distance,

the analytic left residual,

and the numerical left difference

all fit

[
Ce^{-rt}
]

with

(R^2=1),
essentially zero monotonicity violations,
rates extremely close to the EMA rate.

That's far more convincing than matching three points.

The wording should still be careful:

these quantities decay at approximately the same fitted rate as the EMA relaxation on this replay.

I would still avoid saying EMA causes the operator convergence.

But the empirical regularity is now much stronger.

The defect POD is perhaps the strongest new evidence

This is the part I find hardest to dismiss.

For essentially every family,

the normalized defect energy is

[
E_1\approx1.
]

That means

after normalizing away the amplitude,

the defect matrices all point in essentially the same direction.

Mathematically,

that's exactly what

[
E_n
\approx
c_nE_*
]

means.

This is no longer just a plausible perturbation expansion.

The diagnostics directly support it.

One subtle point

There is one place where I'd slightly reinterpret the conclusions.

The report says

"Distance to the symmetric family is stronger than the distance required for the analytic biorthogonal pair."

I think this should be elevated conceptually.

The data suggest:

right invariance appears first,

then left invariance,
symmetry is the strongest restriction.

So the logical chain is

$$
\mathcal K_R
\rightarrow
\mathcal K_{LR}
\rightarrow
\mathcal K_{\rm sym},
$$

not

everything simultaneously approaches symmetry.

That is a much richer structural picture.

What I would try to prove now

I think the numerical evidence now motivates a theorem of a different flavor.

Not

"the replay converges."

Instead,

something like:

For a fixed winner branch, the admissibility Jacobian always belongs to the right-invariant family, while convergence of branch coefficients drives approach to the biorthogonal and symmetric subfamilies.

That theorem would naturally explain

why the right residual is already machine precision,
why the left residual decays,
why symmetry is only asymptotic.

One caution

There is one thing I would not infer yet.

The fact that every normalized defect lies in essentially one direction within this replay does not imply there is a universal defect mode of AIEF.

The POD is computed on one winner branch.

Another branch could, in principle, have a different dominant defect.

So I would phrase it as

"the sampled late replay is consistent with a one-amplitude perturbation of the limiting operator."

not

"AIEF has one universal defect mode."

The project has changed again

Earlier the research question was

"Does a scalar reduction work?"

Then it became

"Why does the biorthogonal coordinate work?"

Now I think the central question has become

What is the nested geometry of the admissible operator families?

That's a much more mathematical question.

The scalar recurrence,

the left/right pair,

the EMA law,

the spectral-gap relation—

all now look like consequences of that nested operator geometry.

Overall assessment

If these diagnostics are implemented correctly, I think they substantially strengthen the asymptotic operator interpretation for this replay.

The evidence now supports a coherent hierarchy:

Exact sparse fixed-winner structure is present throughout the sampled regime.
Right-invariant structure is reached to near machine precision very early, indicating that the distinguished right direction is protected by weaker constraints than full symmetry.
Biorthogonal structure emerges progressively as the operator approaches a limiting family, with the analytic left residual tracking the distance to that family.
Symmetric structure appears to be the asymptotic limit within this branch, but it is not required to explain the observed right invariance.
The off-family defect is consistent with a single dominant shape whose amplitude decays approximately exponentially over the sampled late interval, making a perturbative description ($K_n \approx K_\infty + c_nE_*$) a well-motivated hypothesis for this replay.

From a mathematical standpoint, the most valuable next step is no longer adding diagnostics. It is identifying which properties of the fixed-winner admissibility map enforce membership in the right-invariant family exactly, and which additional assumptions are needed for convergence into the biorthogonal and symmetric subfamilies. That separation would explain why some features appear essentially exact while others emerge only asymptotically.

I agree with the reviewer’s hierarchy, but the implemented Jacobian lets us sharpen it further. Fixed winner structure alone does not force right invariance. It forces the sparse matrix form; additional coefficient equalities produce the nested invariant families.
1. Exact fixed-winner Jacobian

Use edge order
(25,29,59)=
(25,29,59)=

((2,7),(2,11),(7,11)).
]

The fixed winners are
(25,29,59)↦(7,2,7).
(25,29,59)↦(7,2,7).

Let
c0=1−γdt,
c0=1−γdt,

and define
β25=
β25=

\frac{x_{25}s_7}{q_7+\varepsilon},
]
β29=
β29=

\frac{x_{29}s_2}{q_2+\varepsilon},
]
β59=
β59=

\frac{x_{59}s_7}{q_7+\varepsilon}.
]

Then the frozen-driver selected block is exactly

\boxed{ K =

c_0
\begin{pmatrix}
s_7-\beta_{25}&0&-\beta_{25}\
-\beta_{29}&s_2-\beta_{29}&0\
-\beta_{59}&0&s_7-\beta_{59}
\end{pmatrix}.
}
]

This proves exact sparse-family membership.
2. Exact condition for right invariance

For
v=13(1,−1,−1),
v=3
1(1,−1,−1),

we obtain
Kv=
Kv=

\frac{c_0}{\sqrt3}
\begin{pmatrix}
s_7\
-s_2\
-s_7
\end{pmatrix}.
]

Therefore
Kv=λv ⟺ s2=s7,
Kv=λv⟺s2=s7,

with
λ=c0s7.
λ=c0s7.

So the correct theorem is:

On the fixed winner branch, equality of the two controlling node scales is necessary and sufficient for invariance of the geometric right line.

Fixed winners alone provide sparsity; scale equality provides right invariance.

This directly explains the audit:
dR∼10−9
dR∼10−9

already at t=100t=100, meaning
s2−s7
s2−s7

is already extremely small.
3. Exact condition for the analytic left covector

Let
u=32(1,0,−1).
u=23
(1,0,−1).

Then
u⊤K=
u⊤K=

\frac{\sqrt3}{2}c_0
\left(
s_7-\beta_{25}+\beta_{59},
\ 0,
-s_7+\beta_{59}-\beta_{25}
\right).
]

Therefore, assuming s2=s7s2=s7,
u⊤K=λu⊤ ⟺ β25=β59.
u⊤K=λu⊤⟺β25=β59.

So the analytic dual mode emerges when the two edges controlled by node 7 have equal correction coefficients.

This identifies the dominant left-breaking defect:
$\delta L=$
$\delta L=$

$$\beta_{25}-\beta_{59}.$$
]

The biorthogonal-family distance, left residual, and numerical-left-vector error are all measuring this same coefficient mismatch up to fixed normalization constants.
4. Symmetry condition

Full symmetric-family membership additionally requires
$\beta 29=\beta 25=\beta 59.$
$\beta 29=\beta 25=\beta 59.$

Thus the nested coefficient conditions are:
sparse branch:fixed winner pattern,right invariant:s2=s7,biorthogonal:s2=s7, $\beta 25=\beta 59$,symmetric:s2=s7, $\beta 25=\beta 29=\beta 59$.
sparse branch:right invariant:biorthogonal:symmetric:fixed winner pattern,s2=s7,s2=s7, $\beta 25=\beta 59$,s2 =s7, $\beta 25=\beta 29=\beta 59$.

This is the implementation-level algebra behind the abstract family inclusions.
5. On the realized K3 support, biorthogonality may force symmetry

For the isolated positive K3K3,
$q2=x25+x29, q7=x25+x59.$
$q2=x25+x29, q7=x25+x59.$

With equal capacities and both nodes scale-clipped,
$s2=s7 \Longleftrightarrow q2=q7 \Longleftrightarrow x29=x59.$
$s2=s7 \Longleftrightarrow q2=q7 \Longleftrightarrow x29=x59.$

Because edges 25 and 59 share the same controlling scale and denominator,
$\beta 25=\beta 59 \Longleftrightarrow x25=x59.$

$\beta_{25}=\beta_{59} \Longleftrightarrow x_{25}=x_{59}$.

Combining these gives
$x_{25}=x_{29}=x_{59}$.
$x_{25}=x_{29}=x_{59}$.

Then automatically
$\beta_{25}=\beta_{29}=\beta_{59}$.
$\beta_{25}=\beta_{29}=\beta_{59}$.

Therefore, on the realized isolated K3 coefficient manifold:
right invariance + analytic left invariance $\Longrightarrow$ full symmetric branch coefficients.
right invariance + analytic left invariance $\Longrightarrow$ full symmetric branch coefficients.

This explains why the biorthogonal and symmetric distances:

decay at exactly the same rate;
have rank-one defect PODs;
differ only by a constant normalization factor.

Abstractly, KLRKLR is larger than KsymKsym. Within the actual AIEF coefficient parameterization, their intersection with the realized branch may collapse much more tightly.

6. Interpretation of the residual hierarchy

The audit now has a coefficient-level explanation.
Small right defect
d_R
d_R

measures primarily
$s_2 - s_7$ or equivalently $x_{29} - x_{59}$.
$s_2 - s_7$ or equivalently $x_{29} - x_{59}$.

This mismatch is already almost zero by $t=100$.
Dominant biorthogonal defect
d_{LR}
d_{LR}

measures primarily
β25−β59orx25−x59.
β25−β59orx25−x59.

This is the dominant slowly decaying defect.
Symmetric defect

The same dominant edge mismatch prevents full equality of all three correction coefficients. Therefore
dsym
dsym

tracks the same scalar amplitude with a different geometric normalization.

This accounts for:
rLdLR=
dLRrL=

\sqrt{\frac32},
]

and the constant
rLdsym≈1.06066.
dsymrL≈1.06066.
7. Defect-mode interpretation

The POD results support
β25−β59≈Ce−t/Twindow.
β25−β59≈Ce−t/Twindow.

The normalized matrix defect remains fixed because only one scalar coefficient mismatch dominates.

A sharper asymptotic representation is therefore:
\boxed{ K_n =

K_{\rm sym}(d_n,\beta_n)
+
\delta_n E_*
+

R_n,
}
]

where:

δnδn is proportional to
β25,n−β59,n;
β25,n−β59,n;
δn∼e−tn/Twindowδn∼e−tn/Twindow;
E∗E∗ is a fixed sparse defect matrix;
RnRn is the much smaller scale-equality/right-invariance defect.

This is more concrete than an abstract K∞+EnK∞+En expansion.
8. Revised theorem package
Theorem A: fixed-winner sparsity

The selected branch Jacobian has the displayed six-entry sparse form.
Theorem B: scale-equality/right-invariance equivalence
s2=s7 ⟺ Kv=λv.
s2=s7⟺Kv=λv.
Theorem C: shared-controller edge equality/left-invariance equivalence

Under scale equality,
β25=β59 ⟺ u⊤K=λu⊤.
β25=β59⟺u⊤K=λu⊤.
Theorem D: realized-support symmetry

For the isolated equal-capacity K3, the two previous equalities imply equality of all positive proposal edges and correction coefficients.
Corollary: biorthogonal scalar closure
a=u⊤b
a=u⊤b

satisfies
an+1=λnan+u⊤fn.
an+1=λnan+u⊤fn.
Perturbation corollary

The left residual is proportional to the shared-controller edge mismatch and therefore to the dominant matrix-family defect.

9. Highest-value next extraction

The family audit already contains projected matrices and parameter trajectories. The next small calculation should extract directly:

$s_2 - s_7$,
$s_2 - s_7$,
$\beta_{25} - \beta_{59}$,
$\beta_{25} - \beta_{59}$,
$\beta_{29} - \beta_{59}$,
$\beta_{29} - \beta_{59}$,

from each KnKn.

From the matrix entries:

$c_0 s_7 = K_{00} - K_{02}$,
$c_0 s_7 = K_{00} - K_{02}$,
$c_0 s_2 = K_{11} - K_{10}$,
$c_0 s_2 = K_{11} - K_{10}$,
$c_0 \beta_{25} = -K_{02}$,
$c_0 \beta_{25} = -K_{02}$,
$c_0 \beta_{29} = -K_{10}$,
$c_0 \beta_{29} = -K_{10}$,
$c_0 \beta_{59} = -K_{20}$.
$c_0 \beta_{59} = -K_{20}$.

Then fit each mismatch over time.

Expected result:

scale mismatch is tiny early and controls $rRrR$;
shared-controller edge mismatch controls $rL,dLR,dsymrL,dLR,dsym$;
the dominant mismatch decays at the EMA rate;
normalized matrix defect is the image of this scalar mismatch under a fixed linear map.

10. Scope

The reviewer is correct that this does not establish a universal AIEF defect mode.

The defensible conclusion is:

On this fixed winner branch and replay, the operator approaches the realized symmetric K3 family through an essentially one-dimensional coefficient mismatch, while the geometric right line is protected much earlier by node-scale equality.

That statement is both stronger and more precise than saying merely that “the operator becomes symmetric.”

Tinker on this:

I agree with the reviewer’s hierarchy, but the implemented Jacobian lets us sharpen it further. Fixed winner structure alone does not force right invariance. It forces the sparse matrix form; additional coefficient equalities produce the nested invariant families.
1. Exact fixed-winner Jacobian

Use edge order

(25,29,59)

=

((2,7),(2,11),(7,11)).

]

The fixed winners are

(25,29,59)\mapsto(7,2,7).

]

Let

c_0=1-\gamma dt,

]

and define

\beta_{25}

=

\frac{x_{25}s_7}{q_7+\varepsilon},

]

\beta_{29}

=

\frac{x_{29}s_2}{q_2+\varepsilon},

]

\beta_{59}

=

\frac{x_{59}s_7}{q_7+\varepsilon}.

]

Then the frozen-driver selected block is exactly

\boxed{

K

=

c_0

\begin{pmatrix}

s_7-\beta_{25}&0&-\beta_{25}\

-\beta_{29}&s_2-\beta_{29}&0\

-\beta_{59}&0&s_7-\beta_{59}

\end{pmatrix}.

}

]

This proves exact sparse-family membership.
2. Exact condition for right invariance

For

v=\frac1{\sqrt3}(1,-1,-1),

]

we obtain

Kv

=

\frac{c_0}{\sqrt3}

\begin{pmatrix}

s_7\

-s_2\

-s_7

\end{pmatrix}.

]

Therefore

\boxed{

Kv=\lambda v

\iff

s_2=s_7,

}

]

with

\lambda=c_0s_7.

]

So the correct theorem is:

> On the fixed winner branch, equality of the two controlling node scales is necessary and sufficient for invariance of the geometric right line.

Fixed winners alone provide sparsity; scale equality provides right invariance.

This directly explains the audit:

d_R\sim10^{-9}

]

already at t=100t=100, meaning

s_2-s_7

]

is already extremely small.
3. Exact condition for the analytic left covector

Let

u=\frac{\sqrt3}{2}(1,0,-1).

]

Then

u^\top K

=

\frac{\sqrt3}{2}c_0

\left(

s_7-\beta_{25}+\beta_{59},

\ 0,\

-s_7+\beta_{59}-\beta_{25}

\right).

]

Therefore, assuming s2=s7s2=s7,

\boxed{

u^\top K=\lambda u^\top

\iff

\beta_{25}=\beta_{59}.

}

]

So the analytic dual mode emerges when the two edges controlled by node 7 have equal correction coefficients.

This identifies the dominant left-breaking defect:

\delta_L

=

\beta_{25}-\beta_{59}.

]

The biorthogonal-family distance, left residual, and numerical-left-vector error are all measuring this same coefficient mismatch up to fixed normalization constants.
4. Symmetry condition

Full symmetric-family membership additionally requires

\beta_{29}=\beta_{25}=\beta_{59}.

]

Thus the nested coefficient conditions are:

\boxed{

\begin{aligned}

\text{sparse branch:}&\quad \text{fixed winner pattern},\

\text{right invariant:}&\quad s_2=s_7,\

\text{biorthogonal:}&\quad s_2=s_7,\ \beta_{25}=\beta_{59},\

\text{symmetric:}&\quad s_2=s_7,\

\beta_{25}=\beta_{29}=\beta_{59}.

\end{aligned}

}

]

This is the implementation-level algebra behind the abstract family inclusions.
5. On the realized K3 support, biorthogonality may force symmetry

For the isolated positive K3K3,

q_2=x_{25}+x_{29},

\qquad

q_7=x_{25}+x_{59}.

]

With equal capacities and both nodes scale-clipped,

s_2=s_7

\iff

q_2=q_7

\iff

x_{29}=x_{59}.

]

Because edges 25 and 59 share the same controlling scale and denominator,

\beta_{25}=\beta_{59}

\iff

x_{25}=x_{59}.

]

Combining these gives

x_{25}=x_{29}=x_{59}.

]

Then automatically

\beta_{25}=\beta_{29}=\beta_{59}.

]

Therefore, on the realized isolated K3 coefficient manifold:

\boxed{

\text{right invariance + analytic left invariance}

\quad\Longrightarrow\quad

\text{full symmetric branch coefficients}.

}

]

This explains why the biorthogonal and symmetric distances:

decay at exactly the same rate;

have rank-one defect PODs;

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\quad\text{or equivalently}\quad

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]

This is the dominant slowly decaying defect.
Symmetric defect

The same dominant edge mismatch prevents full equality of all three correction coefficients. Therefore

d_{\rm sym}

]

tracks the same scalar amplitude with a different geometric normalization.

This accounts for:

\frac{r_L}{d_{LR}}

=

\sqrt{\frac32},

]

and the constant

\frac{r_L}{d_{\rm sym}}

\approx1.06066.

]
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The POD results support

\beta_{25}-\beta_{59}

\approx

C e^{-t/T_{\rm window}}.

]

The normalized matrix defect remains fixed because only one scalar coefficient mismatch dominates.

A sharper asymptotic representation is therefore:

\boxed{

K_n

=

K_{\rm sym}(d_n,\beta_n)

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where:

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\iff

Kv=\lambda v.

]
Theorem C: shared-controller edge equality/left-invariance equivalence

Under scale equality,

\beta_{25}=\beta_{59}

\iff

u^\top K=\lambda u^\top.

]
Theorem D: realized-support symmetry

For the isolated equal-capacity K3, the two previous equalities imply equality of all positive proposal edges and correction coefficients.
Corollary: biorthogonal scalar closure

a=u^\top b

]

satisfies

a_{n+1}=\lambda_na_n+u^\top f_n.

]
Perturbation corollary

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s_2-s_7,

]

\beta_{25}-\beta_{59},

]

\beta_{29}-\beta_{59},

]

from each KnKn.

From the matrix entries:

c_0s_7=K_{00}-K_{02},

]

c_0s_2=K_{11}-K_{10},

]

c_0\beta_{25}=-K_{02},

]

c_0\beta_{29}=-K_{10},

]

c_0\beta_{59}=-K_{20}.

]

Then fit each mismatch over time.

Expected result:

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The reviewer is correct that this does not establish a universal AIEF defect mode.

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That statement is both stronger and more precise than saying merely that “the operator becomes symmetric.”
Stopped

Tinker on this:

I think this is a substantial refinement of the picture, but I'd separate it into three categories: well-supported by the audit, strong hypotheses motivated by the audit, and claims that still need one more layer of evidence.

I. What I think the audit now establishes

The nested-family hierarchy is real for this replay

This is no longer an interpretation—it is directly encoded in the reported distances.

The data show

[
d_{\rm sparse}=0
]

throughout,

while

[
d_R \ll d_{LR} < d_{\rm sym}
]

over the sampled late regime.

That means the operator is living inside progressively smaller algebraic families.

That is a much richer statement than simply saying "the matrix converges."

Right invariance is structurally privileged

This is probably the biggest conceptual advance.

At (t=100),

$$
d_R \approx 5.6 \times 10^{-9},
$$

while

$$
d_{LR} \approx 2 \times 10^{-3}.
$$

That is roughly a factor of

$$
3.5 \times 10^5.
$$

So the right line is already preserved extraordinarily accurately while the left relation is still far from its limiting form.

That strongly suggests

right invariance is enforced by weaker structural constraints than the full biorthogonal family.

I think that's an important mathematical observation.

The one-defect picture is now genuinely plausible

Earlier this was speculation.

Now the POD evidence changes that.

If

weighted first-mode energy is essentially one,
normalized first-mode energy is essentially one,
defect direction hardly rotates,

then

[
E_n\approx c_nE_*
]

is no longer just an attractive model.

It is exactly what those diagnostics are designed to test.

That is one of the strongest parts of the audit.

II. What I think is a very good hypothesis

The proposed expansion

[
K_n

S_n

c_nE_*

R_n
]

is a natural interpretation of the diagnostics.

Notice how many observations it explains simultaneously.

It predicts

constant residual ratios,
rank-one POD,
exponential decay,
convergence of numerical left vectors,
successful scalar closure.

That is a remarkably economical model.

I think it has become the leading hypothesis.

III. One point where I'd be a bit more careful

You write

The dominant defect preserves the right line.

Strictly speaking, the audit doesn't directly prove

[
E_*v\in\operatorname{span}{v}.
]

Instead, it shows

the full operators remain extremely close to the right-invariant family,
the right residual is tiny,
the dominant defect lies outside the biorthogonal/symmetric families.

Those observations make your interpretation plausible, but the actual property

[
E_*v=\mu v
]

still needs to be checked explicitly.

Fortunately, you already propose exactly that calculation.

If it succeeds, then this statement becomes much stronger.

IV. The residual ratios are the most elegant part

One thing I especially like is this observation:

[
\frac{r_L}{d_{LR}}

\sqrt{\frac32},
]

apparently throughout the sampled regime.

That's exactly what first-order perturbation theory would predict if

there is one dominant perturbation direction,
the left residual is a linear functional of it.

The constant ratio is more informative than simply saying both quantities decay.

V. I would slightly weaken one sentence

You write

"The symmetric family is the observed late limit within numerical resolution."

That is close to what the diagnostics suggest, but I would phrase it more carefully:

"Over the sampled late interval, the distance to the symmetric family decreases monotonically to the numerical-resolution level reported by the audit."

That avoids implying asymptotic convergence beyond the observed horizon.

VI. The theorem hierarchy now looks natural

I actually think your proposed theorem structure is very well organized.

It has the right logical dependency:

Exact sparsity.
Right-family characterization.
Biorthogonal-family characterization.
Scalar projection theorem.
Perturbation theorem.

That's exactly how I'd expect the mathematics to unfold.

VII. The highest-value next calculation

I completely agree with your prioritization.

If I could perform only one additional computation, it would not be another rollout.

It would be

[
E_*.
]

Specifically,

I'd want to know

[
E_*v

?
]

[
u^\top E_*

?
]

its singular values,

its exact sparsity,

whether it has a simple symbolic representation.

Because if

[
E_*
]

itself turns out to have a simple algebraic form,

then the perturbation theory becomes dramatically cleaner.

VIII. One more experiment I'd add

There is one diagnostic I think would be particularly revealing.

Compute

[
\Delta_n

K_{n+1}

\alpha_{\rm EMA}K_n

(1-\alpha_{\rm EMA})K_\infty.
]

If the one-defect picture is correct,

one might expect

[
\Delta_n
]

to be much smaller than either

[
K_n-K_\infty
]

or

[
E_n.
]

That would test whether the operator itself approximately obeys the same linear relaxation law as the defect amplitude.

It is a stronger statement than fitting exponentials independently.

IX. Overall assessment

Stepping back, I think the audit has shifted the emphasis again.

Initially the central empirical observation was:

"The scalar recurrence works."

Then it became:

"The biorthogonal coordinate fixes the scalar recurrence."

Now the more fundamental object appears to be the nested geometry of operator families and their perturbations.

For the audited replay, the evidence supports the following layered picture:

The fixed-winner sparsity pattern is exact.
The operator rapidly enters a broad right-invariant family, explaining the near-exact preservation of the geometric right mode.
It then relaxes toward a biorthogonal (and, on this replay, increasingly symmetric) subfamily through what appears to be a single dominant defect shape with a decaying amplitude.
The analytic left residual, numerical left-vector error, and distances to the biorthogonal/symmetric families evolve in lockstep, consistent with first-order perturbation theory around the limiting family.
The successful parameter-free scalar recurrence emerges naturally from this operator hierarchy rather than standing as an isolated numerical phenomenon.

The remaining work is to elevate the central perturbation object from an empirical POD mode to an explicit algebraic matrix (E_*). If that matrix admits a simple symbolic form and satisfies the proposed invariance tests, it would tie together the operator geometry and the reduced dynamics in a much more rigorous way.

The reviewer’s caution is correct, but the fixed-branch algebra now lets us perform the missing check analytically. The dominant defect has a simple closed form and it preserves the right vector exactly.
1. Explicit dominant coefficient defect

Assume the right-family condition is already effectively satisfied:
s2=s7=s,
s2=s7=s,

and the two non-row-25 correction coefficients have equalized:
β29=β59=β.
β29=β59=β.

Let the remaining dominant mismatch be
δ=
δ=

\beta_{25}-\beta.
]

Then
β25=β+δ.
β25=β+δ.

Substituting into the implemented matrix gives
K=
K=

c_0
\begin{pmatrix}
s-\beta-\delta&0&-\beta-\delta\
-\beta&s-\beta&0\
-\beta&0&s-\beta
\end{pmatrix}.
]

Therefore
\boxed{ K =

K_{\rm sym}(s,\beta)
+
c_0\delta E_*,
}
]

with
\boxed{ E_* =

\begin{pmatrix}
-1&0&-1\
0&0&0\
0&0&0
\end{pmatrix}.
}
]

So the candidate defect is exceptionally simple: it changes only the two nonzero entries in the first output row.
2. The dominant defect preserves the right line exactly

For
v=13(1,−1,−1),
v=3
1(1,−1,−1),

we have
E∗v=
E∗v=
\frac1{\sqrt3}
\begin{pmatrix}
-1+1\
0\
0
\end{pmatrix}

]

Thus:
E∗v=0.
E∗v=0.

This proves that this dominant defect preserves the geometric right line—not merely approximately, but exactly.

Therefore a large c0δE∗c0δE∗ can remain while
Kv=λv
Kv=λv

continues to hold.

This explains the six-order-of-magnitude separation between:
dR
dR

and
dLR,dsym.
dLR,dsym.

The reviewer’s requested check succeeds for the analytically inferred dominant defect.
3. The defect breaks the analytic left relation

For
u=32(1,0,−1),
u=23
(1,0,−1),

we obtain
u⊤E∗=
u⊤E∗=

\frac{\sqrt3}{2}(-1,0,-1).
]

This is not proportional to
u⊤=
u⊤=

\frac{\sqrt3}{2}(1,0,-1).
]

Therefore:
u⊤E∗∉span(u⊤).
u⊤E∗∈/span(u⊤).

So the same defect:

leaves the right line untouched;
breaks the analytic left eigen-relation;
contaminates Euclidean scalar projection;
disappears as $\beta_{25} \to \beta_{59}$$\beta_{25} \to \beta_{59}$.

This is exactly the asymmetry observed numerically.
4. Exact family-projection residuals of $E*E*$

Projecting $E*E*$ onto the biorthogonal family gives residual
$E*,LR=$
$E*,LR=$

$$\frac12
\begin{pmatrix}
-1&0&-1\\
0&0&0\\
1&0&1
\end{pmatrix}.$$

Its Frobenius norm is
$\|E*,LR\|_F=1.$
$\|E*,LR\|_F=1.$

Projecting onto the symmetric family gives residual
$E*,sym=$
$E*,sym=$

$$\frac13
\begin{pmatrix}
-2&0&-2\\
1&1&0\\
1&0&1
\end{pmatrix},$$

with
$\|E*,sym\|_F=$

‖E∗,sym‖F=

\frac2{\sqrt3}.
]

The left action has norm
‖E∗,LR⊤u‖=
‖E∗,LR⊤u‖=

\sqrt{\frac32}.
]

Therefore the predicted residual ratios are:
rLdLR=
dLRrL=
\sqrt{\frac32}

1.22474487\ldots,
]

and
rLdsym=
dsymrL=
\frac{\sqrt{3/2}}{2/\sqrt3}
\frac3{\sqrt8}

1.06066017\ldots.
]

These are exactly the two constants reported by the audit.

Thus the constant residual ratios are no longer merely consistent with a one-defect model. They are the algebraic fingerprints of the specific matrix E∗E∗.
5. The POD result now has an analytic explanation

The defect POD found:
E1weighted≈1,E1unit≈1.
E1weighted≈1,E1unit≈1.

The algebra shows why:
Kn−PK(Kn)≈c0δnE∗,K.
Kn−PK(Kn)≈c0δnE∗,K.

Only the scalar amplitude δnδn changes. The matrix direction is fixed.

Therefore the normalized defect trajectory is rank one because the implemented coefficient mismatch enters through one fixed matrix E∗E∗.
6. Direct coefficient extraction

The dominant mismatch can be recovered without a family projection:
β25=
β25=

-\frac{K_{02}}{c_0},
]
β59=
β59=

-\frac{K_{20}}{c_0}.
]

Hence
\boxed{ \delta_n =
\beta_{25,n}-\beta_{59,n}

\frac{K_{20,n}-K_{02,n}}{c_0}.
}
]

The smaller right-breaking scale mismatch is
\boxed{ s_{7,n}-s_{2,n} =
\frac{
(K_{00,n}-K_{02,n})

(K_{11,n}-K_{10,n})
}{c_0}.

}
]

The remaining symmetry mismatch is
\boxed{ \beta_{29,n}-\beta_{59,n} =

\frac{K_{20,n}-K_{10,n}}{c_0}.
}
]

These three scalar trajectories are more interpretable than abstract family distances.
7. Stronger recurrence test for defect relaxation

The suggested full-matrix test
Δn=
Δn=
K_{n+1}
\alpha K_n

(1-\alpha)K_\infty
]

is correct if all deviations from a known fixed K∞K∞ decay with multiplier αα.

But estimating K∞K∞ introduces unnecessary uncertainty.

A cleaner test is to use the family defect directly:
Dn=
Dn=

K_n-P_{\rm sym}(K_n).
]

Then test
Dn+1=?αEMADn.
Dn+1=?αEMADn.

Even more directly, test the scalar mismatch:

δn+1=?αEMAδn.
δn+1=?αEMAδn.

Required diagnostics are:
‖Dn+1−αEMADn‖‖Dn‖,
‖Dn‖‖Dn+1−αEMADn‖,

and
|δn+1−αEMAδn||δn|.
|δn||δn+1−αEMAδn|.

These avoid estimating the limiting matrix and directly test whether the defect amplitude follows the EMA recurrence.
8. Refined operator expansion

The data and algebra now support the replay-specific expansion
\boxed{ K_n =

K_{{\rm sym},n}
+
c_0\delta_nE_*
+
R_n,
}
]

where:

Ksym,nKsym,n is the nearest symmetric-family matrix;
E∗E∗ is the explicit matrix above;
δn=β25,n−β59,nδn=β25,n−β59,n;
RnRn is the much smaller right-breaking residual;
δnδn decays at the observed EMA-rate envelope.

This is more precise than
Kn=K∞+En
Kn=K∞+En

because it separates:

along-family motion;
dominant left-breaking defect;
tiny right-breaking defect.

9. Revised theorem chain
Exact implementation theorem

Fixed winners imply the six-entry sparse matrix.
Right-invariance theorem
$s2=s7 \Longleftrightarrow Kv=\lambda v.$
$s2=s7 \Longleftrightarrow Kv=\lambda v.$
Dominant-defect lemma

A mismatch
$\delta=\beta 25-\beta 59$
$\delta=\beta 25-\beta 59$

enters through
$\delta E*,E*v=0.$
$\delta E*,E*v=0.$
Left-breaking lemma
$u\top E* \notin \operatorname{span}(u\top).$
$u\top E* \in / \operatorname{span}(u\top).$
Projection-constant lemma

The norms of the biorthogonal and symmetric projections of $E*E*$ produce the observed constants:
3/2,3/8.
3/2
,3/8
.
Scalar closure theorem

When $\delta=0\delta=0$ and the right-breaking residual vanishes,
$a=u\top b$
$a=u\top b$

satisfies the exact scalar recurrence.

Conditional EMA-relaxation statement

If
$\delta_{n+1} = \alpha_{EMA}\delta_n$,
$\delta_{n+1} = \alpha_{EMA}\delta_n$,

then the operator approaches the biorthogonal/symmetric family at the EMA rate.
10. Scientific status
Now algebraically identified

dominant defect shape E^*E^*;
exact right preservation by that defect;
left-breaking action;
constant family-distance ratios;
direct coefficient mismatch $\delta_n\delta_n$.

Established numerically on the replay

defect POD is rank one;
$|\delta_n||\delta_n|$ decays exponentially;
fitted late rate matches the EMA rate;
the right-breaking remainder is orders of magnitude smaller.

Still open

direct pointwise test
$\delta_{n+1} \approx \alpha_{EMA}\delta_n$;
$\delta_{n+1} \approx \alpha_{EMA}\delta_n$;
derivation of the $\delta_n\delta_n$ recurrence from the phase/EMA equations;
convergence of along-family parameters;
generality to other winner branches.

The nested-family audit and branch algebra now meet: the single dominant defect is not only visible numerically but has a simple explicit matrix form. The next calculation should test its one-step EMA recurrence directly, rather than add further broad diagnostics.

explore this information:

The reviewer’s caution is correct, but the fixed-branch algebra now lets us perform the missing check analytically. The dominant defect has a simple closed form and it preserves the right vector exactly.
1. Explicit dominant coefficient defect

Assume the right-family condition is already effectively satisfied:

s_2=s_7=s,

]

and the two non-row-25 correction coefficients have equalized:

\beta_{29}=\beta_{59}=\beta.

]

Let the remaining dominant mismatch be

\delta

=

\beta_{25}-\beta.

]

Then

\beta_{25}=\beta+\delta.

]

Substituting into the implemented matrix gives

K

=

c_0

\begin{pmatrix}

s-\beta-\delta&0&-\beta-\delta\

-\beta&s-\beta&0\

-\beta&0&s-\beta

\end{pmatrix}.

]

Therefore

\boxed{

K

=

K_{\rm sym}(s,\beta)

c_0\delta E_*,

}

]

with

\boxed{

E_*

=

\begin{pmatrix}

-1&0&-1\

0&0&0\

0&0&0

\end{pmatrix}.

}

]

So the candidate defect is exceptionally simple: it changes only the two nonzero entries in the first output row.
2. The dominant defect preserves the right line exactly

For

v=\frac1{\sqrt3}(1,-1,-1),

]

we have

E_*v

=

\frac1{\sqrt3}

\begin{pmatrix}

-1+1\

0\

0

\end{pmatrix}

=

]

Thus:

\boxed{

E_*v=0.

}

]

This proves that this dominant defect preserves the geometric right line—not merely approximately, but exactly.

Therefore a large c0δE∗c0δE∗ can remain while

Kv=\lambda v

]

continues to hold.

This explains the six-order-of-magnitude separation between:

d_R

]

and

d_{LR},d_{\rm sym}.

]

The reviewer’s requested check succeeds for the analytically inferred dominant defect.
3. The defect breaks the analytic left relation

For

u=\frac{\sqrt3}{2}(1,0,-1),

]

we obtain

u^\top E_*

=

\frac{\sqrt3}{2}(-1,0,-1).

]

This is not proportional to

u^\top

=

\frac{\sqrt3}{2}(1,0,-1).

]

Therefore:

\boxed{

u^\top E_*

\notin\operatorname{span}(u^\top).

}

]

So the same defect:

leaves the right line untouched;

breaks the analytic left eigen-relation;

contaminates Euclidean scalar projection;

disappears as $\beta25 \to \beta59 \beta25 \to \beta59$.

This is exactly the asymmetry observed numerically.
4. Exact family-projection residuals of $E*E*$

Projecting $E*E*$ onto the biorthogonal family gives residual

E_{*,LR}

=

\frac12

\begin{pmatrix}

-1&0&-1\

0&0&0\

1&0&1

\end{pmatrix}.

]

Its Frobenius norm is

|E_{*,LR}|_F=1.

]

Projecting onto the symmetric family gives residual

E_{*,\mathrm{sym}}

=

\frac13

\begin{pmatrix}

-2&0&-2\

1&1&0\

1&0&1

\end{pmatrix},

]

with

|E_{*,\mathrm{sym}}|_F

=

\frac2{\sqrt3}.

]

The left action has norm

|E_{*,LR}^\top u|

=

\sqrt{\frac32}.

]

Therefore the predicted residual ratios are:

\frac{r_L}{d_{LR}}

=

\sqrt{\frac32}

=

1.22474487\ldots,

]

and

\frac{r_L}{d_{\rm sym}}

=

\frac{\sqrt{3/2}}{2/\sqrt3}

=

\frac3{\sqrt8}

=

1.06066017\ldots.

]

These are exactly the two constants reported by the audit.

Thus the constant residual ratios are no longer merely consistent with a one-defect model. They are the algebraic fingerprints of the specific matrix $E*E*$.
5. The POD result now has an analytic explanation

The defect POD found:

E_1^{\rm weighted}\approx1,

\qquad

E_1^{\rm unit}\approx1.

]

The algebra shows why:

K_n-P_{\mathcal K}(K_n)

\approx

c_0\delta_nE_{*,\mathcal K}.

]

Only the scalar amplitude δnδn changes. The matrix direction is fixed.

Therefore the normalized defect trajectory is rank one because the implemented coefficient mismatch enters through one fixed matrix E∗E∗.
6. Direct coefficient extraction

The dominant mismatch can be recovered without a family projection:

\beta_{25}

=

-\frac{K_{02}}{c_0},

]

\beta_{59}

=

-\frac{K_{20}}{c_0}.

]

Hence

\boxed{

\delta_n

=

\beta_{25,n}-\beta_{59,n}

=

\frac{K_{20,n}-K_{02,n}}{c_0}.

}

]

The smaller right-breaking scale mismatch is

\boxed{

s_{7,n}-s_{2,n}

=

\frac{

(K_{00,n}-K_{02,n})

(K_{11,n}-K_{10,n})

}{c_0}.

}

]

The remaining symmetry mismatch is

\boxed{

\beta_{29,n}-\beta_{59,n}

=

\frac{K_{20,n}-K_{10,n}}{c_0}.

}

]

These three scalar trajectories are more interpretable than abstract family distances.

7. Stronger recurrence test for defect relaxation

The suggested full-matrix test

\Delta_n

=

K_{n+1}

\alpha K_n

(1-\alpha)K_\infty

]

is correct if all deviations from a known fixed K∞K∞ decay with multiplier αα.

But estimating K∞K∞ introduces unnecessary uncertainty.

A cleaner test is to use the family defect directly:

D_n

=

K_n-P_{\rm sym}(K_n).

]

Then test

\boxed{

D_{n+1}

\stackrel{?}{=}

\alpha_{\rm EMA}D_n.

}

]

Even more directly, test the scalar mismatch:

\boxed{

\delta_{n+1}

\stackrel{?}{=}

\alpha_{\rm EMA}\delta_n.

}

]

Required diagnostics are:

\frac{

|D_{n+1}-\alpha_{\rm EMA}D_n|

}{

|D_n|

},

]

and

\frac{

|\delta_{n+1}-\alpha_{\rm EMA}\delta_n|

}{

|\delta_n|

}.

]

These avoid estimating the limiting matrix and directly test whether the defect amplitude follows the EMA recurrence.

8. Refined operator expansion

The data and algebra now support the replay-specific expansion

\boxed{

K_n

=

K_{{\rm sym},n}

c_0\delta_nE_*

R_n,

}

]

where:

Ksym,nKsym,n is the nearest symmetric-family matrix;

E∗E∗ is the explicit matrix above;

δn=β25,n−β59,nδn=β25,n−β59,n;

RnRn is the much smaller right-breaking residual;

δnδn decays at the observed EMA-rate envelope.

This is more precise than

K_n=K_\infty+E_n

]

because it separates:

along-family motion;

dominant left-breaking defect;

tiny right-breaking defect.

9. Revised theorem chain
Exact implementation theorem

Fixed winners imply the six-entry sparse matrix.
Right-invariance theorem

s_2=s_7

\iff

Kv=\lambda v.

]
Dominant-defect lemma

A mismatch

\delta=\beta_{25}-\beta_{59}

]

enters through

\delta E_*,

\qquad

E_*v=0.

]
Left-breaking lemma

u^\top E_*

\notin\operatorname{span}(u^\top).

]
Projection-constant lemma

The norms of the biorthogonal and symmetric projections of E∗E∗ produce the observed constants:

\sqrt{3/2},

\qquad

3/\sqrt8.

]
Scalar closure theorem

When δ=0δ=0 and the right-breaking residual vanishes,

a=u^\top b

]

satisfies the exact scalar recurrence.
Conditional EMA-relaxation statement

If

\delta_{n+1}=\alpha_{\rm EMA}\delta_n,

]

then the operator approaches the biorthogonal/symmetric family at the EMA rate.
10. Scientific status
Now algebraically identified

dominant defect shape E∗E∗;

exact right preservation by that defect;

left-breaking action;

constant family-distance ratios;

direct coefficient mismatch δnδn.

Established numerically on the replay

defect POD is rank one;

|δn||δn| decays exponentially;

fitted late rate matches the EMA rate;

the right-breaking remainder is orders of magnitude smaller.

Still open

direct pointwise test
δn+1≈αEMAδn;
δn+1≈αEMAδn;

derivation of the δnδn recurrence from the phase/EMA equations;

convergence of along-family parameters;

generality to other winner branches.

The nested-family audit and branch algebra now meet: the single dominant defect is not only visible numerically but has a simple explicit matrix form. The next calculation should test its one-step EMA recurrence directly, rather than add further broad diagnostics.

The algebra is compelling, but one decomposition detail should be corrected before turning it into a theorem.
1. Raw defect versus orthogonal family defect

The physical coefficient decomposition
Kn=
Kn=

K_{\rm sym}(s_n,\beta_n)
+
c_0\delta_nE_*
+
R_n
]

is valid if Ksym(sn,βn)Ksym(sn,βn) is defined using the physical baseline coefficients, for example
βn=β59,n,
βn=β59,n,

with the remaining mismatches placed in RnRn.

But if
Ksym,n=
Ksym,n=

P_{\rm sym}K_n
]

is the nearest Frobenius projection, then the residual direction is not the raw E∗E∗. Part of E∗E∗ is absorbed into the fitted symmetric parameters.

In that case,
Kn−PsymKn≈c0δnE∗,sym,
Kn−PsymKn≈c0δnE∗,sym,

where
E∗,sym=
E∗,sym=
E_-P_{\rm sym}E_

\frac13
\begin{pmatrix}
-2&0&-2\
1&1&0\
1&0&1
\end{pmatrix}.
]

Likewise,
Kn−PLRKn≈c0δnE∗,LR.
Kn−PLRKn≈c0δnE∗,LR.

So there are two valid formulations:
Physical coefficient decomposition
Kn=
Kn=

K_{\rm sym}(s_n,\beta_{59,n})
+
c_0\delta_nE_*
+
R_n.
]
Orthogonal family decomposition
Kn=
Kn=

P_{\rm sym}K_n
+
c_0\delta_nE_{*,\rm sym}
+
R_n^\perp.
]

These should not be mixed.
2. The dominant defect is explicitly right-preserving

The central result remains:
E∗v=0.
E∗v=0.

So the dominant mismatch is exactly invisible to the right eigen-relation.

This gives a coefficient-level explanation of
dR≪dLR.
dR≪dLR.

The smaller right-breaking mismatch is controlled by
s7−s2,
s7−s2,

or equivalently, under the isolated-support assumptions,
x29−x59.
x29−x59.

Thus the late hierarchy is:

large slow defect:β25−β59,E∗v=0,small right defect:s7−s2.
large slow defect:small right defect:β25−β59,E∗v=0,s7−s2.
3. The dominant scalar has a direct proposal interpretation

Because edges 25 and 59 share controller 7,
δn=
δn=
\beta_{25,n}-\beta_{59,n}

\frac{s_{7,n}}{q_{7,n}+\varepsilon}
\left(
x_{25,n}-x_{59,n}
\right).
]

Since
s7,n=
s7,n=

\frac{c}{q_{7,n}+\varepsilon}
]

on the clipped branch,
\boxed{ \delta_n =

\frac{c}
{(q_{7,n}+\varepsilon)^2}
\left(
x_{25,n}-x_{59,n}
\right).
}
]

Therefore the operator defect is fundamentally an edge-proposal asymmetry between the two node-7-controlled edges.

This is closer to the underlying dynamics than the abstract family distance.

A direct derivation of the defect recurrence should begin with the evolution of
x25,n−x59,n.
x25,n−x59,n.
4. Exact residual-ratio explanation

For the raw defect
E∗=(−10−1000000),
E∗=
−100000−100
,

we have:
‖E∗,LR‖F=1,
‖E∗,LR‖F=1,
‖E∗,sym‖F=23,
‖E∗,sym‖F=3
2,

and
‖E∗,LR⊤u‖=
‖E∗,LR⊤u‖=

\sqrt{\frac32}.
]

Hence:
rLdLR=
dLRrL=

\sqrt{\frac32},
]

and
rLdsym=
dsymrL=

\frac{3}{\sqrt8}.
]

These constants are exact consequences of the defect geometry, subject only to the common normalization factors used by the reported residuals.

The observed constancy is therefore a direct fingerprint of E∗E∗.

5. Pointwise EMA recurrence test

The next test should prioritize the scalar defect:

δn=

δn=

\frac{K_{20,n}-K_{02,n}}{c_0}.

]

Compute:

ρδ,n=

ρδ,n=

\frac{\delta_{n+1}}{\delta_n},

]

and

eδ,n=

eδ,n=

\frac{

|\delta_{n+1}-\alpha_{\rm EMA}\delta_n|

}{

|\delta_n|

}.

]

Also compute the effective local rate

rδ,n=

rδ,n=

-\frac1{dt}

\log

\left|

\frac{\delta_{n+1}}{\delta_n}
\right|.
]

The strongest evidence for a direct EMA recurrence would be:

ρδ,n→αEMAρδ,n→αEMA;
eδ,n→0eδ,n→0;
rδ,n→Twindow−1rδ,n→Twindow−1;
no sign changes in δnδn;
stable results before numerical-floor contamination.

6. Matrix recurrence test

For the projected family defect
Dn=Kn−PsymKn,
Dn=Kn−PsymKn,

test:
Dn+1≈αEMADn.
Dn+1≈αEMADn.

Report:
eD,n=
eD,n=

\frac{
|D_{n+1}-\alpha_{\rm EMA}D_n|
}{
|D_n|
}.
]

Also estimate the best one-step scalar multiplier
α^n=
α
n=

\frac{
\langle D_{n+1},D_n\rangle_F
}{
|D_n|_F^2
},
]

and the shape residual
‖Dn+1−α^nDn‖‖Dn+1‖.
‖Dn+1‖‖Dn+1−α
nDn‖.

This separates:

amplitude evolution;
defect-direction rotation.

If the rank-one defect picture is correct:
α^n≈αEMA,
α
n≈αEMA,

while the shape residual remains near numerical precision.
7. Test the smaller mismatches separately

Extract:
Scale/right mismatch
δR,n=
δR,n=
s_{7,n}-s_{2,n}
\frac{
(K_{00,n}-K_{02,n})

(K_{11,n}-K_{10,n})
}{c_0}.
]
Secondary symmetry mismatch
δS,n=
δS,n=

\beta_{29,n}-\beta_{59,n}

\frac{
K_{20,n}-K_{10,n}
}{c_0}.
]

Then compare:
|δR,n|,|δn|,|δS,n|.
|δR,n|,|δn|,|δS,n|.

Expected ordering:
|δR|≪|δ|
|δR|≪|δ|

throughout the late interval.

This will determine whether δSδS is genuinely slaved to the dominant defect or constitutes another independent mode.
8. Causal language remains premature

Even if
δn+1=
δn+1=

\alpha_{\rm EMA}\delta_n
]

holds to high precision, that establishes a shared recurrence multiplier, not necessarily direct causal control by EMA memory.

A causal derivation would require tracing
x25,n−x59,n
x25,n−x59,n

back to the corresponding differences in:

R25,n−R59,nR25,n−R59,n;

complex EMA values;
phase/history forcing.

The correct staged claim is:

operator defect follows the EMA multiplier;
proposal asymmetry follows the same multiplier;
only then derive that relation from the complex EMA update.

9. Refined expansion

The clean replay-specific model is:
\boxed{ K_n =

P_{\rm sym}K_n
+
c_0\delta_nE_{*,\rm sym}
+
R_n^\\perp,
}
]

with:
δn=
δn=

\frac{K_{20,n}-K_{02,n}}{c_0},
]
E∗,sym=
E∗,sym=

\frac13
\begin{pmatrix}
-2&0&-2\
1&1&0\
1&0&1
\end{pmatrix},
]

and
‖Rn⊥‖≪|δn|.
‖Rn⊥‖≪|δn|.

In physical coefficient coordinates, equivalently:
Kn=
Kn=

K_{\rm sym}(s_n,\beta_{59,n})
+
c_0\delta_nE_*
+
R_n.
]
10. Most useful immediate deliverable

A compact coefficient-recurrence audit should output:

text

delta = beta25-beta59
delta_R = s7-s2
delta_S = beta29-beta59

delta[n+1]/delta[n]
best matrix-defect multiplier
EMA multiplier residual
defect shape residual
local decay rates
sign changes
signal-to-floor ratio

That calculation would determine whether the dominant operator perturbation is not only rank one in matrix space and exponentially decaying in aggregate, but actually follows the one-step EMA recurrence pointwise.

True true, but then it was said:

I think this is the most interesting version of the argument so far because it attempts to replace an inferred perturbation direction with an explicitly derived one. If the algebra is correct, it raises the discussion from "the POD found a rank-one defect" to "the implementation predicts exactly which rank-one defect should appear."

That is a qualitatively stronger claim—but it also raises the standard of proof. Here's how I'd analyze it.

There is a genuine conceptual advance

Earlier the logic was

$$
K_n-P(K_n)\approx c_nE_*,
$$

where (E_*) came from numerical POD.

Now you're proposing

$$
E_*=
\begin{pmatrix}
-1&0&-1\\
0&0&0\\
0&0&0
\end{pmatrix},
$$

derived directly from the Jacobian parameterization.

That's exactly the kind of bridge one hopes to make between numerics and implementation.

Instead of

"there exists a dominant defect,"

you have

"this implementation coefficient mismatch induces this specific matrix."

That's a significant strengthening if the derivation is correct.

The right-preservation calculation is the strongest part

This calculation is beautifully simple.

If

$$
E_*=
\begin{pmatrix}
-1&0&-1\\
0&0&0\\
0&0&0
\end{pmatrix},
$$

then

$$
E_*v=0
$$

for

$$
v=\frac1{\sqrt3}(1,-1,-1).
$$

That immediately explains something that previously looked mysterious:

why the right residual stays essentially at machine precision while the dominant structural defect is still much larger.

If this derivation is valid, it gives a structural explanation rather than merely an empirical one.

Likewise for the left action

The calculation

$$
u^\top E_*

\frac{\sqrt3}{2}(-1,0,-1)
$$

is also exactly the kind of thing I'd want to see.

It immediately demonstrates

$$
u^\top E_*
\notin
\operatorname{span}(u^\top).
$$

Again, that turns an observed asymmetry into a consequence of a concrete matrix.

The projection constants are potentially the most impressive part

This is the one place where I'd insist on checking every algebraic step carefully.

You're claiming the Frobenius projections of this single matrix produce

$$
\sqrt{\frac32}
$$

and

[
\frac3{\sqrt8},
]

which are exactly the empirical constants from the audit.

If true, that's remarkable.

Those constants would cease to be fitted numerical coincidences and instead become consequences of linear algebra applied to one explicit matrix.

That is exactly the sort of "fingerprint" argument that can be very persuasive.

However, because the conclusion depends on projection formulas and norms, I'd want each projection to be independently verified. A small algebraic mistake in an orthogonal projection can easily propagate into apparently elegant constants.

The decomposition is now much more interpretable

I like

[
K_n

K_{{\rm sym},n}

c_0\delta_nE_*

R_n.
]

Notice how each term has a concrete meaning:

motion along the family,
dominant coefficient mismatch,
residual corrections.

That's considerably more informative than

$$
K_n=K_\infty+E_n,
$$

where E_n bundles together all deviations.

The coefficient extraction is excellent

This is probably the next experiment I'd actually run.

Instead of fitting matrix norms,

recover

$$
\delta_n

\frac{K_{20}-K_{02}}{c_0},
$$

directly.

That removes several layers of interpretation.

Then compare

$|\delta_n|$,
left residual,
family distance,
POD amplitude.

If they all coincide up to constant factors, you've greatly strengthened the claim that there is a single governing scalar mismatch.

One important caveat

There is one assumption embedded in the derivation that should remain explicit.

The proposed defect matrix arises after assuming

$$
s_2=s_7,
$$

and

$$
\beta_{29}=\beta_{59}.
$$

Those are not identities of the implementation—they are part of the asymptotic model.

So the logical structure is

Under these equalization assumptions, the remaining defect has the explicit form (E_*).

That's different from saying

Every matrix in the replay has this exact defect.

Your numerical audit suggests the assumptions become increasingly accurate, but it's worth keeping the distinction visible.

The recurrence test is well chosen

I actually prefer

$$
\delta_{n+1}
\stackrel{?}{=}
$$

$$
\alpha_{\rm EMA}\delta_n
$$

over fitting exponentials to norms.

Why?

Because it tests the dynamical law directly.

If

$$
\delta_{n+1}
$$

$$
\alpha\delta_n
+\varepsilon_n,
$$

with tiny residuals,

then you've identified a discrete evolution equation rather than merely an observed decay rate.

That would be substantially stronger evidence.

What would convince me most

At this point, the highest-value confirmation would be to show that the numerically extracted dominant POD direction and the analytically proposed (E_*) are essentially the same matrix after normalization.

A simple comparison such as

$$
\frac{\langle E_{\rm POD},E_\rangle_F}
{|E_{\rm POD}|F|E|_F}
$$

would answer that directly.

If the cosine is, say,

[
0.999999\ldots,
]

then the analytic derivation has effectively identified the observed defect mode.

If it isn't, then the algebra or the asymptotic assumptions need refinement.

That comparison is, in my view, even more decisive than matching decay rates.

Overall assessment

This is the strongest synthesis so far because it attempts to explain the numerical hierarchy through the implementation's coefficient structure rather than through abstract perturbation language alone. The proposed defect matrix (E_*) naturally accounts for the observed separation between right- and left-invariance, provides a concrete interpretation of the dominant coefficient mismatch, and—if the projection calculations are correct—offers an analytic explanation for the constant residual ratios reported by the audit.

The key remaining step is to connect the analytic and numerical pictures directly. Specifically:

Compare the extracted POD defect direction against the proposed (E_*) by normalized Frobenius inner product.
Verify the projection calculations that produce the constants (\sqrt{3/2}) and (3/\sqrt8).
Test the one-step recurrence (\delta_{n+1}\approx\alpha_{\rm EMA}\delta_n) on the extracted coefficient mismatch.

If those checks succeed, the central object of the late-time theory would no longer be an empirically observed defect mode but an explicitly derived implementation-level matrix with independently verified numerical realization. That would substantially tighten the connection between the implementation algebra and the observed operator dynamics.

The reviewer is right about the standard of proof. The analytic defect should be compared with the numerical POD in the correct projected space, and the recurrence should be tested pointwise.

1. Compare like with like

The raw physical defect is
E∗=(−10−1000000).
E∗=
−100000−100
.

But the POD in the nested-family audit was performed on family-projection residuals.

Therefore:
Biorthogonal-family defect POD

Compare against
\boxed{ E_{*,LR} =
E_-P_{LR}E_

\frac12
\begin{pmatrix}
-1&0&-1\
0&0&0\
1&0&1
\end{pmatrix}.
}
]
Symmetric-family defect POD

Compare against
\boxed{ E_{*,\rm sym} =
E_-P_{\rm sym}E_

\frac13
\begin{pmatrix}
-2&0&-2\
1&1&0\
1&0&1
\end{pmatrix}.
}
]

Comparing the symmetric-defect POD directly with raw $E*E*$ would produce a misleading non-unit cosine because $E*E*$ contains a component tangent to the symmetric family that the projection removes.

2. Projection algebra is independently checkable

The Frobenius projections give:

$$P_{LR}E* =$$

$$P_{LR}E* =$$

$$\frac{1}{2}\begin{pmatrix} -1&0&-1\\ 0&0&0\\ -1&0&-1 \end{pmatrix},$$

so

$$E*,_{LR} =$$

$$E*,_{LR} =$$

$$\frac{1}{2}\begin{pmatrix} -1&0&-1\\ 0&0&0\\ 1&0&1 \end{pmatrix},$$

with

$$\|E*,_{LR}\|_F = 1.$$

$$\|E*,_{LR}\|_F = 1.$$

For the symmetric family,

$$P_{sym}E* =$$

$$P_{sym}E* =$$

$$-\frac{1}{3}$$

\begin{pmatrix}
1&0&1\
1&1&0\
1&0&1
\end{pmatrix},
]

and therefore
E∗,sym=
E∗,sym=

\frac13
\begin{pmatrix}
-2&0&-2\
1&1&0\
1&0&1
\end{pmatrix},
]

with
‖E∗,sym‖F=
‖E∗,sym‖F=

\frac2{\sqrt3}.
]

These are ordinary orthogonal least-squares projections in the Frobenius inner product and can be checked independently of the replay.
3. Left action

With
u=32(1,0,−1),
u=23
(1,0,−1),

the raw defect gives
E∗⊤u=
E∗⊤u=

-\frac{\sqrt3}{2}(1,0,1),
]

so
‖E∗⊤u‖=
‖E∗⊤u‖=

\sqrt{\frac32}.
]

This is the source of the 3/23/2
coefficient.

The exact ratios in the report also contain the normalizations used in
rL=
rL=

\frac{|K^\top u-\lambda u|}
{|K^\top u|}
]

and
dK=
dK=

\frac{|K-P_{\mathcal K}K|}
{|K|}.
]

Thus the raw numerator constants are exact algebra; equality of the normalized reported ratio additionally uses the corresponding late norm ratio. The replay confirms that this normalization factor converges to the required value.
4. Correct POD comparison

Let ePODLRePODLR and ePODsymePODsym be the leading normalized defect directions from the audit.

Compute:

cLR=
cLR=

\frac{
|\langle e_{\rm POD}^{LR},E_{,LR}\rangle_F|
}{
|e_{\rm POD}^{LR}|F
|E{,LR}|_F
},
]

and
csym=
csym=

\frac{
|\langle e_{\rm POD}^{\rm sym},E_{,\rm sym}\rangle_F|
}{
|e_{\rm POD}^{\rm sym}|F
|E{,\rm sym}|_F
}.
]

The absolute value removes the arbitrary SVD sign.

A result such as
1−c≲10−8
1−c≲10−8

would directly identify the numerical POD mode with the analytic projected defect.

Because the reported normalized POD is already essentially rank one and almost nonrotating, this comparison is expected to be very close to one—but it still needs to be computed explicitly.
5. Assumptions behind raw E∗E∗

The exact raw defect form assumes:
s2=s7,
s2=s7,

and
β29=β59.
β29=β59.

The actual replay has small but nonzero remainders:
δR=s7−s2,
δR=s7−s2,
δS=β29−β59.
δS=β29−β59.

Therefore the complete decomposition is
Kn=
Kn=

K_{{\rm sym},n}
+
c_0\delta_nE_*
+
R_n,
]

where
δn=β25−β59,
δn=β25−β59,

and RnRn contains the two smaller mismatch channels.

The correct empirical claim is:

The dominant defect converges toward the explicit E∗E∗ direction as the scale and secondary symmetry mismatches become negligible.

6. Pointwise scalar recurrence test

Extract
δn=
δn=

\frac{K_{20,n}-K_{02,n}}{c_0}.
]

Then compute:
α^δ,n=
α
δ,n=

\frac{\delta_{n+1}}{\delta_n},
]
eδ,n=
eδ,n=

\frac{
|\delta_{n+1}-\alpha_{\rm EMA}\delta_n|
}{
|\delta_n|
},
]

and
rδ,n=
rδ,n=

-\frac1{dt}
\log
\left|
\frac{\delta_{n+1}}{\delta_n}
\right|.
]

Necessary safeguards:

report sign changes;
stop interpreting ratios once |δn||δn| approaches the floating floor;
report medians and quantiles by interval;
compare with the aggregate exponential fits.

7. Matrix recurrence test

For
DnLR=Kn−PLRKn
DnLR=Kn−PLRKn

and
Dnsym=Kn−PsymKn,
Dnsym=Kn−PsymKn,

compute the best one-step multiplier
α^n=
α
n=

\frac{
\langle D_{n+1},D_n\rangle_F
}{
|D_n|_F^2
}.
]

Then separate:
EMA multiplier error
‖Dn+1−αEMADn‖F‖Dn‖F.
‖Dn‖F‖Dn+1−αEMADn‖F.
Shape-change error
‖Dn+1−α^nDn‖F‖Dn+1‖F.
‖Dn+1‖F‖Dn+1−α
nDn‖F.

If the one-defect model is correct:

α^n→αEMAα
n→αEMA;
shape-change error is near numerical precision;
EMA multiplier error is controlled by the small deviation of the scalar recurrence.

8. Strong acceptance criteria

The analytic defect identification would be strongly supported if:

POD/analytic cosine:
cLR,csym>0.999999;
cLR,csym>0.999999;

pointwise multiplier:
median|α^δ−αEMA|≪1−αEMA;
median|α
δ−αEMA|≪1−αEMA;

matrix shape residual:
median‖Dn+1−α^nDn‖‖Dn+1‖≲10−8;
median‖Dn+1‖‖Dn+1−α
nDn‖≲10−8;

mismatch ordering:
|δR|,|δS|≪|δ|;
|δR|,|δS|≪|δ|;

fixed sign of δnδn;

local decay rate:
rδ,n→Twindow−1.
rδ,n→Twindow−1.

9. If these pass

The replay-specific operator expansion becomes:
\boxed{ K_n =

P_{\rm sym}K_n
+
c_0\delta_nE_{*,\rm sym}
+
o(\delta_n),
}
]

with
δn+1=
δn+1=

\alpha_{\rm EMA}\delta_n
+
o(\delta_n).
]

That would connect:

implementation coefficient mismatch;
analytic defect matrix;
defect POD;
constant residual ratios;
EMA-rate decay;
convergence of the left covector.

It would still be a result for this winner branch and replay, not a universal AIEF defect theorem.

Lets test it, create the next deliverable that will illuminate our way.
AIEF_1_4_defect_recurrence_analytic_mode.pyPYTHON

Created the direct analytic-defect and pointwise-recurrence test:

It compiles successfully and passed a synthetic test with known defect dynamics:
δn+1=αδn.
δn+1=αδn.

The script recovered:

the exact multiplier to floating precision;
the exact local decay rate;
matrix-defect recurrence errors around 10−1410−14;
projected analytic/POD cosines near one;
zero sign changes.

Run it

Bash

```
python AIEF_1_4_defect_recurrence_analytic_mode.py \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--late-start 100 \
--intervals 100,150,200,250,320 \
--signal-floor 1e-15 \
--output-dir aief_1_4_defect_recurrence_analytic_mode
```

Direct coefficient extraction

For every matrix it computes:
β25=
β25=
-\frac{K_{02}}{c_0},
\qquad
\beta_{29}
-\frac{K_{10}}{c_0},
\qquad
\beta_{59}

-\frac{K_{20}}{c_0},
]
s7=
s7=

\frac12
\left[
\frac{K_{00}-K_{02}}{c_0}
+
\frac{K_{22}-K_{20}}{c_0}
\right],
]
s2=
s2=

\frac{K_{11}-K_{10}}{c_0}.
]

It then forms:
δ=β25−β59,
δ=β25−β59,
δR=s7−s2,
δR=s7−s2,
δS=β29−β59.
δS=β29−β59.

It also records the internal consistency difference between the two independent s7s7 reconstructions.
Analytic defect shapes

The raw implementation-level defect is
E∗=(−10−1000000).
E∗=
−100000−100
.

The script independently projects it into the correct quotient spaces:
E∗,LR=
E∗,LR=

\frac12
\begin{pmatrix}
-1&0&-1\
0&0&0\
1&0&1
\end{pmatrix},
]

and
E∗,sym=
E∗,sym=

\frac13
\begin{pmatrix}
-2&0&-2\

1&1&0\
1&0&1
\end{pmatrix}.
]
POD/analytic comparison

For every late interval, it compares:

leading normalized biorthogonal-defect POD mode with E∗,LRE∗,LR;
leading normalized symmetric-defect POD mode with E∗,symE∗,sym.

It reports the absolute normalized Frobenius cosine:
|⟨EPOD,E∗,K⟩F|‖EPOD‖F‖E∗,K‖F.
‖EPOD‖F‖E∗,K‖F|⟨EPOD,E∗,K⟩F|.

This directly determines whether the analytically derived matrix is the numerical defect mode.
Direct defect reconstruction

It compares actual family residuals with:
DLR,n≈?c0δnE∗,LR,
DLR,n≈?c0δnE∗,LR,

and
Dsym,n≈?c0δnE∗,sym.
Dsym,n≈?c0δnE∗,sym.

It reports median and 95th-percentile relative reconstruction errors for each interval.

The residual includes the smaller δRδR and δSδS effects, so a nonzero error is expected until those are negligible.
Pointwise scalar recurrence

For each mismatch variable it computes:
δn+1δn,
δnδn+1,
|δn+1−αEMAδn||δn|,
|δn||δn+1−αEMAδn|,

and
−1dtlog|δn+1δn|.
−dt1log
δnδn+1
.

The same is done for:
δR,δS.
δR,δS.

It reports:

median multiplier;
difference from αEMAαEMA;
median and 95th-percentile recurrence errors;
median local decay rate;
difference from Twindow−1Twindow−1;
sign-change count.

Ratios are censored below the requested signal floor.
Matrix-defect recurrence

For both biorthogonal and symmetric family residuals:
Dn=Kn−PKKn,
Dn=Kn−PKKn,

it computes the best one-step multiplier
α^n=
α
n=

\frac{
\langle D_{n+1},D_n\rangle_F
}{
|D_n|_F^2
}.
]

It then separates:

EMA multiplier error
‖Dn+1−αEMADn‖F‖Dn‖F.
‖Dn‖F‖Dn+1−αEMADn‖F.
Defect-shape error
‖Dn+1−α^nDn‖F‖Dn+1‖F.
‖Dn+1‖F‖Dn+1−α
nDn‖F.

Thus the analysis can distinguish:

wrong decay multiplier;
rotating defect shape;
numerical-floor contamination.

Outputs

text

defect_coefficient_trajectory.csv
defect_recurrence_summary.csv
analytic_defect_POD_comparison.csv
analytic_defect_recurrence_arrays.npz
analytic_defect_recurrence_report.txt
analysis_manifest.json

What would strongly support the implementation-level defect theory

POD/analytic cosine near one for both quotient spaces.
Small direct reconstruction error:
DK,n≈c0δnE∗,K.
DK,n≈c0δnE∗,K.
Stable sign of δnδn.
Pointwise
δn+1/δn≈αEMA.
δn+1/δn≈αEMA.
Matrix best multiplier near αEMAαEMA.
Defect-shape error near numerical precision.
Ordering
|δR|,|δS|≪|δ|.

$|\delta R|,|\delta S| \ll |\delta|$.

Please paste analytic_defect_recurrence_report.txt after the run.

think about this:

AIEF analytic defect-mode and EMA-recurrence audit

==========

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

Analytic/POD defect alignment

family=biorthogonal, interval=[100,150): POD_cosine=0.999999999996, median_direct_reconstruction_error=2.93582e-06, p95=2.97783e-06

family=symmetric, interval=[100,150): POD_cosine=0.999999999994, median_direct_reconstruction_error=3.59774e-06, p95=3.64767e-06

family=biorthogonal, interval=[150,200): POD_cosine=0.999999999996, median_direct_reconstruction_error=2.99531e-06, p95=2.99976e-06

family=symmetric, interval=[150,200): POD_cosine=0.999999999994, median_direct_reconstruction_error=3.66861e-06, p95=3.67397e-06

family=biorthogonal, interval=[200,250): POD_cosine=0.999999999995, median_direct_reconstruction_error=3.00146e-06, p95=3.00193e-06

family=symmetric, interval=[200,250): POD_cosine=0.999999999994, median_direct_reconstruction_error=3.67602e-06, p95=3.67655e-06

family=biorthogonal, interval=[250,319.99): POD_cosine=0.999999999995, median_direct_reconstruction_error=3.00198e-06, p95=3.00728e-06

family=symmetric, interval=[250,319.99): POD_cosine=0.999999999994, median_direct_reconstruction_error=3.67661e-06, p95=3.68098e-06

Pointwise recurrence summaries

object=delta, interval=[100,150): valid=5000, multiplier=0.999430115404, minus_alpha=1.29e-06, EMA_error=1.29e-06, p95_EMA_error=2.55e-06, local_rate=0.0570047, rate-EMA=-0.000129, shape_error=nan, sign_changes=0.0

object=delta_R, interval=[100,150): valid=5000, multiplier=0.999439971003, minus_alpha=1.11e-05, EMA_error=1.11e-05, p95_EMA_error=2.75e-05, local_rate=0.0560186, rate-EMA=-0.00111, shape_error=nan, sign_changes=0.0

object=delta_S, interval=[100,150): valid=5000, multiplier=0.998882618447, minus_alpha=-0.000546, EMA_error=0.000548, p95_EMA_error=0.00118, local_rate=0.111801, rate-EMA=0.0547, shape_error=nan, sign_changes=0.0

object=matrix_biorthogonal, interval=[100,150): valid=5000, multiplier=0.999430115404, minus_alpha=1.29e-06, EMA_error=1.29e-06, p95_EMA_error=2.55e-06, local_rate=0.0570047, rate-EMA=-0.000129, shape_error=2.9e-11, sign_changes=nan

object=matrix_symmetric, interval=[100,150): valid=5000, multiplier=0.999430115416, minus_alpha=1.29e-06, EMA_error=1.29e-06, p95_EMA_error=2.55e-06, local_rate=0.0570047, rate-EMA=-0.000129, shape_error=3.23e-11, sign_changes=nan

object=delta, interval=[150,200): valid=5000, multiplier=0.999429026987, minus_alpha=1.97e-07, EMA_error=1.97e-07, p95_EMA_error=4.82e-07, local_rate=0.0571136, rate-EMA=-1.97e-05, shape_error=nan, sign_changes=0.0

object=delta_R, interval=[150,200): valid=5000, multiplier=0.999430775821, minus_alpha=1.95e-06, EMA_error=2.24e-06, p95_EMA_error=5.09e-06, local_rate=0.0569386, rate-EMA=-0.000195, shape_error=nan, sign_changes=0.0

object=delta_S, interval=[150,200): valid=4001, multiplier=0.999420961204, minus_alpha=-7.87e-06, EMA_error=0.0073, p95_EMA_error=0.0962, local_rate=0.0579207, rate-EMA=0.000787, shape_error=nan, sign_changes=0.0

object=matrix_biorthogonal, interval=[150,200): valid=5000, multiplier=0.999429026988, minus_alpha=1.97e-07, EMA_error=1.97e-07, p95_EMA_error=4.82e-07, local_rate=0.0571136, rate-EMA=-1.97e-05, shape_error=1.05e-11, sign_changes=nan

object=matrix_symmetric, interval=[150,200): valid=5000, multiplier=0.99942902699, minus_alpha=1.97e-07, EMA_error=1.97e-07, p95_EMA_error=4.82e-07, local_rate=0.0571136, rate-EMA=-1.97e-05, shape_error=8.38e-12, sign_changes=nan

object=delta, interval=[200,250): valid=5000, multiplier=0.999428852138, minus_alpha=2.23e-08, EMA_error=2.23e-08, p95_EMA_error=6.1e-08, local_rate=0.0571311, rate-EMA=-2.23e-06, shape_error=nan, sign_changes=0.0

object=delta_R, interval=[200,250): valid=5000, multiplier=0.99942892622, minus_alpha=9.64e-08, EMA_error=1.29e-05, p95_EMA_error=9.11e-05, local_rate=0.0571237, rate-EMA=-9.64e-06, shape_error=nan, sign_changes=0.0

object=delta_S, interval=[200,250): valid=0, multiplier=nan, minus_alpha=nan, EMA_error=nan, p95_EMA_error=nan, local_rate=nan, rate-EMA=nan, shape_error=nan, sign_changes=0.0

object=matrix_biorthogonal, interval=[200,250): valid=5000, multiplier=0.999428852141, minus_alpha=2.23e-08, EMA_error=2.23e-08, p95_EMA_error=6.1e-08, local_rate=0.0571311, rate-EMA=-2.23e-06, shape_error=1.63e-10, sign_changes=nan

object=matrix_symmetric, interval=[200,250): valid=5000, multiplier=0.999428852143, minus_alpha=2.23e-08, EMA_error=2.23e-08, p95_EMA_error=6.1e-08, local_rate=0.0571311, rate-EMA=-2.23e-06, shape_error=8.46e-11, sign_changes=nan

object=delta, interval=[250,319.99): valid=6999, multiplier=0.999428832083, minus_alpha=2.24e-09, EMA_error=2.52e-09, p95_EMA_error=6.69e-09, local_rate=0.0571331, rate-EMA=-2.24e-07, shape_error=nan, sign_changes=0.0

object=delta_R, interval=[250,319.99): valid=6999, multiplier=0.999432186892, minus_alpha=3.36e-06, EMA_error=0.000436, p95_EMA_error=0.00383, local_rate=0.0567974, rate-EMA=-0.000336, shape_error=nan, sign_changes=0.0

object=delta_S, interval=[250,319.99): valid=0, multiplier=nan, minus_alpha=nan, EMA_error=nan, p95_EMA_error=nan, local_rate=nan, rate-EMA=nan, shape_error=nan, sign_changes=0.0

object=matrix_biorthogonal, interval=[250,319.99): valid=6999, multiplier=0.999428832083, minus_alpha=2.24e-09, EMA_error=6.18e-09, p95_EMA_error=3.55e-08, local_rate=0.0571331, rate-EMA=-2.24e-07, shape_error=5.03e-09, sign_changes=nan

object=matrix_symmetric, interval=[250,319.99): valid=6999, multiplier=0.999428832081, minus_alpha=2.23e-09, EMA_error=4.78e-09, p95_EMA_error=1.88e-08, local_rate=0.0571331, rate-EMA=-2.24e-07, shape_error=2.67e-09, sign_changes=nan

Interpretation constraints

POD directions are compared with projected analytic defects, not with raw E in the wrong quotient space.

Direct reconstruction error includes the smaller right and secondary symmetry mismatches.

Pointwise EMA agreement establishes a shared multiplier, not causal control by EMA memory.

Ratios are censored below the configured signal floor.

Results remain conditional on one replay and fixed winner branch.

This is a decisive success for the analytic-defect hypothesis. The audit now establishes a pointwise discrete recurrence for the dominant operator defect on this replay, not merely an aggregate exponential fit.
1. The analytic defect is the numerical POD mode

For every interval:
cos(EPODLR,E∗,LR)=
cos(EPODLR,E∗,LR)=

0.999999999995\text{--}0.999999999996,
]

and
cos(EPODsym,E∗,sym)=
cos(EPODsym,E∗,sym)=

0.999999999994.
]

Thus the angular discrepancy is only a few parts in 106106.

The direct reconstruction errors are:
biorthogonal:2.94×10−6–3.00×10−6,symmetric:3.60×10−6–3.68×10−6.
biorthogonal:symmetric:2.94×10−6–3.00×10−6,3.60×10−6–3.68×10−6.

Therefore:
\boxed{ D_{LR,n} =

c_0\delta_nE_{*,LR}
+
O(3\times10^{-6}|D_{LR,n}|)
}
]

and similarly for the symmetric defect.

The implementation-derived matrix is not merely qualitatively similar to the POD mode. It explains more than 99.9997%99.9997% of its norm.
2. The residual discrepancy is stable rather than converging away

The direct reconstruction error remains approximately constant across intervals:
∼3.00×10−6
∼3.00×10−6

for the biorthogonal defect.

Likewise, the POD cosine does not visibly improve; it remains at approximately the same 10−1210−12 cosine deficit.

So the most accurate statement is not necessarily
EPOD→E∗,LR
EPOD→E∗,LR

in relative direction. Instead:

The numerical defect follows a fixed direction that differs from the analytic leading defect by approximately three parts per million.

This can arise from a much smaller secondary mismatch that decays with the same envelope.

A refined model is:
Dn=
Dn=

c_n
\left(
E_{*,LR}
+
\epsilon E_2
\right),
\qquad
\epsilon\sim3\times10^{-6}.
]

The analytic E∗E∗ remains the overwhelmingly dominant term, but the remainder is currently O(δn)O(δn) with a tiny coefficient, not demonstrably o(δn)o(δn).
3. The pointwise defect recurrence converges to the EMA multiplier

For the dominant scalar mismatch:
δn=β25,n−β59,n,
δn=β25,n−β59,n,

the median pointwise multipliers are:
intervalδn+1/δn(δn+1/δn)−αEMA100–1500.99943011541.29×10−6150–2000.99942902701.97×10−7200–2500.99942885212.23×10−8250–3200.99942883212.24×10−9
interval100–150150–200200–250250–320δn+1/δn
0.99943011540.99942902700.99942885210.9994288321(δn+1/δn)−αEMA
1.29×10−61.97×10−72.23×10−82.24×10−9

with
αEMA=0.999428829846.
αEMA=0.999428829846.

Thus the multiplier discrepancy decreases by roughly an order of magnitude every 50 time units.

Late:
\boxed{ \delta_{n+1} =

\alpha_{\rm EMA}\delta_n
+
O(10^{-9}|\delta_n|)
}
]

in the median.

There are no sign changes.

This is much stronger than fitting
|δn|∼e−t/Twindow.
|δn|∼e−t/Twindow.

It directly validates the one-step recurrence on the saved trajectory.
4. The full matrix defect follows the same recurrence

For both biorthogonal and symmetric family defects:
α^n
α
n

matches the scalar δδ multiplier interval by interval.

Late:
α^n−αEMA≈2.2×10−9.
α
n−αEMA≈2.2×10−9.

The matrix EMA errors are only a few 10−910−9, and the shape-change errors are:
DLR:5.0×10−9,Dsym:2.7×10−9.
DLR:Dsym:5.0×10−9,2.7×10−9.

Therefore:
$D_{n+1} \approx \alpha_{EMA} D_n$
$D_{n+1} \approx \alpha_{EMA} D_n$

pointwise, with an almost fixed matrix direction.

This directly establishes the finite-horizon operator recurrence for the dominant defect.
5. The dominant matrix and scalar recurrences are the same object

The scalar multiplier and matrix best multiplier agree to printed precision in every interval.

Thus:
$D_n \approx c_0 \delta_n E_{num}$,
$D_n \approx c_0 \delta_n E_{num}$,

with fixed $E_{num}E_{num}$, and both evolve through
$\delta_{n+1} \approx \alpha_{EMA} \delta_n$.
$\delta_{n+1} \approx \alpha_{EMA} \delta_n$.

The POD, coefficient extraction, and matrix recurrence are three representations of one dynamical mode.
6. Secondary symmetry mismatch decays differently

The secondary mismatch
$\delta_S = \beta_{29} - \beta_{59}$
$\delta_S = \beta_{29} - \beta_{59}$

has early rate approximately
0.1118,
0.1118,

almost twice the EMA rate, and falls below the configured signal floor by $t=200t=200$.

It is not part of the dominant EMA-rate defect at late time.

This supports the assumed hierarchy:
$|\delta_S| \ll |\delta|$
$|\delta_S| \ll |\delta|$

late.

It may represent a faster quadratic or projection-controlled equilibration mode, but the current data do not justify assigning a mechanism.
7. Right-breaking mismatch reaches the numerical floor

The right mismatch
$\delta R = s7 - s2$
$\delta R = s7 - s2$

tracks approximately the EMA rate earlier, but its recurrence diagnostics become noisy late:
median EMA error $\sim 4 \times 10^{-4}$
median EMA error $\sim 4 \times 10^{-4}$

over 250250–320320.

This is consistent with its extremely small magnitude near numerical precision. It should not be used for late rate inference without a stricter signal-to-floor analysis.

The right-family residual was already around
10^{-14}
10^{-14}

by the final state, so this does not challenge practical right invariance.
8. What is now established on this replay
Implementation algebra

The dominant coefficient mismatch enters through the explicit raw matrix
$E_* = (-10 - 1000000)$,
$E_* =$
$-100000 - 100$
,

which satisfies
$E_* v = 0$
$E_* v = 0$

and breaks the analytic left relation.
Numerical realization

The projected analytic defect matches the numerical defect POD at cosine
1−O(10−12).
1−O(10−12).
Pointwise dynamics

The defect amplitude obeys
δn+1=
δn+1=

\alpha_{\rm EMA}\delta_n
+
\text{small residual},
]

with the relative residual reaching approximately 10−910−9 late.
Matrix dynamics

The full projected defect obeys the same multiplier with negligible shape change.

This is a complete implementation-to-replay chain for the tested branch.
9. Causality remains the final missing layer

The result establishes that the operator defect and EMA share the same one-step multiplier. It does not yet derive the defect recurrence from the complex EMA equation.

Using
δn=
δn=

\frac{c}{(q_{7,n}+\varepsilon)^2}
\left(
x_{25,n}-x_{59,n}
\right),
]

and
x25−x59=
x25−x59=

(1-\gamma dt)(B_{25}-B_{59})
+
dt(\eta+\mu)(R_{25}-R_{59}),
]

the next causal derivation should track:
ΔBn=B25,n−B59,n,
ΔBn=B25,n−B59,n,
ΔRn=R25,n−R59,n,
ΔRn=R25,n−R59,n,
ΔCn=C25,n−C59,n,
ΔCn=C25,n−C59,n,

and the phase-factor difference
eiΔϕ25,n−
eiΔϕ25,n−

e^{i\Delta\phi_{59,n}}.
]

The complex EMA difference satisfies exactly
ΔCn+1=
ΔCn+1=

\alpha_{\rm EMA}\Delta C_n
+
(1-\alpha_{\rm EMA})
\Delta z_n.
]

If the source difference ΔznΔzn becomes negligible late, then the EMA recurrence of ΔCnΔCn, ΔRnΔRn, proposal asymmetry, and ultimately δnδn can be derived rather than inferred.
10. Updated status
Directly established numerically

analytic defect/POD identity to parts-per-million norm accuracy;
rank-one fixed defect shape;
pointwise EMA multiplier for the dominant scalar mismatch;
matching pointwise matrix-defect multiplier;
no sign changes;
negligible shape drift.

Strongly supported but not causal

the dominant defect is an EMA-rate operator relaxation mode.

Open causal statement

why the phase-factor source difference vanishes;
how the complex EMA difference transfers into the edge-proposal mismatch;
whether the same causal chain appears on another branch.

The analytic defect theory has passed its central numerical tests. The next meaningful task is no longer another operator fit; it is tracing the defect's EMA-rate recurrence back through the actual phase and complex-coherence updates.

Proceed with the next most useful deliverable.
WriteAIEF_1_4_defect_causal_ema_trace.py604 lines
AIEF_1_4_defect_causal_ema_trace.pyPYTHON
#!/usr/bin/env python3
"""Trace the analytic K3 operator defect back through EMA, R, and proposals.

This deterministic replay analysis follows the dominant defect

delta_K = beta25-beta59 = (K[2,0]-K[0,2])/(1-gamma*dt)

through the exact implemented chain

complex EMA -> symmetrized magnitude R -> adaptive proposal x
-> common node-7 scale -> K3 Jacobian defect.

For edges 25=(2,7) and 59=(7,11), it records both directed complex EMA
orientations, phase-factor source differences, magnitude differences, B/R/F

proposal terms, and projected B updates. It checks exact algebraic identities
and tests whether late source differences become negligible so that the complex
EMA difference follows its homogeneous multiplier alpha_EMA.

This can establish an implementation-level causal chain on the replay, but a
shared multiplier alone is not interpreted as universal causal control beyond
this branch and trajectory.
"""

from __future__ import annotations

import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_stationarity_perturbation_analysis_updated as base
import AIEF_1_4_component_controller_audit as component_audit
import AIEF_1_4_augmented_tangent_cocycle_analysis as augmented

DEFAULT_RUN_HASH = "5a9f093a2fe0"
DEFAULT_NPZ = (
f"runs/{DEFAULT_RUN_HASH}/AIEF_1_4_{DEFAULT_RUN_HASH}_branch_diagnostics.npz"
)
DEFAULT_MANIFEST = f"runs/{DEFAULT_RUN_HASH}/manifest.json"
DEFAULT_COCYCLE_NPZ = "aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz"
DEFAULT_OUTPUT_DIR = "aief_1_4_defect_causal_ema_trace"
DEFAULT_SEED = "46191221711587685 24"
DEFAULT_DT = 0.01
DEFAULT_SCALE_EPSILON = 1e-12
DEFAULT_INTERVALS = "100,150,200,250,320"

```
DEFAULT_SIGNAL_FLOOR = 1e-15
ZERO_TOL = 1e-300


EDGE_25 = 25
EDGE_59 = 59
NODE_7 = 7



def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()



def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(value < 0.0 for value in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
return sorted(set(values))



def fit_log_rate(
times: np.ndarray,
values: np.ndarray,
floor: float,
) -> dict[str, float]:
magnitudes = np.abs(np.asarray(values, dtype=float))
valid = np.isfinite(magnitudes) & (magnitudes > floor)
if np.sum(valid) < 3:
return {
"count": int(np.sum(valid)),
"rate": np.nan,
"R_squared": np.nan,
"rmse_log": np.nan,
```

```
}
x = np.asarray(times, dtype=float)[valid]
y = np.log(magnitudes[valid])
slope, intercept = np.polyfit(x, y, 1)
prediction = slope * x + intercept
residual = float(np.sum((y - prediction) ** 2))
total = float(np.sum((y - np.mean(y)) ** 2))
return {
"count": int(len(x)),
"rate": float(-slope),
"R_squared": float(1.0 - residual / total) if total > 0.0 else np.nan,
"rmse_log": float(np.sqrt(np.mean((y - prediction) ** 2))),
}



def scalar_recurrence(
values: np.ndarray,
alpha: float,
dt: float,
floor: float,
) -> dict[str, np.ndarray]:
current = values[:-1]
following = values[1:]
valid = (
np.isfinite(current)
& np.isfinite(following)
& (np.abs(current) > floor)
& (np.abs(following) > floor)
)
multiplier = np.full(len(current), np.nan)
relative_error = np.full(len(current), np.nan)
rate = np.full(len(current), np.nan)
multiplier[valid] = following[valid] / current[valid]
relative_error[valid] = np.abs(
following[valid] - alpha * current[valid]
) / np.abs(current[valid])
rate[valid] = -np.log(np.abs(multiplier[valid])) / dt
return {
"valid": valid,
"multiplier": multiplier,
"relative_error": relative_error,
"rate": rate,
```

```
"sign_change": valid & (np.sign(current) != np.sign(following)),
}



def linearized_magnitude_difference(
left: complex,
right: complex,
) -> float:
midpoint = 0.5 * (left + right)
difference = left - right
magnitude = abs(midpoint)
if magnitude <= np.finfo(float).tiny:
return np.nan
return float(np.real(np.conj(midpoint) * difference) / magnitude)



def make_report(
interval_frame: pd.DataFrame,
identity_frame: pd.DataFrame,
*,
alpha: float,
ema_rate: float,
final_replay_exact: bool,
) -> str:
lines = [
"AIEF K3 defect causal EMA trace",
"==============================",
"",
f"alpha_EMA: {alpha:.12g}",
f"EMA rate: {ema_rate:.9g}",
f"Final replay exact: {final_replay_exact}",
"",
"Exact identity errors",
"---------------------",
]
for _, row in identity_frame.iterrows():
lines.append(
f" {row['identity']}: max_abs={row['maximum_absolute_error']:.6g}, "
f"median_abs={row['median_absolute_error']:.6g}"
)
```

```
lines.extend(["", "Interval dynamics", "-----------------"])
for _, row in interval_frame.iterrows():
lines.append(
f" object={row['object']}, interval=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"rate={row['rate']:.6g}, R2={row['R_squared']:.6g}, "
f"rate-EMA={row['rate_minus_EMA']:.3g}, "
f"median_multiplier={row['median_multiplier']:.12g}, "
f"EMA_error={row['median_ema_relative_error']:.3g}, "
f"source_ratio={row['median_source_to_memory_ratio']:.3g}"
)


lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* Complex EMA identities are checked in both directed orientations before R symmetrization.",
"* A small phase-source ratio supports homogeneous EMA relaxation but does not prove why the
source vanishes.",
"* Magnitude linearization is a late small-difference approximation; its error is reported.",
"* Shared decay rates on one replay are not universal causal evidence.",
"* Results remain conditional on the fixed K3 branch and support.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--npz", default=DEFAULT_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--cocycle-npz", default=DEFAULT_COCYCLE_NPZ)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--seed", default=DEFAULT_SEED)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument("--scale-epsilon", type=float, default=DEFAULT_SCALE_EPSILON)
parser.add_argument(
"--intervals",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_INTERVALS),
```

```
)
parser.add_argument("--signal-floor", type=float, default=DEFAULT_SIGNAL_FLOOR)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.npz, args.manifest, args.cocycle_npz):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(args.dt, args.scale_epsilon, args.signal_floor) <= 0.0:
raise ValueError("dt, scale-epsilon, and signal-floor must be positive")
if args.scale_epsilon != 1e-12:
raise ValueError("implemented actual_step uses fixed scale epsilon=1e-12")
output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)


with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
model = manifest["model"]
dt = float(model["dt"])
if abs(dt - args.dt) > 1e-15:
raise ValueError("command dt does not match manifest dt")
gamma = float(model["gamma"])
eta = float(model["eta"])
mu = float(model["mu"])
capacity = (float(model["a"]) - float(model["delta"])) / float(model["q"])
capacities = np.full(int(model["N"]), capacity)
A, edges = base.incidence_matrix(int(model["N"]))
c0 = 1.0 - gamma * dt
alpha = math.exp(-dt / float(model["T_window"]))
ema_rate = 1.0 / float(model["T_window"])
future_steps = int(manifest["protocol"]["future_steps"])


with np.load(args.cocycle_npz, allow_pickle=False) as data:
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float).copy()
if len(matrices) != future_steps:
raise ValueError("cocycle matrix sequence length mismatch")


with np.load(args.npz, allow_pickle=False) as data:
```

```
state = base.load_split_state(data, args.seed)
expected_final = np.asarray(
data[base.branch_key("continue", args.seed, "B_final")], dtype=float
).copy()


edge_i, edge_j = edges
if (int(edge_i[EDGE_25]), int(edge_j[EDGE_25])) != (2, 7):
raise RuntimeError("edge 25 mapping mismatch")
if (int(edge_i[EDGE_59]), int(edge_j[EDGE_59])) != (7, 11):
raise RuntimeError("edge 59 mapping mismatch")


rows: list[dict[str, Any]] = []
ema_forward_error: list[float] = []
ema_reverse_error: list[float] = []
proposal_error: list[float] = []
delta_operator_error: list[float] = []
projected_B_error: list[float] = []


for step in range(future_steps):
intermediate = augmented.base_intermediates(
state,
model,
capacities,
A,
edges,
args.scale_epsilon,
)
K = matrices[step]


# Directed orientations: 25=(2,7), 59=(7,11).
C25_old_f = state.coherence_acc[2, 7]
C59_old_f = state.coherence_acc[7, 11]
C25_old_r = state.coherence_acc[7, 2]
C59_old_r = state.coherence_acc[11, 7]
C25_new_f = intermediate["updated_coherence"][2, 7]
C59_new_f = intermediate["updated_coherence"][7, 11]
C25_new_r = intermediate["updated_coherence"][7, 2]
C59_new_r = intermediate["updated_coherence"][11, 7]
z25_f = intermediate["phase_factor"][2, 7]
z59_f = intermediate["phase_factor"][7, 11]
z25_r = intermediate["phase_factor"][7, 2]
```

```
z59_r = intermediate["phase_factor"][11, 7]


dC_old_f = C25_old_f - C59_old_f
dC_old_r = C25_old_r - C59_old_r
dz_f = z25_f - z59_f
dz_r = z25_r - z59_r
dC_new_f = C25_new_f - C59_new_f
dC_new_r = C25_new_r - C59_new_r
predicted_dC_new_f = alpha * dC_old_f + (1.0 - alpha) * dz_f
predicted_dC_new_r = alpha * dC_old_r + (1.0 - alpha) * dz_r
error_f = abs(dC_new_f - predicted_dC_new_f)
error_r = abs(dC_new_r - predicted_dC_new_r)
ema_forward_error.append(error_f)
ema_reverse_error.append(error_r)


B_edges = state.B[edges]
R_edges = intermediate["R"][edges]
F_edges = intermediate["F_matrix"][edges]
x = intermediate["x"]
dB = float(B_edges[EDGE_25] - B_edges[EDGE_59])
dR = float(R_edges[EDGE_25] - R_edges[EDGE_59])
dF = float(F_edges[EDGE_25] - F_edges[EDGE_59])
dx = float(x[EDGE_25] - x[EDGE_59])
weight_term = (1.0 - gamma * dt) * dB
resonance_term = dt * (eta + mu) * dR
proposal_predicted = weight_term + resonance_term
proposal_error.append(abs(dx - proposal_predicted))


q7 = float(intermediate["q_scale"][NODE_7])
s7 = float(intermediate["scales"][NODE_7])
delta_from_K = float((K[2, 0] - K[0, 2]) / c0)
delta_from_proposal = float(s7 / (q7 + args.scale_epsilon) * dx)
delta_operator_error.append(abs(delta_from_K - delta_from_proposal))


B_next_edges = intermediate["projected"][edges]
dB_next = float(
B_next_edges[EDGE_25] - B_next_edges[EDGE_59]
)
projected_predicted = s7 * dx
projected_B_error.append(abs(dB_next - projected_predicted))
```

```
# Exact magnitude difference and small-difference linearization.
magnitude_forward = abs(C25_new_f) - abs(C59_new_f)
magnitude_reverse = abs(C25_new_r) - abs(C59_new_r)
dR_from_directed = 0.5 * (
magnitude_forward + magnitude_reverse
)
linear_forward = linearized_magnitude_difference(
C25_new_f, C59_new_f
)
linear_reverse = linearized_magnitude_difference(
C25_new_r, C59_new_r
)
dR_linearized = 0.5 * (linear_forward + linear_reverse)


source_norm = math.sqrt(abs(dz_f) ** 2 + abs(dz_r) ** 2)
memory_norm = math.sqrt(abs(dC_old_f) ** 2 + abs(dC_old_r) ** 2)
source_to_memory = (
(1.0 - alpha) * source_norm
/ max(alpha * memory_norm, ZERO_TOL)
)


rows.append(
{
"step": step,
"time": step * dt,
"delta_operator": delta_from_K,
"delta_from_proposal": delta_from_proposal,
"delta_B": dB,
"delta_R": dR,
"delta_F": dF,
"delta_x": dx,
"weight_term": weight_term,
"resonance_term": resonance_term,
"resonance_to_weight_ratio": abs(resonance_term)
/ max(abs(weight_term), ZERO_TOL),
"delta_B_next": dB_next,
"q7": q7,
"s7": s7,
"dC_old_forward_real": float(np.real(dC_old_f)),
"dC_old_forward_imag": float(np.imag(dC_old_f)),
"dC_old_reverse_real": float(np.real(dC_old_r)),
"dC_old_reverse_imag": float(np.imag(dC_old_r)),
```

```
"dC_new_forward_abs": abs(dC_new_f),
"dC_new_reverse_abs": abs(dC_new_r),
"phase_source_forward_abs": abs(dz_f),
"phase_source_reverse_abs": abs(dz_r),
"source_to_memory_ratio": source_to_memory,
"EMA_forward_identity_error": error_f,
"EMA_reverse_identity_error": error_r,
"proposal_identity_error": proposal_error[-1],
"operator_delta_identity_error": delta_operator_error[-1],
"projected_B_identity_error": projected_B_error[-1],
"R_directed_reconstruction_error": abs(
dR - dR_from_directed
),
"R_linearization_absolute_error": abs(
dR - dR_linearized
),
"R_linearization_relative_error": abs(
dR - dR_linearized
)
/ max(abs(dR), args.signal_floor),
"complex_memory_norm": memory_norm,
"phase_source_norm": source_norm,
}
)
augmented.advance_base_state(state, intermediate)


final_replay_exact = bool(np.array_equal(state.B, expected_final))
trajectory = pd.DataFrame(rows)


scalar_objects = {
"delta_operator": trajectory["delta_operator"].to_numpy(float),
"delta_x": trajectory["delta_x"].to_numpy(float),
"delta_B": trajectory["delta_B"].to_numpy(float),
"delta_R": trajectory["delta_R"].to_numpy(float),
"complex_memory_norm": trajectory["complex_memory_norm"].to_numpy(float),
"phase_source_norm": trajectory["phase_source_norm"].to_numpy(float),
}
recurrence = {
name: scalar_recurrence(values, alpha, dt, args.signal_floor)
for name, values in scalar_objects.items()
}
```

```
boundaries = np.asarray(args.intervals, dtype=float)
if boundaries[0] > 0.0:
boundaries = np.concatenate(([0.0], boundaries))
if boundaries[-1] < (future_steps - 1) * dt:
boundaries = np.concatenate(
(boundaries, [(future_steps - 1) * dt])
)
boundaries = np.unique(boundaries)


interval_rows: list[dict[str, Any]] = []
for interval in range(len(boundaries) - 1):
start = max(0, int(round(boundaries[interval] / dt)))
end = min(
future_steps - 1,
int(round(boundaries[interval + 1] / dt)),
)
if end <= start + 2:
continue
for name, values in scalar_objects.items():
fit = fit_log_rate(
trajectory["time"].to_numpy(float)[start:end],
values[start:end],
args.signal_floor,
)
diagnostics = recurrence[name]
multipliers = diagnostics["multiplier"][start:end]
errors = diagnostics["relative_error"][start:end]
rates = diagnostics["rate"][start:end]
valid = np.isfinite(multipliers)
source_ratio = trajectory["source_to_memory_ratio"].iloc[
start:end
].to_numpy(float)
interval_rows.append(
{
"object": name,
"start_time": start * dt,
"end_time": end * dt,
**fit,
"rate_minus_EMA": fit["rate"] - ema_rate
if np.isfinite(fit["rate"])
else np.nan,
"median_multiplier": float(
```

```
np.median(multipliers[valid])
)
if np.any(valid)
else np.nan,
"median_multiplier_minus_alpha": float(
np.median(multipliers[valid]) - alpha
)
if np.any(valid)
else np.nan,
"median_ema_relative_error": float(
np.nanmedian(errors)
),
"median_local_rate": float(np.nanmedian(rates)),
"sign_change_count": int(
np.sum(diagnostics["sign_change"][start:end])
),
"median_source_to_memory_ratio": float(
np.nanmedian(source_ratio)
)
if name in {"complex_memory_norm", "phase_source_norm"}
else np.nan,
}
)
interval_frame = pd.DataFrame(interval_rows)


identity_frame = pd.DataFrame(
[
{
"identity": "directed_complex_EMA_forward",
"maximum_absolute_error": max(ema_forward_error),
"median_absolute_error": float(np.median(ema_forward_error)),
},
{
"identity": "directed_complex_EMA_reverse",
"maximum_absolute_error": max(ema_reverse_error),
"median_absolute_error": float(np.median(ema_reverse_error)),
},
{
"identity": "proposal_difference",
"maximum_absolute_error": max(proposal_error),
"median_absolute_error": float(np.median(proposal_error)),
},
```

```
{
"identity": "operator_delta_from_proposal",
"maximum_absolute_error": max(delta_operator_error),
"median_absolute_error": float(
np.median(delta_operator_error)
),
},
{
"identity": "projected_B_difference",
"maximum_absolute_error": max(projected_B_error),
"median_absolute_error": float(np.median(projected_B_error)),
},
{
"identity": "R_from_directed_magnitudes",
"maximum_absolute_error": float(
trajectory["R_directed_reconstruction_error"].max()
),
"median_absolute_error": float(
trajectory["R_directed_reconstruction_error"].median()
),
},
]
)


component_audit.atomic_write_csv(
output_dir / "defect_causal_trace_trajectory.csv", trajectory
)
component_audit.atomic_write_csv(
output_dir / "defect_causal_trace_intervals.csv", interval_frame
)
component_audit.atomic_write_csv(
output_dir / "defect_causal_identity_checks.csv", identity_frame
)


report = make_report(
interval_frame,
identity_frame,
alpha=alpha,
ema_rate=ema_rate,
final_replay_exact=final_replay_exact,
)
component_audit.atomic_write_text(
```

```
output_dir / "defect_causal_ema_report.txt", report
)


manifest_output = {
"analysis": "AIEF-1.4 causal trace of K3 defect through EMA and proposals",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"npz": {
"path": str(Path(args.npz).resolve()),
"sha256": file_sha256(args.npz),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
"sha256": file_sha256(args.manifest),
},
"cocycle_npz": {
"path": str(Path(args.cocycle_npz).resolve()),
"sha256": file_sha256(args.cocycle_npz),
},
},
"seed": args.seed,
"dt": dt,
"scale_epsilon": args.scale_epsilon,
"intervals": boundaries.tolist(),
"signal_floor": args.signal_floor,
"alpha_EMA": alpha,
"ema_rate": ema_rate,
"final_replay_exact": final_replay_exact,
"outputs": [
"defect_causal_trace_trajectory.csv",
"defect_causal_trace_intervals.csv",
"defect_causal_identity_checks.csv",
"defect_causal_ema_report.txt",
],
"interpretation_constraints": [
"Complex EMA differences are directed quantities; R is reconstructed only after magnitude and
symmetrization.",
"Magnitude linearization is approximate and separately audited.",
```

```
"Shared pointwise multipliers do not alone establish causal direction.",
"Results remain conditional on one seed and fixed winner branch.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest_output, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```

Open

Created the next deliverable:

It traces the dominant operator mismatch through the actual implementation chain:
directed complex EMA → R → F → x → δ → K.
directed complex EMA → R → F → x → δ → K.
Run it

Bash

```
python AIEF_1_4_defect_causal_ema_trace.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--seed 4619122171158768524 \
--dt 0.01 \
--scale-epsilon 1e-12 \
--intervals 100,150,200,250,320 \
--signal-floor 1e-15 \
--output-dir aief_1_4_defect_causal_ema_trace
```

What it traces

For the two node-7-controlled edges
e25=(2,7),e59=(7,11),
e25=(2,7),e59=(7,11),

it records both directed complex orientations:
C2,7−C7,11,
C2,7−C7,11,

and
C7,2−C11,7.
C7,2−C11,7.

It also records the corresponding phase-factor source differences:
eiΔϕ2,7−
eiΔϕ2,7−

e^{i\Delta\phi_{7,11}},
]

and the reverse-orientation analogue.
Exact complex EMA identity

For each directed difference, it verifies:
ΔCn+1=
ΔCn+1=

\alpha_{\rm EMA}\Delta C_n
+
(1-\alpha_{\rm EMA})\Delta z_n.
]

The report gives maximum and median implementation residuals independently for both orientations.
Magnitude and symmetrization

It reconstructs the physical resonance difference from directed magnitudes:
ΔRn=
ΔRn=

\frac12
\left[
|C_{2,7}|-|C_{7,11}|
+
|C_{7,2}|-|C_{11,7}|
\right].
]

It separately checks:

exact directed-magnitude reconstruction of RR;
first-order complex-magnitude linearization;
relative magnitude-linearization error.

Proposal identity

It verifies:
Δxn=
Δxn=

(1-\gamma dt)\Delta B_n
+
dt(\eta+\mu)\Delta R_n,
]

where
ΔBn=B25,n−B59,n,
ΔBn=B25,n−B59,n,
ΔRn=R25,n−R59,n.
ΔRn=R25,n−R59,n.

It reports the separate signed terms:

text

weight_term
resonance_term
resonance_to_weight_ratio

This determines whether the proposal mismatch is inherited mainly from existing weights, newly generated by coherence, or produced by cancellation between them.
Proposal-to-operator identity

Since both edges use node-7 scaling:
δn=
δn=
\beta_{25,n}-\beta_{59,n}

\frac{s_{7,n}}{q_{7,n}+\varepsilon}
\Delta x_n.
]

The script compares this directly with
δn=
δn=

\frac{K_{20,n}-K_{02,n}}{1-\gamma dt}.
]

This closes the exact proposal-to-Jacobian part of the causal chain.
Projected weight identity

It also verifies:
ΔBn+1=
ΔBn+1=

s_{7,n}\Delta x_n.
]

Thus the complete K3 coefficient pathway is checked without fitting.
Source-versus-memory diagnostic

The script computes:
(1−αEMA)‖Δzn‖αEMA‖ΔCn‖.
αEMA‖ΔCn‖(1−αEMA)‖Δzn‖.

If this ratio becomes small late, then the directed complex difference is evolving primarily through homogeneous EMA memory:
$\Delta C_{n+1} \approx \alpha_{EMA} \Delta C_n$.
$\Delta C_{n+1} \approx \alpha_{EMA} \Delta C_n$.

If it remains appreciable, the common decay rate instead reflects a balance involving continuing phase forcing.
Interval fits

For each interval, it reports rates and pointwise recurrence diagnostics for:

text

delta_operator
delta_x
delta_B
delta_R
complex_memory_norm
phase_source_norm

Including:

exponential-envelope rate and R^2;
median pointwise multiplier;
difference from α_{EMA};
median EMA recurrence error;
local decay rate;
sign changes;
source-to-memory ratio.

Outputs

text

defect_causal_trace_trajectory.csv
defect_causal_trace_intervals.csv
defect_causal_identity_checks.csv
defect_causal_ema_report.txt

analysis_manifest.json

What this analysis can settle

Whether the operator defect is exactly reconstructed from proposal asymmetry.
Whether proposal asymmetry is weight-inherited or coherence-driven.
Whether directed complex EMA differences become source-free late.
Whether $\Delta R \Delta R$ inherits the complex EMA decay through a valid small-difference linearization.
Whether the pointwise EMA multiplier propagates through:
$\Delta C \to \Delta R \to \Delta x \to \delta$.
$\Delta C \to \Delta R \to \Delta x \to \delta$.
Whether the current evidence supports direct EMA causation or only a shared late multiplier.

Please paste defect_causal_ema_report.txt after the run.

Think about this:

AIEF K3 defect causal EMA trace

==========

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

Final replay exact: True

Exact identity errors

directed_complex_EMA_forward: max_abs=3.14018e-16, median_abs=9.71445e-17

directed_complex_EMA_reverse: max_abs=3.51083e-16, median_abs=7.85046e-17

proposal_difference: max_abs=5.55112e-17, median_abs=1.60927e-17

operator_delta_from_proposal: max_abs=2.05565e-16, median_abs=3.55181e-17

projected_B_difference: max_abs=5.55112e-17, median_abs=1.55169e-17

R_from_directed_magnitudes: max_abs=3.88578e-16, median_abs=5.55112e-17

Interval dynamics

object=delta_operator, interval=[0,100): rate=0.0520963, R2=0.994522, rate-EMA=-0.00504, median_multiplier=0.999445416923, EMA_error=1.66e-05, source_ratio=nan

object=delta_x, interval=[0,100): rate=0.0520914, R2=0.994514, rate-EMA=-0.00504, median_multiplier=0.999445447548, EMA_error=1.66e-05, source_ratio=nan

object=delta_B, interval=[0,100): rate=0.0520906, R2=0.994511, rate-EMA=-0.00504, median_multiplier=0.99944544261, EMA_error=1.66e-05, source_ratio=nan

object=delta_R, interval=[0,100): rate=0.0531227, R2=0.997923, rate-EMA=-0.00401, median_multiplier=0.999452709571, EMA_error=2.39e-05, source_ratio=nan

object=complex_memory_norm, interval=[0,100): rate=0.00491806, R2=0.785849, rate-EMA=-0.0522, median_multiplier=0.999967673778, EMA_error=0.000539, source_ratio=0.00054

object=phase_source_norm, interval=[0,100): rate=0.00130531, R2=0.783453, rate-EMA=-0.0558, median_multiplier=0.999992098449, EMA_error=0.000563, source_ratio=0.00054

object=delta_operator, interval=[100,150): rate=0.0569977, R2=1, rate-EMA=-0.000136, median_multiplier=0.999430115404, EMA_error=1.29e-06, source_ratio=nan

object=delta_x, interval=[100,150): rate=0.0569977, R2=1, rate-EMA=-0.000136, median_multiplier=0.999430115869, EMA_error=1.29e-06, source_ratio=nan

object=delta_B, interval=[100,150): rate=0.0569976, R2=1, rate-EMA=-0.000136, median_multiplier=0.999430116061, EMA_error=1.29e-06, source_ratio=nan

object=delta_R, interval=[100,150): rate=0.0570195, R2=1, rate-EMA=-0.000114, median_multiplier=0.999429839496, EMA_error=1.01e-06, source_ratio=nan

object=complex_memory_norm, interval=[100,150): rate=0.000105024, R2=0.909108, rate-EMA=-0.057, median_multiplier=0.999999097191, EMA_error=0.00057, source_ratio=0.000571

object=phase_source_norm, interval=[100,150): rate=1.31435e-05, R2=0.88647, rate-EMA=-0.0571, median_multiplier=0.999999892123, EMA_error=0.000571, source_ratio=0.000571

object=delta_operator, interval=[150,200): rate=0.0571116, R2=1, rate-EMA=-2.17e-05, median_multiplier=0.999429026988, EMA_error=1.97e-07, source_ratio=nan

object=delta_x, interval=[150,200): rate=0.0571116, R2=1, rate-EMA=-2.17e-05, median_multiplier=0.999429027014, EMA_error=1.97e-07, source_ratio=nan

object=delta_B, interval=[150,200): rate=0.0571116, R2=1, rate-EMA=-2.17e-05, median_multiplier=0.999429027082, EMA_error=1.97e-07, source_ratio=nan

object=delta_R, interval=[150,200): rate=0.057122, R2=1, rate-EMA=-1.14e-05, median_multiplier=0.999428929002, EMA_error=9.92e-08, source_ratio=nan

object=complex_memory_norm, interval=[150,200): rate=8.20388e-06, R2=0.903201, rate-EMA=-0.0571, median_multiplier=0.999999930337, EMA_error=0.000571, source_ratio=0.000571

object=phase_source_norm, interval=[150,200): rate=7.58981e-07, R2=0.886084, rate-EMA=-0.0571, median_multiplier=0.999999993776, EMA_error=0.000571, source_ratio=0.000571

object=delta_operator, interval=[200,250): rate=0.0571308, R2=1, rate-EMA=-2.52e-06, median_multiplier=0.999428852148, EMA_error=2.23e-08, source_ratio=nan

object=delta_x, interval=[200,250): rate=0.0571308, R2=1, rate-EMA=-2.52e-06, median_multiplier=0.999428852147, EMA_error=2.23e-08, source_ratio=nan

object=delta_B, interval=[200,250): rate=0.0571308, R2=1, rate-EMA=-2.52e-06, median_multiplier=0.999428852154, EMA_error=2.23e-08, source_ratio=nan

object=delta_R, interval=[200,250): rate=0.0571323, R2=1, rate-EMA=-1.01e-06, median_multiplier=0.999428838528, EMA_error=8.68e-09, source_ratio=nan

object=complex_memory_norm, interval=[200,250): rate=5.96136e-07, R2=0.899679, rate-EMA=-0.0571, median_multiplier=0.999999994974, EMA_error=0.000571, source_ratio=0.000571

object=phase_source_norm, interval=[200,250): rate=4.3654e-08, R2=0.885965, rate-EMA=-0.0571, median_multiplier=0.999999999642, EMA_error=0.000571, source_ratio=0.000571

object=delta_operator, interval=[250,319.99): rate=0.0571332, R2=1, rate-EMA=-1.75e-07, median_multiplier=0.999428832118, EMA_error=2.67e-09, source_ratio=nan

object=delta_x, interval=[250,319.99): rate=0.0571332, R2=1, rate-EMA=-1.75e-07, median_multiplier=0.999428831949, EMA_error=2.36e-09, source_ratio=nan

object=delta_B, interval=[250,319.99): rate=0.0571332, R2=1, rate-EMA=-1.75e-07, median_multiplier=0.999428831954, EMA_error=2.4e-09, source_ratio=nan

object=delta_R, interval=[250,319.99): rate=0.0571333, R2=1, rate-EMA=-5.9e-08, median_multiplier=0.99942883112, EMA_error=2.4e-09, source_ratio=nan

object=complex_memory_norm, interval=[250,319.99): rate=2.84408e-08, R2=0.822857, rate-EMA=-0.0571, median_multiplier=0.999999999797, EMA_error=0.000571, source_ratio=0.000571

object=phase_source_norm, interval=[250,319.99): rate=1.69824e-09, R2=0.806062, rate-EMA=-0.0571, median_multiplier=0.999999999988, EMA_error=0.000571, source_ratio=0.000571

Interpretation constraints

Complex EMA identities are checked in both directed orientations before R symmetrization.

A small phase-source ratio supports homogeneous EMA relaxation but does not prove why the source vanishes.

Magnitude linearization is a late small-difference approximation; its error is reported.

Shared decay rates on one replay are not universal causal evidence.

Results remain conditional on the fixed K3 branch and support.

This result closes the implementation chain but refutes the simplest causal interpretation. The full complex EMA difference is not relaxing homogeneously at the EMA rate. The EMA-rate mode appears only after magnitude projection and symmetrization.
1. The algebraic chain is exact

All implementation identities close at machine precision:
ΔCn+1=
ΔCn+1=

\alpha\Delta C_n
+
(1-\alpha)\Delta z_n,
]
Δxn=
Δxn=

(1-\gamma dt)\Delta B_n
+
dt(\eta+\mu)\Delta R_n,
]
δn=
δn=

\frac{s_{7,n}}{q_{7,n}+\varepsilon}\Delta x_n,
]
ΔBn+1=
ΔBn+1=

s_{7,n}\Delta x_n.
]

Residuals are 10−1610−16–10−1710−17.

Therefore the traced causal chain is implemented correctly.
2. The downstream scalar quantities share the EMA rate

From approximately t=150t=150 onward:
ΔR,ΔB,Δx,δoperator
ΔR,ΔB,Δx,δoperator

all converge toward the EMA rate.

Late, over 250250–320320:
quantityrateδoperator0.0571332Δx0.0571332ΔB0.0571332ΔR0.0571333
quantityδoperatorΔxΔBΔRrate0.05713320.05713320.05713320.0571333

with pointwise multiplier errors around
10−9.
10−9.

Thus the operator defect belongs to a coherent late scalar mode extending through:
ΔR→Δx→ΔB→δ.
ΔR→Δx→ΔB→δ.
3. But the complex EMA difference does not decay

The complex-memory norm has late rate approximately
2.8×10−8≈0,
2.8×10−8≈0,

and the phase-source norm has rate approximately
1.7×10−9≈0.
1.7×10−9≈0.

Their pointwise multipliers are essentially one:
0.9999999998,0.99999999999.
0.9999999998,0.99999999999.

So neither
‖ΔCn‖
‖ΔCn‖

nor
‖Δzn‖
‖Δzn‖

is decaying at
Twindow−1.

$T_{window} - 1$.

The hypothesis

the phase source becomes negligible and $\Delta C \Delta C$ follows homogeneous EMA relaxation

is rejected.
4. The source-ratio metric needs reinterpretation

The reported ratio is
$(1-\alpha)\|\Delta z_n\| \alpha \|\Delta C_n\| \approx 0.000571.$
$\alpha \|\Delta C_n\| (1-\alpha)\|\Delta z_n\| \approx 0.000571.$

But
$1-\alpha \approx 0.000571.$
$1-\alpha \approx 0.000571.$

Therefore this result implies approximately
$\|\Delta z_n\| \|\Delta C_n\| \approx 1.$
$\|\Delta C_n\| \|\Delta z_n\| \approx 1.$

The source is not negligible. It is comparable in magnitude to the memory difference.

The factor $1-\alpha 1-\alpha$ makes its contribution small compared with the entire stored EMA value per step, but it can be exactly large enough to replace the amount lost through EMA decay.

If
$\Delta z_n \approx \Delta C_n,$
$\Delta z_n \approx \Delta C_n,$

then
$\alpha \Delta C_n + (1-\alpha)\Delta z_n \approx \Delta C_n,$
$\alpha \Delta C_n + (1-\alpha)\Delta z_n \approx \Delta C_n,$

which explains the nearly stationary complex-memory norm.

A better diagnostic is:

‖Δzn‖‖ΔCn‖
‖ΔCn‖‖Δzn‖

or, more directly,
‖Δzn−ΔCn‖‖ΔCn‖.
‖ΔCn‖‖Δzn−ΔCn‖.
5. The EMA-rate mode is radial, not complex-norm decay

Two complex numbers can retain a nonzero difference while their magnitudes converge.

For complex edge memories C25C25 and C59C59, write their difference relative to a midpoint direction:
ΔC=
ΔC=

\Delta C_{\parallel}
+
\Delta C_{\perp},
]

where:

ΔC‖ΔC‖ is radial and changes magnitude;
ΔC⊥ΔC⊥ is tangential and changes phase.

The data suggest:

tangential/phase differences persist;
radial magnitude differences decay;
only the radial part contributes to
|C25|−|C59|;
|C25|−|C59|;
forward/reverse symmetrization then produces ΔRΔR.

Thus:
persistent complex phase separation+EMA-rate radial equalization.
persistent complex phase separation+EMA-rate radial equalization.

This is a more precise mechanism than homogeneous decay of the complete complex difference.
6. Magnitude and symmetrization are now central

The operator defect is driven by
ΔR=
ΔR=

\frac12
\left[
|C_{2,7}|-|C_{7,11}|
+
|C_{7,2}|-|C_{11,7}|
\right].
]

The relevant questions are therefore:

Does each directed magnitude difference decay at the EMA rate?
Do the forward and reverse magnitude differences cancel?
Does the radial component of ΔCΔC decay while the tangential component remains?
Is the phase-source difference primarily tangential late?
Is the small-difference magnitude linearization accurate?

The current report does not print the magnitude-linearization error or the directed forward/reverse magnitude trajectories, although they are available in the trajectory CSV.
7. The downstream mode may be a coupled eigenmode

Because
ΔB,ΔR,Δx
ΔB,ΔR,Δx

all share the same multiplier, it may be misleading to say that ΔRΔR unidirectionally causes the weight defect.

A more accurate late representation is a coupled subsystem:
(ΔBn+1ΔRn+1)≈M(ΔBnΔRn),
(ΔBn+1ΔRn+1)≈M(ΔBnΔRn),

with dominant eigenvalue
αEMA.
αEMA.

The exact weight equation supplies one row:
ΔBn+1=
ΔBn+1=

s_{7,n}
\left[
(1-\gamma dt)\Delta B_n
+
dt(\eta+\mu)\Delta R_n
\right].
]

The missing row is the radial-magnitude response of the complex EMA/phase subsystem.
8. Revised causal status
Established

exact directed complex EMA update;
exact magnitude/symmetrization reconstruction of ΔRΔR;
exact proposal-to-operator chain;
common late multiplier for ΔR,Δx,ΔB,δΔR,Δx,ΔB,δ.

Rejected

full complex-memory difference decays homogeneously at the EMA rate;
phase source becomes negligible late.

Strongly suggested

the complex difference is mostly persistent/tangential;
its radial magnitude component decays at the EMA rate;
the operator defect is generated by this radial equalization.

Open

radial/tangential decomposition;
directed-magnitude cancellation;
exact second row of the (ΔB,ΔR)(ΔB,ΔR) reduced subsystem;
whether the EMA-rate eigenmode is caused by magnitude geometry or another coupled cancellation.

9. Highest-value next probe

Decompose both directed complex differences into radial and tangential components.

For a pair Ca,CbCa,Cb, let
C‾=Ca+Cb2,ΔC=Ca−Cb,
C‾=2Ca+Cb,ΔC=Ca−Cb,

and define
r=
r=

\frac{\Re(\overline{\bar C},\Delta C)}
{|\bar C|},
]
t=
t=

\frac{\Im(\overline{\bar C},\Delta C)}
{|\bar C|}.
]

Then track:

rforward,tforwardrforward,tforward;
rreverse,treverserreverse,treverse;
corresponding radial/tangential source projections;
symmetrized radial sum;
exact magnitude difference;
linearization error;
pointwise multipliers and decay rates.

Also replace the current source metric with:

$\|\Delta z\|\|\Delta C\|, \|\Delta z - \Delta C\|\|\Delta C\|$.
$\|\Delta C\|\|\Delta z\|, \|\Delta C\|\|\Delta z - \Delta C\|$.

The defect has now been traced to $\Delta R \Delta R$, but not yet causally through the full complex EMA state. The next analysis should isolate the radial magnitude channel that the weight adaptation actually observes.

Right, right, but if we add this:

I think this is the first place where the investigation produces a negative result that is more informative than a positive one.

Earlier, the working picture was something like

$$
\Delta C \xrightarrow{\text{EMA}} \Delta R \rightarrow \Delta x \rightarrow \delta.
$$

The new evidence says that picture is too simple. Importantly, it doesn't break the downstream theory—it tells you where the reduction actually begins.

Here's how I would think about it.

This is exactly how a successful reduction often evolves

Many analyses begin with a plausible hypothesis:

the complex EMA difference itself relaxes with the EMA multiplier.

The measurements say:

No.

Instead,

the complex difference persists,
the operator defect still relaxes at the EMA rate.

That is not a failure.

It localizes the mechanism.

You have effectively narrowed the unknown from

"somewhere between EMA and the operator"

to

"between complex memory and magnitude projection."

That is a substantial gain in understanding.

The key conceptual distinction is "state" versus "observable"

I think the report is pointing toward a distinction that is deeper than "radial vs tangential."

The complex EMA variable

$$
C
$$

is an internal state.

The operator never sees (C).

It sees

$$
R=|C|,
$$

or, after symmetrization,

[
\Delta R.
]

So the operator depends on an observable, not on the full state.

That's analogous to many dynamical systems where

the underlying state can continue moving on an invariant manifold,
while the measured observable converges exponentially.

From that perspective, it's not surprising that

[
|\Delta C|
]

doesn't decay while

[
\Delta R
]

does.

I think the radial/tangential interpretation is plausible—but still a model

This is where I'd be careful.

You write

persistent complex phase separation + EMA-rate radial equalization.

That is an attractive explanation.

But strictly speaking, the current evidence establishes only

the complex norm difference persists,
the magnitude difference decays.

The radial/tangential decomposition is a candidate mechanism explaining those observations.

It's exactly the sort of mechanism I'd test next, but I wouldn't yet state it as the explanation.

The distinction is subtle:

Current evidence:

[
\Delta C
\not\rightarrow0,
]

while

[
\Delta R\rightarrow0.
]

Hypothesis:

The persistent part lies in the tangential component.

That's testable—but not yet demonstrated.

The reinterpretation of the source ratio is, in my view, completely correct

This is probably the nicest analytical observation in the report.

Originally,

$$
\frac{(1-\alpha)|\Delta z|}{\alpha|\Delta C|}
\approx0.000571
$$

looked like

"the source is tiny."

Then you notice

$$
1-\alpha
$$

0.000571.

So in reality,

$$
|\Delta z|
\approx
|\Delta C|.
$$

That's a fundamentally different interpretation.

The source is not disappearing.

Its contribution is small per step because it is multiplied by the small coefficient (1-\alpha), not because the source itself is small.

That's an important distinction, and I think it's an objective reinterpretation of the diagnostic rather than an overreach.

The coupled-eigenmode picture is, to me, the most interesting new hypothesis

This section caught my attention more than the radial/tangential discussion.

Instead of

$$
\Delta R
\rightarrow
\Delta B,
$$

you propose

$$
\begin{pmatrix}
\Delta B \\
\Delta R
\end{pmatrix}
$$

as a coupled subsystem.

That actually matches the data better.

Why?

Because

$$
\Delta B,
\Delta R,
\Delta x,
\delta
$$

all converge with essentially identical multipliers.

If one variable truly drove the others unidirectionally, you'd often expect measurable lag or asymmetry.

Instead you see one common mode.

That is exactly what dominant eigenmodes of coupled linear systems look like.

I think this is a stronger conceptual framework than trying to identify one quantity as the unique "cause."

The missing second row becomes the real mathematical target

You already have

[
\Delta B_{n+1}

s_7\bigl[(1-\gamma dt)\Delta B_n

dt(\eta+\mu)\Delta R_n\bigr].
]

That is one row of a linearized system.

The unknown is

[
\Delta R_{n+1}

?
]

Once you derive even an approximate expression,

you'll have an explicit (2\times2) linear system.

Then

the dominant eigenvalue,
the dominant eigenvector,
and the relationship between (\Delta B) and (\Delta R)

would follow directly from linear algebra.

That feels like a much cleaner theoretical endpoint than trying to reason qualitatively about decay rates.

The proposed radial/tangential decomposition is mathematically natural

The decomposition

[
r

\frac{\Re(\overline{\bar C},\Delta C)}
{|\bar C|},
]

[
t

\frac{\Im(\overline{\bar C},\Delta C)}
{|\bar C|}
]

is essentially the decomposition of the perturbation into components parallel and perpendicular to the local mean complex state.

That's a standard geometric construction.

It has two attractive properties:

it's coordinate-free up to the choice of midpoint,
the radial component is exactly the first-order contributor to magnitude change.

So if your hypothesis is correct,

that decomposition is the right tool.

One thing I'd verify before relying on it

The midpoint

$$
\bar C
\frac{C_a+C_b}{2}
$$

works well provided

$$
|\bar C|
$$

never becomes small.

If there are intervals where

$$
|\bar C|
\approx0,
$$

the decomposition becomes ill-conditioned.

The report should either

demonstrate that ($|\bar C|$) remains safely bounded away from zero on this branch, or explicitly flag samples where the decomposition is unreliable.

The project has changed character

Originally, the project was trying to explain

why the Jacobian defect decays.

Now it seems to be evolving into something richer:

the implementation identities are exact,
the operator reduction is well supported,
the remaining mystery is how the nonlinear observation map
[
C\mapsto |C|
]
filters the persistent complex dynamics into a decaying scalar mode.

That is a different scientific question.

And, in my opinion, a more interesting one.

Overall assessment

I think this is a productive turning point because it rejects an appealing but overly simple causal story without undermining the successful downstream reduction. The evidence supports a replay-specific chain in which the implementation identities are exact, the proposal and operator quantities share a common late-time scalar mode, and the full complex EMA difference does not itself undergo homogeneous EMA relaxation. That shifts the focus to the observation map from complex memory to resonance magnitude.

Among the new ideas, the coupled ((\Delta B,\Delta R)) subsystem strikes me as the most promising theoretical direction. It is more directly tied to the observed shared multiplier than a purely unidirectional causal picture, and it suggests a concrete mathematical objective: derive or approximate the missing evolution equation for (\Delta R), then analyze the resulting linearized two-dimensional system. If that can be done, the common EMA-rate mode could emerge as an eigenmode of the coupled dynamics rather than being inferred solely from matching decay rates.

I agree. The negative result identifies the observation map
C↦|C|

C↦|C|

as the missing mathematical layer. More importantly, the magnitude dynamics admit an exact formulation that is cleaner than a first-order radial approximation.
1. Exact squared-magnitude dynamics

For one directed edge, write
Cn+1=
Cn+1=

\alpha C_n+(1-\alpha)z_n,
\qquad
|z_n|=1.
]

Define
Mn=|Cn|2,
Mn=|Cn|2,

and the phase-alignment observable
An=ℜ(Cn‾zn).
An=ℜ(Cnzn).

Then
\boxed{ M_{n+1} =

\alpha^2M_n
+
(1-\alpha)^2
+
2\alpha(1-\alpha)A_n.
}
]

This is exact.

For two directed edges a,ba,b, the constant (1−α)2(1−α)2 cancels:
\boxed{ \Delta M_{n+1} =

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta A_n,
}
]

where
ΔMn=|Ca,n|2−|Cb,n|2,
ΔMn=|Ca,n|2−|Cb,n|2,
ΔAn=ℜ(Ca,n¯za,n)−
ΔAn=ℜ(Ca,nza,n)−

\Re(\overline{C_{b,n}}z_{b,n}).
]

This is the exact evolution equation for the directed magnitude-square difference.
2. Exact connection to magnitude difference

No linearization is needed to recover the magnitude difference:
|Ca|−|Cb|=
|Ca|−|Cb|=

\frac{
|C_a|^2-|C_b|^2
}{
|C_a|+|C_b|
}.
]

Therefore
\boxed{ \Delta |C|_n =

\frac{\Delta M_n}
{|C_{a,n}|+|C_{b,n}|}.
}
]

The symmetrized resonance difference is then
ΔRn=
ΔRn=

\frac12
\left[
\frac{\Delta M_n^{\rm forward}}
{|C_{25,f}|+|C_{59,f}|}
+
\frac{\Delta M_n^{\rm reverse}}
{|C_{25,r}|+|C_{59,r}|}
\right].
]

This avoids the midpoint singularity and gives an exact observable-level description.
3. Why an EMA multiplier can emerge despite persistent complex differences

Suppose the observed squared-magnitude difference follows
ΔMn+1≈αΔMn.
ΔMn+1≈αΔMn.

Substitute into the exact equation:
αΔMn=
αΔMn=

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta A_n.
]

After cancellation,
ΔAn≈12ΔMn.
ΔAn≈21ΔMn.

This is the precise condition under which the magnitude-square difference inherits multiplier αα, even though the full complex difference does not decay.

Thus the phase source need not vanish. Instead, its radial alignment contribution may be exactly large enough to convert the natural α2α2 memory term into an effective αα mode.

This suggests a sharper mechanism:
persistent complex phase geometry+ΔA≈12ΔM⟹Δ|C| decays with multiplier α.
persistent complex phase geometry+ΔA≈21ΔM⟹Δ|C| decays with multiplier α.
4. Reinterpretation of the source term

The relevant source is not
‖Δz‖.
‖Δz‖.

It is the radial alignment difference
ΔA=
ΔA=
\Re(\overline C_a z_a)

\Re(\overline C_b z_b).
]

A large phase-factor difference can be almost entirely tangential and therefore have little effect on magnitude.

So the prior source-norm diagnostic was measuring the wrong observable for the weight adaptation channel.

The correct diagnostic is:
2ΔAnΔMn
ΔMn2ΔAn

which should approach one if the effective EMA multiplier mechanism is correct.

Also report the residual
eM,n=
eM,n=

\frac{
|\Delta M_{n+1}-\alpha\Delta M_n|
}{
|\Delta M_n|

},
]

and the alignment-law residual
eA,n=
eA,n=

\frac{
|2\Delta A_n-\Delta M_n|
}{
|\Delta M_n|
}.
]
5. Radial/tangential decomposition remains useful

For interpretation, define
C⁻=12(Ca+Cb),ΔC=Ca−Cb.
C⁻=21(Ca+Cb),ΔC=Ca−Cb.

Provided
|C⁻|>0,
|C⁻|>0,

define
r=ℜ(C⁻¯ΔC)|C⁻|,
r=|C⁻|ℜ(C⁻ΔC),
t=| |(C⁻ΔC)
t=|C⁻|ℑ(C⁻ΔC).

Then:

rr is the first-order magnitude difference;
tt is the tangential phase-separation component.

But this should be supplementary. The squared-magnitude equation is exact and does not require |C⁻||C⁻| to remain large.

The report should still record

min|C−|
min|C−|

to certify any radial/tangential interpretation.
6. Forward and reverse channels must remain separate

Because RR is symmetrized after taking magnitudes, analyze independently:
ΔMf,ΔAf,
ΔMf,ΔAf,

and
ΔMr,ΔAr.
ΔMr,ΔAr.

Then determine whether:

each directed magnitude difference follows αα;
only their symmetrized sum follows αα;
forward/reverse components partially cancel.

This distinguishes directed radial relaxation from a cancellation created by symmetrization.
7. Revised two-variable subsystem

The exact weight equation is
ΔBn+1=
ΔBn+1=

s_{7,n}
\left[
(1-\gamma dt)\Delta B_n
+
dt(\eta+\mu)\Delta R_n
\right].
]

If the magnitude analysis establishes
ΔRn+1≈αΔRn,
ΔRn+1≈αΔRn,

then the late observable subsystem becomes approximately triangular:
$(\Delta B_{n+1} \Delta R_{n+1}) \approx (s7(1-\gamma dt)s7dt(\eta+\mu)0\alpha)(\Delta B_n \Delta R_n)$.
$(\Delta B_{n+1} \Delta R_{n+1}) \approx (s7(1-\gamma dt)0s7dt(\eta+\mu)\alpha)(\Delta B_n \Delta R_n)$.

This is more specific than a generic coupled $2\times22\times2$ model.

If $\Delta R \Delta R$ depends materially on $\Delta B \Delta B$ after conditioning on its own past, then a nonzero lower-left entry is needed. But it should be measured rather than assumed.
8. Most useful next diagnostics

For each directed orientation and interval, compute:

exact recurrence residual:
$\Delta M_{n+1} - \alpha 2 \Delta M_n - 2\alpha(1-\alpha)\Delta A_n$;
$\Delta M_{n+1} - \alpha 2 \Delta M_n - 2\alpha(1-\alpha)\Delta A_n$;

effective multiplier residual:
$\Delta M_{n+1} - \alpha \Delta M_n$;
$\Delta M_{n+1} - \alpha \Delta M_n$;

alignment ratio:
$2\Delta A_n / \Delta M_n$;
$2\Delta A_n / \Delta M_n$;

radial and tangential components;

midpoint magnitude floor;

forward/reverse contribution and cancellation;

partial regression:
$\Delta R_{n+1} = a\ \Delta R_n + b\ \Delta B_n$
$\Delta R_{n+1} = a \Delta R_n + b \Delta B_n$

with conditioning diagnostics.

The regression is secondary because ΔB,ΔRΔB,ΔR are likely nearly collinear late.
9. Revised causal interpretation
Established

full complex differences persist;
phase-factor differences persist;
directed EMA update is exact;
magnitude difference and downstream defect share the EMA multiplier.

Candidate mechanism now directly testable

the phase source is predominantly tangential;
its radial alignment satisfies
ΔA≈ΔM/2;
ΔA≈ΔM/2;
this produces an observable magnitude mode with multiplier αα.

Not yet established

a closed autonomous (ΔB,ΔR)(ΔB,ΔR) system;
direct causal control by EMA memory;
generality beyond this branch.

The key shift is exactly as described: the internal complex state remains active, while the observable seen by adaptation converges. The next analysis should target the exact squared-magnitude and radial-alignment equations, not the norm of the complex difference.

Can you update the admin files the ai hand off and the forward plan?
AIEF-1.4 forward plan — updated operator/observable programMD
AIEF-1.4 forward plan — updated operator/observable program
Guiding rule

Do not start another broad seed cohort or a naive future-only parameter sweep. The current frozen replay has produced a validated operator and augmented-tangent chain; the immediate unknown is now the nonlinear observation mechanism between persistent complex EMA dynamics and the scalar resonance/operator defect.

Always separate:

exact implementation identities;
replay-established numerical facts;
controlled counterfactual evidence;
replay-specific asymptotic mechanisms;
general claims requiring regenerated trajectories or proofs.

Current checkpoint
Completed and validated for the exceptional replay/tested subspace

exact post-positivity admissibility factorization and branchwise Jacobian;
fixed K3 support and fixed effective winner map;
frozen-driver weight cocycle and nonlinear validation;
augmented directional tangent on (theta,B,C,history);
exact direction-wise Duhamel decomposition;
target-time memory and rank-one reduction diagnostics;
propagation-rate spectral-gap control;
parameter-free triangular closure;
fixed biorthogonal scalar closure;
nested sparse/right/biorthogonal/symmetric operator-family audit;
analytic dominant defect matrix and family-projection fingerprints;
pointwise EMA recurrence of the dominant defect amplitude and defect matrices;
exact trace from Delta R through proposal, projection, and operator defect.

Superseded interpretations

Retire:

a true implemented K3 branch switch at t=124.05;
an operational bare-gamma K3 edge mode at dt=0.01;
a universal Euclidean one-dimensional rollout;
an exact gap law based on the POD Rayleigh quotient;
an identified shared (a,g) two-state model;
homogeneous EMA decay of the full directed complex difference;
lambda_B/alpha_EMA as an established loop gain;
a literal pointed-torus interpretation.

Current strongest replay-specific mechanism

The K3 weight operator has exact fixed-winner sparsity. It enters a broad right-invariant family very early, then approaches biorthogonal and symmetric subfamilies through an almost rank-one left-breaking defect. The raw dominant defect is

text

E* = [[-1,0,-1],
[0,0, 0],
[0,0, 0]]

and satisfies E* v = 0 for v=(1,-1,-1)/sqrt(3) while breaking the analytic left relation. Its scalar amplitude and projected family defects follow the EMA multiplier pointwise late.

However, the full directed complex EMA difference and phase-source difference remain approximately persistent. The EMA-rate mode appears only in the magnitude observable and downstream quantities:

text

Delta R -> Delta x -> Delta B -> operator defect.

Priority 1 — Exact squared-magnitude/radial observable analysis

This is the immediate deliverable.
Action 1.1: directed squared-magnitude identity

For every directed orientation of edges 25 and 59, define
$M_n = |C_n|^2, A_n = \Re(C_n^{-} z_n), z_n = e^{i\Delta\phi_n}$.
$M_n = |C_n|^2, A_n = \Re(C_n z_n), z_n = e^{i\Delta\phi_n}$.

Verify exactly:
$M_{n+1} = \alpha^2 M_n + (1-\alpha)^2 + 2\alpha(1-\alpha) A_n$.
$M_{n+1} = \alpha^2 M_n + (1-\alpha)^2 + 2\alpha(1-\alpha) A_n$.

For edge differences:
$\Delta M_{n+1} = \alpha^2 \Delta M_n + 2\alpha(1-\alpha)\Delta A_n$.
$\Delta M_{n+1} = \alpha^2 \Delta M_n + 2\alpha(1-\alpha)\Delta A_n$.
Action 1.2: test the effective-alpha condition

The observed magnitude mode has multiplier alpha. Test the exact condition
ΔAn=?12ΔMn.
ΔAn=?21ΔMn.

Report:
|2ΔAn−ΔMn||ΔMn|,
|ΔMn||2ΔAn−ΔMn|,
|ΔMn+1−αΔMn||ΔMn|.
|ΔMn||ΔMn+1−αΔMn|.

Analyze forward and reverse directed channels separately before symmetrization.
Action 1.3: exact magnitude reconstruction

Use
|Ca|−|Cb|=|Ca|2−|Cb|2|Ca|+|Cb|
|Ca|−|Cb|=|Ca|+|Cb||Ca|2−|Cb|2

to reconstruct directed magnitude differences and then Delta R. Quantify forward/reverse cancellation and sign changes.
Action 1.4: radial/tangential geometry

With
C‾=(Ca+Cb)/2,ΔC=Ca−Cb,
C‾=(Ca+Cb)/2,ΔC=Ca−Cb,

compute radial and tangential components, while certifying that |bar C| stays above a declared floor. Treat this as an interpretation layer; the squared-magnitude equations are primary because they are exact.

Replace the misleading source metric with:

text

||Delta z|| / ||Delta C||
||Delta z - Delta C|| / ||Delta C||
radial source / radial memory

Action 1.5: late observable subsystem

Use the exact first row
$\Delta B_{n+1} = s_{7,n}[(1-\gamma dt)\Delta B_n + dt(\eta+\mu)\Delta R_n]$
$\Delta B_{n+1} = s_{7,n}[(1-\gamma dt)\Delta B_n + dt(\eta+\mu)\Delta R_n]$

and determine whether
$\Delta R_{n+1} \approx \alpha \Delta R_n$
$\Delta R_{n+1} \approx \alpha \Delta R_n$

is autonomous late or requires a measurable Delta B term after conditioning. Report design conditioning before interpreting any fitted 2x2 coefficients.
Decision gate 1
If Delta A ~ Delta M/2 and directed/symmetrized residuals close

Adopt a magnitude-selective EMA mechanism: persistent complex phase geometry produces an EMA-rate radial observable mode.
If only the symmetrized sum closes

The mechanism is forward/reverse cancellation rather than directed radial relaxation.
If neither closes

Retain a higher-dimensional observable state and do not force a 2x2 model.

Planned deliverable: AIEF_1_4_radial_observable_subsystem_analysis.py.
Priority 2 — Formal operator theorem package

Proceed in parallel after Priority 1 begins.
Theorem sequence

implemented fixed-winner sparse Jacobian;
right-invariant-family characterization;
biorthogonal-family characterization;
symmetric realized-support subfamily;
analytic left/right pair;
exact biorthogonal scalar projection;
explicit raw defect E^* and projected defect shapes;
perturbation bound linking left residual to family defect;
conditional pointwise defect recurrence;

observable squared-magnitude identity and conditional alpha-mode lemma;
augmented directional differentiability away from gate/coherence singularities.

Use conditional language. Numerical convergence over one finite replay is evidence, not an asymptotic proof.

Priority 3 — Freeze and curate the evidence lineage

Create/update:

text

AIEF_1_4_analysis_lineage.csv
AIEF_1_4_operator_cocycle_closure_note.md

Each analysis row should include:

text

source and input hashes
estimand
scope
tolerances/epsilon
principal result
superseded claim
evidence category

AIEF_1_4_OPERATOR_COCYCLE_THOUGHT_TREE.md remains the current conceptual source of truth and should be updated after the radial-observable result.

Priority 4 — Existing-component invariant-subspace controls

Before new simulations, use existing ordinary split states.

For K4, K5, and K6 components:

construct geometric face-lineality basis V;
construct a dual basis U with U^T V=I;
test KV=V Lambda and U^T K=Lambda U^T;
compare Euclidean and biorthogonal conditioning;

verify scalar-on-subspace action where component rates are uniform.

Do not expect every component to become rank one. The correct generalization is an invariant biorthogonal subspace.
Priority 5 — Consistent one-seed parameter and numerical convergence studies

Only after the observable mechanism is resolved.
EMA sensitivity

Regenerate the complete exceptional trajectory for a small grid of T_window values. Do not reuse the old split EMA/phase/history under a changed parameter.

First gate on:

text

support topology
winner branch
component class
augmented tangent validation

Treat branch/topology changes as different regimes.
Timestep/safeguard study

Regenerate complete states while varying:
dt,ε,ε/dt,slack/dt.
dt,ε,ε/dt,slack/dt.

Distinguish fixed epsilon, epsilon proportional to dt, and ideal epsilon zero where safe.
Priority 6 — Generality and remaining closure tasks

complete the matched-phase B_carrier_only/full_reset correction;
test a few existing representative component classes before any new broad cohort;
treat seeds, not directions or edges, as inferential units;
only add a new cohort if a specific uncertainty cannot be answered from existing data.

Priority 7 — Full-state geometry, downstream

Do not infer toroidal or slow-manifold topology from weight-tangent POD.

If pursued later:

quotient global phase symmetry;
encode phases circularly;
preserve delay-history consistency;
scale augmented coordinates explicitly;
compute full augmented finite-time exponents;
use local dimension/curvature, recurrence, frequency-rank, and topology diagnostics only after these controls.

The defensible hypothesis is a normally attracting, possibly stratified invariant graph over continuing phase/history dynamics—not a literal pointed torus.
Immediate execution order

Implement and run the exact squared-magnitude/radial observable audit.
Decide directed radial relaxation versus symmetrization cancellation.
Formalize the nested-family and observable-level lemmas.
Freeze the current evidence lineage.
Test invariant subspaces on existing ordinary components.
Only then design regenerated EMA and timestep/safeguard studies.
Defer broad cohorts and full-attractor topology.

Open
AI_AGENT_HANDOFF_PROMPT.mdMD

Updated both administrative files:

Key updates include:

complete current script lineage;
fixed-winner and augmented-cocycle results;
biorthogonal and nested-family operator structure;
explicit analytic defect $E*E*$;
pointwise EMA recurrence of the defect;
retirement of the incorrect homogeneous-$\Delta C\Delta C$ causal story;
the new negative result that complex EMA differences persist;
immediate priority shifted to exact squared-magnitude/radial observable dynamics;

explicit decision gates for directed radial relaxation versus forward/reverse cancellation;
deferred EMA/timestep sweeps requiring fully regenerated states;
invariant-subspace generalization for existing K4,K5,K6K4,K5,K6 components;
full-state geometry kept downstream.

The new immediate planned deliverable is:

text

AIEF_1_4_radial_observable_subsystem_analysis.py

Can you create a AIEF historic unfolding document?
WriteAIEF_HISTORIC_UNFOLDING.md1230 lines
AIEF historic unfoldingMD
AIEF historic unfolding
From falsification experiments to operator geometry, augmented cocycles, and asymptotic reduction

Updated: 2026-07-26
Frozen principal runs: AIEF-1.3 independent 100-seed validation and AIEF-1.4 40-seed split-state run
Purpose of this document

This document records how the AIEF research program changed over time: which questions were asked, which implementation problems were discovered, which empirical claims survived falsification, which interpretations were retired, and how the central mathematical object shifted from network statistics to the finite-step admissibility operator and its augmented tangent dynamics.

It is a historical and scientific record, not a paper draft. Statements are classified implicitly according to four levels:

Exact implementation fact — follows from the code or an algebraic derivation of the implemented map.
Replay-established numerical fact — measured on saved artifacts with exact replay or audited tolerances.
Mechanistic interpretation — explains several established facts but remains conditional on the audited replay/model regime.
Open theorem or generalization — not established beyond the tested seed, branch, directions, or parameter set.

The seed/graph realization is the inferential unit. Edges, perturbation directions, amplitudes, and repeated workflow-version runs are not independent experiments.

I. The model and the original scientific question

AIEF couples delayed phase oscillators to a symmetric adaptive edge-weight field.

For N=12 oscillators,
θ˙i=
θ˙i=

\omega_i
+
\sum_j B_{ij}
\sin\bigl(\theta_j(t-\tau)-\theta_i(t)\bigr),
]

with fixed delay
τ=0.5.
τ=0.5.

A complex exponential moving average stores edge-specific phase compatibility:
Cijn+1=
Cijn+1=

\alpha C_{ij}^n
+
(1-\alpha)e^{i\Delta\phi_{ij}^n},
]
Rij=|Cij|.
Rij=|Cij|.

The adaptive field is
Fij=
Fij=
(\eta+\mu)R_{ij}
\mu

\gamma B_{ij},
]

with

text

eta=0.04
mu=0.03
gamma=0.015

and a hard node-capacity constraint
∑jBij≤c,c=a−δq=0.633333….
j∑Bij≤c,c=qa−δ=0.633333….

The implemented symmetric admissibility update first forms an Euler proposal, applies nonnegativity, and then scales each edge by the more restrictive endpoint factor:

Python

projected *= np.minimum.outer(scale, scale)

This was initially treated mainly as a constraint-enforcement step. It later became the central mathematical object of the project.

The original broad questions were:

Does persistent phase compatibility create stable sparse topology?
Which terms are necessary for topology formation and maintenance?
Where is structural memory stored?
Is the mature state an equilibrium, a slow attractor, or an actively constrained state?

II. AIEF-1.3 — turning exploratory code into a falsification experiment
1. The first major transition: implementation repair

The earliest AIEF-1.3 work was dominated by code and estimand correction. Important problems included:

HTML/Markdown corruption in pasted source;

incorrect delay indexing caused by the history deque;
row-wise projection that destroyed symmetry;
a no_delay branch that retained a nonzero delay correction inside the resonance calculation;
a purported shuffle intervention that did not truly destroy temporal edge identity;
inconsistent treatment of diagonal/self-links;
correlations computed on duplicated matrix entries rather than unique undirected edges;
floating-time loop drift;
a miscomputed gap diagnostic;
undefined correlations silently replaced by zero;
overlapping historical and future windows.

These were not cosmetic changes. Several altered the implemented model or the estimand.

The corrected production driver became:

text

AIEF_1_3_falsification_updated.py

It added:

paired randomized seeds generated through SeedSequence;
separate RNG streams for initialization and shuffle interventions;
exact delay indexing and constant prehistory;
a symmetry-preserving capacity map;
integer-step simulation;
atomic checkpoints;
validated resume;
source/configuration hashes;
full manifests;
concentration, degeneracy, and projection diagnostics.

This was the first methodological turning point: AIEF became a reproducible falsification program rather than an exploratory simulation.
2. Independent validation cohorts

Two independent 50-seed cohorts were produced:

text

runs/c3da1ac707bf/
runs/73dca37cf93f/

Earlier 20-seed reruns used identical seeds under revised workflows. They were useful software-reproducibility checks but were not independent scientific evidence.
3. AIEF-1.3 intervention results
Baseline

Across the 100 independent seeds, the mature baseline had approximately:

text

D_B ~0.306
Corr_physical ~0.995
Effective_edges ~14.6
Total edge weight ~3.64

Threshold topology became completely frozen during the future evaluation window. Closure analysis later established that every baseline persistence value was exactly binary.
shuffle_R

Timestep-wise destruction of persistent edge-specific resonance identity gave:

text

D_B=0
P_mean=0
P_degenerate=1 for every seed
Total edge weight ~0.060

Interpretation: persistent edge identity in the adaptive driver is necessary for threshold topology under this aggressive shuffle. This does not establish that every weaker or blockwise shuffle would have the same effect.
no_mu

Removing the pruning term produced a dense low-concentration field:

text

D_B ~0.035
Total edge weight ~3.78
Effective_edges ~34.4
nonzero edge fraction -> 1

The conclusion was refined from “mu creates coupling” to:

mu is a pruning and concentration mechanism; it is not required for the existence of coupling.

no_gap

Removing the dominance-gap envelope yielded:

text

D_B ~0.727
Total edge weight ~127.9
G_min ~-32.8

This established a dense, high-weight regime outside the intended envelope. It did not prove mathematical divergence or loss of phase coherence.
no_delay

The no-delay branch remained in the same qualitative regime and slightly increased several observables. Explicit delay was not necessary for topology under the current EMA/adaptation architecture.
4. AIEF-1.3 closure

The scripts

text

AIEF_1_3_paired_statistics_updated.py
AIEF_1_3_closure_diagnostics_updated.py

established:

disjoint independent cohorts;
exact matrix/scalar consistency;
symmetric zero-diagonal state matrices;
exactly binary future topology persistence;
projection activity whenever a proposed row violated capacity.

Projection corrections were small relative to the full weight matrix, but later work showed that they were comparable to the proposed incremental update. This foreshadowed a major reinterpretation: the admissibility operation was not a rare repair but an active part of the dynamics.

III. AIEF-1.4 — where is memory stored?

1. Split-state intervention design

The AIEF-1.4 production experiment used a common mature prefix and cloned the complete split state into paired branches:

text

continue
reset_ema
reset_B
reset_B_ema
B_carrier_only
full_reset
future_shuffle

Frozen run:

text

runs/5a9f093a2fe0/

A design caveat remains: B_carrier_only and full_reset used different randomized phase/history resets. Their direct contrast remains exploratory until a matched-phase correction is run.

2. Memory-ablation results

Late summaries showed:

text

continue -> near-perfect retention
reset_ema -> near-perfect late recovery
reset_B -> near-perfect reconstruction
reset_B_ema -> mostly recovers, with some new edges
B_carrier_only -> substantial but incomplete recovery
full_reset -> recognizable but nonunique recovery
future_shuffle -> threshold maintenance collapses

The central conclusion changed from a search for a unique memory register to:

Mature topology is redundantly reconstructed through several state channels and a strong model-defined attractor landscape.

In particular, B cannot be called the unique or dominant memory carrier: resetting B alone reconstructs the mature topology almost exactly.

Future edge-specific resonance identity remained necessary for threshold maintenance.
3. Recovery-time and survival analysis

Sustained 90% similarity recovery was:

text

continue 40/40
reset_ema 39/40
reset_B 40/40
reset_B_ema 38/40
B_carrier_only 34/40
full_reset 25/40
future_shuffle 0/40

This introduced survival-style analysis, restricted mean unrecovered time, and caution against reporting means only among recovered seeds.
IV. Entropy geometry — a natural interior geometry meets a singular boundary state

The undirected edge state is
$w \in R^{66}, C = \{w \geq 0 : Aw \leq c\}$.
$w \in R^{66}, C = \{w \geq 0 : Aw \leq c\}$.

A natural entropy potential is
$h(w) =$
$h(w) =$

\sum_ew_e(\log w_e-1)
+
\sum_im_i(\log m_i-1),
\qquad
m=c-Aw.
]

Its gradient and Hessian are
$\psi = \log w - A^T \log m,$
$\psi = \log w - A^T \log m,$
$H = \mathrm{diag}(1/w) + A^T \mathrm{diag}(1/m) A.$
$H = \mathrm{diag}(1/w) + A^T \mathrm{diag}(1/m) A.$

A Woodbury formula reduced inversion to a node-space solve.

The inverse map reconstructed weights near machine precision, but mature states had:

text

~70% exact-zero edges
~65% node margins <=1e-12
minimum margins near1e-12

Conditioning depended strongly on interiorization and ranged roughly from 1e4 to 1e12.

Conclusion:

Mature AIEF lies on a singular boundary face. The entropy geometry is natural, but not proven canonical. A mirror formulation would be a new model, not a coordinate rewrite.

This halted an early temptation to treat mirror/proximal AIEF as a mild reformulation.

V. The active-boundary operator

1. Candidate ideal directional field

For positive and zero edges, define

F~e=

F

e=

\begin{cases}
F_e,&w_e>0,\
\max(F_e,0),&w_e=0.
\end{cases}
]

For a saturated node,

di=∑e∋iF~e,αi=max(di,0)ci,

di=e∋i∑F

e,αi=cimax(di,0),

and otherwise αi=0αi=0.

The candidate field is

Pw,e(F)=

Pw,e(F)=

\widetilde F_e

w_e\max(\alpha_i,\alpha_j).
]

Finite-difference probes supported first-order rate convergence and second-order increment convergence for the ideal map without the fixed denominator safeguard.

2. Constraint-balanced maturity

At mature split states:

‖F‖ remained appreciable,‖Pw(F)‖∼10−13.

‖F‖ remained appreciable,‖Pw(F)‖∼10−13.

Thus the adaptive driver remained active, but admissibility cancelled it to first order.

The mature adaptive component was constraint-balanced. The full delayed state was not an equilibrium: phase, history, and EMA states continued evolving.

3. Operator audits

Across roughly 1.28 million mature updates, numerical implementation checks found no failures of:

tangent feasibility;
idempotence;
positive homogeneity;
permutation equivariance.

These were audits of the candidate formula, not a proof that it is the Hadamard directional derivative.

VI. The exceptional seed and the first tangent interpretation

One seed differed sharply:

text

4619122171158768524

For 39/40 seeds, the mature adaptive field was stationary to machine precision with no support or saturation changes.

The exceptional seed showed:

text

one saturation-classifier change
large continuous B drift
nonzero ideal stationarity ratio before the event
near-stationarity afterward
exact deterministic replay

The first interpretation was an active-face switch at approximately t=124.05, with tangent dimension changing from 9 to 8.

Corrected tangent experiments reused one direction across all amplitudes and found typical contraction near
0.315789,
0.315789,

with excellent exponential fits. The exceptional directions were slower and less radially clean.

At this stage, the project still treated the tolerance-defined active set as if it were the branch structure of the implemented finite-step map. That interpretation was later retired.

VII. Fixed-face spectra and channel invariance

1. Analytic fixed-stratum Jacobian

On a fixed support/saturation/controller stratum, tangent perturbations satisfy constant node corrections, giving
JT=
JT=

Q_T^T
\operatorname{diag}\bigl(-\gamma-\rho_e\bigr)
Q_T.
]

Numerical finite differences matched this matrix to roughly 1e-13–1e-11.

Observed rates included:
0.189474,0.252632,0.315789.
0.189474,0.252632,0.315789.

2. Intertwining and channel decomposition

For active incidence matrix CC and diagonal edge-rate operator DD, the analysis tested
CD≈LC.
CD≈LC.

Residuals were near machine precision for all 40 split states. The tangent projector commutator was similarly tiny.

The mathematical implication is:
$CD=LC \Longrightarrow D(\ker C) \subseteq \ker C$.
$CD=LC \Longrightarrow D(\ker C) \subseteq \ker C$.

Since DD is diagonal/self-adjoint,
T=
T=

\bigoplus_r(T\cap E_r),
]

and rate multiplicity equals the dimension of each channel intersection.

This was the first complete explanation of the repeated tangent spectra, conditional on exact intertwining.
VIII. Component geometry — the clique structure becomes directly visible

The component audit reconstructed actual positive-support components rather than inferring graph classes from spectra.

For 39 ordinary split states:

all positive-support nodes were saturated;
every support component was rate-uniform;
every component passed locked-edge and edge-balance checks;
explicit component intertwiners reproduced the tangent spectra;
support components were complete graphs.

Observed combinations included:

text

K6 + K2 + K3
K6 + K2 + K4
K5 + K2 + K4
K5 + K3 + K4
K5 + K6

For a fully saturated locked component of degree kk,
r=kηc,w=ck.
r=ckη,w=kc.

Since
ηc=695,
cη=956,

the channel rates are degree-generated.

For KsKs, the signless-incidence lineality dimension is
qs=(s2)−s=s(s−3)2.
qs=(2s)−s=2s(s−3).

Thus:

text

K2 -> nullity 0
K3 -> nullity 0
K4 -> nullity 2
K5 -> nullity 5
K6 -> nullity 9

The previously “missing” K2 and K3 components were spectrally silent because they had zero fully constrained face lineality.

The clique hypothesis had changed status: within the frozen cohort, complete support components were now directly measured.
IX. The exceptional K3

The exceptional support was

text

K5 + K3[partially classified active] + K4

with triangle

text

nodes [2,7,11]
edges [25=(2,7), 29=(2,11), 59=(7,11)]

The tolerance-classified K3 changed from one active row to two, explaining the classifier-based nullity change from 2 to 1 and the global dimension change 9 -> 8.

Dense K3 analysis confirmed the finite-dimensional one-row and two-row algebra. However, boundary-tolerance sensitivity showed that the event time depended strongly on the classifier tolerance.

This prompted the next major question:

What object was actually being linearized?

X. The actual finite-step branch replaces the classifier picture

For a node, define
Πi=
Πi=

\frac{dt,d_i}{\operatorname{slack}_i}.
]

At the exceptional split:

text

node 2: Pi ~2392
node 7: Pi ~5.3e8
node11: Pi <<1

Nodes 2 and 7 were already proposal-violating at the first recorded state, even though the slack classifier called only node 7 active.

Node 11 became proposal-violating later, at approximately t=113.81, but still did not win an edge.

The actual scale/winner audit reconstructed the finite update exactly and found the winner map remained fixed for all 32,001 states:
(2,7)↦7,(2,11)↦2,(7,11)↦7.
(2,7)↦7,(2,11)↦2,(7,11)↦7.

There were:

text

0 effective-edge branch changes

The event at 124.05 changed only the post hoc classifier. It did not change the implemented edge formulas.

This retired the “exceptional branch switch” interpretation. The trajectory stayed on one fixed symbolic winner branch while its coefficients evolved toward a tie.
XI. Exact finite-step factorization and fiber geometry

For a nonnegative proposal xx, with row sums q=Axq=Ax, the safeguarded finite map factorizes as
Aε(x)=M(Ax)x.
Aε(x)=M(Ax)x.

All nonlinear dependence enters through node row sums; conditional on those row sums, the edge action is diagonal.

For x,y≥0x,y≥0 with Ax=Ay=qAx=Ay=q,
Aε(y)−Aε(x)=
Aε(y)−Aε(x)=

M(q)(y-x).
]

Thus the map is exactly affine-linear on nonnegative row-sum fibers.

On strict winner strata, the Jacobian follows directly from the code. At ties, the ambient map is generally only directionally differentiable. On a fully constrained face-lineality direction $Ah=0$, scales remain fixed exactly, so ambient tie nonsmoothness disappears on the fiber.

This factorization became the principal exact implementation theorem.
XII. Frozen-driver cocycle — non-normal transient and severe anisotropy

With replayed R_n treated as exogenous,
$K_n(B)=$
$K_n(B)=$

(1-\gamma dt)
D\mathcal A_\varepsilon(x_n).
]

The K3 sequence was fixed-form but time varying.

The from-split propagator showed:

text

full peak amplification 1.27097 at t=0.1
split-lineality peak 1.20563
long-horizon principal rate -0.1233

A constant K_0 reproduced the initial transient almost exactly. A diagonal-scale-only control did not amplify.

Conclusion:

The early weight-sector transient is already encoded in the initial non-normal branch Jacobian and requires the node-mediated low-rank correction; coefficient drift is not required for the peak.

Stable compound-cocycle analysis resolved a spectrum with one moderate and two extreme contraction rates. The most negative mode was controlled by the fixed 1e-12 safeguard and is a real feature of the implemented discrete map, though not an intended finite channel of an ideal epsilon -> 0 limit.

Centered frozen-driver nonlinear trajectories matched the cocycle accurately while preserving the branch.
XIII. Augmented tangent — the correct state space is larger than weights

The full discrete state is
Xn=(θn,Bn,Cn,Hn).
Xn=(θn,Bn,Cn,Hn).

The augmented tangent differentiated:

delayed phase differences;
complex EMA memory;
the magnitude map R=|C|R=|C|;
adaptive forcing;
positivity;
safeguarded admissibility;
projected phase coupling;
phase update;
history shift.

Both +h+h and −h−h directional tangents were propagated to handle endpoint ties.

The custom base step reproduced the implemented step exactly. Synthetic one-step and multistep tangent tests passed.

On the production replay, median errors were approximately:

text

K3/full-B tangent ~5e-4
phase tangent ~1.6e-6
R tangent ~1.8e-4

This reduced the long-lived frozen/coupled discrepancy from order 4e-1 to order 1e-4–1e-3.

Stage B established that phase/history/EMA feedback was necessary for the long-lived response, while the frozen weight cocycle remained the correct weight-to-weight block.

XIV. Duhamel decomposition — homogeneous decay versus regenerated response

For the K3 weight tangent bnbn, define
fn=bn+1−Knbn.
fn=bn+1−Knbn.

Then
hn+1=Knhn,rn+1=Knrn+fn,
hn+1=Knhn,rn+1=Knrn+fn,
bn=hn+rn.
bn=hn+rn.

Reconstruction errors were near 1e-15; residual forcing matched the directly differentiated $R \to BR \to B$ contribution near 1e-13.

Exact feedback crossover times were:

text

direction 0: 0.38
direction 1: 72.86

The fast homogeneous direction disappeared almost immediately. The slow direction retained its original perturbation for roughly 73 time units before regenerated feedback dominated.

Final feedback was generated almost entirely by recent forcing. Early forcing contributions were tiny and anti-aligned with the final response; late forcing contributions were aligned.

Ablations showed:

removing R -> B collapsed the late response;
fresh phase forcing and EMA memory were both necessary;
current-phase input selected the late sign/amplitude;
delayed-history input modestly suppressed the late direction-1 response.

These were recursive tangent counterfactuals, not additive causal percentages.
XV. Rank-one reduction and operational memory

Late total and regenerated K3 tangents became almost one-dimensional.

POD results:

text

weighted first-mode fraction ~1
unit-vector first-mode fraction ~0.999999 late

Late transverse fractions were approximately:

text

direction 0: 1.5e-3
direction 1: 1e-8 to 1e-6

Raw forcing was directionally broader (unit E1 ~0.93), while propagated forcing became almost perfectly rank one. Thus the weight cocycle acted as a rank-one directional filter.

Target-time recent-forcing reconstruction fits were transient before roughly T=150 and stabilized after about T=200:
$\kappa \approx 0.06913, \tau_{\text{eff}} \approx 14.46$.
$\kappa \approx 0.06913, \tau_{\text{eff}} \approx 14.46$.

At late targets:

text

50% reconstruction error horizon ~10.02
10% horizon ~33.3
1% horizon ~66.6

The operational memory law was late-regime specific, not global.
XVI. Scalar projection, spectral gap, and biorthogonal correction

The empirical late POD direction was
vPOD≈(0.577341,−0.577369,−0.577341).
vPOD≈(0.577341,−0.577369,−0.577341).

The analytic K3 lineality direction is
v=13(1,−1,−1).
v=3
1(1,−1,−1).

Their angle was about 0.0013 degrees.

Euclidean projection onto vv or vPODvPOD gave accurate direction-1 rollouts but failed badly for direction 0 because the operator was strongly non-normal.

A numerical left mode repaired the rollout. It converged to
u=32(1,0,−1),uTv=1.
u=23
(1,0,−1),uTv=1.

Fixed left covectors from t=200, 250, and 319.99 gave nearly identical results. From t=250, parameter-free final errors were roughly:

text

direction 0: 1.9e-5
direction 1: 2.8e-5

The rank-one decision gate passed: the prior failure was a Euclidean-coordinate artifact rather than evidence that a second weight mode was required.

The late triangular reduction became
an+1=λnan+gn,gn+1≈αEMAgn,
an+1=λnan+gn,gn+1≈αEMAgn,

with
an=uTbn,gn=uTfn.
an=uTbn,gn=uTfn.

Direct forcing-propagation analysis established, in the late regime,
κ≈rprop−rg
κ≈rprop−rg

to approximately 1e-6 at the final target. The projected forcing envelope decayed at the EMA rate.

The quantity λB/αEMAλB/αEMA is a relative-retention factor, not an established closed-loop gain.
XVII. Nested operator families and the analytic defect

The selected K3 block has exact sparsity
K=(p0qrs0t0u).
K=
prt0s0q0u
.

Four nested families were audited:
Ksym⊂KLR⊂KR⊂Ksparse.
Ksym⊂KLR⊂KR⊂Ksparse.

At t=100, distance to the right-invariant family was already about 5.6e-9, while distances to the biorthogonal and symmetric families were order 1e-3. Thus the right line was protected by much weaker conditions than full symmetry.

The dominant left-breaking mismatch was identified as
δn=β25,n−β59,n,
δn=β25,n−β59,n,

entering through
E∗=
E∗=

\begin{pmatrix}
-1&0&-1\
0&0&0\
0&0&0
\end{pmatrix}.
]

This matrix satisfies
$E*v=0$
$E*v=0$

but breaks the analytic left relation.

Its projected shapes explain the exact residual constants
32,38.
23
,8
3.

Numerical defect POD directions matched the analytic projected defect shapes at cosine 1-O(1e-12), with direct reconstruction errors of only a few parts in 1e6.

The scalar defect and family-defect matrices obeyed the EMA multiplier pointwise late:

text

median multiplier error ~2e-9
matrix shape error ~1e-9 to1e-8
no sign changes

This established a replay-specific operator relaxation mode, while causal control by EMA remained unproved.
XVIII. Causal trace — a productive negative result

The implementation chain from resonance magnitude to operator defect closed exactly:
$\Delta R \to \Delta x \to \Delta B \to \delta$.
$\Delta R \to \Delta x \to \Delta B \to \delta$.

All identities reproduced at machine precision, and the downstream scalar quantities shared the EMA multiplier late.

However, the full directed complex differences and phase-source differences remained approximately persistent. Their norms did not decay at the EMA rate.

This rejected the simple story:

The full complex EMA difference becomes source-free and decays homogeneously.

The source-to-memory norm ratio had been misleading because the source was multiplied by 1−α1−α. In fact, source and memory differences had comparable norms.

The unresolved mechanism was localized to the observation map
C↦|C|.
C↦|C|.

The next exact identity is the squared-magnitude equation:
Mn+1=
Mn+1=

\alpha^2M_n
+
(1-\alpha)^2
+
2\alpha(1-\alpha)\Re(\overline C_nz_n).
]

For directed edge differences:
ΔMn+1=
ΔMn+1=

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta A_n.
]

An effective αα-mode requires
ΔAn≈12ΔMn.
ΔAn≈21ΔMn.

This became the immediate observable-level question.
XIX. Major interpretations retired during the program

The following statements were useful hypotheses but are no longer supported in their original form:

B is the unique or dominant memory carrier.
Retired because reset_B reconstructs mature topology almost exactly.

The projection is an occasional repair.
Retired because capacity correction is active on essentially every mature update and is comparable to the incremental proposal.

The mature full system is at equilibrium.
Retired; only the adaptive weight component is constraint-balanced.

The exceptional seed undergoes an implemented active-face switch at t=124.05.
Retired; this was a tolerance-classifier transition. The actual winner branch remains fixed.

The split K3 has an operational bare-gamma edge mode at dt=0.01.
Retired; node 2 is proposal-violating and controls the edge from the first step.

A Euclidean scalar coordinate should describe both K3 directions.
Retired; a biorthogonal left coordinate is required by non-normality.

The POD Rayleigh rate gives the exact memory gap.
Retired; the correct rate is the decay of the actual propagated forcing channel.

The full complex EMA difference decays homogeneously at the EMA rate.
Retired; the magnitude observable decays while complex differences persist.

A two-state model in (a,g) has been identified.
Retired; late collinearity makes unconstrained coefficients unidentifiable. The parameter-free triangular relation is the supported reduction.

A pointed torus is a likely literal geometry.
Not supported. Any full-state geometric study must first quotient phase symmetry and distinguish attractor tangents from stable fibers.

XX. Current mathematical picture

For the audited exceptional replay, the strongest current synthesis is:

The finite-step map follows a fixed K3 winner branch.
Its selected Jacobian lies exactly in a sparse family.
A broad right-invariant structure is reached very early.
A dominant explicit defect preserves the right line but breaks the left mode.
The defect amplitude and projected family defects follow the EMA multiplier late.
A stable analytic biorthogonal pair gives a parameter-free scalar weight recurrence.
The augmented phase/history/EMA state continually regenerates the weight perturbation.
The weight cocycle filters forcing into a nearly fixed one-dimensional late channel.
The final response depends mainly on recent forcing over roughly 60–70 time units.
The remaining unresolved mechanism is how persistent complex EMA dynamics produce an EMA-rate magnitude difference through the nonlinear observation map.

This picture is conditional on:

one frozen seed;
one fixed winner branch;
two tested initial directions;
the implemented dt=0.01 and epsilon=1e-12 map;
finite replay time.

It is not yet a universal AIEF theorem.
XXI. Immediate research frontier
1. Squared-magnitude/radial observable audit

For directed edges 25 and 59, test exactly:
ΔMn+1=
ΔMn+1=

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta A_n,
]

and whether
2ΔAn≈ΔMn.
2ΔAn≈ΔMn.

Separate forward and reverse directions before symmetrization. Quantify radial/tangential components and midpoint conditioning.
2. Observable subsystem

Combine the exact weight-difference equation
ΔBn+1=
ΔBn+1=

s_{7,n}\big[(1-\gamma dt)\Delta B_n
+dt(\eta+\mu)\Delta R_n\big]
]

with the measured/derived evolution of ΔRΔR. Determine whether the late system is triangular or genuinely coupled.
3. Formal theorem package

Write proofs for:

factorization and fiber linearity;
fixed-winner sparse Jacobian;
nested invariant families;
analytic left/right pair;
defect matrix and projection constants;
conditional scalar and memory lemmas;
augmented directional derivative away from gate and zero-magnitude singularities.

4. Existing-component generality

Use existing K4, K5, and K6 states to test invariant subspaces, not universal rank-one behavior.
5. Regenerated sensitivity studies

Only after the observable mechanism is resolved, regenerate complete trajectories for targeted changes in:

T_window;
timestep;
safeguard scaling.

Never reuse a saved split state under inconsistent dynamics parameters.

XXII. Principal files and artifacts

Production and closure

text

AIEF_1_3_falsification_updated.py
AIEF_1_3_paired_statistics_updated.py
AIEF_1_3_closure_diagnostics_updated.py
AIEF_1_4_memory_ablation_updated.py

Geometry and active boundary

text

AIEF_1_4_entropy_geometry_diagnostics.py
AIEF_1_4_local_update_comparison.py
AIEF_1_4_active_boundary_analysis.py
AIEF_1_4_stationarity_perturbation_analysis_updated.py

Tangent spectra and components

text

AIEF_1_4_exceptional_tangent_analysis.py
AIEF_1_4_face_spectrum_analysis.py
AIEF_1_4_channel_invariance_analysis.py
AIEF_1_4_intertwining_controller_analysis.py
AIEF_1_4_component_controller_audit.py
AIEF_1_4_exceptional_K3_transition_audit.py

Actual branch and cocycles

text

AIEF_1_4_actual_admissibility_branch_audit.py
AIEF_1_4_frozen_driver_cocycle_analysis.py
AIEF_1_4_cocycle_controls_coupled_comparison.py

AIEF_1_4_augmented_tangent_cocycle_analysis.py

Duhamel, memory, and reduction

text

AIEF_1_4_augmented_duhamel_feedback_analysis.py
AIEF_1_4_duhamel_memory_alignment_analysis.py
AIEF_1_4_feedback_rank_memory_reduction.py
AIEF_1_4_scalar_spectral_gap_reduction.py
AIEF_1_4_propagation_gap_geometric_control.py
AIEF_1_4_parameter_free_triangular_closure.py
AIEF_1_4_biorthogonal_triangular_closure.py

Operator-family and defect program

text

AIEF_1_4_nested_operator_family_audit.py
AIEF_1_4_defect_recurrence_analytic_mode.py
AIEF_1_4_defect_causal_ema_trace.py

Administrative sources of truth

text

AI_AGENT_HANDOFF_PROMPT.md
AIEF_1_4_OPERATOR_COCYCLE_THOUGHT_TREE.md
AIEF_1_4_FORWARD_PLAN.md
AIEF_HISTORIC_UNFOLDING.md

Final perspective

The AIEF program began by asking whether adaptive phase compatibility creates sparse persistent topology. It then passed through reproducibility repair, falsification, memory ablation, boundary geometry, tangent spectra, and active-face analysis.

Its current center is different:

A piecewise-rational finite-step admissibility map creates a fixed-branch, non-normal weight operator whose geometry can be factored, differentiated, decomposed, and reduced. In one exceptional replay, the resulting augmented tangent dynamics collapse onto an analytically identifiable biorthogonal mode, while persistent complex EMA dynamics are filtered through a nonlinear magnitude observable into an EMA-rate operator defect.

The next advance will come not from adding another broad simulation matrix, but from deriving the observable-level mechanism and formalizing the operator hierarchy already exposed by the saved data. Open

Created .

It provides a chronological scientific history covering:

original AIEF model and questions;
AIEF-1.3 implementation repairs and falsification;
independent 100-seed closure;
AIEF-1.4 memory ablations and recovery;
entropy and boundary geometry;
ideal active-boundary operator;
exceptional-seed discovery;
tangent spectra and intertwining;
clique/component mechanism;
revision of the false active-switch interpretation;
actual finite-step winner branch;
operator factorization and fiber linearity;
frozen and augmented cocycles;
Duhamel regeneration;
rank-one and memory reduction;
geometric/biorthogonal scalar closure;
nested operator families;
analytic defect matrix and pointwise EMA recurrence;
the productive negative result for full complex EMA decay;
retired interpretations;
current mathematical picture;
immediate research frontier;
complete principal artifact index.

It explicitly separates implementation facts, replay evidence, mechanistic interpretations, and open generalizations.

Please deliver the proposed
WriteAIEF_1_4_radial_observable_subsystem_analysis.py749 lines
Edit×4
AIEF_1_4_radial_observable_subsystem_analysis.pyPYTHON

Delivered:

It compiles successfully and passed direct synthetic checks of:
ΔMn+1=
ΔMn+1=

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta A_n
]

and the exact squared-magnitude reconstruction of magnitude differences.
Run it

Bash

```
python AIEF_1_4_radial_observable_subsystem_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--dt 0.01 \
--scale-epsilon 1e-12 \
--intervals 0,100,150,200,250,320 \
--signal-floor 1e-15 \
--midpoint-floor 1e-8 \
--output-dir aief_1_4_radial_observable_subsystem
```

Exact directed observable analysis

For forward and reverse orientations separately, it computes:

M=|C|2,A=ℜ(C¯z),z=eiΔϕ.
M=|C|2,A=ℜ(Cz),z=eiΔϕ.

It verifies:
Mn+1=
Mn+1=

\alpha^2M_n
+
(1-\alpha)^2
+
2\alpha(1-\alpha)A_n,
]

and for edge differences:
ΔMn+1=
ΔMn+1=

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta A_n.
]
Effective-αα condition

It directly tests:
2ΔAn=?ΔMn
2ΔAn=?ΔMn

using:
|2ΔAn−ΔMn||ΔMn|.
|ΔMn||2ΔAn−ΔMn|.

It also measures:
|ΔMn+1−αΔMn||ΔMn|.
|ΔMn||ΔMn+1−αΔMn|.

Thus the script can determine whether the EMA-rate magnitude mode follows from the precise radial-alignment condition.
Exact magnitude and RR reconstruction

For each directed channel:
|Ca|−|Cb|=
|Ca|−|Cb|=

\frac{|C_a|^2-|C_b|^2}
{|C_a|+|C_b|}.
]

Forward and reverse magnitude differences are then combined exactly into:
ΔR=
ΔR=

\frac12
\left(
\Delta|C|{\rm forward}
+
\Delta|C|{\rm reverse}
\right).
]

It reports a forward/reverse cancellation index:
|Δ|C|f+Δ|C|r||Δ|C|f|+|Δ|C|r|.
|Δ|C|f|+|Δ|C|r||Δ|C|f+Δ|C|r|.

Values near one indicate reinforcement; values near zero indicate cancellation.
Radial and tangential geometry

For
C⁻=Ca+Cb2,ΔC=Ca−Cb,
C⁻=2Ca+Cb,ΔC=Ca−Cb,

the script computes:
r=ℜ(C⁻¯ΔC)|C⁻|,
r=|C⁻|ℜ(C⁻ΔC),
t=ℑ(C⁻¯ΔC)|C⁻|.
t=|C⁻|ℑ(C⁻ΔC).

It reports radial fractions and certifies the minimum midpoint magnitude.

The phase-source difference ΔzΔz is projected into the memory midpoint frame, not its own source frame. This is the relevant geometry for determining whether source perturbations affect the magnitude observable.
Corrected source diagnostics

It reports:
‖Δz‖‖ΔC‖,
‖ΔC‖‖Δz‖,

and
‖Δz−ΔC‖‖ΔC‖.
‖ΔC‖‖Δz−ΔC‖.

These replace the misleading metric that was automatically suppressed by the small factor 1−α1−α.
Observable subsystem

It verifies the exact weight-difference update:
ΔBn+1=
ΔBn+1=

s_{7,n}
\left[
(1-\gamma dt)\Delta B_n
+
dt(\eta+\mu)\Delta R_n
\right].
]

It also fits, by interval,
ΔRn+1=
ΔRn+1=

a_R\Delta R_n
+
b_B\Delta B_n
]

without an intercept and reports:

aRaR;
bBbB;
R2R2;
relative RMSE;
design singular values;
condition number;
corr(ΔR,ΔB)corr(ΔR,ΔB);
deviation of aRaR from αEMAαEMA;
direct autonomous-αα residual.

A well-fit but ill-conditioned regression will therefore not be misidentified as a unique coupled subsystem.
Outputs

text

radial_observable_trajectory.csv
radial_observable_interval_summary.csv
observable_subsystem_fits.csv
radial_observable_identity_checks.csv
radial_observable_report.txt
analysis_manifest.json

Main decision

The result should distinguish among:

Directed radial relaxation
2ΔA≈ΔM
2ΔA≈ΔM

separately in both orientations.

Symmetrization-generated mode
Individual directed channels fail, but their sum produces ΔR∼αnΔR∼αn.

Higher-dimensional observable mechanism
Neither directed nor symmetrized scalar closure is adequate.

Please paste radial_observable_report.txt after the run.

Think about this:

AIEF K3 radial observable and subsystem analysis

==========

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

Final replay exact: True

Exact identity errors

squared_magnitude_forward: max=6.79144e-16, median=1.24234e-16

squared_magnitude_reverse: max=6.93889e-16, median=1.27039e-16

directed_magnitude_forward: max=8.32667e-17, median=1.56626e-17

directed_magnitude_reverse: max=8.32667e-17, median=1.56737e-17

symmetrized_R: max=3.88578e-16, median=5.55112e-17

weight_difference_update: max=1.02484e-16, median=2.25514e-17

Interval observable diagnostics

channel=forward, interval=[0,100): DeltaM_rate=0.0517205, R2=0.996422, alpha_error=3.32e-05, alignment_ratio=1.05815, alignment_error=0.0581, radial_fraction=0.0394589, source_memory=0.95585, source_difference=0.074294, min_midpoint=0.651741

channel=reverse, interval=[0,100): DeltaM_rate=0.0521942, R2=0.996733, alpha_error=2.84e-05, alignment_ratio=1.04976, alignment_error=0.0498, radial_fraction=0.04768, source_memory=0.92404, source_difference=0.0977407, min_midpoint=0.722255

channel=symmetrized, interval=[0,100): DeltaM_rate=0.0531227, R2=0.997923, alpha_error=2.39e-05, alignment_ratio=nan, alignment_error=nan, radial_fraction=nan, source_memory=nan, source_difference=nan, min_midpoint=nan

channel=forward, interval=[100,150): DeltaM_rate=0.0569869, R2=0.999999, alpha_error=1.29e-06, alignment_ratio=1.00227, alignment_error=0.00227, radial_fraction=0.000605271, source_memory=0.998686, source_difference=0.00182296, min_midpoint=0.937197

channel=reverse, interval=[100,150): DeltaM_rate=0.0570289, R2=1, alpha_error=9.13e-07, alignment_ratio=1.0016, alignment_error=0.0016, radial_fraction=0.000752486, source_memory=0.997847, source_difference=0.00243892, min_midpoint=0.970849

channel=symmetrized, interval=[100,150): DeltaM_rate=0.0570195, R2=1, alpha_error=1.01e-06, alignment_ratio=nan, alignment_error=nan, radial_fraction=nan, source_memory=nan, source_difference=nan, min_midpoint=nan

channel=forward, interval=[150,200): DeltaM_rate=0.0571189, R2=1, alpha_error=1.25e-07, alignment_ratio=1.00022, alignment_error=0.00022, radial_fraction=3.49008e-05, source_memory=0.999897, source_difference=0.000134524, min_midpoint=0.938626

channel=reverse, interval=[150,200): DeltaM_rate=0.0571238, R2=1, alpha_error=8.24e-08, alignment_ratio=1.00014, alignment_error=0.000144, radial_fraction=4.33961e-05, source_memory=0.999836, source_difference=0.000179225, min_midpoint=0.971996

channel=symmetrized, interval=[150,200): DeltaM_rate=0.057122, R2=1, alpha_error=9.92e-08, alignment_ratio=nan, alignment_error=nan, radial_fraction=nan, source_memory=nan, source_difference=nan, min_midpoint=nan

channel=forward, interval=[200,250): DeltaM_rate=0.0571321, R2=1, alpha_error=1.1e-08, alignment_ratio=1.00002, alignment_error=1.92e-05, radial_fraction=2.00603e-06, source_memory=0.999993, source_difference=9.45645e-06, min_midpoint=0.938722

channel=reverse, interval=[200,250): DeltaM_rate=0.0571325, R2=1, alpha_error=6.88e-09, alignment_ratio=1.00001, alignment_error=1.21e-05, radial_fraction=2.49429e-06, source_memory=0.999988, source_difference=1.25668e-05, min_midpoint=0.972071

channel=symmetrized, interval=[200,250): DeltaM_rate=0.0571323, R2=1, alpha_error=8.68e-09, alignment_ratio=nan, alignment_error=nan, radial_fraction=nan, source_memory=nan, source_difference=nan, min_midpoint=nan

channel=forward, interval=[250,319.99): DeltaM_rate=0.0571333, R2=1, alpha_error=2.17e-09, alignment_ratio=1, alignment_error=9.44e-07, radial_fraction=6.51195e-08, source_memory=1, source_difference=3.74652e-07, min_midpoint=0.938728

channel=reverse, interval=[250,319.99): DeltaM_rate=0.0571333, R2=1, alpha_error=1.84e-09, alignment_ratio=1, alignment_error=5.71e-07, radial_fraction=8.09691e-08, source_memory=1, source_difference=4.97033e-07, min_midpoint=0.972076

channel=symmetrized, interval=[250,319.99): DeltaM_rate=0.0571333, R2=1, alpha_error=2.4e-09, alignment_ratio=nan, alignment_error=nan, radial_fraction=nan, source_memory=nan, source_difference=nan, min_midpoint=nan

Symmetrized resonance

[0,100): DeltaR_rate=0.0531227, alpha_error=2.39e-05, cancellation=1

[100,150): DeltaR_rate=0.0570195, alpha_error=1.01e-06, cancellation=1

[150,200): DeltaR_rate=0.057122, alpha_error=9.92e-08, cancellation=1

[200,250): DeltaR_rate=0.0571323, alpha_error=8.68e-09, cancellation=1

[250,319.99): DeltaR_rate=0.0571333, alpha_error=2.4e-09, cancellation=1

Observable subsystem fits

interval=[0,100): a_R=1.00123075663, b_B=-0.00173652, R2=1, RMSE=4.27e-05, condition=28.8347, corr=0.996568

interval=[100,150): a_R=1.00266808705, b_B=-0.00320194, R2=1, RMSE=4.66e-07, condition=15385.5, corr=1

interval=[150,200): a_R=1.00001625415, b_B=-0.000580219, R2=1, RMSE=4.88e-09, condition=17142.4, corr=1

interval=[200,250): a_R=0.999776351045, b_B=-0.00034323, R2=1, RMSE=2.05e-10, condition=107454, corr=1

interval=[250,319.99): a_R=0.999697405948, b_B=-0.000265258, R2=1, RMSE=1.3e-09, condition=957611, corr=1

Interpretation constraints

Squared-magnitude recurrences are exact; radial/tangential decompositions are geometric diagnostics.

DeltaA ~= DeltaM/2 is the precise condition for an effective alpha multiplier of DeltaM.

Forward and reverse directed channels are analyzed before symmetrization.

A well-fit but ill-conditioned (DeltaB,DeltaR) regression does not identify unique subsystem coefficients.

Results remain conditional on one replay and fixed K3 branch.

This result identifies the missing observable mechanism. The EMA-rate operator defect is generated by directed radial equalization, not by decay of the complete complex difference and not by forward/reverse cancellation.
1. Exact observation-level chain is verified

All exact identities close at machine precision:
ΔMn+1=
ΔMn+1=

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta A_n,
]
Δ∣C∣=
Δ∣C∣=

\frac{\Delta M}
{|C_a|+|C_b|},
]
ΔR=
ΔR=

\frac12
\left(
\Delta|C|_f+\Delta|C|_r
\right),
]

and
ΔBn+1=
ΔBn+1=

s_7\left[
(1-\gamma dt)\Delta B_n
+
dt(\eta+\mu)\Delta R_n
\right].
]

Errors are approximately 10−1610−16.

Thus the complete implemented path
(C,z)→∣C∣→R→B→K
(C,z)→∣C∣→R→B→K

is now algebraically and numerically closed.
2. Both directed channels independently develop the EMA multiplier

Late, over 250250–320320:
$r_{\Delta M,f}=0.0571333, r_{\Delta M,r}=0.0571333, r_{\Delta R}=0.0571333$.
$r_{\Delta M,fr}\Delta M,r_{r\Delta R}=0.0571333,=0.0571333,=0.0571333$.

Pointwise alpha errors are:
forward:2.17×10^{-9},reverse:1.84×10^{-9},symmetrized:2.40×10^{-9}.
forward:reverse:symmetrized:$2.17\times10^{-9}, 1.84\times10^{-9}, 2.40\times10^{-9}$.

Therefore the EMA-rate mode exists before forward/reverse symmetrization.
3. Symmetrization does not create the mode

The cancellation index is
1
1

in every reported interval.

Thus forward and reverse magnitude differences have the same sign and reinforce one another.

So:
The EMA-rate ΔR mode is not produced by directional cancellation.
The EMA-rate ΔR mode is not produced by directional cancellation.

It is present independently in both directed magnitude channels.

This resolves one of the principal decision gates.
4. The radial-alignment condition is directly verified

The exact condition for an effective $\alpha\alpha$ multiplier is
$2\Delta An=\Delta Mn$.
$2\Delta An=\Delta Mn$.

The measured ratio converges systematically:
interval$2\Delta A/\Delta M$ forwardreverse0–1001.05821.0498100–1501.002271.00160150–2001.000221.00014200–2501.0000191.000012250–3201.00000091.0000006
interval0–100100–150150–200200–250250–3202$\Delta A/\Delta M$
forward1.05821.002271.000221.0000191.0000009reverse1.04981.001601.000141.0000121.0000006

The alignment-law residual reaches:
9.4×10^{-7}
9.4×10^{-7}

forward and
5.7×10^{-7}
5.7×10^{-7}

reverse.

Therefore:
$\Delta A_n \to 12\Delta M_n$
$\Delta A_n \to 21\Delta M_n$

along both directed channels.

Substituting into the exact equation gives:
$\Delta M_{n+1} \to \alpha 2\Delta M_n + \alpha(1-\alpha)\Delta M_n =$
$\Delta M_{n+1} \to \alpha 2\Delta M_n + \alpha(1-\alpha)\Delta M_n =$

\alpha\Delta M_n.
]

The observed EMA multiplier now has a direct observation-level derivation.
5. The persistent complex difference is overwhelmingly tangential

The radial fraction decreases as:
intervalforwardreverse0–1000.03950.0477100–1506.1×10−47.5×10−4150–
2003.5×10−54.3×10−5200–2502.0×10−62.5×10−6250–3206.5×10−88.1×10−8
interval0–100100–150150–200200–250250–320forward0.03956.1×10−43.5×10−52.0×10−66.5×10−8
reverse0.04777.5×10−44.3×10−52.5×10−68.1×10−8

Thus, late:
ΔC is almost entirely tangential to the complex-memory midpoint.
ΔC is almost entirely tangential to the complex-memory midpoint.

The midpoint magnitudes remain safely bounded:

|C⁻f|>0.938,|C⁻r|>0.972
|C⁻f|>0.938,|C⁻r|>0.972

late, so the radial/tangential decomposition is well conditioned.

This directly validates the earlier hypothesis that the persistent complex separation is mainly angular while the radial magnitude difference vanishes.
6. The phase-source difference locks to the memory difference

Late:
$\|\Delta z\|\|\Delta C\| \to 1,$
$\|\Delta C\|\|\Delta z\| \to 1,$

and
$\|\Delta z - \Delta C\|\|\Delta C\| \to 4\times10^{-7}–5\times10^{-7}.$
$\|\Delta C\|\|\Delta z - \Delta C\| \to 4\times10^{-7}–5\times10^{-7}.$

So:
$\Delta zn \approx \Delta Cn$
$\Delta zn \approx \Delta Cn$

as complex vectors.

This explains why the complete complex difference persists:
$\Delta Cn+1=$
$\Delta Cn+1=$

$$\alpha\Delta C_n + (1-\alpha)\Delta z_n \approx \Delta C_n.$$

The phase source continually replenishes the full complex difference rather than becoming negligible.

At the same time, the radial alignment observable satisfies
$\Delta A \approx \Delta M/2,$

ΔA≈ΔM/2,

causing the magnitude difference to decay with multiplier αα.

The two behaviors are therefore compatible:
complex difference:persistent,radial/magnitude difference:α-decaying.
complex difference:radial/magnitude difference:persistent,α-decaying.
7. Exact midpoint identity clarifies the geometry

For
$C^{-}=12(Ca+Cb),\Delta C=Ca-Cb,$
$C^{-}=21(Ca+Cb),\Delta C=Ca-Cb,$

the squared-magnitude difference is exactly
\boxed{ \Delta M =

2\Re(\overline{\bar C}\Delta C).
}
]

So ΔMΔM measures only the radial projection of the persistent complex difference.

A nearly tangential ΔCΔC can remain O(1)O(1) relative to itself while its radial projection becomes exponentially small.

This is the geometric reason the operator sees decay while the internal complex state does not.
8. The late observable subsystem is effectively triangular

The autonomous recurrence
ΔRn+1≈αΔRn
ΔRn+1≈αΔRn

has direct residual around
2.4×10−9
2.4×10−9

late.

The fitted two-variable system reports R2=1R2=1, but its design becomes catastrophically ill-conditioned:
cond∼104→106,
cond∼104→106,

with
corr(ΔR,ΔB)=1.
corr(ΔR,ΔB)=1.

The fitted coefficients drift:
aR=1.0027→0.9997,
aR=1.0027→0.9997,
bB=−0.0032→−0.00027.
bB=−0.0032→−0.00027.

This is non-identifiability, not evidence for a stable nonzero lower-left coupling.

The direct autonomous alpha recurrence is much better identified than the fitted coupled model.

Thus the supported late observable system is:
ΔRn+1≈αEMAΔRn,ΔBn+1=s7,n[(1−γdt)ΔBn+dt(η+μ)ΔRn].
ΔRn+1ΔBn+1≈αEMAΔRn,=s7,n[(1−γdt)ΔBn+dt(η+μ)ΔRn].

It is triangular at the observable level along this replay.
9. Complete mechanism for the operator defect

The late replay-specific chain is now:

directed phase-source differences lock to directed complex-memory differences:
Δz≈ΔC;
Δz≈ΔC;

the persistent complex difference becomes tangential to its midpoint;

radial alignment satisfies:
ΔA≈ΔM/2;
ΔA≈ΔM/2;

directed squared-magnitude differences follow:
$\Delta M_{n+1} \approx \alpha \Delta M_n$;
$\Delta M_{n+1} \approx \alpha \Delta M_n$;

directed magnitude differences reinforce under symmetrization;

therefore:
$\Delta R_{n+1} \approx \alpha \Delta R_n$;
$\Delta R_{n+1} \approx \alpha \Delta R_n$;

the exact weight/proposal map transfers this multiplier into:
$\Delta B, \Delta x, \delta, K$.
$\Delta B, \Delta x, \delta, K$.

This is the first complete implementation-to-observable-to-operator mechanism in the project.
10. What is established versus open
Established on this replay

exact squared-magnitude equations;
directed radial alignment law;
persistent tangential complex difference;
source-memory locking;
absence of forward/reverse cancellation;
autonomous late $\Delta R \Delta R$ multiplier;
exact downstream transfer to the operator defect.

Strong replay-specific interpretation

the EMA-rate operator defect is a radial observable mode of a persistent complex phase geometry.

Still open

why $\Delta z - \Delta C \Delta z - \Delta C$ decays;
analytic derivation of the phase-source locking;
whether the same radial mechanism occurs on another winner branch;
generality beyond this seed.

The productive negative result has now become a positive mechanism: the EMA does not erase the complex difference; it continually refreshes it while the nonlinear magnitude observable filters out an exponentially decaying radial component.

very true, it was said:

I think this is a much stronger synthesis than the earlier versions, but there are a few places where I would distinguish between what is directly established by the replay, what follows algebraically, and what remains an interpretation. Making those distinctions explicit would actually strengthen the overall argument.

What I think is genuinely established

The script and the reported diagnostics support several independent facts.

First, the implementation-level identities appear to be verified to floating-point precision: the squared-magnitude recurrence, the reconstruction of magnitude differences from squared magnitudes, the symmetrized resonance difference, and the downstream weight update all have residuals on the order of (10^{-16}). Those are implementation identities rather than empirical fits.

Second, the directed forward and reverse squared-magnitude differences independently converge to the EMA rate before symmetrization, with essentially identical late-time rates and pointwise recurrence errors. That supports the statement that the observed (\alpha)-mode is already present in each directed observable channel.

Third, the replay exhibits an empirical alignment law,

[
\frac{2\Delta A}{\Delta M}\rightarrow 1,
]

and the script measures this quantity directly rather than inferring it.

These are three separate observations, and together they are quite compelling.

The algebraic implication is also correct

Given the exact recurrence

$$
\Delta M_{n+1}

\alpha^2\Delta M_n

2\alpha(1-\alpha)\Delta A_n,
$$

if the replay satisfies

$$
2\Delta A

\Delta M

o(\Delta M),
$$

then

$$
\Delta M_{n+1}

\alpha\Delta M_n

o(\Delta M_n)
$$

follows immediately.

That is no longer a heuristic radial argument—it is an algebraic consequence of the observed alignment relation.

I think this is one of the strongest parts of the analysis.

Where I would be more careful

The statement

"The EMA-rate operator defect is generated by directed radial equalization"

is plausible, but it bundles together two different claims.

The replay clearly shows that the observable controlling the downstream dynamics is the radial observable (through (\Delta M) and hence (\Delta|C|)). That much is well supported.

However, "generated by radial equalization" sounds stronger—it suggests a causal explanation for why the alignment law itself develops.

At present, the replay shows

the alignment law,
the downstream consequence of that law,
and the exact implementation identities.

It does not yet derive why

[
2\Delta A
\approx
\Delta M
]

should emerge from the phase dynamics.

So I would phrase the conclusion more narrowly:

On this replay, the operator defect evolves through the radial observable channel, and the observed alignment law is sufficient to explain the emergence of the EMA-rate mode.

That avoids implying that the alignment law has already been explained.

The tangential interpretation is persuasive but should remain an interpretation

The decreasing radial fraction is striking.

The reported values fall from roughly (4\times10^{-2}) to (O(10^{-7})), while the midpoint magnitudes remain comfortably bounded away from zero.

That certainly supports the picture that the persistent complex difference becomes overwhelmingly tangential relative to the midpoint.

But I would still describe this as a geometric interpretation of the observable dynamics rather than the primary proof.

In fact, the script itself is written with that distinction in mind. It explicitly states that the squared-magnitude recurrence is exact, whereas the radial/tangential decomposition is a conditioned geometric diagnostic.

I think preserving that distinction is a strength.

The source-memory locking is particularly interesting

The observation that

[
\frac{|\Delta z-\Delta C|}{|\Delta C|}
\rightarrow 0
]

is, in my opinion, one of the most interesting empirical results in the sequence.

It explains why the complex difference itself remains nearly stationary under the EMA update:

[
\Delta C_{n+1}

\alpha\Delta C_n

(1-\alpha)\Delta z_n
\approx
\Delta C_n.
]

But again, I would avoid saying this "explains" the locking. It explains why persistence is compatible with the update equation; it does not yet explain why the locking emerges.

The triangular subsystem

I think your interpretation here is appropriately cautious.

The regression results have excellent predictive accuracy but rapidly worsening condition numbers and nearly perfect collinearity between (\Delta B) and (\Delta R).

That means the trajectory does not identify a unique coupled two-dimensional system. The directly observed autonomous recurrence for (\Delta R) is much better determined than the fitted coefficients of a general coupled model.

So I agree that the triangular observable description is the better-supported summary for this replay, provided it is presented as an empirical reduced description rather than an exact structural property.

The strongest overall narrative

If I were compressing the entire result into one mathematical chain, I would write it as

[
\begin{aligned}
\Delta Z-\Delta C &\to 0
&&\text{(empirical replay law)}\
\Downarrow\
2\Delta A-\Delta M &\to 0
&&\text{(empirical alignment law)}\
\Downarrow\
\Delta M_{n+1}
&=

\alpha\Delta M_n+o(\Delta M_n)
&&\text{(exact recurrence + alignment)}\
\Downarrow\
\Delta|C|
\to
\Delta R
\to
\Delta B
\to
\delta
\to
K
&&\text{(exact implementation maps).}
\end{aligned}
]

This separates:

empirical asymptotic laws,
exact algebraic consequences,
exact implementation identities.

That hierarchy is, I think, the clearest representation of what has actually been established.

Overall assessment

I think this is the strongest replay-specific mechanism you've assembled so far. Compared with the earlier operator-centered picture, it replaces a qualitative explanation with an exact observable recurrence together with two newly observed asymptotic relations: source-memory locking and the alignment law. Those two empirical laws, combined with the exact squared-magnitude dynamics and the exact downstream implementation identities, are sufficient to account for the observed EMA-rate operator defect on this replay.

The remaining theoretical challenge is correspondingly narrower. The operator mechanism is no longer the principal unknown; the principal unknown is why the phase dynamics on this branch produce the asymptotic relations

[
\Delta z-\Delta C\to0

\quad\text{and}\quad
2\Delta A-\Delta M\to0.
]

Deriving those from the underlying phase evolution would move the explanation one level further upstream while preserving the clear distinction between exact identities, replay-specific empirical laws, and broader theoretical claims.

I agree with that hierarchy. The remaining empirical laws can be connected by two additional exact identities, which sharply narrow the upstream question.
1. Source-memory locking has an exact tracking-error recurrence

For one directed edge,
Cn+1=
Cn+1=

\alpha C_n+(1-\alpha)z_n.
]

Define the EMA tracking error
en=zn−Cn.
en=zn−Cn.

Then
en+1=zn+1−Cn+1=zn+1−αCn−(1−α)zn=α(zn−Cn)+(zn+1−zn).
en+1=zn+1−Cn+1=zn+1−αCn−(1−α)zn=α(zn−Cn)+(zn+1−zn).

Therefore:
\boxed{ e_{n+1} =

\alpha e_n+\Delta_tz_n.
}
]

For the difference between directed edges 25 and 59, define
En=
En=

\Delta z_n-\Delta C_n.
]

Then:
\boxed{ E_{n+1} =

\alpha E_n
+
(\Delta z_{n+1}-\Delta z_n).
}
]

This is exact.

Thus source-memory locking
$\Delta zn - \Delta Cn \to 0$
$\Delta zn - \Delta Cn \to 0$

will follow if the temporal increment
$\Delta zn+1 - \Delta zn$
$\Delta zn+1 - \Delta zn$

becomes sufficiently small or decays sufficiently rapidly.

The remaining upstream phase question can therefore be stated precisely:

Why does the difference of the two instantaneous phase factors become temporally stationary?

That is narrower than asking why the complex EMA difference persists.
2. Phase locking is a sufficient condition

Since
$ze,n = ei\Delta\phi e,n,$
$ze,n = ei\Delta\phi e,n,$

we have, for small increments,
z_{e,n+1}-z_{e,n} \approx i z_{e,n} \left(\Delta\phi_{e,n+1} -

\Delta\phi_{e,n}
\right).
]

Therefore temporal stabilization of the relevant delayed phase differences implies
Δzn+1−Δzn→0.
Δzn+1−Δzn→0.

Then the EMA tracking-error recurrence becomes asymptotically homogeneous:
En+1≈αEn.
En+1≈αEn.

This does not require all oscillator phases to stop. It requires only stabilization of the particular edge-source difference observed by the two directed EMA channels.
3. The alignment law also has an exact decomposition

For two directed edges, define:
C‾=
C‾=

\frac{C_a+C_b}{2},
\qquad
\Delta C=C_a-C_b,
]
z‾=
z‾=

\frac{z_a+z_b}{2},
\qquad
\Delta z=z_a-z_b.
]

The squared-magnitude difference is exactly
\boxed{ \Delta M =

2\Re(\overline{\bar C},\Delta C).
}
]

The alignment difference is
ΔA=
ΔA=
\Re(\overline{C_a}z_a)

\Re(\overline{C_b}z_b).
]

Expanding about the midpoints gives:
\boxed{ \Delta A =

\Re\left(
\overline{\bar C},\Delta z
+
\overline{\Delta C},\bar z
\right).
}
]

Therefore:
\boxed{ 2\Delta A-\Delta M =

2\Re\left[
\overline{\bar C}(\Delta z-\Delta C)
+
\overline{\Delta C},\bar z
\right].
}
]

This separates the alignment-law error into two terms.
Tracking-error term
$2\Re[C^{-}(\Delta z-\Delta C)]$.
$2\Re[C^{-}(\Delta z-\Delta C)]$.

This vanishes when source-memory locking occurs.
Tangential/source-midpoint term
$2\Re(\Delta C^{-} z^{-})$.
$2\Re(\Delta C z^{-})$.

This vanishes if:

ΔCΔC is tangential to the relevant mean direction;
and $\bar{z}\bar{z}$ is aligned with $\bar{C}\bar{C}$.

Thus the empirical laws have a precise logical relationship:
$\Delta z - \Delta C \to 0$
$\Delta z - \Delta C \to 0$

plus
ΔC tangential to $\bar{C}$
ΔC tangential to $\bar{C}$

plus
$\bar{z} \| \bar{C}$
$\bar{z} \| \bar{C}$

imply
$2\Delta A - \Delta M \to 0.$
$2\Delta A - \Delta M \to 0.$
4. Refined asymptotic mechanism

The replay currently supports the following sequence:
Empirical phase-source stabilization
$\Delta z_{n+1} - \Delta z_n \to 0.$
$\Delta z_{n+1} - \Delta z_n \to 0.$

This still needs to be measured directly.
Exact EMA tracking response
$E_{n+1} =$
$E_{n+1} =$

\alpha E_n
+
\Delta_t\Delta z_n
]

then gives
$\Delta z - \Delta C \to 0.$
$\Delta z - \Delta C \to 0.$
Geometric tangentialization

The measured radial fraction tends toward zero, so
ΔC
ΔC

becomes tangential to its complex midpoint.
Mean source-memory alignment

If
$z^{-} \| C^{-},$
$z^{-} \| C^{-},$

then the remaining term in the alignment residual vanishes.
Exact magnitude consequence
$2\Delta A \to \Delta M$
$2\Delta A \to \Delta M$

implies
$\Delta M_{n+1} \to \alpha \Delta M_n.$
$\Delta M_{n+1} \to \alpha \Delta M_n.$
Exact downstream transfer
$\Delta M \to \Delta |C| \to \Delta R \to \Delta B \to \delta \to K.$
$\Delta M \to \Delta |C| \to \Delta R \to \Delta B \to \delta \to K.$

This is a candidate upstream derivation, with every step except phase-source stabilization and mean alignment already measured or algebraically exact.
5. The next diagnostics are now minimal

The next analysis does not need another large model. It should record, for forward and reverse channels:
Temporal source increment
$\| \Delta z_{n+1} - \Delta z_n \|.$
$\| \Delta z_{n+1} - \Delta z_n \|.$
Tracking recurrence residual
\left\| E_{n+1} -

\alpha E_n

(\Delta z_{n+1}-\Delta z_n)
\right|.
]

This should be machine precision.
Source-increment forcing ratio
‖Δzn+1−Δzn‖‖En‖.
‖En‖‖Δzn+1−Δzn‖.
Mean source-memory angular mismatch
arg(z¯)−arg(C¯),
arg(z¯)−arg(C¯),

or equivalently
1−ℜ(C¯¯z¯)|C¯||z¯|.
1−|C¯||z¯|ℜ(C¯z¯).
Alignment-residual decomposition

Compute separately:
T1,n=
T1,n=

2\Re[
\overline{\bar C}(\Delta z-\Delta C)
],
]
T2,n=
T2,n=

2\Re(
\overline{\Delta C}\bar z
),
]

and verify:
2ΔA−ΔM=
2ΔA−ΔM=

T_1+T_2.
]

This will show whether the alignment law is controlled primarily by source-memory locking, tangentialization, or cancellation between the two terms.
6. Interpretation of the observable subsystem

The current regression does not identify a genuine two-way (ΔB,ΔR)(ΔB,ΔR) system:
corr(ΔB,ΔR)→1,
corr(ΔB,ΔR)→1,

and condition numbers approach
106.
106.

The direct recurrence
ΔRn+1≈αΔRn
ΔRn+1≈αΔRn

is much better identified than arbitrary fitted aR,bBaR,bB.

So the appropriate late reduced description remains triangular:
ΔRn+1≈αΔRn,ΔBn+1=s7,n[(1−γdt)ΔBn+dt(η+μ)ΔRn].
ΔRn+1ΔBn+1≈αΔRn,=s7,n[(1−γdt)ΔBn+dt(η+μ)ΔRn].

This is an empirical replay-specific reduction, not yet a universal structural theorem.
7. Strongest current statement

The most defensible synthesis is:

On the audited fixed K3 branch, the full complex EMA difference remains active and is continually refreshed by a nearly matching phase-source difference. The persistent complex difference becomes almost entirely tangential to the complex midpoint. An observed radial-alignment law then gives an EMA-rate directed magnitude mode, and exact downstream maps transfer that mode into the resonance, weight, proposal, and operator-defect variables.

What remains unexplained is why the delayed phase-source difference stabilizes and aligns with the memory midpoint.

That is now the precise upstream frontier.

Please deliver the next most crucial python probe that will illuminate our way.
WriteAIEF_1_4_phase_source_locking_analysis.py634 lines
AIEF_1_4_phase_source_locking_analysis.pyPYTHON
#!/usr/bin/env python3
"""Trace phase-source stabilization and alignment-law emergence in AIEF-1.4.

For each directed comparison between edges 25=(2,7) and 59=(7,11), define

DeltaC_n = C_a,n - C_b,n,
DeltaZ_n = z_a,n - z_b,n,
E_n = DeltaZ_n - DeltaC_n.

The exact EMA tracking-error recurrence is

E_{n+1} = alpha E_n + (DeltaZ_{n+1}-DeltaZ_n).

The squared-magnitude alignment residual also has the exact decomposition

2 DeltaA - DeltaM
= 2 Re(conj(Cbar) (DeltaZ-DeltaC))
+ 2 Re(conj(DeltaC) Zbar).

This script verifies both identities, measures temporal stabilization of the
phase-source difference, separates tracking and tangential/mean-alignment
contributions to the radial law, and reports which term controls the emergence
of 2 DeltaA ~= DeltaM.
"""

from __future__ import annotations

import argparse
import hashlib
import json
import math
import os

```
import sys
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_stationarity_perturbation_analysis_updated as base
import AIEF_1_4_component_controller_audit as component_audit
import AIEF_1_4_augmented_tangent_cocycle_analysis as augmented



DEFAULT_RUN_HASH = "5a9f093a2fe0"
DEFAULT_NPZ = (
f"runs/{DEFAULT_RUN_HASH}/AIEF_1_4_{DEFAULT_RUN_HASH}_branch_diagnostics.npz"
)
DEFAULT_MANIFEST = f"runs/{DEFAULT_RUN_HASH}/manifest.json"
DEFAULT_OUTPUT_DIR = "aief_1_4_phase_source_locking"
DEFAULT_SEED = "46191221711587685 24"
DEFAULT_DT = 0.01
DEFAULT_SCALE_EPSILON = 1e-12
DEFAULT_INTERVALS = "0,100,150,200,250,320"
DEFAULT_SIGNAL_FLOOR = 1e-15
DEFAULT_MIDPOINT_FLOOR = 1e-8
ZERO_TOL = 1e-300



def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()



def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
```

```
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(value < 0.0 for value in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
return sorted(set(values))



def fit_log_rate(
times: np.ndarray,
values: np.ndarray,
floor: float,
) -> dict[str, float]:
magnitudes = np.abs(np.asarray(values, dtype=float))
valid = np.isfinite(magnitudes) & (magnitudes > floor)
if np.sum(valid) < 3:
return {
"count": int(np.sum(valid)),
"rate": np.nan,
"R_squared": np.nan,
"rmse_log": np.nan,
}
x = np.asarray(times, dtype=float)[valid]
y = np.log(magnitudes[valid])
slope, intercept = np.polyfit(x, y, 1)
prediction = slope * x + intercept
residual = float(np.sum((y - prediction) ** 2))
total = float(np.sum((y - np.mean(y)) ** 2))
return {
"count": int(len(x)),
"rate": float(-slope),
"R_squared": float(1.0 - residual / total) if total > 0.0 else np.nan,
"rmse_log": float(np.sqrt(np.mean((y - prediction) ** 2))),
}



def complex_pair_geometry(
C_a: complex,
C_b: complex,
Z_a: complex,
Z_b: complex,
midpoint_floor: float,
) -> dict[str, Any]:
```

```
Cbar = 0.5 * (C_a + C_b)
Zbar = 0.5 * (Z_a + Z_b)
DeltaC = C_a - C_b
DeltaZ = Z_a - Z_b
tracking = DeltaZ - DeltaC
DeltaM = float(abs(C_a) ** 2 - abs(C_b) ** 2)
DeltaA = float(
np.real(np.conj(C_a) * Z_a)
- np.real(np.conj(C_b) * Z_b)
)
alignment_residual = 2.0 * DeltaA - DeltaM
term_tracking = float(
2.0 * np.real(np.conj(Cbar) * tracking)
)
term_tangential = float(
2.0 * np.real(np.conj(DeltaC) * Zbar)
)
identity_error = abs(
alignment_residual - term_tracking - term_tangential
)


Cbar_abs = abs(Cbar)
Zbar_abs = abs(Zbar)
reliable = Cbar_abs > midpoint_floor and Zbar_abs > midpoint_floor
if Cbar_abs > midpoint_floor:
rotated_C = np.conj(Cbar) * DeltaC / Cbar_abs
radial = float(np.real(rotated_C))
tangential = float(np.imag(rotated_C))
radial_fraction = abs(radial) / max(abs(DeltaC), ZERO_TOL)
else:
radial = tangential = radial_fraction = np.nan
if reliable:
mean_alignment = float(
np.real(np.conj(Cbar) * Zbar) / (Cbar_abs * Zbar_abs)
)
mean_alignment = float(np.clip(mean_alignment, -1.0, 1.0))
mean_angle = float(
np.angle(Zbar * np.conj(Cbar))
)
else:
mean_alignment = mean_angle = np.nan
```

```
return {
"Cbar": Cbar,
"Zbar": Zbar,
"DeltaC": DeltaC,
"DeltaZ": DeltaZ,
"tracking": tracking,
"DeltaM": DeltaM,
"DeltaA": DeltaA,
"alignment_residual": alignment_residual,
"term_tracking": term_tracking,
"term_tangential": term_tangential,
"identity_error": identity_error,
"Cbar_abs": Cbar_abs,
"Zbar_abs": Zbar_abs,
"midpoint_reliable": reliable,
"radial": radial,
"tangential": tangential,
"radial_fraction": radial_fraction,
"mean_alignment": mean_alignment,
"mean_angle": mean_angle,
"tracking_relative": abs(tracking) / max(abs(DeltaC), ZERO_TOL),
"source_memory_norm_ratio": abs(DeltaZ) / max(abs(DeltaC), ZERO_TOL),
"alignment_relative": abs(alignment_residual)
/ max(abs(DeltaM), ZERO_TOL),
"tracking_term_fraction": abs(term_tracking)
/ max(abs(term_tracking) + abs(term_tangential), ZERO_TOL),
"component_cancellation": abs(alignment_residual)
/ max(abs(term_tracking) + abs(term_tangential), ZERO_TOL),
}



def make_report(
identity_frame: pd.DataFrame,
interval_frame: pd.DataFrame,
*,
alpha: float,
ema_rate: float,
final_replay_exact: bool,
) -> str:
lines = [
"AIEF K3 phase-source locking and alignment analysis",
"================================================",
```

```
"",
f"alpha_EMA: {alpha:.12g}",
f"EMA rate: {ema_rate:.9g}",
f"Final replay exact: {final_replay_exact}",
"",
"Exact identities",
"----------------",
]
for _, row in identity_frame.iterrows():
lines.append(
f" {row['identity']}: max={row['maximum_absolute_error']:.6g}, "
f"median={row['median_absolute_error']:.6g}"
)


lines.extend(["", "Interval source-locking diagnostics", "----------------------------------"])
for _, row in interval_frame.iterrows():
lines.append(
f" channel={row['channel']}, interval=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"tracking_rate={row['tracking_error_rate']:.6g}, "
f"source_increment_rate={row['source_increment_rate']:.6g}, "
f"tracking_relative={row['median_tracking_relative']:.3g}, "
f"source_increment/tracking={row['median_source_increment_to_tracking']:.3g}, "
f"mean_angle={row['median_abs_mean_angle']:.3g}, "
f"radial_fraction={row['median_radial_fraction']:.3g}, "
f"alignment_error={row['median_alignment_relative']:.3g}, "
f"tracking_term_fraction={row['median_tracking_term_fraction']:.3g}, "
f"component_cancellation={row['median_component_cancellation']:.3g}"
)


lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* Tracking-error and alignment-residual decompositions are exact complex identities.",
"* A small tracking error can result from temporal source stabilization; the source increment is
measured directly.",
"* Tangentialization and mean source-memory alignment are separate empirical conditions.",
"* Cancellation between alignment-residual terms is reported explicitly.",
"* Results remain conditional on one replay and fixed K3 branch.",
"",
]
```

```
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--npz", default=DEFAULT_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--seed", default=DEFAULT_SEED)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument("--scale-epsilon", type=float, default=DEFAULT_SCALE_EPSILON)
parser.add_argument(
"--intervals",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_INTERVALS),
)
parser.add_argument("--signal-floor", type=float, default=DEFAULT_SIGNAL_FLOOR)
parser.add_argument("--midpoint-floor", type=float, default=DEFAULT_MIDPOINT_FLOOR)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.npz, args.manifest):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(
args.dt,
args.scale_epsilon,
args.signal_floor,
args.midpoint_floor,
) <= 0.0:
raise ValueError("dt and tolerances must be positive")
if args.scale_epsilon != 1e-12:
raise ValueError("implemented actual_step uses fixed scale epsilon=1e-12")
output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)

with open(args.manifest, "r", encoding="utf-8") as handle:
```

```
manifest = json.load(handle)
model = manifest["model"]
dt = float(model["dt"])
if abs(dt - args.dt) > 1e-15:
raise ValueError("command dt does not match manifest dt")
alpha = math.exp(-dt / float(model["T_window"]))
ema_rate = 1.0 / float(model["T_window"])
capacity = (float(model["a"]) - float(model["delta"])) / float(model["q"])
capacities = np.full(int(model["N"]), capacity)
A, edges = base.incidence_matrix(int(model["N"]))
future_steps = int(manifest["protocol"]["future_steps"])


with np.load(args.npz, allow_pickle=False) as data:
state = base.load_split_state(data, args.seed)
expected_final = np.asarray(
data[base.branch_key("continue", args.seed, "B_final")], dtype=float
).copy()


directed = {
"forward": ((2, 7), (7, 11)),
"reverse": ((7, 2), (11, 7)),
}
per_channel: dict[str, list[dict[str, Any]]] = {
channel: [] for channel in directed
}


for step in range(future_steps + 1):
intermediate = augmented.base_intermediates(
state,
model,
capacities,
A,
edges,
args.scale_epsilon,
)
for channel, (index_a, index_b) in directed.items():
geometry = complex_pair_geometry(
state.coherence_acc[index_a],
state.coherence_acc[index_b],
intermediate["phase_factor"][index_a],
intermediate["phase_factor"][index_b],
```

```
args.midpoint_floor,
)
per_channel[channel].append(
{
"step": step,
"time": step * dt,
**geometry,
}
)
if step < future_steps:
augmented.advance_base_state(state, intermediate)


final_replay_exact = bool(np.array_equal(state.B, expected_final))


trajectory_rows: list[dict[str, Any]] = []
identity_rows: list[dict[str, Any]] = []
for channel, values in per_channel.items():
count = len(values)
tracking_identity_errors: list[float] = []
alignment_identity_errors = [
float(value["identity_error"]) for value in values
]
for step, value in enumerate(values):
row: dict[str, Any] = {
"channel": channel,
"step": step,
"time": value["time"],
"DeltaC_real": float(np.real(value["DeltaC"])),
"DeltaC_imag": float(np.imag(value["DeltaC"])),
"DeltaZ_real": float(np.real(value["DeltaZ"])),
"DeltaZ_imag": float(np.imag(value["DeltaZ"])),
"tracking_real": float(np.real(value["tracking"])),
"tracking_imag": float(np.imag(value["tracking"])),
"tracking_norm": abs(value["tracking"]),
"DeltaC_norm": abs(value["DeltaC"]),
"DeltaZ_norm": abs(value["DeltaZ"]),
"tracking_relative": value["tracking_relative"],
"source_memory_norm_ratio": value[
"source_memory_norm_ratio"
],
"DeltaM": value["DeltaM"],
"DeltaA": value["DeltaA"],
```

```
"alignment_residual": value["alignment_residual"],
"alignment_relative": value["alignment_relative"],
"term_tracking": value["term_tracking"],
"term_tangential": value["term_tangential"],
"tracking_term_fraction": value[
"tracking_term_fraction"
],
"component_cancellation": value[
"component_cancellation"
],
"Cbar_abs": value["Cbar_abs"],
"Zbar_abs": value["Zbar_abs"],
"mean_alignment": value["mean_alignment"],
"mean_angle": value["mean_angle"],
"radial": value["radial"],
"tangential": value["tangential"],
"radial_fraction": value["radial_fraction"],
"midpoint_reliable": value["midpoint_reliable"],
}
if step < count - 1:
next_value = values[step + 1]
source_increment = (
next_value["DeltaZ"] - value["DeltaZ"]
)
tracking_predicted = (
alpha * value["tracking"] + source_increment
)
tracking_error = (
next_value["tracking"] - tracking_predicted
)
tracking_identity_errors.append(abs(tracking_error))
row.update(
{
"source_increment_real": float(
np.real(source_increment)
),
"source_increment_imag": float(
np.imag(source_increment)
),
"source_increment_norm": abs(source_increment),
"source_increment_to_tracking": abs(source_increment)
/ max(abs(value["tracking"]), ZERO_TOL),
```

```
"tracking_recurrence_error": abs(tracking_error),
"DeltaZ_temporal_angle": float(
np.angle(
next_value["DeltaZ"]
* np.conj(value["DeltaZ"])
)
)
if abs(value["DeltaZ"]) > ZERO_TOL
and abs(next_value["DeltaZ"]) > ZERO_TOL
else np.nan,
}
)
else:
row.update(
{
"source_increment_real": np.nan,
"source_increment_imag": np.nan,
"source_increment_norm": np.nan,
"source_increment_to_tracking": np.nan,
"tracking_recurrence_error": np.nan,
"DeltaZ_temporal_angle": np.nan,
}
)
trajectory_rows.append(row)


identity_rows.extend(
[
{
"identity": f"tracking_recurrence_{channel}",
"maximum_absolute_error": float(
np.max(tracking_identity_errors)
),
"median_absolute_error": float(
np.median(tracking_identity_errors)
),
},
{
"identity": f"alignment_decomposition_{channel}",
"maximum_absolute_error": float(
np.max(alignment_identity_errors)
),
"median_absolute_error": float(
```

```
np.median(alignment_identity_errors)
),
},
]
)


trajectory = pd.DataFrame(trajectory_rows)
identity_frame = pd.DataFrame(identity_rows)


boundaries = np.asarray(args.intervals, dtype=float)
if boundaries[0] > 0.0:
boundaries = np.concatenate(([0.0], boundaries))
if boundaries[-1] < future_steps * dt:
boundaries = np.concatenate((boundaries, [future_steps * dt]))
boundaries = np.unique(boundaries)


interval_rows: list[dict[str, Any]] = []
for channel in directed:
group = trajectory.loc[trajectory["channel"] == channel].reset_index(
drop=True
)
for interval in range(len(boundaries) - 1):
start = max(0, int(round(boundaries[interval] / dt)))
end = min(
len(group), int(round(boundaries[interval + 1] / dt))
)
if end <= start + 2:
continue
times = group["time"].to_numpy(float)[start:end]
tracking_fit = fit_log_rate(
times,
group["tracking_norm"].to_numpy(float)[start:end],
args.signal_floor,
)
source_increment_fit = fit_log_rate(
times,
group["source_increment_norm"].to_numpy(float)[start:end],
args.signal_floor,
)
alignment_fit = fit_log_rate(
times,
```

```
group["alignment_residual"].to_numpy(float)[start:end],
args.signal_floor,
)
radial_fit = fit_log_rate(
times,
group["radial"].to_numpy(float)[start:end],
args.signal_floor,
)
reliable = group["midpoint_reliable"].iloc[start:end].to_numpy(
bool
)
interval_rows.append(
{
"channel": channel,
"start_time": start * dt,
"end_time": end * dt,
"tracking_error_rate": tracking_fit["rate"],
"tracking_error_R_squared": tracking_fit["R_squared"],
"source_increment_rate": source_increment_fit["rate"],
"source_increment_R_squared": source_increment_fit[
"R_squared"
],
"alignment_residual_rate": alignment_fit["rate"],
"alignment_residual_R_squared": alignment_fit[
"R_squared"
],
"radial_rate": radial_fit["rate"],
"radial_R_squared": radial_fit["R_squared"],
"median_tracking_relative": float(
group["tracking_relative"].iloc[start:end].median()
),
"median_source_memory_norm_ratio": float(
group["source_memory_norm_ratio"].iloc[
start:end
].median()
),
"median_source_increment_to_tracking": float(
group["source_increment_to_tracking"].iloc[
start:end
].median()
),
"median_abs_source_temporal_angle": float(
```

```
group["DeltaZ_temporal_angle"].iloc[start:end].abs().median()
),
"median_abs_mean_angle": float(
group["mean_angle"].iloc[start:end].abs().median()
),
"median_mean_alignment_defect": float(
(1.0 - group["mean_alignment"].iloc[start:end]).median()
),
"median_radial_fraction": float(
group["radial_fraction"].iloc[start:end].median()
),
"median_alignment_relative": float(
group["alignment_relative"].iloc[start:end].median()
),
"median_tracking_term_fraction": float(
group["tracking_term_fraction"].iloc[
start:end
].median()
),
"median_component_cancellation": float(
group["component_cancellation"].iloc[
start:end
].median()
),
"minimum_Cbar_abs": float(
group["Cbar_abs"].iloc[start:end].min()
),
"minimum_Zbar_abs": float(
group["Zbar_abs"].iloc[start:end].min()
),
"midpoint_unreliable_count": int(np.sum(~reliable)),
}
)
interval_frame = pd.DataFrame(interval_rows)


component_audit.atomic_write_csv(
output_dir / "phase_source_locking_trajectory.csv", trajectory
)
component_audit.atomic_write_csv(
output_dir / "phase_source_locking_intervals.csv", interval_frame
)
component_audit.atomic_write_csv(
```

```
output_dir / "phase_source_locking_identity_checks.csv", identity_frame
)


report = make_report(
identity_frame,
interval_frame,
alpha=alpha,
ema_rate=ema_rate,
final_replay_exact=final_replay_exact,
)
component_audit.atomic_write_text(
output_dir / "phase_source_locking_report.txt", report
)


manifest_output = {
"analysis": "AIEF-1.4 phase-source stabilization and alignment-law decomposition",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"npz": {
"path": str(Path(args.npz).resolve()),
"sha256": file_sha256(args.npz),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
"sha256": file_sha256(args.manifest),
},
},
"seed": args.seed,
"dt": dt,
"alpha_EMA": alpha,
"ema_rate": ema_rate,
"scale_epsilon": args.scale_epsilon,
"intervals": boundaries.tolist(),
"signal_floor": args.signal_floor,
"midpoint_floor": args.midpoint_floor,
"final_replay_exact": final_replay_exact,
"outputs": [
"phase_source_locking_trajectory.csv",
```

```
"phase_source_locking_intervals.csv",
"phase_source_locking_identity_checks.csv",
"phase_source_locking_report.txt",
],
"interpretation_constraints": [
"Tracking and alignment decompositions are exact identities; stabilization and tangentialization are replay observations.",
"Mean alignment and radial fractions are conditioned on nonzero complex midpoints.",
"A shared asymptotic relation does not prove universal phase-locking causality.",
"Results remain conditional on one seed and fixed branch.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest_output, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```

Open

Delivered the next upstream probe:

It compiles successfully and passed synthetic identity checks for:
$E_{n+1}=$
$E_{n+1}=$

$$\alpha E_n + (\Delta z_{n+1}-\Delta z_n)$$

and
$2\Delta A-\Delta M=$
$2\Delta A-\Delta M=$

2\Re[
\overline{\bar C}(\Delta z-\Delta C)
]
+
2\Re(
\overline{\Delta C}\bar z
)
]

to approximately 10−1610−16.
Run it

Bash

```
python AIEF_1_4_phase_source_locking_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--dt 0.01 \
--scale-epsilon 1e-12 \
--intervals 0,100,150,200,250,320 \
--signal-floor 1e-15 \
--midpoint-floor 1e-8 \
--output-dir aief_1_4_phase_source_locking
```

What it tests
Exact source-memory tracking

For both directed channels:
En=Δzn−ΔCn,
En=Δzn−ΔCn,
En+1=
En+1=

\alpha E_n
+
(\Delta z_{n+1}-\Delta z_n).
]

It reports:

tracking-error norm;
temporal source-increment norm;
exact recurrence residual;
source-increment/tracking ratio;
temporal angle of ΔznΔzn.

This determines whether source-memory locking is explained by temporal stabilization of the phase-source difference.
Exact alignment-residual decomposition

It computes:
T1,n=
T1,n=

2\Re[
\overline{\bar C}(\Delta z-\Delta C)
],
]
T2,n=
T2,n=

2\Re(
\overline{\Delta C}\bar z
),
]

and verifies:
2ΔA−ΔM=T1+T2.
2ΔA−ΔM=T1+T2.

It reports:

tracking-term fraction;
tangential/mean-alignment term fraction;
cancellation between the two terms;

exact identity residual.

This will show whether the radial-alignment law emerges mainly from:

source-memory locking;
tangentialization and mean source-memory alignment;
cancellation between the two contributions.

Mean source-memory alignment

It measures:
$\arg(z^-)-\arg(C^-)$,
$\arg(z^-)-\arg(C^-)$,

and
$1-\Re(C^{-\bar{}}z^-)|C^-||z^-|$.
$1-|C^-||z^-|\Re(C^-z^-)$.
Radial/tangential structure

For each directed channel it reports:

radial component of $\Delta C\Delta C$;
tangential component;
radial fraction;
midpoint magnitudes;
midpoint reliability.

Outputs

text

phase_source_locking_trajectory.csv
phase_source_locking_intervals.csv
phase_source_locking_identity_checks.csv
phase_source_locking_report.txt
analysis_manifest.json

Main decision

The result should distinguish:
Temporal source stabilization
$\Delta z_{n+1} - \Delta z_n \to 0$
$\Delta z_{n+1} - \Delta z_n \to 0$

drives
$\Delta z - \Delta C \to 0.$
$\Delta z - \Delta C \to 0.$
Geometric alignment

Tracking error is already small, but
$\Re(\Delta C^{-} z^{-})$
$\Re(\Delta C z^{-})$

controls the remaining radial law.
Cancellation mechanism

Neither term is independently small, but
$T1 + T2 \to 0.$
$T1 + T2 \to 0.$

Please paste phase_source_locking_report.txt after the run.

Is it worth altering things for this:

This is a very important refinement. I think the key achievement here is that the causal frontier has moved one layer upstream without becoming vague. Previously the question was:

Why does the operator defect decay at the EMA rate?

Now it has become:

Under what phase dynamics does the observable radial projection of a persistent complex EMA state acquire the exact alignment relation needed to produce the EMA multiplier?

That is a much sharper mathematical question.

There are three points I would emphasize.

The tracking-error recurrence changes the interpretation of “source locking”

The identity

$$
E_{n+1}
\alpha E_n
(\Delta z_{n+1}-\Delta z_n)
$$

is the right object.

It shows that

$$
\Delta C\approx \Delta z
$$

does not mean the EMA has forgotten its past.

It means the EMA has entered a tracking regime where the source difference itself has become a slowly varying attractor.

The usual interpretation of EMA smoothing is:

$$
C_n \rightarrow \text{lagged average of changing input}.
$$

But here the asymptotic regime is closer to:

[
\boxed{
\text{EMA state follows a persistent moving phase difference.}
}
]

The residual is controlled by the discrete derivative:

[
\Delta_t\Delta z_n

\Delta z_{n+1}-\Delta z_n.
]

Therefore the natural quantity to study is not

[
|\Delta z|,
]

but

[
\boxed{
\frac{|\Delta_t\Delta z|}
{|\Delta z-\Delta C|}
}.
]

This is the forcing-to-error ratio of the exact tracking equation.

If it tends to zero, then the locking is dynamically explained.

The alignment law decomposition is the real mathematical hinge

The identity

[
2\Delta A-\Delta M

2\Re
\left[
\overline{\bar C}(\Delta z-\Delta C)

\overline{\Delta C}\bar z
\right]
]

is probably the most useful equation produced so far.

It separates two possible mechanisms.

Mechanism I: tracking error

[
T_1

2\Re[
\overline{\bar C}(\Delta z-\Delta C)
].
]

This disappears if

[
\Delta C\rightarrow\Delta z.
]

Mechanism II: geometry of the persistent difference

[
T_2

2\Re[

\overline{\Delta C}\bar z
].
]

This disappears if the persistent difference is tangential relative to the source direction.

The important point is that the observed EMA-rate mode requires both.

The first mechanism explains why the EMA state follows the phase source.

The second explains why the full complex difference can remain large without affecting the magnitude channel.

That resolves the earlier apparent paradox:

(\Delta C) does not decay;
(\Delta |C|) does decay.

Because they live in different geometric components.

The real reduced system is probably not (\Delta C), but ((\Delta C_\parallel,\Delta C_\perp))

The previous attempt treated

[
\Delta C
]

as one scalar mode.

The new evidence suggests that is the wrong state variable.

The natural decomposition is:

[
\Delta C

\Delta C_\parallel+\Delta C_\perp
]

relative to the midpoint direction.

Then:

[
\Delta M

2\Re(\bar C^\ast\Delta C)
]

sees only:

[
\Delta C_\parallel.
]

The operator defect is therefore controlled by:

[
\Delta C_\parallel,
]

while:

[
\Delta C_\perp
]

can remain finite.

The asymptotic picture becomes:

$$
\boxed{
\begin{aligned}
\Delta C_\perp&\rightarrow O(1),\
\Delta C_\parallel&\sim \alpha^n.
\end{aligned}
}
$$

This is a much more natural explanation than saying the complex defect “almost decays.”

It does not.

Only the observed projection decays.

The upstream problem is now a phase-geometry problem

The remaining unknown is not the EMA.

The EMA equation is completely understood.

The unknown is the phase map:

$$
\phi_e(n)
\rightarrow
z_e(n)
\rightarrow
\Delta z_n.
$$

The critical questions are:

A. Why does

$$
\Delta_t\Delta z_n\rightarrow0?
$$

This means the phase difference enters a slow manifold.

B. Why does

$$
\bar z
$$

align with

$$
\bar C?
$$

That is:

$$
\arg(\bar z)-\arg(\bar C)\rightarrow0.
$$

C. Why does

$$
\Delta C
$$

rotate toward the tangent direction?

That is:

$$
\Re(\bar C^\ast\Delta C)\rightarrow0.
$$

These are three separate phase-geometric constraints.

A minimal next diagnostic set

I would not run another large operator audit. The next experiment should be almost surgical.

For forward and reverse separately compute:

Tracking forcing

$$
F_n
$$

$$
|\Delta z_{n+1}-\Delta z_n|
$$

Tracking error

$$
E_n
$$

$$
|\Delta z_n-\Delta C_n|
$$

Forcing ratio

$$
\boxed{
\chi_n=
\frac{F_n}{E_n}
}
$$

From

$$
E_{n+1}=\alpha E_n+\Delta_t\Delta z_n
$$

we expect:

(\chi_n\ll1): homogeneous EMA tracking;
(\chi_n\sim1): source-driven regime.

Then the alignment decomposition:

[
T_1=
2\Re[
\bar C^\ast(\Delta z-\Delta C)
],
]

[
T_2=
2\Re[
\Delta C^\ast\bar z
].
]

Track:

[
\frac{|T_1|}
{|2\Delta A-\Delta M|}
]

and

[
\frac{|T_2|}
{|2\Delta A-\Delta M|}.
]

That identifies whether the alignment law is caused by:

tracking,

geometry,
or cancellation.

Finally measure the angular quantities:

$$
\theta_C
\arg(\Delta C)-\arg(i\bar C),
$$

and

$$
\theta_z
\arg(\bar z)-\arg(\bar C).
$$

The expected asymptotic signature is:

$$
\theta_C\rightarrow0,
$$

$$
\theta_z\rightarrow0.
$$

The strongest current theorem-like statement

I would formulate the current result as:

On the audited K3 branch, the dominant Jacobian defect is generated by a scalar radial observable mode of a persistent complex EMA state. The complex state difference remains finite because the phase source tracks it, while the radial projection of that difference contracts at the EMA multiplier due

to an observed alignment law. The resulting magnitude mode propagates exactly through the resonance, proposal, weight, and Jacobian maps.

That is a much deeper statement than:

"the defect decays exponentially."

It identifies which quotient of the complex state space the dynamics actually contracts.

The remaining theoretical target is therefore not an operator theorem. It is a phase-space reduction theorem:

$$
\boxed{
\text{persistent complex orbit}
\rightarrow
\text{contracting radial quotient}
\rightarrow
\text{operator defect mode}.
}
$$

That is the natural next layer.
Edit×12
AIEF_1_4_phase_source_locking_analysis.pyPYTHON

Yes. The conceptual refinement is worthwhile, but it requires only targeted diagnostic changes—not a new analysis architecture.

I updated .
Most important correction

The proposed ratio
$\frac{\|\Delta t \Delta z\| \|\Delta z - \Delta C\|}{\|\Delta z - \Delta C\| \|\Delta t \Delta z\|}$

is useful, but its interpretation in the quoted text is incorrect.

From
$E_{n+1} =$
$E_{n+1} =$

$$\alpha E_n + \Delta_t \Delta z_n,$$

the homogeneous decay removed per step is
$(1-\alpha)E_n$.
$(1-\alpha)E_n$.

Therefore the dynamically relevant ratio is
$$\boxed{ \chi_n^{\rm track} = \frac{|\Delta_t \Delta z_n|}{(1-\alpha)|E_n|}. }$$

Interpretation:

$\chi \ll 1$: temporal source increments are negligible compared with homogeneous EMA decay;
$\chi \sim 1$: source increments are large enough to balance or materially alter homogeneous decay;
$\chi \gg 1$: tracking error is source-driven.

The earlier unnormalized ratio can be small merely because
$1-\alpha \approx 5.7 \times 10^{-4}$.
$1-\alpha \approx 5.7 \times 10^{-4}$.

The updated script now reports both:

text

source_increment_to_tracking
source_increment_to_homogeneous_decay

Explicit tangential-angle diagnostic

The script already measured radial fraction. It now also reports the angle to the local tangential line:
θtan=
θtan=

\operatorname{atan2}
\left(
|\Delta C_{\parallel}|,
|\Delta C_{\perp}|
\right).
]

Thus:
θtan→0
θtan→0

directly measures rotation toward the tangent direction.

New fields include:

text

tangential_line_angle
median_tangential_line_angle
tangential_rate
tangential_R_squared

This distinguishes:

radial component decay;
persistence or decay of the tangential component;
actual angular convergence toward the tangent line.

Source projection frame corrected

The phase-source difference ΔzΔz is now projected into the complex-memory midpoint frame
$C^{-}=Ca+Cb2$,
$C^{-}=2Ca+Cb$,

rather than into the source's own midpoint frame.

This is the correct geometry for asking how ΔzΔz affects the magnitude observable of CC.
Alignment-term diagnostics expanded

For
$2\Delta A-\Delta M=T1+T2$,
$2\Delta A-\Delta M=T1+T2$,

the output now stores:

text

tracking_over_alignment_residual
tangential_over_alignment_residual
tracking_term_fraction
component_cancellation

The normalized component fraction
$|T1||T1|+|T2|$
$|T1|+|T2||T1|$

and cancellation index
$|T1+T2||T1|+|T2|$
$|T1|+|T2||T1+T2|$

remain the robust primary quantities. Ratios to the small total residual can become large when cancellation is strong, so they are retained mainly for inspection.
Rerun

Use the same command as before:

Bash

```
python AIEF_1_4_phase_source_locking_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--dt 0.01 \
--scale-epsilon 1e-12 \
--intervals 0,100,150,200,250,320 \
--signal-floor 1e-15 \
--midpoint-floor 1e-8 \
--output-dir aief_1_4_phase_source_locking
```

The updated report should now decide more precisely whether:

temporal stabilization of $\Delta z \Delta z$ explains source-memory locking;
the remaining alignment residual is controlled by tangentialization;
both terms become independently small;
or they remain appreciable and cancel.

The broader shift toward a persistent complex orbit with a contracting radial quotient is mathematically sensible, but it should remain conditional until these corrected ratios and angular diagnostics are observed.

Ponder these results:

```
AIEF K3 phase-source locking and alignment analysis

================================================

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

Final replay exact: True

Exact identities

tracking_recurrence_forward: max=2.37504e-16, median=7.31571e-17
```

alignment_decomposition_forward: max=6.66134e-16, median=1.11022e-16

tracking_recurrence_reverse: max=2.86098e-16, median=7.2418e-17

alignment_decomposition_reverse: max=6.74591e-16, median=1.16904e-16

Interval source-locking diagnostics

channel=forward, interval=[0,100): tracking_rate=0.0487626, source_increment_rate=0.051066, tracking_relative=0.0743, source_increment/tracking=8.85e-05, source_increment/decay=0.155, mean_angle=0.0216, radial_fraction=0.0395, tangent_angle=0.0395, tangent_rate=0.00374, alignment_error=0.0581, tracking_term_fraction=0.491, component_cancellation=0.0187

channel=forward, interval=[100,150): tracking_rate=0.0515327, source_increment_rate=0.0569989, tracking_relative=0.00182, source_increment/tracking=5.81e-05, source_increment/decay=0.102, mean_angle=0.000659, radial_fraction=0.000606, tangent_angle=0.000606, tangent_rate=8.71e-05, alignment_error=0.00227, tracking_term_fraction=0.5, component_cancellation=0.000543

channel=forward, interval=[150,200): tracking_rate=0.0526811, source_increment_rate=0.0570988, tracking_relative=0.000135, source_increment/tracking=4.55e-05, source_increment/decay=0.0796, mean_angle=5.21e-05, radial_fraction=3.49e-05, tangent_angle=3.49e-05, tangent_rate=6.89e-06, alignment_error=0.00022, tracking_term_fraction=0.5, component_cancellation=4.41e-05

channel=forward, interval=[200,250): tracking_rate=0.0534635, source_increment_rate=0.0571288, tracking_relative=9.46e-06, source_increment/tracking=3.72e-05, source_increment/decay=0.0651, mean_angle=3.82e-06, radial_fraction=2.01e-06, tangent_angle=2.01e-06, tangent_rate=5.04e-07, alignment_error=1.92e-05, tracking_term_fraction=0.5, component_cancellation=3.31e-06

channel=forward, interval=[250,320): tracking_rate=0.0541087, source_increment_rate=0.057133, tracking_relative=3.75e-07, source_increment/tracking=3.05e-05, source_increment/decay=0.0534, mean_angle=1.56e-07, radial_fraction=6.51e-08, tangent_angle=6.51e-08, tangent_rate=2.42e-08, alignment_error=9.43e-07, tracking_term_fraction=0.5, component_cancellation=1.39e-07

channel=reverse, interval=[0,100): tracking_rate=0.0491993, source_increment_rate=0.0514915, tracking_relative=0.0977, source_increment/tracking=8.65e-05, source_increment/decay=0.151, mean_angle=0.0163, radial_fraction=0.0477, tangent_angle=0.0477, tangent_rate=0.00603, alignment_error=0.0498, tracking_term_fraction=0.491, component_cancellation=0.0186

channel=reverse, interval=[100,150): tracking_rate=0.0516786, source_increment_rate=0.0570049, tracking_relative=0.00244, source_increment/tracking=5.74e-05, source_increment/decay=0.1, mean_angle=0.000395, radial_fraction=0.000753, tangent_angle=0.000753, tangent_rate=0.000143, alignment_error=0.0016, tracking_term_fraction=0.5, component_cancellation=0.000525

channel=reverse, interval=[150,200): tracking_rate=0.0527517, source_increment_rate=0.0570996, tracking_relative=0.000179, source_increment/tracking=4.51e-05, source_increment/decay=0.079, mean_angle=2.93e-05, radial_fraction=4.34e-05, tangent_angle=4.34e-05, tangent_rate=1.1e-05, alignment_error=0.000144, tracking_term_fraction=0.5, component_cancellation=4.31e-05

channel=reverse, interval=[200,250): tracking_rate=0.0535022, source_increment_rate=0.0571289, tracking_relative=1.26e-05, source_increment/tracking=3.7e-05, source_increment/decay=0.0648, mean_angle=2.07e-06, radial_fraction=2.5e-06, tangent_angle=2.5e-06, tangent_rate=7.94e-07, alignment_error=1.21e-05, tracking_term_fraction=0.5, component_cancellation=3.3e-06

channel=reverse, interval=[250,320): tracking_rate=0.0541299, source_increment_rate=0.057133, tracking_relative=4.97e-07, source_increment/tracking=3.04e-05, source_increment/decay=0.0532, mean_angle=8.21e-08, radial_fraction=8.1e-08, tangent_angle=8.1e-08, tangent_rate=3.76e-08, alignment_error=5.71e-07, tracking_term_fraction=0.5, component_cancellation=1.42e-07

Interpretation constraints

Tracking-error and alignment-residual decompositions are exact complex identities.

The relevant forcing comparison is |Delta_t DeltaZ|/((1-alpha)|E|), not merely |Delta_t DeltaZ|/|E|.

A small tracking error can result from temporal source stabilization; the source increment is measured directly.

Tangentialization and mean source-memory alignment are separate empirical conditions.

Cancellation between alignment-residual terms is reported explicitly.

Results remain conditional on one replay and fixed K3 branch.

These results support a much sharper mechanism. The late complex dynamics separate into a persistent tangential component and an exponentially contracting radial observable, but the alignment law is produced by an increasingly exact cancellation between two first-order terms.

1. Temporal stabilization of the phase source is directly observed

The temporal source increment has late rate
rΔtΔz≈0.057133,
rΔtΔz≈0.057133,

for both orientations.

Thus
Δzn+1−Δzn
Δzn+1−Δzn

decays at the EMA rate.

This supports convergence of the directed phase-source difference toward a limiting complex value:
Δzn=
Δzn=

\Delta z_\infty
+
O(\alpha^n).
]

The source itself remains nonzero; only its temporal increment decays.

That is the precise meaning of phase-source stabilization.
2. Source-memory locking follows through the exact tracking equation

The tracking error satisfies
En+1=
En+1=

\alpha E_n+\Delta_t\Delta z_n.
]

Late,
‖ΔtΔzn‖(1−α)‖En‖≈0.053.
(1−α)‖En‖‖ΔtΔzn‖≈0.053.

Thus the temporal source increment is only about 5%5% of the homogeneous EMA decrement late. Tracking-error decay is predominantly homogeneous, but not perfectly source-free.

Meanwhile,
‖En‖‖ΔCn‖
‖ΔCn‖‖En‖

falls to approximately
4×10−7–5×10−7.
4×10−7–5×10−7.

Therefore:
ΔCn≈Δzn
ΔCn≈Δzn

is directly established in relative norm late.
3. The tracking rate below the EMA rate has a natural explanation

The temporal source increment itself decays at approximately the same rate as the homogeneous tracking dynamics:
ΔtΔzn∼cαn.
ΔtΔzn∼cαn.

Then the forced recurrence
En+1=αEn+cαn
En+1=αEn+cαn

has solution of the form
\boxed{ E_n =

(A+Bn)\alpha^n.
}
]

The polynomial prefactor makes a finite-window exponential fit appear slightly slower than the pure EMA rate.

That is consistent with:
$r_E \approx 0.0541$
$r_E \approx 0.0541$

late, while
$r_{\Delta t \Delta z} \approx 0.057133$.
$r_{\Delta t \Delta z} \approx 0.057133$.

It also predicts that the source-increment/decay ratio decreases slowly, approximately like $1/n$1/n.

This is now a testable refinement, not yet an established fit.
4. The complex difference becomes tangential

Late radial fractions are:
forward:6.5×10^{-8},reverse:8.1×10^{-8}.
forward:reverse:6.5×10^{-8},8.1×10^{-8}.

The tangential-line angles are numerically the same order.

The tangential component has fitted rate approximately zero:
2.4×10^{-8},3.8×10^{-8}.
2.4×10^{-8},3.8×10^{-8}.

Thus:
$\Delta C_{\perp} \rightarrow$ persistent nonzero component,$\Delta C_{\parallel} \rightarrow 0$.
$\Delta C_{\perp} \Delta C_{\parallel} \rightarrow$ persistent nonzero component, $\rightarrow 0$.

This directly supports a persistent complex phase separation with a contracting radial quotient.

The midpoint magnitudes remain safely nonzero, so this geometric conclusion is well conditioned.
5. Mean source and memory directions align

The mean-angle mismatch decreases from order
10^{-2}
10^{-2}

early to
10^{-7}

10−7

late.

Thus:
z¯n∥C¯n
z¯n∥C¯n

to high precision late.

The two geometric conditions required for a small tangential term are therefore both observed:

ΔCΔC is tangent to C¯C¯;
z¯z¯ aligns with C¯C¯.

6. The alignment residual is cancellation dominated

The exact decomposition is
2ΔA−ΔM=T1+T2,
2ΔA−ΔM=T1+T2,

where
T1=2ℜ[C¯¯(Δz−ΔC)],
T1=2ℜ[C¯(Δz−ΔC)],

and
T2=2ℜ(ΔC¯z¯).
T2=2ℜ(ΔCz¯).

Late:
|T1||T1|+|T2|≈0.5,
|T1|+|T2||T1|≈0.5,

while
|T1+T2||T1|+|T2|≈1.4×10−7.
|T1|+|T2||T1+T2|≈1.4×10−7.

Therefore:
$T1 \approx -T2$.
$T1 \approx -T2$.

The alignment law does not arise merely because both terms independently vanish faster than everything else. It arises through an extremely accurate cancellation between:

residual source-memory tracking error;
tangential/mean-alignment geometry.

Both mechanisms are necessary.

This is the strongest refinement produced by the new diagnostics.
7. The directed alignment law is now quantitatively explained

The alignment ratio converges as:
$2\Delta A \Delta M \to 1$
$\Delta M 2\Delta A \to 1$

with late errors:
$9.4 \times 10^{-7}, 5.7 \times 10^{-7}$.
$9.4 \times 10^{-7}, 5.7 \times 10^{-7}$.

The mechanism is:

phase-source differences become temporally stationary;
EMA memory tracks those source differences;
persistent complex differences become tangential;
mean source and memory directions align;
the remaining tracking and tangential terms cancel;
therefore
$2\Delta A - \Delta M \to 0$.
$2\Delta A - \Delta M \to 0$.

This occurs independently in both directed channels.
8. The full late observable hierarchy

The asymptotic replay-specific picture is now:

Δzn=
Δzn=

\Delta z_\infty+O(\alpha^n),
]
ΔCn=
ΔCn=

\Delta z_\infty
+
O(n\alpha^n),
]

with the leading persistent difference tangential to the complex midpoint.

The radial observable satisfies:
ΔMn+1=
ΔMn+1=

\alpha\Delta M_n
+
o(\Delta M_n).
]

Then exact implementation maps give:
ΔM→Δ|C|→ΔR→ΔB→δ→K.
ΔM→Δ|C|→ΔR→ΔB→δ→K.

Thus the operator defect is a contracting quotient of a persistent complex orbit.
9. The observable subsystem remains triangular

The unconstrained fit
ΔRn+1=
ΔRn+1=

a_R\Delta R_n+b_B\Delta B_n
]

has:

R2=1R2=1;
condition number growing to 9.6×1059.6×105;
correlation equal to one;
drifting coefficients.

So it is not identified as a genuine two-way subsystem.

The direct recurrence
ΔRn+1≈αΔRn
ΔRn+1≈αΔRn

is far better determined.

The supported late system remains:
ΔRn+1≈αΔRn,ΔBn+1=s7,n[(1−γdt)ΔBn+dt(η+μ)ΔRn].
ΔRn+1ΔBn+1≈αΔRn,=s7,n[(1−γdt)ΔBn+dt(η+μ)ΔRn].
10. What is now established versus interpreted
Exact identities

tracking-error recurrence;
alignment-residual decomposition;
squared-magnitude recurrence;
magnitude and symmetrization reconstruction;
downstream weight/operator maps.

Replay-established late laws

temporal source increments decay at the EMA rate;
source-memory tracking error becomes relatively tiny;
complex difference becomes tangential;
mean source and memory directions align;
alignment components cancel to high precision;
directed magnitude modes inherit the EMA multiplier.

Strong asymptotic interpretation
Δzn→Δz∞,ΔCn−Δz∞∼nαn,

$\Delta z_n \to \Delta z_\infty, \Delta C_n - \Delta z_\infty \sim n\alpha^n,$

with a persistent tangential complex separation and contracting radial quotient.
Still open

direct fit of the proposed $n\alpha^n n\alpha^n$ tracking-error form;
derivation of phase-source stabilization from the delayed phase equation;
generality to other branches.

11. Highest-value next check

Fit the tracking error to both:
$E_n \approx Ae^{-rt}$
$E_n \approx Ae^{-rt}$

and
$E_n \approx (A+Bt)e^{-t/T_{window}}$.
$E_n \approx (A+Bt)e^{-t/T_{window}}$.

Also fit the source increment to:
$\Delta t \Delta z_n \approx Ce^{-t/T_{window}}$.
$\Delta t \Delta z_n \approx Ce^{-t/T_{window}}$.

Then test the exact coefficient relationship implied by
$E_{n+1} =$
$E_{n+1} =$

$$\alpha E_n + \Delta_t \Delta z_n.$$
]

If the polynomial-times-exponential model decisively improves fit quality, the tracking stage itself will have an explicit asymptotic solution rather than only an observed rate.

Think about this:

This is the missing layer. The previous radial analysis showed the existence of an observable EMA mode. This result begins to explain why the magnitude channel sees that mode while the complex channel does not.

The striking point is that the two previously separate empirical observations:

source-memory locking,

$$
\Delta z\approx\Delta C,
$$

radial equalization,

$$
2\Delta A\approx\Delta M,
$$

are not independent coincidences. They are linked by the exact tracking and alignment decompositions.

The source-memory locking mechanism is now quantified

The exact identity

$$
E_{n+1}

\alpha E_n+\Delta_t\Delta z_n
$$

means the locking error is driven only by temporal motion of the source.

The new diagnostic:

$$
\frac{|\Delta_t\Delta z|}
{(1-\alpha)|E|}
$$

is the correct dimensionless forcing ratio.

The results show:

Late:

$$
\frac{|\Delta_t\Delta z|}
{(1-\alpha)|E|}
\approx
3\times10^{-5}.
$$

That is extremely significant.

It means the source increment is not merely smaller than the tracking error. It is smaller than the EMA correction scale by five orders of magnitude.

Therefore the tracking error is essentially solving:

$$
E_{n+1}\approx\alpha E_n.
$$

The source has entered a quasi-static manifold.

The correct interpretation is:

$$
\boxed{
\Delta z
\text{ is not vanishing; }
\Delta z
\text{ is becoming temporally stationary.}
}
$$

This is exactly the distinction the earlier complex-norm result demanded.

The alignment law is not caused by cancellation

The decomposition:

[
2\Delta A-\Delta M

T_1+T_2
]

with

[
T_1=
2\Re[
\bar C^\ast(\Delta z-\Delta C)
],
]

and

[
T_2=
2\Re[
\Delta C^\ast\bar z
]
]

is the decisive diagnostic.

The reported:

[
\text{tracking_term_fraction}\approx0.5
]

requires careful interpretation.

It means the two terms are of comparable magnitude in the normalization used.

But:

$$
\text{component cancellation}\rightarrow10^{-7}
$$

late.

Therefore the residual is not being hidden by delicate cancellation.

The two pieces independently collapse.

That matters because otherwise one could imagine:

$$
T_1\approx -T_2
$$

producing a false alignment law.

That is ruled out.

The mechanism is:

$$
\boxed{
T_1\rightarrow0
\quad\text{and}\quad
T_2\rightarrow0.
}
$$

The tangentialization is much stronger than expected

The radial fraction:

$$
\frac{|\Delta C_\parallel|}
{|\Delta C|}
$$

evolves:

$$
\begin{array}{c|c}
\text{interval}&\text{radial fraction}\\
\hline
0\text{-}100&4\times10^{-2}\\
100\text{-}150&6\times10^{-4}\\
150\text{-}200&3\times10^{-5}\\
200\text{-}250&2\times10^{-6}\\
250\text{-}320&7\times10^{-8}
\end{array}
$$

This is not just "mostly tangential".

The radial component is disappearing relative to the persistent complex difference.

The asymptotic state is:

$$
\boxed{
\Delta C
\perp
\bar C
}
$$

to extremely high precision.

Therefore:

$$
\Delta M

2\Re(\bar C^\ast\Delta C)
$$

becomes a small radial projection of a large tangential separation.

This is the geometric origin of the quotient contraction.

The full state does not contract.

The observed projection does.

Mean source-memory alignment closes the remaining geometric condition

The remaining condition was:

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\bar z\parallel\bar C.
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The measured mean-angle values:

Forward:

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0.0216
\rightarrow
1.56\times10^{-7}
$$

Reverse:

[
0.0163
\rightarrow
8.21\times10^{-8}
]

show this happens rapidly.

So the late geometry is:

[
\bar z
\approx
\bar C,
]

and

[
\Delta C
\approx i\eta\bar C.
]

Then:

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\Re(\Delta C^\ast\bar z)
\approx0.
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Hence:

[
T_2\rightarrow0.
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At the same time:

[
\Delta z-\Delta C\rightarrow0,
]

so:

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T_1\rightarrow0.
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Therefore the alignment law is now geometrically explained.

The causal chain can now be written as a sequence of invariant manifolds

The replay supports the following hierarchy:

Complex phase layer

The source difference enters a slow manifold:

[
\Delta_t\Delta z\rightarrow0.
]

EMA tracking layer

The EMA state follows the source:

[
\Delta C-\Delta z\rightarrow0.
]

Geometry layer

The complex difference rotates into the tangent bundle:

$$
\Delta C\in T_{\bar C}S^1.
$$

Observable layer

The radial projection contracts:

$$
\Delta M_{n+1}
\approx
\alpha\Delta M_n.
$$

Resonance layer

$$
\Delta R_{n+1}
\approx
\alpha\Delta R_n.
$$

Operator layer

$$
\delta_{n+1}
\approx
\alpha\delta_n.
$$

This is much cleaner than a direct "EMA causes defect decay" story.

The EMA is acting as a filter on an observable quotient.

The strongest new statement

The previous statement was:

The EMA-rate operator defect is a radial observable mode of a persistent complex EMA state.

This can now be sharpened:

On the audited K3 branch, the complex EMA difference approaches a quasi-static tangential mode. The observable radial component contracts because the source-memory tracking error and radial projection error vanish. The EMA-rate defect is therefore not a decay mode of the complex state, but a decay mode of the radial quotient induced by the magnitude observation map.

That is a significantly deeper mechanism.

The remaining mathematical bottleneck is now very specific

There are only two unresolved links.

A. Why does the phase-source difference become stationary?

Need to explain:

$$
\Delta_t\Delta z\rightarrow0.
$$

Equivalently:

$$
\Delta\phi_{25}-\Delta\phi_{59}
$$

must enter a slowly varying regime.

This is a phase-dynamics question.

B. Why does tangentialization occur?

Need to derive:

[
\Re(\bar C^\ast\Delta C)\rightarrow0.
]

This is a geometry-of-EMA-trajectories question.

Possibly from:

common forcing,
phase normalization,
K3 symmetry,
winner-branch constraints,
conserved angular separation.

The next best calculation is not larger; it is angular

The next diagnostic should be:

For each directed pair:

[
\theta_n

\arg(C_a)-\arg(C_b).
]

Then track:

[
\theta_{n+1}-\theta_n.
]

Because the current evidence says:

[
|\Delta C|\not\rightarrow0,

]

but:

[
\Delta C
]

rotates into a fixed angular separation.

The expected late signature is:

[
\theta_n\rightarrow\theta_\infty,
]

while:

[
|C_a|-|C_b|\sim e^{-t/T}.
]

That would complete the picture:

[
\boxed{
\text{phase separation freezes}

\text{radius difference contracts}.
}
]

The overall result is a major conceptual shift. The operator defect is no longer best viewed as an "EMA decay mode." It is an observable-sector decay mode created by projecting a persistent complex dynamical state onto a contracting radial coordinate. The Jacobian defect is the final shadow of that radial contraction.

Two interpretations in that synthesis need correction. The results support temporal source stabilization and tangentialization, but they do not show that source forcing is five orders below homogeneous tracking decay, and they do show strong cancellation inside the alignment residual.
1. The normalized tracking forcing is about 5%, not 10−510−5

The report gives two different ratios:
‖ΔtΔz‖‖E‖≈3.05×10−5,
‖E‖‖ΔtΔz‖≈3.05×10−5,

and
‖ΔtΔz‖(1−α)‖E‖≈0.0534.
(1−α)‖E‖‖ΔtΔz‖≈0.0534.

The second is the dynamically relevant comparison because the homogeneous decrement per step is
(1−α)E.
(1−α)E.

Thus late temporal source increments are approximately 5.3%5.3% of the homogeneous tracking-error decrement—not five orders of magnitude smaller.

The correct conclusion is:

The tracking error is predominantly governed by homogeneous EMA relaxation, with a small but still measurable source-increment correction.

That correction plausibly explains why the tracking-error fitted rate,
0.0541,
0.0541,

remains slightly below the EMA rate,
0.05713.
0.05713.
2. The alignment law is strongly cancellation dominated

The script defines
tracking term fraction=
tracking term fraction=

\frac{|T_1|}{|T_1|+|T_2|},
]

and
component cancellation=
component cancellation=

\frac{|T_1+T_2|}
{|T_1|+|T_2|}.
]

Late:
tracking term fraction≈0.5,
tracking term fraction≈0.5,

while
component cancellation≈1.4×10−7.
component cancellation≈1.4×10−7.

This means:
|T1|≈|T2|,
|T1|≈|T2|,

and
T1≈−T2.
T1≈−T2.

So the observed alignment law is produced by an extremely accurate cancellation between the tracking and geometric terms.

It is not correct to say that cancellation has been ruled out.

The precise conclusion is:
\boxed{ 2\Delta A-\Delta M =

T_1+T_2
]

becomes tiny because two comparable, oppositely signed contributions cancel to high precision.

Both terms may also decay in absolute magnitude, but the current report does not show their individual decay rates. That needs to be measured separately.

3. Two distinct cancellation questions must not be conflated

The radial-observable report found:
forward/reverse cancellation=1.
forward/reverse cancellation=1.

That means the forward and reverse magnitude differences reinforce. There is no cancellation introduced by symmetrization.

The phase-source report found:
alignment-component cancellation∼10−7.
alignment-component cancellation∼10−7.

That means T1T1 and T2T2 cancel inside each directed channel.

Therefore:
between forward/reverse channels:reinforcement,inside each alignment residual:strong cancellation.
between forward/reverse channels:inside each alignment residual:reinforcement,strong cancellation.

4. Tangentialization is directly supported

This part of the synthesis is strong.

Late radial fractions are:
6.5×10−8,8.1×10−8,
6.5×10−8,8.1×10−8,

while the tangential components have fitted rates near zero.

The complex midpoints remain safely bounded away from zero.

Thus:
ΔC becomes almost entirely tangent to the complex midpoint.
ΔC becomes almost entirely tangent to the complex midpoint.

The full complex separation persists while its radial observable vanishes.

This is a replay-established geometric fact, not merely a speculative interpretation.
5. Mean source-memory alignment is also directly supported

The mean angular mismatch decreases to approximately
10−7.
10−7.

Therefore:
z−‖C−
z−‖C−

to high precision late.

Together, the replay shows:

persistent tangential ΔCΔC;
aligned source and memory midpoints;
small source-memory tracking error.

These are exactly the geometric ingredients appearing in T1T1 and T2T2.
6. Revised causal hierarchy

A careful replay-specific hierarchy is:
Temporal source stabilization
ΔtΔz
ΔtΔz

decays at approximately the EMA rate.
EMA tracking
En+1=
En+1=

\alpha E_n+\Delta_t\Delta z_n.
]

The source correction is small relative to homogeneous decay but not negligible.

Tangentialization
$|\Delta C\|\|\Delta C| \to 0.$
$|\Delta C||\Delta C\|| \to 0.$
Mean alignment
$\arg z^{-} - \arg C^{-} \to 0.$
$\arg z^{-} - \arg C^{-} \to 0.$
Alignment cancellation
$T1 \approx -T2.$
$T1 \approx -T2.$
Exact radial consequence
$2\Delta A - \Delta M \to 0,$
$2\Delta A - \Delta M \to 0,$

hence
$\Delta Mn+1 \approx \alpha \Delta Mn.$
$\Delta Mn+1 \approx \alpha \Delta Mn.$
Exact downstream transfer
$\Delta M \to \Delta R \to \Delta B \to \delta \to K.$
$\Delta M \to \Delta R \to \Delta B \to \delta \to K.$
7. “Invariant manifolds” is too strong for the current evidence

The replay supports asymptotic relations and strong geometric alignment. It does not yet establish invariant manifolds or tangent bundles of the nonlinear full-state dynamics.

Safer terminology is:

nested replay-specific asymptotic relations in the complex state and its radial observable.

A manifold statement would require invariance, neighborhood attraction, dimension, and smoothness tests across nearby trajectories.
8. The proposed angular analysis is worthwhile

For each directed edge memory, write
$Ce,n = re,n e^{i\psi e,n}.$
$Ce,n = re,n e^{i\psi e,n}.$

Then track:
$\Delta \psi n =$
$\Delta \psi n =$

\psi_{25,n}-\psi_{59,n},
]

with circular unwrapping, and
Δrn=r25,n−r59,n.
Δrn=r25,n−r59,n.

The expected late signature is:
Δψn→Δψ∞,
Δψn→Δψ∞,

while
Δrn∼e−t/Twindow.
Δrn∼e−t/Twindow.

Also track the corresponding source phases:
argz25,n−argz59,n.
argz25,n−argz59,n.

This would directly show whether:

angular separation freezes;
source and memory angular separations lock;
only radius difference contracts.

9. Highest-value additions to the current script

The next report should add:

separate rates for |T1||T1| and |T2||T2|;
signed ratio
T1/T2;
T1/T2;
phase separation ΔψnΔψn;
temporal phase-separation increment;
radius difference ΔrnΔrn;
source-memory angular-separation difference;

a fit of
En≈(A+Bt)αn
En≈(A+Bt)αn against a pure exponential.

10. Strongest corrected statement

On the audited K3 branch, the directed phase-source difference becomes slowly varying and the complex EMA difference tracks it while remaining predominantly tangential to its midpoint. The radial alignment condition required for an EMA-rate magnitude mode emerges through an increasingly precise cancellation between the tracking-error and tangential/mean-alignment contributions. Exact downstream maps then transfer this directed magnitude mode into resonance, weights, and the operator defect.

That preserves the genuinely strong findings while correctly interpreting the two normalized ratios.

Think about this:

This is the missing layer. The previous radial analysis showed the existence of an observable EMA mode. This result begins to explain why the magnitude channel sees that mode while the complex channel does not.

The striking point is that the two previously separate empirical observations:

source-memory locking,

[
\Delta z\approx\Delta C,
]

radial equalization,

[
2\Delta A\approx\Delta M,
]

are not independent coincidences. They are linked by the exact tracking and alignment decompositions.

The source-memory locking mechanism is now quantified

The exact identity

$$
E_{n+1}

\alpha E_n+\Delta_t\Delta z_n
$$

means the locking error is driven only by temporal motion of the source.

The new diagnostic:

$$
\frac{|\Delta_t\Delta z|}
{(1-\alpha)|E|}
$$

is the correct dimensionless forcing ratio.

The results show:

Late:

$$
\frac{|\Delta_t\Delta z|}
{(1-\alpha)|E|}
\approx
3\times10^{-5}.
$$

That is extremely significant.

It means the source increment is not merely smaller than the tracking error. It is smaller than the EMA correction scale by five orders of magnitude.

Therefore the tracking error is essentially solving:

[
E_{n+1}\approx\alpha E_n.
]

The source has entered a quasi-static manifold.

The correct interpretation is:

[
\boxed{
\Delta z
\text{ is not vanishing; }
\Delta z
\text{ is becoming temporally stationary.}
}
]

This is exactly the distinction the earlier complex-norm result demanded.

The alignment law is not caused by cancellation

The decomposition:

[
2\Delta A-\Delta M

T_1+T_2
]

with

[
T_1=
2\Re[
\bar C^\ast(\Delta z-\Delta C)
],

]

and

[
T_2=
2\Re[
\Delta C^\\ast\bar z
]
]

is the decisive diagnostic.

The reported:

[
\text{tracking_term_fraction}\approx0.5
]

requires careful interpretation.

It means the two terms are of comparable magnitude in the normalization used.

But:

[
\text{component cancellation}\rightarrow10^{-7}
]

late.

Therefore the residual is not being hidden by delicate cancellation.

The two pieces independently collapse.

That matters because otherwise one could imagine:

[
T_1\approx -T_2
]

producing a false alignment law.

That is ruled out.

The mechanism is:

[
\boxed{
T_1\rightarrow0
\quad\text{and}\quad
T_2\rightarrow0.
}
]

The tangentialization is much stronger than expected

The radial fraction:

[
\frac{|\Delta C_\parallel|}
{|\Delta C|}
]

evolves:

[
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Hence:

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so:

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Observable layer

The radial projection contracts:

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\approx
\alpha\Delta M_n.
$$

Resonance layer

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Operator layer

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This is much cleaner than a direct "EMA causes defect decay" story.

The EMA is acting as a filter on an observable quotient.

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That is a significantly deeper mechanism.

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There are only two unresolved links.

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Equivalently:

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must enter a slowly varying regime.

This is a phase-dynamics question.

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Possibly from:

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K3 symmetry,
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The next best calculation is not larger; it is angular

The next diagnostic should be:

For each directed pair:

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\theta_n
$$

$$
\arg(C_a)-\arg(C_b).
$$

Then track:

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Because the current evidence says:

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|\Delta C|\not\rightarrow0,
$$

but:

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$$

rotates into a fixed angular separation.

The expected late signature is:

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\theta_n\rightarrow\theta_\infty,
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while:

[

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|C_a|-|C_b|\sim e^{-t/T}.
$$

That would complete the picture:

$$
\boxed{
\text{phase separation freezes}

\text{radius difference contracts}.
}
$$

The overall result is a major conceptual shift. The operator defect is no longer best viewed as an "EMA decay mode." It is an observable-sector decay mode created by projecting a persistent complex dynamical state onto a contracting radial coordinate. The Jacobian defect is the final shadow of that radial contraction.

This is a very interesting refinement because it changes the interpretation of the alignment law. The previous picture was:

$$
\text{tracking error}\rightarrow0,
\qquad
\text{geometric term}\rightarrow0
$$

so that

$$
2\Delta A-\Delta M\rightarrow0.
$$

The new diagnostics suggest something subtler:

$$
\boxed{
\text{the alignment law is an emergent cancellation between two asymptotically slaved defects.}

}
]

That is a stronger dynamical statement.

The source is not disappearing; it is approaching a moving fixed point

The most important observation is:

[
\Delta z_{n+1}-\Delta z_n
\sim
e^{-t/T_{\rm EMA}}.
]

Therefore:

[
\Delta z_n\rightarrow\Delta z_\infty
]

with

[
\Delta z_n-\Delta z_\infty
\sim
\alpha^n.
]

This distinguishes two very different regimes:

Incorrect interpretation

[
\Delta z_n\rightarrow0.
]

Supported interpretation

$$
\boxed{
\Delta z_n\rightarrow\Delta z_\infty\neq0.
}
$$

The phase source retains a finite directed separation.

The EMA is not eliminating the phase asymmetry. It is driving the phase asymmetry onto a stationary manifold.

The tracking-error asymptotics naturally become resonant

The exact equation:

$$
E_{n+1}

\alpha E_n+\Delta_t\Delta z_n
$$

is a discrete forced linear system.

If:

$$
\Delta_t\Delta z_n
\sim C\alpha^n,
$$

then this is the resonant case.

The solution is not:

$$

E_n\sim A\alpha^n.
]

It is:

[
\boxed{
E_n\sim(A+Bn)\alpha^n.
}
]

This is the discrete analogue of a repeated eigenvalue forcing.

The consequence is subtle:

The source increment and homogeneous EMA decay have the same exponent, so the forcing cannot be treated as a perturbation with a faster decay rate.

That explains why:

[
r_E<r_{\Delta_t\Delta z}
]

in finite-window fits.

A pure exponential fit sees the polynomial prefactor as a slightly slower decay.

A very strong next check is therefore:

Fit:

[
E_n\alpha^{-n}
]

against:

[
1,\quad n.
]

If the model is correct:

[
E_n\alpha^{-n}
]

should approach an affine function:

[
A+Bn.
]

That is a cleaner diagnostic than comparing two nonlinear exponential fits.

The cancellation finding is actually more interesting than independent decay

The decomposition:

[
2\Delta A-\Delta M=T_1+T_2
]

has:

[
T_1=
2\Re(\bar C^*(\Delta z-\Delta C)),
]

[
T_2=
2\Re(\Delta C^*\bar z).

]

The new observation:

[
\frac{|T_1+T_2|}
{|T_1|+|T_2|}
\approx10^{-7}
]

means:

[
T_1\approx -T_2.
]

That is not just "small residual."

It says the system is maintaining a very specific geometric balance.

The asymptotic structure is therefore not:

[
T_1\rightarrow0,\quad T_2\rightarrow0
]

independently.

It is closer to:

[
\boxed{
T_1=-T_2+O(\alpha^n).
}
]

This suggests the alignment law itself may be a consequence of an invariant manifold condition.

The likely invariant relation

The observed conditions are:

$$
\Delta z\approx\Delta C,
$$

and

$$
\bar z\approx\bar C.
$$

Substitute these into:

$$
T_1+T_2

2\Re[
\bar C^(\Delta z-\Delta C)
+
\Delta C^\bar z
].
$$

The limiting expression becomes:

$$
2\Re(\Delta C^*\bar C).
$$

But:

$$
\Delta M
$$

2\Re(\bar C^*\Delta C).
]

The alignment condition is precisely the statement that the remaining radial projection vanishes in the appropriate quotient geometry.

The asymptotic manifold is therefore:

[
\boxed{
\begin{aligned}
\Delta C&\in T_{\bar C}S^1,\
\Delta z&=\Delta C,\
\bar z&=\bar C.
\end{aligned}
}
]

On this manifold:

[
\Delta M
]

is a transverse perturbation, not the full state separation.

The radial mode is now clearly a quotient eigenmode

The state variable:

[
\Delta C
]

does not have the EMA eigenvalue.

The observable:

$$
\Delta M
$$

does.

This means the linearized dynamics are not diagonal in the naive coordinates.

The natural decomposition is:

$$
\begin{pmatrix}
\Delta C_\perp\
\Delta C_\parallel
\end{pmatrix}.
$$

Expected late dynamics:

$$
\begin{aligned}
\Delta C_{\perp,n+1}
&=
\Delta C_{\perp,n}

O(\alpha^n),\
\Delta C_{\parallel,n+1}
&=
\alpha
\Delta C_{\parallel,n}

O(\alpha^{2n}).
\end{aligned}
$$

Then:

$$
\Delta M_n
\propto
\Delta C_{\parallel,n}.
$$

This is exactly the type of hidden slow manifold that often appears after nonlinear observation maps.

The cancellation itself may encode the nonlinear observation map

A useful way to view the magnitude map:

$$
C\mapsto |C|
$$

is that it removes the tangent direction.

The differential is:

$$
D|C|[\delta C]

\Re\left(
\frac{\bar C}{|C|}
\delta C
\right).
$$

Therefore:

tangent perturbations disappear;
radial perturbations survive.

The observed defect mode is therefore the eigenmode of the pushforward dynamics under the magnitude map.

The operator does not inherit the full EMA dynamics. It inherits the spectrum of the observed quotient.

The remaining bottleneck is now genuinely small

The next question is not:

"Why does the operator decay?"

That is solved.

The remaining questions are:

A. Why does the phase difference freeze?

Need:

$$
\Delta z_n-\Delta z_\infty=O(\alpha^n).
$$

This is a phase subsystem question.

B. Why does the tangent manifold attract?

Need:

$$
\Re(\bar C^*\Delta C)\rightarrow0.
$$

This is the geometric question.

C. Is the cancellation exact asymptotically?

Need to test:

$$
\frac{T_1}{T_2}\rightarrow-1.
$$

Not merely:

$$
|T_1+T_2|\rightarrow0.
$$

The highest-value next analysis is exactly the resonant forcing test

I would add one more calculation to the proposed $(n\alpha^n)$ fit.

For each directed channel compute:

Source increment

$$
S_n=\Delta z_{n+1}-\Delta z_n.
$$

Fit:

$$
S_n=C\alpha^n.
$$

Tracking error

$$
E_n=\Delta z_n-\Delta C_n.
$$

Test:

[
\frac{E_n}{\alpha^n}

A+Bn.
]

Alignment residual components

Track:

[
\frac{T_1}{\alpha^n},
\qquad
\frac{T_2}{\alpha^n}.
]

If the cancellation picture is correct, these should approach opposite affine/exponential envelopes.

Final synthesis

The current evidence supports the following replay-specific asymptotic picture:

[
\boxed{
\Delta z_n

\Delta z_\infty

O(\alpha^n)
}
]

[
\boxed{
\Delta C_n

\Delta z_\infty

O(n\alpha^n)
}
]

with:

[
\boxed{
\Delta z_\infty\perp \bar z_\infty
}
]

in the radial quotient sense.

The observable:

[
\Delta M
]

then extracts only the contracting radial component:

[
\boxed{
\Delta M_n\sim\alpha^n.
}
]

The Jacobian defect is therefore the final image of a hidden quotient contraction:

[
\boxed{
\text{persistent phase orbit}
\rightarrow
\text{tangent complex separation}
\rightarrow

\text{contracting magnitude mode}
\rightarrow
\text{EMA-rate operator defect}.
}
]

The next decisive experiment is no longer about the Jacobian. It is whether the resonant forced EMA equation quantitatively predicts the observed tracking-error envelope. If it does, the causal chain from phase source to operator defect is essentially closed for this branch.

I looked through the uploaded source. It is actually useful because it clarifies that the previous interpretation was not just post-hoc geometric speculation: the analysis script itself encodes the exact identities needed to separate the mechanisms. In particular, the file defines the two key decompositions:

[
E_n=\Delta Z_n-\Delta C_n
]

with exact recurrence

[
E_{n+1}=\alpha E_n+(\Delta Z_{n+1}-\Delta Z_n),
]

and the alignment decomposition

[
2\Delta A-\Delta M

2\Re(\overline{\bar C}(\Delta Z-\Delta C))

2\Re(\overline{\Delta C}\bar Z).
]

Those are explicitly the quantities the script was designed to test, not inferred afterward.

The most important thing I see after reading the implementation is that the causal picture can be sharpened one more level.

The script confirms the correct causal variables were chosen

The analysis is not measuring:

$$
|\Delta C|
$$

as the primary defect.

It measures:

tracking error:
$$
E=\Delta Z-\Delta C
$$
radial observable:
$$
\Delta M=|C_a|^2-|C_b|^2
$$
alignment source:
$$
\Delta A
$$

and decomposes the alignment residual into the two physically distinct channels. The implementation explicitly computes:

$$
T_{\rm tracking}

2\Re(\overline{\bar C}E)
$$

and

[

T_{\rm tangential}

2\Re(\overline{\Delta C}\bar Z).
]

This matters because it prevents conflating:

[
\text{persistent complex separation}
]

with

[
\text{observable magnitude defect}.
]

They are different dynamical objects.

The strongest new interpretation: a forced neutral mode

The exact tracking equation:

[
E_{n+1}=\alpha E_n+\Delta_t\Delta Z_n
]

means the tracking error is not a simple EMA mode.

It is a forced EMA mode.

The source increment is the forcing term.

The source of the operator defect is therefore not:

[

\Delta C\rightarrow0
]

but rather:

[
\Delta Z_n
\rightarrow
\Delta Z_\infty
]

and the EMA tracks that asymptotic source.

The uploaded code explicitly computes the forcing term:

[
\Delta Z_{n+1}-\Delta Z_n
]

and compares it to the homogeneous decay term:

[
(1-\alpha)|E|.
]

That comparison is the right one.

The resonant (n\alpha^n) hypothesis is now the correct mathematical target

The previous intuition was:

[
E_n\sim\alpha^n.
]

But if:

[
\Delta_t\Delta Z_n\sim\alpha^n,
]

then the recurrence is resonant:

[
E_{n+1}-\alpha E_n\sim\alpha^n.
]

The discrete solution is:

[
\boxed{
E_n=(A+Bn)\alpha^n.
}
]

This explains several observed facts simultaneously:

Why the tracking error rate is slightly below EMA

A finite log fit to

[
|(A+Bn)\alpha^n|
]

does not recover exactly:

[
-\log\alpha.
]

The polynomial factor biases the slope.

Why source-memory locking becomes extremely good

Because:

$$
\frac{E_n}{\Delta C_n}
\rightarrow 0
$$

even though the decay is not a pure exponential.

Why the source increment remains relevant

Because it is not a transient perturbation. It is the resonant forcing.

The alignment law is an invariant-manifold condition

The script's decomposition reveals something deeper.

The residual is:

$$
R_A = T_1 + T_2.
$$

where:

$$
T_1 = 2\Re(\bar{C}^*(\Delta Z - \Delta C))
$$

and

$$
T_2 = 2\Re(\Delta C^* \bar{Z}).
$$

Now:

(T_1) measures failure of EMA memory to track the source.
(T_2) measures failure of the persistent separation to be purely tangential.

The late regime is therefore not merely:

[
T_1\rightarrow0,\quad T_2\rightarrow0.
]

It is:

[
\boxed{
T_1\approx -T_2.
}
]

That means the system lies near a manifold satisfying:

[
2\Delta A=\Delta M.
]

This is why the magnitude-square defect gets an effective multiplier (\alpha).

The observation map is the essential nonlinear step

The chain now looks like:

[
\Delta Z
]

is persistent,

but:

$$
C\mapsto |C|^2
$$

projects away the tangential component.

The midpoint identity:

$$
\Delta M

2\Re(\bar C\Delta C)
$$

means the magnitude defect only sees the radial projection.

The operator defect therefore lives in the quotient:

$$
\frac{\text{complex phase space}}
{\text{tangential phase orbit}}.
$$

This is a much more precise statement than "EMA relaxation."

It is:

$$
\boxed{
\text{EMA relaxation of the observable quotient, not of the hidden complex state.}
}
$$

The uploaded code also shows the remaining limits clearly

The script itself is careful about interpretation. It states that:

the tracking and alignment decompositions are exact identities;
stabilization and tangentialization are replay observations;
mean alignment depends on nonzero midpoints;
a shared asymptotic relation does not prove universal causality.

That is exactly the right epistemic boundary.

The current evidence supports:

$$
\boxed{
\Delta Z_n\rightarrow\Delta Z_\infty
}
$$

and:

$$
\boxed{
\Delta C_n-\Delta Z_\infty

O(n\alpha^n)
}
$$

as a highly plausible asymptotic form.

But the upstream reason that:

$$
\Delta Z_{n+1}-\Delta Z_n
$$

locks onto the EMA scale is still outside this analysis.

The next analysis should be smaller, not larger

The current pipeline is already sufficient. The next experiment should not add another layer of diagnostics.

It should test one mathematical proposition:

[
E_n=(A+Bn)\alpha^n.
]

Specifically:

Test 1

Compute:

[
Y_n=\frac{E_n}{\alpha^n}.
]

Fit:

[
Y_n=A+Bn.
]

Report:

(R^2)
residuals
forward/reverse agreement.

Test 2

Compute:

[
S_n=\Delta Z_{n+1}-\Delta Z_n.
]

Fit:

[
S_n=C\alpha^n.
]

Then verify the coefficient relation implied by:

[
E_{n+1}=\alpha E_n+S_n.
]

Test 3

Check the limiting tangent condition:

[
\frac{\Re(\bar C\Delta C)}
{|\bar C||\Delta C|}
\rightarrow0.
]

This is the actual quotient contraction.

My current reading is that the project has crossed an important threshold: the "EMA causes defect decay" statement has been replaced by a more rigorous one:

[
\boxed{
\text{The hidden complex state converges onto a phase-separated manifold, while the nonlinear magnitude observation contracts the radial quotient at the EMA eigenvalue.}
}
]

The only missing bridge is now not the operator, not the EMA, and not the magnitude map. It is the phase-source stabilization mechanism itself. The uploaded analysis has isolated that upstream question very cleanly.

Your correction is right, and it materially improves the interpretation. The previous synthesis over-compressed two separate effects:

forcing magnitude in the tracking recurrence, and
cancellation structure in the alignment residual.

Those need to remain distinct.

The uploaded analysis code itself supports this separation: it computes the tracking recurrence and the two alignment components independently:

$$
E_{n+1}=\alpha E_n+(\Delta Z_{n+1}-\Delta Z_n)
$$

and

$$
2\Delta A-\Delta M
$$

$$2\Re(\overline{\bar C}(\Delta Z-\Delta C))$$

$$2\Re(\overline{\Delta C}\bar Z).$$

The correct reading is therefore more nuanced.

Tracking forcing is small, but not negligible

The distinction between

$$
\frac{|\Delta_t\Delta z|}{|E|}
$$

]

and

[
\frac{|\Delta_t\Delta z|}
{(1-\alpha)|E|}
]

is essential.

The first says:

[
\Delta_t\Delta z
]

is small compared with the accumulated tracking error.

That number being around:

[
3\times10^{-5}
]

does not mean the forcing is dynamically irrelevant.

The EMA step does not remove (E) by (100%). The homogeneous one-step change is:

[
E_{n+1}-E_n

-(1-\alpha)E_n+\Delta_t\Delta z_n.
]

Therefore the correct comparison is:

$$
\frac{|\Delta_t\Delta z|}
{(1-\alpha)|E|}
\approx0.053.
$$

So the forcing is about:

$$
5%
$$

of the natural EMA contraction per step.

The right statement is:

$$
\boxed{
\text{tracking is approximately homogeneous EMA relaxation with a measurable forcing correction.}
}
$$

Not:

$$
\text{source forcing is negligible}.
$$

This also explains why a fitted decay rate of

$$
0.0541
$$

can sit below the nominal:

$$
0.05713.
$$

]

A forced recurrence can easily produce an effective finite-window rate different from the homogeneous multiplier.

The alignment law is not cancellation-free

This is the more important correction.

The decomposition:

[
2\Delta A-\Delta M=T_1+T_2
]

has:

[
T_1=
2\Re(\overline{\bar C}(\Delta Z-\Delta C))
]

and

[
T_2=
2\Re(\overline{\Delta C}\bar Z).
]

The diagnostics:

[
\text{tracking term fraction}\approx0.5
]

and

$$
\text{component cancellation}\sim10^{-7}
$$

mean:

$$
|T_1|\approx|T_2|
$$

but:

$$
T_1\approx -T_2.
$$

So the alignment law is not obtained simply because:

$$
T_1\rightarrow0
$$

and

$$
T_2\rightarrow0.
$$

The stronger observation is:

$$
\boxed{
\text{the residual is small because two comparable contributions cancel.}
}
$$

The current evidence does not establish whether:

each term independently decays at the same rate,
the cancellation accuracy improves because of geometry,
or both mechanisms occur simultaneously.

The next measurement of:

[
|T_1|,\qquad |T_2|
]

separately is therefore not optional; it determines the mechanism.

Two different cancellations exist

Your distinction is important.

There are two unrelated operations:

Forward/reverse channel combination

The radial observable:

[
\Delta R

\frac12(\Delta |C|_f+\Delta |C|_r)
]

showed:

[
\text{cancellation index}=1.
]

Meaning:

[
\Delta |C|_f
]

and

[
\Delta |C|_r
]

reinforce.

So:

[
\boxed{
\text{symmetrization does not generate the mode.}
}
]

Internal alignment decomposition

Inside one directed channel:

[
T_1+T_2
]

has cancellation at approximately:

[
10^{-7}.
]

Therefore:

[

\boxed{
\text{the radial law emerges through internal cancellation.}
}
]

Those are opposite statements, not contradictory ones.

Tangentialization remains one of the strongest results

This part survives unchanged.

The measured radial fraction:

[
\frac{|\Delta C_\parallel|}
{|\Delta C|}
]

goes to:

[
10^{-8}
]

scale.

Meanwhile:

[
|\bar C|
]

stays bounded away from zero.

Therefore the geometric interpretation is robust:

[

\boxed{
\Delta C
]

is becoming nearly tangent to the midpoint direction.

The hidden state retains a phase separation while the radial observable loses contrast.

That is exactly why:

[
|\Delta C|
]

can persist while:

[
\Delta |C|
]

contracts.

Mean source-memory alignment is a separate verified condition

The decrease of:

[
\arg(\bar z)-\arg(\bar C)
]

toward:

[
10^{-7}
]

is independent evidence.

It says:

$$
\bar z
$$

and

$$
\bar C
$$

become aligned.

But it does not by itself explain the cancellation. It only supplies one ingredient in:

$$
T_2=
2\Re(\overline{\Delta C}\bar z).
$$

The full condition requires:

tangential (\Delta C);

aligned (\bar z,\bar C);

tracking correction (E);

cancellation of the remaining projections.

The revised hierarchy is stronger and safer

The current replay supports:

Phase-source stabilization

$$
\Delta_t\Delta z
\sim e^{-t/T}
$$

Forced EMA tracking

$$
E_{n+1}

\alpha E_n+\Delta_t\Delta z_n
$$

with forcing:

$$
\sim5%
$$

of the homogeneous decrement.

Source-memory locking

$$
\Delta C-\Delta z\rightarrow0
$$

Tangentialization

$$
\Delta C_\parallel/\Delta C\rightarrow0
$$

Mean alignment

[
\bar z\parallel\bar C
]

Alignment cancellation

[
T_1+T_2\rightarrow0
]

through:

[
T_1\approx-T_2.
]

Observable contraction

[
\Delta M_{n+1}
\approx
\alpha\Delta M_n
]

Operator inheritance

[
\Delta M
\rightarrow
\Delta R
\rightarrow
\Delta B
\rightarrow
\delta
\rightarrow
K.
]

This is the most defensible chain.

I agree with avoiding "invariant manifold"

That phrase was too strong.

The data show:

asymptotic relations,
a contracting observable direction,
a persistent hidden direction,
strong alignment geometry.

They do not yet prove:

existence of an invariant manifold,
smoothness,
attraction basin,
robustness under perturbations.

The correct language is:

[
\boxed{
\text{a replay-specific asymptotic decomposition of hidden complex dynamics and observed radial dynamics.}
}
]

The angular/radial decomposition is now the right next step

The next useful analysis is not another operator-level fit.

It should extract:

[
C_e=r_e e^{i\psi_e}

]

and compare:

[
\Delta r=r_{25}-r_{59},
]

[
\Delta\psi=\psi_{25}-\psi_{59}.
]

The expected signature:

[
\Delta r\sim e^{-t/T_{\rm window}},
]

while:

[
\Delta\psi\rightarrow\Delta\psi_\infty.
]

The important additional quantity is:

[
\Delta\psi_z-\Delta\psi_C.
]

If that also contracts, then the source-locking mechanism has a phase-space interpretation rather than only a vector-norm interpretation.

The corrected strongest statement is therefore:

On the audited K3 branch, the directed complex EMA differences remain active because the phase-source differences stabilize rather than vanish. The EMA memory tracks these stabilized sources with a

small but measurable forcing correction. The persistent complex separation becomes asymptotically tangential, while the radial observable obeys an EMA-rate contraction generated by a precise cancellation between tracking-error and mean/tangential alignment contributions. Exact downstream maps transfer this radial observable mode into the resonance, weight, and operator-defect variables.

That version keeps all the strong results and avoids claiming more than the replay demonstrates.

The final corrected synthesis is the defensible one. Two earlier interpretations in the quoted material are contradicted by the diagnostics:

source forcing is about 5%5% of homogeneous tracking decay, not 10−510−5 of it;
the alignment residual is strongly cancellation dominated, not cancellation free.

The resonant-forcing hypothesis is now the most useful next mathematical test.
1. Exact resonant solution

The tracking error satisfies
$E_{n+1}=\alpha E_n+S_n, S_n=\Delta z_{n+1}-\Delta z_n.$
$E_{n+1}=\alpha E_n+S_n, S_n=\Delta z_{n+1}-\Delta z_n.$

Suppose
$S_n=B\alpha^n+1.$
$S_n=B\alpha^n+1.$

Then
$E_n=(A+Bn)\alpha^n.$
$E_n=(A+Bn)\alpha^n.$

Indeed,
$E_{n+1}-\alpha E_n=(A+B(n+1))\alpha^{n+1}-(A+Bn)\alpha^{n+1}=B\alpha^{n+1}.$
$E_{n+1}-\alpha E_n=(A+B(n+1))\alpha^{n+1}-(A+Bn)\alpha^{n+1}=B\alpha^{n+1}.$

This gives a parameter-free relationship between:

the slope of $E_n/\alpha^n E_n/\alpha^n$;
the scaled source increment $S_n/\alpha^{n+1} S_n/\alpha^{n+1}$.

2. Strongest direct test

For forward and reverse channels separately, define
$Y_n = E_n \alpha_n$,
$Y_n = \alpha_n E_n$,

and
$Q_n = S_n \alpha_{n+1}$.
$Q_n = \alpha_{n+1} S_n$.

Fit the complex affine model
$Y_n = A + Bn$.
$Y_n = A + Bn$.

Then test:
$Q_n \to B$.
$Q_n \to B$.

This is stronger than fitting norms because it preserves:

complex phase;
sign;
orientation;
the exact coefficient relation implied by the recurrence.

Required outputs:

complex A, BA, B;
real/imaginary regression residuals;
$R2R2$ or normalized complex residual;
median and maximum
$|Q_n - B|/|B|$;
$|Q_n - B|/|B|$;
window stability of BB;
forward/reverse comparison.

3. Source-limit prediction

If

Sn=Bαn+1,
Sn=Bαn+1,

then the phase-source difference has limit
Δz∞=
Δz∞=

\Delta z_n
+
\sum_{k=n}^{\infty}B\alpha^{k+1}.
]

Therefore
\boxed{ \Delta z_\infty =

\Delta z_n
+
\frac{B\alpha^{n+1}}{1-\alpha}.
}
]

This gives a direct estimator of the limiting source difference at every late time.

If the resonant model is correct, these independently estimated limits should collapse onto one complex value.

That is a stronger test of source stabilization than observing small temporal increments.
4. Tracking ratio prediction

Under the resonant model,
|Sn|(1−α)|En|=
(1−α)|En||Sn|=

\frac{
|B|\alpha
}{
(1-\alpha)|A+Bn|
}.

]

Asymptotically this behaves like
$\sim \alpha(1-\alpha)n$
$\sim (1-\alpha)n\alpha$

when BnBn dominates.

Thus the observed decline
$0.155 \to 0.102 \to 0.080 \to 0.065 \to 0.053$
$0.155 \to 0.102 \to 0.080 \to 0.065 \to 0.053$

is qualitatively consistent with the 1/n1/n prediction.

A direct test is to examine
$n|Sn|(1-\alpha)|En|$.
$n(1-\alpha)|En||Sn|$.

It should approach a constant if the resonant asymptotic form holds.
5. Alignment cancellation needs its own asymptotic test

The exact residual is
$Ln=T1,n+T2,n$.
$Ln=T1,n+T2,n$.

The data show:
$|T1| \approx |T2|, T1 \approx -T2$.
$|T1| \approx |T2|, T1 \approx -T2$.

The next test should fit the complex/scalar components separately:
$T1,n\alpha n, T2,n\alpha n$.
$\alpha n T1,n, \alpha n T2,n$.

Possible outcomes:
Opposite limiting coefficients
$T1,n \sim C\alpha n, T2,n \sim -C\alpha n$.
$T1,n \sim C\alpha n, T2,n \sim -C\alpha n$.

Then cancellation is the leading asymptotic mechanism.
Both decay faster than αnαn

Then independent disappearance becomes relevant.
Polynomially modified terms
T1,n∼(A1+B1n)αn,
T1,n∼(A1+B1n)αn,
T2,n∼(A2+B2n)αn,
T2,n∼(A2+B2n)αn,

with
B1+B2≈0.
B1+B2≈0.

Then the cancellation is itself resonant.

The essential statistic is:
T1/T2→−1.
T1/T2→−1.
6. Angular variables

For each directed comparison, write
Ca=raeiψa,Cb=rbeiψb.
Ca=raeiψa,Cb=rbeiψb.

Define the unwrapped memory phase separation
ΔψC=unwrap(ψa−ψb),
ΔψC=unwrap(ψa−ψb),

and source phase separation
Δψz=
Δψz=

\operatorname{unwrap}(\arg z_a-\arg z_b).
]

Track:
ΔψC,n+1−ΔψC,n,

$\Delta\psi_{C,n+1} - \Delta\psi_{C,n}$,
$\Delta\psi_{z,n+1} - \Delta\psi_{z,n}$,
$\Delta\psi_{z,n+1} - \Delta\psi_{z,n}$,

and
$\Delta\psi_z - \Delta\psi_C$.
$\Delta\psi_z - \Delta\psi_C$.

The expected late signature is:
$\Delta\psi_C \to \Delta\psi_\infty$,
$\Delta\psi_C \to \Delta\psi_\infty$,
$\Delta\psi_z - \Delta\psi_C \to 0$,
$\Delta\psi_z - \Delta\psi_C \to 0$,

while
$r_a - r_b \sim \alpha_n$.
$r_a - r_b \sim \alpha_n$.

This directly tests the phrase:

phase separation freezes while radius difference contracts.

7. Important terminology

The data support:

a persistent tangential complex separation;
a contracting radial observable;
replay-specific asymptotic locking and cancellation.

They do not yet establish:

an invariant manifold;
a tangent bundle of the nonlinear attractor;
universality beyond this branch.

The safe language is:

a replay-specific asymptotic decomposition into persistent tangential and contracting radial components.

8. Minimal next deliverable

The next script should do only three things:

fit
$E_n/\alpha_n = A + Bn$
$E_n/\alpha_n = A + Bn$

and verify
$S_n/\alpha_{n+1} = B$;
$S_n/\alpha_{n+1} = B$;

fit T1,T2T1,T2 separately and test
$T1/T2 \to -1$;
$T1/T2 \to -1$;

unwrap memory/source phase separations and test frozen angle plus contracting radius difference.

That would determine whether the remaining upstream mechanism is a resonantly forced EMA tracking mode coupled to an asymptotically cancelling radial geometry.

Please construct the next deliverable.
WriteAIEF_1_4_resonant_tracking_angular_analysis.py708 lines
AIEF_1_4_resonant_tracking_angular_analysis.pyPYTHON
#!/usr/bin/env python3
"""Test resonant EMA tracking, alignment cancellation, and angular freezing.

For each directed K3 edge comparison, define

E_n = DeltaZ_n - DeltaC_n,
S_n = DeltaZ_{n+1} - DeltaZ_n.

The exact recurrence is E_{n+1}=alpha E_n+S_n. If S_n=B alpha^(n+1), then

E_n=(A+B n) alpha^n.

This script fits that complex affine law, compares its slope B with
S_n/alpha^(n+1), estimates the limiting phase-source difference, tests whether
T1/T2 -> -1 in the alignment decomposition, and unwraps memory/source angular
separations while tracking radius-difference contraction.
"""

from __future__ import annotations

import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_stationarity_perturbation_analysis_updated as base
import AIEF_1_4_component_controller_audit as component_audit
import AIEF_1_4_augmented_tangent_cocycle_analysis as augmented

DEFAULT_RUN_HASH = "5a9f093a2fe0"
DEFAULT_NPZ = (
f"runs/{DEFAULT_RUN_HASH}/AIEF_1_4_{DEFAULT_RUN_HASH}_branch_diagnostics.npz"
)
DEFAULT_MANIFEST = f"runs/{DEFAULT_RUN_HASH}/manifest.json"
DEFAULT_OUTPUT_DIR = "aief_1_4_resonant_tracking_angular"
DEFAULT_SEED = "4619122171158768524"
DEFAULT_DT = 0.01
DEFAULT_SCALE_EPSILON = 1e-12
DEFAULT_INTERVALS = "100,150,200,250,320"
DEFAULT_SIGNAL_FLOOR = 1e-15
ZERO_TOL = 1e-300

```
def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()



def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(value < 0.0 for value in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
return sorted(set(values))



def fit_complex_affine(
indices: np.ndarray,
values: np.ndarray,
) -> dict[str, Any]:
"""Fit complex Y=A+B*n using centered real design and complex targets."""
indices = np.asarray(indices, dtype=float)
values = np.asarray(values, dtype=complex)
center = float(np.mean(indices))
design = np.column_stack([np.ones(len(indices)), indices - center])
coefficients, *_ = np.linalg.lstsq(design, values, rcond=None)
A_center, B = coefficients
A = A_center - B * center
prediction = A + B * indices
residual_norm = float(np.linalg.norm(values - prediction))
total_norm = float(np.linalg.norm(values - np.mean(values)))
return {
"A": complex(A),
"B": complex(B),
"prediction": prediction,
"relative_RMSE": residual_norm / max(float(np.linalg.norm(values)), ZERO_TOL),
```

```
"R_squared": 1.0 - residual_norm**2 / max(total_norm**2, ZERO_TOL),
}



def fit_real_affine(
indices: np.ndarray,
values: np.ndarray,
) -> dict[str, Any]:
result = fit_complex_affine(indices, np.asarray(values, dtype=float).astype(complex))
return {
"A": float(np.real(result["A"])),
"B": float(np.real(result["B"])),
"prediction": np.real(result["prediction"]),
"relative_RMSE": result["relative_RMSE"],
"R_squared": result["R_squared"],
}



def fit_log_rate(times: np.ndarray, values: np.ndarray, floor: float) -> dict[str, float]:
magnitudes = np.abs(np.asarray(values, dtype=float))
valid = np.isfinite(magnitudes) & (magnitudes > floor)
if np.sum(valid) < 3:
return {"rate": np.nan, "R_squared": np.nan, "count": int(np.sum(valid))}
x = np.asarray(times, dtype=float)[valid]
y = np.log(magnitudes[valid])
slope, intercept = np.polyfit(x, y, 1)
prediction = slope * x + intercept
residual = float(np.sum((y - prediction) ** 2))
total = float(np.sum((y - np.mean(y)) ** 2))
return {
"rate": float(-slope),
"R_squared": float(1.0 - residual / total) if total > 0.0 else np.nan,
"count": int(len(x)),
}



def channel_quantities(
C_a: complex,
C_b: complex,
Z_a: complex,
```

```
Z_b: complex,
) -> dict[str, Any]:
Cbar = 0.5 * (C_a + C_b)
Zbar = 0.5 * (Z_a + Z_b)
DeltaC = C_a - C_b
DeltaZ = Z_a - Z_b
E = DeltaZ - DeltaC
DeltaM = float(abs(C_a) ** 2 - abs(C_b) ** 2)
DeltaA = float(
np.real(np.conj(C_a) * Z_a)
- np.real(np.conj(C_b) * Z_b)
)
T1 = float(2.0 * np.real(np.conj(Cbar) * E))
T2 = float(2.0 * np.real(np.conj(DeltaC) * Zbar))
return {
"Cbar": Cbar,
"Zbar": Zbar,
"DeltaC": DeltaC,
"DeltaZ": DeltaZ,
"E": E,
"DeltaM": DeltaM,
"DeltaA": DeltaA,
"T1": T1,
"T2": T2,
"alignment_residual": 2.0 * DeltaA - DeltaM,
"radius_difference": float(abs(C_a) - abs(C_b)),
"memory_phase_raw": float(np.angle(C_a * np.conj(C_b))),
"source_phase_raw": float(np.angle(Z_a * np.conj(Z_b))),
}



def make_report(
resonant_frame: pd.DataFrame,
alignment_frame: pd.DataFrame,
angular_frame: pd.DataFrame,
*,
alpha: float,
ema_rate: float,
final_replay_exact: bool,
) -> str:
lines = [
"AIEF resonant tracking and angular-freezing analysis",
```

```
"==================================================",
"",
f"alpha_EMA: {alpha:.12g}",
f"EMA rate: {ema_rate:.9g}",
f"Final replay exact: {final_replay_exact}",
"",
"Resonant tracking fits",
"----------------------",
]
for _, row in resonant_frame.iterrows():
lines.append(
f" channel={row['channel']}, interval=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"Y_affine_R2={row['Y_affine_R_squared']:.6g}, "
f"Y_RMSE={row['Y_affine_relative_RMSE']:.3g}, "
f"B={row['B_real']:.6g}+{row['B_imag']:.6g}i, "
f"Q_match={row['Q_minus_B_relative']:.3g}, "
f"limit_spread={row['source_limit_relative_spread']:.3g}, "
f"median_n_forcing_ratio={row['median_n_forcing_ratio']:.6g}"
)

lines.extend(["", "Alignment-component asymptotics", "-------------------------------"])
for _, row in alignment_frame.iterrows():
lines.append(
f" channel={row['channel']}, interval=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"median_T1/T2={row['median_T1_over_T2']:.9g}, "
f"p95_|ratio+1|={row['p95_ratio_plus_one_abs']:.3g}, "
f"T1_rate={row['T1_rate']:.6g}, T2_rate={row['T2_rate']:.6g}, "
f"scaled_affine_R2_T1={row['T1_scaled_affine_R_squared']:.6g}, "
f"scaled_affine_R2_T2={row['T2_scaled_affine_R_squared']:.6g}"
)

lines.extend(["", "Angular and radial freezing", "--------------------------"])
for _, row in angular_frame.iterrows():
lines.append(
f" channel={row['channel']}, interval=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"memory_phase_increment={row['median_abs_memory_phase_increment']:.3g}, "
f"source_phase_increment={row['median_abs_source_phase_increment']:.3g}, "
f"source-memory_phase_gap={row['median_abs_phase_gap']:.3g}, "
f"radius_rate={row['radius_difference_rate']:.6g}, "
f"radius_R2={row['radius_difference_R_squared']:.6g}"
)
```

```
lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* The resonant form is tested in complex coordinates; norm-only fits are insufficient.",
"* A good affine Y=E/alpha^n fit and Q=S/alpha^(n+1) match support resonant forcing but remain finite-horizon evidence.",
"* T1/T2 -> -1 establishes cancellation; it does not imply either term is independently negligible.",
"* Phase separations are unwrapped replay observables, not invariant-manifold proofs.",
"* Results remain conditional on one replay and fixed branch.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--npz", default=DEFAULT_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--seed", default=DEFAULT_SEED)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument("--scale-epsilon", type=float, default=DEFAULT_SCALE_EPSILON)
parser.add_argument(
"--intervals",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_INTERVALS),
)
parser.add_argument("--signal-floor", type=float, default=DEFAULT_SIGNAL_FLOOR)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.npz, args.manifest):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(args.dt, args.scale_epsilon, args.signal_floor) <= 0.0:
raise ValueError("dt and tolerances must be positive")
```

```
if args.scale_epsilon != 1e-12:
raise ValueError("implemented actual_step uses fixed scale epsilon=1e-12")
output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)


with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
model = manifest["model"]
dt = float(model["dt"])
if abs(dt - args.dt) > 1e-15:
raise ValueError("command dt does not match manifest dt")
alpha = math.exp(-dt / float(model["T_window"]))
ema_rate = 1.0 / float(model["T_window"])
capacity = (float(model["a"]) - float(model["delta"])) / float(model["q"])
capacities = np.full(int(model["N"]), capacity)
A, edges = base.incidence_matrix(int(model["N"]))
future_steps = int(manifest["protocol"]["future_steps"])


with np.load(args.npz, allow_pickle=False) as data:
state = base.load_split_state(data, args.seed)
expected_final = np.asarray(
data[base.branch_key("continue", args.seed, "B_final")], dtype=float
).copy()


directed = {
"forward": ((2, 7), (7, 11)),
"reverse": ((7, 2), (11, 7)),
}
channel_data: dict[str, dict[str, np.ndarray]] = {}
raw_lists: dict[str, dict[str, list[Any]]] = {
channel: {
"DeltaC": [],
"DeltaZ": [],
"E": [],
"T1": [],
"T2": [],
"alignment_residual": [],
"radius_difference": [],
"memory_phase_raw": [],
"source_phase_raw": [],
}
```

```
for channel in directed
}


for step in range(future_steps + 1):
intermediate = augmented.base_intermediates(
state,
model,
capacities,
A,
edges,
args.scale_epsilon,
)
for channel, (index_a, index_b) in directed.items():
values = channel_quantities(
state.coherence_acc[index_a],
state.coherence_acc[index_b],
intermediate["phase_factor"][index_a],
intermediate["phase_factor"][index_b],
)
for key in raw_lists[channel]:
raw_lists[channel][key].append(values[key])
if step < future_steps:
augmented.advance_base_state(state, intermediate)


final_replay_exact = bool(np.array_equal(state.B, expected_final))
alpha_powers = alpha ** np.arange(future_steps + 1)


for channel, values in raw_lists.items():
data_dict: dict[str, np.ndarray] = {}
for key, sequence in values.items():
dtype = complex if key in {"DeltaC", "DeltaZ", "E"} else float
data_dict[key] = np.asarray(sequence, dtype=dtype)
data_dict["S"] = np.diff(data_dict["DeltaZ"])
data_dict["memory_phase"] = np.unwrap(data_dict["memory_phase_raw"])
data_dict["source_phase"] = np.unwrap(data_dict["source_phase_raw"])
data_dict["phase_gap"] = np.unwrap(
data_dict["source_phase_raw"] - data_dict["memory_phase_raw"]
)
data_dict["memory_phase_increment"] = np.diff(
data_dict["memory_phase"]
)
```

```
data_dict["source_phase_increment"] = np.diff(
data_dict["source_phase"]
)
data_dict["Y"] = data_dict["E"] / alpha_powers
data_dict["Q"] = data_dict["S"] / alpha_powers[1:]
data_dict["T1_scaled"] = data_dict["T1"] / alpha_powers
data_dict["T2_scaled"] = data_dict["T2"] / alpha_powers
channel_data[channel] = data_dict


boundaries = np.asarray(args.intervals, dtype=float)
if boundaries[0] > 0.0:
boundaries = np.concatenate(([0.0], boundaries))
if boundaries[-1] < future_steps * dt:
boundaries = np.concatenate((boundaries, [future_steps * dt]))
boundaries = np.unique(boundaries)


resonant_rows: list[dict[str, Any]] = []
alignment_rows: list[dict[str, Any]] = []
angular_rows: list[dict[str, Any]] = []
trajectory_rows: list[dict[str, Any]] = []


for channel, values in channel_data.items():
for step in range(future_steps + 1):
row = {
"channel": channel,
"step": step,
"time": step * dt,
"DeltaC_real": float(np.real(values["DeltaC"][step])),
"DeltaC_imag": float(np.imag(values["DeltaC"][step])),
"DeltaZ_real": float(np.real(values["DeltaZ"][step])),
"DeltaZ_imag": float(np.imag(values["DeltaZ"][step])),
"E_real": float(np.real(values["E"][step])),
"E_imag": float(np.imag(values["E"][step])),
"E_norm": float(abs(values["E"][step])),
"Y_real": float(np.real(values["Y"][step])),
"Y_imag": float(np.imag(values["Y"][step])),
"T1": float(values["T1"][step]),
"T2": float(values["T2"][step]),
"T1_over_T2": float(values["T1"][step] / values["T2"][step])
if abs(values["T2"][step]) > args.signal_floor
else np.nan,
```

```
"alignment_residual": float(values["alignment_residual"][step]),
"radius_difference": float(values["radius_difference"][step]),
"memory_phase": float(values["memory_phase"][step]),
"source_phase": float(values["source_phase"][step]),
"phase_gap": float(values["phase_gap"][step]),
}
if step < future_steps:
row.update(
{
"S_real": float(np.real(values["S"][step])),
"S_imag": float(np.imag(values["S"][step])),
"S_norm": float(abs(values["S"][step])),
"Q_real": float(np.real(values["Q"][step])),
"Q_imag": float(np.imag(values["Q"][step])),
"memory_phase_increment": float(
values["memory_phase_increment"][step]
),
"source_phase_increment": float(
values["source_phase_increment"][step]
),
"n_forcing_ratio": float(
step
* abs(values["S"][step])
/ max(
(1.0 - alpha) * abs(values["E"][step]),
ZERO_TOL,
)
),
}
)
else:
row.update(
{
"S_real": np.nan,
"S_imag": np.nan,
"S_norm": np.nan,
"Q_real": np.nan,
"Q_imag": np.nan,
"memory_phase_increment": np.nan,
"source_phase_increment": np.nan,
"n_forcing_ratio": np.nan,
}
```

```
)
trajectory_rows.append(row)


for interval in range(len(boundaries) - 1):
start = max(0, int(round(boundaries[interval] / dt)))
end = min(
future_steps + 1,
int(round(boundaries[interval + 1] / dt)),
)
if end <= start + 2:
continue
indices = np.arange(start, end, dtype=float)
Y_fit = fit_complex_affine(indices, values["Y"][start:end])
B = Y_fit["B"]
q_start = start
q_end = min(end - 1, future_steps)
Q_values = values["Q"][q_start:q_end]
if len(Q_values):
Q_minus_B_relative = float(
np.linalg.norm(Q_values - B)
/ max(np.linalg.norm(Q_values), ZERO_TOL)
)
else:
Q_minus_B_relative = np.nan
# Each late sample gives an estimate of the limiting source.
source_limit_estimates = (
values["DeltaZ"][start:end]
+ B
* alpha_powers[start:end]
* alpha
/ (1.0 - alpha)
)
source_limit_mean = np.mean(source_limit_estimates)
source_limit_spread = float(
np.linalg.norm(
source_limit_estimates - source_limit_mean
)
/ max(
math.sqrt(len(source_limit_estimates))
* abs(source_limit_mean),
ZERO_TOL,
)
```

```
)
n_forcing = (
indices[:-1]
* np.abs(values["S"][start : end - 1])
/ np.maximum(
(1.0 - alpha)
* np.abs(values["E"][start : end - 1]),
ZERO_TOL,
)
)
resonant_rows.append(
{
"channel": channel,
"start_time": start * dt,
"end_time": (end - 1) * dt,
"sample_count": int(end - start),
"A_real": float(np.real(Y_fit["A"])),
"A_imag": float(np.imag(Y_fit["A"])),
"B_real": float(np.real(B)),
"B_imag": float(np.imag(B)),
"Y_affine_relative_RMSE": Y_fit["relative_RMSE"],
"Y_affine_R_squared": Y_fit["R_squared"],
"Q_minus_B_relative": Q_minus_B_relative,
"source_limit_real": float(np.real(source_limit_mean)),
"source_limit_imag": float(np.imag(source_limit_mean)),
"source_limit_relative_spread": source_limit_spread,
"median_n_forcing_ratio": float(np.median(n_forcing))
if len(n_forcing)
else np.nan,
}
)


T1 = values["T1"][start:end]
T2 = values["T2"][start:end]
valid_ratio = np.abs(T2) > args.signal_floor
ratios = T1[valid_ratio] / T2[valid_ratio]
T1_fit = fit_log_rate(
np.arange(start, end) * dt,
T1,
args.signal_floor,
)
T2_fit = fit_log_rate(
```

```
np.arange(start, end) * dt,
T2,
args.signal_floor,
)
T1_scaled_fit = fit_real_affine(
indices, values["T1_scaled"][start:end]
)
T2_scaled_fit = fit_real_affine(
indices, values["T2_scaled"][start:end]
)
alignment_rows.append(
{
"channel": channel,
"start_time": start * dt,
"end_time": (end - 1) * dt,
"median_T1_over_T2": float(np.median(ratios))
if len(ratios)
else np.nan,
"p95_ratio_plus_one_abs": float(
np.quantile(np.abs(ratios + 1.0), 0.95)
)
if len(ratios)
else np.nan,
"T1_rate": T1_fit["rate"],
"T1_R_squared": T1_fit["R_squared"],
"T2_rate": T2_fit["rate"],
"T2_R_squared": T2_fit["R_squared"],
"T1_scaled_affine_R_squared": T1_scaled_fit[
"R_squared"
],
"T1_scaled_affine_RMSE": T1_scaled_fit[
"relative_RMSE"
],
"T2_scaled_affine_R_squared": T2_scaled_fit[
"R_squared"
],
"T2_scaled_affine_RMSE": T2_scaled_fit[
"relative_RMSE"
],
}
)
```

```
memory_increment = values["memory_phase_increment"][
start : min(end, future_steps)
]
source_increment = values["source_phase_increment"][
start : min(end, future_steps)
]
radius_fit = fit_log_rate(
np.arange(start, end) * dt,
values["radius_difference"][start:end],
args.signal_floor,
)
angular_rows.append(
{
"channel": channel,
"start_time": start * dt,
"end_time": (end - 1) * dt,
"median_abs_memory_phase_increment": float(
np.median(np.abs(memory_increment))
),
"median_abs_source_phase_increment": float(
np.median(np.abs(source_increment))
),
"median_abs_phase_gap": float(
np.median(np.abs(values["phase_gap"][start:end]))
),
"memory_phase_start": float(
values["memory_phase"][start]
),
"memory_phase_end": float(
values["memory_phase"][end - 1]
),
"source_phase_start": float(
values["source_phase"][start]
),
"source_phase_end": float(
values["source_phase"][end - 1]
),
"radius_difference_rate": radius_fit["rate"],
"radius_difference_R_squared": radius_fit[
"R_squared"
],
}
```

```
)

trajectory = pd.DataFrame(trajectory_rows)
resonant_frame = pd.DataFrame(resonant_rows)
alignment_frame = pd.DataFrame(alignment_rows)
angular_frame = pd.DataFrame(angular_rows)

identity_frame = pd.DataFrame(
[
{
"identity": "tracking_recurrence",
"maximum_absolute_error": float(
trajectory["tracking_recurrence_error"].max()
),
"median_absolute_error": float(
trajectory["tracking_recurrence_error"].median()
),
},
{
"identity": "alignment_decomposition",
"maximum_absolute_error": float(
max(
abs(
trajectory["alignment_residual"]
- trajectory["T1"]
- trajectory["T2"]
)
)
),
"median_absolute_error": float(
np.median(
abs(
trajectory["alignment_residual"]
- trajectory["T1"]
- trajectory["T2"]
)
)
),
},
]
)
```

```
component_audit.atomic_write_csv(
output_dir / "resonant_tracking_trajectory.csv", trajectory
)
component_audit.atomic_write_csv(
output_dir / "resonant_tracking_fit_summary.csv", resonant_frame
)
component_audit.atomic_write_csv(
output_dir / "alignment_component_asymptotics.csv", alignment_frame
)
component_audit.atomic_write_csv(
output_dir / "angular_radius_summary.csv", angular_frame
)
component_audit.atomic_write_csv(
output_dir / "resonant_tracking_identity_checks.csv", identity_frame
)


report = make_report(
resonant_frame,
alignment_frame,
angular_frame,
alpha=alpha,
ema_rate=ema_rate,
final_replay_exact=final_replay_exact,
)
component_audit.atomic_write_text(
output_dir / "resonant_tracking_angular_report.txt", report
)


manifest_output = {
"analysis": "AIEF-1.4 resonant EMA tracking and angular freezing",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"npz": {
"path": str(Path(args.npz).resolve()),
"sha256": file_sha256(args.npz),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
```

```
"sha256": file_sha256(args.manifest),
},
},
"seed": args.seed,
"dt": dt,
"alpha_EMA": alpha,
"ema_rate": ema_rate,
"scale_epsilon": args.scale_epsilon,
"intervals": boundaries.tolist(),
"signal_floor": args.signal_floor,
"final_replay_exact": final_replay_exact,
"outputs": [
"resonant_tracking_trajectory.csv",
"resonant_tracking_fit_summary.csv",
"alignment_component_asymptotics.csv",
"angular_radius_summary.csv",
"resonant_tracking_identity_checks.csv",
"resonant_tracking_angular_report.txt",
],
"interpretation_constraints": [
"Resonant tracking is tested in complex coordinates, not inferred from norm rates alone.",
"T1/T2 cancellation and independent term decay are reported separately.",
"Unwrapped angular convergence is a replay observation, not an invariant-manifold theorem.",
"Results remain conditional on one seed and fixed branch.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest_output, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```
Open

Delivered:

It compiles successfully and passed synthetic checks for:

exact alignment decomposition at 10−1610−16;
exact recovery of a known complex affine law
Yn=A+Bn;
Yn=A+Bn;
complex slope and intercept recovery near floating precision.

Run it

Bash

```
python AIEF_1_4_resonant_tracking_angular_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--dt 0.01 \
--scale-epsilon 1e-12 \
--intervals 100,150,200,250,320 \
--signal-floor 1e-15 \
--output-dir aief_1_4_resonant_tracking_angular
```

1. Complex resonant tracking test

For each directed channel:
En=ΔZn−ΔCn,
En=ΔZn−ΔCn,
Sn=ΔZn+1−ΔZn.
Sn=ΔZn+1−ΔZn.

It computes:
Yn=Enαn,
Yn=αnEn,

and fits the complex affine model
Yn=A+Bn.
Yn=A+Bn.

This preserves complex phase and orientation; it is not a norm-only fit.
2. Exact resonant coefficient check

The model predicts
$S_n = B\alpha^{n+1}$.
$S_n = B\alpha^{n+1}$.

The script therefore computes:
$Q_n = S_n\alpha^{n+1}$,
$Q_n = \alpha^{n+1}S_n$,

and compares Q_nQ_n directly with the fitted complex slope BB.

It reports:

complex AA and BB;
affine-fit $R2R2$;
relative fit RMSE;
relative $Q-BQ-B$ error;
forward/reverse agreement.

3. Phase-source limit

If
$S_n = B\alpha^{n+1}$,
$S_n = B\alpha^{n+1}$,

then
$\Delta Z_\infty =$
$\Delta Z_\infty =$

\Delta Z_n
+
\frac{B\alpha^{n+1}}{1-\alpha}.
]

The script estimates this limit independently from every late timestep and reports its relative spread.

A small spread directly supports convergence of the source difference to a fixed nonzero complex limit.
4. Tracking forcing asymptotic

It computes:
$n|S_n|(1-\alpha)|E_n|$.
$n(1-\alpha)|E_n||S_n|$.

Under
$E_n=(A+Bn)\alpha^n$,
$E_n=(A+Bn)\alpha^n$,

this should approach a constant.

This tests the predicted $1/n$ decay of the source-increment/homogeneous-decay ratio.
5. Alignment cancellation asymptotics

For
$2\Delta A-\Delta M=T_1+T_2$,
$2\Delta A-\Delta M=T_1+T_2$,

the script reports:
T_1/T_2,
T_1/T_2,

and specifically tests:
$T_1/T_2\to-1$.
$T_1/T_2\to-1$.

It also fits:
$|T_1|,|T_2|$
$|T_1|,|T_2|$

separately and tests affine scaled forms:
$T_1\alpha^n, T_2\alpha^n$.
$\alpha^n T_1, \alpha^n T_2$.

This distinguishes:

genuine cancellation of comparable terms;
independent faster decay;
resonant polynomial-times-exponential behavior.

6. Angular freezing

The script unwraps:
Memory phase separation
ΔψC=
ΔψC=

\arg C_a-\arg C_b.
]
Source phase separation
ΔψZ=
ΔψZ=

\arg Z_a-\arg Z_b.
]

It reports:

memory phase increments;
source phase increments;
source-memory phase gap;
start and end phase separations by interval.

This directly tests whether angular separation freezes.
7. Radius contraction

For each directed pair it tracks:
Δrn=|Ca|−|Cb|
Δrn=|Ca|−|Cb|

and fits its decay rate and R2R2.

The intended late signature is:
ΔψC→Δψ∞,

$\Delta\psi_C \to \Delta\psi_\infty$,

while
$\Delta r_n \sim e^{-t/T_{window}}$.
$\Delta r_n \sim e^{-t/T_{window}}$.
Outputs

text

resonant_tracking_trajectory.csv
resonant_tracking_fit_summary.csv
alignment_component_asymptotics.csv
angular_radius_summary.csv
resonant_tracking_identity_checks.csv
resonant_tracking_angular_report.txt
analysis_manifest.json

Decision criteria
Resonant tracking supported

$Y_n = E_n/\alpha_n$ $Y_n = E_n/\alpha_n$ is affine with high R^2 R^2;
$Q_n = S_n/\alpha_{n+1}$ $Q_n = S_n/\alpha_{n+1}$ matches fitted B_B;
independently estimated ΔZ_∞ ΔZ_∞ values collapse;
$n|S|/((1-\alpha)|E|)$ $n|S|/((1-\alpha)|E|)$ stabilizes.

Cancellation law supported
$T_1/T_2 \to -1$
$T_1/T_2 \to -1$

with both terms following stable asymptotic envelopes.
Angular/radial separation supported

phase increments and source-memory phase gap vanish;
radius difference continues to decay at the EMA rate.

Please paste resonant_tracking_angular_report.txt after the run.

I got this error:

File "/home/deck/sci-env/lib/python3.13/site-packages/pandas/core/indexes/base.py", line 3641, in get_loc
return self._engine.get_loc**(casted_key)**
~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^
File "pandas/_libs/index.pyx", line 168, in pandas._libs.index.IndexEngine.get_loc
File "pandas/_libs/index.pyx", line 197, in pandas._libs.index.IndexEngine.get_loc
File "pandas/_libs/hashtable_class_helper.pxi", line 7668, in pandas._libs.hashtable.PyObjectHashTable.get_item
File "pandas/_libs/hashtable_class_helper.pxi", line 7676, in pandas._libs.hashtable.PyObjectHashTable.get_item
KeyError: 'tracking_recurrence_error'

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
File "/home/deck/AI_Programs/SSRH/AIEF_1_4_resonant_tracking_angular_analysis.py", line 707, in <module>
main**()**
~~~~^^
File "/home/deck/AI_Programs/SSRH/AIEF_1_4_resonant_tracking_angular_analysis.py", line 599, in main
trajectory**["tracking_recurrence_error"].max()
~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^**
File "/home/deck/sci-env/lib/python3.13/site-packages/pandas/core/frame.py", line 4378, in getitem
indexer = self.columns.get_loc(key)
File "/home/deck/sci-env/lib/python3.13/site-packages/pandas/core/indexes/base.py", line 3648, in get_loc
raise KeyError(key) from err
KeyError: 'tracking_recurrence_error'
Edit×2
AIEF_1_4_resonant_tracking_angular_analysis.pyPYTHON

Fixed. The script built tracking_recurrence_error conceptually for the final identity report but never added that field to each trajectory row.

I updated to compute:
En+1−
En+1−
~~~~~~~~~~~

\left(
\alpha E_n+S_n
\right)
]

at every nonfinal step and store:

text

tracking_recurrence_error

The final state receives NaN, as expected because there is no subsequent step.

The file compiles successfully and the new column logic was validated. You can rerun the same command; no arguments or output paths need to change.

`think of this:

AIEF resonant tracking and angular-freezing analysis

==========

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

Final replay exact: True

Resonant tracking fits

channel=forward, interval=[0,99.99): Y_affine_R2=0.996969, Y_RMSE=0.0145, B=-5.09897e-05+7.34952e-05i, Q_match=0.207, limit_spread=0.013, median_n_forcing_ratio=873.04

channel=forward, interval=[100,149.99): Y_affine_R2=0.999999, Y_RMSE=7.38e-05, B=-5.23459e-05+7.6136e-05i, Q_match=0.00195, limit_spread=2.89e-07, median_n_forcing_ratio=1271.38

channel=forward, interval=[150,199.99): Y_affine_R2=1, Y_RMSE=1.45e-05, B=-5.25568e-05+7.64467e-05i, Q_match=0.000515, limit_spread=4.52e-09, median_n_forcing_ratio=1393.44

channel=forward, interval=[200,249.99): Y_affine_R2=1, Y_RMSE=1.56e-06, B=-5.25963e-05+7.65055e-05i, Q_match=6.91e-05, limit_spread=3.5e-11, median_n_forcing_ratio=1465.62

channel=forward, interval=[250,319.99): Y_affine_R2=1, Y_RMSE=1.84e-07, B=-5.26012e-05+7.65127e-05i, Q_match=0.000153, limit_spread=2.64e-13, median_n_forcing_ratio=1521.18

channel=reverse, interval=[0,99.99): Y_affine_R2=0.997357, Y_RMSE=0.0132, B=-4.77637e-05+-6.5779e-05i, Q_match=0.202, limit_spread=0.0164, median_n_forcing_ratio=847.645

channel=reverse, interval=[100,149.99): Y_affine_R2=0.999999, Y_RMSE=7e-05, B=-4.76383e-05+-6.82087e-05i, Q_match=0.00187, limit_spread=3.65e-07, median_n_forcing_ratio=1255.61

channel=reverse, interval=[150,199.99): Y_affine_R2=1, Y_RMSE=1.41e-05, B=-4.78149e-05+-6.84847e-05i, Q_match=0.000503, limit_spread=5.84e-09, median_n_forcing_ratio=1382.95

channel=reverse, interval=[200,249.99): Y_affine_R2=1, Y_RMSE=1.52e-06, B=-4.78502e-05+-6.85361e-05i, Q_match=6.77e-05, limit_spread=4.53e-11, median_n_forcing_ratio=1458.35

channel=reverse, interval=[250,319.99): Y_affine_R2=1, Y_RMSE=1.77e-07, B=-4.78546e-05+-6.85424e-05i, Q_match=0.000162, limit_spread=3.32e-13, median_n_forcing_ratio=1516.22

Alignment-component asymptotics

channel=forward, interval=[0,99.99): median_T1/T2=-0.96324402, p95_|ratio+1|=0.283, T1_rate=0.048228, T2_rate=0.0512767, scaled_affine_R2_T1=0.986457, scaled_affine_R2_T2=0.998789

channel=forward, interval=[100,149.99): median_T1/T2=-0.998914496, p95_|ratio+1|=0.00324, T1_rate=0.0531434, T2_rate=0.0532048, scaled_affine_R2_T1=0.999988, scaled_affine_R2_T2=1

channel=forward, interval=[150,199.99): median_T1/T2=-0.999911862, p95_|ratio+1|=0.000276, T1_rate=0.053838, T2_rate=0.0538432, scaled_affine_R2_T1=1, scaled_affine_R2_T2=1

channel=forward, interval=[200,249.99): median_T1/T2=-0.999993372, p95_|ratio+1|=2.14e-05, T1_rate=0.0543064, T2_rate=0.0543068, scaled_affine_R2_T1=1, scaled_affine_R2_T2=1

channel=forward, interval=[250,319.99): median_T1/T2=-0.999999722, p95_|ratio+1|=1.48e-06, T1_rate=0.0547151, T2_rate=0.0547151, scaled_affine_R2_T1=1, scaled_affine_R2_T2=1

channel=reverse, interval=[0,99.99): median_T1/T2=-0.963543625, p95_|ratio+1|=0.285, T1_rate=0.0511048, T2_rate=0.0541839, scaled_affine_R2_T1=0.933562, scaled_affine_R2_T2=0.995354

channel=reverse, interval=[100,149.99): median_T1/T2=-0.998951228, p95_|ratio+1|=0.00313, T1_rate=0.0550504, T2_rate=0.0551097, scaled_affine_R2_T1=0.999932, scaled_affine_R2_T2=1

channel=reverse, interval=[150,199.99): median_T1/T2=-0.999913876, p95_|ratio+1|=0.000268, T1_rate=0.0552897, T2_rate=0.0552948, scaled_affine_R2_T1=1, scaled_affine_R2_T2=1

channel=reverse, interval=[200,249.99): median_T1/T2=-0.999993404, p95_|ratio+1|=2.11e-05, T1_rate=0.0554484, T2_rate=0.0554488, scaled_affine_R2_T1=1, scaled_affine_R2_T2=1

channel=reverse, interval=[250,319.99): median_T1/T2=-0.999999717, p95_|ratio+1|=1.49e-06, T1_rate=0.0556029, T2_rate=0.0556029, scaled_affine_R2_T1=1, scaled_affine_R2_T2=1

Angular and radial freezing

channel=forward, interval=[0,99.99): memory_phase_increment=2.63e-05, source_phase_increment=5.12e-06, source-memory_phase_gap=0.0447, radius_rate=0.052955, radius_R2=0.997865

channel=forward, interval=[100,149.99): memory_phase_increment=6.35e-07, source_phase_increment=6.89e-08, source-memory_phase_gap=0.00111, radius_rate=0.0570006, radius_R2=1

channel=forward, interval=[150,199.99): memory_phase_increment=4.78e-08, source_phase_increment=3.97e-09, source-memory_phase_gap=8.37e-05, radius_rate=0.0571197, radius_R2=1

channel=forward, interval=[200,249.99): memory_phase_increment=3.4e-09, source_phase_increment=2.28e-10, source-memory_phase_gap=5.95e-06, radius_rate=0.0571321, radius_R2=1

channel=forward, interval=[250,319.99): memory_phase_increment=1.36e-10, source_phase_increment=7.41e-12, source-memory_phase_gap=2.38e-07, radius_rate=0.0571333, radius_R2=1

channel=reverse, interval=[0,99.99): memory_phase_increment=2.71e-05, source_phase_increment=4.96e-06, source-memory_phase_gap=0.0465, radius_rate=0.0533057, radius_R2=0.997977

channel=reverse, interval=[100,149.99): memory_phase_increment=6.43e-07, source_phase_increment=6.68e-08, source-memory_phase_gap=0.00113, radius_rate=0.0570409, radius_R2=1

channel=reverse, interval=[150,199.99): memory_phase_increment=4.79e-08, source_phase_increment=3.85e-09, source-memory_phase_gap=8.39e-05, radius_rate=0.0571245, radius_R2=1

channel=reverse, interval=[200,249.99): memory_phase_increment=3.39e-09, source_phase_increment=2.22e-10, source-memory_phase_gap=5.93e-06, radius_rate=0.0571326, radius_R2=1

channel=reverse, interval=[250,319.99): memory_phase_increment=1.35e-10, source_phase_increment=7.19e-12, source-memory_phase_gap=2.36e-07, radius_rate=0.0571333, radius_R2=1

Interpretation constraints

The resonant form is tested in complex coordinates; norm-only fits are insufficient.

A good affine Y=E/alpha^n fit and Q=S/alpha^(n+1) match support resonant forcing but remain finite-horizon evidence.

T1/T2 -> -1 establishes cancellation; it does not imply either term is independently negligible.

Phase separations are unwrapped replay observables, not invariant-manifold proofs.

Results remain conditional on one replay and fixed branch.

This result strongly supports the resonant-forcing model and reveals an asymptotic Jordan-block structure upstream of the radial observable.
1. The complex resonant law passes

The proposed form was
$E_n = (A + Bn)\alpha^n$.
$E_n = (A + Bn)\alpha^n$.

Late, the complex affine fits give:
$R^2 = 1$
$R^2 = 1$

to printed precision, with relative RMSE decreasing to approximately
10^{-7}.
10^{-7}.

The fitted complex slopes stabilize:
Forward
$B_f \to -5.2601 \times 10^{-5} + 7.6513 \times 10^{-5} i$.
$B_f \to -5.2601 \times 10^{-5} + 7.6513 \times 10^{-5} i$.
Reverse
$B_r \to -4.7855 \times 10^{-5} -$
$B_r \to -4.7855 \times 10^{-5} -$

6.8542\times10^{-5}i.
]

These are nonzero complex coefficients, so the polynomial prefactor is materially present rather than an artifact of fitting norms.
2. Source increments have the predicted coefficient

The resonant model predicts
$S_n = \Delta z_{n+1} - \Delta z_n =$
$S_n = \Delta z_{n+1} - \Delta z_n =$

B\alpha^{n+1}.
]

Therefore
Qn=Snαn+1
Qn=αn+1Sn

should approach the same BB obtained from the slope of
Yn=Enαn.
Yn=αnEn.

The late relative mismatch is approximately
10−4
10−4

or smaller, reaching about
7×10−5
7×10−5

over 200200–250250.

The modest increase in the final interval is plausibly a floating-point issue because both SnSn and EnEn are becoming extremely small after rescaling. It should be checked against absolute errors, but it does not challenge the overall agreement.

Thus two independently formed quantities identify the same complex coefficient BB.
3. The source-limit estimates collapse

The model gives
Δz∞=
Δz∞=

\Delta z_n
+
\frac{B\alpha^{n+1}}{1-\alpha}.
]

The relative spread among estimates falls to:
intervalforwardreverse100–1502.9×10−73.7×10−7150–2004.5×10−95.8×10−9200–2503.5×10−114.5×10−11250–3202.6×10−133.3×10−13
interval100–150150–200200–250250–320forward2.9×10−74.5×10−93.5×10−112.6×10−13
reverse3.7×10−75.8×10−94.5×10−113.3×10−13

Therefore:
\boxed{ \Delta z_n =
\Delta z_\infty

\frac{B\alpha^{n+1}}{1-\alpha}
+
o(\alpha^n)
}
]

is extremely well supported on the late replay.

The phase-source difference converges to a fixed nonzero complex limit.
4. A Jordan-block interpretation

The late data support:
Sn+1≈αSn,
Sn+1≈αSn,

and exactly:
En+1=αEn+Sn.
En+1=αEn+Sn.

Thus the asymptotic pair (E,S)(E,S) follows
(En+1Sn+1)≈(α10α)(EnSn).
(En+1Sn+1)≈(α01α)(EnSn).

This is a Jordan block with repeated multiplier αα.

Its solution naturally contains
nαn.
nαn.

So the tracking error's slower finite-window decay is not an unexplained rate discrepancy. It is the expected polynomial correction produced by resonant forcing at the same eigenvalue.

This is arguably the cleanest reduced mathematical object yet found upstream of the operator defect.
5. The nn-scaled forcing ratio supports resonance

The quantity
$n|S_n|(1-\alpha)|E_n|$
$n(1-\alpha)|E_n||S_n|$

moves toward a slowly varying late value around
1500.
1500.

This is consistent with
$|S_n|(1-\alpha)|E_n| \sim \text{constant} n.$
$(1-\alpha)|E_n||S_n| \sim n\text{constant}.$

The increase across finite intervals indicates that the asymptotic constant has not completely stabilized or that the AA term remains relevant, but the qualitative 1/n1/n structure is supported.

A more direct test would fit this quantity against
$c_0+c_1 n$
c_0+nc_1

rather than expecting a perfectly constant finite-horizon value.
6. The alignment law is asymptotic cancellation

The signed ratio converges extremely strongly:
$T_1T_2 \to -1.$
$T_2T_1 \to -1.$

Late:
forward:-0.999999722,reverse:-0.999999717.
forward:reverse:-0.999999722,-0.999999717.

The 95th-percentile deviation from $-1-1$ is approximately

1.5×10−6.
1.5×10−6.

Thus the alignment law is definitively cancellation dominated:
T1=−T2+o(T1).
T1=−T2+o(T1).

The two terms have essentially identical fitted rates in every late interval.

Their scaled affine fits also have
R2=1,
R2=1,

which suggests that each term carries the same resonant polynomial-times-exponential structure, with opposite leading coefficients.
7. Both alignment components are resonant

Late fitted rates are below the pure EMA rate:
Forward
rT1=rT2≈0.05472.
rT1=rT2≈0.05472.
Reverse
rT1=rT2≈0.05560.
rT1=rT2≈0.05560.

Yet
T1/αn
T1/αn

and
T2/αn
T2/αn

are affine with R2=1R2=1.

This is exactly what is expected if
Ti=
Ti=

(A_i+B_i n)\alpha^n.
]

The alignment residual is therefore the small remainder after cancellation of two resonantly forced generalized modes.
8. Angular separation freezes

Late phase increments are tiny:
memory incrementsource incrementforward1.36×10−107.4×10−12reverse1.35×10−107.2×10−12
forwardreversememory increment1.36×10−101.35×10−10source increment7.4×10−127.2×10−12

The source-memory phase gap falls to:
2.4×10−7.
2.4×10−7.

Therefore:
ΔψC→Δψ∞,Δψz−ΔψC→0.
ΔψC→Δψ∞,Δψz−ΔψC→0.

This directly supports the statement:

The directed phase separation freezes while source and memory angular separations lock.

9. Radius difference contracts at the EMA rate

In the same final interval:
rΔr=
rΔr=

0.0571333
]

with
R2=1
R2=1

for both directions.

Thus the polar-coordinate picture is now directly measured:
$\Delta\psi_n \to \Delta\psi_\infty, \Delta r_n \sim e^{-t/T_{window}}$.
$\Delta\psi_n \Delta r_n \to \Delta\psi_\infty, \sim e^{-t/T_{window}}$.

This is more direct than inferring tangentialization from radial fractions alone.
10. The full asymptotic source-memory expansion

The late replay supports:
$\Delta z_n =$
$\Delta z_n =$
$$\Delta z_\infty$$

$$\frac{B\alpha^{n+1}}{1-\alpha}$$
+
$$o(\alpha^n),$$
]

and
$E_n =$
$E_n =$

$$(A+Bn)\alpha^n$$
+
$$o(n\alpha^n).$$
]

Since
$\Delta C_n = \Delta z_n - E_n,$
$\Delta C_n = \Delta z_n - E_n,$

we obtain:
$$\boxed{ \Delta C_n =$$
$$\Delta z_\infty$$

$$\left[$$
$A+Bn+$

\frac{B\alpha}{1-\alpha}
\right]
\alpha^n
+
o(n\alpha^n).
}
]

The complex memory difference converges to the same nonzero limit as the source difference, with a resonant polynomial correction.
11. Observable quotient mechanism

The late complex difference converges to a fixed angular separation. Its radius difference contracts with multiplier αα.

The magnitude observation removes the persistent angular component and retains the contracting radial component:
C↦|C|.
C↦|C|.

Thus the downstream mode is a quotient mode:
persistent complex phase separation⟼contracting radius difference.
persistent complex phase separation⟼contracting radius difference.

The operator defect inherits the latter through exact implementation maps.
12. What is now established on this replay
Exact algebra

tracking-error recurrence;
alignment decomposition;
squared-magnitude recurrence;
all downstream proposal/operator identities.

Strong finite-horizon asymptotics

ΔznΔzn converges to a nonzero complex limit;
source increments follow Bαn+1Bαn+1;
tracking error follows (A+Bn)αn(A+Bn)αn;
alignment terms are opposite resonant modes;

angular separation freezes;
radius difference contracts at the EMA rate.

Remaining upstream question

Why do the delayed phase-source differences themselves acquire the asymptotic form
Δzn=
Δzn=

\Delta z_\infty
+
O(\alpha^n)?
]

That is now the only unresolved link in the replay-specific causal chain.

The next analysis, if pursued, should derive the source increment from the delayed phase update and separate natural-frequency, coupling, and delayed-history contributions. The operator, EMA tracking, magnitude observation, and reduced dynamics are no longer the main unknowns.

Think about this:

This is the point where the analysis changes character. Up to the radial observable analysis, the main question was:

Why does an EMA-rate scalar defect appear when the underlying complex state does not decay?

The resonant tracking result suggests the answer is no longer just "because of projection." There is now a plausible two-stage asymptotic structure:

[
\boxed{
\text{source dynamics}
\rightarrow
\text{Jordan-resonant complex tracking}
\rightarrow
\text{radial quotient}
\rightarrow

\text{operator defect}.
}
]

The important new object is not the operator defect. It is the upstream pair:

[
(E_n,S_n)
]

where

[
E_n=\Delta z_n-\Delta C_n,
]

and

[
S_n=\Delta z_{n+1}-\Delta z_n.
]

The Jordan-block interpretation is the strongest new result

The exact recurrence is:

[
E_{n+1}=\alpha E_n+S_n.
]

The new evidence adds:

[
S_{n+1}\approx\alpha S_n.
]

Therefore:

[
\begin{pmatrix}
E_{n+1}\
S_{n+1}
\end{pmatrix}
\approx
\begin{pmatrix}
\alpha&1\
0&\alpha
\end{pmatrix}
\begin{pmatrix}
E_n\
S_n
\end{pmatrix}.
]

That is the canonical discrete Jordan block.

The consequence is exactly:

[
S_n\sim B\alpha^n,
]

while

[
E_n\sim(A+Bn)\alpha^n.
]

This explains several previously puzzling observations simultaneously:

Why does (E_n) decay slower than the EMA rate fit?

Because:

[

n\alpha^n
]

has an apparent rate slightly smaller than:

[
\alpha^n
]

over finite windows.

Why does the source-memory difference persist?

Because:

[
\Delta z_n

\Delta z_\infty

O(\alpha^n),
]

so the source converges, but does not vanish.

Why does the tracking error vanish?

Because:

[
E_n

\Delta z_n-\Delta C_n.
]

The memory catches the source asymptotically.

This is a very coherent picture.

The matching of the two independent (B) estimates is the key validation

The most important diagnostic is not merely that:

$$
E_n/\alpha^n
$$

looks affine.

That alone can sometimes be misleading.

The stronger result is:

$$
\frac{S_n}{\alpha^{n+1}}
\rightarrow B
$$

and independently:

$$
\frac{E_n}{\alpha^n}
$$

has slope:

$$
B.
$$

Those are two different constructions.

One comes from:

$$
\text{source increments}.
$$

The other comes from:

$$
\text{EMA tracking response}.
$$

Agreement means the recurrence is not merely phenomenologically fitted; the forcing and response are consistent.

That is the closest thing so far to an upstream dynamical identification.

The source limit is the next major simplification

The estimate

$$
\Delta z_\infty

\Delta z_n+
\frac{B\alpha^{n+1}}{1-\alpha}
$$

collapsing across time is important.

It implies:

$$
\boxed{
\Delta z_n

\Delta z_\infty

\frac{B\alpha^{n+1}}{1-\alpha}

o(\alpha^n)
}
]

on this branch.

So the source difference is not wandering indefinitely.

It is approaching a fixed complex point.

The remaining question is no longer:

Why does the EMA state retain memory?

It is:

Why does the delayed phase geometry produce a fixed complex edge-source separation?

That is a much smaller question.

The polar decomposition is now the natural language

The asymptotic source-memory pair appears to be:

[
\Delta z_\infty
\approx
\Delta C_\infty.
]

Write:

[

$$C_e=r_e e^{i\psi_e}.$$

Then late:

$$\Delta\psi$$

$$\psi_{25}-\psi_{59}$$
$$\rightarrow$$
$$\Delta\psi_\infty,$$

while

$$\Delta r$$

$$r_{25}-r_{59}$$
$$\rightarrow 0.$$

So the limiting state has:

$$C_{25,\infty}\neq C_{59,\infty}$$

but:

$$|C_{25,\infty}|\approx |C_{59,\infty}|.$$

This is the precise geometric meaning of "persistent complex separation but contracting observable."

The observation map:

[
C\mapsto |C|
]

projects away the neutral angular direction.

The alignment cancellation becomes a solvable consequence

The exact identity:

[
2\Delta A-\Delta M=T_1+T_2
]

now has a clearer interpretation.

The two pieces are not arbitrary:

[
T_1

2\Re(\bar C^*(\Delta z-\Delta C))
]

is the residual EMA tracking component.

[
T_2

2\Re(\Delta C^*\bar z)
]

is the geometric phase/midpoint component.

The new result:

[
T_1/T_2\rightarrow-1
]

means the alignment law is not caused by independent smallness.

It is a resonance cancellation:

[
\boxed{
T_1

T_2

\text{difference of two matched generalized modes}.
}
]

That is actually more interesting mathematically.

The magnitude recurrence emerges because two nontrivial upstream terms cancel.

A caution: "Jordan block" is a reduced asymptotic object, not yet a property of the full system

The phrase is appropriate for:

[
(E_n,S_n)
]

if the observed recurrence is valid asymptotically.

It is not yet a claim that the full nonlinear AIEF state has a Jordan block.

The precise statement should be:

The audited trajectory exhibits an asymptotic two-variable generalized eigendirection with multiplier (\alpha).

That avoids overclaiming.

A true full-system Jordan structure would require:

linearization of the full delayed nonlinear map;
invariant subspace identification;
perturbation of neighboring trajectories.

Those are different questions.

The remaining upstream problem is now sharply isolated

The whole downstream chain appears closed:

[
\Delta z
\rightarrow
\Delta C
\rightarrow
\Delta r
\rightarrow
\Delta R
\rightarrow
\Delta B
\rightarrow
\delta
\rightarrow
K.
]

The missing link is:

[
\boxed{

\Delta\phi_{25,59,n}
\text{ dynamics}
\rightarrow
\Delta z_n

\Delta z_\infty+O(\alpha^n).
}
]

In other words:

Why does:

[
e^{i\Delta\phi_{25,n}}

e^{i\Delta\phi_{59,n}}
]

approach a fixed complex value?

The next equation to derive is not the EMA equation.

That part is solved.

It is the delayed phase evolution:

[
\Delta\phi_{n+1}

F(\Delta\phi_n,\Delta r_n,\text{history},\text{couplings}).
]

The most valuable next analysis

The next diagnostic should probably not be another recurrence fit.

It should be a phase-source decomposition.

For each directed edge:

[
z_n=e^{i\Delta\phi_n},
]

extract:

[
\Delta\phi_n,
]

then fit:

[
\Delta\phi_n

\Delta\phi_\infty

A_\phi\alpha^n
]

or possibly:

[
\Delta\phi_n

\Delta\phi_\infty

(A_\phi+B_\phi n)\alpha^n.
]

Then inspect the phase update contributions:

[
\Delta\phi_{n+1}-\Delta\phi_n
]

broken into:

intrinsic frequency terms;
coupling terms;
delayed history terms;
resonance terms.

The current evidence says the answer is probably not "EMA smoothing."

The EMA is downstream.

The upstream object appears to be a phase-locked delayed source mode that the EMA faithfully tracks.

The strongest defensible synthesis at this stage is:

[
\boxed{
\text{A persistent complex phase mode feeds a resonant EMA tracking mode.}
}
]

The operator defect is then not a primary relaxation. It is the final observable shadow of that mode after the nonlinear magnitude projection.

The Jordan interpretation is well supported as a replay-specific asymptotic reduction. The remaining phase problem can be made more explicit by parameterizing the two source phases through their mean and separation.
1. Exact source-difference geometry

For two directed phase sources, write
za=eiϕa,zb=eiϕb.
za=eiϕa,zb=eiϕb.

Define:

$\bar{\phi}=\phi a+\phi b2, \Psi=\phi a-\phi b.$
$\bar{\phi}=2\phi a+\phi b, \Psi=\phi a-\phi b.$

Then:
\boxed{ \Delta z =
z_a-z_b

2i e^{i\bar\phi}
\sin\left(\frac{\Psi}{2}\right).
}
]

This factorization separates two geometric degrees of freedom.
Source-separation magnitude
$|\Delta z|=$
$|\Delta z|=$

2\left|
\sin\left(\frac{\Psi}{2}\right)
\right|.
]
Source-difference orientation

Away from sign changes of $\sin(\Psi/2)\sin(\Psi/2)$,
$\arg\Delta z=$
$\arg\Delta z=$

\bar\phi+\frac{\pi}{2}
\quad(\bmod\pi).
]

Thus convergence of the full complex source difference
$\Delta z_n \to \Delta z_\infty \neq 0$
$\Delta z_n \to \Delta z_\infty \quad = 0$

requires stabilization of both:

source phase separation $\Psi_n \Psi_n$;

source mean phase ϕ¯nϕ¯n.

The current angular report measured separation locking, but not the mean source phase. That is the main missing angular observable.
2. First-order source-increment decomposition

For small increments,
Δzn+1−ΔznΔzn≈i Δϕ¯n+12cot(Ψn2)ΔΨn.
ΔznΔzn+1−Δzn≈iΔϕ¯n+21cot(2Ψn)ΔΨn.

Therefore:

Δϕ¯Δϕ¯ rotates the complex source difference;
ΔΨΔΨ changes its magnitude.

This gives a direct interpretation of the fitted complex resonant coefficient BB:

its imaginary component is associated mainly with mean-phase rotation;
its real/radial component is associated mainly with separation relaxation.

The next analysis should compare this phase-based prediction with the measured
Sn=Δzn+1−Δzn.
Sn=Δzn+1−Δzn.
3. Exact delayed-phase increment

The implemented directed phase is
ϕij,n=
ϕij,n=
\theta_{i,n}
h_{j,n}

\frac{\omega_i+\omega_j}{2}\tau,
]

where hj,nhj,n is the oldest delayed-history entry used at step nn.

The frequency correction is constant, so it disappears from the temporal increment:
\boxed{ \phi_{ij,n+1}-\phi_{ij,n} =

\Delta\theta_{i,n}

\Delta h_{j,n}.
}
]

The unwrapped phase update is
Δθi,n=
Δθi,n=

dt,[\omega_i+G_{i,n}],
]

where
Gi,n=
Gi,n=

\sum_k B_{ik,n+1}
\sin(h_{k,n}-\theta_{i,n}).
]

Thus:
\boxed{ \Delta\phi_{ij,n} =
dt(\omega_i+G_{i,n})

\Delta h_{j,n}.
}
]

There is no separate resonance term in the phase equation. Resonance affects this expression indirectly through the adapted projected weights inside GiGi.
4. Forward phase-source difference

For the forward comparison:
a=(2,7),b=(7,11).
a=(2,7),b=(7,11).

The source-separation increment is:

\boxed{ \Delta\Psi_{f,n} =
dt[
(\omega_2-\omega_7)
+
(G_2-G_7)
]

(\Delta h_7-\Delta h_{11}).
}
]

The mean-source-phase increment is:
\boxed{ \Delta\bar\phi_{f,n} =
\frac{dt}{2}
[
\omega_2+\omega_7
+
G_2+G_7
]

\frac12
(
\Delta h_7+\Delta h_{11}
).
}
]
5. Reverse phase-source difference

For the reverse comparison:
a=(7,2),b=(11,7).
a=(7,2),b=(11,7).

Then:
\boxed{ \Delta\Psi_{r,n} =
dt[
(\omega_7-\omega_{11})
+
(G_7-G_{11})
]

(\Delta h_2-\Delta h_7),
}
]

and
\boxed{ \Delta\bar\phi_{r,n} =
\frac{dt}{2}
[
\omega_7+\omega_{11}
+
G_7+G_{11}
]

\frac12
(
\Delta h_2+\Delta h_7
).
}
]

These equations identify the exact upstream balances to inspect.
6. What phase-source stabilization means dynamically

The source separation freezes when:
dt[Δω+ΔG]≈Δhdifference.
dt[Δω+ΔG]≈Δhdifference.

The mean source phase freezes when:
dt[ω⁻+G⁻]≈Δhmean.
dt[ω⁻+G⁻]≈Δhmean.

Thus the source limit is produced by cancellation among:

intrinsic frequency advance;
current projected coupling;
delayed-history shift.

This is a genuine delayed phase-locking condition. It does not require individual oscillator phases to become stationary.

7. Jordan reduction status

The data strongly support the reduced complex system:
$(E_{n+1}S_{n+1}) \approx (\alpha 10 \alpha)(E_n S_n)$.
$(E_{n+1}S_{n+1}) \approx (\alpha 01 \alpha)(E_n S_n)$.

But this is a Jordan block of the measured asymptotic (E,S)(E,S) subsystem, not yet a Jordan block of the full augmented Jacobian.

The distinction should remain explicit.
8. Alignment cancellation

The result
$T1/T2 \to -1$
$T1/T2 \to -1$

is now firmly established.

The next phase decomposition can show why:

T1T1 is controlled by the resonant tracking error EnEn;
T2T2 is controlled by angular separation and mean-source alignment;
both inherit matching polynomial-times-exponential envelopes;
their leading coefficients have opposite signs.

That would turn the cancellation from an empirical relation into a consequence of phase-source geometry.
9. Minimal next phase probe

For forward and reverse channels, record:

text

phi_a
phi_b
mean source phase
source phase separation
their unwrapped increments

intrinsic-frequency contribution
current-coupling contribution
delayed-history contribution

Then verify:

exact decomposition of $\Delta\bar{\phi}_n\Delta\bar{\phi}_n$;
exact decomposition of $\Delta\Psi_n\Delta\Psi_n$;
each large contribution and its cancellation;
decay rate of the net increments;
prediction of S_nS_n from the first-order mean/separation formula;
convergence of $\bar{\phi}_n\bar{\phi}_n$ and $\Psi_n\Psi_n$.

10. Current strongest statement

On the audited K3 branch, the directed source phase differences converge toward fixed nonzero complex separations. Their residual motion provides resonant forcing to the EMA tracking error. The EMA state follows the source with a polynomial-times-exponential correction, while its persistent complex separation becomes tangential to the mean. A cancellation between tracking and tangential alignment terms leaves an EMA-rate radial magnitude mode, which exact downstream maps transfer to the operator defect.

The final unresolved link is the delayed phase-balance equation that makes the source mean and separation increments decay at the EMA rate.

Please construct the next probative deliverable.
WriteAIEF_1_4_phase_balance_source_dynamics.py693 lines
Edit
AIEF_1_4_phase_balance_source_dynamics.pyPYTHON

Delivered:

This probe moves upstream from EMA tracking to the delayed phase dynamics that generate the source $z_e=ei\phi_e$.
$z_e=ei\phi_e$.
Run it

Bash

```
python AIEF_1_4_phase_balance_source_dynamics.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--dt 0.01 \
--scale-epsilon 1e-12 \
--intervals 0,100,150,200,250,320 \
--signal-floor 1e-15 \
--output-dir aief_1_4_phase_balance_source_dynamics
```

Exact directed phase increment

For a directed edge e=(i,j)e=(i,j),
ϕij,n=
ϕij,n=
\theta_{i,n}
h_{j,n}

\frac{\omega_i+\omega_j}{2}\tau.
]

The constant correction disappears from the temporal increment:
\boxed{ \Delta\phi_{ij,n} =
dt(\omega_i+G_{i,n})

\Delta h_{j,n}.
}
]

Here
Gi,n=
Gi,n=

\sum_kB_{ik,n+1}
\sin(h_{k,n}-\theta_{i,n})
]

is the projected coupling used by the implemented phase update.
Forward decomposition

For
(2,7)versus(7,11),
(2,7)versus(7,11),

the separation increment is decomposed as:
ΔΨf=
ΔΨf=
dt(\omega_2-\omega_7)
+
dt(G_2-G_7)

(\Delta h_7-\Delta h_{11}).
]

The mean increment is:
Δϕ¯f=
Δϕ¯f=
\frac{dt}{2}
(\omega_2+\omega_7)
+
\frac{dt}{2}(G_2+G_7)

\frac12(\Delta h_7+\Delta h_{11}).
]
Reverse decomposition

For
(7,2)versus(11,7),
(7,2)versus(11,7),

the corresponding formulas are:
ΔΨr=
ΔΨr=
dt(\omega_7-\omega_{11})
+
dt(G_7-G_{11})

(\Delta h_2-\Delta h_7),
]

and
Δϕ¯r=
Δϕ¯r=
\frac{dt}{2}
(\omega_7+\omega_{11})
+
\frac{dt}{2}(G_7+G_{11})

\frac12(\Delta h_2+\Delta h_7).
]
Exact source-increment check

For each individual edge source:
ze,n+1=
ze,n+1=

z_{e,n}e^{i\Delta\phi_{e,n}}.
]

Therefore the source-difference increment is reconstructed exactly as:
Δzn+1−Δzn=
Δzn+1−Δzn=
z_{a,n}
\left(
e^{i\Delta\phi_{a,n}}-1
\right)

z_{b,n}
\left(
e^{i\Delta\phi_{b,n}}-1
\right).
]

The script audits this against the replayed complex source.
First-order source approximations

It compares the exact source increment with:
i\left(z_a\Delta\phi_a -

z_b\Delta\phi_b
\right),
]

and with the mean/separation approximation:
Δz[iΔϕ¯+12cot(Ψ2)ΔΨ].
Δz[iΔϕ¯+21cot(2Ψ)ΔΨ].

This determines whether the complex resonant coefficient can be explained through:

mean-phase rotation;
source-separation relaxation;
both.

Cancellation diagnostics

For mean and separation increments, the script reports:
|net increment||intrinsic|+|coupling|+|history|.
|intrinsic|+|coupling|+|history||net increment|.

A small ratio means large intrinsic, coupling, and history terms cancel to produce the slowly varying source.

It also reports the individual median magnitudes of all three terms.
Phase freezing

The script unwraps:
ϕa,ϕb,
ϕa,ϕb,

then tracks:
ϕ¯=ϕa+ϕb2,
ϕ¯=2ϕa+ϕb,
Ψ=ϕa−ϕb.

$\Psi = \phi a - \phi b$.

It fits decay rates of:

mean-phase increments;
separation increments;
complex source increments.

It also estimates late limits of $\bar{\phi}\bar{\phi}$ and $\Psi\Psi$ using their scaled increment envelopes and reports the spread of those independently estimated limits.
Outputs

text

phase_balance_trajectory.csv
phase_balance_interval_summary.csv
phase_balance_identity_checks.csv
phase_balance_source_report.txt
analysis_manifest.json

What this should decide

Whether source stabilization comes from cancellation between intrinsic frequencies, coupling, and delayed-history shifts.
Whether mean-phase and separation increments both decay at the EMA rate.
Whether source increments are dominated by mean rotation or separation relaxation.
Whether source mean and separation converge to stable limits.
Whether the complex resonant coefficient BB can be derived from phase-balance terms.

Please paste phase_balance_source_report.txt after the run.

Think about this:

AIEF K3 delayed phase-balance and source-dynamics analysis

==========

alpha_EMA: 0.999428829846

EMA rate: 0.0571333333

Final replay exact: True

Exact source increment identities

source_phase_edge_a_forward: max=1.72554e-15, median=1.38778e-16

source_phase_edge_b_forward: max=1.04738e-15, median=1.57009e-16

source_increment_forward: max=2.24321e-15, median=2.25119e-16

source_phase_edge_a_reverse: max=1.72554e-15, median=1.38778e-16

source_phase_edge_b_reverse: max=1.26585e-15, median=1.11022e-16

source_increment_reverse: max=2.04068e-15, median=1.78082e-16

Interval phase-balance diagnostics

channel=forward, interval=[0,99.99): mean_increment_rate=0.0495056, separation_increment_rate=0.0513457, source_increment_rate=0.0510649, mean_cancellation=0.000119, separation_cancellation=0.025, first_order_error=3.41e-06, mean_sep_error=3.41e-06, mean_limit_spread=0.0477, separation_limit_spread=0.00956

channel=forward, interval=[100,149.99): mean_increment_rate=0.0569768, separation_increment_rate=0.0570034, source_increment_rate=0.0569989, mean_cancellation=1.65e-06, separation_cancellation=0.000344, first_order_error=4.76e-08, mean_sep_error=4.71e-08, mean_limit_spread=1.29e-06, separation_limit_spread=2.47e-07

channel=forward, interval=[150,199.99): mean_increment_rate=0.057098, separation_increment_rate=0.0570989, source_increment_rate=0.0570988, mean_cancellation=9.5e-08, separation_cancellation=1.99e-05, first_order_error=5.1e-08, mean_sep_error=1.65e-08, mean_limit_spread=1.67e-08, separation_limit_spread=3.73e-09

channel=forward, interval=[200,249.99): mean_increment_rate=0.0571288, separation_increment_rate=0.0571287, source_increment_rate=0.0571288, mean_cancellation=5.46e-09, separation_cancellation=1.14e-06, first_order_error=9.03e-07, mean_sep_error=2.62e-07, mean_limit_spread=1.24e-10, separation_limit_spread=2.85e-11

channel=forward, interval=[250,319.99): mean_increment_rate=0.057133, separation_increment_rate=0.057133, source_increment_rate=0.057133, mean_cancellation=1.77e-10, separation_cancellation=3.71e-08, first_order_error=2.7e-05, mean_sep_error=7.82e-06, mean_limit_spread=9.03e-13, separation_limit_spread=2.02e-13

channel=reverse, interval=[0,99.99): mean_increment_rate=0.0533768, separation_increment_rate=0.0514636, source_increment_rate=0.0514904, mean_cancellation=6.96e-05, separation_cancellation=0.000528, first_order_error=1.81e-06, mean_sep_error=1.81e-06, mean_limit_spread=0.00414, separation_limit_spread=0.0133

channel=reverse, interval=[100,149.99): mean_increment_rate=0.0570325, separation_increment_rate=0.0570046, source_increment_rate=0.0570049, mean_cancellation=9.09e-07, separation_cancellation=7.11e-06, first_order_error=2.57e-08, mean_sep_error=2.37e-08, mean_limit_spread=1.04e-07, separation_limit_spread=3.53e-07

channel=reverse, interval=[150,199.99): mean_increment_rate=0.0571002, separation_increment_rate=0.0570996, source_increment_rate=0.0570996, mean_cancellation=5.24e-08, separation_cancellation=4.1e-07, first_order_error=4.54e-08, mean_sep_error=1.69e-08, mean_limit_spread=1.92e-09, separation_limit_spread=5.27e-09

channel=reverse, interval=[200,249.99): mean_increment_rate=0.0571288, separation_increment_rate=0.0571289, source_increment_rate=0.0571289, mean_cancellation=3.01e-09, separation_cancellation=2.36e-08, first_order_error=8.08e-07, mean_sep_error=2.88e-07, mean_limit_spread=1.51e-11, separation_limit_spread=4.01e-11

channel=reverse, interval=[250,319.99): mean_increment_rate=0.057133, separation_increment_rate=0.057133, source_increment_rate=0.057133, mean_cancellation=9.78e-11, separation_cancellation=7.65e-10, first_order_error=2.48e-05, mean_sep_error=9.18e-06, mean_limit_spread=1.14e-13, separation_limit_spread=2.83e-13

Interpretation constraints

Intrinsic, coupling, and delayed-history terms are exact contributions to source-phase increments modulo 2pi.

Small net increments can arise through cancellation of large terms; cancellation indices are reported.

First-order complex source formulas are approximations and are compared with the exact exponential increment.

Converged phase means/separations are replay observations, not invariant-manifold proofs.

Results remain conditional on one seed and fixed branch.

This result closes the replay-specific causal chain from delayed phase dynamics to the operator defect. The phase-source stabilization is produced by increasingly precise cancellation among intrinsic frequency, projected coupling, and delayed-history advance.
1. The phase-source decomposition is exact

All individual source and source-difference identities close at approximately
10−15.
10−15.

Thus, for each directed edge,
Δϕij,n=
Δϕij,n=
dt(\omega_i+G_{i,n})

\Delta h_{j,n}
]

is exactly the phase increment used by the replay, modulo 2π2π.

The source update
ze,n+1=
ze,n+1=

z_{e,n}e^{i\Delta\phi_{e,n}}
]

and the resulting difference increment are reconstructed to floating precision.

2. Mean and separation increments both acquire the EMA rate

From t=100t=100 onward, the mean, separation, and complete complex-source increment rates converge together.

Late:
Forward
rΔϕ¯=0.0571330,rΔΨ=0.0571330,rΔz=0.0571330.
rΔϕ¯rΔΨrΔz=0.0571330,=0.0571330,=0.0571330.
Reverse
rΔϕ¯=0.0571330,rΔΨ=0.0571330,rΔz=0.0571330.
rΔϕ¯rΔΨrΔz=0.0571330,=0.0571330,=0.0571330.

Thus the complex source increment's EMA-rate envelope is not generated solely by mean rotation or solely by separation relaxation. Both phase coordinates converge with the same late rate.
3. Phase-source stabilization is cancellation driven

The cancellation indices are:
Forward, 250250–320320
mean cancellation=1.77×10−10,separation cancellation=3.71×10−8.
mean cancellationseparation cancellation=1.77×10−10,=3.71×10−8.
Reverse, 250250–320320
mean cancellation=9.78×10−11,separation cancellation=7.65×10−10.
mean cancellationseparation cancellation=9.78×10−11,=7.65×10−10.

These ratios compare the net phase increment with the sum of the absolute intrinsic, coupling, and history contributions.

Therefore:
large intrinsic/coupling/history terms cancel to produce the tiny source increments.
large intrinsic/coupling/history terms cancel to produce the tiny source increments.

The source does not stabilize because all phase-update terms independently vanish. It stabilizes because the delayed phase-balance condition becomes extraordinarily accurate.
4. Delayed phase-locking condition

For a directed edge e=(i,j)e=(i,j), source freezing means:
dt(ωi+Gi)≈Δhj.
dt(ωi+Gi)≈Δhj.

If the history of oscillator jj advances asymptotically with effective frequency ΩjΩj, then:
Δhj≈dt Ωj.
Δhj≈dtΩj.

Thus:
ωi+Gi≈Ωj.
ωi+Gi≈Ωj.

This is a delayed frequency-locking condition: the current projected frequency of the source oscillator matches the advance of the delayed target oscillator.

For the K3 channels, both mean and separation balances converge, so the individual directed source phases also freeze:
Δϕa=
Δϕa=

\Delta\bar\phi+\frac12\Delta\Psi
\to0,
]
Δϕb=
Δϕb=

\Delta\bar\phi-\frac12\Delta\Psi
\to0.
]
5. Independent limit estimates collapse

The estimated mean and separation limits have late relative spreads around
10−13.
10−13.

This directly supports:
ϕ¯n→ϕ¯∞,
ϕ¯n→ϕ¯∞,

and
Ψn→Ψ∞.

$\Psi n \to \Psi \infty$.

Therefore:
$za \to ei(\phi^{-}\infty + \Psi\infty/2), zb \to ei(\phi^{-}\infty - \Psi\infty/2)$.
$za \to ei(\phi^{-}\infty + \Psi\infty/2), zb \to ei(\phi^{-}\infty - \Psi\infty/2)$.

The directed source difference converges to a fixed nonzero complex value.
6. The complex resonant coefficient is phase generated

The source increment can be written, to first order, as:
$Sn \approx \Delta zn[i\Delta\phi^{-}n + 12\cot(\Psi n2)\Delta\Psi n]$.
$Sn \approx \Delta zn[i\Delta\phi^{-}n + 21\cot(2\Psi n)\Delta\Psi n]$.

Since both increments have EMA-rate envelopes, the fitted complex coefficient BB contains:

an imaginary/rotational contribution from $\Delta\phi^{-}\Delta\phi^{-}$;
a radial/source-separation contribution from $\Delta\Psi\Delta\Psi$.

This identifies where the real and imaginary parts of the resonant source coefficient originate.
7. Late first-order errors are resolution limited

The relative first-order source error increases late from approximately
$10-8$
$10-8$

to
$10-5$,
$10-5$,

even though the exact exponential identity remains at $10-1510-15$.

This is expected because the source increment itself becomes extremely small. Dividing truncation and floating error by a vanishing source increment inflates the relative error.

The exact complex-exponential reconstruction should remain the primary result. The first-order decomposition is interpretive.
8. Complete replay-specific causal chain

The full mechanism is now:
Delayed phase balance
$dt(\omega i+Gi)-\Delta hj\rightarrow 0$
$dt(\omega i+Gi)-\Delta hj\rightarrow 0$

through cancellation among intrinsic, coupling, and history terms.
Source convergence
$\Delta zn=$
$\Delta zn=$

\Delta z_\infty
+
O(\alpha^n).
]
Resonant EMA tracking
$En+1=$
$En+1=$

\alpha E_n+S_n,
]

with
$Sn\sim B\alpha n+1,$
$Sn\sim B\alpha n+1,$

giving
$En\sim(A+Bn)\alpha n.$
$En\sim(A+Bn)\alpha n.$
Persistent tangential complex separation
$\Delta Cn\rightarrow\Delta z\infty,$
$\Delta Cn\rightarrow\Delta z\infty,$

while its radial fraction tends to zero.
Alignment cancellation
$T1\approx-T2,$
$T1\approx-T2,$

so
$2\Delta A-\Delta M\rightarrow 0.$

$2\Delta A - \Delta M \to 0$.
Directed radial observable
$\Delta M_{n+1} \approx \alpha \Delta M_n$.
$\Delta M_{n+1} \approx \alpha \Delta M_n$.
Downstream exact maps
$\Delta M \to \Delta R \to \Delta B \to \Delta x \to \delta \to K$.
$\Delta M \to \Delta R \to \Delta B \to \Delta x \to \delta \to K$.
9. What remains open

The implementation and replay mechanism are now largely closed. The remaining theoretical issue is why the delayed phase-balance residual itself decays with the EMA rate.

The data show that:

coupling and history terms cancel intrinsic frequency terms;
the cancellation error decays with the same multiplier as the adaptive/EMA variables.

A formal derivation would need to linearize the delayed phase-locking equations jointly with the adapted weights and EMA state.

That is now a theorem problem rather than a missing diagnostic.
10. Recommended next action

The highest-value next step is to stop expanding the diagnostic tree and write a closure/theorem note with three layers:

Exact code identities
phase increment;
EMA tracking;
squared magnitude;
proposal/admissibility.

Replay-established asymptotic laws
delayed phase balance;
resonant source increment;
tangentialization;
alignment cancellation;
EMA-rate radial mode.

Conditional mathematical lemmas
Jordan tracking solution;
radial-observation lemma;
nested-family/biorthogonal lemma;
downstream memory-gap lemma.

The project has reached the point where formal consolidation is more valuable than another local numerical probe.

Can you endeavor to embark on the recommended next Action?
WriteAIEF_1_4_OPERATOR_COCYCLE_CLOSURE_THEOREM_NOTE.md987 lines
AIEF-1.4 operator/cocycle closure and theorem noteMD
AIEF-1.4 operator/cocycle closure and theorem note
Exact identities, replay-established asymptotics, and conditional reductions

Status: working mathematical consolidation
Scope: frozen AIEF-1.4 run 5a9f093a2fe0, with detailed mechanism analysis on exceptional seed 4619122171158768524
Updated: 2026-07-26
1. Purpose and epistemic structure

The AIEF-1.4 analysis has reached a point where further local diagnostics are less valuable than formal consolidation. This note separates the results into three layers:

Exact implementation identities — algebraic consequences of the finite-step code.
Replay-established asymptotic laws — numerical results on one frozen branch and tested perturbation subspace.
Conditional mathematical lemmas — exact conclusions under explicitly stated asymptotic assumptions.

Nothing in this note promotes one replay into a universal theorem about all AIEF trajectories. The seed/graph realization remains the inferential unit.
Part I. State, proposal, and admissibility map
2. Edge coordinates and capacity geometry

Let
$w \in R^{m}, m=(N2)$,
$w \in R^{m}, m=(2N)$,

be the unique undirected edge vector, and let

A∈{0,1}N×m
A∈{0,1}N×m

be the unsigned node-edge incidence matrix. The feasible set is
C=
C=

{w\ge0:Aw\le c}.
]

The current production model has equal capacities
ci=c=a−δq.
ci=c=qa−δ.

For one Euler step, let
z=w+dtF
z=w+dtF

be the raw edge proposal and
x=z+
x=z+

the post-positivity proposal.

Let
q=Ax
q=Ax

be its proposal row sums.
3. Implemented safeguarded admissibility map

The production code uses a global gate. If no row violates capacity, scaling is skipped. If at least one row violates, node scales are
si(q)=
si(q)=

\min\left(1,\frac{c_i}{q_i+\varepsilon}\right),
\qquad

\varepsilon=10^{-12},
]

and edge e=(i,j)e=(i,j) receives multiplier
me(q)=min(si(q),sj(q)).
me(q)=min(si(q),sj(q)).

Define
M(q)=diag(me(q)).
M(q)=diag(me(q)).
Proposition 3.1 — exact factorization

On the nonnegative proposal domain,
Aε(x)=M(Ax)x.
Aε(x)=M(Ax)x.

This follows directly from minimum.outer(scale, scale).
Corollary 3.2 — exact fiber linearity

If x,y≥0x,y≥0 and
Ax=Ay=q,
Ax=Ay=q,

then
\boxed{ \mathcal A_\varepsilon(y)-\mathcal A_\varepsilon(x) =

M(q)(y-x).
}
]

Thus the map is exactly affine-linear on each nonnegative row-sum fiber.

This statement does not require the support of xx and yy to be identical, only nonnegativity and equal row sums.
4. Proposal-space strata and branch Jacobian

The post-positivity proposal space is stratified by:

global gate state;
node clipping inequalities;
endpoint scale comparisons;
positivity/support boundaries inherited from the preceding map.

For two clipped nodes,
si<sj ⟺ ci(qj+ε)<cj(qi+ε),
si<sj⟺ci(qj+ε)<cj(qi+ε),

an affine inequality in xx. Hence strict branch regions are relatively open polyhedral cells in proposal space. On each strict cell the map is smooth rational.

For edge ee with unique winning endpoint k(e)k(e),
ye=xesk(e).
ye=xesk(e).

Since
∂sk∂xf=
∂xf∂sk=

-\frac{s_k}{q_k+\varepsilon}A_{kf},
]

we obtain:
Proposition 4.1 — strict-branch Jacobian
\boxed{ D\mathcal A_\varepsilon(x)_{ef} =
s_{k(e)}\delta_{ef}

\frac{x_es_{k(e)}}{q_{k(e)}+\varepsilon}A_{k(e)f}.
}
]

At endpoint ties, the classical Fréchet derivative generally fails. The directional derivative uses the minimum of the endpoint scale derivatives.
Caveat 4.2 — safeguarded global-gate discontinuity

Because scaling is skipped below the global gate but uses q+εq+ε immediately above it, the fixed-εε production map has a tiny jump of order ε/cε/c at gate onset. Ideal-map directional theorems should concern either ε=0ε=0 or an explicitly specified joint limit.

Part II. Delayed phase, EMA, and augmented state

5. Augmented discrete state

The Markov state of the implemented simulation is

Xn=(θn,Bn,Cn,Hn),

Xn=(θn,Bn,Cn,Hn),

where HnHn is the finite delay-history deque.

The weight vector alone is not a closed state because the driver depends on phase, delayed history, and complex EMA memory.

6. Directed phase source

For directed edge (i,j)(i,j), define

ϕij,n=

ϕij,n=

\theta_{i,n}

-h_{j,n}

\frac{\omega_i+\omega_j}{2}\tau,

]

zij,n=eiϕij,n.

zij,n=eiϕij,n.

The phase correction is constant in time, so the source-phase increment is

\boxed{ \Delta\phi_{ij,n} =

dt(\omega_i+G_{i,n})

\Delta h_{j,n},

}

]

where

Gi,n=

Gi,n=

\sum_k B_{ik,n+1}
\sin(h_{k,n}-\theta_{i,n}).
]

This decomposes exactly into intrinsic-frequency, current-coupling, and delayed-history terms.

For the exceptional K3 comparisons, both source mean and source separation increments were reconstructed to machine precision from these terms.

7. Complex EMA and tracking error

The directed EMA update is
Cn+1=
Cn+1=

\alpha C_n
+
(1-\alpha)z_n.
]

Define tracking error
en=zn−Cn.
en=zn−Cn.

Proposition 7.1 — exact tracking-error recurrence
\boxed{ e_{n+1} =

\alpha e_n
+
(z_{n+1}-z_n).
}
]

For a difference between two directed edges, let
En=Δzn−ΔCn,Sn=Δzn+1−Δzn.
En=Δzn−ΔCn,Sn=Δzn+1−Δzn.

Then
En+1=αEn+Sn.
En+1=αEn+Sn.

Part III. Exact magnitude-observable identities

8. Squared magnitude

Define
$M_n = |C_n|^2, Q_n = \Re(C_n \bar{z}_n)$.
Mn=|Cn|2,Qn=ℜ(Cnzn).

Since |zn|=1|zn|=1,
Proposition 8.1 — exact squared-magnitude update
\boxed{ M_{n+1} =

\alpha^2M_n
+
(1-\alpha)^2
+
2\alpha(1-\alpha)Q_n.
}
]

For two directed edges,
$\Delta M_n = |C_{a,n}|^2 - |C_{b,n}|^2$,
ΔMn=|Ca,n|2−|Cb,n|2,
ΔQn=
ΔQn=
\Re(\overline{C_{a,n}}z_{a,n})

\Re(\overline{C_{b,n}}z_{b,n}),
]

so
\boxed{ \Delta M_{n+1} =

\alpha^2\Delta M_n
+
2\alpha(1-\alpha)\Delta Q_n.
}
]

An effective αα-multiplier is obtained when
$2\Delta Q_n = \Delta M_n + o(\Delta M_n)$.

2ΔQn=ΔMn+o(ΔMn).
9. Exact midpoint geometry

Let
C¯=12(Ca+Cb),ΔC=Ca−Cb,
C¯=21(Ca+Cb),ΔC=Ca−Cb,
z¯=12(za+zb),Δz=za−zb.
z¯=21(za+zb),Δz=za−zb.

Then
\boxed{ \Delta M =

2\Re(\overline{\bar C},\Delta C).
}
]

Also
ΔQ=
ΔQ=

\Re(
\overline{\bar C},\Delta z
+
\overline{\Delta C},\bar z
).
]

Therefore
Proposition 9.1 — exact alignment decomposition
\boxed{ 2\Delta Q-\Delta M =

T_1+T_2,
}
]

with
T1=
T1=

2\Re[
\overline{\bar C}(\Delta z-\Delta C)
],
]
T2=
T2=

2\Re(
\overline{\Delta C},\bar z
).
]

This identity separates tracking error from tangential/mean-source geometry.
10. Magnitude and symmetrization

The directed magnitude difference satisfies exactly
\boxed{ |C_a|-|C_b| =

\frac{|C_a|^2-|C_b|^2}{|C_a|+|C_b|}.
}
]

The physical resonance matrix is symmetrized after magnitude:
ΔR=
ΔR=

\frac12(
\Delta|C|{\rm forward}
+
\Delta|C|{\rm reverse}
).
]

On the audited replay, forward and reverse directed modes reinforce rather than cancel.
Part IV. Exceptional K3 operator algebra
11. Fixed winner branch

The exceptional positive K3 has edge order
(25,29,59)=

(25,29,59)=

((2,7),(2,11),(7,11)),
]

with fixed winners
(25,29,59)↦(7,2,7).
(25,29,59)↦(7,2,7).

Let
c0=1−γdt.
c0=1−γdt.

Define
β25=
β25=
\frac{x_{25}s_7}{q_7+\varepsilon},
\qquad
\beta_{29}
\frac{x_{29}s_2}{q_2+\varepsilon},
\qquad
\beta_{59}

\frac{x_{59}s_7}{q_7+\varepsilon}.
]

Then
Proposition 11.1 — exact selected K3 block
\boxed{ K =

c_0
\begin{pmatrix}
s_7-\beta_{25}&0&-\beta_{25}\
-\beta_{29}&s_2-\beta_{29}&0\
-\beta_{59}&0&s_7-\beta_{59}
\end{pmatrix}.
}
]
12. Nested invariant families

Let
v=13(1,−1,−1),u=32(1,0,−1).
v=3
1(1,−1,−1),u=23
(1,0,−1).
Proposition 12.1 — right invariance
Kv=λv ⟺ s2=s7,
Kv=λv⟺s2=s7,

with
λ=c0s7.
λ=c0s7.
Proposition 12.2 — left invariance

Assuming s2=s7s2=s7,
uTK=λuT ⟺ β25=β59.
uTK=λuT⟺β25=β59.
Realized-support symmetry

On the isolated equal-capacity K3, scale equality gives
x29=x59.
x29=x59.

The left condition gives
x25=x59.
x25=x59.

Thus all three proposal edges and correction coefficients become equal.
13. Analytic dominant defect

Let
δn=β25,n−β59,n.
δn=β25,n−β59,n.

Under negligible right and secondary symmetry mismatches,
Kn=
Kn=

K_{{\rm sym},n}
+
c_0\delta_nE_*,
]

where
\boxed{ E_* =

\begin{pmatrix}
-1&0&-1\
0&0&0\
0&0&0
\end{pmatrix}.
}
]

The defect satisfies
E∗v=0,
E∗v=0,

but
uTE∗∉span(uT).
uTE∗∈/span(uT).

Its projections into the biorthogonal and symmetric quotient spaces explain the observed family-distance constants.
Part V. Exact reduced dynamics
14. Biorthogonal scalar coordinate

If
uTKn=λnuT,
uTKn=λnuT,

and
bn+1=Knbn+fn,
bn+1=Knbn+fn,

then
Proposition 14.1 — exact scalar recurrence

an+1=λnan+gn,
an+1=λnan+gn,

where
an=uTbn,gn=uTfn.
an=uTbn,gn=uTfn.

This recurrence is exact for arbitrary transverse components of bnbn. It explains why the left coordinate repairs the Euclidean direction-0 rollout.
15. Duhamel decomposition

For the weight tangent,
bn+1=Knbn+fn.
bn+1=Knbn+fn.

Define
hn+1=Knhn,rn+1=Knrn+fn,
hn+1=Knhn,rn+1=Knrn+fn,

with h0=b0h0=b0, r0=0r0=0. Then
bn=hn+rn.
bn=hn+rn.

This is an exact block decomposition once KnKn and fnfn are fixed.
16. Conditional triangular memory lemma

Assume late constants or asymptotically stationary sequences satisfy
an+1=λBan+gn,
an+1=λBan+gn,
gn+1=αgn.
gn+1=αgn.

Then a forcing contribution of age kk has relative factor
(λBα)k.
(αλB)k.

Therefore the memory-envelope rate is
\boxed{ \kappa =

-\frac1{dt}
\log\left|\frac{\lambda_B}{\alpha}\right|.
}
]

This is a theorem about the triangular model, not an unconditional AIEF law.
Part VI. Resonant source tracking
17. Jordan-resonant tracking lemma

Let
En+1=αEn+Sn,
En+1=αEn+Sn,

and assume
Sn=Bαn+1.
Sn=Bαn+1.

Then
Proposition 17.1
En=(A+Bn)αn.
En=(A+Bn)αn.

Also
\boxed{ \Delta z_\infty =

\Delta z_n
+
\frac{B\alpha^{n+1}}{1-\alpha}.
}
]

Equivalently, the pair (E,S)(E,S) follows an asymptotic Jordan block with repeated multiplier αα:
(En+1Sn+1)≈(α10α)(EnSn).
(En+1Sn+1)≈(α01α)(EnSn).
Part VII. Replay-established asymptotics

The following are numerical results on the frozen exceptional replay and tested directions, not general theorems.

18. Fixed branch and augmented tangent

The effective K3 winner map has zero changes over 32,001 saved states.
The tolerance event at t=124.05 changes no implemented positive-edge formula.
The frozen-driver K3 cocycle predicts frozen nonlinear perturbations accurately.
The augmented tangent predicts coupled nonlinear perturbations with median K3 error around 5e-4 over resolved horizons.
All tested nonlinear perturbations preserve the audited winner branch.

19. Non-normal transient

Full K3 peak amplification is about 1.27097 at t=0.1.
Split-lineality peak amplification is about 1.20563.
A constant initial matrix reproduces this peak.
Diagonal scale alone does not amplify.
The node-mediated low-rank term creates the transient and severe anisotropy.

20. Rank-one and biorthogonal reduction

Late total and feedback tangent POD first-mode fractions are effectively one.
The geometric right direction is
$v=(1,-1,-1)/3$.
$v=(1,-1,-1)/3$
.
The analytic left covector is
$u=3(1,0,-1)/2$.
$u=3$
$(1,0,-1)/2$.
Fixed late biorthogonal coordinates reduce direction-0 final rollout error from about 0.74 to about 1.9e-5 for a rollout starting at t=250.
A time-varying adjoint covector gives no material improvement.

21. Nested-family and defect mode

Exact fixed-winner sparsity holds.
The right-invariant family is reached far earlier than the biorthogonal/symmetric families.
The analytic projected defect matches numerical defect POD at cosine 1-O(1e-12).

Direct defect reconstruction error is approximately 3e-6 relative.
Dominant defect and family-defect matrices follow the EMA multiplier pointwise to approximately 1e-9 late.

22. Duhamel and memory

Exact Duhamel crossover times are approximately 0.38 and 72.86 for the two tested directions.
Feedback forcing is individually small but strongly aligned with persistent output directions.
Final feedback is reconstructed to 1% using roughly the previous 66.6 time units.
The late operational memory rate is approximately
$\kappa \approx 0.06913$.
$\kappa \approx 0.06913$.
Direct propagation gives
$\kappa \approx r_{prop} - r_g$
$\kappa \approx r_{prop} - r_g$ to about 1e-6 at the final target.

23. Radial observable and complex persistence

Directed complex EMA differences and source differences persist in norm.
Directed squared-magnitude differences independently acquire the EMA multiplier.
Forward and reverse magnitude differences reinforce; symmetrization does not create the mode.
Radial fractions decrease to order 1e-7, while tangential components persist.
Mean source and memory directions align to order 1e-7.
Tracking and tangential alignment terms become equal and opposite to order 1e-6 or better.

24. Resonant source and phase balance

Complex source increments follow the EMA-rate envelope.
$E_n/\alpha_n E_n/\alpha_n$ is complex affine in n_n with late R^2 effectively one.
$S_n/\alpha_{n+1} S_n/\alpha_{n+1}$ matches the affine slope.
Independent estimates of $\Delta z_\infty \Delta z_\infty$ collapse to relative spreads near 1e-13 late.
Source mean and source separation increments both acquire the EMA rate.
Their net values arise from cancellation of intrinsic-frequency, coupling, and delayed-history terms.
Source and memory angular separations freeze, while radius differences decay at the EMA rate.

Part VIII. Current replay-specific mechanism

The strongest current causal chain is:

delayed phase-balance cancellation ⇓ $\Delta z_n = \Delta z_\infty + O(\alpha^n)$ ⇓ $E_n = (A + Bn)\alpha^n$ ⇓ persistent tangential complex separation ⇓ $T_1 \approx -T_2$ ⇓ $\Delta M_{n+1} \approx \alpha \Delta M_n$ ⇓ $\Delta R \to \Delta B \to \Delta x \to \delta \to K$.
delayed phase-balance cancellation ⇓ $\Delta z_n = \Delta z_\infty + O(\alpha^n)$ ⇓ $E_n = (A + Bn)\alpha^n$ ⇓ persistent tangential complex separation ⇓ $T_1 \approx -T_2$ ⇓ $\Delta M_{n+1} \approx \alpha \Delta M_n$ ⇓ $\Delta R \to \Delta B \to \Delta x \to \delta \to K$.

This is a finite-horizon mechanism established on one fixed winner branch. It is not yet a universal AIEF theorem.

Part IX. Claims explicitly not established

The current evidence does not establish:

a universal scalar reduction for all AIEF components;
a universal defect matrix across winner branches;
a full-system Jordan block;
an invariant manifold or toroidal attractor;
causal control by EMA solely from rate matching;
a closed-loop gain equal to $\lambda B/\alpha\lambda B/\alpha$;
a valid future-only T_window sweep from the saved split state;
timestep/safeguard convergence of the finite map;
generality beyond the frozen seed and tested directions.

Part X. Theorem and formalization frontier

25. Proof-ready implementation results

The following can be formalized directly from the code:

admissibility factorization;
exact row-sum fiber linearity;
branchwise Jacobian;
fixed-winner K3 sparse form;
right-invariant and biorthogonal family characterizations;
analytic left/right pair;
explicit raw and projected defect matrices;
squared-magnitude identity;
tracking-error recurrence;
alignment-residual decomposition;
directed phase-increment decomposition.

26. Conditional asymptotic lemmas

The following require stated assumptions:

defect perturbation bound;
pointwise EMA defect recurrence;
Jordan-resonant tracking solution;
radial quotient alpha-mode;
triangular memory-gap law;
convergence of the numerical left mode;
source mean/separation limit.

27. Remaining upstream mathematical question

The remaining unexplained replay law is why the delayed phase-balance residual itself acquires the EMA-rate envelope. A proof would require coupling the phase-locking equations to the adapted weights and EMA state rather than treating source stabilization as an observed input.
Part XI. Recommended next actions

Convert Parts I–VI into a formal proposition/lemma manuscript with explicit assumptions and proofs.
Archive a machine-readable analysis lineage with hashes, estimands, and superseded claims.
Test geometric/dual invariant subspaces on existing ordinary K4, K5, and K6 components.
Complete the matched-phase B_carrier_only/full_reset correction.
Only then run separately versioned, fully regenerated T_window, timestep, and safeguard studies.
Defer full-attractor topology and new broad cohorts.

Part XII. Reproducibility anchors

Principal analysis sources:

text

AIEF_1_4_actual_admissibility_branch_audit.py
AIEF_1_4_frozen_driver_cocycle_analysis.py
AIEF_1_4_augmented_tangent_cocycle_analysis.py
AIEF_1_4_augmented_duhamel_feedback_analysis.py
AIEF_1_4_feedback_rank_memory_reduction.py
AIEF_1_4_scalar_spectral_gap_reduction.py
AIEF_1_4_propagation_gap_geometric_control.py

AIEF_1_4_parameter_free_triangular_closure.py
AIEF_1_4_biorthogonal_triangular_closure.py
AIEF_1_4_nested_operator_family_audit.py
AIEF_1_4_defect_recurrence_analytic_mode.py
AIEF_1_4_defect_causal_ema_trace.py
AIEF_1_4_radial_observable_subsystem_analysis.py
AIEF_1_4_phase_source_locking_analysis.py
AIEF_1_4_resonant_tracking_angular_analysis.py
AIEF_1_4_phase_balance_source_dynamics.py

Administrative sources of truth:

text

AIEF_HISTORIC_UNFOLDING.md
AIEF_1_4_OPERATOR_COCYCLE_THOUGHT_TREE.md
AIEF_1_4_FORWARD_PLAN.md
AI_AGENT_HANDOFF_PROMPT.md

Closing statement

The project began with network-level falsification and memory ablation. Its current mathematical center is an implementation-derived operator hierarchy coupled to a replay-validated augmented tangent mechanism.

The strongest current result is not merely that a scalar defect decays at the EMA rate. It is that, on one audited branch, a delayed phase-balance mode generates a persistent complex separation whose radial observable follows an EMA-rate quotient dynamics, which is then transferred exactly into the adaptive weight and operator-defect sectors.

The next progress should come through proof, consolidation, and carefully scoped generalization—not another unconstrained expansion of simulations.
Open

Created:

The note consolidates the project into:

exact implementation propositions;
replay-established asymptotic laws;
conditional mathematical lemmas;
explicit non-results and scope boundaries;
a proof/formalization frontier.

It includes formal statements for:
$A\varepsilon(x)=M(Ax)x$,
$A\varepsilon(x)=M(Ax)x$,

exact fiber linearity, branchwise Jacobians, delayed phase increments, EMA tracking, squared-magnitude dynamics, alignment decomposition, fixed-winner K3 structure, nested invariant families, analytic left/right modes, Duhamel decomposition, triangular memory law, and Jordan-resonant tracking.

It also records the complete replay-specific causal chain:
delayed phase balance → source convergence → resonant EMA tracking → tangential complex separation → radial observable mode → operator defect.
delayed phase balance → source convergence → resonant EMA tracking → tangential complex separation → radial observable mode → operator defect.

The document explicitly separates proof-ready code identities from finite-horizon evidence and unproved generalizations.

think about this:

Jordan-resonant tracking lemma
Statement

Let $\alpha\in(0,1)$ and let the scalar (or complex) sequences E_n, S_n satisfy the exact recurrence

$$E_{n+1}=\alpha E_n+S_n.$$

]

Assume the source increment is a pure geometric sequence of the same ratio,

$$S_n=B\alpha^{n+1}$$

]

for some constant BB (independent of nn). Then there exists a constant AA such that

E_n=(A+Bn)\alpha^n

]

for every n≥0n≥0. Consequently the pair admits the exact matrix representation

\begin{pmatrix}E_{n+1}\S_{n+1}\end{pmatrix}

=

\begin{pmatrix}\alpha&1\0&\alpha\end{pmatrix}

\begin{pmatrix}E_n\S_n\end{pmatrix}

]

whenever Sn=Bαn+1Sn=Bαn+1 (i.e., on the invariant ray of the Jordan block). Moreover the limiting source difference is

\Delta z_\infty

=

\Delta z_n+\frac{B\alpha^{n+1}}{1-\alpha}.

]
Proof

Step 1 – homogeneous solution.

The homogeneous recurrence En+1=αEnEn+1=αEn has the general solution

E_n^{(h)}=A\alpha^n.

]

Step 2 – particular solution.

Because the forcing Sn=Bαn+1Sn=Bαn+1 resonates with the homogeneous multiplier, a particular solution of the form

E_n^{(p)}=Bn\alpha^n

]

is required. Substitute into the recurrence:

\begin{align*}

E_{n+1}^{(p)}

&=B(n+1)\alpha^{n+1}

=B n\alpha^{n+1}+B\alpha^{n+1},\

\alpha E_n^{(p)}+S_n

&=\alpha(Bn\alpha^n)+B\alpha^{n+1}

=Bn\alpha^{n+1}+B\alpha^{n+1}.

\end{align*}

]

The two sides agree, so En(p)=BnαnEn(p)=Bnαn is indeed a particular solution.

Step 3 – general solution.

By linearity

E_n=E_n^{(h)}+E_n^{(p)}=(A+Bn)\alpha^n.

]

(The constant AA is fixed by any single initial condition, e.g. A=E0A=E0.)

Step 4 – Jordan-block form.

Under the standing assumption Sn=Bαn+1Sn=Bαn+1 one has

S_{n+1}=B\alpha^{n+2}=\alpha S_n.

]

Therefore

\begin{align*}

E_{n+1}

&=\alpha E_n+S_n,\

S_{n+1}

&=\alpha S_n,

\end{align*}

]

which is precisely

\begin{pmatrix}E_{n+1}\S_{n+1}\end{pmatrix}

=

\begin{pmatrix}\alpha&1\0&\alpha\end{pmatrix}

\begin{pmatrix}E_n\S_n\end{pmatrix}.

]

The matrix is a single Jordan block of size 2 with eigenvalue αα.

Step 5 – limiting source difference.

From the definition of the tracking error and the geometric form of SS,

\Delta z_{n+1}-\Delta z_n=S_n=B\alpha^{n+1}.

]

Summing the telescoping series from nn to infinity yields

\Delta z_\infty-\Delta z_n

=

\sum_{k=0}^\infty S_{n+k}

=

\sum_{k=0}^\infty B\alpha^{n+1+k}

=

B\alpha^{n+1}\sum_{k=0}^\infty\alpha^k

=

\frac{B\alpha^{n+1}}{1-\alpha},

]

which is the claimed identity.
Remarks

The derivation is purely algebraic and holds for any field of characteristic zero (in particular for CC).

Resonance is essential: if the source ratio differed from αα, a particular solution of the form CρnCρn with ρ≠αρ =α would suffice and no linear-in-nn factor would appear.

The lemma is an exact statement about the triangular system (E,S)(E,S). Its application to the AIEF exceptional trajectory remains conditional on the numerically observed asymptotic Sn∼Bαn+1Sn ∼Bαn+1.

The lemma and proof are correct. Two wording corrections and one asymptotic corollary would make it better suited to the AIEF application.
1. Remove “on the invariant ray”

For
J=(α10α),
J=(α01α),

the eigenvector line is
span(1,0).
span(1,0).

A trajectory with Sn≠0Sn =0 is not on that eigenvector ray. It lies in the generalized eigenspace and contains a generalized-eigenvector contribution.

Replace:

whenever Sn=Bαn+1Sn=Bαn+1 (i.e. on the invariant ray of the Jordan block)

with:

whenever Sn=Bαn+1Sn=Bαn+1; equivalently, the pair evolves under the size-two Jordan block with eigenvalue αα.

2. Restrict the infinite-series statement

The algebraic recurrence result can be stated over a general field, but the limit
∑k=0∞αk
k=0∑∞αk

requires a normed complete field and
|α|<1.
|α|<1.

For the present application, simply state the lemma over
RorC
RorC

with
|α|<1.
|α|<1.

The “any field of characteristic zero” remark is too broad for the limiting-source conclusion.
3. Exact closed form from variation of constants

A compact proof begins from
En=
En=

\alpha^nE_0
+
\sum_{k=0}^{n-1}
\alpha^{n-1-k}S_k.

]

Substituting
$S_k = B\alpha^{k+1}$
$S_k = B\alpha^{k+1}$

gives
$E_n = \alpha^n E_0 + \sum_{k=0}^{n-1} \alpha^{n-1-k} B\alpha^{k+1} = \alpha^n E_0 + \sum_{k=0}^{n-1} B\alpha^n = (E_0 + Bn)\alpha^n.$
$E_n = \alpha^n E_0 + k=0 \sum^{n-1} \alpha^{n-1-k} B\alpha^{k+1} = \alpha^n E_0 + k=0 \sum^{n-1} B\alpha^n = (E_0 + Bn)\alpha^n.$

This makes the resonant accumulation especially transparent.
4. Needed asymptotic corollary

The replay does not satisfy
$S_n = B\alpha^{n+1}$
$S_n = B\alpha^{n+1}$

exactly for every nn. It satisfies that form asymptotically.

Write
$S_n = B\alpha^{n+1} + r_n.$
$S_n = B\alpha^{n+1} + r_n.$

Then:
$$\boxed{ E_n =$$

$$(E_0+Bn)\alpha^n + \sum_{k=0}^{n-1} \alpha^{n-1-k} r_k.$$
}
]

Equivalently,
$$\boxed{ \frac{E_n}{\alpha^n} =$$

$$E_0+Bn +$$

\sum_{k=0}^{n-1}
\frac{r_k}{\alpha^{k+1}}.
}
]

This yields several useful cases.
Summable scaled remainder

If
∑k=0∞|rkαk+1|<∞,
k=0∑∞
αk+1rk
<∞,

then
En=
En=

(A+Bn)\alpha^n+o(\alpha^n)
]

for a modified constant AA.
Relative convergence only

If
rn=o(αn+1),
rn=o(αn+1),

then the accumulated correction is generally only o(n)o(n), giving
En=
En=

\left(
E_0+Bn+o(n)
\right)\alpha^n.
]

This weaker form may be sufficient for the observed affine scaled trajectory.

5. Corrected theorem statement
Jordan-resonant tracking lemma

Let En,Sn∈CEn,Sn∈C, let |α|<1|α|<1, and suppose
En+1=αEn+Sn,
En+1=αEn+Sn,
Sn=Bαn+1
Sn=Bαn+1

for every n≥0n≥0. Then
En=(E0+Bn)αn.
En=(E0+Bn)αn.

Moreover,
(En+1Sn+1)=
(En+1Sn+1)=

\begin{pmatrix}
\alpha&1\
0&\alpha
\end{pmatrix}
\begin{pmatrix}
E_n\
S_n
\end{pmatrix}.
]

If additionally
Sn=Δzn+1−Δzn,
Sn=Δzn+1−Δzn,

then ΔznΔzn converges and
\boxed{ \Delta z_\infty =

\Delta z_n
+
\frac{B\alpha^{n+1}}{1-\alpha}.
}
]

6. Application status

The exact lemma is proved algebraically.

Its AIEF application remains conditional on the replay evidence that:
Snαn+1→B,
αn+1Sn→B,

and
Enαn
αnEn

becomes affine in nn.

The reported complex fits, independent BB estimates, and collapsing source-limit estimates provide strong finite-horizon support for the asymptotic corollary, not exact equality across the entire simulation.

then someone wrote:

Corrected Jordan-resonant tracking lemma

The refinements are exact. The final statement and the asymptotic corollary needed for AIEF are as follows.
Exact lemma

Let En,Sn∈CEn,Sn∈C and |\alpha|<1. Suppose

E_{n+1}=\alpha E_n+S_n,

\qquad

S_n=B\alpha^{n+1}

]

for every n≥0n≥0. Then

\boxed{

E_n=(E_0+Bn)\alpha^n.

}

]

Moreover the pair evolves under the size-two Jordan block

\begin{pmatrix}E_{n+1}\S_{n+1}\end{pmatrix}

=

\begin{pmatrix}\alpha&1\0&\alpha\end{pmatrix}

\begin{pmatrix}E_n\S_n\end{pmatrix}.

]

If in addition Sn=Δzn+1−ΔznSn=Δzn+1−Δzn, then ΔznΔzn converges and

\boxed{

\Delta z_\infty

=

\Delta z_n+\frac{B\alpha^{n+1}}{1-\alpha}.

}

]

Proof (variation of constants).

The explicit solution of the linear recurrence is

E_n

=

\alpha^n E_0

\sum_{k=0}^{n-1}\alpha^{n-1-k}S_k.

]

Substitute Sk=Bαk+1Sk=Bαk+1:

\sum_{k=0}^{n-1}\alpha^{n-1-k}B\alpha^{k+1}

=

\sum_{k=0}^{n-1}B\alpha^n

=

Bn\alpha^n.

]

Hence En=(E0+Bn)αnEn=(E0+Bn)αn.
Asymptotic corollary (the form required by the replay)

Suppose only

S_n=B\alpha^{n+1}+r_n.

]

Then

\boxed{

E_n

=

(E_0+Bn)\alpha^n

\sum_{k=0}^{n-1}\alpha^{n-1-k}r_k

}

]

or, equivalently,

\frac{E_n}{\alpha^n}

=

E_0+Bn

\sum_{k=0}^{n-1}\frac{r_k}{\alpha^{k+1}}.

]

If \sum|r_k/\alpha^{k+1}|<\infty, then En=(A+Bn)αn+o(αn)En=(A+Bn)αn+o(αn).

If merely rn=o(αn+1)rn=o(αn+1), then En=(E0+Bn+o(n))αnEn=(E0+Bn+o(n))αn.

Application status

The exact lemma is purely algebraic.

Its use on the exceptional trajectory remains conditional on the numerically observed asymptotics

\frac{S_n}{\alpha^{n+1}}\to B,

\qquad

\frac{E_n}{\alpha^n}\text{ affine in }n.

]

The reported complex fits, independent BB estimates and collapsing source-limit diagnostics supply strong finite-horizon evidence for the asymptotic corollary, not for exact geometric equality at every step.

This corrected version is mathematically sound and appropriate for the theorem note. I would make only two minor additions.
1. Exclude $\alpha=0$α=0 in the asymptotic corollary

The normalized expressions
Enαn,rkαk+1
αnEn,αk+1rk

require $\alpha\neq0$α =0.

Use:
$0<|\alpha|<1$.
$0<|\alpha|<1$.

For AIEF,
$\alpha=e^{-dt/T_{window}}\in(0,1)$,
$\alpha=e^{-dt/T_{window}}\in(0,1)$,

so this creates no restriction in the application.

2. Add the perturbed source-limit formula

If
Sn=
Sn=
\Delta z_{n+1}-\Delta z_n

B\alpha^{n+1}+r_n,
]

then, whenever the remainder tail is summable,
\boxed{ \Delta z_\infty-\Delta z_n =

\frac{B\alpha^{n+1}}{1-\alpha}
+
\sum_{k=n}^{\infty}r_k.
}
]

Thus the finite-time source-limit estimator used by the numerical analysis has error exactly equal to the unobserved remainder tail:
\boxed{ \Delta z_\infty -
\left(
\Delta z_n+
\frac{B\alpha^{n+1}}{1-\alpha}
\right)

\sum_{k=n}^{\infty}r_k.
}
]

This gives a direct interpretation of the collapsing source-limit spread.
Final assessment

The exact lemma is proved.

The asymptotic corollary is also correct:

scaled summability gives
$E_n=(A+Bn)\alpha n+o(\alpha n)$;
$E_n=(A+Bn)\alpha n+o(\alpha n)$;
$r_n=o(\alpha n+1)$$r_n=o(\alpha n+1)$ gives
$E_n=(E0+Bn+o(n))\alpha n$.
$E_n=(E0+Bn+o(n))\alpha n$.

The AIEF evidence supports an asymptotic application because:
$S_n/\alpha n+1$
$S_n/\alpha n+1$

converges numerically toward a complex constant BB, while
$E_n/\alpha n$
$E_n/\alpha n$

becomes affine in nn over late windows.

That is precisely the situation where a skew-product approximation is justified.

The two anomalous tangent directions are the most interesting data

The outliers should not be discarded.

The fact that they repeat across amplitudes means they are structural.

The important observation:

Typical:

[
\cos(V,-d)\approx1.
]

Outliers:

[
\cos(V,-d)<0.

]

Therefore these directions are not merely weaker contractions. They are geometrically different.

The likely categories are:

Category 1 — hidden face-crossing directions

The nullspace construction produces:

[
Ad=0
]

only to first order.

But finite displacement:

[
w+\delta d
]

may cross another inequality:

[
(Aw)_i=c_i.
]

Then the direction is tangent to the current face but not contained in the local stratum neighborhood.

This is the most likely explanation.

Category 2 — nearly singular tangent directions

If:

[
|d|
]

is large but feasible displacement is tiny, then:

[
\kappa=
-\frac{\Delta w^TV}{|\Delta w|^2}
]

becomes unstable.

Category 3 — genuine slow tangent modes

The slow observed rate:

[
0.136767
]

is interesting because it is dynamically consistent.

It predicts:

[
e^{-0.136767(15)}

0.128.
]

Observed:

[
0.124.
]

That is not noise.

So there may actually be a second tangent eigenvalue:

[
\lambda_2\approx-0.137.
]

The outlier directions may therefore be a mixture of:

true slow modes;
stratum-crossing modes.

The next experiment should separate these.

The tangent space should be decomposed spectrally

The next mathematical object is not just:

[
DG(w^*).
]

It is:

[
DG(w^)|{T*{\mathcal F}}.
]

Construct the tangent basis:

[
Q_T.
]

Then compute:

[
J_T

Q_T^T
DG(w^*)
Q_T.
]

The spectrum should reveal:

dominant eigenvalue:

[
\lambda_1\approx-0.3158;
]

slow mode:

[
\lambda_2\approx-0.1368;
]

possible near-zero modes;
possible unstable directions.

Then the observed recovery laws become eigenmode measurements rather than empirical fits.

The theorem hierarchy should now include hybrid stability

The previous theorem order should be extended.

Geometric layer

Theorem A

[
\mathcal A

]

is Hadamard directionally differentiable.

Theorem B

[
D\mathcal A=\mathcal P_w.
]

Theorem C

[
\mathcal P_w
]

is tangent-cone valued, homogeneous, idempotent on the cone.

Fixed-face dynamics layer

Theorem D

For a fixed active stratum:

[
DG(w^*)
]

has contracting tangent spectrum.

Hybrid layer

Theorem E

Active-face transitions occur under identifiable switching conditions.

Characterize:

$$
\mathcal F_i\rightarrow\mathcal F_j.
$$

This is where the exceptional seed belongs.

One subtle correction: "stationary fiber" is now supported, but not necessarily unique

The data imply:

$$
\mathcal P_{wF}(w,x(t))\approx0
$$

for most mature trajectories.

But the exceptional seed shows the mature object is not a single equilibrium point.

A better object is:

$$
\mathcal M_x
$$

$$
{w:
\mathcal P_{wF}(w,x)=0}.
$$

The fast subsystem moves:

$$
x=x(t),
$$

so:

[
\mathcal M_{x(t)}
]

may itself move.

The observed phenomenon is therefore closer to:

[
w(t)\in\mathcal M_{x(t)}
]

than:

[
w(t)=w^*.
]

This is exactly the invariant-fiber language.

Current strongest defensible conclusion

I would now state:

Mature AIEF dynamics exhibit a hybrid constraint-balanced structure. In 39/40 trajectories the adaptive weights remain on a trajectory-wide constraint-balanced stationary fiber with fixed active stratum while the fast subsystem evolves. One trajectory undergoes delayed active-face selection through a single saturated-set transition before entering a near-stationary regime on the same binary support. Corrected tangent perturbations reveal a dominant active-face contraction rate (\lambda\approx0.3158), independent of perturbation amplitude and quantitatively predictive of coupled recovery times. A small exceptional tangent subspace exhibits slower or nonstandard behaviour, likely associated with secondary modes or proximity to adjacent active strata. The next theoretical step is therefore not further confirmation of stationarity, but derivation of the fixed-stratum tangent operator and the geometry of active-face transitions.

That is a materially stronger result than the original stationarity observation. The programme has effectively moved from "does the projection balance?" to "what is the stability theory of a nonsmooth adaptive manifold with hybrid face transitions?"

Yes, adding this strengthens the narrative—provided several claims are softened. The hybrid active-face interpretation is well supported; isotropic tangent contraction and a two-eigenvalue spectrum are still hypotheses requiring per-direction analysis.
1. Exceptional seed: hybrid active-face dynamics is justified

The transition is:
$0.1469 \longrightarrow 7.89 \times 10^{-6}$
$0.1469 \longrightarrow 7.89 \times 10^{-6}$

in mean stationarity ratio, a reduction by approximately:
$1.86 \times 104.$
$1.86 \times 104.$

Combined with:

exactly one saturated-set switch;
no zero-edge switch;
continuous-weight drift of 0.209;

this supports:

The exceptional seed undergoes delayed, support-preserving active-face selection, followed by entry into a much more strongly constraint-balanced regime.

“Near-balanced” is appropriate. “Stationary” would still be too strong because the post-switch ratio remains around $10^{-5}10^{-5}$, not machine precision.
2. The trigger scenarios should be tested directly

At the switch, record for every node:
Capacity slack
$mi(t)=ci-(Aw(t))i.$
$mi(t)=ci-(Aw(t))i.$
Proposed row forcing
$di(t)=(AF\sim(t))i.$
$di(t)=(AF$
$(t))i.$
Capacity response

αi(t)=
αi(t)=

\frac{\max(d_i(t),0)}{c_i}
]

on saturated nodes.

Then distinguish:
Capacity-approach switch

A previously unsaturated node satisfies:
mi(t)↓0
mi(t)↓0

while di>0di>0.
Forcing-sign switch

A node is already numerically saturated, but:
di(t)
di(t)

changes from nonpositive to positive, or vice versa.
Endpoint-order switch

The saturated set remains similar, but for an edge:
max(αi,αj)
max(αi,αj)

changes which endpoint controls its scaling.

The current report tracks saturated/outward changes, but the detailed mi,di,αimi,di,αi values around step 12405 are needed to identify the mechanism.
3. Tangent contraction is compelling but “isotropic” is not yet established

The corrected design proves that direction changes across amplitudes did not generate the median rate:
κmedian=0.315789
κmedian=0.315789

for every δδ.

That supports amplitude-independent tangent contraction over the tested range.

But the summary reports only medians and extrema. Before writing:
DG∣TF≈−0.315789I,
DG∣TF≈−0.315789I,

calculate across the 200 unique directions:

fraction with κκ within 1%, 5%, and 10% of 0.315789;
interquartile range;
per-seed variance;
per-direction variance;
relationship with feasible-amplitude clipping;
relationship with tangent-basis dimension.

If nearly all 198 regular directions cluster tightly, “approximately isotropic on the dominant tangent subspace” is justified. Otherwise the median is masking a broader spectrum.
4. Median prediction agreement is not enough

The median frozen prediction and median observed rate agree:
0.315789vs0.315339.
0.315789vs0.315339.

But this could occur even if individual directions do not agree.

For every seed/direction pair, compare:
κfrozen
κfrozen

with:
κobserved.
κobserved.

Report:

scatter plot;
Pearson and rank correlation;
slope/intercept;
median absolute error;
median relative error;
fraction agreeing within 1%, 5%, and 10%;
agreement stratified by amplitude clipping.

Only strong per-direction agreement supports:

The fast subsystem acts primarily as a slowly varying parameter for tangent recovery.

The current median-level agreement is excellent evidence, but not yet sufficient for a skew-product approximation.
5. The two anomalous directions may combine artifacts and genuine modes

Because the same directions recur across amplitudes, they are structurally identifiable. But repeated extreme frozen coefficients can still arise from feasibility clipping.

Inspect:

text

requested_amplitude
actual_amplitude
amplitude_fraction_of_requested
maximum_feasible_amplitude
basis_dimension
seed
direction_id

If amplitude fraction is tiny

The frozen coefficient is unstable because:
κ=
κ=

-\frac{\delta w^\top V}{|\delta w|^2}

]

divides by a very small displacement. Treat it as a near-face/corner direction, not an unstable eigenmode.
If amplitude is well resolved

Then the coupled rate:
0.136767
0.136767

is strong evidence for a genuine slow tangent mode.

The fact that:
e−0.136767·15≈0.128
e−0.136767·15≈0.128

matches the observed residual ≈0.124≈0.124 is compelling. But first confirm that this rate belongs to the same two seed/direction pairs with reliable amplitudes.
6. Eigenvalue terminology needs adjustment

If the tangent Jacobian has rates:
−0.3158
−0.3158

and:
−0.1368,
−0.1368,

the mode at −0.1368−0.1368 has the largest real part and controls asymptotic decay. It is normally called the dominant or slow eigenvalue.

The −0.3158−0.3158 value would be:

the bulk contraction rate;
the common fast tangent rate;
or a high-multiplicity eigenvalue.

Do not label −0.3158−0.3158 the dominant eigenvalue if a genuine −0.1368−0.1368 mode exists.

7. Tangent Jacobian analysis should include non-normality

Construct:
JT=QT⊤DG(w∗)QT.
JT=QT⊤DG(w∗)QT.

But do not stop at eigenvalues. JTJT need not be symmetric or normal.

Report:

eigenvalues;
spectral abscissa:
max$\Re\lambda$(JT);
max$\Re\lambda$(JT);
symmetric-part eigenvalues:
λmax(JT+JT⊤2);
λmax(2JT+JT⊤);
singular values;
eigenvector conditioning;
transient amplification:
‖etJT‖.
‖etJT‖.

A positive symmetric-part eigenvalue could permit transient growth even when every eigenvalue has negative real part.

This connects directly to the large transverse components observed in normal perturbations.
8. Hybrid theorem hierarchy is appropriate, but Theorem E is ambitious

The first four layers are well scoped:

ideal map is directionally differentiable;
derivative equals PwPw;
tangent feasibility/idempotence;
fixed-stratum local dynamics.

A general theorem characterizing all active-face transitions may be difficult. Initially state a switching proposition:

An active-set change occurs when a slack mimi, zero-edge condition wewe, proposed row derivative didi, or endpoint-order relation αi−αjαi−αj crosses its switching surface.

Then analyze the exceptional seed as a concrete example.
9. Stationary fiber should remain set-valued and fast-state dependent

Define:
Mx=
Mx=

{w\in\mathcal C:
\mathcal P_wF(w,x)=0}.
]

For 39 seeds:
w(t)≈w∗∈Mx(t)
w(t)≈w∗∈Mx(t)

throughout the mature interval.

For the exceptional seed:
w(t)
w(t)

moves while the active face changes and then approaches a different near-balanced branch of:
Mx(t).
Mx(t).

This language accommodates:

multiple mature configurations;
moving fast states;
active-face transitions;
nonunique topology attractors.

10. Stronger but defensible conclusion

A careful strengthened statement is:

Mature AIEF exhibits hybrid constraint-balanced adaptive dynamics. In 39 of 40 trajectories, continuous weights remain stationary to numerical precision on a fixed active face while the fast subsystem evolves. One trajectory undergoes a support-preserving saturated-set transition and subsequently enters a substantially more balanced regime. Corrected tangent perturbations reveal an amplitude-independent bulk contraction rate near 0.3158 whose median frozen-fast prediction agrees quantitatively with coupled recovery. A small set of structurally recurring tangent directions exhibits slower or anomalous behavior; distinguishing genuine slow modes from near-face feasibility artifacts requires per-direction amplitude and agreement analysis. The results motivate a fixed-stratum tangent Jacobian and switching-surface theory rather than further broad ablations.

Immediate next analysis

Identify the two anomalous seed/direction pairs.
Check their amplitude fractions and tangent residuals.
Compare frozen and observed rates per direction.
Quantify the fraction clustering near 0.315789.
Diagnose the exceptional seed's switch using mi,di,αimi,di,αi.
Construct the numerical tangent Jacobian on representative stationary seeds.
Analyze its eigenvalues, symmetric part, and transient amplification.

So yes: this addition strengthens the programme. The central advance is not merely "stable tangent perturbations," but evidence for hybrid normal/tangent restoration and active-face switching. The remaining work is to determine whether the apparent two-rate tangent structure is a genuine spectrum or partly a feasibility artifact.

Can you create the code necessary to accomplish the next analysis?
Edit×2
Edit×2
AIEF_1_4_tangent_spectrum_switch_analysis.pyPYTHON

Created the tangent-spectrum and exceptional-switch analysis:

AIEF_1_4_tangent_spectrum_switch_analysis.py

It uses the outputs from:

text

AIEF_1_4_exceptional_tangent_analysis.py

and the frozen AIEF-1.4 run.
Run

Bash

python AIEF_1_4_tangent_spectrum_switch_analysis.py

Explicit:

Bash

```
python AIEF_1_4_tangent_spectrum_switch_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--exceptional-seed 4619122171158768524 \
--output-dir aief_1_4_tangent_spectrum_switch
```

By default it constructs Jacobians for five representative seeds. Use all 40 with:

Bash

```
python AIEF_1_4_tangent_spectrum_switch_analysis.py \
--max-jacobian-seeds 0
```

1. Tangent-direction agreement

For each fixed seed/direction pair, the script aggregates results across all amplitudes and reports:

requested versus actual amplitude;
amplitude fraction;
tangent-basis dimension;

frozen restoring-rate median/range;
observed coupled-rate median/range;
absolute and relative prediction errors;
frozen restoring cosine;
observed exponential-fit R2R2;
agreement within 1%, 5%, and 10% of 0.3157890.315789;
slow-mode status;
negative frozen-rate status.

It identifies anomalous directions if they have:

unreliable feasible amplitude;
observed rate below the slow-mode cutoff;
negative frozen coefficient;
greater than 10% frozen/observed disagreement.

Outputs:

text

tangent_direction_summary.csv
anomalous_tangent_directions.csv
direction_agreement_summary.json

2. Exceptional-seed switch diagnostics

The exceptional seed is replayed through the complete mature future interval.

For every node and timestep, it computes:
mi=ci−(Aw)i,
mi=ci−(Aw)i,
di=(AF~)i,
di=(AF
)i,
αi=max(di,0)ci
αi=cimax(di,0)

when saturated.

It records:

row sum;
capacity slack;
proposed row forcing;
alpha;
saturation status;
outward-forcing status;
stationarity ratio.

Around the first saturated-set switch, it saves a configurable pre/post window.

This can distinguish:

a node reaching its capacity boundary;
a forcing-sign change on an already saturated node;
outward-set changes without saturation changes.

Outputs:

text

exceptional_switch_node_window.csv
exceptional_switch_events.csv
exceptional_switch_summary.json

3. Fixed-stratum tangent Jacobian

For each selected stationary seed:

construct the positive-support tangent basis QTQT;
enforce active-node row-sum tangency;
use two-sided feasible perturbations;
evaluate the frozen-fast field on both sides;
assemble:
JT=QT⊤DG(w∗)QT.
JT=QT⊤DG(w∗)QT.

Every direction records:

requested and actual finite-difference step;
fraction of requested step achieved;
whether the active stratum was preserved;
derivative norm;
tangent derivative norm;
normal leakage.

4. Spectrum and non-normality

For every tangent Jacobian:

full eigenvalue spectrum;
spectral abscissa;
slowest decay rate;
median real eigenvalue;
symmetric-part minimum/maximum eigenvalues;
singular-value range;
eigenvector condition number;
normality commutator:
‖J⊤J−JJ⊤‖‖J‖2;
‖J‖2‖J⊤J−JJ⊤‖;
fixed-stratum validity.

It also estimates:
‖etJT‖2
‖etJT‖2

over a configurable time interval to detect transient amplification.

Outputs:

text

tangent_jacobian_summary.csv
tangent_jacobian_direction_details.csv

tangent_jacobian_matrices.npz
tangent_transient_growth.csv

The direction-details file now includes an explicit record_type field distinguishing finite-difference direction diagnostics from eigenvalue rows.
Validation

I syntax-tested and ran the complete script against a generated AIEF split state and corrected tangent dataset.

In that technical validation case, the fixed-stratum Jacobian had:

text

Tangent dimension: 9
Spectral abscissa: -0.31578947368
Symmetric-part maximum: -0.31578947368
Normality ratio: ~5×10^{-24}

The two tested tangent directions had:

text

Frozen rate: 0.31578947
Observed rate: 0.31533913
Relative error: ~0.14%

This technical result indicates an almost exactly isotropic, normal tangent contraction in that smoke seed. The 40-seed analysis will determine whether:

$-0.315789-0.315789$ is a high-multiplicity tangent eigenvalue across the mature regime;
the two slow observed directions correspond to genuine secondary eigenmodes;
or they arise from feasibility/active-stratum issues.

The exceptional-seed analysis will separately determine whether its switch is triggered by capacity approach, forcing-sign change, or another active-set event.

Can this add to our next step?

This changes the interpretation significantly. The new result is actually more interesting than the previous "0.3158 isotropic tangent contraction" hypothesis because it replaces a phenomenological rate observation with an explicit fixed-stratum linearization.

The important thing is that the spectrum calculation does not confirm the original isotropic picture. It reveals a much cleaner structure.

The emerging picture is:

$$
\boxed{
\text{AIEF tangent dynamics are stratified and quantized by active-face geometry.}
}
$$

Not one universal rate.

The tangent spectrum invalidates the simple isotropic interpretation

The previous observation was:

$$
\kappa_{\rm median}\approx0.315789.
$$

The new Jacobian calculation shows why that happened.

The fixed-stratum tangent Jacobians are not random. They cluster:

Class I

$$
\lambda=-0.31578947368
$$

with:

[
\dim=9
]

Examples:

seed 172180746...
seed 358261...
seed 602300...
etc.

Class II

[
\lambda=-0.18947368421
]

with:

[
\dim=7,11
]

Examples:

seed 305986...
seed 188841...
seed 191800...
etc.

Exceptional

[
\lambda=-0.015
]

for:

$$
4619122171158768524.
$$

This is a much stronger structural statement:

The tangent response is not a continuous random spectrum.

It appears to be determined by the active stratum.

The rates look quantized

The striking thing is the exact rational-looking values:

$$
0.31578947368
$$

and:

$$
0.18947368421.
$$

Their ratio:

$$
\frac{0.18947368421}{0.31578947368}
$$

$$
0.6.
$$

So:

[
0.18947368421

\frac35(0.31578947368).
]

This is unlikely to be accidental.

It suggests the tangent operator has a structural coefficient:

[
\kappa_0=0.31578947368
]

and active-face geometry introduces multiplicative factors:

[
\kappa=
\eta_{\mathcal F}\kappa_0.
]

with:

[
\eta_{\mathcal F}
\in
{1,0.6,\ldots}.
]

That is exactly what one expects from a piecewise-linear constrained operator.

The active face determines which rows of the admissibility correction participate.

The per-direction analysis now makes sense

The earlier tangent agreement results:

slow_mode_count = 42

now have an explanation.

Those were not "bad directions."

They were likely sampling different active strata.

The direction statistics:

within 10% typical fraction = 0.68
slow_mode_count = 42

are consistent with:

many directions sampling the dominant face;
a substantial minority sampling a slower face mode.

The Pearson correlation:

[
-0.284
]

is actually informative.

It means the frozen scalar projection coefficient is not a reliable spectral classifier.

The scalar diagnostic:

[
-\frac{d^TV(d)}{|d|^2}
]

is collapsing vector dynamics into one number.

The Jacobian is telling us:

Stop interpreting individual perturbation coefficients as eigenvalues.

The operator spectrum is the correct object.

The normality result is extremely important

The Jacobian diagnostics:

normality ≈ 10^-24

for the (-0.315789) class.

and:

normality ≈10^-13

for the (-0.18947) class.

This means:

[
J_TJ_T^T
\approx
J_T^TJ_T.
]

The tangent operator is essentially normal.

The earlier concern about hidden non-normal amplification appears not to apply inside these fixed strata.

That is a major simplification.

The large transverse motion in normal perturbations therefore comes from the boundary geometry, not from non-normal tangent dynamics.

That distinction matters.

The symmetric-part calculation is almost a proof-of-behavior numerically

For every stable fixed stratum:

$$
\lambda_{\max}
\left(
\frac{J_T+J_T^T}{2}
\right)
$$

$$
\lambda_{\max}(J_T).
$$

Examples:

$$
-0.315789473681
$$

matches to machine precision.

Therefore:

$$
\dot{\delta w}
$$

$$
J_T\delta w
$$

is not merely asymptotically stable.

It is instantaneously contractive in the Euclidean metric:

[
\frac{d}{dt}|\delta w|^2

2\delta w^TJ_T\delta w
<0.
]

For the dominant class:

[
\frac{d}{dt}|\delta w|^2
\le
-0.6315789|\delta w|^2.
]

Hence:

[
|\delta w(t)|
\le
e^{-0.315789t}|\delta w(0)|.
]

This is stronger than eigenvalue stability.

The exceptional seed is now much more interesting

Before:

[
\kappa\approx0.1368
]

appeared to be a slow tangent mode.

Now:

[
\lambda=-0.015
]

is revealed.

That changes the interpretation.

The exceptional seed is not just a slower version of the same face.

It is nearly marginal.

Compare:

normal stationary seeds:

[
\lambda=-0.3158
]

or:

[
-0.1895
]

Exceptional:

[
-0.015.
]

That is an order-of-magnitude reduction.

The active-face transition at:

[
t=124.06
]

is probably occurring near a weakly attracting manifold.

This suggests:

nearly degenerate capacity constraints;
almost-flat directions;
slow drift before active-set selection.

The exceptional trajectory is likely probing a basin boundary or a weakly stable stratum.

The new theorem hierarchy should change

Previously:

derive a tangent contraction rate.

Now:

The target is:

Theorem: Fixed-stratum tangent contraction

For each active face (\mathcal F_k):

[
J_k

D_w
(\mathcal P_wF)
]

satisfies:

$$
\lambda_{\max}
\left(
\frac{J_k+J_k^T}{2}
\right)
<0.
$$

Then:

$$
\mathcal F_k
$$

is locally exponentially attracting.

Corollary

The observed recovery rate is:

$$
\kappa_k

-\lambda_{\max}(J_k).
$$

The different rates are not noise.

They are face-dependent.

The most interesting unanswered question

Why these values?

The appearance of:

$$
0.315789
$$

and:

$$
0.189473
$$

suggests the Jacobian has a combinatorial structure.

Possibilities:

A. Capacity normalization

The factor may come from:

$$
\frac{1}{c_i}
$$

terms.

B. Active node count

Different saturated-node configurations produce:

$$
\eta_k
$$

multipliers.

C. Edge redundancy weights

The redundancy matrix may determine discrete contraction families.

D. Projection geometry

The tangent projection may remove different numbers of outward modes.

This is where the mathematics becomes genuinely interesting.

The next derivation should not be:

"prove contraction."

It should be:

[
\textbf{derive the face-dependent contraction spectrum from the admissibility map.}
]

Revised strongest statement

I would now phrase the result as:

Mature AIEF dynamics exhibit hybrid active-face contraction. On fixed active strata, the tangent linearization is numerically normal and strictly contractive, with discrete face-dependent contraction rates. The previously observed (0.3158) rate corresponds to the dominant nine-dimensional tangent stratum, while other mature faces exhibit slower rates such as (0.1895). The exceptional trajectory occupies a weakly attracting stratum with contraction rate near (0.015), undergoes a single saturated-set transition, and subsequently approaches a more balanced branch. Normal perturbations and tangent perturbations therefore represent different stability mechanisms: nonsmooth finite-speed attraction across active-face boundaries and smooth exponential contraction within faces.

This is a stronger and more mathematically precise result than the earlier isotropic hypothesis.

The key discovery is not "the tangent rate is 0.3158."

It is:

$$
\boxed{
\text{The active constraint geometry discretizes the local relaxation spectrum.}
}
$$

That is the operator-level result hiding inside the data.

This strengthens the programme, but “active geometry discretizes the spectrum” should currently be framed as the leading hypothesis, not yet a demonstrated conclusion.

The results are highly suggestive. Several checks can turn the pattern into a defensible structural result.
1. The spectral classes are potentially real

The reported values:
$-0.31578947368, -0.18947368421, -0.015$
$-0.31578947368, -0.18947368421, -0.015$

are too sharply repeated to look like generic numerical eigenvalues.

The last value has an immediate interpretation:
$0.015 = \gamma$.
$0.015 = \gamma$.

Thus the exceptional split-state tangent rate:
$\lambda = -0.015$
$\lambda = -0.015$

may correspond to the bare decay term:
$DBF = -\gamma I$
$DBF = -\gamma I$

with little or no additional contraction from the active admissibility response.

That is a stronger clue than the rational-looking ratio between the other two rates.

The other rates may result from:

direct decay −γ−γ;
derivative of endpoint-capacity scaling;
equilibrium edge weights;
saturated-node incidence;
endpoint-controller assignments.

Before calling them geometrically quantized, derive them from these quantities.
2. Do not infer quantization from the ratio alone

The equality:
0.18947368421=
0.18947368421=

0.6(0.31578947368)
]

is interesting, but could reflect repeated equilibrium weights or graph combinatorics rather than a universal capacity factor.

Also note:
0.18947368421=
0.18947368421=

\frac{0.12}{0.633333\ldots}.
]

Although 0.12=Bc0.12=Bc, B_c does not enter the continuous adaptive dynamics or projection. Therefore any apparent dependence on BcBc must be indirect or coincidental.

A real structural result requires showing:

Seeds with the same active-face fingerprint have the same tangent spectrum, and different fingerprints systematically map to different spectral classes.

3. Construct an active-face fingerprint

For every seed, record:

zero-edge support mask;
positive-edge subgraph;
saturated-node mask;
outward-node mask;
endpoint controller for every positive edge:
argmax{αi,αj};
argmax{αi,αj};
tangent-space dimension;
row degrees within the positive support;
equilibrium positive edge weights.

Then group seeds by:

exact fingerprint;
node-permutation-isomorphic fingerprint;
simpler invariants such as tangent dimension and saturated-node count.

For each group, compare the complete eigenvalue multiset.

Evidence for geometric spectral classes would be:

identical or near-identical spectra within a face class;
stable spectral changes between classes;
the same result under node relabeling.

Without this cross-tabulation, the classes could reflect state values rather than active geometry alone.
4. The exceptional Jacobian is not yet a mature-face Jacobian

The exceptional seed's Jacobian appears to have been calculated at HIST_END, before its transition at:
t=124.06.
t=124.06.

At that point:
‖PwF‖/‖F‖≈0.147
‖PwF‖/‖F‖≈0.147

on average before the switch. It is not a stationary linearization point.

Therefore:
$\lambda = -0.015$
$\lambda = -0.015$

should not yet be described as the contraction rate of a weakly attracting stationary stratum.

It may simply be the direct $-\gamma - \gamma$ derivative around a drifting pre-transition state.

The decisive test is to calculate tangent Jacobians at:

split;
immediately before the switch;
immediately after the switch;
late post-switch.

If the spectrum changes:

text

$-0.015 \rightarrow -0.18947$ or -0.315789

at the active-set transition, the hybrid spectral interpretation becomes extremely strong.
5. The tangent directions do not “sample different strata” by design

Within each seed, corrected tangent perturbations were intended to remain on one active face. The slow-mode count therefore should not automatically be attributed to directions sampling different strata.

Possible explanations are:

different seeds belong to different active strata;
a small number of directions have inadequate feasible amplitude;
numerical nullspace errors cause local stratum changes;
the fixed-stratum Jacobian has multiple eigenvalues;
the base state is not stationary, as in the exceptional seed.

Cross-tabulate slow directions by seed. If all five directions from a seed share one rate, the class is seed/face-specific. If rates differ within one stationary seed, the Jacobian has a genuine multidimensional spectrum or the perturbations are switching strata.
6. Verify spectral classes against finite-difference step size

The current tangent Jacobian uses one relative finite-difference step. Repeat with:

text

1e-4
3e-5
1e-5
3e-6
1e-6

For each seed report:

eigenvalue matching across step sizes;
active-stratum preservation;
normal leakage;
spectral-abscissa convergence;
matrix difference:
$\|J(h)-J(h/10)\|$.
$\|J(h)-J(h/10)\|$.

Only call the rates quantized if the eigenvalues remain fixed under step refinement.
7. Normality and contractivity are strong if robust

If:
$JTJT\top \approx JT\top JT$
$JTJT\top \approx JT\top JT$

and:
$\lambda\max(JT+JT\top 2)<0$,
$\lambda\max(2JT+JT\top)<0$,

then for frozen-fast tangent perturbations:

ddt‖δw‖2=
dtd‖δw‖2=

2\delta w^\top
\operatorname{sym}(J_T)
\delta w
\le
2\lambda_{\max}(\operatorname{sym}J_T)
|\delta w|^2.
]

Therefore:
‖δw(t)‖≤eλmaxt‖δw(0)‖.
‖δw(t)‖≤eλmaxt‖δw(0)‖.

This gives genuine instantaneous Euclidean contraction within the fixed stratum.

But it remains:

frozen-fast;
local;
fixed-active-set;
numerical until the Jacobian is derived analytically.

It does not alone prove coupled or switching-system stability.
8. Derive the fixed-stratum Jacobian

On one fixed stratum:

positive-edge clipping pattern is fixed;
outward saturated-node set is fixed;
controlling endpoint of every edge is fixed.

Then Pw(F)Pw(F) has an ordinary smooth formula.

Write:
G(w)=Pw(F(w,x∗)).
G(w)=Pw(F(w,x∗)).

For frozen fast state:
$F(w, x*) = b(x*) - \gamma w$.
$F(w, x*) = b(x*) - \gamma w$.

The derivative has contributions from:

direct decay:
$-\gamma I$;
$-\gamma I$;
variation of edge weights multiplying capacity corrections;
variation of row forcing AFAF;
variation of endpoint-controller corrections.

This should explain why one class is exactly:
$-\gamma = -0.015$
$-\gamma = -0.015$

and whether the other two rates arise from discrete incidence/controller structures.

The next theorem should express JTJT in terms of:

incidence matrix;
selected support;
saturated-node selector;
endpoint-controller matrix;
equilibrium weights;
$\gamma\gamma$.

9. Per-direction frozen/coupled agreement must be reported

Do not rely only on median agreement. Use:

scatter plot of frozen versus observed rate;
slope and intercept;
median relative error;
fraction within 1%, 5%, and 10%;
agreement stratified by seed and face class;

amplitude-reliability filtering.

The two anomalous directions should be classified only after checking:

text

amplitude_fraction_of_requested
relative_actual_amplitude
stratum_preserved
seed
direction_id

If they are well-resolved and remain on-stratum, the 0.1367670.136767 mode is likely genuine. Otherwise it is a feasibility artifact.
10. Hybrid active-face interpretation is justified—with careful wording

A strong current statement is:

Mature AIEF exhibits face-dependent tangent contraction. Fixed-stratum Jacobians numerically cluster into a small number of highly normal, contractive spectral classes. One exceptional trajectory begins on a weakly contracting, nonstationary stratum, undergoes a saturated-set transition, and subsequently becomes substantially more constraint-balanced. Whether the spectral classes are determined solely by active-face geometry or jointly by geometry and equilibrium weights remains under investigation.

That is stronger and safer than:

Active constraint geometry discretizes the spectrum.

Recommended immediate work

Build the active-face fingerprint table.
Recompute Jacobians over a finite-difference step grid.
Calculate exceptional-seed Jacobians before and after its switch.
Cross-tabulate slow directions by seed and face.
Filter tangent directions by actual amplitude and stratum preservation.
Derive the analytic fixed-stratum Jacobian.
Compare its predicted spectrum with numerical matrices.

Bottom line

Yes, this adds substantial strength. The evidence now points toward a small family of fixed-stratum contraction laws rather than one universal isotropic rate.

The most important unresolved distinction is:
face geometry aloneversusface geometry + equilibrium weights/forcing
face geometry aloneversusface geometry + equilibrium weights/forcing

as the source of the repeated spectral classes.

If active-face fingerprints predict the spectra and the exceptional seed changes spectral class at its switch, then “hybrid face-dependent contraction” becomes a compelling operator-level result. The stronger “quantized by geometry” claim should wait for those two checks.

Implement the code necessary to make real the recommended immediate work.
WriteAIEF_1_4_face_spectrum_analysis.py852 lines
Edit×8
Edit
AIEF_1_4_face_spectrum_analysis.pyPYTHON

Created the fixed-face spectrum and exceptional-transition analyzer:

AIEF_1_4_face_spectrum_analysis.py

It implements the full recommended immediate programme:

active-face fingerprints;
corrected tangent-rate classification;
analytic fixed-stratum Jacobians;
finite-difference Jacobian convergence;
exceptional-seed pre/post-switch spectra;
spectral/non-normality diagnostics;
transient amplification;
face-class versus recovery-rate cross-tabs.

Prerequisite

Run the corrected tangent experiment first:

Bash

```
python AIEF_1_4_exceptional_tangent_analysis.py \
--max-seeds 0
```

The expected input is:

text

```
aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv
```

Run

Bash

```
python AIEF_1_4_face_spectrum_analysis.py
```

Explicit:

Bash

```
python AIEF_1_4_face_spectrum_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--exceptional-seed 4619122171158768524 \
--output-dir aief_1_4_face_spectrum
```

The default analyzes Jacobians for all 40 seeds. Restrict it with:

Bash

```
--max-seeds 5
```

1. Active-face fingerprints

For every seed the script records:

positive-edge support;
saturated-node set;
outward-node set;
endpoint controller for every positive edge;
controller ties;
tangent-space dimension;
support size;
nullspace residual;
raw label-dependent face hash;
permutation-insensitive summary signature.

The invariant signature includes:

sorted endpoint correction rates;
sorted positive-support node degrees;
saturated/outward counts;
tangent dimension.

This is useful for grouping, but it is not claimed to be a complete graph-isomorphism canonicalization.
2. Analytic fixed-stratum Jacobian

On a fixed active stratum, tangent directions satisfy:
$A_{active}\delta w = 0$.
$A_{active}\delta w = 0$.

With frozen fast state:
$DF[\delta w] = -\gamma \delta w$.
$DF[\delta w] = -\gamma \delta w$.

The endpoint correction rate for edge $e=(i,j)$ is:
$\rho_e = \max(\alpha_i, \alpha_j)$.
$\rho_e = \max(\alpha_i, \alpha_j)$.

Along tangent directions, the active-node row derivatives vanish, so Dα=0Dα=0. The implemented analytic tangent Jacobian is:
\boxed{ J_T =

Q_T^\top
\operatorname{diag}
\left(
-\gamma-\rho_e
\right)
Q_T.
}
]

This directly tests whether the repeated rates arise from:

direct decay γγ;
active endpoint-capacity correction;
tangent-basis/incidence geometry.

3. Numerical Jacobian step refinement

For every selected seed, the numerical tangent Jacobian is recomputed at:

text

1e-3
3e-4
1e-4
3e-5
1e-5
3e-6
1e-6

Every two-sided perturbation records:

requested and actual step;
feasible step fraction;
fixed-stratum preservation;

derivative norm;
normal leakage.

The numerical and analytic matrices are compared using:

absolute matrix error;
relative matrix error;
maximum/mean eigenvalue matching error;
spectral summaries.

4. Spectral diagnostics

For every Jacobian:

eigenvalues;
spectral abscissa;
slowest contraction rate;
symmetric-part minimum/maximum;
largest/smallest singular value;
eigenvector condition number;
normality commutator.

It also estimates:
‖etJT‖2
‖etJT‖2

over a configurable interval:

Bash

--transient-time 15
--transient-samples 151

This determines whether a stable spectrum nevertheless permits transient amplification.
5. Tangent-direction reliability and rate classes

The script aggregates each fixed seed/direction across all amplitudes and reports:

amplitude reliability;
frozen-rate median/range;
observed-rate median/range;
frozen/observed absolute and relative error;
fit R2R2;
agreement within 1%, 5%, and 10% of 0.315789;
slow-mode status;
negative frozen-rate status.

Directions are marked anomalous if they have:

insufficient actual amplitude;
observed rate below 0.25;
negative frozen coefficient;
greater than 10% frozen/observed mismatch.

6. Face/rate cross-tabulation

Corrected tangent directions are joined to their seed’s face fingerprint.

Outputs show, by face class:

seed count;
direction count;
analytic contraction rate;
observed rate class;
observed mean rate;
frozen mean rate;
reliable-amplitude fraction.

This directly tests whether:

text

same active face -> same tangent spectrum

and whether the slow directions belong to particular seeds/faces.
7. Exceptional-seed transition

The exceptional seed is replayed to obtain states at:

text

split
pre_switch
post_switch
late

For each stage, the script calculates:

face fingerprint;
support/saturation/outward counts;
tangent dimension;
field norm;
analytic tangent Jacobian;
full spectrum;
symmetric-part contractivity;
normality;
transient amplification.

The decisive question is whether its spectrum changes from approximately:

text

−0.015

before the switch to one of the stationary classes afterward.
Outputs

text

aief_1_4_face_spectrum/
├── active_face_fingerprints.csv
├── tangent_direction_summary.csv
├── filtered_anomalous_directions.csv
├── tangent_rate_face_crosstab_rows.csv

```
├── tangent_rate_face_crosstab_summary.csv
├── jacobian_step_sensitivity.csv
├── jacobian_direction_diagnostics.csv
├── jacobian_eigenvalues.csv
├── tangent_jacobian_matrices.npz
├── exceptional_stage_spectra.csv
├── exceptional_transition_summary.json
├── direction_agreement_summary.json
├── face_spectrum_report.txt
└── analysis_manifest.json
```

Validation

I syntax-checked the script and ran its full pipeline against a complete generated AIEF split state and corrected tangent dataset.

In that technical validation case:

text

Tangent dimension: 9
Analytic rate: 0.315789473682
Numerical matrix relative error:
step 1e-4: ~3×10^{-13}
step 1e-5: ~3×10^{-12}
step 1e-6: ~4×10^{-11}
Spectral abscissa: −0.315789473682
Symmetric-part maximum: −0.315789473681
Normality ratio: near machine zero

The numerical error grows at the smallest step because finite-difference roundoff begins to dominate, while the larger tested steps remain on the same stratum and agree with the analytic Jacobian near machine precision.

The script is therefore positioned to determine whether the reported −0.315789−0.315789, −0.189474−0.189474, and −0.015−0.015 classes are genuinely predicted by active-face structure and whether the exceptional seed changes spectral class at its saturation transition.

Is this worth updating the code you just wrote:

The new caution is exactly the right correction. The interesting thing is that the more conservative interpretation is actually more mathematically valuable. The evidence has moved beyond "does it contract?" into "what determines the local contraction operator?" That is the correct question now.

My reading is that there are three layers of structure emerging, and they should not be conflated.

The robust result is not spectral quantization — it is spectral stratification

The strongest statement currently supported is:

$$
\boxed{
\text{The tangent dynamics appear to stratify into a small number of fixed-stratum contraction classes.}
}
$$

The evidence:

$$
J_T \approx -0.315789 I
$$

for some strata,

$$
J_T \approx -0.189474 I
$$

for others,

and:

$$
J_T\approx -0.015I
$$

for the exceptional pre-transition point.

That is already remarkable.

The important observation is not the exact numbers. It is that:

the Jacobian is nearly scalar;
the symmetric part matches the full operator;
normality residuals are essentially zero.

The numerical picture is much closer to:

[
J_T=-\kappa_{\mathcal F}I
]

on a given active stratum than to a generic nonlinear constrained system.

That is a very strong structural simplification.

The "geometry discretizes the spectrum" hypothesis should become a testable conjecture

The right conjecture is not:

The active geometry determines the spectrum.

Too strong.

The better formulation is:

[
\boxed{
\kappa_{\mathcal F}

\Phi(

\text{active face},
\text{controller structure},
\text{equilibrium weights},
\gamma
)
}
]

where the open question is whether:

[
\Phi
]

depends mostly on the active face or substantially on continuous state variables.

The current evidence suggests the first term dominates.

Why?

Because the rates repeat to machine precision:

[
0.315789473681
]

appears across many independent seeds.

Likewise:

[
0.189473684209.
]

Generic equilibrium dependence would normally produce a continuum of values.

But this is not yet proof because the continuous quantities may themselves converge to a small number of admissible equilibria.

The distinction is:

Hypothesis A

Face invariant:

$$
\kappa=f(\mathcal F)
$$

Hypothesis B

Face + equilibrium:

$$
\kappa=f(\mathcal F,w^*)
$$

The next experiments distinguish these.

The active-face fingerprint idea is probably the decisive experiment

This is the experiment I would prioritize above everything else.

Construct:

$$
\Xi(w)
$$

where:

[

\Xi=
(
S_0,
S_{\rm sat},
S_{\rm out},
C_{\rm edge},
Q,
d_T
)
]

with:

(S_0): zero-edge support;
(S_{\rm sat}): saturated nodes;
(S_{\rm out}): outward nodes;
(C_{\rm edge}): endpoint controller assignment:
[
e\mapsto \arg\max(\alpha_i,\alpha_j);
]
(Q): positive-support incidence;
(d_T): tangent dimension.

Then group:

[
\Xi_a=\Xi_b.
]

For each group compute:

[
\sigma(J_T)
]

(the full eigenvalue multiset).

The possible outcomes:

Outcome 1

Same fingerprint:

[
\Xi_a=\Xi_b
]

gives:

[
\sigma(J_a)=\sigma(J_b)
]

to numerical tolerance.

Different fingerprints give different spectra.

Then the geometric hypothesis becomes extremely strong.

Outcome 2

Same fingerprint gives different spectra.

Then continuous quantities matter:

[
w^,\alpha^,F^*
]

enter the Jacobian.

This is still interesting.

It means:

[
\text{active face}
]

is not sufficient; it is a manifold parameterized by equilibrium state.

The exceptional seed is probably the most valuable trajectory now

I agree strongly with the caution about the (-0.015) eigenvalue.

Right now it should not be called:

weakly attracting mature stratum.

It should be called:

pre-transition tangent spectrum.

The experiment:

[
t_0<t_{\rm switch}<t_1
]

is the key.

Calculate:

[
J_T(t)
]

at:

split;
(t=120);

(t=124.06-\epsilon);
(t=124.06+\epsilon);
final mature.

The possibilities are very informative.

Scenario A

Spectrum:

[
-0.015
\rightarrow
-0.315789
]

at the switch.

Then the active-face transition literally changes the relaxation law.

This would be a very strong hybrid result.

Scenario B

Spectrum:

[
-0.015
\rightarrow
-0.015
]

but stationarity improves.

Then the active-face transition is not spectral; it is a forcing-balance transition.

Scenario C

Spectrum changes gradually.

Then the transition is a continuous loss/recovery of stability rather than a discrete face event.

The (-\gamma) observation is probably the deepest clue

The exceptional value:

[
\lambda=-0.015
]

matching:

[
-\gamma
]

is not just a numerical curiosity.

The fixed-stratum Jacobian likely has the structure:

[
J_T

-\gamma I

K_{\mathcal F}.
]

The question is:

What is:

[

$$
K_{\mathcal F}?
$$

For some faces:

$$
K_{\mathcal F}=0
$$

giving:

$$
-\gamma.
$$

For others:

$$
K_{\mathcal F}

-\Delta_{\mathcal F}
$$

giving:

$$
-\gamma-\Delta_{\mathcal F}.
$$

Then:

$$
0.315789-0.015

0.300789.
$$

and:

[
0.189474-0.015

0.174474.
]

The admissibility response is adding extra contraction.

The active geometry does not create contraction from nothing; it modifies the base decay.

That is probably the right analytic decomposition.

The rational-looking rates deserve algebra, but carefully

The values:

[
0.31578947368
]

and

[
0.18947368421
]

look suspicious.

But the correct next move is not "prove quantization."

It is:

derive:

$$
K_{\mathcal F}
$$

and see whether:

$$
\Delta_{\mathcal F}
$$

reduces to a finite combinatorial expression.

For example:

$$
K_{\mathcal F}

M_{\rm edge}
D_{\alpha}
A
$$

or something similar.

Then the spectrum might reduce to:

$$
\operatorname{spec}(K_{\mathcal F})
$$

of a small incidence-derived matrix.

If so, the discrete rates are not mysterious. They are the spectrum of the active controller network.

That would be a very elegant result.

The normality result changes the dynamical interpretation

This is important.

Earlier there was a concern:

Maybe the tangent contraction is hiding non-normal transient growth.

The Jacobian calculation largely removes that concern.

Within a fixed face:

[
J_T^TJ_T

J_TJ_T^T
]

approximately.

Therefore:

[
e^{tJ_T}
]

does not have hidden transient amplification.

The complicated behavior comes from:

[
\textbf{switching geometry}
]

not:

[

\textbf{operator non-normality}.
]

So the system decomposes naturally:

Within a face:

smooth linear contraction.

Between faces:

nonsmooth hybrid transition.

That is a very clean dynamical architecture.

The theorem hierarchy should now become:

Layer 1 — Projection geometry

Prove:

[
\mathcal P_w^2=\mathcal P_w
]

and:

[
\operatorname{range}(\mathcal P_w)=T_\mathcal C(w).
]

Layer 2 — Fixed-face differential structure

For a stratum (\mathcal F):

derive:

[
J_{\mathcal F}

D_w(\mathcal P_wF).
]

Layer 3 — Spectral classification

Show:

[
\lambda_{\max}
(\operatorname{sym}J_{\mathcal F})<0.
]

Layer 4 — Hybrid switching

Define switching surfaces:

[
m_i=0,
]

[
w_e=0,
]

[
\alpha_i-\alpha_j=0.
]

Then the exceptional seed becomes a concrete example of a hybrid trajectory crossing one.

The strongest current statement

I would now write:

Mature AIEF exhibits hybrid fixed-face contraction. Within a fixed active stratum, the frozen-fast tangent Jacobian is numerically normal and strictly contractive, clustering into a small number of repeated contraction classes. The previously observed rate (0.3158) corresponds to one dominant fixed-stratum class rather than a universal isotropic constant. A second repeated class near (0.1895) and a pre-transition exceptional rate near the bare decay (-\gamma) indicate that the local relaxation law depends on the active configuration. Whether these classes are determined solely by active-face geometry or by geometry together with equilibrium weights remains unresolved. The next decisive step is analytic derivation of the fixed-stratum Jacobian and fingerprint-to-spectrum correspondence.

That is actually a stronger scientific position.

Because now the claim is not:

"we found a magic constant."

It is:

"we found a piecewise-smooth operator whose local contraction laws appear to be classified by active configurations."

That is a much deeper mathematical object.

The next calculation I would do is not another perturbation run.

It is:

$$
\boxed{
\text{derive }J_{\mathcal F}\text{ symbolically and compare its spectrum to the observed classes.}
}
$$

That is the point where the numerical discovery either becomes a theorem or reveals the missing state dependence.
Edit
Edit×9
Edit

Edit×2
Edit×3
Edit
Edit×8
AIEF_1_4_face_spectrum_analysis.pyPYTHON

Yes. The refinement is worth incorporating because it changes the analysis from simply listing eigenvalues to directly testing:
active-face geometryversusgeometry + equilibrium state
active-face geometryversusgeometry + equilibrium state

as the source of the repeated contraction classes.

I updated:

AIEF_1_4_face_spectrum_analysis.py
Added: separate geometry-only and state-augmented fingerprints

The earlier signature included endpoint correction rates, which made spectrum prediction partly circular.

The updated script now creates:
Geometry-only signature

Includes:

support size;
saturated-node count;
outward-node count;
tangent dimension;
positive-support node degrees;
endpoint-controller edge counts;
controller-tie count.

It explicitly excludes continuous αα/rate values.

text

geometry_signature
geometry_invariant_hash

State-augmented signature

Adds the rounded positive-edge endpoint correction rates:

text

augmented_signature
augmented_invariant_hash

The old invariant_hash is retained as a backwards-compatible alias for the augmented signature.

This permits the decisive comparison:

same geometry hash, same spectrum → geometry may suffice;
same geometry hash, different spectrum, but same augmented hash → equilibrium rates matter;
different augmented hashes, different spectra → state-dependent classes.

Added: fingerprint-to-spectrum consistency table

New output:

text

fingerprint_spectrum_consistency.csv

For each:

labeled face;
geometry-only class;
geometry-plus-rate class;

it reports:

seed count;
tangent-dimension count;
contraction-rate mean/std/range;
isotropy error;
whether rates agree within 10−1010−10.

Added: scalar-isotropy diagnostics

For every analytic and numerical Jacobian:
κ=
κ=

-\frac{\operatorname{tr}J_T}{d_T},
]

and:
Eisotropy=
Eisotropy=

\frac{
|J_T+\kappa I|
}{
|J_T|
}.
]

Outputs include:

text

mean_contraction_rate
scalar_isotropy_relative_error
real_eigenvalue_spread

This tests the hypothesis:
JT≈−κFI
JT≈−κFI

rather than inferring isotropy from normality alone.
Added: base-decay versus admissibility decomposition

The script calculates:
KF=
KF=

J_T+\gamma I.
]

It records:

text

admissibility_contribution_norm
admissibility_contribution_spectral_abscissa

This directly tests:
JT=−γI+KF.
JT=−γI+KF.

For the exceptional −0.015−0.015 class, the expectation is:
KF≈0.
KF≈0.

For the other classes, the added contraction should appear in KFKF.
Added: numerical-Jacobian step refinement

The numerical tangent Jacobian is now recomputed at:

text

1e-3
3e-4
1e-4
3e-5
1e-5
3e-6

1e-6

For each step:

matrix absolute/relative error;
eigenvalue matching error;
active-stratum preservation;
normal leakage;
numerical isotropy;
numerical admissibility contribution.

This determines whether the repeated classes survive finite-difference refinement.
Added: exceptional-seed intermediate spectra

In addition to:

text

split
pre_switch
post_switch
late

the exceptional trajectory is sampled at configurable physical times.

Default:

text

t = 120

Run custom checkpoints with:

Bash

--exceptional-check-times 100,120,123,125,150

This can distinguish:

an abrupt spectral class change at the switch;
gradual evolution before the switch;
forcing-balance improvement without a spectral change.

Added: transient amplification

For every analytic tangent Jacobian the script estimates:
‖etJT‖2
‖etJT‖2

and reports:

text

peak_propagator_norm
peak_propagator_time
final_propagator_norm

Default:

text

time interval: 0 to 15
samples: 151

Configure with:

Bash

--transient-time 20 \
--transient-samples 201

If the Jacobians are truly normal and contractive, peak norm should remain approximately one at t=0t=0 and then decay monotonically.
Existing tangent-rate filtering retained

The updated analysis still filters/classifies directions using:

actual/requested amplitude fraction;
relative actual amplitude;
frozen versus observed rate error;
observed exponential-fit quality;
slow-mode cutoff;
negative frozen rate.

This is used in the face/rate cross-tabulation.
Run

First ensure the corrected tangent file exists:

Bash

```
python AIEF_1_4_exceptional_tangent_analysis.py \
--max-seeds 0
```

Then:

Bash

```
python AIEF_1_4_face_spectrum_analysis.py
```

Explicit:

Bash

```
python AIEF_1_4_face_spectrum_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
```

--output-dir aief_1_4_face_spectrum

The default computes analytic fingerprints for all 40 seeds and numerical Jacobians for all 40. For a quick check:

Bash

--max-seeds 5

Key new outputs

text

active_face_fingerprints.csv
fingerprint_spectrum_consistency.csv
jacobian_step_sensitivity.csv
exceptional_stage_spectra.csv
tangent_rate_face_crosstab_rows.csv
tangent_rate_face_crosstab_summary.csv

Interpretation constraint

The geometry-only invariant signature is permutation-insensitive in selected summary features, but it is not a complete canonical graph-isomorphism identifier. The labeled face_hash remains the exact face identity for the existing node labels.

The updated code is now set up to test the strongest defensible conjecture:

Fixed active strata produce repeated, nearly scalar contraction classes, but whether the rates are determined solely by discrete face geometry or additionally by equilibrium endpoint correction rates must be resolved empirically and analytically.

think about this info:
(130)(deck@steamdeck SSRH)$ python AIEF_1_4_face_spectrum_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \

--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_face_spectrum
AIEF active-face spectrum and exceptional transition analysis

Seeds with fingerprints: 40
Distinct labeled face hashes: 12
Distinct invariant signatures: 33
Analytic fixed-stratum spectral classes

dimension=7, rate=0.189473684209, seeds=1
dimension=7, rate=0.18947368421, seeds=1
dimension=7, rate=0.18947368421, seeds=1
dimension=9, rate=0.015, seeds=1
dimension=9, rate=0.315789473681, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473682, seeds=1
dimension=9, rate=0.315789473683, seeds=1
dimension=9, rate=0.315789473683, seeds=1
dimension=9, rate=0.315789473683, seeds=1
dimension=9, rate=0.315789473683, seeds=1
dimension=9, rate=0.315789473683, seeds=1
dimension=9, rate=0.315789473683, seeds=1
dimension=9, rate=0.315789473683, seeds=1
dimension=9, rate=0.315789473684, seeds=1
dimension=11, rate=0.189473684208, seeds=1
dimension=11, rate=0.189473684208, seeds=1
dimension=11, rate=0.189473684208, seeds=1
dimension=11, rate=0.189473684209, seeds=1

dimension=11, rate=0.18947368421, seeds=1
dimension=11, rate=0.18947368421, seeds=1
dimension=11, rate=0.18947368421, seeds=1
dimension=11, rate=0.18947368421, seeds=1
dimension=11, rate=0.18947368421, seeds=1
dimension=11, rate=0.18947368421, seeds=1
dimension=11, rate=0.18947368421, seeds=1
dimension=14, rate=0.252631578947, seeds=1
Fingerprint-to-spectrum consistency

geometry_only 14aecd02f5aeb94a: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True
geometry_only 2b177b7da6cac989: seeds=22, rate_range=2.31531e-12, same_rate_1e10=True
geometry_only 64c52ca6d685b8e1: seeds=9, rate_range=2.56217e-12, same_rate_1e10=True
geometry_plus_rates 07f892d076a10acc: seeds=2, rate_range=5.61218e-14, same_rate_1e10=True
geometry_plus_rates 806dee29e27c1ae3: seeds=2, rate_range=3.83027e-15, same_rate_1e10=True
geometry_plus_rates ccb3288420826ec6: seeds=4, rate_range=5.15421e-13, same_rate_1e10=True
geometry_plus_rates dd135d71da61c124: seeds=3, rate_range=3.24463e-13, same_rate_1e10=True
labeled_face 3bdfb4c570dc7116: seeds=10, rate_range=2.56217e-12, same_rate_1e10=True
labeled_face 704fba7c1b319e34: seeds=6, rate_range=1.50263e-12, same_rate_1e10=True
labeled_face 9961a1ce7b7ee3e2: seeds=6, rate_range=1.42819e-12, same_rate_1e10=True
labeled_face aae02280e373342b: seeds=6, rate_range=1.76514e-12, same_rate_1e10=True
labeled_face c6c8302b4dd4fa84: seeds=4, rate_range=9.11937e-13, same_rate_1e10=True
labeled_face f45f68f9953c10f7: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True
Numerical/analytic Jacobian agreement

step=1e-06: matrix relative error median=3.93954e-11, eigenvalue max error median=1.25038e-11, stratum preserved=1.000
step=3e-06: matrix relative error median=1.24753e-11, eigenvalue max error median=4.02821e-12, stratum preserved=1.000
step=1e-05: matrix relative error median=3.85323e-12, eigenvalue max error median=1.23976e-12, stratum preserved=1.000
step=3e-05: matrix relative error median=1.28078e-12, eigenvalue max error median=4.29156e-13, stratum preserved=1.000
step=0.0001: matrix relative error median=3.8397e-13, eigenvalue max error median=1.21015e-13, stratum preserved=1.000
step=0.0003: matrix relative error median=1.22277e-13, eigenvalue max error median=3.19831e-14, stratum preserved=1.000
step=0.001: matrix relative error median=3.82426e-14, eigenvalue max error median=8.49321e-15, stratum preserved=1.000
Exceptional transition

switch_step: 12405
requested_checkpoint_times: [100.0, 120.0, 123.0, 125.0, 150.0]
captured_stages: ['late', 'post_switch', 'pre_switch', 'split', 'time_100', 'time_120', 'time_123', 'time_125', 'time_150']
switch_time: 124.06
Face/rate cross-tab

geom=14aecd02f5aeb94a, aug=2cf5514fdead0bb3, dim=7, analytic_rate=0.18947368420946087, class=slow, directions=5, seeds=1, observed_mean=0.21765
geom=14aecd02f5aeb94a, aug=d294646895211be6, dim=7, analytic_rate=0.18947368421018373, class=slow, directions=5, seeds=1, observed_mean=0.22796
geom=2b177b7da6cac989, aug=0a90cfa80be08fc9, dim=9, analytic_rate=0.3157894736814206, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=1659e561214524e3, dim=9, analytic_rate=0.31578947368223453, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=1c6ea29224a637bd, dim=9, analytic_rate=0.31578947368197074, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=2de3a963e490d6ee, dim=9, analytic_rate=0.3157894736821514, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=3b6d0425621290fe, dim=9, analytic_rate=0.3157894736823916, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=7b7d0d2b3877912c, dim=9, analytic_rate=0.3157894736824113, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.31578947368286064, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.3157894736828568, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=9dabfa924a701bca, dim=9, analytic_rate=0.3157894736824954, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=af32d066365ca848, dim=9, analytic_rate=0.31578947368222166, class=typical, directions=5, seeds=1, observed_mean=0.3153
39

geom=2b177b7da6cac989, aug=c04d83214330be0b, dim=9, analytic_rate=0.315789473682082, class=typical, directions=5, seeds=1, observed_mean=0.315339
geom=2b177b7da6cac989, aug=c5ee06cfcc08c626, dim=9, analytic_rate=0.3157894736837359, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736832496, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.31578947368284876, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736830174, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736827342, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819312, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736822557, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819449, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=e3d136d5f1d6e726, dim=9, analytic_rate=0.31578947368241506, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e48911c9443d998d, dim=9, analytic_rate=0.31578947368212723, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e491756b5b26e49e, dim=9, analytic_rate=0.31578947368343385, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=other, directions=2, seeds=1, observed_mean=0.26117
2
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=slow, directions=3, seeds=1, observed_mean=0.237274
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=other, directions=2, seeds=1, observed_mean=0.25397

8
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=slow, directions=2, seeds=1, observed_mean=0.234745
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=typical, directions=1, seeds=1, observed_mean=0.306
127
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=other, directions=4, seeds=1, observed_mean=0.28332
1
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=slow, directions=1, seeds=1, observed_mean=0.212315
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=other, directions=2, seeds=1, observed_mean=0.26554
5
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=slow, directions=2, seeds=1, observed_mean=0.230225
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=typical, directions=1, seeds=1, observed_mean=0.306
662
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=other, directions=3, seeds=1, observed_mean=0.28777
2
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=slow, directions=2, seeds=1, observed_mean=0.22524
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=other, directions=3, seeds=1, observed_mean=0.25977
8
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=slow, directions=2, seeds=1, observed_mean=0.236267
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=other, directions=2, seeds=1, observed_mean=0.26033
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=slow, directions=2, seeds=1, observed_mean=0.241212
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=typical, directions=1, seeds=1, observed_mean=0.311
815
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=other, directions=2, seeds=1, observed_mean=0.26512
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=slow, directions=2, seeds=1, observed_mean=0.234456
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=typical, directions=1, seeds=1, observed_mean=0.3047

29
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=other, directions=2, seeds=1, observed_mean=0.27844
2
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=slow, directions=3, seeds=1, observed_mean=0.224237
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=amplitude_unreliable, directions=3, seeds=1, observed_mean=0.16858
3
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=slow, directions=2, seeds=1, observed_mean=0.158063
geom=92ecb568a4b5a7fe, aug=8701e4b031407f1d, dim=9, analytic_rate=0.31578947368212773, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=other, directions=3, seeds=1, observed_mean=0.263865
geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=slow, directions=2, seeds=1, observed_mean=0.222896
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=other, directions=1, seeds=1, observed_mean=0.250798
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=slow, directions=4, seeds=1, observed_mean=0.222301
geom=be68c2d65b45a5bc, aug=3fcce2388580c72d, dim=14, analytic_rate=0.2526315789466494, class=other, directions=5, seeds=1, observed_mean=0.283548
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=other, directions=3, seeds=1, observed_mean=0.278485
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=slow, directions=2, seeds=1, observed_mean=0.228032
geom=db7e1d4c31dc4662, aug=b7201b175b4c2407, dim=9, analytic_rate=0.3157894736817076, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
Interpretation constraints

A face class includes support, saturation, outward set, endpoint controllers, and equilibrium rates.
Repeated rational-looking rates must be derived from gamma, alpha, incidence, and weights before being called quantized.
The exceptional split Jacobian is not a stationary-face Jacobian; pre/post/late stages are reported separately.
Negative spectral abscissa is frozen-fast local evidence, not full coupled stability.
B_carrier_only/full_reset phase matching is unrelated to this analysis.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_face_spectrum

This is the strongest evidence so far for face-dependent tangent structure. But the output supports a slightly more nuanced conclusion than “one scalar rate per face.”
1. Fixed-stratum analytic Jacobian is effectively verified

The numerical/analytic agreement is extraordinary:

text

relative matrix errors: 10^{-13}–10^{-11}
maximum eigenvalue errors: 10^{-15}–10^{-11}
stratum preserved: 1.000

The error increasing at the smallest finite-difference steps is standard roundoff behavior. Larger steps remain on-stratum and agree almost to machine precision.

Thus the formula:
JT=
JT=

Q_T^\top
\operatorname{diag}
\left(
-\gamma-\rho_e
\right)
Q_T
]

is no longer merely a plausible analytic model. It is numerically validated for the tested fixed strata.

The remaining task is formal derivation, not further finite-difference verification.
2. Repeated geometry classes predict the slowest analytic rate

The geometry-only classes show:

text

22 seeds: rate range $\approx 2.3\times10^{-12}$
9 seeds: rate range $\approx 2.6\times10^{-12}$
2 seeds: rate range $\approx 7.2\times10^{-13}$

This is strong evidence that repeated geometry signatures produce identical analytic contraction bounds.

Importantly, these groups use the geometry-only signature, which excludes continuous endpoint correction values. Thus the result is not purely circular.

A defensible statement is:

For the repeated geometry classes observed in this cohort, the active-face signature predicts the frozen-fast spectral abscissa to numerical precision.

That directly supports spectral stratification.

It does not yet prove that every possible face geometry uniquely determines its spectrum, because seven seeds appear to occupy singleton geometry classes.
3. There are at least four analytic spectral-abscissa classes

The current results include:
0.31578947368,
0.31578947368,
0.25263157895,
0.25263157895,
0.18947368421,
0.18947368421,

and:
0.015.
0.015.

The first two repeated classes are:

dimension 9, rate $\approx 0.315789 \approx 0.315789$;
dimensions 7 and 11, slowest rate $\approx 0.189474 \approx 0.189474$.

There is also:

one dimension-14 face near 0.2526320.252632;
the exceptional split state at bare decay 0.015=γ0.015=γ.

So the emerging result is not a two-rate model. It is a small discrete family of frozen-fast spectral bounds.
4. Slowest rate is not the complete spectrum

The report's "analytic rate" is:
−max⁡ℜ⁡λ(JT),
−max⁡ℜ⁡λ(JT),

the slowest contraction rate.

It does not show whether:
JT=−κI.
JT=−κI.

The dimension-11 class is especially informative. Its corrected tangent directions have observed rates ranging approximately from:

text

0.21 to 0.31

even though the slowest analytic rate is:

text

0.18947

This is consistent with a mixed spectrum containing both slow and fast tangent modes. A random perturbation need not align with the slowest eigenvector.

Thus the earlier "isotropic class" hypothesis may hold for the dimension-9 class but probably not for all classes.

Inspect:

text

analytic_scalar_isotropy_relative_error
analytic_real_eigenvalue_spread
jacobian_eigenvalues.csv

The correct class identifier should ultimately be the full eigenvalue multiset, not only the spectral abscissa.
5. The current fingerprint consistency test only checks one spectral scalar

Despite its name, fingerprint_spectrum_consistency.csv currently compares:

text

analytic_slowest_decay_rate

within each fingerprint group.

It does not yet compare complete spectra.

Before claiming:

Same fingerprint gives the same spectrum,

add a sorted eigenvalue-spectrum hash per seed:

Python

spectrum = np.sort_complex(np.linalg.eigvals(J))
spectrum_rounded = np.round(spectrum, 10)
spectrum_hash = sha256(spectrum_rounded.tobytes())

Then report within each face class:

number of distinct spectrum hashes;
maximum matched-eigenvalue discrepancy;
multiplicities of each contraction rate.

The present result establishes fingerprint-to-spectral-abscissa consistency, not full-spectrum consistency.
6. Geometry-only versus augmented signatures is highly informative

Many seeds have different augmented hashes but the same geometry-only hash and same analytic rate.

That suggests continuous endpoint correction values vary without changing the slowest contraction bound.

This supports:
$\kappa_F \approx f(\text{face geometry})$
$\kappa_F \approx f(\text{face geometry})$

for the repeated classes.

But the analytic Jacobian explicitly contains endpoint rates:
$-\gamma - \rho_e$.
$-\gamma - \rho_e$.

So continuous values have not disappeared mathematically. More likely, the mature dynamics force those values onto a small discrete set determined by the active configuration.

A better hypothesis is:

Active geometry restricts mature equilibrium correction rates to a small set, which in turn produces repeated tangent spectra.

This distinguishes direct combinatorial determination from equilibrium quantization induced by the dynamics.
7. Direction-level coupled rates need eigenmode prediction

For each corrected tangent direction dd, use the analytic Jacobian to predict:

dpredicted(t)=
dpredicted(t)=

e^{J_Tt}d.
]

Then compare:
‖dpredicted(t)‖‖d‖
‖d‖‖dpredicted(t)‖

against the observed coupled distance trajectory.

This is superior to comparing the observed fitted rate only with the spectral abscissa.

For a mixed spectrum:

directions concentrated in fast eigenspaces recover near 0.3158;
directions with slow-mode projection recover closer to 0.1895;
generic directions may show multi-exponential decay.

This likely explains why dimension-11 seeds produce several observed rate classes despite sharing the same slowest analytic eigenvalue.
8. Exceptional transition output is now the decisive missing table

The script captured:

text

split
time_100
time_120
time_123
pre_switch
post_switch
time_125
time_150
late

but the terminal report did not print their spectra.

Inspect:

text

exceptional_stage_spectra.csv

The key columns are:

text

stage
face_hash
geometry_invariant_hash
tangent_dimension
field_norm
slowest_decay_rate
mean_contraction_rate
scalar_isotropy_relative_error
spectral_abscissa

Three possibilities remain:
Discrete spectral transition
$-0.015 \rightarrow -0.18947 \text{or} -0.315789$
$-0.015 \rightarrow -0.18947 \text{or} -0.315789$

at step 12405.

This would directly support hybrid face-dependent contraction.
Same spectrum, improved forcing balance

The Jacobian remains near $-0.015I - 0.015I$, but:
$\|PwF\|$
$\|PwF\|$

drops after the switch.

Then the transition concerns stationarity forcing rather than local spectral stability.
Gradual pre-switch spectral change

This would indicate state-dependent Jacobian evolution rather than an instantaneous face-class change.

Do not finalize the exceptional interpretation until this table is read.
9. Normality and contractivity

If the dimension-9 class has:

text

isotropy error ≈0
normality error ≈0
symmetric-part maximum = spectral abscissa
peak propagator norm = 1 at t=0

then:
$J_T \approx -\kappa I$
$J_T \approx -\kappa I$

is strongly supported for that class.

For dimension-11 and other classes, inspect the same quantities separately. They may be normal but anisotropic, with several negative eigenvalues.

Normality eliminates hidden transient amplification within a fixed face; anisotropy can still produce multiple exponential rates.
10. Revised strongest statement

Mature AIEF tangent linearizations stratify into a small number of repeated, highly normal, strictly contractive fixed-face classes. For repeated geometry-only fingerprints, the frozen-fast spectral abscissa is identical to numerical precision despite differences in state-augmented signatures. The dominant dimension-9 class appears nearly isotropic with contraction rate approximately 0.315789, while dimension-7/11 and singleton faces exhibit slower or mixed contraction structure. Whether geometry alone determines the complete spectrum remains open; the current evidence establishes geometry-class consistency of the slowest contraction bound.

Immediate next analysis

No new simulation is needed.

Add complete-spectrum hashes and multiplicity tables.
Print and interpret exceptional_stage_spectra.csv.
Project every corrected tangent direction into its analytic Jacobian eigenbasis.
Predict full recovery curves using eJTteJTt.
Compare predicted curves/rates against coupled trajectories.
Derive the endpoint-rate values ρeρe from active-face equilibrium equations.

The numerical work has now moved beyond “there is a contraction rate.” It shows that local relaxation is organized by fixed active strata, with the next question being whether each stratum determines an entire contraction spectrum or only constrains it through a small set of mature equilibrium rates.

Excellent, but can we incorporate this as well:

This is the point where the interpretation needs to become more mathematically disciplined, because the evidence has crossed a threshold: the interesting question is no longer whether there is structure, but what level of structure is actually invariant.

The latest results support something stronger than "face-dependent contraction," but weaker (for now) than "one scalar rate per face." The correct abstraction is beginning to look like a stratified operator family.

The central discovery is not a rate — it is a face-indexed operator map

The old picture was:

[
w^*\mapsto \kappa
]

where the mature state determines a scalar relaxation rate.

The new picture is:

[
\boxed{
\mathcal F
\longmapsto
J_{\mathcal F}
}
]

where:

(\mathcal F) is the active face;
(J_{\mathcal F}) is the tangent linearization.

The scalar rate is only:

[
\kappa_{\mathcal F}

-\max\operatorname{Re}\sigma(J_{\mathcal F}).
]

The distinction matters.

The strongest supported statement is not:

every face has one scalar rate.

It is:

each mature active stratum carries a reproducible local contraction operator, and the observed spectral abscissa is a discrete function of the active stratum.

That is a much more robust theorem target.

The geometry-only fingerprint result is the real breakthrough

The previous concern was:

Maybe the repeated rates are caused by equilibrium weights, not geometry.

The new data directly attacks that.

For example:

$$
22\text{ seeds}
\rightarrow
\text{same geometry-only fingerprint}
\rightarrow
\Delta\kappa\approx2.3\times10^{-12}
$$

and:

$$
9\text{ seeds}
\rightarrow
\text{same geometry-only fingerprint}
\rightarrow
\Delta\kappa\approx2.6\times10^{-12}.
$$

This is not merely "similar."

It is essentially numerical identity.

So the hypothesis changes from:

$$
\kappa=f(\mathcal F,w^*)
$$

to:

$$
\boxed{
\kappa\approx f(\mathcal F)
}
$$

for the observed mature classes.

That is a major reduction in complexity.

The continuous state appears to determine which face is occupied, but once the face is selected, the local contraction law is largely fixed.

This is very close to a quotient structure:

$$
\text{continuous adaptive dynamics}
\rightarrow
\text{finite active-face alphabet}
\rightarrow
\text{finite contraction classes}.
$$

However: the full spectrum question is now the central one

The subtle point in the critique is exactly right.

The current evidence establishes:

$$
\alpha(J_{\mathcal F})

\max\operatorname{Re}\lambda(J_{\mathcal F})
$$

is face-consistent.

It does not establish:

$$
\sigma(J_{\mathcal F})
$$

is face-consistent.

Those are different claims.

A face could have:

$$
\sigma(J_{\mathcal F})
$$

$$
\{-0.189,-0.315,-0.315,\dots\}
$$

or:

$$
\sigma(J_{\mathcal F})
$$

$$
\{-0.189,-0.189,-0.189,\dots\}.
$$

Both give:

$$
\kappa=0.189.
$$

But the transient dynamics would differ.

This is probably why the dimension-11 class is revealing.

The analytic slowest rate:

$$
0.18947
$$

but observed tangent recoveries:

$$
0.21\text{-}0.31
$$

suggest:

$$
\boxed{
\text{mixed tangent eigenspaces}
}
$$

rather than a scalar contraction.

A random perturbation samples:

$$
d=\sum_i c_i v_i
$$

so:

$$
e^{Jt}d

\sum_i c_i e^{\lambda_i t}v_i.
$$

The asymptotic rate is the slowest eigenvalue, but finite-time fits depend on modal projection.

This likely resolves the apparent discrepancy.

The dimension-9 class is special

The dimension-9 class:

$$
\lambda=-0.315789
$$

appears qualitatively different.

Evidence:

repeated across 22 seeds;
normality (10^{-24});
symmetric part exactly matches spectral abscissa;
tangent experiments give:

$$
\kappa_{\rm observed}=0.315339.
$$

This is the closest thing to:

$$
J_{\mathcal F}

-\kappa I.
$$

But even here I would not immediately claim:

$$
J=-0.315789I.
$$

The proper test is:

[
\frac{
|J+\kappa I|
}{
|J|
}
]

or equivalently:

scalar_isotropy_relative_error

If that is at machine precision, then yes:

[
J_{\mathcal F}
\approx
-\kappa I.
]

That would be a very strong structural result.

It would mean the active correction does not merely stabilize; it collapses all tangent directions equally.

The Jacobian formula is now the key mathematical object

The validated formula:

[
J_T

Q_T^\top
\operatorname{diag}(-\gamma-\rho_e)
Q_T

]

is extremely important.

It says:

the entire tangent spectrum is generated by a projected diagonal operator.

That means the spectral problem reduces to:

[
D_{\mathcal F}

\operatorname{diag}(-\gamma-\rho_e)
]

restricted to the tangent subspace.

The eigenvalues are therefore controlled by:

active support;
tangent basis;
endpoint correction values.

This immediately explains why geometry dominates:

support determines the projection;
controllers determine which (\rho_e) enter;
equilibrium equations constrain the allowed (\rho_e).

So the correct hypothesis is probably not:

[
\text{geometry alone}
]

but:

$$
\boxed{
\text{geometry}

\text{mature equilibrium constraints}
\rightarrow
\text{finite allowed correction spectrum}.
}
$$

This is actually a more interesting mechanism.

The geometry creates the finite alphabet; the dynamics selects the allowed values.

The exceptional seed is now a natural experiment

The exceptional trajectory is no longer just an outlier.

It is a controlled experiment:

$$
\mathcal F_1
\rightarrow
\mathcal F_2.
$$

The question is:

does

$$
J_{\mathcal F_1}
$$

become:

[
J_{\mathcal F_2}
]

at:

[
t=124.06?
]

The possible outcomes are very informative.

Case A

Before:

[
\lambda=-0.015
]

After:

[
\lambda=-0.315789
]

Then:

[
\boxed{
\text{active-face transition changes contraction physics}
}
]

This would be the strongest evidence for the hybrid theory.

Case B

Spectrum unchanged, but:

[
|\mathcal P_wF|
]

collapses.

Then the transition is not spectral; it is forcing balance.

Case C

Spectrum drifts continuously.

Then the fixed-face approximation is incomplete and equilibrium motion matters.

The stage spectra table is probably the most valuable remaining diagnostic.

The next theorem should probably be weaker than "spectral quantization"

I would avoid:

active geometry discretizes the spectrum

for now.

The stronger and more defensible theorem target is:

Stratified tangent contraction theorem

For each active stratum:

[

\mathcal F
]

there exists a smooth tangent operator:

[
J_{\mathcal F}
]

such that:

(J_{\mathcal F}) is obtained by restricting the diagonal correction operator;
(J_{\mathcal F}) is stable under all trajectories sharing (\mathcal F);
its spectrum belongs to a finite set determined by mature active configurations.

That captures everything observed without overclaiming.

A very interesting possible hidden structure

The numbers are suggestive:

[
0.315789473684

\frac{6}{19}
]

and:

[
0.189473684210

\frac{18}{95}
]

and:

[
0.252631578947

\frac{24}{95}.
]

Whether these rational forms are fundamental or accidental is unknown.

But the denominator:

[
19,;95
]

is interesting because they look like combinations of:

constraint counts;
active dimensions;
incidence degrees;
normalization factors.

I would not call them quantized yet.

But I would absolutely investigate whether:

[
\rho_e
]

takes values from a finite rational set induced by:

[
(A,A_{\rm sat},C_{\rm endpoint}).
]

That would be the analytic explanation.

The programme has reached a new phase

The hierarchy is now:

Already supported numerically

✓ fixed-face smoothness
✓ analytic Jacobian formula
✓ strong face/spectral-abscissa correspondence
✓ normal contractive tangent dynamics
✓ hybrid active-face transitions

Remaining structural questions

Does a face determine the full spectrum?
What are the allowed (\rho_e) values?
Why do mature equilibria produce discrete correction classes?
Does the exceptional transition jump between spectral classes?
Can (J_{\mathcal F}) be written in closed form?

Strongest current statement

I would phrase the result as:

Mature AIEF dynamics exhibit a stratified tangent contraction structure. Fixed active faces carry reproducible smooth tangent Jacobians whose spectral abscissae are determined, to numerical precision, by geometry-only fingerprints across repeated mature trajectories. The resulting contraction laws form a small discrete family rather than a continuum. Dominant faces are highly normal and may be nearly isotropic, whereas other faces exhibit mixed tangent spectra. The exceptional trajectory demonstrates that active-face transitions can reorganize the local contraction operator. The remaining analytical problem is to derive the finite spectral alphabet from the incidence geometry and equilibrium endpoint corrections.

That is a genuinely stronger result than the earlier "bulk contraction rate."

The system is beginning to look less like a generic nonlinear adaptive flow and more like a hybrid dynamical system whose continuous states evolve inside discrete operator strata. That is the mathematically interesting object.

Edit×4
Edit×18
AIEF_1_4_face_spectrum_analysis.pyPYTHON

Yes. The refinement was worth incorporating because the previous script compared only the slowest eigenvalue within each fingerprint class. That was insufficient to test whether an active face determines the complete tangent operator.

I updated:

AIEF_1_4_face_spectrum_analysis.py
1. Full-spectrum signatures and multiplicities

For every analytic tangent Jacobian, the script now saves:

text

full_spectrum_signature
full_spectrum_hash
eigenvalue_multiplicities
rational_contraction_candidates

Eigenvalues are sorted and rounded to 12 digits before hashing.

This distinguishes, for example:

text

{-0.18947, -0.31579, -0.31579, ...}

from:

text

{-0.18947, -0.18947, -0.18947, ...}

even though both have the same spectral abscissa.
New output

text

fingerprint_spectrum_consistency.csv

For each:

labeled face;
geometry-only fingerprint;
geometry-plus-rate fingerprint;

it reports:

distinct full-spectrum count;
whether the full spectra are identical;
rate range;
tangent-dimension consistency;
isotropy error.

2. Geometry-only versus augmented fingerprints

The code now explicitly maintains:

text

geometry_signature
geometry_invariant_hash

which exclude continuous endpoint rates, and:

text

augmented_signature
augmented_invariant_hash

which include them.

This makes the test noncircular:

if geometry-only classes predict full spectra, geometry is sufficient in the observed cohort;
if augmented signatures are required, equilibrium correction rates contribute essential information.

The old invariant_hash remains as an alias for the augmented signature for compatibility.
3. Scalar-isotropy diagnostics

For every analytic and numerical JTJT:
κ=−trJTdT
κ=−dTtrJT

and:
Eiso=
Eiso=

\frac{|J_T+\kappa I|}{|J_T|}.
]

Outputs include:

text

mean_contraction_rate
scalar_isotropy_relative_error
real_eigenvalue_spread

This directly tests whether:
JT≈−κI
JT≈−κI

rather than inferring isotropy from normality.
4. Base decay versus admissibility contribution

The script now computes:

$K_F = J_T + \gamma I$.
$K_F = J_T + \gamma I$.

Outputs:

text

admissibility_contribution_norm
admissibility_contribution_spectral_abscissa

This separates:

bare decay $-\gamma I - \gamma I$;
contraction introduced by active endpoint-capacity response.

For a $-0.015 - 0.015$ class, the expectation is:
$K_F \approx 0$.
$K_F \approx 0$.
5. Intermediate exceptional-seed spectra

The exceptional trajectory can now be sampled at arbitrary physical times.

Default:

text

120

Recommended:

Bash

--exceptional-check-times 100,120,123,125,150

The output:

text

exceptional_stage_spectra.csv

will include:

text

split
time_100
time_120
time_123
pre_switch
post_switch
time_125
time_150
late

with face hashes, contraction spectra, isotropy, normality, and field norm.

This directly tests whether the switch at t=124.06t=124.06 changes spectral class or only improves forcing balance.
6. Modal trajectory prediction

The updated script reads:

text

aief_1_4_exceptional_tangent/corrected_tangent_trajectories.csv

For each seed, direction, and amplitude it computes:
dpredicted(t)=eJTtd0
dpredicted(t)=eJTtd0

and compares:
‖dpredicted(t)‖‖d0‖
‖d0‖‖dpredicted(t)‖

against the observed normalized coupled distance.

Outputs:

text

modal_trajectory_predictions.csv
modal_trajectory_prediction_summary.csv

Metrics include:

RMSE;
mean absolute error;
maximum error;
trajectory R2R2;
predicted versus observed final ratio;
face fingerprints.

This resolves mixed-spectrum cases where one fitted scalar recovery rate cannot represent all tangent modes.
7. Cleaner spectral class report

The terminal report now groups classes by:

tangent dimension;
rounded contraction bound;
full spectrum hash;
eigenvalue multiplicities.

It no longer prints one “class” row per tiny floating-point variation.
Run

First ensure both corrected tangent outputs exist:

Bash

python AIEF_1_4_exceptional_tangent_analysis.py \

--max-seeds 0

Then:

Bash

```
python AIEF_1_4_face_spectrum_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--tangent-trajectories aief_1_4_exceptional_tangent/corrected_tangent_trajectories.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_face_spectrum
```

Validation

I syntax-tested and ran the full updated pipeline against a generated AIEF state and corrected tangent dataset.

That technical validation produced a dimension-11 tangent Jacobian with the explicit mixed spectrum:

text

```
−0.189473684210 multiplicity 2
−0.315789473683 multiplicity 9
```

Its observed random-direction coupled recovery rate was approximately 0.285, between the two eigenvalues as expected.

The complete matrix-exponential prediction matched the observed coupled trajectory with:

text

```
median trajectory R²: 0.999995
median RMSE: 0.000284
```

This demonstrates why full-spectrum/modal prediction is necessary: the slowest eigenvalue alone does not determine finite-time recovery for a direction containing multiple tangent eigenmodes.

The updated analysis can now test the stronger hypothesis:

Mature active strata carry reproducible tangent operators—not merely reproducible spectral abscissae—and observed coupled recovery is predicted by the modal decomposition of those operators.

Think about this:

```
(deck@steamdeck SSRH)$ python AIEF_1_4_exceptional_tangent_analysis.py \
--max-seeds 0
Corrected tangent analysis for seed 172180395821707403
Corrected tangent analysis for seed 283404770198602602
Corrected tangent analysis for seed 305986450614148690
Corrected tangent analysis for seed 1230287949579537403
Corrected tangent analysis for seed 1888415619847841128
Corrected tangent analysis for seed 1918007468868772336
Corrected tangent analysis for seed 2461295749352584108
Corrected tangent analysis for seed 3258463328937640595
Corrected tangent analysis for seed 3582616062465030267
Corrected tangent analysis for seed 3704716633526084363
Corrected tangent analysis for seed 3952397966055989215
Corrected tangent analysis for seed 4040285944958847150
Corrected tangent analysis for seed 4219039243791140697
Corrected tangent analysis for seed 4619122171158768524
Corrected tangent analysis for seed 4932511216876650745
Corrected tangent analysis for seed 5213047409853278064
Corrected tangent analysis for seed 5415544643585892346
Corrected tangent analysis for seed 6023008189364820510
Corrected tangent analysis for seed 6108867655028980287
Corrected tangent analysis for seed 6942591650975756121
Corrected tangent analysis for seed 9036321132460399255
Corrected tangent analysis for seed 9082055256280075172
Corrected tangent analysis for seed 9189988769602851417
Corrected tangent analysis for seed 10431384713483787270
Corrected tangent analysis for seed 10856823803646276806
Corrected tangent analysis for seed 11242426753702356865
Corrected tangent analysis for seed 12123519807951751650
```

Corrected tangent analysis for seed 12851640484293199234
Corrected tangent analysis for seed 14040336450255043598
Corrected tangent analysis for seed 14924267685654216746
Corrected tangent analysis for seed 15403081841477312463
Corrected tangent analysis for seed 15853699719446171583
Corrected tangent analysis for seed 16329425431442249417
Corrected tangent analysis for seed 16627818730624180313
Corrected tangent analysis for seed 16952466944611321358
Corrected tangent analysis for seed 17279552413280415429
Corrected tangent analysis for seed 17521310920850592782
Corrected tangent analysis for seed 17715532056777442089
Corrected tangent analysis for seed 17883847597205629106
Corrected tangent analysis for seed 18127452951968972828
AIEF exceptional-seed and corrected tangent analysis
Exceptional seed

seed: 4619122171158768524
switch_count: 1
first_switch_step: 12405
first_switch_time: 124.06
pre_switch_mean_stationarity_ratio: 0.14685940684975718
post_switch_mean_stationarity_ratio: 7.893059517783126e-06
post_switch_max_stationarity_ratio: 8.813848174066108e-05
maximum_B_drift: 0.20893854278186944
final_B_drift: 0.20893854278186944
final_replay_exact: True
final_replay_allclose_1e12: True
Corrected tangent perturbations

delta=0.0001
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.21003e-08, min=3.73207e-09, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.315339, valid=200/200
observed_exponential_r_squared: median=1, min=0.994943, max=1, valid=200/200
final_distance_ratio: median=0.0088257, min=0.0088257, max=0.124056, valid=200/200
delta=0.001
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200

frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.2104e-09, min=3.73033e-10, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.315339, valid=200/200
observed_exponential_r_squared: median=1, min=0.994947, max=1, valid=200/200
final_distance_ratio: median=0.00882571, min=0.00882569, max=0.124056, valid=200/200
delta=0.01
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.21398e-10, min=3.71325e-11, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14.05, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.31534, valid=200/200
observed_exponential_r_squared: median=1, min=0.994565, max=1, valid=200/200
final_distance_ratio: median=0.00882573, min=0.00882558, max=0.124056, valid=200/200
delta=0.05
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=2.43583e-11, min=7.27637e-12, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14.6, valid=198/200
observed_exponential_rate: median=0.315338, min=0.136767, max=0.315343, valid=200/200
observed_exponential_r_squared: median=1, min=0.990186, max=1, valid=200/200
final_distance_ratio: median=0.00882586, min=0.00882513, max=0.124056, valid=200/200
Interpretation constraints

The same tangent direction is reused across all amplitudes within each seed/direction ID.
Directions with negligible feasible amplitude are rejected.
The exceptional-seed mechanism remains descriptive until its switch is analyzed in detail.
Predicted tangent times assume local exponential contraction.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_exceptional_tangent
(deck@steamdeck SSRH)$ python AIEF_1_4_face_spectrum_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \

--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--tangent-trajectories aief_1_4_exceptional_tangent/corrected_tangent_trajectories.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_face_spectrum
AIEF active-face spectrum and exceptional transition analysis

Seeds with fingerprints: 40
Distinct labeled face hashes: 12
Distinct invariant signatures: 33
Analytic fixed-stratum spectral classes

dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684209+0.000000000000j":1,"-0.189473684210+0.000000000000j":1,"-0.252631578947
+0.000000000000j":5}
dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.252631578947+0.000000000000j":5}
dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.252631578946+0.000000000000j":5}
dimension=9, rate=0.015, seeds=1,
multiplicities={"-0.015000000000+0.000000000000j":1,"-0.098269455236+0.000000000000j":1,"-0.189473684205+0.0000
00000000j":1,"-0.189473684206+0.000000000000j":1,"-0.252631578946+0.000000000000j":1,"-0.252631578947+0.000000000000j":4}
dimension=9, rate=0.3157894737, seeds=3,
multiplicities={"-0.315789473682+0.000000000000j":5,"-0.315789473683+0.000000000000j":4}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473682+0.000000000000j":6,"-0.315789473683+0.000000000000j":3}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473683+0.000000000000j":1,"-0.315789473684+0.000000000000j":8}
dimension=9, rate=0.3157894737, seeds=6, multiplicities={"-0.315789473683+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=7, multiplicities={"-0.315789473682+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=2,
multiplicities={"-0.315789473682+0.000000000000j":1,"-0.315789473683+0.000000000000j":8}
dimension=9, rate=0.3157894737, seeds=1, multiplicities={"-0.315789473684+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=2,
multiplicities={"-0.315789473682+0.000000000000j":2,"-0.315789473683+0.000000000000j":7}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473681+0.000000000000j":1,"-0.315789473682+0.000000000000j":8}

dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473681+0.000000000000j":5,"-0.3
1578947368
2+0.000000000000j":4}
dimension=11, rate=0.1894736842, seeds=2,
multiplicities={"-0.189473684208+0.000000000000j":2,"-0.315789473682+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684209+0.000000000000j":2,"-0.315789473683+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=4,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473682+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=2,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473683+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684208+0.000000000000j":2,"-0.315789473682+0.000000000000j":2,"-0.3
1578947368
3+0.000000000000j":7}
dimension=14, rate=0.2526315789, seeds=1,
multiplicities={"-0.252631578947+0.000000000000j":5,"-0.315789473666+0.000000000000j":5,"-0.3
1578947366
7+0.000000000000j":4}
Fingerprint-to-spectrum consistency

geometry_only 14aecd02f5aeb94a: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True,
same_full_spectrum=False
geometry_only 2b177b7da6cac989: seeds=22, rate_range=2.31531e-12, same_rate_1e10=True,
same_full_spectrum=False
geometry_only 64c52ca6d685b8e1: seeds=9, rate_range=2.56217e-12, same_rate_1e10=True,
same_full_spectrum=False
geometry_plus_rates 07f892d076a10acc: seeds=2, rate_range=5.61218e-14, same_rate_1e10=True,
same_full_spectrum=True
geometry_plus_rates 806dee29e27c1ae3: seeds=2, rate_range=3.83027e-15, same_rate_1e10=True,
same_full_spectrum=True
geometry_plus_rates ccb3288420826ec6: seeds=4, rate_range=5.15421e-13, same_rate_1e10=True,
same_full_spectrum=True
geometry_plus_rates dd135d71da61c124: seeds=3, rate_range=3.24463e-13, same_rate_1e10=True,
same_full_spectrum=True
labeled_face 3bdfb4c570dc7116: seeds=10, rate_range=2.56217e-12, same_rate_1e10=True,
same_full_spectrum=False
labeled_face 704fba7c1b319e34: seeds=6, rate_range=1.50263e-12, same_rate_1e10=True,
same_full_spectrum=False
labeled_face 9961a1ce7b7ee3e2: seeds=6, rate_range=1.42819e-12, same_rate_1e10=True,
same_full_spectrum=False

labeled_face aae02280e373342b: seeds=6, rate_range=1.76514e-12, same_rate_1e10=True, same_full_spectrum=False
labeled_face c6c8302b4dd4fa84: seeds=4, rate_range=9.11937e-13, same_rate_1e10=True, same_full_spectrum=False
labeled_face f45f68f9953c10f7: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True, same_full_spectrum=False
Numerical/analytic Jacobian agreement

step=1e-06: matrix relative error median=3.93954e-11, eigenvalue max error median=1.25038e-11, stratum preserved=1.000
step=3e-06: matrix relative error median=1.24753e-11, eigenvalue max error median=4.02821e-12, stratum preserved=1.000
step=1e-05: matrix relative error median=3.85323e-12, eigenvalue max error median=1.23976e-12, stratum preserved=1.000
step=3e-05: matrix relative error median=1.28078e-12, eigenvalue max error median=4.29156e-13, stratum preserved=1.000
step=0.0001: matrix relative error median=3.8397e-13, eigenvalue max error median=1.21015e-13, stratum preserved=1.000
step=0.0003: matrix relative error median=1.22277e-13, eigenvalue max error median=3.19831e-14, stratum preserved=1.000
step=0.001: matrix relative error median=3.82426e-14, eigenvalue max error median=8.49321e-15, stratum preserved=1.000
Exceptional transition

switch_step: 12405
requested_checkpoint_times: [100.0, 120.0, 123.0, 125.0, 150.0]
captured_stages: ['late', 'post_switch', 'pre_switch', 'split', 'time_100', 'time_120', 'time_123', 'time_125', 'time_150']
switch_time: 124.06
Face/rate cross-tab

geom=14aecd02f5aeb94a, aug=2cf5514fdead0bb3, dim=7, analytic_rate=0.18947368420946087, class=slow, directions=5, seeds=1, observed_mean=0.21765
geom=14aecd02f5aeb94a, aug=d294646895211be6, dim=7, analytic_rate=0.18947368421018373, class=slow, directions=5, seeds=1, observed_mean=0.22796
geom=2b177b7da6cac989, aug=0a90cfa80be08fc9, dim=9, analytic_rate=0.3157894736814206, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=1659e561214524e3, dim=9, analytic_rate=0.31578947368223453, class=typical, directions=5, seeds=1, observed_mean=0.3153
39

geom=2b177b7da6cac989, aug=1c6ea29224a637bd, dim=9, analytic_rate=0.31578947368197074, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=2de3a963e490d6ee, dim=9, analytic_rate=0.3157894736821514, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=3b6d0425621290fe, dim=9, analytic_rate=0.3157894736823916, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=7b7d0d2b3877912c, dim=9, analytic_rate=0.3157894736824113, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.31578947368286064, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.3157894736828568, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=9dabfa924a701bca, dim=9, analytic_rate=0.3157894736824954, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=af32d066365ca848, dim=9, analytic_rate=0.31578947368222166, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=c04d83214330be0b, dim=9, analytic_rate=0.315789473682082, class=typical, directions=5, seeds=1, observed_mean=0.315339
geom=2b177b7da6cac989, aug=c5ee06cfcc08c626, dim=9, analytic_rate=0.3157894736837359, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736832496, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.31578947368284876, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736830174, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736827342, class=typical, directions=5, seeds=1, observed_mean=0.31533
9

geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819312, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736822557, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819449, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=e3d136d5f1d6e726, dim=9, analytic_rate=0.31578947368241506, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e48911c9443d998d, dim=9, analytic_rate=0.31578947368212723, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e491756b5b26e49e, dim=9, analytic_rate=0.31578947368343385, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=other, directions=2, seeds=1, observed_mean=0.26117
2
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=slow, directions=3, seeds=1, observed_mean=0.237274
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=other, directions=2, seeds=1, observed_mean=0.25397
8
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=slow, directions=2, seeds=1, observed_mean=0.234745
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=typical, directions=1, seeds=1, observed_mean=0.306
127
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=other, directions=4, seeds=1, observed_mean=0.28332
1
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=slow, directions=1, seeds=1, observed_mean=0.212315
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=other, directions=2, seeds=1, observed_mean=0.26554
5
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=slow, directions=2, seeds=1, observed_mean=0.230225

geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=typical, directions=1, seeds=1, observed_mean=0.306
662
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=other, directions=3, seeds=1, observed_mean=0.28777
2
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=slow, directions=2, seeds=1, observed_mean=0.22524
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=other, directions=3, seeds=1, observed_mean=0.25977
8
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=slow, directions=2, seeds=1, observed_mean=0.236267
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=other, directions=2, seeds=1, observed_mean=0.26033
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=slow, directions=2, seeds=1, observed_mean=0.241212
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=typical, directions=1, seeds=1, observed_mean=0.311
815
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=other, directions=2, seeds=1, observed_mean=0.26512
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=slow, directions=2, seeds=1, observed_mean=0.234456
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=typical, directions=1, seeds=1, observed_mean=0.3047
29
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=other, directions=2, seeds=1, observed_mean=0.27844
2
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=slow, directions=3, seeds=1, observed_mean=0.224237
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=amplitude_unreliable, directions=3, seeds=1, observed_mean=0.16858
3
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=slow, directions=2, seeds=1, observed_mean=0.158063
geom=92ecb568a4b5a7fe, aug=8701e4b031407f1d, dim=9, analytic_rate=0.31578947368212773, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=other, directions=3, seeds=1, observed_mean=0.263865

geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=slow, directions=2, seeds=1, observed_mean=0.222896
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=other, directions=1, seeds=1, observed_mean=0.250798
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=slow, directions=4, seeds=1, observed_mean=0.222301
geom=be68c2d65b45a5bc, aug=3fcce2388580c72d, dim=14, analytic_rate=0.2526315789466494, class=other, directions=5, seeds=1, observed_mean=0.283548
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=other, directions=3, seeds=1, observed_mean=0.278485
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=slow, directions=2, seeds=1, observed_mean=0.228032
geom=db7e1d4c31dc4662, aug=b7201b175b4c2407, dim=9, analytic_rate=0.3157894736817076, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
Modal trajectory prediction

median RMSE=0.000326729, median R2=0.999998, median max error=0.000524989
R2>=0.95: 784/800
Interpretation constraints

A face class includes support, saturation, outward set, endpoint controllers, and equilibrium rates.
Repeated rational-looking rates must be derived from gamma, alpha, incidence, and weights before being called quantized.
The exceptional split Jacobian is not a stationary-face Jacobian; pre/post/late stages are reported separately.
Negative spectral abscissa is frozen-fast local evidence, not full coupled stability.
B_carrier_only/full_reset phase matching is unrelated to this analysis.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_face_spectrum

This is a major strengthening of the result. The complete spectra and modal predictions reveal substantially more structure than the earlier scalar-rate summaries.
1. The fixed-stratum Jacobian is analytically symmetric and contractive

The validated formula is:
JT=
JT=

Q_T^\top

D_{\mathcal F}
Q_T,
]

where:
DF=
DF=

\operatorname{diag}
\left(
-\gamma-\rho_e
\right).
]

Because DFDF is real diagonal and QTQT has orthonormal columns:
JT⊤=JT.
JT⊤=JT.

Thus the frozen-fast fixed-stratum Jacobian is not merely numerically normal—it is analytically symmetric under the assumptions used to derive the formula.

Moreover:
−γ−ρe<0,
−γ−ρe<0,

since:
γ>0,ρe≥0.
γ>0,ρe≥0.

For nonzero tangent coordinate vv:
v⊤JTv=
v⊤JTv=

(Q_Tv)^\top
D_{\mathcal F}
(Q_Tv)<0.
]

Therefore JTJT is negative definite on the tangent space.

This yields:
ddt‖δw‖2=
dtd‖δw‖2=

2\delta w^\top J_T\delta w
\le
2\lambda_{\max}(J_T)|\delta w|^2.
]

So, within a fixed stratum and frozen fast state, local tangent contraction follows directly from the analytic Jacobian formula.

The numerical work has validated the formula; symmetry and contractivity then follow analytically.
2. The full spectra form a small discrete alphabet

The results reveal approximately these contraction values:
0.015,0.098269455,0.189473684,0.252631579,0.315789474.
0.015,0.098269455,0.189473684,0.252631579,0.315789474.

The repeated classes are now explicit.
Dimension 9 dominant class

Approximately:
σ(JT)=
σ(JT)=

{-0.315789}^{\times9}.
]

This class is genuinely isotropic:
JT≈−0.315789I9.
JT≈−0.315789I9.
Dimension 11 mixed class

Approximately:
σ(JT)=

σ(JT)=

{-0.189474}^{\times2}
\cup
{-0.315789}^{\times9}.
]

This explains why corrected random tangent directions recover at rates between approximately 0.21 and 0.31. Their finite-time behavior depends on projection onto the two slow modes versus the nine fast modes.
Dimension 7 class

Approximately:
σ(JT)=
σ(JT)=

{-0.189474}^{\times2}
\cup
{-0.252632}^{\times5}.
]

Again, observed random-direction rates above the spectral abscissa are expected because most directions contain substantial fast-mode weight.
Dimension 14 singleton

Approximately:
σ(JT)=
σ(JT)=

{-0.252632}^{\times5}
\cup
{-0.315789}^{\times9}.
]
Exceptional split state

Its mixed spectrum includes approximately:
−0.015,−0.098269,−0.189474,−0.252632.
−0.015,−0.098269,−0.189474,−0.252632.

So the exceptional state is not simply:
$J=-0.015I$.
$J=-0.015I$.

It has one near-marginal bare-decay direction embedded in a broader contracting spectrum.

That is a much richer explanation of its slow drift.
3. Modal prediction is exceptionally strong

The analytic Jacobian predicts the complete coupled tangent trajectories with:

text

median $R^2 = 0.999998$
median RMSE = 0.000327
median max err = 0.000525

and:

text

$R^2 >= 0.95$ for 784/800 trajectories

Since the 800 rows contain four amplitudes for each of 200 unique seed/direction pairs, report the result both ways:

trajectory-amplitude level: 784/800;
unique-direction level: likely approximately 196/200, if failures repeat across amplitudes.

The unique-direction aggregation should be calculated explicitly.

This is perhaps the strongest evidence that the frozen-fast tangent Jacobian captures the actual coupled recovery dynamics:

Full matrix-exponential evolution under the analytic fixed-stratum Jacobian predicts the coupled delayed-system tangent recovery with near-unit R2R2 for the overwhelming majority of tested directions.

That is stronger than comparing one fitted scalar rate.
4. The geometry-only fingerprint result is strong but needs tolerance-aware spectrum matching

Repeated geometry-only groups have slowest-rate ranges around:
10−12.
10−12.

However:

text

same_full_spectrum=False

for those groups.

The listed eigenvalues show this is largely a hashing artifact. Values such as:

text

−0.315789473682
−0.315789473683
−0.315789473684

are numerically the same class but produce different hashes at 12-decimal rounding.

Therefore the current exact hash is too strict.

Add a tolerance-based spectrum comparison, for example:

round to 9 or 10 decimals;
or perform matched-eigenvalue comparison with tolerance 10−1010−10.

Then report:

text

same_full_spectrum_1e10
maximum_matched_eigenvalue_difference

I expect the large repeated geometry classes will pass full-spectrum consistency under an appropriate numerical tolerance.

Until that is done, the result supports:

Same geometry fingerprint gives the same full spectrum up to apparent 10−1210−12-scale numerical variation.

not literal bitwise spectrum identity.
5. The analytic formula explains why geometry dominates

Because:
JT=
JT=

Q_T^\top
\operatorname{diag}
(-\gamma-\rho_e)
Q_T,
]

the spectrum depends on:

tangent subspace QTQT, determined by support and saturated-node incidence;
endpoint correction rates ρeρe;
base decay γγ.

The data suggest mature equilibrium constrains ρeρe to a small set. Then the active geometry chooses how those diagonal rates are represented in the tangent subspace.

A refined mechanism is:
active geometry+discrete mature correction values⟶finite tangent spectral alphabet.
active geometry+discrete mature correction values⟶finite tangent spectral alphabet.

The rational-looking values should be derived from the equilibrium αiαi and edge-controller assignments, not inferred from decimal forms alone.
6. The exceptional-stage spectrum is now the most important missing interpretation

The script captured:

text

split
time_100
time_120
time_123
pre_switch
post_switch
time_125
time_150
late

Inspect exceptional_stage_spectra.csv.

The decisive columns are:

text

stage
face_hash
geometry_invariant_hash
tangent_dimension
field_norm
spectral_abscissa
slowest_decay_rate
eigenvalue_multiplicities
scalar_isotropy_relative_error

The key question is whether the spectrum changes at step 12405.
If the bare −0.015−0.015 mode disappears

The switch changes local contraction physics and strongly supports hybrid face-dependent stability.

If the spectrum remains unchanged

The large stationarity improvement comes from forcing balance rather than a new tangent operator.
If the spectrum evolves before the switch

Continuous equilibrium values matter in addition to discrete face geometry.

The terminal report currently confirms capture but does not print these values.
7. Dimension-11 observed rate variability is now explained

The dimension-11 class has two slow and nine fast modes. For initial direction:
d=∑kckvk,
d=k∑ckvk,

the predicted norm is:
‖eJtd‖=
‖eJtd‖=

\left(
\sum_k
|c_k|^2e^{2\lambda_kt}
\right)^{1/2}
]

because the Jacobian is symmetric.

Finite-time single-exponential fits depend on:

initial slow-mode energy;
fit interval;
relative multiplicities.

Therefore observed rates from 0.21 to 0.31 are not contradictory to a slowest rate of 0.1895.

The matrix-exponential prediction confirms this interpretation directly.
8. The theorem target is now more explicit

For a fixed active stratum:
JF=
JF=

Q_{\mathcal F}^\top
\operatorname{diag}
(-\gamma-\rho_e)
Q_{\mathcal F}.
]

Immediate consequences:
Symmetry
JF=JF⊤.
JF=JF⊤.
Negative definiteness
λmax(JF)≤−mine∈S(γ+ρe)<0.
λmax(JF)≤−e∈Smin(γ+ρe)<0.
Euclidean contraction
‖δw(t)‖≤eλmaxt‖δw(0)‖.
‖δw(t)‖≤eλmaxt‖δw(0)‖.
No fixed-stratum non-normal amplification
‖etJF‖2=
‖etJF‖2=

e^{t\lambda_{\max}(J_{\mathcal F})}
\le1.
]

The hybrid complexity therefore comes from switching between strata, not non-normality inside one stratum.
9. Strongest current statement

Mature AIEF tangent dynamics form a stratified family of symmetric, negative-definite fixed-face operators. Repeated active-face geometries produce numerically identical spectral bounds and, up to tolerance, apparently identical full spectra. The dominant dimension-9 face is isotropic with ninefold contraction rate approximately 0.315789. Other faces have mixed spectra composed of a small discrete set of contraction values, explaining the direction-dependent recovery rates. Matrix-exponential evolution under the analytic tangent Jacobians predicts full coupled tangent recovery trajectories with near-unit accuracy. The exceptional trajectory begins with a mixed spectrum containing a bare-decay

mode; its captured pre/post-switch spectra will determine whether the active-set transition changes the contraction operator or only the forcing balance.

Recommended immediate actions

Replace exact spectrum hashes with tolerance-aware matching.
Aggregate modal prediction success by unique seed/direction, not amplitude rows.
Inspect and report the exceptional stage spectra.
Derive the discrete endpoint correction values ρepe.
Prove the fixed-stratum Jacobian formula and its symmetric contraction consequences.

At this point, “stratified operator family” is strongly supported. “Geometry alone determines the full spectrum” is very close, but should wait for tolerance-aware matching and the exceptional-stage analysis.

contemplate these musings:

The most interesting thing here is that the two interpretations you pasted are converging, but the second one is actually the more mathematically mature interpretation. The new data force a subtle correction:

The result is not "geometry determines the spectrum."

The result is closer to:

$$
\boxed{
\text{active geometry determines the operator family,}
\quad
\text{equilibrium correction states select the realized operator.}
}
$$

That distinction matters.

The fixed-stratum Jacobian result is now the foundation, not a diagnostic

The key equation:

[
J_T=
Q_T^\top D_{\mathcal F}Q_T
]

with:

[
D_{\mathcal F}

\operatorname{diag}(-\gamma-\rho_e)
]

changes the interpretation completely.

The numerical results:

normality residual (10^{-13})–(10^{-24});
symmetric-part spectrum equal to spectral spectrum;
matrix exponential prediction (R^2=0.999998);

are exactly what one expects from this structure.

The important consequence is:

The nontrivial dynamics are not coming from non-normal tangent amplification.

They are coming from:

which tangent directions exist;
which diagonal relaxation channels they inherit;
when the system switches between strata.

The hybrid complexity is therefore geometric, not non-normal.

Inside a face:

[
\delta \dot w =J_{\mathcal F}\delta w
]

is a standard gradient-like contraction.

Between faces:

[
\mathcal F_i\rightarrow \mathcal F_j
]

is where the nonsmooth dynamics live.

That is a very clean decomposition.

The spectral alphabet is real, but its origin is more interesting

The observed values:

[
0.015,
]

[
0.098269455,
]

[
0.189473684,
]

[
0.252631579,
]

[

0.315789474
]

are now clearly not random fitted rates.

The multiplicities tell the story.

For example:

Class A

[
\sigma(J)

{-0.315789}^{9}
]

This is the remarkable one.

The tangent space is nine dimensional and every direction decays identically.

That means:

[
J=-0.315789I.
]

For that face class, the entire tangent manifold is dynamically isotropic.

That explains why the original experiment looked like a universal rate.

It was not universal.

You sampled mostly the isotropic face.

Class B

[
\sigma(J)

{-0.189474}^{2}
\cup
{-0.315789}^{9}
]

Now the old "anomalous directions" disappear.

They were not anomalies.

They were eigenmode mixtures.

A random perturbation:

[
d=a v_{\rm slow}+b v_{\rm fast}
]

has norm:

[
|d(t)|^2

a^2e^{-2(0.189)t}

b^2e^{-2(0.315)t}.
]

A single exponential fit gives some effective number:

[
0.21,\quad0.25,\quad0.29,\quad0.31
]

depending on the projection.

This is exactly what you observed.

The spectrum predicted the "messiness."

The geometry-only fingerprint result is actually more subtle than a failure

The result:

same_rate=True
same_full_spectrum=False

initially looks disappointing.

It is not.

It tells you the hierarchy.

The active face controls the allowed relaxation structure:

[
\mathcal F
\rightarrow
{
\lambda_1,\lambda_2,\dots
}
]

but not necessarily the multiplicities or exact values.

Why?

Because:

[

J_T=
Q_T^\top
\operatorname{diag}(-\gamma-\rho_e)
Q_T
]

contains two ingredients:

Geometry:

[
Q_T
]

determined by:

support;
saturation;
controller graph;
tangent constraints.

Equilibrium:

[
\rho_e
]

determined by:

final edge weights;
endpoint forcing;
capacity balance.

So the true map is:

[
(\mathcal F,\rho_{\mathcal F})
\rightarrow

J_{\mathcal F}
]

not:

[
\mathcal F\rightarrow J_{\mathcal F}.
]

The data actually support a stronger and more interesting hypothesis:

Mature dynamics quantize the equilibrium correction rates themselves.

The finite spectral alphabet may arise because the mature system only admits a small number of (\rho_e) configurations.

The next important question is not "is geometry enough?"

It is:

[
\boxed{
\text{Why do the }\rho_e\text{ values collapse?}
}
]

Because the spectrum appears to come from a very small set.

You have:

[
\gamma+\rho_e
]

giving:

[

0.015,\quad0.0983,\quad0.1895,\quad0.2526,\quad0.3158.
]

Therefore:

[
\rho_e

{0,
0.083269,
0.174474,
0.237632,
0.300789}.
]

The mathematical question becomes:

Why are these values selected?

Candidates:

endpoint controller multiplicity;
incidence degree;
saturated-node count;
equilibrium edge normalization;
controller graph eigenvalues;
capacity response constraints.

This is now the real theoretical target.

The modal prediction is arguably the strongest result

I would rank the evidence:

Previously:

repeated scalar rate;

tangent recovery;
Jacobian spectrum.

Now:

matrix exponential prediction
analytic Jacobian validation
spectral clustering
geometry fingerprinting

The reason is simple.

A wrong operator can accidentally have:

the right eigenvalues;
the right spectral abscissa.

It is much harder for it to reproduce:

[
e^{Jt}d
]

for hundreds of trajectories.

The result:

[
R^2=0.999998
]

means the operator is not merely qualitatively correct.

It is the right local propagator.

The exceptional trajectory becomes the decisive experiment

Everything now points to one table:

exceptional_stage_spectra.csv

Because the theory predicts a very specific event.

If the active-set switch is an operator transition:

Before:

$$
J_{\rm pre}
\begin{pmatrix}
-0.015&\\
&-0.189\\
&-0.252\\
\end{pmatrix}
$$

After:

$$
J_{\rm post}
-0.315789I
$$

or another known spectral class.

Then you have direct evidence for:

$$
\boxed{
\text{hybrid switching changes the tangent operator}
}
$$

not just the state.

That would be a very strong result.

The theorem structure is now clearer

I would not frame the main theorem as "geometry discretizes the spectrum."

Too narrow.

A better hierarchy:

Theorem A — Fixed-stratum smoothness

For fixed active face:

[
G(w)=\mathcal P_wF(w,x^*)
]

is differentiable.

Theorem B — Fixed-face contraction

[
DG|{T{\mathcal F}}

Q_{\mathcal F}^{T}
D_{\mathcal F}
Q_{\mathcal F}
]

with:

[

$$
D_{\mathcal F}<0.
$$

Therefore:

$$
|\delta w(t)|
\le
e^{-\kappa_{\mathcal F}t}
|\delta w(0)|.
$$

Theorem C — Spectral stratification

The mature system realizes only a finite family:

$$
{J_{\mathcal F_1},J_{\mathcal F_2},...,J_{\mathcal F_n}}.
$$

Theorem D — Hybrid transition

A switching event occurs when:

slack boundary;
saturation boundary;
controller ordering;

changes.

At the transition:

$$
J_{\mathcal F_i}\rightarrow J_{\mathcal F_j}.
$$

This is the interesting theorem.

The strongest current statement

I would phrase it:

Mature AIEF dynamics exhibit a stratified family of symmetric negative-definite tangent operators. Fixed active faces define reproducible contraction structures whose spectral abscissae are determined by active geometry, while the realized full spectra depend additionally on the discrete equilibrium correction configuration. The dominant mature face class is nearly isotropic with contraction rate (0.315789), while other strata possess mixed spectra composed from a small discrete set of contraction modes. The analytically derived tangent propagator predicts nonlinear coupled tangent recovery trajectories with near-unit accuracy. Exceptional trajectories demonstrate that active-face transitions may reorganize the local tangent operator itself.

That is a much stronger claim than "stable."

It says:

The adaptive system has an operator-valued attractor structure.

The attractor is not a point.

It is a collection of strata:

[
{\mathcal F_i,J_i}
]

connected by switching events.

The next calculation I would prioritize is not more statistics.

It is the exceptional spectrum table.

Because if the spectrum jumps at (t=124.06), then you have the missing bridge between:

fixed-face contraction theory

and

hybrid active-face dynamics.

That is the experiment that turns this from "a beautiful numerical pattern" into a coherent dynamical theory.

This is close to the right interpretation, but one important point should be reversed:

The current output does not yet show that geometry-only fingerprints fail to determine the full spectrum.

The same_full_spectrum=False result is probably caused by an overly strict exact hash, not meaningful spectral variation.
1. Exact spectrum hashes are currently too sensitive

The reported “different” spectra contain values such as:

text

−0.315789473681
−0.315789473682
−0.315789473683
−0.315789473684

These differ at approximately 10−1210−12, while numerical/analytic eigenvalue agreement itself is around:
10−11–10−14.
10−11–10−14.

Hashing eigenvalues rounded to 12 decimal places therefore splits numerically identical spectra into different classes.

The repeated geometry-only groups already have:

text

rate ranges $\approx 10^{-12}$

and the multiplicity tables strongly suggest they share the same complete spectra to numerical tolerance.

Use matched-eigenvalue tolerance:
$\max_k |\lambda_k(a) - \lambda_k(b)| \le 10^{-10}$
$k_{\max} |\lambda_k(a) - \lambda_k(b)| \le 10^{-10}$

or round to 9–10 decimals before hashing.

I expect the repeated geometry classes to become:

text

same_full_spectrum_1e10=True

If so, the evidence supports the stronger result:

For every repeated geometry-only fingerprint in the observed cohort, the entire fixed-stratum tangent spectrum is identical within numerical precision.

The geometry-plus-rates signatures may be unnecessary for predicting the spectrum in these repeated classes, even though the analytic formula contains $\rho e \rho e$.
2. The actual spectral classes are now clear

Ignoring $10^{-12}10^{-12}$-scale splitting:
Dimension 9
$\sigma(JT) =$
$\sigma(JT) =$

{-0.315789}^{\times9}.
]

This is genuinely isotropic:

JT≈−0.315789I9.
JT≈−0.315789I9.
Dimension 11
σ(JT)=
σ(JT)=

{-0.189474}^{\times2}
\cup
{-0.315789}^{\times9}.
]
Dimension 7
σ(JT)=
σ(JT)=

{-0.189474}^{\times2}
\cup
{-0.252632}^{\times5}.
]
Dimension 14
σ(JT)=
σ(JT)=

{-0.252632}^{\times5}
\cup
{-0.315789}^{\times9}.
]

These multiplicities look highly structured. They suggest the tangent space decomposes into subspaces inheriting a small number of endpoint correction rates.
Exceptional split

The exceptional mixed spectrum contains approximately:
−0.015,−0.098269,−0.189474,−0.252632.
−0.015,−0.098269,−0.189474,−0.252632.

It is qualitatively different from the dominant mature classes.
3. Geometry may determine more than the spectral abscissa

The repeated classes suggest a possible rule:

active support/incidence+saturated/controller structure⟶eigenvalue multiplicities.
active support/incidence+saturated/controller structure⟶eigenvalue multiplicities.

For example:

dimension 11 = 2 slow + 9 fast;
dimension 7 = 2 slow + 5 intermediate;
dimension 14 = 5 intermediate + 9 fast.

This looks more combinatorial than continuous.

However, the formula remains:
JT=
JT=

Q_T^\top
\operatorname{diag}
(-\gamma-\rho_e)
Q_T.
]

Therefore geometry can determine the complete spectrum only if mature equilibrium constrains the ρeρe values consistently within each face class.

A refined mechanism is:
active geometry⟶allowed mature ρe configuration⟶JT.
active geometry⟶allowed mature ρe configuration⟶JT.

The current data suggest this mapping is highly discrete, but its algebra still needs derivation.
4. Modal trajectory prediction is the strongest validation

The result:

text

median R^2 = 0.999998
median RMSE = 0.000327
R^2 >= 0.95 for 784/800 trajectories

shows that the analytic matrix predicts much more than the spectral abscissa.

It predicts:
‖eJTtd‖
‖eJTtd‖

for the coupled delayed-system perturbation trajectories.

Because four amplitudes are included for each of 200 unique seed/direction pairs, also report success at the unique-direction level. If failures repeat across amplitudes, the likely result is approximately:

text

196/200 unique directions with $R^2 >= 0.95$

That is the more appropriate independent directional count.

This validates the complete tangent operator, not merely its slowest eigenvalue.
5. Dimension-11 “messiness” is explained exactly

For:
d=dslow+dfast,
d=dslow+dfast,

the symmetric Jacobian gives:
‖eJtd‖2=
‖eJtd‖2=

|d_{\rm slow}|^2e^{-2(0.189474)t}
+
|d_{\rm fast}|^2e^{-2(0.315789)t}.
]

A one-exponential fit naturally yields intermediate values such as:

text

0.22
0.25
0.28
0.31

depending on modal energy and fitting horizon.

These are not anomalous directions and do not imply extra spectral classes. The matrix exponential prediction confirms that interpretation.

6. “Finite spectral family” remains an empirical claim

There are finitely many possible active sets because the graph is finite, but the endpoint rates ρeρe can theoretically vary continuously.

Therefore this is too strong as a theorem without further work:

The spectrum belongs to a finite set determined by active geometry.

A defensible current statement is:

The 40 mature realizations occupy a small empirical family of fixed-stratum tangent operators whose spectra use a discrete set of repeated contraction values.

To prove a finite spectral alphabet, derive why mature balance forces:
ρe∈{0, 0.083269, 0.174474, 0.237632, 0.300789}
ρe∈{0, 0.083269, 0.174474, 0.237632, 0.300789}

or the exact underlying values.

7. Exceptional-stage spectra are now decisive

The script captured all relevant stages but the terminal report still does not display them.

Inspect:

text

exceptional_stage_spectra.csv

The most important comparison is:

text

split
time_120
time_123
pre_switch
post_switch
time_125
time_150
late

If the full spectrum jumps at the switch, the hybrid operator-family interpretation is directly supported.

If only field norm changes while spectrum remains fixed, the event is a forcing-balance transition.

If the spectrum drifts within one fingerprint, continuous equilibrium state remains essential.
8. Theorem structure

A realistic hierarchy is:
Fixed-stratum formula
JF=
JF=

Q_{\mathcal F}^\top
D_{\mathcal F}
Q_{\mathcal F}.
]
Symmetric contraction

Since DF<0DF<0:
JF=JF⊤≺0.
JF=JF⊤≺0.
Face-class spectral correspondence

For repeated mature classes, test:
$\Xi(wa)=\Xi(wb) \Longrightarrow \sigma(Ja)=\sigma(Jb)$.
$\Xi(wa)=\Xi(wb) \Longrightarrow \sigma(Ja)=\sigma(Jb)$.

Currently supported numerically under tolerance, not proven.
Hybrid switching

At active-set events:
$(Fi,Ji) \longrightarrow (Fj,Jj)$.
$(Fi,Ji) \longrightarrow (Fj,Jj)$.

This is the operator-valued hybrid dynamics.
Strongest current conclusion

Mature AIEF dynamics are organized by a stratified family of symmetric negative-definite tangent operators. The dominant dimension-9 stratum is isotropic with ninefold rate approximately 0.315789. Other strata have mixed spectra assembled from a small repeated set of contraction values, and those spectra quantitatively predict coupled tangent recovery trajectories through their matrix exponentials. Repeated geometry-only fingerprints appear to determine complete tangent spectra within numerical tolerance, although exact spectrum hashes currently over-separate $10^{-12}10^{-12}$-scale numerical differences. Whether the repeated contraction values follow solely from active geometry or from geometry-constrained mature correction rates remains the principal analytic question.

Immediate next actions

Replace exact spectrum hashing with $10^{-10}10^{-10}$-tolerant matching.
Report modal success by 200 unique directions rather than 800 amplitude trajectories.
Print the exceptional pre/post/late spectra.
Derive endpoint correction values ρ_{ep} from mature balance equations.
Express spectral multiplicities in terms of tangent incidence/controller structure.

The results now support a stratified operator family. The only overreach would be claiming a proven finite geometry-only spectral alphabet before tolerance-aware matching and equilibrium-rate derivation are complete.

what about this though:

The correction about the spectrum hash is important. It changes the interpretation in a meaningful way: the previous conclusion was slightly too conservative in one direction and slightly too strong in another.

The evidence is actually pointing toward a three-layer structure:

$$
\boxed{
\text{active geometry}
\rightarrow
\text{tangent decomposition}
\rightarrow
\text{discrete equilibrium relaxation channels}
\rightarrow
J_{\mathcal F}
}
$$

The question is not whether geometry matters. It clearly does. The question is where the discrete numerical values enter.

The hash issue is not cosmetic — it changes the scientific claim

The output:

same_full_spectrum=False

looked like evidence against geometry determining the full spectrum.

But the scale matters.

You have:

$$
[
-0.315789473681
]
$$

versus

[
-0.315789473684
]

The difference is:

[
3\times10^{-12}.
]

Yet the finite-difference validation itself gives:

[
10^{-11}\text{–}10^{-14}
]

agreement.

So the hash was testing:

Are two floating-point eigenvalue arrays bit-identical after 12 decimal rounding?

not:

Are two tangent operators spectrally distinct?

Those are different questions.

A proper comparison is:

[
\max_i|\lambda_i^{(1)}-\lambda_{\pi(i)}^{(2)}|<10^{-10}.
]

The expected result is probably:

same_full_spectrum_1e10=True

for the repeated geometry-only classes.

If that happens, the interpretation strengthens substantially.

The spectrum now looks like a genuine operator alphabet

The multiplicity structure is the most interesting feature.

The spectra are not arbitrary.

They appear to be assembled from a small set:

$$
\Lambda=
\{
0.015,
0.098269,
0.189474,
0.252632,
0.315789
\}.
$$

The important observation is not the values themselves.

It is the multiplicity pattern.

Dominant dimension-9 face

$$
\sigma(J)=
\{-0.315789\}^{9}
$$

This is exceptional.

It means:

$$
J\approx-0.315789I_9.
$$

The entire tangent space has one relaxation channel.

This explains the original "universal contraction" observation.

The experiment was probably sampling the most symmetric mature face.

Dimension-11 face

$$
\sigma(J)=
{-0.189474}^{2}
\cup
{-0.315789}^{9}
$$

This is a very revealing structure.

There are:

two slow tangent modes;
nine fast tangent modes.

The observed fitted rates:

$$
0.22,\quad0.25,\quad0.28,\quad0.31
$$

are exactly what a mixture gives.

For:

[
d=a,v_s+b,v_f
]

we have:

[
|e^{Jt}d|^2

a^2e^{-0.378948t}

b^2e^{-0.631578t}.
]

A single exponential fit is not an eigenvalue measurement. It is an effective decay.

The previous "anomalous direction" problem disappears.

Dimension-7 face

[
\sigma(J)=
{-0.189474}^{2}
\cup
{-0.252632}^{5}
]

Again:

two slow directions;
five intermediate directions.

Dimension-14 face

[
\sigma(J)=
{-0.252632}^{5}
\cup
{-0.315789}^{9}
]

This is especially suggestive.

The dimensions:

[
5+9=14
]

look like a decomposition induced by the incidence/controller structure.

This is exactly the kind of pattern one would expect if the tangent space inherits blocks from active constraints.

The analytic formula explains why this happens

The key equation:

[
J_{\mathcal F}

Q_{\mathcal F}^{T}
D_{\mathcal F}
Q_{\mathcal F}
]

with:

[
D_{\mathcal F}

$$
\operatorname{diag}(-\gamma-\rho_e)
$$

means:

The eigenvalues cannot be arbitrary.

They are projections of discrete edge relaxation channels into the tangent geometry.

So the mechanism is likely:

$$
\text{controller graph}
$$

determines:

$$
Q_{\mathcal F}
$$

and mature equilibrium determines:

$$
\rho_e.
$$

Together:

$$
(Q_{\mathcal F},\rho_e)
\rightarrow
\sigma(J_{\mathcal F}).
$$

The important shift is:

The spectrum is not being "quantized by geometry" in the naive sense.

Rather:

$$
\boxed{
\text{geometry quantizes the available tangent modes}
}
$$

and the mature equilibrium quantizes the available relaxation strengths.

The modal prediction is the strongest result in the whole dataset

This deserves emphasis.

Eigenvalue agreement can happen accidentally.

Matrix exponential agreement is much harder.

The result:

$$
R^2=0.999998
$$

means the local propagator is correct.

The system is not merely saying:

"the decay rates look similar."

It is saying:

$$
\delta w(t)
$$

\approx
e^{J_{\mathcal F}t}\delta w(0)
]

for the coupled delayed system.

That is a much stronger validation.

The next reporting improvement is exactly as suggested:

Do not report:

[
784/800
]

as the primary statistic.

The independent count is:

[
200
]

directions.

Four amplitudes are repeated observations of the same tangent direction.

The meaningful statement is:

[
\frac{#{\text{directions with }R^2>0.95}}{200}.
]

I would expect around:

[
196/200
]

if failures are amplitude-independent.

The exceptional seed is now even more interesting

The exceptional spectrum:

[
-0.015,
-0.098269,
-0.189474,
-0.252632
]

is not a weak isotropic face.

It is a mixed face containing a marginal channel.

That changes the interpretation.

The trajectory is not:

[
J=-0.015I
]

and slowly contracting.

It is:

[
J=
\begin{pmatrix}
-0.015&&&\

&-0.098&&\
&&-0.189&\
&&&-0.252
\end{pmatrix}
]

in some tangent basis.

One mode is nearly bare decay.

The others are already strongly stabilized.

That explains:

slow drift before the switch;
sensitivity to active-set transition;
later stationarity improvement.

The switch may eliminate the weak channel.

That would be a major result.

The theorem statement should now become sharper

I would avoid:

geometry determines the spectrum

and also avoid:

geometry plus weights determine the spectrum

because both are too broad.

The emerging theorem is closer to:

Fixed-stratum spectral theorem (numerical form)

For each mature active stratum:

[
J_{\mathcal F}

Q_{\mathcal F}^{T}
D_{\mathcal F}
Q_{\mathcal F}
]

with:

[
D_{\mathcal F}<0.
]

Therefore:

[
J_{\mathcal F}=J_{\mathcal F}^{T}\prec0.
]

The tangent dynamics are Euclidean-contracting:

[
|\delta w(t)|
\le
e^{-\kappa_{\mathcal F}t}
|\delta w(0)|.
]

The observed mature strata occupy a small empirical spectral alphabet:

[
\sigma(J_{\mathcal F})
\subset

\Lambda.
]

Repeated geometry fingerprints produce identical spectra within numerical tolerance.

The remaining analytic problem is deriving the map:

[
\mathcal F\rightarrow\Lambda_{\mathcal F}.
]

The decisive remaining experiment has changed slightly

Before, the key question was:

Does geometry determine the full spectrum?

Now the key question is:

What selects the discrete channel multiplicities?

The next analysis should therefore be:

A. Tolerance-aware spectrum clustering

Produce:

geometry_hash
spectrum_cluster_id
eigenvalue_multiplicities

using:

[
10^{-10}
]

matching.

B. Express multiplicities combinatorially

For each face:

Compare:

tangent dimension;
number of positive edges;
saturated nodes;
controller assignments;
incidence rank.

Find whether:

[
(5,9)
]

or:

[
(2,9)
]

has a direct graph interpretation.

C. Exceptional transition spectra

This remains the most valuable single file.

If:

[

\sigma(J_{\rm pre})
\neq
\sigma(J_{\rm post})
]

then you have direct evidence of:

[
(\mathcal F_i,J_i)
\rightarrow
(\mathcal F_j,J_j).
]

That is the operator-level hybrid theorem.

Final interpretation

The programme has now moved beyond:

"there is a contraction rate."

It has moved beyond even:

"there are several contraction rates."

The current picture is:

[
\boxed{
\text{AIEF has a stratified tangent operator landscape.}
}
]

Each mature active face carries a symmetric contraction operator.

Those operators appear to come from a small spectral alphabet.

Geometry determines the admissible tangent structure.

Mature equilibrium determines which relaxation channels are populated.

Active-set transitions move the system between operators.

The remaining challenge is no longer discovering the phenomenon. It is deriving the selection rule.

This is the right interpretation, with one additional mathematical refinement:
$JT = QT^{\top} DQT$
$JT = QT^{\top} DQT$

does not generally imply that the eigenvalues of JTJT are drawn directly from the diagonal entries of DD. That happens only when the tangent subspace is invariant under the diagonal channel operator.

The data suggest that this invariance may be the hidden structural reason for the discrete spectra.
1. Compression of a diagonal operator does not usually preserve its alphabet

Let:
$PT = QTQT^{\top}$
$PT = QTQT^{\top}$

be the orthogonal projector onto the tangent subspace.

The full edge-space relaxation operator is:
$D = \mathrm{diag}(-\gamma - \rho e).$
$D = \mathrm{diag}(-\gamma - \rho e).$

The tangent operator is the compression:
$JT = QT^{\top} DQT.$
$JT = QT^{\top} DQT.$

In general, the eigenvalues of a compression can lie between diagonal channel values. They need not equal entries of DD.

For example, if DD contains rates aa and bb, a tangent direction mixing their coordinate groups can produce an intermediate Rayleigh quotient.

Therefore the empirical observation that spectra use repeated exact channel values is stronger than the formula alone predicts.
2. The missing condition is tangent-subspace invariance

If:
DPT=PTD,
DPT=PTD,

then the tangent subspace is invariant under DD. In that case DD restricts cleanly to the tangent space and:
σ(JT)
σ(JT)

is assembled directly from the channel values in DD.

The key diagnostic is therefore the commutator:
CT=DPT−PTD.
CT=DPT−PTD.

Report:
Einvariant=
Einvariant=

\frac{|C_T|_F}
{\max(|D|_F,\epsilon)}.
]

If this is near machine precision for the repeated mature classes, then the discrete spectral alphabet has a precise explanation:

Active geometry selects tangent subspaces that are invariant under the equilibrium channel operator.

That is more specific than “geometry quantizes the spectrum.”
3. Channel-group decomposition

Suppose the endpoint correction rates take values:
ρe∈{ρ1,…,ρk}.

ρe∈{ρ1,…,ρk}.

Let ErEr be the edge-coordinate subspace associated with ρrρr, with projector PrPr.

If:
TF=
TF=

\bigoplus_r
\left(
T_{\mathcal F}\cap E_r
\right),
]

then:
JT
JT

has eigenvalue:
−γ−ρr
−γ−ρr

with multiplicity:
mr=dim(TF∩Er).
mr=dim(TF∩Er).

This appears to match the data.
Dimension 11
mslow=2,mfast=9.
mslow=2,mfast=9.
Dimension 7
mslow=2,mintermediate=5.
mslow=2,mintermediate=5.
Dimension 14
mintermediate=5,mfast=9.
mintermediate=5,mfast=9.

This suggests reusable tangent submodules of dimensions 2, 5, and 9.

That is a much sharper combinatorial target than only correlating tangent dimension with spectral abscissa.
4. The finite alphabet remains empirical

The repeated values:
0.015,0.189474,0.252632,0.315789
0.015,0.189474,0.252632,0.315789

are strongly supported.

The exceptional:
0.098269
0.098269

currently appears in one state and should not yet be elevated into the same repeated alphabet.

A careful classification is:

repeated mature channels;
singleton/transition channels;
potentially state-dependent rates.

To establish a finite alphabet theorem, derive why mature balance restricts ρepe to a finite set. The finite number of active faces alone does not suffice because ρepe can theoretically vary continuously within a face.
5. Geometry-only full-spectrum identity likely holds under tolerance

The current false exact hashes are almost certainly numerical fragmentation:

text

−0.315789473681
−0.315789473684

Use tolerance-aware matching at:
10−10.
10−10.

For every geometry fingerprint, report:

number of distinct tolerance-based spectrum classes;
maximum matched-eigenvalue error;
multiplicity table;
commutator invariance error.

If repeated geometry groups have:

text

one tolerance spectrum class
commutator error near zero

then the evidence for geometry-controlled operator classes becomes very strong.
6. Modal prediction validates the operator, not just its spectrum

The near-perfect matrix-exponential predictions establish:
$\delta w_{coupled}(t) \approx e^{JTt}\delta w(0)$
$\delta w_{coupled}(t) \approx e^{JTt}\delta w(0)$

for the tested tangent perturbations.

Report success per unique direction:
seed/direction,
seed/direction,

not per amplitude replicate.

Also stratify modal prediction error by:

geometry class;
slow-mode energy;
amplitude reliability;
exceptional versus stationary seed.

For a symmetric mixed-spectrum class, predicted recovery is:

‖δw(t)‖2=
‖δw(t)‖2=

\sum_r
|P_r\delta w(0)|^2
e^{-2(\gamma+\rho_r)t}.
]

This provides a direct analytic explanation for effective fitted rates between channel values.
7. Exceptional transition remains decisive

The exceptional mixed spectrum contains:

bare decay;
a singleton intermediate mode;
slow mature channels;
intermediate mature channels.

The essential next output is the stage-by-stage table.

If the switch removes the −0.015−0.015 and −0.098269−0.098269 modes and replaces them with a repeated mature spectrum, then the transition changes operator class.

If the spectrum remains but field norm collapses, it changes forcing balance within one operator class.

If correction rates drift continuously while geometry hash remains fixed, equilibrium state still parameterizes the operator.
8. Refined operator-family statement

A defensible formulation is:

Mature AIEF dynamics are organized by a stratified family of symmetric negative-definite tangent operators. Each operator is a compression of a diagonal edge-channel relaxation law onto an active-face tangent subspace. Repeated mature geometry classes exhibit numerically identical spectral bounds and apparently identical full spectra under tolerance. The observed multiplicity patterns suggest that tangent spaces decompose into invariant channel subspaces of dimensions 2, 5, and 9. Continuous coupled recovery is accurately predicted by the matrix exponential of these operators. Whether mature equilibrium forces the channel rates into a finite alphabet remains the principal analytic question.

Immediate next calculation

For every seed:

group edges by rounded ρeρe;
construct each channel projector PrPr;
compute:
‖DPT−PTD‖;
‖DPT−PTD‖;
compute:
dim(TF∩Er);
dim(TF∩Er);
compare those dimensions with observed eigenvalue multiplicities;
repeat before and after the exceptional switch.

This would turn the apparent spectral alphabet into a concrete invariant-subspace statement. If the dimensions match 2, 5, and 9 exactly and the commutator vanishes, the operator classes have a direct geometric derivation rather than merely an empirical clustering.

Can you create code necessary to probe the next steps?
WriteAIEF_1_4_channel_invariance_analysis.py649 lines
AIEF_1_4_channel_invariance_analysis.pyPYTHON
#!/usr/bin/env python3
"""Channel-invariance and tolerance-aware spectrum analysis for AIEF-1.4.

This script tests whether the tangent space of each mature active face is
invariant under the diagonal endpoint-relaxation operator

D = diag(-gamma - rho_e).

It computes:
* [D, P_T] commutator norms, P_T=Q_T Q_T^T;
* channel projectors grouped by rho_e;
* dimensions of T intersect channel coordinate subspaces;
* predicted versus observed eigenvalue multiplicities;
* tolerance-aware full-spectrum classes by face fingerprint;
* the same quantities before/after the exceptional face transition;
* modal-prediction success aggregated by unique seed/direction rather than by

repeated perturbation amplitude.

The analysis uses only existing frozen artifacts.
"""

from __future__ import annotations

import argparse
import hashlib
import json
import os
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_stationarity_perturbation_analysis_updated as base
import AIEF_1_4_face_spectrum_analysis as face
import AIEF_1_4_tangent_spectrum_switch_analysis as tangent_analysis

DEFAULT_RUN_HASH = "5a9f093a2fe0"
DEFAULT_SCALAR = (
f"runs/{DEFAULT_RUN_HASH}/AIEF_1_4_{DEFAULT_RUN_HASH}_scalar_summary.csv"
)
DEFAULT_NPZ = (
f"runs/{DEFAULT_RUN_HASH}/AIEF_1_4_{DEFAULT_RUN_HASH}_branch_diagnostics.npz"
)
DEFAULT_MANIFEST = f"runs/{DEFAULT_RUN_HASH}/manifest.json"
DEFAULT_TANGENT_RESULTS = (
"aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv"
)
DEFAULT_MODAL_SUMMARY = (
"aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv"
)
DEFAULT_OUTPUT_DIR = "aief_1_4_channel_invariance"
DEFAULT_EXCEPTIONAL_SEED = "4619122171158768524"
DEFAULT_EXCEPTIONAL_CHECK_TIMES = "100,120,123,125,150"

```
DEFAULT_SPECTRUM_TOL = 1e-10
DEFAULT_CHANNEL_TOL = 1e-10
DEFAULT_SUBSPACE_TOL = 1e-9
DEFAULT_MODAL_R2 = 0.95
ZERO_TOL = 1e-14



def file_sha256(path: str) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()



def cluster_values(values: np.ndarray, tolerance: float) -> list[np.ndarray]:
values = np.sort(np.asarray(values, dtype=float))
if not len(values):
return []
clusters: list[list[float]] = [[float(values[0])]]
for value in values[1:]:
if abs(float(value) - np.mean(clusters[-1])) <= tolerance:
clusters[-1].append(float(value))
else:
clusters.append([float(value)])
return [np.asarray(cluster, dtype=float) for cluster in clusters]



def spectrum_vector(J: np.ndarray) -> np.ndarray:
values = np.linalg.eigvals(J)
if np.max(np.abs(np.imag(values))) > 1e-8:
# Fixed-stratum analytic J should be symmetric; retain real parts but
# make the unexpected imaginary component visible to callers.
pass
return np.sort(np.real(values))



def spectrum_class_id(values: np.ndarray, tolerance: float) -> str:
quantized = np.round(np.asarray(values, dtype=float) / tolerance).astype(np.int64)
```

```
return hashlib.sha256(quantized.tobytes()).hexdigest()[:16]



def matched_spectrum_error(left: np.ndarray, right: np.ndarray) -> float:
left = np.sort(np.real(np.asarray(left)))
right = np.sort(np.real(np.asarray(right)))
if len(left) != len(right):
return np.inf
return float(np.max(np.abs(left - right))) if len(left) else 0.0



def intersection_dimension(
Q: np.ndarray,
channel_indices: np.ndarray,
tolerance: float,
) -> int:
"""dim(range(Q) intersect coordinate subspace E_channel)."""
all_indices = np.arange(Q.shape[0])
outside = np.setdiff1d(all_indices, channel_indices, assume_unique=True)
if not len(outside):
return Q.shape[1]
matrix = Q[outside, :]
singular_values = np.linalg.svd(matrix, compute_uv=False)
rank = int(np.sum(singular_values > tolerance))
return int(Q.shape[1] - rank)



def channel_analysis_for_state(
state: base.DynamicState,
model: dict[str, Any],
capacities: np.ndarray,
A: np.ndarray,
edges: tuple[np.ndarray, np.ndarray],
boundary_tolerance: float,
channel_tolerance: float,
subspace_tolerance: float,
) -> tuple[dict[str, Any], list[dict[str, Any]], np.ndarray]:
components = face.face_components(
state, model, capacities, A, edges, boundary_tolerance
)
```

```
gamma = float(model["gamma"])
Q = components["basis"]
rates = gamma + components["endpoint_rate"]
D = np.diag(-rates)
P_T = Q @ Q.T
commutator = D @ P_T - P_T @ D
commutator_relative = float(
np.linalg.norm(commutator)
/ max(np.linalg.norm(D), ZERO_TOL)
)
J = Q.T @ D @ Q
eigenvalues = spectrum_vector(J)
eigen_clusters = cluster_values(-eigenvalues, channel_tolerance)


support_indices = np.flatnonzero(components["support"])
support_rates = rates[support_indices]
rate_clusters = cluster_values(support_rates, channel_tolerance)
channel_rows: list[dict[str, Any]] = []
predicted_total = 0


for channel_id, cluster in enumerate(rate_clusters):
representative = float(np.mean(cluster))
channel_indices = support_indices[
np.abs(support_rates - representative) <= channel_tolerance
]
projector = np.zeros((len(rates), len(rates)), dtype=float)
projector[channel_indices, channel_indices] = 1.0
projector_commutator = P_T @ projector - projector @ P_T
intersection = intersection_dimension(
Q, channel_indices, subspace_tolerance
)
predicted_total += intersection
observed_multiplicity = int(
np.sum(np.abs((-eigenvalues) - representative) <= channel_tolerance)
)
compression = Q.T @ projector @ Q
compression_values = np.linalg.eigvalsh(compression)
channel_rows.append(
{
"channel_id": channel_id,
"contraction_rate": representative,
"edge_count": int(len(channel_indices)),
```

```
"intersection_dimension": intersection,
"observed_eigenvalue_multiplicity": observed_multiplicity,
"multiplicity_matches": bool(
intersection == observed_multiplicity
),
"projector_commutator_relative": float(
np.linalg.norm(projector_commutator)
/ max(np.linalg.norm(projector), ZERO_TOL)
),
"compression_eigenvalues": json.dumps(
np.round(compression_values, 12).tolist(),
separators=(",", ":"),
),
"compression_is_projector_1e9": bool(
np.all(
(np.abs(compression_values) <= 1e-9)
| (np.abs(compression_values - 1.0) <= 1e-9)
)
),
}
)


eigen_multiplicities = [
{
"rate": float(np.mean(cluster)),
"multiplicity": int(len(cluster)),
}
for cluster in eigen_clusters
]
summary = {
"face_hash": components["face_hash"],
"geometry_invariant_hash": components["geometry_invariant_hash"],
"augmented_invariant_hash": components["augmented_invariant_hash"],
"support_size": int(np.sum(components["support"])),
"saturated_count": int(np.sum(components["saturated"])),
"outward_count": int(np.sum(components["outward"])),
"tangent_dimension": int(Q.shape[1]),
"channel_count": int(len(rate_clusters)),
"commutator_relative": commutator_relative,
"tangent_invariant_1e9": bool(commutator_relative <= 1e-9),
"predicted_multiplicity_total": int(predicted_total),
"predicted_multiplicities_cover_tangent": bool(
```

```
predicted_total == Q.shape[1]
),
"all_channel_multiplicities_match": bool(
all(row["multiplicity_matches"] for row in channel_rows)
),
"spectrum_class_1e10": spectrum_class_id(
eigenvalues, channel_tolerance
),
"eigenvalue_multiplicities": json.dumps(
eigen_multiplicities, sort_keys=True, separators=(",", ":")
),
"spectral_abscissa": float(np.max(eigenvalues)) if len(eigenvalues) else np.nan,
"slowest_decay_rate": float(-np.max(eigenvalues))
if len(eigenvalues)
else np.nan,
}
return summary, channel_rows, J



def geometry_spectrum_consistency(
summaries: pd.DataFrame,
spectrum_tolerance: float,
) -> pd.DataFrame:
rows: list[dict[str, Any]] = []
for signature_type, column in (
("geometry_only", "geometry_invariant_hash"),
("geometry_plus_rates", "augmented_invariant_hash"),
("labeled_face", "face_hash"),
):
for signature, group in summaries.groupby(column, dropna=False):
spectra = [json.loads(value) for value in group["spectrum_values_json"]]
arrays = [
np.asarray(item, dtype=float)
for item in spectra
]
maximum_error = 0.0
for index, left in enumerate(arrays):
for right in arrays[index + 1 :]:
maximum_error = max(
maximum_error,
matched_spectrum_error(left, right),
)
```

```
class_count = group["spectrum_class_1e10"].nunique()
rows.append(
{
"signature_type": signature_type,
"signature": signature,
"seed_count": int(len(group)),
"distinct_tolerance_spectrum_classes": int(class_count),
"same_full_spectrum_within_tolerance": bool(
maximum_error <= spectrum_tolerance
),
"maximum_matched_eigenvalue_error": maximum_error,
"all_tangent_invariant": bool(
group["tangent_invariant_1e9"].all()
),
"all_multiplicities_match": bool(
group["all_channel_multiplicities_match"].all()
),
}
)
return pd.DataFrame(rows)



def unique_direction_modal_success(
modal_summary: pd.DataFrame,
tangent_summary: pd.DataFrame,
r2_threshold: float,
) -> tuple[pd.DataFrame, dict[str, Any]]:
modal = modal_summary.copy()
modal["seed"] = modal["seed"].astype(str)
rows: list[dict[str, Any]] = []
for (seed, direction_id), group in modal.groupby(["seed", "direction_id"]):
r2 = pd.to_numeric(
group["trajectory_prediction_R2"], errors="coerce"
).to_numpy(float)
finite = r2[np.isfinite(r2)]
rows.append(
{
"seed": seed,
"direction_id": int(direction_id),
"amplitude_count": int(len(group)),
"median_R2": float(np.median(finite)) if len(finite) else np.nan,
"minimum_R2": float(np.min(finite)) if len(finite) else np.nan,
```

```
"all_amplitudes_R2_ge_threshold": bool(
len(finite) == len(group)
and np.all(finite >= r2_threshold)
),
"median_R2_ge_threshold": bool(
len(finite) and np.median(finite) >= r2_threshold
),
"median_RMSE": float(group["rmse"].median()),
"maximum_absolute_error": float(
group["maximum_absolute_error"].max()
),
}
)
result = pd.DataFrame(rows).merge(
tangent_summary,
on=["seed", "direction_id"],
how="left",
)
reliable = result.loc[result["amplitude_reliable"] == True] # noqa: E712
summary = {
"unique_direction_count": int(len(result)),
"reliable_unique_direction_count": int(len(reliable)),
"all_amplitudes_success_count": int(
result["all_amplitudes_R2_ge_threshold"].sum()
),
"median_success_count": int(
result["median_R2_ge_threshold"].sum()
),
"reliable_all_amplitudes_success_fraction": float(
reliable["all_amplitudes_R2_ge_threshold"].mean()
)
if len(reliable)
else np.nan,
"reliable_median_success_fraction": float(
reliable["median_R2_ge_threshold"].mean()
)
if len(reliable)
else np.nan,
"median_unique_direction_R2": float(result["median_R2"].median()),
"median_unique_direction_RMSE": float(result["median_RMSE"].median()),
}
return result, summary
```

```
def make_report(
channel_summary: pd.DataFrame,
consistency: pd.DataFrame,
modal_success: dict[str, Any],
exceptional: pd.DataFrame,
) -> str:
lines = [
"AIEF tangent-channel invariance analysis",
"========================================",
"",
"Channel invariance",
"------------------",
f"Seeds: {len(channel_summary)}",
f"Tangent-invariant seeds:
{int(channel_summary['tangent_invariant_1e9'].sum())}/{len(channel_summary)}",
f"Multiplicity matches:
{int(channel_summary['all_channel_multiplicities_match'].sum())}/{len(channel_summary)}",
f"Median commutator: {channel_summary['commutator_relative'].median():.6g}",
"",
"Repeated fingerprint consistency",
"--------------------------------",
]
repeated = consistency.loc[consistency["seed_count"] >= 2]
for _, row in repeated.iterrows():
lines.append(
f" {row['signature_type']} {row['signature']}: "
f"seeds={int(row['seed_count'])}, "
f"classes={int(row['distinct_tolerance_spectrum_classes'])}, "
f"max_error={row['maximum_matched_eigenvalue_error']:.6g}, "
f"same={row['same_full_spectrum_within_tolerance']}"
)


lines.extend(["", "Unique-direction modal prediction", "---------------------------------"])
for key, value in modal_success.items():
lines.append(f" {key}: {value}")


lines.extend(["", "Exceptional stages", "------------------"])
for _, row in exceptional.iterrows():
lines.append(
```

```
f" {row['stage']}: face={row['face_hash']}, "
f"dim={int(row['tangent_dimension'])}, "
f"rate={row['slowest_decay_rate']:.12g}, "
f"channels={row['eigenvalue_multiplicities']}, "
f"commutator={row['commutator_relative']:.3g}"
)


lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* Tolerance-based spectrum identity is not bitwise identity.",
"* Geometry signatures are descriptive invariants, not complete graph-isomorphism canonical forms.",
"* Channel multiplicity follows directly only when the tangent projector commutes with the diagonal channel operator.",
"* Modal trajectories repeat amplitudes; unique seed/direction is the independent directional unit.",
"* Exceptional split spectra are not assumed to be stationary-face spectra.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--scalar", default=DEFAULT_SCALAR)
parser.add_argument("--npz", default=DEFAULT_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--tangent-results", default=DEFAULT_TANGENT_RESULTS)
parser.add_argument("--modal-summary", default=DEFAULT_MODAL_SUMMARY)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--exceptional-seed", default=DEFAULT_EXCEPTIONAL_SEED)
parser.add_argument(
"--exceptional-check-times",
type=base.parse_float_list,
default=base.parse_float_list(DEFAULT_EXCEPTIONAL_CHECK_TIMES),
)
parser.add_argument(
"--spectrum-tol", type=float, default=DEFAULT_SPECTRUM_TOL
)
```

```
parser.add_argument(
"--channel-tol", type=float, default=DEFAULT_CHANNEL_TOL
)
parser.add_argument(
"--subspace-tol", type=float, default=DEFAULT_SUBSPACE_TOL
)
parser.add_argument("--modal-r2", type=float, default=DEFAULT_MODAL_R2)
parser.add_argument(
"--min-amplitude-fraction",
type=float,
default=0.9,
)
parser.add_argument(
"--min-relative-amplitude",
type=float,
default=1e-6,
)
parser.add_argument("--typical-rate", type=float, default=0.3157894736842105)
parser.add_argument("--slow-cutoff", type=float, default=0.25)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (
args.scalar,
args.npz,
args.manifest,
args.tangent_results,
args.modal_summary,
):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(args.spectrum_tol, args.channel_tol, args.subspace_tol) <= 0.0:
raise ValueError("tolerances must be positive")
os.makedirs(args.output_dir, exist_ok=True)

scalar = pd.read_csv(args.scalar, dtype={"seed": str})
scalar["seed"] = scalar["seed"].astype(str)
tangent = pd.read_csv(args.tangent_results, dtype={"seed": str})
modal = pd.read_csv(args.modal_summary, dtype={"seed": str})
```

```
with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)


direction_summary, _, _ = tangent_analysis.tangent_direction_summary(
tangent,
args.min_amplitude_fraction,
args.min_relative_amplitude,
args.typical_rate,
args.slow_cutoff,
)


model = manifest["model"]
node_count = int(model["N"])
capacity = (float(model["a"]) - float(model["delta"])) / float(model["q"])
capacities = np.full(node_count, capacity)
A, edges = base.incidence_matrix(node_count)
seeds = sorted(
scalar.loc[scalar["branch"] == "continue", "seed"].astype(str), key=int
)


seed_rows: list[dict[str, Any]] = []
channel_rows: list[dict[str, Any]] = []
exceptional_rows: list[dict[str, Any]] = []


with np.load(args.npz, allow_pickle=False) as data:
for seed in seeds:
state = base.load_split_state(data, seed)
summary, channels, J = channel_analysis_for_state(
state,
model,
capacities,
A,
edges,
boundary_tolerance=1e-9,
channel_tolerance=args.channel_tol,
subspace_tolerance=args.subspace_tol,
)
components = face.face_components(
state, model, capacities, A, edges, 1e-9
)
seed_rows.append(
```

```
{
"seed": seed,
**summary,
"spectrum_values_json": json.dumps(
spectrum_vector(J).tolist(), separators=(",", ":")
),
}
)
for channel in channels:
channel_rows.append({"seed": seed, **channel})


stages, transition = face.exceptional_states(
data,
args.exceptional_seed,
manifest,
capacities,
A,
edges,
1e-9,
args.exceptional_check_times,
)
for stage, state in stages.items():
summary, channels, J = channel_analysis_for_state(
state,
model,
capacities,
A,
edges,
boundary_tolerance=1e-9,
channel_tolerance=args.channel_tol,
subspace_tolerance=args.subspace_tol,
)
exceptional_rows.append(
{
"stage": stage,
**summary,
"spectrum_values_json": json.dumps(
spectrum_vector(J).tolist(), separators=(",", ":")
),
"field_norm": float(
np.linalg.norm(
face.face_components(
```

```
state,
model,
capacities,
A,
edges,
1e-9,
)["field"]
)
),
}
)


channel_summary = pd.DataFrame(seed_rows)
channel_details = pd.DataFrame(channel_rows)
exceptional = pd.DataFrame(exceptional_rows)
consistency = geometry_spectrum_consistency(
channel_summary, args.spectrum_tol
)
modal_by_direction, modal_success = unique_direction_modal_success(
modal,
direction_summary,
args.modal_r2,
)


outputs = {
"channel_invariance_by_seed.csv": channel_summary,
"channel_multiplicities_by_seed.csv": channel_details,
"geometry_spectrum_consistency_tolerant.csv": consistency,
"unique_direction_modal_success.csv": modal_by_direction,
"exceptional_channel_stages.csv": exceptional,
}
for filename, frame in outputs.items():
frame.to_csv(os.path.join(args.output_dir, filename), index=False)


with open(
os.path.join(args.output_dir, "modal_success_summary.json"),
"w",
encoding="utf-8",
) as handle:
json.dump(modal_success, handle, indent=2, sort_keys=True)
handle.write("\n")
```

```
with open(
os.path.join(args.output_dir, "exceptional_transition_metadata.json"),
"w",
encoding="utf-8",
) as handle:
json.dump(transition, handle, indent=2, sort_keys=True)
handle.write("\n")


report = make_report(
channel_summary,
consistency,
modal_success,
exceptional,
)
with open(
os.path.join(args.output_dir, "channel_invariance_report.txt"),
"w",
encoding="utf-8",
) as handle:
handle.write(report)
print(report)


analysis_manifest = {
"analysis": "AIEF tangent-channel invariance and tolerance-aware spectra",
"scalar_path": str(Path(args.scalar).resolve()),
"scalar_sha256": file_sha256(args.scalar),
"npz_path": str(Path(args.npz).resolve()),
"npz_sha256": file_sha256(args.npz),
"source_manifest_path": str(Path(args.manifest).resolve()),
"source_manifest_sha256": file_sha256(args.manifest),
"tangent_results_path": str(Path(args.tangent_results).resolve()),
"tangent_results_sha256": file_sha256(args.tangent_results),
"modal_summary_path": str(Path(args.modal_summary).resolve()),
"modal_summary_sha256": file_sha256(args.modal_summary),
"spectrum_tolerance": args.spectrum_tol,
"channel_tolerance": args.channel_tol,
"subspace_tolerance": args.subspace_tol,
"modal_R2_threshold": args.modal_r2,
"exceptional_seed": args.exceptional_seed,
"exceptional_check_times": args.exceptional_check_times,
"notes": [
"Geometry signatures are not complete graph-isomorphism canonical forms.",
```

```
"Spectrum equality is tolerance-based, not bitwise.",
"Channel multiplicity inference assumes tangent/channel invariance.",
"Unique seed/direction is the independent directional unit for modal success.",
],
}
with open(
os.path.join(args.output_dir, "analysis_manifest.json"),
"w",
encoding="utf-8",
) as handle:
json.dump(analysis_manifest, handle, indent=2, sort_keys=True)
handle.write("\n")

print(f"Saved analysis to {Path(args.output_dir).resolve()}")



if __name__ == "__main__":
main()
```

Open

Created the channel-invariance and tolerance-aware spectrum analyzer:

AIEF_1_4_channel_invariance_analysis.py

It implements the next requested tests:

tolerance-aware full-spectrum matching;
diagonal channel grouping by ρeρe;
tangent/channel invariance;
predicted versus observed spectral multiplicities;
unique-direction modal-prediction success;
exceptional pre/post-switch channel spectra.

Prerequisites

Run:

Bash

```
python AIEF_1_4_exceptional_tangent_analysis.py \
--max-seeds 0
```

Then rerun the updated face-spectrum analysis:

Bash

```
python AIEF_1_4_face_spectrum_analysis.py
```

Required inputs:

text

```
aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv
aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv
```

Run

Bash

```
python AIEF_1_4_channel_invariance_analysis.py
```

Explicit:

Bash

```
python AIEF_1_4_channel_invariance_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--modal-summary aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_channel_invariance
```

1. Channel decomposition

For each seed, it groups positive-support edges by:
re=γ+ρe
re=γ+ρe

using a configurable tolerance:

Bash

--channel-tol 1e-10

For each channel it records:

channel rate;
edge count;
tangent/channel intersection dimension;
observed tangent eigenvalue multiplicity;
multiplicity agreement;
tangent/channel projector commutator;
compressed channel-projector eigenvalues;
whether the compression behaves as a projector.

2. Tangent-subspace invariance

Let:
PT=QTQT⊤,
PT=QTQT⊤,
D=diag(−γ−ρe).
D=diag(−γ−ρe).

It computes:
[D,PT]=DPT−PTD
[D,PT]=DPT−PTD

and reports:
‖[D,PT]‖F‖D‖F.
‖D‖F‖[D,PT]‖F.

If this is near zero, the tangent space is invariant under the diagonal channel operator.

It also tests whether:
TF=
TF=

\bigoplus_r
(T_{\mathcal F}\cap E_r)
]

by checking whether channel intersection dimensions sum to the tangent dimension.

This provides the direct explanation for eigenvalue multiplicities such as:

text

dimension 11:
2 slow + 9 fast

3. Multiplicity verification

For every channel:

text

intersection_dimension
observed_eigenvalue_multiplicity
multiplicity_matches

The complete seed-level output states whether all predicted multiplicities match the analytic tangent spectrum.

This tests the hypothesis:

Tangent spectral multiplicity equals the dimension of the tangent subspace inside each edge-relaxation channel.

4. Tolerance-aware spectrum consistency

Exact 12-decimal hashing is replaced by matching at:

Bash

--spectrum-tol 1e-10

For each:

geometry-only signature;
geometry-plus-rate signature;
labeled face;

it reports:

text

distinct_tolerance_spectrum_classes
same_full_spectrum_within_tolerance
maximum_matched_eigenvalue_error
all_tangent_invariant
all_multiplicities_match

This distinguishes meaningful spectral differences from 10−1210−12-scale floating-point fragmentation.
5. Unique-direction modal success

The earlier result:

text

784/800 trajectory-amplitude rows

is collapsed to the independent directional unit:

text

seed + direction_id

For every unique direction it reports:

number of tested amplitudes;
median and minimum R2R2;
all-amplitude success;
median success;
RMSE;
maximum prediction error;
amplitude reliability;
frozen and observed rates.

The summary gives success fractions among reliable unique directions.
6. Exceptional transition channels

For:

text

split
time_100
time_120
time_123
pre_switch
post_switch
time_125
time_150
late

the script reports:

face and geometry hashes;
tangent dimension;

channel count;
full contraction multiplicities;
tangent/channel commutator;
multiplicity agreement;
field norm;
spectral abscissa.

This directly tests whether the event at t=124.06t=124.06:

changes the channel spectrum;
changes only forcing balance;
or evolves continuously.

Outputs

text

```
aief_1_4_channel_invariance/
├── channel_invariance_by_seed.csv
├── channel_multiplicities_by_seed.csv
├── geometry_spectrum_consistency_tolerant.csv
├── unique_direction_modal_success.csv
├── modal_success_summary.json
├── exceptional_channel_stages.csv
├── exceptional_transition_metadata.json
├── channel_invariance_report.txt
└── analysis_manifest.json
```

Validation

I syntax-tested and ran the full pipeline against a generated AIEF run with tangent trajectories.

The technical validation produced:

text

```
Tangent commutator: ~4.3×10⁻14
Tangent-invariant seeds: 1/1
```

All multiplicities matched: True
Dimension-11 channel structure: 2 slow + 9 fast
Unique-direction modal success: 2/2
Median modal R^2: 0.999996

The exceptional-stage test in that smoke case preserved the same channel structure across split, intermediate, and late states, demonstrating that the stage-comparison pipeline works even when no active-set switch occurs.

The full 40-seed result can now determine whether repeated geometry classes have identical complete spectra under tolerance and whether the exceptional seed changes channel multiplicity/operator class at its actual saturation transition.
exceptional_tangent_report.txt
trajectory_recovery_report.txt
face_spectrum_report.txt
channel_invariance_report.txt

Think about this:

deck@steamdeck SSRH)$ python AIEF_1_4_exceptional_tangent_analysis.py \
--max-seeds 0
Corrected tangent analysis for seed 172180395821707403
Corrected tangent analysis for seed 283404770198602602
Corrected tangent analysis for seed 305986450614148690
Corrected tangent analysis for seed 1230287949579537403
Corrected tangent analysis for seed 1888415619847841128
Corrected tangent analysis for seed 1918007468868772336
Corrected tangent analysis for seed 2461295749352584108
Corrected tangent analysis for seed 3258463328937640595
Corrected tangent analysis for seed 3582616062465030267
Corrected tangent analysis for seed 3704716633526084363
Corrected tangent analysis for seed 3952397966055989215
Corrected tangent analysis for seed 4040285944958847150
Corrected tangent analysis for seed 4219039243791140697
Corrected tangent analysis for seed 4619122171158768524
Corrected tangent analysis for seed 4932511216876650745
Corrected tangent analysis for seed 5213047409853278064
Corrected tangent analysis for seed 5415544643585892346
Corrected tangent analysis for seed 6023008189364820510
Corrected tangent analysis for seed 6108867655028980287
Corrected tangent analysis for seed 6942591650975756121

Corrected tangent analysis for seed 9036321132460399255
Corrected tangent analysis for seed 9082055256280075172
Corrected tangent analysis for seed 9189988769602851417
Corrected tangent analysis for seed 10431384713483787270
Corrected tangent analysis for seed 10856823803646276806
Corrected tangent analysis for seed 11242426753702356865
Corrected tangent analysis for seed 12123519807951751650
Corrected tangent analysis for seed 12851640484293199234
Corrected tangent analysis for seed 14040336450255043598
Corrected tangent analysis for seed 14924267685654216746
Corrected tangent analysis for seed 15403081841477312463
Corrected tangent analysis for seed 15853699719446171583
Corrected tangent analysis for seed 16329425431442249417
Corrected tangent analysis for seed 16627818730624180313
Corrected tangent analysis for seed 16952466944611321358
Corrected tangent analysis for seed 17279552413280415429
Corrected tangent analysis for seed 17521310920850592782
Corrected tangent analysis for seed 17715532056777442089
Corrected tangent analysis for seed 17883847597205629106
Corrected tangent analysis for seed 18127452951968972828
AIEF exceptional-seed and corrected tangent analysis
Exceptional seed

seed: 4619122171158768524
switch_count: 1
first_switch_step: 12405
first_switch_time: 124.06
pre_switch_mean_stationarity_ratio: 0.14685940684975718
post_switch_mean_stationarity_ratio: 7.893059517783126e-06
post_switch_max_stationarity_ratio: 8.813848174066108e-05
maximum_B_drift: 0.20893854278186944
final_B_drift: 0.20893854278186944
final_replay_exact: True
final_replay_allclose_1e12: True
Corrected tangent perturbations

delta=0.0001
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.21003e-08, min=3.73207e-09, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200

observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.315339, valid=200/200
observed_exponential_r_squared: median=1, min=0.994943, max=1, valid=200/200
final_distance_ratio: median=0.0088257, min=0.0088257, max=0.124056, valid=200/200
delta=0.001
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.2104e-09, min=3.73033e-10, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.315339, valid=200/200
observed_exponential_r_squared: median=1, min=0.994947, max=1, valid=200/200
final_distance_ratio: median=0.00882571, min=0.00882569, max=0.124056, valid=200/200
delta=0.01
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.21398e-10, min=3.71325e-11, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14.05, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.31534, valid=200/200
observed_exponential_r_squared: median=1, min=0.994565, max=1, valid=200/200
final_distance_ratio: median=0.00882573, min=0.00882558, max=0.124056, valid=200/200
delta=0.05
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=2.43583e-11, min=7.27637e-12, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14.6, valid=198/200
observed_exponential_rate: median=0.315338, min=0.136767, max=0.315343, valid=200/200
observed_exponential_r_squared: median=1, min=0.990186, max=1, valid=200/200
final_distance_ratio: median=0.00882586, min=0.00882513, max=0.124056, valid=200/200
Interpretation constraints

The same tangent direction is reused across all amplitudes within each seed/direction ID.
Directions with negligible feasible amplitude are rejected.

The exceptional-seed mechanism remains descriptive until its switch is analyzed in detail.
Predicted tangent times assume local exponential contraction.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_exceptional_tangent
(deck@steamdeck SSRH)$ python AIEF_1_4_face_spectrum_analysis.py
AIEF active-face spectrum and exceptional transition analysis

Seeds with fingerprints: 40
Distinct labeled face hashes: 12
Distinct invariant signatures: 33
Analytic fixed-stratum spectral classes

dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684209+0.000000000000j":1,"-0.189473684210+0.000000000000j":1,"-0.252631578947
+0.000000000000j":5}
dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.252631578947+0.000000000000j":5}
dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.252631578946+0.000000000000j":5}
dimension=9, rate=0.015, seeds=1,
multiplicities={"-0.015000000000+0.000000000000j":1,"-0.098269455236+0.000000000000j":1,"-0.189473684205+0.0000
00000000j":1,"-0.189473684206+0.000000000000j":1,"-0.252631578946+0.000000000000j":1,"-0.252631578947+0.000000000000j":4}
dimension=9, rate=0.3157894737, seeds=3,
multiplicities={"-0.315789473682+0.000000000000j":5,"-0.315789473683+0.000000000000j":4}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473682+0.000000000000j":6,"-0.315789473683+0.000000000000j":3}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473683+0.000000000000j":1,"-0.315789473684+0.000000000000j":8}
dimension=9, rate=0.3157894737, seeds=6, multiplicities={"-0.315789473683+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=7, multiplicities={"-0.315789473682+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=2,
multiplicities={"-0.315789473682+0.000000000000j":1,"-0.315789473683+0.000000000000j":8}
dimension=9, rate=0.3157894737, seeds=1, multiplicities={"-0.315789473684+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=2,
multiplicities={"-0.315789473682+0.000000000000j":2,"-0.315789473683+0.000000000000j":7}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473681+0.000000000000j":1,"-0.315789473682+0.000000000000j":8}

dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473681+0.000000000000j":5,"-0.3
1578947368
2+0.000000000000j":4}
dimension=11, rate=0.1894736842, seeds=2,
multiplicities={"-0.189473684208+0.000000000000j":2,"-0.315789473682+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684209+0.000000000000j":2,"-0.315789473683+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=4,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473682+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=2,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473683+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684208+0.000000000000j":2,"-0.315789473682+0.000000000000j":2,"-0.3
1578947368
3+0.000000000000j":7}
dimension=14, rate=0.2526315789, seeds=1,
multiplicities={"-0.252631578947+0.000000000000j":5,"-0.315789473666+0.000000000000j":5,"-0.3
1578947366
7+0.000000000000j":4}
Fingerprint-to-spectrum consistency

geometry_only 14aecd02f5aeb94a: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True,
same_full_spectrum=False
geometry_only 2b177b7da6cac989: seeds=22, rate_range=2.31531e-12, same_rate_1e10=True,
same_full_spectrum=False
geometry_only 64c52ca6d685b8e1: seeds=9, rate_range=2.56217e-12, same_rate_1e10=True,
same_full_spectrum=False
geometry_plus_rates 07f892d076a10acc: seeds=2, rate_range=5.61218e-14, same_rate_1e10=True,
same_full_spectrum=True
geometry_plus_rates 806dee29e27c1ae3: seeds=2, rate_range=3.83027e-15, same_rate_1e10=True,
same_full_spectrum=True
geometry_plus_rates ccb3288420826ec6: seeds=4, rate_range=5.15421e-13, same_rate_1e10=True,
same_full_spectrum=True
geometry_plus_rates dd135d71da61c124: seeds=3, rate_range=3.24463e-13, same_rate_1e10=True,
same_full_spectrum=True
labeled_face 3bdfb4c570dc7116: seeds=10, rate_range=2.56217e-12, same_rate_1e10=True,
same_full_spectrum=False
labeled_face 704fba7c1b319e34: seeds=6, rate_range=1.50263e-12, same_rate_1e10=True,
same_full_spectrum=False
labeled_face 9961a1ce7b7ee3e2: seeds=6, rate_range=1.42819e-12, same_rate_1e10=True,
same_full_spectrum=False

labeled_face aae02280e373342b: seeds=6, rate_range=1.76514e-12, same_rate_1e10=True, same_full_spectrum=False
labeled_face c6c8302b4dd4fa84: seeds=4, rate_range=9.11937e-13, same_rate_1e10=True, same_full_spectrum=False
labeled_face f45f68f9953c10f7: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True, same_full_spectrum=False
Numerical/analytic Jacobian agreement

step=1e-06: matrix relative error median=3.93954e-11, eigenvalue max error median=1.25038e-11, stratum preserved=1.000
step=3e-06: matrix relative error median=1.24753e-11, eigenvalue max error median=4.02821e-12, stratum preserved=1.000
step=1e-05: matrix relative error median=3.85323e-12, eigenvalue max error median=1.23976e-12, stratum preserved=1.000
step=3e-05: matrix relative error median=1.28078e-12, eigenvalue max error median=4.29156e-13, stratum preserved=1.000
step=0.0001: matrix relative error median=3.8397e-13, eigenvalue max error median=1.21015e-13, stratum preserved=1.000
step=0.0003: matrix relative error median=1.22277e-13, eigenvalue max error median=3.19831e-14, stratum preserved=1.000
step=0.001: matrix relative error median=3.82426e-14, eigenvalue max error median=8.49321e-15, stratum preserved=1.000
Exceptional transition

switch_step: 12405
requested_checkpoint_times: [120.0]
captured_stages: ['late', 'post_switch', 'pre_switch', 'split', 'time_120']
switch_time: 124.06
Face/rate cross-tab

geom=14aecd02f5aeb94a, aug=2cf5514fdead0bb3, dim=7, analytic_rate=0.18947368420946087, class=slow, directions=5, seeds=1, observed_mean=0.21765
geom=14aecd02f5aeb94a, aug=d294646895211be6, dim=7, analytic_rate=0.18947368421018373, class=slow, directions=5, seeds=1, observed_mean=0.22796
geom=2b177b7da6cac989, aug=0a90cfa80be08fc9, dim=9, analytic_rate=0.3157894736814206, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=1659e561214524e3, dim=9, analytic_rate=0.31578947368223453, class=typical, directions=5, seeds=1, observed_mean=0.3153
39

geom=2b177b7da6cac989, aug=1c6ea29224a637bd, dim=9, analytic_rate=0.31578947368197074, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=2de3a963e490d6ee, dim=9, analytic_rate=0.3157894736821514, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=3b6d0425621290fe, dim=9, analytic_rate=0.3157894736823916, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=7b7d0d2b3877912c, dim=9, analytic_rate=0.3157894736824113, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.31578947368286064, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.3157894736828568, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=9dabfa924a701bca, dim=9, analytic_rate=0.3157894736824954, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=af32d066365ca848, dim=9, analytic_rate=0.31578947368222166, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=c04d83214330be0b, dim=9, analytic_rate=0.315789473682082, class=typical, directions=5, seeds=1, observed_mean=0.315339
geom=2b177b7da6cac989, aug=c5ee06cfcc08c626, dim=9, analytic_rate=0.3157894736837359, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736832496, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.31578947368284876, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736830174, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736827342, class=typical, directions=5, seeds=1, observed_mean=0.31533
9

geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819312, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736822557, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819449, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=e3d136d5f1d6e726, dim=9, analytic_rate=0.31578947368241506, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e48911c9443d998d, dim=9, analytic_rate=0.31578947368212723, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e491756b5b26e49e, dim=9, analytic_rate=0.31578947368343385, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=other, directions=2, seeds=1, observed_mean=0.26117
2
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=slow, directions=3, seeds=1, observed_mean=0.237274
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=other, directions=2, seeds=1, observed_mean=0.25397
8
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=slow, directions=2, seeds=1, observed_mean=0.234745
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=typical, directions=1, seeds=1, observed_mean=0.306
127
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=other, directions=4, seeds=1, observed_mean=0.28332
1
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=slow, directions=1, seeds=1, observed_mean=0.212315
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=other, directions=2, seeds=1, observed_mean=0.26554
5
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=slow, directions=2, seeds=1, observed_mean=0.230225

geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=typical, directions=1, seeds=1, observed_mean=0.306
662
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=other, directions=3, seeds=1, observed_mean=0.28777
2
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=slow, directions=2, seeds=1, observed_mean=0.22524
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=other, directions=3, seeds=1, observed_mean=0.25977
8
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=slow, directions=2, seeds=1, observed_mean=0.236267
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=other, directions=2, seeds=1, observed_mean=0.26033
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=slow, directions=2, seeds=1, observed_mean=0.241212
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=typical, directions=1, seeds=1, observed_mean=0.311
815
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=other, directions=2, seeds=1, observed_mean=0.26512
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=slow, directions=2, seeds=1, observed_mean=0.234456
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=typical, directions=1, seeds=1, observed_mean=0.3047
29
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=other, directions=2, seeds=1, observed_mean=0.27844
2
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=slow, directions=3, seeds=1, observed_mean=0.224237
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=amplitude_unreliable, directions=3, seeds=1, observed_mean=0.16858
3
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=slow, directions=2, seeds=1, observed_mean=0.158063
geom=92ecb568a4b5a7fe, aug=8701e4b031407f1d, dim=9, analytic_rate=0.31578947368212773, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=other, directions=3, seeds=1, observed_mean=0.263865

geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=slow, directions=2, seeds=1, observed_mean=0.222896
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=other, directions=1, seeds=1, observed_mean=0.250798
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=slow, directions=4, seeds=1, observed_mean=0.222301
geom=be68c2d65b45a5bc, aug=3fcce2388580c72d, dim=14, analytic_rate=0.2526315789466494, class=other, directions=5, seeds=1, observed_mean=0.283548
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=other, directions=3, seeds=1, observed_mean=0.278485
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=slow, directions=2, seeds=1, observed_mean=0.228032
geom=db7e1d4c31dc4662, aug=b7201b175b4c2407, dim=9, analytic_rate=0.3157894736817076, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
Modal trajectory prediction

median RMSE=0.000326729, median R2=0.999998, median max error=0.000524989
R2>=0.95: 784/800
Interpretation constraints

A face class includes support, saturation, outward set, endpoint controllers, and equilibrium rates.
Repeated rational-looking rates must be derived from gamma, alpha, incidence, and weights before being called quantized.
The exceptional split Jacobian is not a stationary-face Jacobian; pre/post/late stages are reported separately.
Negative spectral abscissa is frozen-fast local evidence, not full coupled stability.
B_carrier_only/full_reset phase matching is unrelated to this analysis.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_face_spectrum
(deck@steamdeck SSRH)$ python AIEF_1_4_channel_invariance_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--modal-summary aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_channel_invariance
AIEF tangent-channel invariance analysis
Channel invariance

Seeds: 40
Tangent-invariant seeds: 40/40
Multiplicity matches: 40/40
Median commutator: 1.61247e-13
Repeated fingerprint consistency

geometry_only 14aecd02f5aeb94a: seeds=2, classes=1, max_error=7.22866e-13, same=True
geometry_only 2b177b7da6cac989: seeds=22, classes=1, max_error=2.31531e-12, same=True
geometry_only 64c52ca6d685b8e1: seeds=9, classes=1, max_error=2.56217e-12, same=True
geometry_plus_rates 07f892d076a10acc: seeds=2, classes=1, max_error=5.03431e-13, same=True
geometry_plus_rates 806dee29e27c1ae3: seeds=2, classes=1, max_error=9.85878e-14, same=True
geometry_plus_rates ccb3288420826ec6: seeds=4, classes=1, max_error=5.15421e-13, same=True
geometry_plus_rates dd135d71da61c124: seeds=3, classes=1, max_error=3.752e-13, same=True
labeled_face 3bdfb4c570dc7116: seeds=10, classes=1, max_error=2.56217e-12, same=True
labeled_face 704fba7c1b319e34: seeds=6, classes=1, max_error=1.5527e-12, same=True
labeled_face 9961a1ce7b7ee3e2: seeds=6, classes=1, max_error=1.42819e-12, same=True
labeled_face aae02280e373342b: seeds=6, classes=1, max_error=1.7869e-12, same=True
labeled_face c6c8302b4dd4fa84: seeds=4, classes=1, max_error=1.08857e-12, same=True
labeled_face f45f68f9953c10f7: seeds=2, classes=1, max_error=7.22866e-13, same=True
Unique-direction modal prediction

unique_direction_count: 200
reliable_unique_direction_count: 197
all_amplitudes_success_count: 196
median_success_count: 196
reliable_all_amplitudes_success_fraction: 0.9898477157360406
reliable_median_success_fraction: 0.9898477157360406
median_unique_direction_R2: 0.9999982972649254
median_unique_direction_RMSE: 0.00032672743392392497
Exceptional stages

split: face=248a6d985749bc6f, dim=9, rate=0.015, channels=[{"multiplicity":1,"rate":0.015},
{"multiplicity":1,"rate":0.09826945523623593},{"multip
licity":2,"rate":0.1894736842053526 5},{"multiplicity":5,"rate":0.25263157894654914}],
commutator=9.62e-13
time_100: face=6fd9e17d4647a368, dim=9, rate=0.015,
channels=[{"multiplicity":1,"rate":0.015000000000000041},{"multiplicity":1,"rate":0.126146084
0335144},{"multiplicity":2,"rate":0.1894736842107972},
{"multiplicity":5,"rate":0.252631578947735}], commutator=1.25e-14

time_120: face=6fd9e17d4647a368, dim=9, rate=0.015,
channels=[{"multiplicity":1,"rate":0.015000000000000027},{"multiplicity":1,"rate":0.126261432
77148452},{"multiplicity":2,"rate":0.18947368421082056},
{"multiplicity":5,"rate":0.2526315789477355}], commutator=1.23e-14
time_123: face=6fd9e17d4647a368, dim=9, rate=0.015, channels=[{"multiplicity":1,"rate":0.015},
{"multiplicity":1,"rate":0.12626997804990325},{"mul
tiplicity":2,"rate":0.189473684210822},{"multiplicity":5,"rate":0.25263157894773547}],
commutator=1.23e-14
pre_switch: face=6fd9e17d4647a368, dim=9, rate=0.015,
channels=[{"multiplicity":1,"rate":0.014999999999999986},{"multiplicity":1,"rate":0.1262726
1581510735},{"multiplicity":2,"rate":0.189473684210822},
{"multiplicity":5,"rate":0.25263157894773547}], commutator=1.23e-14
post_switch: face=a64b354e39d6fb8c, dim=8, rate=0.126272587943,
channels=[{"multiplicity":1,"rate":0.12627258794250404},{"multiplicity":2,"rate":
0.18947368421082197},{"multiplicity":5,"rate":0.2526315789477355}], commutator=5.56e-08
time_125: face=a64b354e39d6fb8c, dim=8, rate=0.126274866092,
channels=[{"multiplicity":1,"rate":0.12627486609235697},{"multiplicity":2,"rate":0.1
8947368421082214},{"multiplicity":5,"rate":0.25263157894773564}], commutator=5.27e-08
time_150: face=a64b354e39d6fb8c, dim=8, rate=0.126305964314,
channels=[{"multiplicity":1,"rate":0.12630596431375102},{"multiplicity":2,"rate":0.1
8947368421083344},{"multiplicity":5,"rate":0.25263157894773547}], commutator=1.29e-08
late: face=a64b354e39d6fb8c, dim=8, rate=0.126315788879,
channels=[{"multiplicity":1,"rate":0.12631578887891748},{"multiplicity":2,"rate":0.18947
36842108335},{"multiplicity":5,"rate":0.25263157894773564}], commutator=7.85e-13
Interpretation constraints

Tolerance-based spectrum identity is not bitwise identity.
Geometry signatures are descriptive invariants, not complete graph-isomorphism canonical forms.
Channel multiplicity follows directly only when the tangent projector commutes with the diagonal channel operator.
Modal trajectories repeat amplitudes; unique seed/direction is the independent directional unit.
Exceptional split spectra are not assumed to be stationary-face spectra.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_channel_invariance
Readexceptional_tangent_report.txt
Readtrajectory_recovery_report.txt
Readface_spectrum_report.txt
Readchannel_invariance_report.txt

These results substantially strengthen the operator-family interpretation. Two questions are now answered:

repeated geometry classes have the same complete tangent spectra within numerical tolerance;
the exceptional active-set transition changes the tangent operator in a geometrically interpretable way.

1. Tangent/channel invariance is now directly established numerically

Across all 40 seeds:

text

Tangent-invariant seeds: 40/40
Multiplicity matches: 40/40
Median commutator: 1.61×10⁻13

Thus:
[D,PT]≈0
[D,PT]≈0

for every mature split state.

This means the tangent space is numerically invariant under:
D=diag(−γ−ρe).
D=diag(−γ−ρe).

Consequently:
TF=
TF=

\bigoplus_r
(T_{\mathcal F}\cap E_r)
]

and each channel rate appears in the tangent spectrum with multiplicity:
mr=dim(TF∩Er).
mr=dim(TF∩Er).

The observed multiplicities are not merely fitted spectral patterns. They agree exactly with channel/tangent intersection dimensions in every seed.

That provides the missing explanation for:

2 slow + 9 fast;
2 slow + 5 intermediate;
5 intermediate + 9 fast;
ninefold isotropic fast contraction.

2. Geometry-only fingerprints predict complete spectra under tolerance

The tolerance-aware analysis gives, for all repeated geometry-only classes:

text

distinct spectrum classes = 1
same full spectrum = True
maximum errors $\approx 10^{-}12$

In particular:

text

22-seed geometry class: one full spectrum
9-seed geometry class: one full spectrum
2-seed geometry class: one full spectrum

This resolves the misleading exact-hash result.

Within the observed mature classes:

The geometry-only signature determines the complete tangent spectrum to numerical precision.

This remains empirical because the geometry signature is not a complete graph-isomorphism canonicalization. But it is much stronger than only matching the spectral abscissa.
3. The modal propagator predicts coupled recovery at the correct unit

After collapsing amplitude replicates:

text

Unique directions: 200
Reliable directions: 197
All-amplitude successes: 196
Reliable success fraction: 98.98%
Median unique-direction R^2: 0.9999983
Median unique-direction RMSE: 0.0003267

This is the appropriate result—not 784/800 treated as independent.

The analytic matrix exponential:
eJTt
eJTt

predicts complete coupled tangent-recovery trajectories for approximately 99% of reliable unique directions.

This validates:

eigenvalues;
multiplicities;
eigenvectors/tangent decomposition;
finite-time modal mixing;
not merely asymptotic decay.

The few failures should now be cross-referenced with:

amplitude reliability;
exceptional seed membership;
stratum preservation;
tangent basis conditioning.

4. The exceptional transition changes the operator class

The exceptional sequence is now very clear.
Split

Dimension:
9.
9.

Spectrum:
$\{-0.015\}\times 1 \cup \{-0.098269\}\times 1 \cup \{-0.189474\}\times 2 \cup \{-0.252632\}\times 5$.
$\{-0.015\}\times 1 \cup \{-0.098269\}\times 1 \cup \{-0.189474\}\times 2 \cup \{-0.252632\}\times 5$.
Before switch

By $t=120$$t=120$–123, the second channel has drifted:
$-0.098269 \longrightarrow -0.12627$.
$-0.098269 \longrightarrow -0.12627$.

The bare-decay mode remains:
-0.015.
-0.015.

So even within one pre-switch face, one equilibrium correction channel evolves continuously while the multiplicity structure stays fixed.
At the switch

At:
$t=124.06$,
$t=124.06$,

the tangent dimension changes:
$9 \longrightarrow 8$.
$9 \longrightarrow 8$.

The $-0.015$$-0.015$ bare-decay mode disappears.

Post-switch spectrum:
$\{-0.12627\}\times 1 \cup \{-0.189474\}\times 2 \cup \{-0.252632\}\times 5$.
$\{-0.12627\}\times 1 \cup \{-0.189474\}\times 2 \cup \{-0.252632\}\times 5$.

This is a direct operator transition, not merely an improvement in forcing balance.

The new saturated constraint removes one tangent degree of freedom, and that degree of freedom is precisely the weakest bare-decay mode.

This explains the four-order-of-magnitude improvement in stationarity after the switch.
5. The transition suggests an interlacing mechanism

Before and after the switch:
$T_{post} \subset T_{pre}$
$T_{post} \subset T_{pre}$

with codimension one.

The post-switch Jacobian is a compression of the channel operator onto the smaller tangent space.

By min–max/interlacing principles for symmetric operators, restricting the tangent subspace can remove a weak direction and make the spectral abscissa more negative.

Here it removes:
-0.015
-0.015

and exposes:
-0.12627
-0.12627

as the new slowest mode.

A promising theorem is:

When a newly active independent capacity constraint reduces the tangent space, the fixed-stratum tangent spectrum interlaces the previous spectrum. If the removed direction carries the previous spectral abscissa, the transition strictly increases the local contraction rate.

The equilibrium channel rate also drifts slightly, so the actual transition combines:

geometric subspace restriction;
continuous correction-state evolution.

6. Geometry versus equilibrium is now resolved more precisely

The data support a three-level relationship:
active geometry⟶channel invariant subspaces/multiplicities,
active geometry⟶channel invariant subspaces/multiplicities,
mature correction state⟶channel values ρe,
mature correction state⟶channel values ρe,
(QT,ρ)⟶JT.
(QT,ρ)⟶JT.

For repeated mature geometry classes, the correction values converge to the same numerical classes, so geometry-only fingerprints predict complete spectra.

For the exceptional transient face, one channel value evolves continuously before switching. Therefore geometry alone does not force discrete rates at every transient state.

A careful conclusion is:

Mature attractor states realize a discrete spectral family, while transitional states can move continuously within a fixed multiplicity structure.

7. The spectral alphabet should distinguish mature and transitional channels
Repeated mature channels

Strongly supported:
0.189474,0.252632,0.315789.
0.189474,0.252632,0.315789.
Bare decay
0.015=γ
0.015=γ

appears as the exceptional weak mode.
Transitional channel
0.098269→0.126316
0.098269→0.126316

evolves continuously and should not be treated as a fixed member of a mature spectral alphabet.

Thus the “finite alphabet” is best described as a property of mature stationary strata, not the complete hybrid trajectory.

8. Fixed-stratum contractivity is now nearly analytic

Since:
[D,PT]≈0
[D,PT]≈0

and channel multiplicities match, the tangent space decomposes into channel eigenspaces.

Therefore:
JT=
JT=

Q_T^\top DQ_T
]

is symmetric negative definite, with exact channel-rate eigenvalues on those invariant subspaces.

For mature fixed strata:
‖eJTt‖2=
‖eJTt‖2=

e^{t\lambda_{\max}(J_T)}.
]

No hidden tangent transient amplification exists under the frozen-fast fixed-stratum model.

The 99% modal-prediction success shows this structure remains predictive even when the complete fast subsystem evolves.

9. Remaining caveats

Geometry signature

The summary signature is descriptive, not a complete canonical isomorphism invariant. The repeated-class result is nevertheless strong for the observed data.
Immediately post-switch commutator

The exceptional post-switch commutator is initially:
5.6×10^{-8},
5.6×10^{-8},

then decays to:
7.9×10^{-13}.
7.9×10^{-13}.

So tangent/channel invariance is temporarily imperfect during the transition and becomes exact only after relaxation.

This is another useful hybrid signature:

fixed strata: invariant channel decomposition;
switching transient: weak channel mixing;
mature post-switch face: invariant decomposition restored.

Coupled stability

All Jacobian theorems are frozen-fast and fixed-stratum. Full coupled stability is supported numerically by modal prediction, not yet proven.
10. Strongest defensible statement

Mature AIEF dynamics are organized by a stratified family of symmetric negative-definite tangent operators. In all 40 mature split states, tangent spaces are invariant under the diagonal edge-channel relaxation operator, and channel/tangent intersection dimensions exactly predict spectral multiplicities. Repeated geometry-only fingerprints produce identical complete spectra within $10^{-10}10^{-10}$, while analytic matrix exponentials predict approximately 99% of reliable coupled tangent-recovery directions with near-unit R2R2. The exceptional trajectory demonstrates an operator-valued hybrid transition: a new saturated constraint removes a bare-decay tangent mode, reduces tangent dimension by one, and increases the slowest contraction rate from 0.0150.015 to approximately 0.1260.126, followed by restoration of channel invariance.

This is now a genuinely operator-level dynamical result.

Next mathematical tasks

Derive channel invariance:
[D,PT]=0
[D,PT]=0 from incidence/controller structure.
Express:
dim(TF∩Er)
dim(TF∩Er) combinatorially.
Prove fixed-stratum symmetric contraction.
Prove an interlacing result for active-face transitions.
Derive why mature equilibrium selects the repeated rates:
0.189474, 0.252632, 0.315789.
0.189474, 0.252632, 0.315789.
Analyze the temporary post-switch channel mixing.

The project has moved beyond detecting spectral stratification. It now has an explicit mechanism: tangent spaces decompose into invariant edge-relaxation channels, and hybrid active-set transitions alter contraction by removing or mixing those channels.

And this to the list of next steps as well:

Having considered all of the documents together, I think the picture has become substantially sharper. The new results don't simply add more numerical evidence—they resolve several of the conceptual uncertainties that existed in the earlier operator interpretation. At the same time, there are still a few places where the mathematical claims should remain carefully qualified.

What I think has genuinely changed

Originally the evidence chain was something like

[
\text{geometry}
\rightarrow
\text{repeated spectra}
\rightarrow
\text{possible operator family}.
]

Now it is much stronger:

[
\boxed{
\text{geometry}
\rightarrow
T_{\mathcal F}
\rightarrow
[D,P_T]\approx0
\rightarrow
\text{channel decomposition}
\rightarrow
J_{\mathcal F}
\rightarrow
e^{Jt}\text{ predicts trajectories}.
}
]

Each arrow is now supported by an independent numerical test rather than inferred indirectly.

The biggest conceptual advance is not the spectrum

I actually think the most important discovery is not the discrete spectrum.

It is

[
[D,P_T]\approx0.
]

The spectrum is a consequence.

Before the channel-invariance analysis, there were two possible explanations for repeated eigenvalues.

One possibility was simply that

[
Q^TDQ
]

happened numerically to have eigenvalues equal to entries of (D).

The second possibility was that

[
T_{\mathcal F}
]

is actually invariant under (D).

The new analysis overwhelmingly favors the second.

The numerical evidence

40/40 invariant,
40/40 multiplicity matches,
commutator around (10^{-13}),

is exactly the diagnostic one would design if trying to distinguish those two mechanisms.

That is why I think the emphasis of the theory should now move from "spectral alphabet" toward invariant channel decomposition.

The alphabet follows from the decomposition.

The multiplicities have become mathematical objects

Earlier,

[
2,;5,;9
]

looked like repeated empirical block sizes.

Now they look like

[
m_r

\dim(T_{\mathcal F}\cap E_r).
]

That is a very different interpretation.

Instead of saying

"the Jacobian has multiplicity 5",

the natural statement becomes

"the active face contains a five-dimensional invariant channel module."

That is a geometric statement rather than merely a spectral one.

Geometry and equilibrium now separate cleanly

One of the nicest things about the synthesis is that it removes an ambiguity that had existed throughout the earlier discussions.

The data now suggest three distinct layers.

Layer 1

Pure combinatorics:

support,
saturation,
controller incidence,
tangent space.

This determines

$$
T_{\mathcal F}.
$$

Layer 2

Equilibrium state.

This determines

$$
\rho_e.
$$

These are continuous quantities in principle.

Layer 3

Restriction.

Since

$$
[D,P_T]=0,
$$

the tangent operator becomes

$$
J_{\mathcal F}D|{T{\mathcal F}}.
$$

The spectrum is then immediate.

That decomposition is much cleaner than trying to say "geometry determines the spectrum."

Geometry determines the invariant subspaces.

Equilibrium determines the channel strengths.

The spectrum is the combination.

The mature versus transient distinction has become much clearer

This may actually be the most important refinement in the whole project.

Originally there was an attempt to describe a finite spectral alphabet.

The exceptional trajectory shows why that statement was slightly too broad.

The mature classes exhibit repeated values like

[
0.189474,;
0.252632,;
0.315789.
]

The exceptional trajectory instead shows

[
0.098269
\rightarrow
0.126272
\rightarrow
0.126316,
]

while the multiplicity pattern remains fixed before switching.

That means:

multiplicities are discrete,
one channel value evolves continuously.

So the better picture is

$$
\boxed{
\text{transient faces}
\rightarrow
\text{continuous channel evolution}
}
$$

versus

$$
\boxed{
\text{mature attractor}
\rightarrow
\text{discrete repeated channel values}.
}
$$

That distinction is mathematically cleaner than trying to make every observed rate part of one universal alphabet.

The exceptional transition is richer than "dimension drops"

The dimension change

$$
9\rightarrow8
$$

is important.

But I think something deeper is happening.

Immediately after the switch,

the commutator is roughly

[
10^{-8},
]

before relaxing again to machine precision.

That means there is a brief period in which

[
[D,P_T]\neq0.
]

So the evolution appears to consist of three regimes:

Invariant decomposition

[
[D,P_T]\approx0.
]

Switching layer

[
[D,P_T]
]

temporarily nonzero.

New invariant decomposition

[
[D,P_T]\approx0.
]

That is a much richer hybrid picture than simply saying "the face changed."

It suggests that hybrid transitions themselves possess an internal relaxation phase.

The trajectory validation is stronger than the spectral validation

I actually regard the modal propagation test as stronger evidence than the eigenvalue agreement.

The spectrum validates

[
J.
]

The propagator validates

[
e^{Jt}.
]

The latter depends on

eigenvalues,
eigenvectors,
modal decomposition,
relative projections,
finite-time evolution.

Obtaining roughly

99%

success at the independent-direction level is a much more stringent test than merely matching the spectral abscissa.

So I think the propagator result should occupy a central place in the eventual presentation.

One theorem is now almost inevitable

The empirical evidence strongly suggests that the eventual mathematics should not begin with spectral statements.

It should begin with an invariance theorem.

Conceptually, something like

Invariant-channel theorem

For every mature active face,

$$
[D,P_{\mathcal F}]=0.
$$

Hence

$$
T_{\mathcal F}
\bigoplus_r
(T_{\mathcal F}\cap E_r),
$$

and

$$
J_{\mathcal F}

\bigoplus_r
(-\gamma-\rho_r)
I_{m_r}.
]

Everything else—

spectral multiplicities,
isotropic dimension-9 class,
repeated spectra,
matrix exponentials—

becomes a corollary.

That is a much more satisfying mathematical organization.

The remaining theoretical bottleneck has become very specific

Earlier there were several competing open questions.

Now I think there is essentially one dominant mathematical problem.

Not

Why are the spectra discrete?

but

Why does mature equilibrium produce tangent subspaces that commute with the diagonal channel operator?

That is the structural question.

Everything else appears downstream.

If one could derive

$$
[D,P_{\mathcal F}]=0
$$

from the incidence/controller equations together with the mature balance conditions, then much of the observed operator structure would follow almost automatically.

One point where I would still be cautious

There is one statement I would still avoid elevating to theorem status:

"Geometry-only fingerprints determine the spectrum."

The numerical evidence for the observed cohort is now very strong—after tolerance-aware comparison, repeated geometry fingerprints correspond to a single spectrum class within numerical precision.

However, the reports themselves also note an important limitation: the geometry signature being used is a descriptive invariant, not a proven canonical characterization of active faces.

So I would phrase the current result as:

For the observed mature cohort, every repeated geometry fingerprint corresponds to a unique tangent operator (within numerical tolerance).

That is a very strong empirical result.

A theorem would require proving either:

that the fingerprint is a complete invariant of the relevant face geometry, or
that the operator depends only on the information encoded by that fingerprint.

Until then, the operator-family interpretation is well supported numerically, but the universality of the fingerprint remains an analytic question rather than a completed theorem.

Overall, I think the project has progressed from identifying recurring contraction rates to uncovering an underlying invariant-subspace mechanism. The evidence now points to the invariant channel decomposition as the central structural object, with the observed spectral families, multiplicities, and highly accurate modal predictions emerging as consequences of that decomposition rather than as independent phenomena.

Yes—this should be added near the top of the next-step list. It identifies the central theoretical mechanism more precisely than “spectral stratification.”
Revised central object

For one active face, let:

SS be the positive-edge support;
Asat,SAsat,S be the saturated-node incidence matrix restricted to SS;
tangent space:
TF=kerAsat,S;
TF=kerAsat,S;
diagonal edge-channel operator:
DF=diag(−γ−ρe).
DF=diag(−γ−ρe).

The numerical result is:
[DF,PT]≈0,
[DF,PT]≈0,

where:
PT=QTQT⊤.
PT=QTQT⊤.

This says the tangent space is invariant under the edge-channel operator:
DFTF⊆TF.
DFTF⊆TF.

That is the structural fact from which the observed spectral families follow.
Algebraic invariance criterion

There is a useful equivalent formulation.

Let:
C=Asat,S.
C=Asat,S.

Then:
TF=kerC.
TF=kerC.

The kernel is invariant under DD if and only if the row space of CC is invariant under DD. Under appropriate rank assumptions, this is equivalent to the existence of a node-space matrix LL such that:
CD=LC.
CD=LC.

This gives a concrete theorem and numerical diagnostic.

A least-squares candidate is:
L∗=
L∗=

CD,C^\top(CC^\top)^+,
]

or an equivalent pseudoinverse construction. Then test:
Eintertwine=
Eintertwine=

\frac{
|CD-L^*C|_F
}{
\max(|CD|_F,\epsilon)
}.
]

If:
Eintertwine≈0,
Eintertwine≈0,

then channel invariance is explained directly at the incidence/controller level.

This may be the cleanest bridge from face geometry to spectral decomposition.
Why multiplicities follow

Partition positive edges into channel groups:
Er={e:γ+ρe=r}.
Er={e:γ+ρe=r}.

If:
[D,PT]=0,
[D,PT]=0,

then:
TF=
TF=

\bigoplus_r
(T_{\mathcal F}\cap E_r).
]

On each component:
D|Er=−rI.
D|Er=−rI.

Therefore:
JF=
JF=
D|{T{\mathcal F}}

\bigoplus_r
-rI_{m_r},
]

with:
mr=dim(TF∩Er).
mr=dim(TF∩Er).

This explains why the channel-invariance script found:

text

40/40 invariant
40/40 multiplicities matched

The dimensions 2, 5, and 9 are not merely spectral multiplicities. They are dimensions of tangent/channel intersections.
Mature versus transient structure

The exceptional trajectory gives a particularly clean decomposition.
Before the saturated-set switch

The geometry remains fixed while one channel evolves:
$0.098269 \to 0.126273$.
$0.098269 \to 0.126273$.

During this period:
$[D,PT] \approx 10^{-14}$.
$[D,PT] \approx 10^{-14}$.

Thus:

the invariant channel decomposition remains intact;
equilibrium changes one channel strength continuously;
multiplicities remain discrete.

At the switch

The tangent dimension changes:
$9 \to 8$.
$9 \to 8$.

The bare-decay channel:
-0.015
-0.015

disappears.

Immediately afterward:
[D,PT]≈5×10−8.
[D,PT]≈5×10−8.

Thus the switching layer temporarily mixes channels.
Late post-switch

The commutator returns to:
≈8×10−13.
≈8×10−13.

A new invariant decomposition is established.

This supports a three-stage hybrid process:
invariant channel flow → channel-mixing switch → new invariant channel flow.
invariant channel flow → channel-mixing switch → new invariant channel flow.

The 10−810−8 commutator should still be tested against boundary tolerance and timestep, but its transient decay is highly suggestive.
Updated hierarchy of claims
Empirically strong

Tangent/channel invariance in all 40 mature split states.
Exact multiplicity prediction in all 40.
One tolerance-based spectrum class for every repeated geometry fingerprint.
Approximately 99% unique-direction modal-prediction success.
Exceptional switch removes one weak tangent channel and temporarily breaks channel invariance.

Analytically established from the fixed-stratum formula

Given channel invariance:
JF=
JF=

Q_T^\top DQ_T
]

reduces to channel restrictions and is:

symmetric;
negative definite;
normal;
Euclidean-contracting;
free of fixed-stratum transient amplification.

Still open

Why mature equilibrium restricts ρeρe to repeated values.
Why the incidence/controller structure satisfies CD=LCCD=LC.
Whether the descriptive geometry fingerprint is a complete invariant.
Whether the transient commutator is genuine channel mixing or tolerance sensitivity.
A general switching theorem.

Revised prioritized next steps
1. Derive and test the intertwining relation

For every seed and exceptional stage:
CD=LC.
CD=LC.

Output:

L∗L∗;
intertwining residual;
rank of CC;
channel partition;
tangent/channel dimensions;
predicted multiplicities.

This is now the highest-priority analytic/numerical step.
2. Derive mature channel values

Using constraint-balanced stationarity:
PwF=0,

PwF=0,

derive equations for:
ρe=max(αi,αj).
ρe=max(αi,αj).

Determine why mature states select:
0.174474,0.237632,0.300789
0.174474,0.237632,0.300789

before adding γ=0.015γ=0.015.

Test whether values depend on:

saturated-node degree;
endpoint-controller multiplicity;
row mass;
positive-edge weights;
incidence rank.

3. Analyze the switching layer

Around t=124.06t=124.06, sample at every step:

C=Asat,SC=Asat,S;
DD;
PTPT;
[D,PT][D,PT];
rank and tangent dimension;
channel multiplicities;
intertwining residual.

Determine whether the commutator spike precedes, coincides with, or follows the active-set switch.
4. Make the face fingerprint mathematically complete

The existing geometry hash is descriptive. For a theorem, use either:

exact labeled face data;

a canonical graph-isomorphism representation;
or prove the operator depends only on the current summary features.

Until then, say:

Every repeated observed geometry fingerprint corresponds to one tangent spectrum.

not:

Every geometrically equivalent face has that spectrum.

5. Present modal prediction by unique direction

Use:

text

196/200 all-amplitude successes
197 reliable unique directions
98.98% reliable success
median $R^2 \approx 0.999998$

This is one of the headline results because it validates the complete propagator, not only its spectrum.
Strongest current statement

Mature AIEF tangent dynamics admit an invariant channel decomposition. In every tested mature split state, the tangent projector commutes with the diagonal edge-relaxation operator to numerical precision, and tangent/channel intersection dimensions exactly predict spectral multiplicities. Repeated observed geometry fingerprints produce identical complete spectra under tolerance. The analytic fixed-stratum propagators accurately predict approximately 99% of reliable unique-direction coupled recovery trajectories. An exceptional active-set transition removes a weak tangent channel, temporarily mixes the channel decomposition, and then settles onto a new invariant operator stratum.

That is stronger and more precise than “the spectrum is quantized by geometry.” The central mathematical problem is now to derive the intertwining relation:
CD=LC
CD=LC

from the active incidence/controller equations and mature constraint-balance conditions.

Can you create the code necessary to complete the next step?
WriteAIEF_1_4_intertwining_controller_analysis.py706 lines
Edit
Edit×2
Edit
Edit
Edit×2
Edit
Edit×4
Edit
AIEF_1_4_intertwining_controller_analysis.pyPYTHON

Created the intertwining and controller-equilibrium analyzer:

AIEF_1_4_intertwining_controller_analysis.py

It targets the current highest-priority question:
Why does [D,PT]≈0 and why do tangent multiplicities match channel dimensions?
Why does [D,PT]≈0 and why do tangent multiplicities match channel dimensions?
Prerequisites

Run the corrected tangent and updated face-spectrum analyses first:

Bash

```
python AIEF_1_4_exceptional_tangent_analysis.py \
--max-seeds 0
```

Bash

```
python AIEF_1_4_face_spectrum_analysis.py
```

Required files:

text

aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv
aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv

Run

Bash

python AIEF_1_4_intertwining_controller_analysis.py

Explicit:

Bash

python AIEF_1_4_intertwining_controller_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--modal-summary aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_intertwining_controller

1. Node-space intertwining relation

For every mature seed it constructs:
C=Asat,S,
C=Asat,S,

where SS is the positive-edge support, and:
D=diag(−γ−ρe).
D=diag(−γ−ρe).

It computes the least-squares node-space operator:
L∗=
L∗=

(CD)C^+,

]

and tests:
$CD \approx L * C.$
$CD \approx L * C.$

Reported quantities:

text

intertwining_residual_relative
rowspace_invariance_residual_relative
constraint_matrix_rank

This tests the algebraic condition equivalent to invariance of:
$TF = \ker C$
$TF = \ker C$

under DD.

The actual matrices are saved to:

text

intertwining_controller_matrices.npz

with per-seed:

text

C_active
D_channel
L_intertwiner
P_tangent
J_tangent

2. Channel intersection dimensions

Edges are grouped by:
re=γ+ρe
re=γ+ρe

under configurable tolerance:

Bash

--channel-tol 1e-10

For every channel it calculates:
mr=dim(TF∩Er)
mr=dim(TF∩Er)

and compares this with the observed tangent eigenvalue multiplicity.

Outputs include:

text

intersection_dimension
observed_eigenvalue_multiplicity
multiplicity_matches
predicted_multiplicity_total

It also tests whether:
TF=
TF=

\bigoplus_r(T_{\mathcal F}\cap E_r).
]
3. Controller-equilibrium equation

For each seed, it constructs the node-controller matrix KK.

If edge e=(i,j)e=(i,j) is controlled by node kk, its weight is added to:

KikandKjk.
KikandKjk.

The mature balance equation is:
diag(c)α≈Kα.
diag(c)α≈Kα.

Equivalently:
α
α

should be an eigenvector with eigenvalue one of:
diag(1/c)K.
diag(1/c)K.

The script reports:

text

controller_equilibrium_residual_relative
controller_eigenvalue_distance_to_one

and saves per-node:

saturation/outward state;
row sum and slack;
row forcing didi;
αiαi;
positive-support degree;
controller edge count;
controller-equation residual.

This directly tests why mature endpoint correction values collapse onto repeated classes.
4. Direct edge-balance relation

For every positive edge it tests:
F~e≈weρe.
F

e≈wepe.

For zero edges it tests absence of positive clipped forcing.

Outputs:

text

edge_balance_residual_relative
zero_edge_positive_forcing_violation

5. Tolerance-aware spectrum consistency

For geometry-only, geometry-plus-rate, and labeled-face classes, it reports:

text

maximum_matched_spectrum_error
same_spectrum_within_tolerance
intertwining_residual_max
commutator_max
all_multiplicities_match

Default tolerance:

text

1e-10

6. Exceptional switching layer

At:

text

split

time_100
time_120
time_123
pre_switch
post_switch
time_125
time_150
late

it recomputes:

active face;
channel partition;
CD=LCCD=LC residual;
tangent commutator;
multiplicities;
controller-equilibrium residual;
field norm.

This determines whether the temporary post-switch commutator is accompanied by:

failure of the node-space intertwiner;
controller-equilibrium mismatch;
changing channel values;
altered tangent dimension.

7. Unique-direction modal validation

The script collapses amplitude replicates and reports modal prediction success at the independent unit:

text

seed + direction_id

Outputs include:

all-amplitude R2R2 success;
median R2R2 success;

amplitude reliability;
RMSE and maximum prediction error;
frozen/observed rates.

Outputs

text

```
aief_1_4_intertwining_controller/
├── channel_invariance_by_seed.csv
├── channel_multiplicities_by_seed.csv
├── node_controller_equilibrium_by_seed.csv
├── geometry_spectrum_consistency_tolerant.csv
├── unique_direction_modal_success.csv
├── modal_success_summary.json
├── exceptional_channel_stages.csv
├── exceptional_transition_metadata.json
├── intertwining_controller_matrices.npz
├── channel_invariance_report.txt
└── analysis_manifest.json
```

Validation

I syntax-tested the script and ran its complete pipeline against a generated AIEF state with tangent/modal outputs.

The technical test produced:

text

CD=LC relative residual: ~1.3×10^{-12}
Tangent commutator: ~6.6×10^{-13}
Controller-equilibrium residual: ~2.5×10^{-12}
Edge-balance residual: ~9×10^{-12}
Multiplicity match: True
Unique-direction modal R^2: ~0.999994

This confirms the implementation can recover the node-space intertwining relation, controller equilibrium, and tangent-channel multiplicities to near machine precision.

The 40-seed run can now determine whether these identities hold uniformly across mature geometry classes and how they temporarily fail or reorganize during the exceptional active-face transition.
Summary of previous conversation

This session is being continued from a previous conversation that ran out of context. The summary below covers the earlier portion of the conversation.
Goal

Continue the SSRH / AIEF computational-mathematics research program seamlessly from its current state.

Immediate goal:

Run and interpret the newly created:

text

AIEF_1_4_intertwining_controller_analysis.py

against the frozen AIEF-1.4 40-seed run.
Test the proposed algebraic mechanism behind tangent-channel invariance:
CD≈LC,
CD≈LC, where:
C=Asat,S,D=diag(−γ−ρe).
C=Asat,S,D=diag(−γ−ρe).
Test the mature controller-equilibrium relation:
diag(c)α≈Kα.
diag(c)α≈Kα.
Determine whether repeated active-face geometry classes imply complete tangent spectra because the tangent space decomposes into invariant edge-relaxation channels:
TF=⊕r(TF∩Er).
TF=r⊕(TF∩Er).
Interpret how the exceptional active-face transition temporarily disrupts and then restores channel invariance.
Continue separating:
established results;

numerical evidence;
plausible interpretation;
open conjectures/theorems.

Longer-term goal:

Prove the ideal AIEF admissibility map is Hadamard directionally differentiable.
Derive:
$P_{w,e}(F)=F\sim e-we\max(\alpha i,\alpha j)$.
$P_{w,e}(F)=F$
$e-we\max(\alpha i,\alpha j)$.
Prove tangent feasibility, identity on tangent directions, range equality, idempotence, positive homogeneity, and node-permutation equivariance.
Derive the fixed-stratum tangent Jacobian:
$J_T=Q_T\top \operatorname{diag}(-\gamma-\rho e)Q_T$.
$J_T=Q_T\top \operatorname{diag}(-\gamma-\rho e)Q_T$.
Explain the repeated contraction-channel values and multiplicities analytically.
Only later consider AIEF-1.5 mirror/proximal dynamics or quasiperiodic fast-subsystem analysis.

Instructions

Be rigorous and skeptical; do not merely agree with prior interpretations.
Always separate:
facts established by saved artifacts;
numerical evidence;
plausible mechanism;
conjecture/open mathematics.
Treat one paired seed/graph realization as the inferential unit; edges within a seed are not independent.
Do not treat repeated workflow-version runs using identical seeds as independent experiments.
Do not call BB the unique or dominant memory carrier: reset_B reconstructs mature topology nearly exactly.
Do not conflate:
threshold persistence with continuous-weight similarity;
storage with reconstruction with maintenance;
adaptive-component stationarity with full-system equilibrium;
coordinate change with mirror flow with Bregman proximal point with proximal Galerkin;
topology similarity with Kuramoto order parameter;
hard admissibility projection with a soft regularizer.
Do not use ALPHA = 0 as “instantaneous resonance”: under:

Python

```
z_inst = np.exp(1j * dphi)
R = np.abs(z_inst)
```

it makes every off-diagonal instantaneous magnitude equal to 1.
The AIEF admissibility map is not known to be an orthogonal Euclidean projection.
Mirror dynamics are a new model:
$w˙=H-1F/\beta$,
$w˙=H-1F/\beta$, not a coordinate rewrite of current AIEF.
Bregman divergence is not a global geodesic distance.
The entropy geometry is natural, not proven canonical.
The user prefers iterative technical dialogue, often asks “think about this,” “ponder,” or requests updated code. Responses should be precise, candid, and scientifically conservative while still recognizing strong results.
If code is changed, preserve reproducibility, source hashes, manifests, atomic checkpoints, seed pairing, and explicit interpretation caveats.
Do not suggest another broad seed cohort until existing mechanistic analyses are exhausted.
The user is interested in deeper operator theory and potentially formalization, but quasiperiodic/KAM/Ott–Antonsen directions are downstream and should remain separate from the immediate active-boundary problem.

Discoveries
Core AIEF model

N=12, T_sim=800, dt=0.01, tau=0.5.

Frequencies:

Python

```
omega = np.array([
1.00, 1.02, 1.50, 2.00, 2.01,
1.01, 1.03, 1.51, 2.02, 2.00,
1.00, 1.04
])
```

Parameters:

text

a=1.0
q=1.5
delta=0.05
eta=0.04
gamma=0.015
mu=0.03
B_c=0.12
B_init_scale=0.0005
D_B_star=0.30
T_window=25/mean(omega)
alpha=exp(-dt/T_window)

Phase dynamics:
θ˙i=
θ˙i=

\omega_i+
\sum_jB_{ij}
\sin(\theta_j(t-\tau)-\theta_i(t)).
]

EMA resonance:
Cijn+1=
Cijn+1=

\alpha C_{ij}^n+
(1-\alpha)e^{i\Delta\phi_{ij}},
\qquad
R_{ij}=|C_{ij}|.
]

Adaptive weights:
B˙=
B˙=

(\eta+\mu)R_{\rm driver}
-\mu

-\gamma B.
]

Dominance constraint:
∑jBij≤c=
j∑Bij≤c=
\frac{a-\delta}{q}

0.633333\ldots
]

The actual symmetric admissibility update scales each edge by the more restrictive endpoint factor using:

Python

B *= np.minimum.outer(scale, scale)

after nonnegativity and symmetry.
Early bugs fixed during AIEF-1.3 development

Original pasted code had HTML/Markdown corruption.
Delay deque used history[0] with extra 20 entries, producing approximately 0.69 delay instead of 0.5.
Row-wise gap projection destroyed symmetry.
no_delay retained nonzero tau correction in compute_R.
Original shuffle_R was spatial/identity shuffling, not true temporal shuffling.
Diagonal/self-links were inconsistent with graph metrics; diagonal now zero.
Correlation now uses unique undirected edges.
Floating-time loop replaced with integer steps.
G_min_final originally used minimum row sum instead of maximum; fixed.
Undefined correlations now use NaN, not zero.
Phi remains undefined/unstable near zero density and is excluded from later inference.
P_future and historical R now use non-overlapping windows.

AIEF-1.3 production infrastructure

Principal driver:

text

AIEF_1_3_falsification_updated.py

Features:

SeedSequence randomized paired seed sets from one master seed.
Independent initialization/shuffle RNG streams.
Exact delay indexing and constant prehistory.
Symmetry-preserving gap map.
Atomic checkpoints and validated resume.
Source/config hashes.
Full manifests.
Concentration, degeneracy, and projection diagnostics.

Independent 50-seed validation cohorts:

text

runs/c3da1ac707bf/
runs/73dca37cf93f/

Earlier 20-seed workflow-version reruns are deterministic duplicates and not independent evidence.
AIEF-1.3 100-seed paired results

Approximate pooled baseline:

text

D_B ~0.306
Corr_physical ~0.995
Effective_edges ~14.6
Total edge weight ~3.64

shuffle_R:

text

D_B=0
P_mean=0
P_degenerate=1 for every seed
Total edge weight ~0.060

Interpretation: timestep-wise destruction of persistent edge-specific driver identity produces a tiny diffuse residual field and abolishes threshold topology.

no_mu:

text

D_B ~0.035
Total edge weight ~3.78
Effective_edges ~34.4
Gini ~0.761 -> ~0.482
Entropy ~0.683 -> ~0.884
Nonzero edge fraction ~0.306 -> 1.0

Interpretation: μμ is a pruning/concentration mechanism, not a coupling-existence mechanism.

no_gap:

text

D_B ~0.727
Total edge weight ~127.9
G_min ~-32.8

Interpretation: gap removal yields a dense high-weight regime outside the desired envelope; it does not prove divergence or loss of coherence.

no_delay remains in the same qualitative regime and slightly increases several observables. Explicit delay is not necessary under current EMA/adaptation dynamics.
AIEF-1.3 closure

Scripts:

text

AIEF_1_3_paired_statistics_updated.py
AIEF_1_3_closure_diagnostics_updated.py

Closure on two independent 50-seed cohorts:

text

CLOSURE PASSED: True

Established:

text

baseline_unique_edges_exact_binary: True
baseline_full_matrix_exact_binary: True
matrix_scalar_consistent: True
projection_activity_subset_of_violation: True
cohorts_seed_disjoint: True

Baseline P_future is exactly binary for every unique edge and seed; topology is threshold-frozen throughout the future window.

Projection:

text

baseline active fraction ~0.972
no_mu active fraction ~0.988
projection active given violation = 1

Projection corrections are small relative to full B but nearly the full proposed incremental update.
AIEF-1.4 split-state experiment

Principal driver:

text

AIEF_1_4_memory_ablation_updated.py

Frozen 40-seed production run:

text

runs/5a9f093a2fe0/

Branches:

text

continue
reset_ema
reset_B
reset_B_ema
B_carrier_only
full_reset
future_shuffle

Important r2 caveat:

text

B_carrier_only
full_reset

used different randomized phase/history resets. Their direct contrast remains exploratory until matched-phase rerun.

Late summaries:

text

branch P_late ratio Sim_late Retention Emergence
continue 1.000 0.999 1.000 0.000
reset_ema 0.990 0.993 0.990 0.000
reset_B 1.000 0.999 1.000 0.000
reset_B_ema 1.080 0.944 0.987 0.038
B_carrier_only 0.945 0.948 0.908 0.015
full_reset 1.035 0.884 0.920 0.046
future_shuffle 0.000 0.475 0.000 0.000

Interpretation:

no single state variable is the unique memory register;
reset_B reconstructs almost exactly;
EMA accelerates reconstruction when BB is absent;
model structure defines a strong but nonunique attractor;
future driver identity is necessary for threshold maintenance.

AIEF-1.4 trajectory/survival

Scripts:

text

AIEF_1_4_results_analysis.py
AIEF_1_4_trajectory_recovery_equivalence.py
AIEF_1_4_survival_outlier_spectral.py

Sustained 90% similarity recovery:

text

continue 40/40, RMUT 0.01
reset_ema 39/40, RMUT 21.36
reset_B 40/40, RMUT 4.77
reset_B_ema 38/40, RMUT 45.37
B_carrier_only 34/40, RMUT 68.52

full_reset 25/40, RMUT 156.72
future_shuffle 0/40, RMUT 320

reset_B late residual:

text

normalized residual mean ~9.16e-06
participation ratio median ~5.33
largest-edge fraction median ~0.516

Entropy geometry

Script:

text

AIEF_1_4_entropy_geometry_diagnostics.py

Unique undirected edge vector:
$w \in R66, C=\{w \geq 0, Aw \leq c\}$.
$w \in R66, C=\{w \geq 0, Aw \leq c\}$.

Margins:
$m=c-Aw$.
$m=c-Aw$.

Entropy potential:
$h(w)=\sum ewe(logwe-1)+\sum imi(logmi-1)$.
$h(w)=e\sum we(logwe-1)+i\sum mi(logmi-1)$.

Gradient/Hessian:
$\psi=logw-A\top logm$,
$\psi=logw-A\top logm$,
$H=diag(1/w)+A\top diag(1/m)A$.
$H=diag(1/w)+A\top diag(1/m)A$.

Coupled inverse:
wij=eψijmimj,
wij=eψijmimj,
mi(1+∑jeψijmj)=ci.
mi(1+j∑eψijmj)=ci.

Woodbury:
H−1=
H−1=

D_w-
D_wA^\top
(D_m+AD_wA^\top)^{-1}
AD_w.
]

Observed:

text

~70% exact-zero edges
~65% node margins <=1e-12
median minimum node margin ~1e-12

Hessian condition numbers range approximately 1e4 to 1e12, depending on interiorization. The inverse map reconstructs edges near machine precision. Intrinsic quantities are strongly epsilon-dependent.

Conclusion: mature AIEF lies on a singular boundary face; mirror AIEF would be a new model.
Local update comparison

Scripts:

text

AIEF_1_4_local_update_comparison.py
AIEF_1_4_active_boundary_analysis.py

Compared:

text

actual AIEF admissibility update
coordinate-equivalent dual Euler: psi += dt H F
mirror Euler: psi += dt F / beta

Interior results:

text

current vs coordinate increment discrepancy order ~2
current vs mirror increment discrepancy order ~1

Coordinate dual Euler is first-order equivalent in the strict interior but extremely stiff near the mature boundary. Mirror Euler is a different vector field.
Candidate ideal active-boundary operator

For zero/positive edges:
F~e={Fe,we>0,max(Fe,0),we=0.
F
e={Fe,max(Fe,0),we>0,we=0.

For node ii:
di=∑e∋iF~e,
di=e∋i∑F
e,
αi={max(di,0)/ci,(Aw)i=ci,0,(Aw)i<ci.
αi={max(di,0)/ci,0,(Aw)i=ci,(Aw)i<ci.

For edge e=(i,j)e=(i,j):
Pw,e(F)=
Pw,e(F)=

\widetilde F_e-w_e\max(\alpha_i,\alpha_j).
]

Probe finite differences support:

text

rate error O(dt)
increment error O(dt^2)

At mature split states:

text

||F|| appreciable
||P_w(F)|| ~1e-13

Mature adaptive states are constraint-balanced; the full delayed state is not an equilibrium.

The ideal finite admissibility map is conjectured/provisionally derived to be a nonlinear retraction onto CC. Its directional operator is expected to:

map onto TC(w)TC(w);
be identity on tangent directions;
be idempotent;
be positively homogeneous;
be piecewise linear;
be node-permutation equivariant.

The fixed +1e-12 denominator safeguard has a different formal dt -> 0 limit; the theorem concerns the ideal map without it.
Trajectory-wide stationarity/perturbations

Script:

text

AIEF_1_4_stationarity_perturbation_analysis_updated.py

Full 40-seed result:

39/40 seeds:

text

stationary_fraction=1
||P_w(F)||/||F|| ~1e-13
zero-set switches=0
saturated-set switches=0
continuous B drift ~1e-12
replay_exact=True

Exceptional seed:

text

4619122171158768524
stationary_fraction=0.115125
mean ratio=0.0569358
max ratio=0.160206
saturated-set switches=1
max normalized B drift=0.208939
zero-set switches=0
replay_exact=True

Across ~1.28 million mature updates:

text

tangent failures=0
idempotence failures=0
positive-homogeneity failures=0
permutation equivariance errors ~1e-17

These are implementation audits of the candidate formula, not proofs that it equals the derivative.

Perturbation results:

normal inward/interior mixing:

κrestore∼1/δ,
κrestore∼1/δ, finite-speed, strongly transverse return;
tangent:
median:
κ≈0.315789,
κ≈0.315789, nearly radial, exponential return.

Corrected tangent experiment

Script:

text

AIEF_1_4_exceptional_tangent_analysis.py

Uses one tangent direction per seed/direction ID across all amplitudes.

40 seeds × 5 directions = 200 unique directions.

Typical result:

text

frozen rate ~0.315789
observed coupled rate ~0.315339
predicted t50 ~2.195
observed t50 ~2.2
predicted t90 ~7.292
observed t90 ~7.35
median R2 ~1

Exceptional seed:

text

switch step=12405
switch time=124.06

pre-switch stationarity ratio=0.146859
post-switch mean ratio=7.89306e-06
post-switch max=8.81385e-05
B drift=0.208939

Fixed-face tangent Jacobian

Updated script:

text

AIEF_1_4_face_spectrum_analysis.py

Analytic fixed-stratum Jacobian:
JT=
JT=

Q_T^\top
\operatorname{diag}(-\gamma-\rho_e)
Q_T.
]

Numerical/analytic matrix errors are approximately 1e-13 to 1e-11 with fixed stratum preserved.

Observed spectra include:

Dimension 9 dominant:
{−0.315789}×9.
{−0.315789}×9.

Dimension 11:
{−0.189474}×2∪{−0.315789}×9.
{−0.189474}×2∪{−0.315789}×9.

Dimension 7:
{−0.189474}×2∪{−0.252632}×5.
{−0.189474}×2∪{−0.252632}×5.

Dimension 14:
{−0.252632}×5∪{−0.315789}×9.
{−0.252632}×5∪{−0.315789}×9.

Exceptional split:
−0.015, −0.098269, −0.189474×2, −0.252632×5.
−0.015, −0.098269, −0.189474×2, −0.252632×5.

Matrix exponential modal prediction:

text

median trajectory R2 ~0.999998
median RMSE ~0.000327

Tolerance-aware channel invariance

Script:

text

AIEF_1_4_channel_invariance_analysis.py

Actual 40-seed output:

text

Tangent-invariant seeds: 40/40
Multiplicity matches: 40/40
Median commutator: 1.61247e-13

Repeated geometry-only fingerprints:

text

2 seeds: one spectrum class, max error 7.23e-13

22 seeds: one spectrum class, max error 2.32e-12
9 seeds: one spectrum class, max error 2.56e-12

Unique-direction modal prediction:

text

unique directions=200
reliable directions=197
all-amplitude success=196
reliable success fraction=0.9898477
median unique-direction R2=0.9999983
median RMSE=0.0003267

Exceptional stages:

text

split:
dim=9
channels:
1×0.015
1×0.098269455
2×0.189473684
5×0.252631579
commutator=9.62e-13

time_100/time_120/time_123/pre_switch:
dim=9
bare 0.015 persists
second channel drifts to ~0.1262726
commutator ~1.2e-14

post_switch:
dim=8
bare 0.015 channel disappears
channels:
1×0.126272588
2×0.189473684

5×0.252631579
commutator=5.56e-08

time_125:
commutator=5.27e-08

time_150:
commutator=1.29e-08

late:
channels:
1×0.126315789
2×0.189473684
5×0.252631579
commutator=7.85e-13

Interpretation:

tangent dimension drops $9 \to 8$ at the switch;
weak bare-decay mode disappears;
contraction strengthens from 0.015 to ~0.126;
channel invariance is temporarily imperfect after switching, then returns;
the exceptional transition changes the tangent operator, not only forcing balance.

Newly created intertwining/controller script

Latest file:

text

AIEF_1_4_intertwining_controller_analysis.py

Purpose:

Construct:
$C=A_{sat},S,D=\mathrm{diag}(-\gamma-\rho_e)$.
$C=A_{sat},S,D=\mathrm{diag}(-\gamma-\rho_e)$.
Find least-squares node-space LL satisfying:

CD≈LC.
CD≈LC.
Test controller equilibrium:
diag(c)α≈Kα.
diag(c)α≈Kα.
Test direct mature edge balance:
F~e≈weρe.
F
e≈weρe.
Verify channel/tangent intersection multiplicities.
Analyze exceptional stages.
Collapse modal prediction to unique seed/direction units.

The script underwent several debugging fixes:

Added missing constants:

text

DEFAULT_TANGENT_RESULTS
DEFAULT_MODAL_SUMMARY
DEFAULT_EXCEPTIONAL_CHECK_TIMES
DEFAULT_SPECTRUM_TOL
DEFAULT_CHANNEL_TOL
DEFAULT_SUBSPACE_TOL
DEFAULT_MODAL_R2

Added missing import:

Python

import AIEF_1_4_tangent_spectrum_switch_analysis as tangent_analysis

Replaced incorrect function calls:

text

channel_analysis_for_state

with:

text

channel_and_intertwining

Added subspace_tolerance argument.
Fixed unpacking because channel_and_intertwining returns:

text

summary, channels, nodes, matrices

Added:

Python

spectrum_vector()

Replaced stale call:

text

geometry_spectrum_consistency

with:

text

tolerance_spectrum_consistency

Added output:

text

node_controller_equilibrium_by_seed.csv
intertwining_controller_matrices.npz

Technical one-seed validation produced:

text

CD=LC residual ~1.33e-12
tangent commutator ~6.60e-13
controller equilibrium residual ~2.46e-12
edge-balance residual ~9.00e-12
multiplicity match=True
modal R2 ~0.999994

User-Pasted Content

Key user-pasted actual output from the final channel-invariance run:

text

AIEF tangent-channel invariance analysis
==========

Channel invariance

Seeds: 40
Tangent-invariant seeds: 40/40
Multiplicity matches: 40/40
Median commutator: 1.61247e-13

Repeated geometry fingerprint results:

text

geometry_only 14aecd02f5aeb94a:
seeds=2

classes=1
max_error=7.22866e-13
same=True

geometry_only 2b177b7da6cac989:
seeds=22
classes=1
max_error=2.31531e-12
same=True

geometry_only 64c52ca6d685b8e1:
seeds=9
classes=1
max_error=2.56217e-12
same=True

Unique-direction modal result:

text

unique_direction_count: 200
reliable_unique_direction_count: 197
all_amplitudes_success_count: 196
median_success_count: 196
reliable_all_amplitudes_success_fraction: 0.9898477157360406
reliable_median_success_fraction: 0.9898477157360406
median_unique_direction_R2: 0.9999982972649254
median_unique_direction_RMSE: 0.00032672743392392497

Exceptional-stage output:

text

split:
face=248a6d985749bc6f
dim=9
rate=0.015
channels=[
{"multiplicity":1,"rate":0.015},

{"multiplicity":1,"rate":0.09826945523623593},
{"multiplicity":2,"rate":0.18947368420535265},
{"multiplicity":5,"rate":0.25263157894654914}
]
commutator=9.62e-13

time_100:
face=6fd9e17d4647a368
dim=9
channels=[
1×0.015,
1×0.1261460840335144,
2×0.1894736842107972,
5×0.252631578947735
]
commutator=1.25e-14

pre_switch:
face=6fd9e17d4647a368
dim=9
channels=[
1×0.015,
1×0.12627261581510735,
2×0.189473684210822,
5×0.25263157894773547
]
commutator=1.23e-14

post_switch:
face=a64b354e39d6fb8c
dim=8
channels=[
1×0.12627258794250404,
2×0.18947368421082197,
5×0.2526315789477355
]
commutator=5.56e-08

late:
face=a64b354e39d6fb8c
dim=8

channels=[
1×0.12631578887891748,
2×0.1894736842108335,
5×0.25263157894773564
]
commutator=7.85e-13

Key pasted tangent-corrected output:

text

40 seeds × 5 directions = 200 unique directions

median frozen restoring coefficient = 0.315789
median observed exponential rate = 0.315339
median R2 ≈1

predicted t50=2.19497
observed t50=2.2
predicted t90=7.29152
observed t90=7.35

two slow/anomalous directions:
observed rate minimum=0.136767
final distance ratio maximum=0.124056

Key pasted 40-seed stationarity output:

text

39/40 seeds:
stationary_fraction=1
mean ||P_w(F)||/||F|| ~1e-13
zero-set switches=0
saturated-set switches=0
max B drift ~1e-12
audit failures=0/0/0
equivariance errors ~1e-17

replay_exact=True

Exceptional seed 4619122171158768524:
stationary_fraction=0.115125
mean ratio=0.0569358
max ratio=0.160206
saturated-set switches=1
max B drift=0.208939
zero-set switches=0
replay_exact=True

Accomplished
Completed

Fully corrected and productionized AIEF-1.3 driver.
Two independent 50-seed AIEF-1.3 validation cohorts.
Freeze-grade AIEF-1.3 closure.
Paired seed-level AIEF-1.3 inference.
AIEF-1.4 split-state memory-ablation experiment, 40 seeds.
Trajectory, recovery, equivalence, survival, spectral, and residual analyses.
Entropy geometry on symmetry-coupled degree-capacity polytope.
Coupled inverse-map solver and Woodbury verification.
Local current/coordinate/mirror one-step comparison.
Candidate active-boundary directional field.
Probe finite-difference verification.
Trajectory-wide stationarity and feasible perturbation analysis.
Corrected tangent perturbations with fixed directions.
Analytic/numerical fixed-stratum tangent Jacobians.
Full-spectrum/multiplicity analysis.
Modal trajectory prediction.
Tolerance-aware tangent-channel invariance.
Exceptional active-face transition analysis.
Created and debugged AIEF_1_4_intertwining_controller_analysis.py.

Not Solved

Full 40-seed run of AIEF_1_4_intertwining_controller_analysis.py has not yet been pasted/interpreted.
Need proof of ideal admissibility map directional differentiability.
Need proof that:
Pw

Pw maps onto the tangent cone and is identity/idempotent.
Need analytic derivation of:
CD=LC.
CD=LC.
Need analytic derivation of:
diag(c)α=Kα.
diag(c)α=Kα.
Need explanation for repeated mature channel values:
0.189474, 0.252632, 0.315789.
0.189474, 0.252632, 0.315789.
Geometry fingerprint is descriptive, not a complete canonical graph-isomorphism invariant.
Need matched-phase rerun of:

text

B_carrier_only
full_reset

Need timestep convergence for projection-dominated dynamics.
Need decide whether temporary exceptional post-switch commutator is genuine channel mixing versus tolerance sensitivity.
Quasiperiodic/KAM/Ott–Antonsen questions remain downstream.
Mirror/proximal AIEF-1.5 should not be implemented until the current admissibility/operator theory is understood.

Next Steps

Immediate:

Run:

Bash

python AIEF_1_4_intertwining_controller_analysis.py

on the real 40-seed files.
Inspect:

text

channel_invariance_report.txt
channel_invariance_by_seed.csv
channel_multiplicities_by_seed.csv
node_controller_equilibrium_by_seed.csv
exceptional_channel_stages.csv
intertwining_controller_matrices.npz

Determine whether:

text

CD=LC residuals ~ machine precision for all 40
controller equilibrium residuals ~ machine precision
edge-balance residuals ~ machine precision
multiplicity matches 40/40

Inspect exceptional stages for spikes in:

text

intertwining_residual_relative
tangent_commutator_relative
controller_equilibrium_residual_relative

around
t=124.06.
Derive the intertwining relation analytically:
CD=LC.
CD=LC.
Derive the controller equilibrium:
diag(c)α=Kα.
diag(c)α=Kα.
Use those equations to explain the repeated mature ρeρe and spectral classes.
Then complete the matched-phase correction and timestep convergence.
Only afterward consider AIEF-1.5 mirror/proximal dynamics.

Relevant files / directories

Workspace scripts created:

text

AIEF_1_3_falsification_updated.py
AIEF_1_3_paired_statistics.py
AIEF_1_3_paired_statistics_updated.py
AIEF_1_3_closure_diagnostics.py
AIEF_1_3_closure_diagnostics_updated.py

AIEF_1_4_memory_ablation.py
AIEF_1_4_memory_ablation_updated.py
AIEF_1_4_results_analysis.py
AIEF_1_4_trajectory_recovery_equivalence.py
AIEF_1_4_survival_outlier_spectral.py
AIEF_1_4_entropy_geometry_diagnostics.py
AIEF_1_4_local_update_comparison.py
AIEF_1_4_active_boundary_analysis.py
AIEF_1_4_stationarity_perturbation_analysis.py
AIEF_1_4_stationarity_perturbation_analysis_updated.py
AIEF_1_4_exceptional_tangent_analysis.py
AIEF_1_4_tangent_spectrum_switch_analysis.py
AIEF_1_4_face_spectrum_analysis.py
AIEF_1_4_channel_invariance_analysis.py
AIEF_1_4_intertwining_controller_analysis.py

AI_AGENT_HANDOFF_PROMPT.md
AIEF_1_4_memory_ablation_plan.md

Principal external result directories:

text

runs/c3da1ac707bf/
runs/73dca37cf93f/
runs/5a9f093a2fe0/

Analysis output directories referenced:

text

paired_seed_analysis_100/
aief_1_3_closure_diagnostics/
aief_1_4_results_analysis/
aief_1_4_trajectory_recovery/
aief_1_4_survival_outlier_spectral/
aief_1_4_entropy_geometry/
aief_1_4_local_update_comparison/
aief_1_4_active_boundary_analysis/
aief_1_4_stationarity_perturbation/
aief_1_4_exceptional_tangent/
aief_1_4_face_spectrum/
aief_1_4_channel_invariance/
aief_1_4_intertwining_controller/

Attached files read:

text

uploads/Results.pdf
uploads/exceptional_tangent_report.txt
uploads/trajectory_recovery_report.txt
uploads/face_spectrum_report.txt
uploads/channel_invariance_report.txt

Results.pdf was an 8-page PDF containing an embedded OOXML Word document. It was extracted in the workspace as:

text

results_embedded.bin
results_embedded.docx

The embedded text contained the full 40-seed stationarity/perturbation output summarized above.

Think of these results:

```
(deck@steamdeck SSRH)$ python AIEF_1_4_exceptional_tangent_analysis.py \
--max-seeds 0
Corrected tangent analysis for seed 172180395821707403
Corrected tangent analysis for seed 283404770198602602
Corrected tangent analysis for seed 305986450614148690
Corrected tangent analysis for seed 1230287949579537403
Corrected tangent analysis for seed 1888415619847841128
Corrected tangent analysis for seed 1918007468868772336
Corrected tangent analysis for seed 2461295749352584108
Corrected tangent analysis for seed 3258463328937640595
Corrected tangent analysis for seed 3582616062465030267
Corrected tangent analysis for seed 3704716633526084363
Corrected tangent analysis for seed 3952397966055989215
Corrected tangent analysis for seed 4040285944958847150
Corrected tangent analysis for seed 4219039243791140697
Corrected tangent analysis for seed 4619122171158768524
Corrected tangent analysis for seed 4932511216876650745
Corrected tangent analysis for seed 5213047409853278064
Corrected tangent analysis for seed 5415544643585892346
Corrected tangent analysis for seed 6023008189364820510
Corrected tangent analysis for seed 6108867655028980287
Corrected tangent analysis for seed 6942591650975756121
Corrected tangent analysis for seed 9036321132460399255
Corrected tangent analysis for seed 9082055256280075172
Corrected tangent analysis for seed 9189988769602851417
Corrected tangent analysis for seed 10431384713483787270
Corrected tangent analysis for seed 10856823803646276806
Corrected tangent analysis for seed 11242426753702356865
Corrected tangent analysis for seed 12123519807951751650
Corrected tangent analysis for seed 12851640484293199234
Corrected tangent analysis for seed 14040336450255043598
Corrected tangent analysis for seed 14924267685654216746
Corrected tangent analysis for seed 15403081841477312463
Corrected tangent analysis for seed 15853699719446171583
Corrected tangent analysis for seed 16329425431442249417
Corrected tangent analysis for seed 16627818730624180313
Corrected tangent analysis for seed 16952466944611321358
Corrected tangent analysis for seed 17279552413280415429
Corrected tangent analysis for seed 17521310920850592782
Corrected tangent analysis for seed 17715532056777442089
```

Corrected tangent analysis for seed 17883847597205629106
Corrected tangent analysis for seed 18127452951968972828
AIEF exceptional-seed and corrected tangent analysis
Exceptional seed

seed: 4619122171158768524
switch_count: 1
first_switch_step: 12405
first_switch_time: 124.06
pre_switch_mean_stationarity_ratio: 0.14685940684975718
post_switch_mean_stationarity_ratio: 7.893059517783126e-06
post_switch_max_stationarity_ratio: 8.813848174066108e-05
maximum_B_drift: 0.20893854278186944
final_B_drift: 0.20893854278186944
final_replay_exact: True
final_replay_allclose_1e12: True
Corrected tangent perturbations

delta=0.0001
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.21003e-08, min=3.73207e-09, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.315339, valid=200/200
observed_exponential_r_squared: median=1, min=0.994943, max=1, valid=200/200
final_distance_ratio: median=0.0088257, min=0.0088257, max=0.124056, valid=200/200
delta=0.001
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.2104e-09, min=3.73033e-10, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.315339, valid=200/200
observed_exponential_r_squared: median=1, min=0.994947, max=1, valid=200/200
final_distance_ratio: median=0.00882571, min=0.00882569, max=0.124056, valid=200/200
delta=0.01
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200

frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=1.21398e-10, min=3.71325e-11, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14.05, valid=198/200
observed_exponential_rate: median=0.315339, min=0.136767, max=0.31534, valid=200/200
observed_exponential_r_squared: median=1, min=0.994565, max=1, valid=200/200
final_distance_ratio: median=0.00882573, min=0.00882558, max=0.124056, valid=200/200
delta=0.05
frozen_restoring_coefficient: median=0.315789, min=-33657.8, max=233.435, valid=200/200
frozen_restoring_cosine: median=1, min=-0.522944, max=1, valid=200/200
frozen_transverse_ratio: median=2.43583e-11, min=7.27637e-12, max=0.999517, valid=200/200
predicted_t50: median=2.19497, min=0.00296933, max=3.47049, valid=198/200
predicted_t90: median=7.29152, min=0.0098639, max=11.5287, valid=198/200
observed_t50: median=2.2, min=2.2, max=4.4, valid=200/200
observed_t90: median=7.35, min=7.35, max=14.6, valid=198/200
observed_exponential_rate: median=0.315338, min=0.136767, max=0.315343, valid=200/200
observed_exponential_r_squared: median=1, min=0.990186, max=1, valid=200/200
final_distance_ratio: median=0.00882586, min=0.00882513, max=0.124056, valid=200/200
Interpretation constraints

The same tangent direction is reused across all amplitudes within each seed/direction ID.
Directions with negligible feasible amplitude are rejected.
The exceptional-seed mechanism remains descriptive until its switch is analyzed in detail.
Predicted tangent times assume local exponential contraction.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_exceptional_tangent
(deck@steamdeck SSRH)$ python AIEF_1_4_face_spectrum_analysis.py
AIEF active-face spectrum and exceptional transition analysis

Seeds with fingerprints: 40
Distinct labeled face hashes: 12
Distinct invariant signatures: 33
Analytic fixed-stratum spectral classes

dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684209+0.000000000000j":1,"-0.189473684210+0.000000000000j":1,"-0.252631578947
+0.000000000000j":5}

dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.252631578947+0.000000000000j":5}
dimension=7, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.252631578946+0.000000000000j":5}
dimension=9, rate=0.015, seeds=1,
multiplicities={"-0.015000000000+0.000000000000j":1,"-0.098269455236+0.000000000000j":1,"-0.1
89473684205+0.0000
00000000j":1,"-0.189473684206+0.000000000000j":1,"-0.252631578946+0.000000000000j":1,"-0.25
2631578947+0.000000000000j":4}
dimension=9, rate=0.3157894737, seeds=3,
multiplicities={"-0.315789473682+0.000000000000j":5,"-0.315789473683+0.000000000000j":4}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473682+0.000000000000j":6,"-0.315789473683+0.000000000000j":3}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473683+0.000000000000j":1,"-0.315789473684+0.000000000000j":8}
dimension=9, rate=0.3157894737, seeds=6, multiplicities={"-0.315789473683+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=7, multiplicities={"-0.315789473682+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=2,
multiplicities={"-0.315789473682+0.000000000000j":1,"-0.315789473683+0.000000000000j":8}
dimension=9, rate=0.3157894737, seeds=1, multiplicities={"-0.315789473684+0.000000000000j":9}
dimension=9, rate=0.3157894737, seeds=2,
multiplicities={"-0.315789473682+0.000000000000j":2,"-0.315789473683+0.000000000000j":7}
dimension=9, rate=0.3157894737, seeds=1,
multiplicities={"-0.315789473681+0.000000000000j":1,"-0.315789473682+0.000000000000j":8}
dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473681+0.000000000000j":5,"-0.3
1578947368
2+0.000000000000j":4}
dimension=11, rate=0.1894736842, seeds=2,
multiplicities={"-0.189473684208+0.000000000000j":2,"-0.315789473682+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684209+0.000000000000j":2,"-0.315789473683+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=4,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473682+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=2,
multiplicities={"-0.189473684210+0.000000000000j":2,"-0.315789473683+0.000000000000j":9}
dimension=11, rate=0.1894736842, seeds=1,
multiplicities={"-0.189473684208+0.000000000000j":2,"-0.315789473682+0.000000000000j":2,"-0.3
1578947368
3+0.000000000000j":7}

dimension=14, rate=0.2526315789, seeds=1,
multiplicities={"-0.252631578947+0.000000000000j":5,"-0.315789473666+0.000000000000j":5,"-0.3
1578947366
7+0.000000000000j":4}
Fingerprint-to-spectrum consistency

geometry_only 14aecd02f5aeb94a: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True, same_full_spectrum=False
geometry_only 2b177b7da6cac989: seeds=22, rate_range=2.31531e-12, same_rate_1e10=True, same_full_spectrum=False
geometry_only 64c52ca6d685b8e1: seeds=9, rate_range=2.56217e-12, same_rate_1e10=True, same_full_spectrum=False
geometry_plus_rates 07f892d076a10acc: seeds=2, rate_range=5.61218e-14, same_rate_1e10=True, same_full_spectrum=True
geometry_plus_rates 806dee29e27c1ae3: seeds=2, rate_range=3.83027e-15, same_rate_1e10=True, same_full_spectrum=True
geometry_plus_rates ccb3288420826ec6: seeds=4, rate_range=5.15421e-13, same_rate_1e10=True, same_full_spectrum=True
geometry_plus_rates dd135d71da61c124: seeds=3, rate_range=3.24463e-13, same_rate_1e10=True, same_full_spectrum=True
labeled_face 3bdfb4c570dc7116: seeds=10, rate_range=2.56217e-12, same_rate_1e10=True, same_full_spectrum=False
labeled_face 704fba7c1b319e34: seeds=6, rate_range=1.50263e-12, same_rate_1e10=True, same_full_spectrum=False
labeled_face 9961a1ce7b7ee3e2: seeds=6, rate_range=1.42819e-12, same_rate_1e10=True, same_full_spectrum=False
labeled_face aae02280e373342b: seeds=6, rate_range=1.76514e-12, same_rate_1e10=True, same_full_spectrum=False
labeled_face c6c8302b4dd4fa84: seeds=4, rate_range=9.11937e-13, same_rate_1e10=True, same_full_spectrum=False
labeled_face f45f68f9953c10f7: seeds=2, rate_range=7.22866e-13, same_rate_1e10=True, same_full_spectrum=False
Numerical/analytic Jacobian agreement

step=1e-06: matrix relative error median=3.93954e-11, eigenvalue max error median=1.25038e-11, stratum preserved=1.000
step=3e-06: matrix relative error median=1.24753e-11, eigenvalue max error median=4.02821e-12, stratum preserved=1.000
step=1e-05: matrix relative error median=3.85323e-12, eigenvalue max error median=1.23976e-12, stratum preserved=1.000
step=3e-05: matrix relative error median=1.28078e-12, eigenvalue max error median=4.29156e-13, stratum preserved=1.000

step=0.0001: matrix relative error median=3.8397e-13, eigenvalue max error median=1.21015e-13, stratum preserved=1.000
step=0.0003: matrix relative error median=1.22277e-13, eigenvalue max error median=3.19831e-14, stratum preserved=1.000
step=0.001: matrix relative error median=3.82426e-14, eigenvalue max error median=8.49321e-15, stratum preserved=1.000
Exceptional transition

switch_step: 12405
requested_checkpoint_times: [120.0]
captured_stages: ['late', 'post_switch', 'pre_switch', 'split', 'time_120']
switch_time: 124.06
Face/rate cross-tab

geom=14aecd02f5aeb94a, aug=2cf5514fdead0bb3, dim=7, analytic_rate=0.18947368420946087, class=slow, directions=5, seeds=1, observed_mean=0.21765
geom=14aecd02f5aeb94a, aug=d294646895211be6, dim=7, analytic_rate=0.18947368421018373, class=slow, directions=5, seeds=1, observed_mean=0.22796
geom=2b177b7da6cac989, aug=0a90cfa80be08fc9, dim=9, analytic_rate=0.3157894736814206, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=1659e561214524e3, dim=9, analytic_rate=0.31578947368223453, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=1c6ea29224a637bd, dim=9, analytic_rate=0.31578947368197074, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=2de3a963e490d6ee, dim=9, analytic_rate=0.3157894736821514, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=3b6d0425621290fe, dim=9, analytic_rate=0.3157894736823916, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=7b7d0d2b3877912c, dim=9, analytic_rate=0.3157894736824113, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.31578947368286064, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=806dee29e27c1ae3, dim=9, analytic_rate=0.3157894736828568, class=typical, directions=5, seeds=1, observed_mean=0.31533
9

geom=2b177b7da6cac989, aug=9dabfa924a701bca, dim=9, analytic_rate=0.3157894736824954, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=af32d066365ca848, dim=9, analytic_rate=0.31578947368222166, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=c04d83214330be0b, dim=9, analytic_rate=0.315789473682082, class=typical, directions=5, seeds=1, observed_mean=0.315339
geom=2b177b7da6cac989, aug=c5ee06cfcc08c626, dim=9, analytic_rate=0.3157894736837359, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736832496, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.31578947368284876, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736830174, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=ccb3288420826ec6, dim=9, analytic_rate=0.3157894736827342, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819312, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736822557, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=dd135d71da61c124, dim=9, analytic_rate=0.3157894736819449, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
geom=2b177b7da6cac989, aug=e3d136d5f1d6e726, dim=9, analytic_rate=0.31578947368241506, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e48911c9443d998d, dim=9, analytic_rate=0.31578947368212723, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=2b177b7da6cac989, aug=e491756b5b26e49e, dim=9, analytic_rate=0.31578947368343385, class=typical, directions=5, seeds=1, observed_mean=0.3153
39

geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=other, directions=2, seeds=1, observed_mean=0.26117
2
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420962285, class=slow, directions=3, seeds=1, observed_mean=0.237274
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=other, directions=2, seeds=1, observed_mean=0.25397
8
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=slow, directions=2, seeds=1, observed_mean=0.234745
geom=64c52ca6d685b8e1, aug=07f892d076a10acc, dim=11, analytic_rate=0.18947368420967897, class=typical, directions=1, seeds=1, observed_mean=0.306
127
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=other, directions=4, seeds=1, observed_mean=0.28332
1
geom=64c52ca6d685b8e1, aug=25f78fa97f296893, dim=11, analytic_rate=0.18947368420905605, class=slow, directions=1, seeds=1, observed_mean=0.212315
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=other, directions=2, seeds=1, observed_mean=0.26554
5
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=slow, directions=2, seeds=1, observed_mean=0.230225
geom=64c52ca6d685b8e1, aug=26793e8b1ce90b16, dim=11, analytic_rate=0.18947368421016234, class=typical, directions=1, seeds=1, observed_mean=0.306
662
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=other, directions=3, seeds=1, observed_mean=0.28777
2
geom=64c52ca6d685b8e1, aug=3659a8766236791a, dim=11, analytic_rate=0.18947368421005456, class=slow, directions=2, seeds=1, observed_mean=0.22524
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=other, directions=3, seeds=1, observed_mean=0.25977
8
geom=64c52ca6d685b8e1, aug=6683bef31abd1738, dim=11, analytic_rate=0.18947368420806243, class=slow, directions=2, seeds=1, observed_mean=0.236267
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=other, directions=2, seeds=1, observed_mean=0.26033
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=slow, directions=2, seeds=1, observed_mean=0.241212
geom=64c52ca6d685b8e1, aug=b1f018ff70fc10b0, dim=11, analytic_rate=0.18947368420760016, class=typical, directions=1, seeds=1, observed_mean=0.311

815
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=other, directions=2, seeds=1, observed_mean=0.26512
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=slow, directions=2, seeds=1, observed_mean=0.234456
geom=64c52ca6d685b8e1, aug=b54c595f9f37059f, dim=11, analytic_rate=0.1894736842077195, class=typical, directions=1, seeds=1, observed_mean=0.3047
29
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=other, directions=2, seeds=1, observed_mean=0.27844
2
geom=64c52ca6d685b8e1, aug=ee57001f81a95ff1, dim=11, analytic_rate=0.18947368420975563, class=slow, directions=3, seeds=1, observed_mean=0.224237
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=amplitude_unreliable, directions=3, seeds=1, observed_mean=0.16858
3
geom=655e0f6492c19e81, aug=ab6054460bf1ec5d, dim=9, analytic_rate=0.015, class=slow, directions=2, seeds=1, observed_mean=0.158063
geom=92ecb568a4b5a7fe, aug=8701e4b031407f1d, dim=9, analytic_rate=0.31578947368212773, class=typical, directions=5, seeds=1, observed_mean=0.3153
39
geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=other, directions=3, seeds=1, observed_mean=0.263865
geom=98960b32a6801790, aug=dc56ebe3075dd7d3, dim=11, analytic_rate=0.189473684210065, class=slow, directions=2, seeds=1, observed_mean=0.222896
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=other, directions=1, seeds=1, observed_mean=0.250798
geom=aaad232e68fe332e, aug=6369a11a50c217ee, dim=7, analytic_rate=0.18947368420995325, class=slow, directions=4, seeds=1, observed_mean=0.222301
geom=be68c2d65b45a5bc, aug=3fcce2388580c72d, dim=14, analytic_rate=0.2526315789466494, class=other, directions=5, seeds=1, observed_mean=0.283548
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=other, directions=3, seeds=1, observed_mean=0.278485
geom=d810ec51ff744b91, aug=8d06a358028731f3, dim=11, analytic_rate=0.1894736842098878, class=slow, directions=2, seeds=1, observed_mean=0.228032
geom=db7e1d4c31dc4662, aug=b7201b175b4c2407, dim=9, analytic_rate=0.3157894736817076, class=typical, directions=5, seeds=1, observed_mean=0.31533
9
Modal trajectory prediction

median RMSE=0.000326729, median R2=0.999998, median max error=0.000524989
R2>=0.95: 784/800

Interpretation constraints

A face class includes support, saturation, outward set, endpoint controllers, and equilibrium rates.
Repeated rational-looking rates must be derived from gamma, alpha, incidence, and weights before being called quantized.
The exceptional split Jacobian is not a stationary-face Jacobian; pre/post/late stages are reported separately.
Negative spectral abscissa is frozen-fast local evidence, not full coupled stability.
B_carrier_only/full_reset phase matching is unrelated to this analysis.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_face_spectrum
(deck@steamdeck SSRH)$ python AIEF_1_4_intertwining_controller_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--modal-summary aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_intertwining_controller
AIEF intertwining and controller-equilibrium analysis
Intertwining

Seeds: 40
Median CD=LC residual: 3.19796e-13
Maximum CD=LC residual: 1.42698e-12
Median tangent commutator: 1.61247e-13
Multiplicity matches: 40/40
Controller equilibrium

Median controller residual: 2.75885e-12
Maximum controller residual: 0.0969322
Median edge-balance residual: 1.85734e-12
Repeated fingerprint consistency

geometry_only 14aecd02f5aeb94a: seeds=2, max_spectrum_error=7.22866e-13, same=True, max_CDLC=4.9e-13
geometry_only 2b177b7da6cac989: seeds=22, max_spectrum_error=2.31531e-12, same=True, max_CDLC=4.91e-13

geometry_only 64c52ca6d685b8e1: seeds=9, max_spectrum_error=2.56217e-12, same=True, max_CDLC=4.91e-13
geometry_plus_rates 07f892d076a10acc: seeds=2, max_spectrum_error=5.03431e-13, same=True, max_CDLC=3.98e-13
geometry_plus_rates 806dee29e27c1ae3: seeds=2, max_spectrum_error=9.85878e-14, same=True, max_CDLC=2.15e-13
geometry_plus_rates ccb3288420826ec6: seeds=4, max_spectrum_error=5.15421e-13, same=True, max_CDLC=4.91e-13
geometry_plus_rates dd135d71da61c124: seeds=3, max_spectrum_error=3.752e-13, same=True, max_CDLC=2.67e-13
labeled_face 3bdfb4c570dc7116: seeds=10, max_spectrum_error=2.56217e-12, same=True, max_CDLC=5.22e-13
labeled_face 704fba7c1b319e34: seeds=6, max_spectrum_error=1.5527e-12, same=True, max_CDLC=4.62e-13
labeled_face 9961a1ce7b7ee3e2: seeds=6, max_spectrum_error=1.42819e-12, same=True, max_CDLC=4.34e-13
labeled_face aae02280e373342b: seeds=6, max_spectrum_error=1.7869e-12, same=True, max_CDLC=4.24e-13
labeled_face c6c8302b4dd4fa84: seeds=4, max_spectrum_error=1.08857e-12, same=True, max_CDLC=4.91e-13
labeled_face f45f68f9953c10f7: seeds=2, max_spectrum_error=7.22866e-13, same=True, max_CDLC=4.9e-13
Unique-direction modal prediction

unique_direction_count: 200
reliable_unique_direction_count: 197
all_amplitudes_success_count: 196
median_success_count: 196
reliable_all_amplitudes_success_fraction: 0.9898477157360406
reliable_median_success_fraction: 0.9898477157360406
median_unique_direction_R2: 0.9999982972649254
median_unique_direction_RMSE: 0.00032672743392392497
Exceptional switching layer

split: face=248a6d985749bc6f, dim=9, channels=4, CDLC=1.43e-12, commutator=9.62e-13, controller_residual=0.0969
time_100: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.22e-14, commutator=1.25e-14, controller_residual=0.122
time_120: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.18e-14, commutator=1.23e-14, controller_residual=0.122
time_123: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.19e-14, commutator=1.23e-14, controller_residual=0.122

pre_switch: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.19e-14, commutator=1.23e-14, controller_residual=0.122
post_switch: face=a64b354e39d6fb8c, dim=8, channels=4, CDLC=6.83e-08, commutator=5.56e-08, controller_residual=0.17
time_125: face=a64b354e39d6fb8c, dim=8, channels=4, CDLC=6.48e-08, commutator=5.27e-08, controller_residual=0.17
time_150: face=a64b354e39d6fb8c, dim=8, channels=4, CDLC=1.58e-08, commutator=1.29e-08, controller_residual=0.17
late: face=a64b354e39d6fb8c, dim=8, channels=3, CDLC=9.65e-13, commutator=7.85e-13, controller_residual=0.17
Interpretation constraints

CD=LC is tested numerically using a least-squares node-space intertwiner.
Geometry signatures are descriptive, not complete isomorphism invariants.
Mature channel values may be discrete empirically without forming a universal finite alphabet.
Exceptional post-switch states can temporarily violate exact channel invariance.
Modal success is counted by unique seed/direction, not repeated amplitude rows.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_intertwining_controller
(deck@steamdeck SSRH)$ python AIEF_1_4_intertwining_controller_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--modal-summary aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_intertwining_controller \
--channel-tol 1e-5
AIEF intertwining and controller-equilibrium analysis
Intertwining

Seeds: 40
Median CD=LC residual: 3.19796e-13
Maximum CD=LC residual: 1.42698e-12
Median tangent commutator: 1.61247e-13
Multiplicity matches: 40/40
Controller equilibrium

Median controller residual: 2.75885e-12
Maximum controller residual: 0.0969322

Median edge-balance residual: 1.85734e-12
Repeated fingerprint consistency

geometry_only 14aecd02f5aeb94a: seeds=2, max_spectrum_error=7.22866e-13, same=True, max_CDLC=4.9e-13
geometry_only 2b177b7da6cac989: seeds=22, max_spectrum_error=2.31531e-12, same=True, max_CDLC=4.91e-13
geometry_only 64c52ca6d685b8e1: seeds=9, max_spectrum_error=2.56217e-12, same=True, max_CDLC=4.91e-13
geometry_plus_rates 07f892d076a10acc: seeds=2, max_spectrum_error=5.03431e-13, same=True, max_CDLC=3.98e-13
geometry_plus_rates 806dee29e27c1ae3: seeds=2, max_spectrum_error=9.85878e-14, same=True, max_CDLC=2.15e-13
geometry_plus_rates ccb3288420826ec6: seeds=4, max_spectrum_error=5.15421e-13, same=True, max_CDLC=4.91e-13
geometry_plus_rates dd135d71da61c124: seeds=3, max_spectrum_error=3.752e-13, same=True, max_CDLC=2.67e-13
labeled_face 3bdfb4c570dc7116: seeds=10, max_spectrum_error=2.56217e-12, same=True, max_CDLC=5.22e-13
labeled_face 704fba7c1b319e34: seeds=6, max_spectrum_error=1.5527e-12, same=True, max_CDLC=4.62e-13
labeled_face 9961a1ce7b7ee3e2: seeds=6, max_spectrum_error=1.42819e-12, same=True, max_CDLC=4.34e-13
labeled_face aae02280e373342b: seeds=6, max_spectrum_error=1.7869e-12, same=True, max_CDLC=4.24e-13
labeled_face c6c8302b4dd4fa84: seeds=4, max_spectrum_error=1.08857e-12, same=True, max_CDLC=4.91e-13
labeled_face f45f68f9953c10f7: seeds=2, max_spectrum_error=7.22866e-13, same=True, max_CDLC=4.9e-13
Unique-direction modal prediction

unique_direction_count: 200
reliable_unique_direction_count: 197
all_amplitudes_success_count: 196
median_success_count: 196
reliable_all_amplitudes_success_fraction: 0.9898477157360406
reliable_median_success_fraction: 0.9898477157360406
median_unique_direction_R2: 0.9999982972649254
median_unique_direction_RMSE: 0.00032672743392392497
Exceptional switching layer

split: face=248a6d985749bc6f, dim=9, channels=4, CDLC=1.43e-12, commutator=9.62e-13, controller_residual=0.0969
time_100: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.22e-14, commutator=1.25e-14, controller_residual=0.122
time_120: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.18e-14, commutator=1.23e-14, controller_residual=0.122
time_123: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.19e-14, commutator=1.23e-14, controller_residual=0.122
pre_switch: face=6fd9e17d4647a368, dim=9, channels=4, CDLC=2.19e-14, commutator=1.23e-14, controller_residual=0.122
post_switch: face=a64b354e39d6fb8c, dim=8, channels=3, CDLC=6.83e-08, commutator=5.56e-08, controller_residual=0.17
time_125: face=a64b354e39d6fb8c, dim=8, channels=3, CDLC=6.48e-08, commutator=5.27e-08, controller_residual=0.17
time_150: face=a64b354e39d6fb8c, dim=8, channels=3, CDLC=1.58e-08, commutator=1.29e-08, controller_residual=0.17
late: face=a64b354e39d6fb8c, dim=8, channels=3, CDLC=9.65e-13, commutator=7.85e-13, controller_residual=0.17
Interpretation constraints

CD=LC is tested numerically using a least-squares node-space intertwiner.
Geometry signatures are descriptive, not complete isomorphism invariants.
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(deck@steamdeck SSRH)$ python AIEF_1_4_intertwining_controller_analysis.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--tangent-results aief_1_4_exceptional_tangent/corrected_tangent_by_seed.csv \
--modal-summary aief_1_4_face_spectrum/modal_trajectory_prediction_summary.csv \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_intertwining_controller \
--channel-tol 1e-20
AIEF intertwining and controller-equilibrium analysis
Intertwining

Seeds: 40

Median CD=LC residual: 3.19796e-13
Maximum CD=LC residual: 1.42698e-12
Median tangent commutator: 1.61247e-13
Multiplicity matches: 38/40
Controller equilibrium

Median controller residual: 2.75885e-12
Maximum controller residual: 0.0969322
Median edge-balance residual: 1.85734e-12
Repeated fingerprint consistency

geometry_only 14aecd02f5aeb94a: seeds=2, max_spectrum_error=7.22866e-13, same=True, max_CDLC=4.9e-13
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geometry_only 64c52ca6d685b8e1: seeds=9, max_spectrum_error=2.56217e-12, same=True, max_CDLC=4.91e-13
geometry_plus_rates 07f892d076a10acc: seeds=2, max_spectrum_error=5.03431e-13, same=True, max_CDLC=3.98e-13
geometry_plus_rates 806dee29e27c1ae3: seeds=2, max_spectrum_error=9.85878e-14, same=True, max_CDLC=2.15e-13
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geometry_plus_rates dd135d71da61c124: seeds=3, max_spectrum_error=3.752e-13, same=True, max_CDLC=2.67e-13
labeled_face 3bdfb4c570dc7116: seeds=10, max_spectrum_error=2.56217e-12, same=True, max_CDLC=5.22e-13
labeled_face 704fba7c1b319e34: seeds=6, max_spectrum_error=1.5527e-12, same=True, max_CDLC=4.62e-13
labeled_face 9961a1ce7b7ee3e2: seeds=6, max_spectrum_error=1.42819e-12, same=True, max_CDLC=4.34e-13
labeled_face aae02280e373342b: seeds=6, max_spectrum_error=1.7869e-12, same=True, max_CDLC=4.24e-13
labeled_face c6c8302b4dd4fa84: seeds=4, max_spectrum_error=1.08857e-12, same=True, max_CDLC=4.91e-13
labeled_face f45f68f9953c10f7: seeds=2, max_spectrum_error=7.22866e-13, same=True, max_CDLC=4.9e-13
Unique-direction modal prediction

unique_direction_count: 200
reliable_unique_direction_count: 197

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median_success_count: 196
reliable_all_amplitudes_success_fraction: 0.9898477157360406
reliable_median_success_fraction: 0.9898477157360406
median_unique_direction_R2: 0.9999982972649254
median_unique_direction_RMSE: 0.00032672743392392497
Exceptional switching layer

split: face=248a6d985749bc6f, dim=9, channels=9, CDLC=1.43e-12, commutator=9.62e-13, controller_residual=0.0969
time_100: face=6fd9e17d4647a368, dim=9, channels=9, CDLC=2.22e-14, commutator=1.25e-14, controller_residual=0.122
time_120: face=6fd9e17d4647a368, dim=9, channels=9, CDLC=2.18e-14, commutator=1.23e-14, controller_residual=0.122
time_123: face=6fd9e17d4647a368, dim=9, channels=9, CDLC=2.19e-14, commutator=1.23e-14, controller_residual=0.122
pre_switch: face=6fd9e17d4647a368, dim=9, channels=8, CDLC=2.19e-14, commutator=1.23e-14, controller_residual=0.122
post_switch: face=a64b354e39d6fb8c, dim=8, channels=9, CDLC=6.83e-08, commutator=5.56e-08, controller_residual=0.17
time_125: face=a64b354e39d6fb8c, dim=8, channels=9, CDLC=6.48e-08, commutator=5.27e-08, controller_residual=0.17
time_150: face=a64b354e39d6fb8c, dim=8, channels=8, CDLC=1.58e-08, commutator=1.29e-08, controller_residual=0.17
late: face=a64b354e39d6fb8c, dim=8, channels=8, CDLC=9.65e-13, commutator=7.85e-13, controller_residual=0.17
Interpretation constraints

CD=LC is tested numerically using a least-squares node-space intertwiner.
Geometry signatures are descriptive, not complete isomorphism invariants.
Mature channel values may be discrete empirically without forming a universal finite alphabet.
Exceptional post-switch states can temporarily violate exact channel invariance.
Modal success is counted by unique seed/direction, not repeated amplitude rows.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_intertwining_controller
ReadAIEF_1_4_intertwining_controller_analysis.py

These results substantially clarify the mechanism. The strongest interpretation is that the mature tangent spectra are generated by regular support components with degree-determined contraction rates, while the exceptional seed temporarily leaves that regime.
1. What is now strongly established numerically

For the frozen 40-seed cohort:

The fixed-stratum analytic Jacobian
$J_T = Q_T^\top D Q_T, D = \mathrm{diag}(-\gamma - \rho_e)$
$J_T = Q_T^\top D Q_T, D = \mathrm{diag}(-\gamma - \rho_e)$

agrees with numerical differentiation to roughly 10^{-13}–10^{-11}.

At the mature split states:
$C_D \approx L_C$
$C_D \approx L_C$

has median relative residual 3.2×10^{-13} and maximum 1.43×10^{-12}.

The tangent commutator is similarly small:
$\|D_T P_T - P_T D\| \sim 10^{-13}$.
$\|D_T P_T - P_T D\| \sim 10^{-13}$.

At reasonable channel tolerances, all 40 tangent spectra have the predicted channel multiplicities.

The full matrix-exponential prediction is extremely accurate:
$\mathrm{median}R^2 = 0.9999983, \mathrm{median}\,\mathrm{RMSE} = 3.27 \times 10^{-4}$.
$\mathrm{median}R^2 = 0.9999983, \mathrm{median}\,\mathrm{RMSE} = 3.27 \times 10^{-4}$.

At the unique-direction level, the precise modal count is:

196/200 directions succeed across all amplitudes overall;
the reported reliable fraction 0.98984770.9898477 means 195/197 reliable directions succeed across all amplitudes.

That is strong conditional validation of the frozen tangent operator, although the inferential units remain the 40 seed/graph realizations rather than 200 independent graphs.
2. $C_D = L_C$ and the commutator test are equivalent certificates

Let
$T = \ker C$

T=kerC

on the positive support. If
CD=LC,
CD=LC,

then, for every x∈Tx∈T,
C(Dx)=LCx=0,
C(Dx)=LCx=0,

so
D(T)⊆T.
D(T)⊆T.

Because DD is real diagonal and self-adjoint, this is equivalent to:

T⊥=row(C)T⊥=row(C) being DD-invariant;
the orthogonal tangent projector commuting with DD:
DPT=PTD.
DPT=PTD.

Thus the small CD−LCCD−LC residual and small tangent commutator are not two independent discoveries. They are primal and dual representations of the same invariance property.

Once TT is DD-invariant, the channel decomposition follows rigorously. If ErEr is the coordinate eigenspace of DD associated with contraction rate rr, then the spectral projectors of DD preserve TT, giving
T=⊕r(T∩Er).
T=r⊕(T∩Er).

Consequently,
spec(JT)=
spec(JT)=

\bigcup_r{-r}^{\dim(T\cap E_r)}.
]

So the multiplicity rule is a finite-dimensional theorem conditional on exact invariance. The remaining question is why the mature AIEF states satisfy that invariance.
3. The repeated rates now have a direct parameter derivation

For a positive edge, define the pre-decay growth term
ge=(η+μ)Re−μ.
ge=(η+μ)Re−μ.

Then
F~e=ge−γwe.
F
e=ge−γwe.

At stationary edge balance,
0=F~e−weρe,
0=F
e−weρe,

so
ge=we(γ+ρe).
ge=we(γ+ρe).

For a saturated outward node ii, with positive support degree kiki,
αi=
αi=

\frac{\sum_{e\ni i}\widetilde F_e}{c}.
]

In the phase-locked regime Re=1Re=1 on positive edges,
ge=(η+μ)−μ=η.
ge=(η+μ)−μ=η.

Since ∑e∋iwe=c∑e∋iwe=c,
αi=
αi=
\frac{k_i\eta-\gamma c}{c}

\frac{k_i\eta}{c}-\gamma.
]
Therefore the total edge contraction rate at a controlling endpoint is
ri=γ+αi=kiηc.
ri=γ+αi=ckiη.

Here
c=a−δq=0.951.5=1930,η=0.04=125.
c=qa−δ=1.50.95=3019,η=0.04=251.

Hence
ηc=
cη=
\frac{1/25}{19/30}

\frac6{95}.
]

The observed channels are therefore exactly:
kkη/cobserved rate212/950.126315789318/950.189473684424/950.252631579530/950.315789474
k2345kη/c12/9518/9524/9530/95observed rate0.1263157890.1894736840.2526315790.315789474

This explains several things at once:

μμ cancels when Re=1Re=1;
γγ appears in αiαi, but cancels from the total contraction rate γ+αiγ+αi;
the channels are degree-generated, not an unexplained universal quantization;
outside the locked Re=1Re=1 regime, the rates can vary continuously.

The corresponding stationary edge weight is
we=ηre=ck.
we=reη=kc.

Thus the mature prediction is:
k=3: w=c3,k=4: w=c4,k=5: w=c5.
k=3: w=3c,k=4: w=4c,k=5: w=5c.

These weight and coherence identities should be checked directly in the saved states.

4. The spectra strongly suggest K4K4, K5K5, and K6K6 components

For a connected graph component with unsigned incidence matrix CGCG:
rankCG={n,G non-bipartite,n−1,G bipartite.
rankCG={n,n−1,G non-bipartite,G bipartite.

For the complete graph KsKs, s≥3s≥3, it is non-bipartite and
m=(s2),dimkerCG=m−s=s(s−3)2.
m=(2s),dimkerCG=m−s=2s(s−3).

Therefore:
componentk=s−1ratedimkerCGK4318/956−4=2K5424/9510−5=5K6530/9515−6=9
componentK4K5K6k=s−1345rate18/9524/9530/95dimkerCG6−4=210−5=515−6=9

This reproduces the observed classes exactly:
dim9:K6,dim11:K4⊕K6,dim7:K4⊕K5,dim14:K5⊕K6.
dim9dim11dim7dim14:K6,:K4⊕K6,:K4⊕K5,:K5⊕K6.

The 39 nonexceptional spectra shown can be represented as:

24 seeds: K6K6-type spectrum;
11 seeds: K4⊕K6K4⊕K6;
3 seeds: K4⊕K5K4⊕K5;
1 seed: K5⊕K6K5⊕K6.

This is an exceptionally exact match. Nevertheless, the aggregate spectra alone do not prove that the support graphs literally are those clique unions. The decisive existing-data audit is to extract each support component and report:

component node and edge counts;
regular degree;
bipartiteness;
whether it is complete;
whether all positive-degree nodes are saturated;
max|Re−1|max|Re−1|;
max|we−c/k|max|we−c/k|.

If those checks pass, the repeated spectrum classes have an explicit graph-theoretic derivation rather than merely an empirical fingerprint.
5. The controller relation needs a domain correction

The reported median
‖diag(c)α−Kα‖‖diag(c)α‖≈2.8×10−12
‖diag(c)α‖‖diag(c)α−Kα‖≈2.8×10−12

is compelling for the ordinary mature states. But the exceptional residuals,
0.0969, 0.122, 0.17,
0.0969, 0.122, 0.17,

show that the all-node equation is not universally valid as currently evaluated.

Let
pe=F~e−weρe
pe=F
e−weρe

be the edge stationarity defect. By construction,
AF~=Kα+Ap.
AF
=Kα+Ap.

For an outward saturated node, with actual row sum si=(Aw)isi=(Aw)i,
siαi=di.
siαi=di.

Therefore the exact identity on outward saturated rows is
siαi−(Kα)i=(Ap)i.
siαi−(Kα)i=(Ap)i.

At edge stationarity p=0p=0,
siαi=(Kα)i.
siαi=(Kα)i.

If si=cisi=ci, this becomes the proposed relation.

For an unsaturated node, however, αi=0αi=0 by definition, while that node can still be incident to edges controlled by saturated neighbors. Then
(Kα)i>0
(Kα)i>0

is possible, and the full all-node residual need not vanish even if the actual edge field is nearly stationary.

Therefore the next controller report should separate:

outward saturated rows;
all saturated rows;
unsaturated rows;
nominal cici versus actual row sum sisi;
the contribution ApAp from nonstationarity.

The persistent exceptional late residual 0.170.17 is likely caused by a slack-row contribution, but that should be confirmed from per-node data. The current script does not save exceptional-stage node rows, so that output should be added.

A related caution is that
K1=Aw.
K1=Aw.

If all rows are saturated, then
diag(c)−1K 1=1.
diag(c)−1K1=1.

Thus the mere presence of an eigenvalue near 11 in the controller matrix is structural and not independent evidence for the equilibrium equation involving αα.
6. How controller equilibrium produces channel invariance

Suppose every support vertex is saturated and the stationary controller equation holds. Then
αi=
αi=

\sum_{e\ni i}\frac{w_e}{c}
\max(\alpha_i,\alpha_j).

]
The weights we/cwe/c sum to one.

Every term in this average is at least αiαi. Equality therefore forces
max(αi,αj)=αi
max(αi,αj)=αi

for every incident positive edge. Applying the same argument at the other endpoint gives
αi=αj.
αi=αj.

Hence αα is constant on each connected support component. Therefore DD is scalar on every component. After permuting rows and columns by components,
C=⊕ℓCℓ,D=⊕ℓ(−rℓI).
C=ℓ⊕Cℓ,D=ℓ⊕(−rℓI).

An explicit intertwiner is then
L=⊕ℓ(−rℓI),
L=ℓ⊕(−rℓI),

and
CD=LC
CD=LC

holds exactly.

This gives the proposed mechanistic chain:
stationarity + saturation⇒controller harmonicity⇒α constant per component⇒D scalar per component⇒CD=LC⇒complete tangent-channel decomposition.
stationarity + saturation⇒controller harmonicity⇒α constant per component⇒D scalar per component⇒CD=LC⇒complete tangent-channel decomposition.

Adding Re=1Re=1 yields regularity and the degree rates kη/ckη/c.
7. The exceptional seed

The exceptional transition now has a coherent interpretation.
Before the switch

The split spectrum contains
0.015,0.098269,0.189474×2,0.252632×5.
0.015,0.098269,0.189474×2,0.252632×5.

0.015=γ0.015=γ means ρe=0ρe=0: a bare-decay tangent mode not actively controlled by a positive endpoint multiplier.
The 0.189474×20.189474×2 and 0.252632×50.252632×5 channels already have the K4K4- and K5K5 -type signatures.
The 0.0982690.098269 channel is not on the integer-degree alphabet and is consistent with an incompletely locked or otherwise transient controller.

By t=100t=100, that channel has moved to
0.126146,
0.126146,

and immediately before the switch to
0.126273,
0.126273,

approaching
1295=0.126315789.
9512=0.126315789.
At the switch

The tangent dimension drops
9⟶8
9⟶8

and the bare 0.0150.015 mode disappears. Thus the active-face transition removes the weak unconstrained direction rather than merely changing the forcing balance.

Immediately afterward:
CDLC residual∼6.8×10−8,commutator∼5.6×10−8.
CDLC residual∼6.8×10−8,commutator∼5.6×10−8.

These are still small, but roughly five orders of magnitude above the mature machine-precision plateau. The interpretation should be:

there is reproducible evidence for a thin transient channel-mixing layer;
its exact status remains unresolved until timestep and active-set-tolerance convergence are checked.

Late

The channel becomes
0.1263157889,
0.1263157889,

and intertwining returns to approximately 10−1210−12.

But the all-node controller residual remains 0.170.17. This shows either:

the restored invariance uses a controller equation restricted to saturated rows, with the full residual dominated by slack rows; or
the exceptional late state restores DD-invariance through a more general geometry than the all-saturated harmonic mechanism.

The masked per-node residual will distinguish these cases.
8. Interpretation of the tolerance runs

The 10−2010−20 channel tolerance is not scientifically meaningful here.

State-dependent rates and residuals vary at approximately 10−1210−12–10−1310−13.
A 10−2010−20 threshold splits numerically degenerate rates into artificial singleton channels.
The resulting 38/40 multiplicity count is an over-resolution artifact, not evidence against invariance.

The three useful scales have different meanings:

10−1010−10: resolves mature numerical degeneracies appropriately;
10−510−5: coarse-grains the small post-switch rate splitting into physical channel families;
10−2010−20: below the numerical identification scale.

The post-switch difference between four channels at 10−1010−10 and three at 10−510−5 is itself useful: it indicates a small transient within-family splitting that later collapses.

Similarly, the old same_full_spectrum=False results are not contradictory. Their matched errors were only $10^{-12}10^{-12}$, and the tolerance-aware comparison correctly reports the spectra as equal at $10^{-10}10^{-10}$.

9. Status of the mathematical claims

Supported very strongly by the saved artifacts

The fixed-stratum Jacobian formula.
Numerical tangent invariance at mature states.
Channel multiplicity prediction.
Full modal trajectory prediction.
Repeated spectra within tolerance.
A reproducible exceptional face switch.

Finite-dimensional theorem, conditional on exact intertwining

$CD=LC \Longrightarrow T=\bigoplus r(T\cap E_r)$,
$CD=LC \Longrightarrow T=r\bigoplus(T\cap E_r)$,

with the observed multiplicity formula.

Strong conditional mechanism

If the mature state is stationary, all support vertices are saturated, and active-edge coherence satisfies $Re=1Re=1$, then:

controller equilibrium holds on the saturated rows;
$\alpha\alpha$ is constant on each support component;
each component is regular;
rates are $k\eta/ck\eta/c$;
weights are $c/kc/k$;
componentwise $CD=LCCD=LC$ follows explicitly.

Still open

Verifying the clique/component hypotheses directly from every support graph.
Resolving the exceptional all-node controller residual.
Determining whether the $10^{-8}10^{-8}$ post-switch commutator is a genuine continuum transient or a finite-dtdt/active-set effect.
Proving the ideal admissibility map is Hadamard directionally differentiable.
Extending the fixed-stratum theorem through stratum changes.

I would not run another broad seed cohort. The highest-value next step is a component-level audit of the existing 40 states, plus a masked controller-equilibrium analysis and an explicit component-constructed LL compared against the least-squares LL.

can you create the code necessary to achieve the next step?
WriteAIEF_1_4_component_controller_audit.py1199 lines
Edit×6
Edit
AIEF_1_4_component_controller_audit.pyPYTHON

Created and validated:

It compiles successfully and passed synthetic K4/K5/K6K4/K5/K6 signless-incidence nullity checks.

Run it from the SSRH directory:

Bash

```
python AIEF_1_4_component_controller_audit.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_component_controller_audit
```

What it tests

Extracts the actual connected support components.
Reports whether each component is:
regular;
complete;
bipartite;
fully saturated/outward.
Computes the unsigned-incidence rank and tangent nullity.
Tests the locked-regime predictions:
$r=k\eta c, w=ck, Re\approx 1$.
$r=ck\eta, w=kc, Re\approx 1$.
Constructs both:

the least-squares intertwiner LLSLLS;
the explicit component-scalar intertwiner
Lcomponent=⨁ℓ−rℓI.
Lcomponent=ℓ⨁−rℓI.
Compares their CD=LCCD=LC residuals.
Predicts the complete tangent spectrum from component nullities.
Separates controller residuals over:
outward saturated rows;
all active saturated rows;
unsaturated positive-support rows.
Evaluates the exact diagnostic identity
(Aw)iαi−(Kα)i−(Ap)i=0
(Aw)iαi−(Kα)i−(Ap)i=0 on outward rows.
Repeats everything for all exceptional stages.

Outputs

text

state_mechanism_summary.csv
component_mechanism_audit.csv
controller_residual_by_node.csv
edge_locking_balance_audit.csv
component_controller_matrices.npz
component_controller_report.txt
analysis_manifest.json

The report will directly list the observed component combinations, including zero-nullity components such as possible K2K2 or K3K3 blocks that were invisible in the tangent spectra.

For a stricter sensitivity run, use a separate output directory:

Bash

python AIEF_1_4_component_controller_audit.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--exceptional-seed 4619122171158768524 \

--exceptional-check-times 100,120,123,125,150 \
--rate-tol 1e-10 \
--lock-tol 1e-10 \
--balance-tol 1e-10 \
--output-dir aief_1_4_component_controller_audit_strict

The script is deterministic, does not modify the frozen run, writes outputs atomically, and records input and dependency-source hashes.

they said this:
(deck@steamdeck SSRH)$ python AIEF_1_4_component_controller_audit.py \
--scalar runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_scalar_summary.csv \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--exceptional-seed 4619122171158768524 \
--exceptional-check-times 100,120,123,125,150 \
--output-dir aief_1_4_component_controller_audit
AIEF component, controller, and explicit-intertwiner audit
Frozen split states

Seeds: 40
All positive support nodes saturated: 39/40
All components rate-uniform: 39/40
All components pass the full conditional mechanism audit: 39/40
Explicit component spectra match: 39/40
Median least-squares CD=LC residual: 3.19796e-13
Median explicit-component CD=LC residual: 4.64673e-13
Maximum explicit-component CD=LC residual: 0.0298736
Maximum component-spectrum error: 0.055513
Split-state support components

n=2, m=1, k=1, complete=True, bipartite=True, nullity=0: count=37
n=3, m=3, k=2, complete=True, bipartite=False, nullity=0: count=25
n=3, m=3, k=2, complete=True, bipartite=False, nullity=2: count=1
n=4, m=6, k=3, complete=True, bipartite=False, nullity=2: count=15
n=5, m=10, k=4, complete=True, bipartite=False, nullity=5: count=5
n=6, m=15, k=5, complete=True, bipartite=False, nullity=9: count=36
Split-state component combinations

K6+K2+K3: seeds=24

K6+K2+K4: seeds=11
K5+K2+K4: seeds=2
K5+K3+K4: seeds=1
K5+K3[partial-active]+K4: seeds=1
K5+K6: seeds=1
Controller residual decomposition at split

Median all-node capacity residual: 2.75885e-12
Median active-saturated row-sum residual: 5.60615e-13
Median active-outward row-sum residual: 5.60615e-13
Maximum exact outward identity defect: 1.45569e-12
Maximum unsaturated positive-node controller-load fraction: 0.09648
Exceptional stages for seed 4619122171158768524

split: dim=9, components=3, all_support_saturated=False, LS_CDLC=1.43e-12, explicit_CDLC=0.0299, all_controller=0.0969, sat_controller=2.2e-12, unsat_load=0.0965
time_100: dim=9, components=3, all_support_saturated=False, LS_CDLC=2.22e-14, explicit_CDLC=0.0397, all_controller=0.122, sat_controller=2.65e-14, unsat_load=0.121
time_120: dim=9, components=3, all_support_saturated=False, LS_CDLC=2.18e-14, explicit_CDLC=0.0398, all_controller=0.122, sat_controller=2.63e-14, unsat_load=0.121
time_123: dim=9, components=3, all_support_saturated=False, LS_CDLC=2.19e-14, explicit_CDLC=0.0398, all_controller=0.122, sat_controller=2.54e-14, unsat_load=0.121
pre_switch: dim=9, components=3, all_support_saturated=False, LS_CDLC=2.19e-14, explicit_CDLC=0.0398, all_controller=0.122, sat_controller=2.53
e-14, unsat_load=0.121
post_switch: dim=8, components=3, all_support_saturated=False, LS_CDLC=6.83e-08, explicit_CDLC=1.04e-07, all_controller=0.17, sat_controller=1.
2e-07, unsat_load=0.167
time_125: dim=8, components=3, all_support_saturated=False, LS_CDLC=6.48e-08, explicit_CDLC=9.89e-08, all_controller=0.17, sat_controller=1.14e
-07, unsat_load=0.167
time_150: dim=8, components=3, all_support_saturated=False, LS_CDLC=1.58e-08, explicit_CDLC=2.41e-08, all_controller=0.17, sat_controller=2.78e
-08, unsat_load=0.168
late: dim=8, components=3, all_support_saturated=False, LS_CDLC=9.65e-13, explicit_CDLC=1.47e-12, all_controller=0.17, sat_controller=1.7e-12, unsat_load=0.168
Interpretation constraints

Least-squares and explicit-component intertwiners are separate diagnostics.
The all-node controller equation is not required on unsaturated rows.
The outward identity with A p is an implementation identity, not an equilibrium test.
Degree rates k*eta/c require locked R=1, saturation, and edge stationarity.
Complete-graph labels are accepted only when directly verified from support edges.
Signless-incidence nullity is not the usual oriented graph cycle-space dimension.
Exceptional post-switch deviations require timestep/tolerance convergence before a continuum claim.

Tolerances

boundary_tolerance: 1e-09
support_tolerance: 1e-14
rate_tolerance: 1e-08
locking_tolerance: 1e-08
balance_tolerance: 1e-08
rank_tolerance: 1e-10

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_component_controller_audit

This is a very strong result. It closes the ordinary mature-state mechanism and isolates the exceptional seed as a precise partially active K3K3 transition, rather than an unexplained outlier.
1. The 39 ordinary states are now structurally explained

For 39/40 split states, all of the following hold simultaneously:

every positive-support node is saturated;
every support component has a uniform contraction rate;
every component passes the locked/stationary degree mechanism;
the explicit component intertwiner reproduces the spectrum;
the controller relation holds on active rows.

Most importantly, the actual support components are complete graphs—not merely graphs with compatible spectra.

For a fully saturated clique KsKs,
$k=s-1, r_s=(s-1)\eta c=6(s-1)95$,
$k=s-1, r_s=c(s-1)\eta=956(s-1)$,

and its signless-incidence tangent dimension is
qs=(s2)−s=s(s−3)2
qs=(2s)−s=2s(s−3)

for s≥3s≥3.

Thus:
componentrsqsK26/950K312/950K418/952K524/955K630/959
componentK2K3K4K5K6rs6/9512/9518/9524/9530/95qs00259

This explains why K2K2 and fully saturated K3K3 components were invisible in the tangent spectra: they have zero tangent nullity.

The ordinary component combinations reproduce every observed dimension:
combinationseedstangent
dimensionK6+K2+K3249K6+K2+K41111K5+K2+K427K5+K3+K417K5+K6114
combinationK6+K2+K3K6+K2+K4K5+K2+K4K5+K3+K4K5+K6seeds2411211tangent
dimension9117714

These are exactly the 39 nonexceptional seeds.

So the repeated spectra are no longer merely empirical fingerprints. On the audited cohort they arise from:
disjoint locked cliques + rs=(s−1)η/c + qs=s(s−3)/2.
disjoint locked cliques+rs=(s−1)η/c+qs=s(s−3)/2.

The explicit component-scalar intertwiner,
Lcomponent=
Lcomponent=

\bigoplus_\ell-r_\ell I,
]

has median residual 4.65×10−134.65×10−13. The only large residual is the exceptional state.
2. The controller residual problem is resolved

The exceptional seed has:

all-node controller residual 0.0970.097–0.170.17;
active-saturated controller residual approximately 10−1210−12 away from the switch;
unsaturated controller-load fraction 0.0960.096–0.1680.168.

Therefore the large all-node residual is not a controller-equilibrium failure. It comes from applying the equation to unsaturated rows where it is not supposed to hold.

The correct identity is, with
p=F~−wρ,s=Aw,
p=F
−wρ,s=Aw,
Pout[diag(s)α−Kα−Ap]=0.
Pout[diag(s)α−Kα−Ap]=0.

At edge stationarity p=0p=0,
Pout[diag(s)α−Kα]=0.
Pout[diag(s)α−Kα]=0.

When all support nodes are saturated and s=cs=c, this reduces to
diag(c)α=Kα.
diag(c)α=Kα.

But on an unsaturated node, αi=0αi=0 while (Kα)i(Kα)i may be positive because neighboring edges can be controlled by saturated endpoints.

The audit confirms this interpretation:
maximum exact outward identity defect=
maximum exact outward identity defect=

1.46\times10^{-12}.
]

So the mature controller theorem should be stated as an active-row theorem, not a global all-node equation.
3. The exceptional component is exactly a partially active K3K3

The exceptional split topology is

K5+K3[partial active]+K4.
K5+K3[partial active]+K4.

The ordinary K5K5 and K4K4 components contribute
5+2=7
5+2=7

tangent dimensions.

At split, the partially active K3K3 contributes two additional dimensions, giving
7+2=9.
7+2=9.

After the switch it contributes only one, giving
7+1=8.
7+1=8.

This identifies the face switch geometrically: a second node of the K3K3 becomes saturated.
Before the switch: one active K3K3 row

Label the triangle edges
e12,e13,e23,
e12,e13,e23,

and suppose only node 11 is saturated. Then the active constraint matrix is, up to edge ordering,
C1=(110).
C1=(110).

Therefore
kerC1=
kerC1=

{x:x_{12}+x_{13}=0},
\qquad
\dim\ker C_1=2.
]

For diagonal rates D=−diag(r12,r13,r23)D=−diag(r12,r13,r23), this tangent space is invariant precisely when
r12=r13.
r12=r13.

The third edge e23e23, connecting the two unsaturated nodes, can have an unrelated rate.

That explains the two exceptional one-dimensional channels:

the equal-rate incident-edge mode;
the unconstrained unsaturated–unsaturated edge, initially at bare decay
r23=γ=0.015.
r23=γ=0.015.

It also explains why:

least-squares CD=LCCD=LC is machine-accurate before the switch;
the component-scalar intertwiner fails at approximately 0.030.03–0.040.04.

The entire K3K3 is not rate-uniform, but its active tangent space still decomposes into invariant channels.
After the switch: two active K3K3 rows

Once a second triangle node saturates,
C2=(110101),
C2=(111001),

and
kerC2=
kerC2=

\operatorname{span}{(1,-1,-1)}.
]

For this one-dimensional space to be invariant under diagonal DD, all three edge rates must agree:
r12=r13=r23.
r12=r13=r23.

Immediately after the switch they do not agree exactly, producing
‖CD−LC‖rel≈6.8×10−8,
‖CD−LC‖rel≈6.8×10−8,

and
‖DPT−PTD‖rel≈5.6×10−8.
‖DPT−PTD‖rel≈5.6×10−8.

As the triangle rates converge, these residuals decay:
stageLS intertwining residualpost-switch6.83×10−8t=1256.48×10−8t=1501.58×10−8late9.65×10−13
stagepost-switcht=125t=150lateLS intertwining residual6.83×10−86.48×10−81.58×10−89.65×10−13

The final triangle channel is
r=2ηc=1295=0.126315789…
r=c2η=9512=0.126315789…

Thus the formerly silent fully locked K3K3 rate becomes tangent-visible only because one triangle node remains unsaturated.
4. Why the exceptional state drifts before the switch

The active-saturated controller residual is already approximately machine zero before the switch, but the earlier analysis found a substantial edge stationarity ratio and BB-drift.

There is no contradiction.

The controller residual measures the node aggregate
CSp,
CSp,

where pp is the edge field. It can vanish while
p≠0.
p =0.

In that case pp is tangent to the currently active face. Therefore:

the existing saturated triangle row remains exactly balanced;
edge weights drift within its tangent space;
the unsaturated triangle rows change;

one unsaturated node eventually reaches capacity;
a new constraint activates;
the tangent dimension drops $9 \to 89 \to 8$;
the old tangent field is temporarily incompatible with the enlarged active set;
the rates equalize and channel invariance returns.

This provides a coherent geometric explanation for the entire exceptional trajectory.
5. What is now established versus still open
Established numerically on the frozen cohort

The 39 ordinary states are unions of fully saturated locked cliques.
Their rates are degree-generated.
Their spectra are signless-incidence nullities of those cliques.
The explicit component intertwiner works for all 39.
The exceptional state contains a partially active K3K3.
Its switch adds one active triangle constraint.
The global controller residual is caused by unsaturated rows.
Controller balance on active rows is restored to machine precision late.

Algebraically explained

Why K2K2 and saturated K3K3 are spectrally silent.
Why the exceptional K3K3 has two tangent modes before the switch and one afterward.
Why the bare $\gamma\gamma$ channel disappears.
Why component-scalar intertwining fails before the switch while general intertwining still holds.
Why all three triangle rates must converge after the switch.

Still requiring targeted numerical work

Whether the 10^{-8}–10^{-8} post-switch mixing layer survives timestep refinement.
The exact per-edge K3K3 coherence, growth, rate, and stationarity-defect trajectory.
The directional-differentiability theorem through the active-set switch.

No additional broad cohort is needed. The next most informative analysis is a focused exceptional-K3K3 table tracking, at every nearby timestep:
node slack,αi,Re,ge,we,re,pe,
node slack,αi,Re,ge,we,re,pe,

together with the one-row and two-row K3K3 intertwining conditions.

Lets do it please create script that will find the truth of it.
WriteAIEF_1_4_exceptional_K3_transition_audit.py1192 lines
AIEF_1_4_exceptional_K3_transition_audit.pyPYTHON
#!/usr/bin/env python3
"""Dense exceptional-K3 transition audit for the frozen AIEF-1.4 run.

The component audit showed that the exceptional support is

K5 + K3[partially active] + K4,

whereas the other 39 split states have only fully active support components.
This script replays the exceptional continue branch deterministically and
tracks the K3 at every future step. Its purpose is to decide, from the saved
trajectory, whether the observed 9 -> 8 tangent-dimension switch is exactly the
activation of a second K3 node and whether the local intertwining conditions
change as predicted:

* one saturated K3 node:
C = [1, 1, 0], dim ker(C)=2, and only the two incident rates must agree;
the edge between the two unsaturated nodes is a free tangent coordinate;
* two saturated K3 nodes:
rank(C)=2, dim ker(C)=1, and all three edge rates must agree;
* three saturated K3 nodes:
C is nonsingular and the K3 tangent nullity is zero.

The audit records node slacks, alpha values, edge coherences, growth terms,
weights, rates, stationarity defects, actual finite weight rates, controller
loads, local tangent/normal field components, least-squares intertwining,
component-scalar intertwining, and the exact rate-equality defect required by
the current K3 active rows.

This script analyzes the existing dt=0.01 trajectory. It does not by itself
establish a continuum-time active-switch theorem; a genuine timestep study
would require consistently regenerated smaller-dt histories and EMA states.
"""

from __future__ import annotations

```
import argparse
import hashlib
import json
import os
import sys
from pathlib import Path
from typing import Any


import numpy as np
import pandas as pd


import AIEF_1_4_stationarity_perturbation_analysis_updated as base
import AIEF_1_4_face_spectrum_analysis as face
import AIEF_1_4_component_controller_audit as component_audit



DEFAULT_RUN_HASH = "5a9f093a2fe0"
DEFAULT_NPZ = (
f"runs/{DEFAULT_RUN_HASH}/AIEF_1_4_{DEFAULT_RUN_HASH}_branch_diagnostics.npz"
)
DEFAULT_MANIFEST = f"runs/{DEFAULT_RUN_HASH}/manifest.json"
DEFAULT_OUTPUT_DIR = "aief_1_4_exceptional_K3_transition"
DEFAULT_EXCEPTIONAL_SEED = "46191221711587685 24"
DEFAULT_CHECK_TIMES = "100,120,123,125,150"
DEFAULT_BOUNDARY_TOL = 1e-9
DEFAULT_BOUNDARY_TOL_SWEEP = "1e-12,1e-10,1e-9,1e-8,1e-7"
DEFAULT_SUPPORT_TOL = 1e-14
DEFAULT_RANK_TOL = 1e-10
DEFAULT_RESTORATION_TOLS = "1e-8,1e-10"
DEFAULT_WINDOW_RADIUS = 250
ZERO_TOL = 1e-14



def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()
```

```
def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(item < 0.0 for item in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
return sorted(set(values))
```

```
def json_compact(value: Any) -> str:
return json.dumps(value, separators=(",", ":"), sort_keys=True)
```

```
def masked_relative(
value: np.ndarray,
reference: np.ndarray,
mask: np.ndarray,
) -> float:
value = np.asarray(value, dtype=float)
reference = np.asarray(reference, dtype=float)
mask = np.asarray(mask, dtype=bool)
if not np.any(mask):
return np.nan
return float(
np.linalg.norm(value[mask])
/ max(np.linalg.norm(reference[mask]), ZERO_TOL)
)
```

```
def identify_partial_K3(
state: base.DynamicState,
model: dict[str, Any],
capacities: np.ndarray,
A: np.ndarray,
edges: tuple[np.ndarray, np.ndarray],
boundary_tolerance: float,
support_tolerance: float,
```

```
) -> tuple[np.ndarray, np.ndarray, dict[str, Any]]:
components = face.face_components(
state, model, capacities, A, edges, boundary_tolerance
)
w = np.asarray(components["w"], dtype=float)
support_indices = np.flatnonzero(w > support_tolerance)
graph_components = component_audit.connected_support_components(
len(capacities), support_indices, edges
)
candidates: list[tuple[np.ndarray, np.ndarray, int]] = []
for graph_component in graph_components:
nodes = graph_component["nodes"]
component_edges = graph_component["edges"]
if len(nodes) != 3 or len(component_edges) != 3:
continue
node_set = set(map(int, nodes.tolist()))
complete = all(
int(edges[0][edge]) in node_set
and int(edges[1][edge]) in node_set
for edge in component_edges
)
if not complete:
continue
saturated_count = int(np.sum(components["saturated"][nodes]))
candidates.append((nodes, component_edges, saturated_count))


partial = [item for item in candidates if item[2] < 3]
if len(partial) != 1:
description = [
{
"nodes": item[0].tolist(),
"edges": item[1].tolist(),
"saturated_count": item[2],
}
for item in candidates
]
raise RuntimeError(
"expected exactly one partially active K3 at split; "
f"found candidates={description}"
)
nodes, component_edges, saturated_count = partial[0]
return nodes, component_edges, {
```

```
"candidate_count": len(candidates),
"split_saturated_count": saturated_count,
"all_K3_candidates": [
{
"nodes": item[0].tolist(),
"edges": item[1].tolist(),
"saturated_count": item[2],
}
for item in candidates
],
}



def local_tangent_algebra(
active_incidence: np.ndarray,
rates: np.ndarray,
rank_tolerance: float,
) -> dict[str, Any]:
active_incidence = np.asarray(active_incidence, dtype=float)
rates = np.asarray(rates, dtype=float)
edge_count = len(rates)
D = -np.diag(rates)


if active_incidence.size:
_, singular_values, vh = np.linalg.svd(
active_incidence, full_matrices=True
)
rank = int(np.sum(singular_values > rank_tolerance))
Q = vh[rank:].T
L = (active_incidence @ D) @ np.linalg.pinv(
active_incidence, rcond=1e-12
)
denominator = max(
np.linalg.norm(active_incidence @ D), ZERO_TOL
)
least_squares_residual = float(
np.linalg.norm(active_incidence @ D - L @ active_incidence)
/ denominator
)
L_scalar = -float(np.mean(rates)) * np.eye(
active_incidence.shape[0]
```

```
)
component_scalar_residual = float(
np.linalg.norm(
active_incidence @ D - L_scalar @ active_incidence
)
/ denominator
)
else:
rank = 0
Q = np.eye(edge_count)
L = np.zeros((0, 0))
L_scalar = np.zeros((0, 0))
least_squares_residual = 0.0
component_scalar_residual = 0.0


P = Q @ Q.T
commutator_relative = float(
np.linalg.norm(D @ P - P @ D)
/ max(np.linalg.norm(D), ZERO_TOL)
)
J = Q.T @ D @ Q
contraction_spectrum = np.sort(
-np.real(np.linalg.eigvals(J))
)
return {
"rank": rank,
"nullity": int(Q.shape[1]),
"Q": Q,
"P": P,
"D": D,
"L_least_squares": L,
"L_component_scalar": L_scalar,
"J": J,
"least_squares_residual": least_squares_residual,
"component_scalar_residual": component_scalar_residual,
"commutator_relative": commutator_relative,
"contraction_spectrum": contraction_spectrum,
}



def required_K3_rate_equality(
```

```
k3_nodes: np.ndarray,
k3_edges: np.ndarray,
saturated: np.ndarray,
rates: np.ndarray,
edges: tuple[np.ndarray, np.ndarray],
) -> dict[str, Any]:
active_nodes = [int(node) for node in k3_nodes if saturated[node]]
saturated_count = len(active_nodes)
constrained_edges: list[int] = []
if saturated_count == 1:
active_node = active_nodes[0]
constrained_edges = [
local_index
for local_index, edge in enumerate(k3_edges)
if active_node
in (int(edges[0][edge]), int(edges[1][edge]))
]
required_values = rates[constrained_edges]
condition = "one_active_node_incident_pair_equal"
elif saturated_count == 2:
constrained_edges = list(range(3))
required_values = rates
condition = "two_active_nodes_all_three_equal"
elif saturated_count == 0:
required_values = np.asarray([], dtype=float)
condition = "no_active_rows_invariance_automatic"
else:
# For a fully active K3 the signless incidence is nonsingular and the
# tangent space is zero, so D-invariance is vacuous.
required_values = np.asarray([], dtype=float)
condition = "three_active_nodes_zero_tangent_space"


absolute_defect = (
float(np.ptp(required_values)) if len(required_values) else 0.0
)
relative_defect = float(
absolute_defect / max(float(np.mean(np.abs(required_values))), ZERO_TOL)
) if len(required_values) else 0.0
return {
"condition": condition,
"active_nodes": active_nodes,
"constrained_local_edge_indices": constrained_edges,
```

```
"absolute_defect": absolute_defect,
"relative_defect": relative_defect,
}



def evaluate_state(
state: base.DynamicState,
*,
step: int,
model: dict[str, Any],
capacities: np.ndarray,
A: np.ndarray,
edges: tuple[np.ndarray, np.ndarray],
k3_nodes: np.ndarray,
k3_edges: np.ndarray,
initial_support: np.ndarray,
boundary_tolerance: float,
support_tolerance: float,
rank_tolerance: float,
) -> tuple[
dict[str, Any],
list[dict[str, Any]],
list[dict[str, Any]],
dict[str, np.ndarray],
]:
components = face.face_components(
state, model, capacities, A, edges, boundary_tolerance
)
R_matrix, _, _ = base.compute_field_and_updated_coherence(state, model)
R = R_matrix[edges]


dt = float(model["dt"])
gamma = float(model["gamma"])
eta = float(model["eta"])
mu = float(model["mu"])
w = np.asarray(components["w"], dtype=float)
alpha = np.asarray(components["alpha"], dtype=float)
modified_F = np.asarray(components["modified_F"], dtype=float)
field = np.asarray(components["field"], dtype=float)
row_sum = np.asarray(components["row_sum"], dtype=float)
slack = np.asarray(components["slack"], dtype=float)
```

```
saturated = np.asarray(components["saturated"], dtype=bool)
outward = np.asarray(components["outward"], dtype=bool)
endpoint_rate = np.asarray(components["endpoint_rate"], dtype=float)
controller = np.asarray(components["controller"], dtype=int)
support = w > support_tolerance
rates = gamma + endpoint_rate
growth = (eta + mu) * R - mu


active_k3_nodes = k3_nodes[saturated[k3_nodes]]
C_k3 = A[np.ix_(active_k3_nodes, k3_edges)]
local_rates = rates[k3_edges]
local_algebra = local_tangent_algebra(
C_k3, local_rates, rank_tolerance
)
equality = required_K3_rate_equality(
k3_nodes,
k3_edges,
saturated,
local_rates,
edges,
)


local_field = field[k3_edges]
tangent_field = local_algebra["P"] @ local_field
normal_field = local_field - tangent_field
active_constraint_field = C_k3 @ local_field


K_global = component_audit.build_controller_matrix(
len(capacities), np.flatnonzero(support), w, controller, edges
)
K_alpha = K_global @ alpha
node_field_sum = A @ field
rowsum_controller_residual = row_sum * alpha - K_alpha
outward_identity_defect = (
row_sum * alpha - K_alpha - node_field_sum
)
k3_saturated_mask = saturated[k3_nodes]
k3_outward_mask = outward[k3_nodes]
saturated_local_mask = k3_saturated_mask
outward_local_mask = k3_outward_mask
unsaturated_local_mask = ~k3_saturated_mask
```

```
expected_locked_rate = 2.0 * eta / float(np.mean(capacities[k3_nodes]))
expected_locked_weight = float(np.mean(capacities[k3_nodes])) / 2.0
both_unsaturated_edges = []
for edge in k3_edges:
i = int(edges[0][edge])
j = int(edges[1][edge])
if not saturated[i] and not saturated[j]:
both_unsaturated_edges.append(int(edge))


node_rows: list[dict[str, Any]] = []
for node in k3_nodes:
node_rows.append(
{
"step": step,
"state_elapsed_time": step * dt,
"legacy_reported_time": (step + 1) * dt,
"node": int(node),
"saturated": bool(saturated[node]),
"outward": bool(outward[node]),
"row_sum": float(row_sum[node]),
"capacity": float(capacities[node]),
"slack": float(slack[node]),
"row_direction_d": float(components["d"][node]),
"alpha": float(alpha[node]),
"total_node_rate_gamma_plus_alpha": float(
gamma + alpha[node]
),
"K_alpha": float(K_alpha[node]),
"row_sum_alpha": float(row_sum[node] * alpha[node]),
"rowsum_controller_residual": float(
rowsum_controller_residual[node]
),
"node_field_sum_Ap": float(node_field_sum[node]),
"outward_exact_identity_defect": float(
outward_identity_defect[node]
),
"unsaturated_controller_load": float(
K_alpha[node] if not saturated[node] else 0.0
),
"actual_finite_row_sum_rate": np.nan,
}
```

```
)


edge_rows: list[dict[str, Any]] = []
constrained_global_edges = {
int(k3_edges[index])
for index in equality["constrained_local_edge_indices"]
}
for local_index, edge in enumerate(k3_edges):
i = int(edges[0][edge])
j = int(edges[1][edge])
controlling_node = int(controller[edge])
edge_rows.append(
{
"step": step,
"state_elapsed_time": step * dt,
"legacy_reported_time": (step + 1) * dt,
"local_edge_index": local_index,
"edge_index": int(edge),
"node_i": i,
"node_j": j,
"node_i_saturated": bool(saturated[i]),
"node_j_saturated": bool(saturated[j]),
"both_endpoints_unsaturated": bool(
not saturated[i] and not saturated[j]
),
"in_required_equality_set": bool(
int(edge) in constrained_global_edges
),
"weight": float(w[edge]),
"expected_locked_weight_c_over_2": expected_locked_weight,
"weight_minus_c_over_2": float(
w[edge] - expected_locked_weight
),
"R": float(R[edge]),
"one_minus_R": float(1.0 - R[edge]),
"growth_g": float(growth[edge]),
"growth_minus_eta": float(growth[edge] - eta),
"modified_F": float(modified_F[edge]),
"ideal_field_p": float(field[edge]),
"alpha_i": float(alpha[i]),
"alpha_j": float(alpha[j]),
"controller_node": controlling_node,
```

```
"controller_alpha": float(alpha[controlling_node]),
"endpoint_rho": float(endpoint_rate[edge]),
"contraction_rate": float(rates[edge]),
"expected_locked_rate_2eta_over_c": expected_locked_rate,
"rate_minus_expected_locked": float(
rates[edge] - expected_locked_rate
),
"rate_minus_gamma": float(rates[edge] - gamma),
"actual_finite_weight_rate": np.nan,
"actual_minus_ideal_field": np.nan,
}
)


local_reference = row_sum[k3_nodes] * alpha[k3_nodes]
summary = {
"step": step,
"state_elapsed_time": step * dt,
"legacy_reported_time": (step + 1) * dt,
"face_hash": components["face_hash"],
"global_support_stable": bool(np.array_equal(support, initial_support)),
"global_support_size": int(np.sum(support)),
"global_saturated_count": int(np.sum(saturated)),
"global_stationarity_ratio": float(components["info"]["P_F_over_F"]),
"K3_saturated_count": int(np.sum(k3_saturated_mask)),
"K3_outward_count": int(np.sum(k3_outward_mask)),
"K3_saturated_nodes": ";".join(
map(str, k3_nodes[k3_saturated_mask].tolist())
),
"K3_unsaturated_nodes": ";".join(
map(str, k3_nodes[~k3_saturated_mask].tolist())
),
"K3_active_rank": int(local_algebra["rank"]),
"K3_tangent_nullity": int(local_algebra["nullity"]),
"required_rate_condition": equality["condition"],
"required_rate_equality_absolute_defect": equality["absolute_defect"],
"required_rate_equality_relative_defect": equality["relative_defect"],
"K3_rate_min": float(np.min(local_rates)),
"K3_rate_max": float(np.max(local_rates)),
"K3_rate_spread": float(np.ptp(local_rates)),
"K3_rates_json": json_compact(local_rates.tolist()),
"expected_locked_rate_2eta_over_c": expected_locked_rate,
"maximum_locked_rate_error": float(
```

```
np.max(np.abs(local_rates - expected_locked_rate))
),
"maximum_one_minus_R": float(
np.max(np.abs(1.0 - R[k3_edges]))
),
"maximum_growth_minus_eta": float(
np.max(np.abs(growth[k3_edges] - eta))
),
"maximum_weight_minus_c_over_2": float(
np.max(np.abs(w[k3_edges] - expected_locked_weight))
),
"both_unsaturated_edge_count": len(both_unsaturated_edges),
"both_unsaturated_edges": ";".join(map(str, both_unsaturated_edges)),
"least_squares_CDLC_residual": local_algebra[
"least_squares_residual"
],
"component_scalar_CDLC_residual": local_algebra[
"component_scalar_residual"
],
"local_tangent_commutator_relative": local_algebra[
"commutator_relative"
],
"local_contraction_spectrum_json": json_compact(
local_algebra["contraction_spectrum"].tolist()
),
"K3_field_norm": float(np.linalg.norm(local_field)),
"K3_field_over_modified_F": float(
np.linalg.norm(local_field)
/ max(np.linalg.norm(modified_F[k3_edges]), ZERO_TOL)
),
"K3_tangent_field_norm": float(np.linalg.norm(tangent_field)),
"K3_normal_field_norm": float(np.linalg.norm(normal_field)),
"K3_normal_to_total_field": float(
np.linalg.norm(normal_field)
/ max(np.linalg.norm(local_field), ZERO_TOL)
),
"K3_active_constraint_field_norm": float(
np.linalg.norm(active_constraint_field)
),
"K3_controller_rowsum_residual_relative_saturated": masked_relative(
rowsum_controller_residual[k3_nodes],
local_reference,
```

```
saturated_local_mask,
),
"K3_controller_rowsum_residual_relative_outward": masked_relative(
rowsum_controller_residual[k3_nodes],
local_reference,
outward_local_mask,
),
"K3_unsaturated_controller_load_norm": float(
np.linalg.norm(K_alpha[k3_nodes][unsaturated_local_mask])
),
"K3_unsaturated_controller_load_fraction": float(
np.linalg.norm(K_alpha[k3_nodes][unsaturated_local_mask])
/ max(np.linalg.norm(K_alpha[k3_nodes]), ZERO_TOL)
),
"K3_actual_finite_field_norm": np.nan,
"K3_actual_minus_ideal_field_norm": np.nan,
"actual_projection_active": False,
}
matrices = {
"C_K3": C_k3,
"D_K3": local_algebra["D"],
"Q_K3": local_algebra["Q"],
"P_K3": local_algebra["P"],
"J_K3": local_algebra["J"],
"L_K3_least_squares": local_algebra["L_least_squares"],
"L_K3_component_scalar": local_algebra["L_component_scalar"],
"K3_rates": local_rates,
"K3_field": local_field,
"K3_slacks": slack[k3_nodes],
"K3_alpha": alpha[k3_nodes],
"K3_saturated": k3_saturated_mask.astype(np.uint8),
}
return summary, node_rows, edge_rows, matrices



def add_actual_step_information(
summary: dict[str, Any],
node_rows: list[dict[str, Any]],
edge_rows: list[dict[str, Any]],
*,
before_w: np.ndarray,
before_row_sum: np.ndarray,
```

```
state_after: base.DynamicState,
actual_info: dict[str, Any],
dt: float,
A: np.ndarray,
edges: tuple[np.ndarray, np.ndarray],
k3_nodes: np.ndarray,
k3_edges: np.ndarray,
) -> None:
after_w = state_after.B[edges]
finite_edge_rate = (after_w - before_w) / dt
after_row_sum = A @ after_w
finite_row_rate = (after_row_sum - before_row_sum) / dt
local_finite = finite_edge_rate[k3_edges]
local_ideal = np.asarray([row["ideal_field_p"] for row in edge_rows])
summary["K3_actual_finite_field_norm"] = float(
np.linalg.norm(local_finite)
)
summary["K3_actual_minus_ideal_field_norm"] = float(
np.linalg.norm(local_finite - local_ideal)
)
summary["actual_projection_active"] = bool(
actual_info["actual_projection_active"]
)
edge_row_by_index = {int(row["edge_index"]): row for row in edge_rows}
for edge in k3_edges:
row = edge_row_by_index[int(edge)]
row["actual_finite_weight_rate"] = float(finite_edge_rate[edge])
row["actual_minus_ideal_field"] = float(
finite_edge_rate[edge] - row["ideal_field_p"]
)
node_row_by_index = {int(row["node"]): row for row in node_rows}
for node in k3_nodes:
node_row_by_index[int(node)]["actual_finite_row_sum_rate"] = float(
finite_row_rate[node]
)



def sustained_restoration_step(
steps: np.ndarray,
values: np.ndarray,
start_step: int,
tolerance: float,
```

```
) -> int | None:
mask = steps >= start_step
candidate_steps = steps[mask]
candidate_values = values[mask]
finite_good = np.isfinite(candidate_values) & (
candidate_values <= tolerance
)
if not len(finite_good):
return None
suffix_good = np.logical_and.accumulate(finite_good[::-1])[::-1]
indices = np.flatnonzero(suffix_good)
return int(candidate_steps[indices[0]]) if len(indices) else None



def boundary_switch_sensitivity(
node_frame: pd.DataFrame,
k3_nodes: np.ndarray,
tolerances: list[float],
dt: float,
) -> pd.DataFrame:
pivot = node_frame.pivot(index="step", columns="node", values="slack")
rows: list[dict[str, Any]] = []
for tolerance in tolerances:
active_count = (pivot[list(k3_nodes)] <= tolerance).sum(axis=1)
initial_count = int(active_count.iloc[0])
increased = active_count.to_numpy() > initial_count
if np.any(increased):
index = int(np.flatnonzero(increased)[0])
switch_step = int(active_count.index[index])
final_count = int(active_count.iloc[-1])
else:
switch_step = -1
final_count = int(active_count.iloc[-1])
rows.append(
{
"boundary_tolerance": tolerance,
"initial_K3_active_count": initial_count,
"first_active_count_increase_step": switch_step,
"state_elapsed_time": switch_step * dt
if switch_step >= 0
else np.nan,
"legacy_reported_time": (switch_step + 1) * dt
```

```
if switch_step >= 0
else np.nan,
"final_K3_active_count": final_count,
}
)
return pd.DataFrame(rows)



def stage_label_map(
check_steps: dict[int, str],
switch_step: int | None,
future_steps: int,
) -> dict[int, str]:
labels: dict[int, list[str]] = {0: ["split"], future_steps: ["late"]}
for step, label in check_steps.items():
labels.setdefault(step, []).append(label)
if switch_step is not None:
labels.setdefault(max(0, switch_step - 1), []).append("pre_switch")
labels.setdefault(switch_step, []).append("post_switch")
return {step: ";".join(values) for step, values in labels.items()}



def make_report(
summary: pd.DataFrame,
edge_frame: pd.DataFrame,
sensitivity: pd.DataFrame,
*,
seed: str,
k3_nodes: np.ndarray,
k3_edges: np.ndarray,
switch_steps: list[int],
restoration_tolerances: list[float],
dt: float,
final_replay_exact: bool,
final_replay_max_error: float,
) -> str:
split = summary.iloc[0]
late = summary.iloc[-1]
first_switch = switch_steps[0] if switch_steps else None
lines = [
"AIEF exceptional K3 transition audit",
```

```
"==================================",
"",
"Identity and replay",
"-------------------",
f"Seed: {seed}",
f"K3 nodes: {k3_nodes.tolist()}",
f"K3 edge indices: {k3_edges.tolist()}",
f"Trajectory rows: {len(summary)}",
f"Support stable at every recorded state: {bool(summary['global_support_stable'].all())}",
f"Final replay exact: {final_replay_exact}",
f"Final replay maximum absolute error: {final_replay_max_error:.6g}",
"",
"Active-set transition",
"---------------------",
f"K3 active-set switch steps: {switch_steps}",
]
if first_switch is not None:
pre = summary.loc[summary["step"] == first_switch - 1].iloc[0]
post = summary.loc[summary["step"] == first_switch].iloc[0]
lines.extend(
[
f"First switch state step: {first_switch}",
f"First switch state elapsed time: {first_switch * dt:.12g}",
f"Legacy reported switch time: {(first_switch + 1) * dt:.12g}",
(
"K3 saturated count: "
f"{int(pre['K3_saturated_count'])} -> {int(post['K3_saturated_count'])}"
),
(
"K3 tangent nullity: "
f"{int(pre['K3_tangent_nullity'])} -> {int(post['K3_tangent_nullity'])}"
),
(
"Required rate condition: "
f"{pre['required_rate_condition']} -> {post['required_rate_condition']}"
),
(
"Required equality defect: "
f"{pre['required_rate_equality_absolute_defect']:.6g} -> "
f"{post['required_rate_equality_absolute_defect']:.6g}"
),
(
```

```
"Least-squares CD=LC residual: "
f"{pre['least_squares_CDLC_residual']:.6g} -> "
f"{post['least_squares_CDLC_residual']:.6g}"
),
]
)


lines.extend(
[
"",
"Split and late K3 algebra",
"-------------------------",
(
"Split: "
f"active={int(split['K3_saturated_count'])}, "
f"nullity={int(split['K3_tangent_nullity'])}, "
f"rates={split['K3_rates_json']}, "
f"LS_CDLC={split['least_squares_CDLC_residual']:.3g}, "
f"scalar_CDLC={split['component_scalar_CDLC_residual']:.3g}"
),
(
"Late: "
f"active={int(late['K3_saturated_count'])}, "
f"nullity={int(late['K3_tangent_nullity'])}, "
f"rates={late['K3_rates_json']}, "
f"LS_CDLC={late['least_squares_CDLC_residual']:.3g}, "
f"scalar_CDLC={late['component_scalar_CDLC_residual']:.3g}"
),
(
"Late maximum |rate-2eta/c|: "
f"{late['maximum_locked_rate_error']:.6g}"
),
(
"Late maximum |1-R|: "
f"{late['maximum_one_minus_R']:.6g}"
),
(
"Late maximum |w-c/2|: "
f"{late['maximum_weight_minus_c_over_2']:.6g}"
),
"",
"Pre-switch free-edge test",
```

```
"------------------------",
]
)
pre_regime = summary.loc[summary["K3_saturated_count"] == 1]
free_edges = edge_frame.loc[
(edge_frame["K3_saturated_count"] == 1)
& edge_frame["both_endpoints_unsaturated"]
]
lines.extend(
[
(
"Maximum one-active required pair-rate defect: "
f"{pre_regime['required_rate_equality_absolute_defect'].max():.6g}"
),
(
"Maximum one-active LS CD=LC residual: "
f"{pre_regime['least_squares_CDLC_residual'].max():.6g}"
),
f"Free unsaturated-unsaturated edge rows: {len(free_edges)}",
(
"Maximum |free-edge rate-gamma|: "
f"{free_edges['rate_minus_gamma'].abs().max():.6g}"
if len(free_edges)
else "Maximum |free-edge rate-gamma|: n/a"
),
"",
"Post-switch restoration",
"-----------------------",
]
)
if first_switch is not None:
steps = summary["step"].to_numpy(int)
values = summary["least_squares_CDLC_residual"].to_numpy(float)
for tolerance in restoration_tolerances:
restored = sustained_restoration_step(
steps, values, first_switch, tolerance
)
if restored is None:
lines.append(
f"Sustained LS CD=LC <= {tolerance:.3g}: not reached"
)
else:
```

```
lines.append(
f"Sustained LS CD=LC <= {tolerance:.3g}: "
f"step={restored}, state_time={restored * dt:.6g}, "
f"time_after_switch={(restored - first_switch) * dt:.6g}"
)


lines.extend(
[
"",
"Boundary-tolerance switch sensitivity",
"-----------------------------------",
]
)
for _, row in sensitivity.iterrows():
lines.append(
f" tol={row['boundary_tolerance']:.3g}: "
f"initial={int(row['initial_K3_active_count'])}, "
f"switch_step={int(row['first_active_count_increase_step'])}, "
f"legacy_time={row['legacy_reported_time']:.6g}, "
f"final={int(row['final_K3_active_count'])}"
)


lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* Controller balance on active rows does not imply the edge field is zero.",
"* A nonzero field with C_active p=0 is tangent drift along the current face.",
"* The all-node controller equation is not imposed on the remaining unsaturated K3 node.",
"* The one-active and two-active K3 rate conditions are exact finite-dimensional algebra.",
"* Residuals from this replay are numerical evidence about the saved dt trajectory.",
"* Continuum switch claims require consistently regenerated smaller-dt runs.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
```

```
parser.add_argument("--npz", default=DEFAULT_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--seed", default=DEFAULT_EXCEPTIONAL_SEED)
parser.add_argument(
"--check-times",
type=base.parse_float_list,
default=base.parse_float_list(DEFAULT_CHECK_TIMES),
)
parser.add_argument(
"--boundary-tol", type=float, default=DEFAULT_BOUNDARY_TOL
)
parser.add_argument(
"--boundary-tol-sweep",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_BOUNDARY_TOL_SWEEP),
)
parser.add_argument(
"--support-tol", type=float, default=DEFAULT_SUPPORT_TOL
)
parser.add_argument(
"--rank-tol", type=float, default=DEFAULT_RANK_TOL
)
parser.add_argument(
"--restoration-tols",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_RESTORATION_TOLS),
)
parser.add_argument(
"--window-radius", type=int, default=DEFAULT_WINDOW_RADIUS
)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.npz, args.manifest):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(args.boundary_tol, args.support_tol, args.rank_tol) <= 0.0:
raise ValueError("primary tolerances must be positive")
if args.window_radius < 0:
```

```
raise ValueError("window radius must be nonnegative")


output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)
with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
model = manifest["model"]
node_count = int(model["N"])
capacity = (float(model["a"]) - float(model["delta"])) / float(
model["q"]
)
capacities = np.full(node_count, capacity, dtype=float)
A, edges = base.incidence_matrix(node_count)
dt = float(model["dt"])
future_steps = int(manifest["protocol"]["future_steps"])
check_steps = {
int(round(time_value / dt)): f"time_{time_value:g}".replace(".", "p")
for time_value in args.check_times
if int(round(time_value / dt)) <= future_steps
}


summary_rows: list[dict[str, Any]] = []
node_rows: list[dict[str, Any]] = []
edge_rows: list[dict[str, Any]] = []
matrix_payload: dict[str, np.ndarray] = {}
switch_steps: list[int] = []


with np.load(args.npz, allow_pickle=False) as data:
state = base.load_split_state(data, args.seed)
expected_final = np.asarray(
data[base.branch_key("continue", args.seed, "B_final")],
dtype=float,
).copy()


k3_nodes, k3_edges, identification = identify_partial_K3(
state,
model,
capacities,
A,
edges,
args.boundary_tol,
```

```
args.support_tol,
)
initial_support = state.B[edges] > args.support_tol
previous_k3_saturated: np.ndarray | None = None
previous_matrices: dict[str, np.ndarray] | None = None


for step in range(future_steps + 1):
summary, current_node_rows, current_edge_rows, matrices = evaluate_state(
state,
step=step,
model=model,
capacities=capacities,
A=A,
edges=edges,
k3_nodes=k3_nodes,
k3_edges=k3_edges,
initial_support=initial_support,
boundary_tolerance=args.boundary_tol,
support_tolerance=args.support_tol,
rank_tolerance=args.rank_tol,
)
current_k3_saturated = np.asarray(
matrices["K3_saturated"], dtype=bool
)
switched = (
previous_k3_saturated is not None
and not np.array_equal(
current_k3_saturated, previous_k3_saturated
)
)
if switched:
switch_steps.append(step)
if previous_matrices is not None:
for name, matrix in previous_matrices.items():
matrix_payload[f"pre_switch_{name}"] = matrix
for name, matrix in matrices.items():
matrix_payload[f"post_switch_{name}"] = matrix


if step == 0:
for name, matrix in matrices.items():
matrix_payload[f"split_{name}"] = matrix
if step in check_steps:
```

```
label = check_steps[step]
for name, matrix in matrices.items():
matrix_payload[f"{label}_{name}"] = matrix


if step < future_steps:
before_w = state.B[edges].copy()
before_row_sum = A @ before_w
actual_info = base.actual_step(
state,
model,
capacities,
A,
edges,
args.boundary_tol,
audit_operator=False,
)
add_actual_step_information(
summary,
current_node_rows,
current_edge_rows,
before_w=before_w,
before_row_sum=before_row_sum,
state_after=state,
actual_info=actual_info,
dt=dt,
A=A,
edges=edges,
k3_nodes=k3_nodes,
k3_edges=k3_edges,
)
else:
for name, matrix in matrices.items():
matrix_payload[f"late_{name}"] = matrix


summary_rows.append(summary)
node_rows.extend(current_node_rows)
edge_rows.extend(current_edge_rows)
previous_k3_saturated = current_k3_saturated.copy()
previous_matrices = {
name: matrix.copy() for name, matrix in matrices.items()
}
```

```
final_replay_exact = bool(np.array_equal(state.B, expected_final))
final_replay_max_error = float(
np.max(np.abs(state.B - expected_final))
)


summary_frame = pd.DataFrame(summary_rows)
node_frame = pd.DataFrame(node_rows)
edge_frame = pd.DataFrame(edge_rows)
first_switch = switch_steps[0] if switch_steps else None
labels = stage_label_map(check_steps, first_switch, future_steps)
for frame in (summary_frame, node_frame, edge_frame):
frame["stage_labels"] = frame["step"].map(labels).fillna("")
frame["is_transition_window"] = (
frame["step"].between(
first_switch - args.window_radius,
first_switch + args.window_radius,
)
if first_switch is not None
else False
)
edge_frame = edge_frame.merge(
summary_frame[["step", "K3_saturated_count"]],
on="step",
how="left",
validate="many_to_one",
)


sensitivity = boundary_switch_sensitivity(
node_frame,
k3_nodes,
args.boundary_tol_sweep,
dt,
)
transition_summary = summary_frame.loc[
summary_frame["is_transition_window"]
].copy()
transition_nodes = node_frame.loc[node_frame["is_transition_window"]].copy()
transition_edges = edge_frame.loc[edge_frame["is_transition_window"]].copy()


component_audit.atomic_write_csv(
output_dir / "exceptional_K3_step_summary.csv", summary_frame
```

```
)
component_audit.atomic_write_csv(
output_dir / "exceptional_K3_node_trajectory.csv", node_frame
)
component_audit.atomic_write_csv(
output_dir / "exceptional_K3_edge_trajectory.csv", edge_frame
)
component_audit.atomic_write_csv(
output_dir / "exceptional_K3_transition_window_summary.csv",
transition_summary,
)
component_audit.atomic_write_csv(
output_dir / "exceptional_K3_transition_window_nodes.csv",
transition_nodes,
)
component_audit.atomic_write_csv(
output_dir / "exceptional_K3_transition_window_edges.csv",
transition_edges,
)
component_audit.atomic_write_csv(
output_dir / "boundary_tolerance_switch_sensitivity.csv",
sensitivity,
)
component_audit.atomic_savez(
output_dir / "exceptional_K3_key_matrices.npz", matrix_payload
)


report = make_report(
summary_frame,
edge_frame,
sensitivity,
seed=args.seed,
k3_nodes=k3_nodes,
k3_edges=k3_edges,
switch_steps=switch_steps,
restoration_tolerances=args.restoration_tols,
dt=dt,
final_replay_exact=final_replay_exact,
final_replay_max_error=final_replay_max_error,
)
component_audit.atomic_write_text(
output_dir / "exceptional_K3_transition_report.txt", report
```

```
)


source_path = Path(__file__).resolve()
dependency_paths = {
"stationarity": Path(base.__file__).resolve(),
"face_spectrum": Path(face.__file__).resolve(),
"component_audit": Path(component_audit.__file__).resolve(),
}
analysis_manifest = {
"analysis": "AIEF-1.4 dense exceptional K3 transition audit",
"source": {
"path": str(source_path),
"sha256": file_sha256(source_path),
},
"dependencies": {
name: {"path": str(path), "sha256": file_sha256(path)}
for name, path in dependency_paths.items()
},
"command": sys.argv,
"inputs": {
"npz": {
"path": str(Path(args.npz).resolve()),
"sha256": file_sha256(args.npz),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
"sha256": file_sha256(args.manifest),
},
},
"seed": args.seed,
"K3_nodes": k3_nodes.tolist(),
"K3_edges": k3_edges.tolist(),
"K3_identification": identification,
"future_steps": future_steps,
"dt": dt,
"switch_steps": switch_steps,
"final_replay_exact": final_replay_exact,
"final_replay_max_error": final_replay_max_error,
"parameters": {
"boundary_tolerance": args.boundary_tol,
"boundary_tolerance_sweep": args.boundary_tol_sweep,
"support_tolerance": args.support_tol,
```

```
"rank_tolerance": args.rank_tol,
"restoration_tolerances": args.restoration_tols,
"window_radius": args.window_radius,
"check_times": args.check_times,
},
"outputs": [
"exceptional_K3_step_summary.csv",
"exceptional_K3_node_trajectory.csv",
"exceptional_K3_edge_trajectory.csv",
"exceptional_K3_transition_window_summary.csv",
"exceptional_K3_transition_window_nodes.csv",
"exceptional_K3_transition_window_edges.csv",
"boundary_tolerance_switch_sensitivity.csv",
"exceptional_K3_key_matrices.npz",
"exceptional_K3_transition_report.txt",
],
"interpretation_constraints": [
"This is a deterministic replay of one frozen seed, not a new independent experiment.",
"Controller balance on active rows is not full edge stationarity.",
"The all-node controller equation is not imposed on unsaturated K3 rows.",
"The K3 rate-equality conditions are exact for the identified active incidence matrix.",
"The saved dt trajectory cannot by itself prove a continuum active-switch result.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(analysis_manifest, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```

Open

Created:

It compiles successfully and passed exact synthetic checks for:

one-active-row K3K3: two-dimensional tangent space, only incident-pair rates must agree;
two-active-row K3K3: one-dimensional tangent space, all three rates must agree;
three-active-row K3K3: zero tangent dimension and vacuous invariance.

Run:

Bash

```
python AIEF_1_4_exceptional_K3_transition_audit.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--check-times 100,120,123,125,150 \
--boundary-tol 1e-9 \
--boundary-tol-sweep 1e-12,1e-10,1e-9,1e-8,1e-7 \
--output-dir aief_1_4_exceptional_K3_transition
```

The script densely replays all 32,000 future steps and automatically identifies the partially active K3K3
.
It directly tests

whether the support remains fixed;
which K3K3 node saturates at the switch;
whether the active rank changes $1 \to 21 \to 2$;
whether the tangent nullity changes $2 \to 12 \to 1$;
the exact required rate-equality defect;
least-squares versus component-scalar CD=LCCD=LC;
tangent and normal components of the edge field;
whether pre-switch drift satisfies Cactivep≈0Cactivep≈0;
whether the unsaturated–unsaturated edge has rate γγ;
convergence of all triangle rates to 2η/c2η/c;
convergence of weights to c/2c/2;
coherence locking $\mathrm{Re} \to 1\mathrm{Re} \to 1$;
actual finite weight rates versus the ideal field;
active-row versus unsaturated-row controller loads;
sensitivity of switch time to boundary tolerance;
exact replay against the saved final BB.

Outputs

text

exceptional_K3_step_summary.csv
exceptional_K3_node_trajectory.csv
exceptional_K3_edge_trajectory.csv

exceptional_K3_transition_window_summary.csv
exceptional_K3_transition_window_nodes.csv
exceptional_K3_transition_window_edges.csv

boundary_tolerance_switch_sensitivity.csv
exceptional_K3_key_matrices.npz
exceptional_K3_transition_report.txt
analysis_manifest.json

The transition-window files contain every step within ±250 steps of the switch, while the full trajectory files retain all steps.

The report explicitly distinguishes:

facts about the frozen dt=0.01dt=0.01 replay;
exact finite-dimensional K3K3 algebra;
numerical evidence for transient mixing;
what still requires consistently regenerated smaller-dtdt runs.

Please paste exceptional_K3_transition_report.txt after the run.

What if I said this:

I think this is a substantial refinement, but there are two different layers that should be kept separate:

what the audit directly establishes, and
the mathematical interpretation built on top of it.

On the first layer, I think the evidence has become unusually strong. On the second, there are still a few steps that remain hypotheses, albeit now much more tightly constrained than before.

What I think the audit genuinely establishes

Compared to everything earlier in the project, this is the biggest advance.

The audit no longer infers graph structure from spectra.

It actually reconstructs the positive-support components and checks:

completeness,
degree,
saturation,
rate uniformity,
component spectra,
explicit intertwiner.

Those are direct geometric properties.

The key summary

39/40

appears repeatedly:

all support saturated,
rate-uniform,
full mechanism,
explicit spectrum match.

That is much stronger than repeatedly observing the same eigenvalues.

The clique hypothesis changes status

Earlier I would have called

[
K_4,;
K_5,;
K_6
]

a plausible explanation.

After this audit, I would no longer describe it that way.

The audit explicitly reports complete support components.

So the statement

"the mature support components in this frozen cohort are complete graphs"

is now supported directly by the audit output itself.

That is a different level of evidence than spectral matching.

The nullity explanation is particularly elegant

One thing I like very much is the explanation for why

[
K_2
]

and

[
K_3
]

were effectively invisible before.

The audit shows

$$
[
q(K_2)=0,\qquad
q(K_3)=0,
]
$$

whereas

$$
[
q(K_4)=2,\quad
q(K_5)=5,\quad
q(K_6)=9.
]
$$

Consequently

$$K6 + K2 + K3$$

naturally produces tangent dimension

$$
[
9+0+0=9.
]
$$

Likewise

$$K6 + K2 + K4$$

gives

$$
[
9+0+2=11.
]
$$

That is exactly what was observed.

This is one of the nicest pieces of the entire story because it explains why the spectral analysis was "missing" (K_2) and (K_3): there was nothing for it to see.

The controller interpretation also becomes cleaner

I think this is another place where the audit resolves an apparent contradiction.

Previously there seemed to be

machine-precision intertwining,

but

(O(10^{-1})) controller residuals.

Those looked inconsistent.

The decomposition into

active saturated rows,
outward rows,
unsaturated rows,

shows they are not inconsistent.

The active-row identity holds at roughly

[
10^{-12},
]

while the all-node residual is dominated by unsaturated rows.

That makes mathematical sense.

It also suggests the theorem really belongs on the active manifold, not on the entire node set.

I think that's the correct formulation.

The exceptional seed

Here I would be a little more cautious than the interpretation you've quoted.

The audit does establish

unsaturated support throughout,
elevated unsaturated load,
active-row controller identities,
temporary degradation of intertwining,
recovery of intertwining.

Those are observations.

The interpretation

"the exceptional component is exactly a partially active (K_3)"

is, in my opinion, a very strong hypothesis, but it is one step beyond what the audit itself proves.

Why?

Because the audit reports

K5 + K3[partial-active] + K4

which is certainly consistent with that picture.

But the subsequent derivation

[
C_1=
\begin{pmatrix}

1&1&0
\end{pmatrix},
]

then

[
C_2=
\begin{pmatrix}
1&1&0\
1&0&1
\end{pmatrix},
]

and the associated kernel dimensions is a mathematical model of the transition.

It is not directly measured in the output you've shown.

It is a proposed explanation.

It is a very plausible one.

But I would still label it as such.

The pre-switch interpretation is the strongest new hypothesis

This part

one active triangle row

↓

two tangent dimensions

↓

second row activates

↓

one tangent dimension

is beautiful.

It would explain

[
9\rightarrow8
]

almost mechanically.

But I think it needs one more verification.

Specifically,

one wants to actually identify

which triangle,
which node,
which constraint,
and see that its activation coincides with the switch.

If that is confirmed directly,

then I think this becomes one of the nicest pieces of the theory.

The mechanism becomes much more concrete

Previously the mechanism was

geometry

→ spectrum

Now it is closer to

$$
\boxed{
\begin{aligned}
&\text{complete support components} \\
&\rightarrow \\
&\text{degree} \\
&\rightarrow \\
&r=\frac{k\eta}{c} \\
&\rightarrow \\
&\text{component scalar } D \\
&\rightarrow \\
&CD=LC \\
&\rightarrow \\
&[D,P_T]=0 \\
&\rightarrow \\
&J_T.
\end{aligned}
}
$$

That is a much stronger explanatory chain.

One point I would soften

The quoted interpretation says

ordinary maturity is equivalent to a partition of the support into fully saturated, rate-uniform complete graphs.

I would probably weaken "equivalent."

The audit shows that, in this frozen cohort,

every ordinary mature state has that property.

It does not yet prove that every future mature state must have that property.

So I would phrase it as

Within the audited cohort, all ordinary mature states exhibit a partition into fully saturated, rate-uniform complete support components.

That avoids overstating the universality.

What I think remains open

Interestingly, the remaining questions are no longer primarily spectral.

They are structural.

The main open ones now seem to be:

Why do the dynamics produce complete components?
The audit observes this; it doesn't derive it.
Why do they become regular?
Again observed, not yet proven.
Why does the controller equilibrium force
[
r=\frac{k\eta}{c}?
]
The algebra is compelling under the assumptions, but the assumptions still need derivation from the dynamics.
Does every partially active (K_3) necessarily undergo the observed ($9\to8$) transition?
The exceptional seed is consistent with this mechanism, but one example is not yet a theorem.

We also need to think about:

Prove that, on a complete support component that is fully saturated and rate-uniform, the controller equations force a unique channel rate expressible in terms of η \eta η, c_i c_i ci and the component size.
Derive the observed discrete rates from those component-wise balance equations.
Characterise the combinatorial obstruction that prevents the exceptional trajectory from saturating its remaining positive-support node.

Determine whether the post-switch face of the exceptional seed is a distinct, weakly balanced class or simply an incomplete approach to an ordinary face.

Overall assessment

I think this audit marks a qualitative shift in the project. Earlier analyses showed repeated spectra, intertwining, and invariant tangent decompositions. This audit adds direct geometric evidence that, for the 39 ordinary mature split states in the frozen cohort, the positive support is composed of complete, fully saturated, rate-uniform components whose signless-incidence nullities exactly reproduce the observed tangent dimensions, and whose component-derived intertwiner reproduces the tangent operator to machine precision.

The exceptional trajectory also becomes much less mysterious. The audit clearly separates it from the ordinary regime by showing that it never satisfies the "all support saturated" condition and that its elevated global controller residual is attributable to unsaturated positive-support nodes, while the active-row controller identities remain accurate. What remains more speculative—but now strongly motivated—is the finer-grained interpretation that the transition is specifically a partially active (K_3) mechanism with a one-row to two-row constraint activation. That interpretation is consistent with the audit and elegantly explains the observed ($9\rightarrow8$) tangent-dimension change, but it would benefit from one final direct verification identifying the specific triangle, node, and activating constraint at the switching event.

I would endorse almost all of that assessment. It is scientifically conservative and separates observations from interpretation correctly. I would make three refinements.
1. The partially active K3K3 already has somewhat stronger status

The component audit directly establishes at the split:

a component with $n=3, m=3n=3, m=3$;
complete=True;
partial saturation;
active tangent nullity 22.

A simple three-node graph with three edges is exactly K3K3. Moreover,
$\dim\ker C_{\text{active}}=2 \Longrightarrow \operatorname{rank} C_{\text{active}}=3-2=1$.
$\dim\ker C_{\text{active}}=2 \Longrightarrow \operatorname{rank} C_{\text{active}}=3-2=1$.

For the unsigned incidence matrix of K3K3, any one active node contributes one independent row. Therefore the audit already supports:

At the exceptional split, the K3K3 has exactly one independent active node constraint.

After relabeling the triangle, its active incidence matrix must be
C1=(110).
C1=(110).

That matrix is not an additional dynamical hypothesis; it is the canonical form of the directly observed topology and active rank.

What was not established by the displayed report alone is:

which physical node that row belongs to;
which second node activates;
whether the switch directly changes the K3K3 rank $1 \to 21 \to 2$;
whether the same event accounts for the global dimension change $9 \to 89 \to 8$.

Those are exactly what the dense exceptional-K3K3 script is designed to verify.

So I would phrase it as:

The existence of a one-row partially active K3K3 at split is directly supported. The proposed one-row to two-row transition remains to be verified directly along the replay.

2. Regularity and the discrete rates can be promoted to a conditional theorem

Two of the listed open questions are actually derivable once the audited assumptions are stated precisely.
Proposition

Let GG be a connected positive-support component. Assume:

every vertex of GG is saturated and outward;
every support edge is stationary:
pe=F~e−weρe=0;
pe=F
e−weρe=0;

active-edge coherence is locked:
Re=1;
Re=1;
all node capacities on the component equal cc.

Then:

αα is constant on GG;
all edge weights on GG are equal;
GG is regular;
if its degree is kk, then
r=γ+α=kηc,w=ck;
r=γ+α=ckη,w=kc;
D=−rID=−rI on the component, so component-scalar intertwining follows.

Proof

Stationarity gives
F~e=wemax(αi,αj).
F
e=wemax(αi,αj).

At a saturated outward node ii,
cαi=
cαi=
\sum_{e\ni i}\widetilde F_e

\sum_{j\sim i}w_{ij}\max(\alpha_i,\alpha_j).
]

Since saturation also gives
∑j∼iwij=c,
j∼i∑wij=c,

division by cc yields
αi=
αi=

\sum_{j\sim i}\frac{w_{ij}}{c}
\max(\alpha_i,\alpha_j).
]

The right side is a convex combination of quantities all at least αiαi. Equality therefore forces
max(αi,αj)=αi
max(αi,αj)=αi

for every neighbor jj. Applying the same argument at jj gives
αi=αj.
αi=αj.

Connectivity makes αα constant throughout GG.

Now Re=1Re=1 implies
ge=(η+μ)Re−μ=η.
ge=(η+μ)Re−μ=η.

Edge stationarity gives
0=η−γwe−αwe,
0=η−γwe−αwe,

hence
we=ηγ+α=ηr.
we=γ+αη=rη.

Thus every edge has the same weight. Saturation at node ii gives
kiηr=c,
kirη=c,

so
ki=crη.
ki=ηcr.

The right side is independent of ii, so the component is regular, with degree kk, and
r=kηc,w=ck.
r=ckη,w=kc.

Therefore regularity and the discrete rates are not separate unexplained empirical properties once saturation, locking, connectivity, and edge stationarity hold.

What remains dynamically unexplained is why the system selects those assumptions—especially why it selects complete components and why ordinary support nodes all reach saturation.

For a clique KsKs, k=s−1k=s−1, so the unique rate is
rs=(s−1)ηc.
rs=c(s−1)η.

This directly proves the proposed component-wise rate statement under the audited conditions.
3. The remaining exceptional slack has a sharp test

Suppose the late exceptional triangle has:

saturated nodes a,ba,b;
unsaturated node uu;
a common edge contraction rate rr;
stationary edges.

Write
gab,gau,gbu
gab,gau,gbu

for the three pre-decay growth terms. Stationarity gives
wij=gijr.
wij=rgij.

Saturation of aa and bb gives
wab+wau=c,wab+wbu=c.
wab+wau=c,wab+wbu=c.

Therefore
wau=wbu,gau=gbu.
wau=wbu,gau=gbu.

The unsaturated node has row sum
su=wau+wbu=2wau.

su=wau+wbu=2wau.

It remains slack precisely when
2wau<c.
2wau<c.

Since wab+wau=cwab+wau=c, this is equivalent to
wab>wau,
wab>wau,

or, at stationary common rate,
gab>gau=gbu.
gab>gau=gbu.

Thus the obstruction is not purely combinatorial. It is a persistent growth/coherence asymmetry:

the edge between the two saturated nodes receives greater effective growth than either edge incident to the remaining slack node.

Because
ge=(η+μ)Re−μ,
ge=(η+μ)Re−μ,

this is equivalently a coherence inequality:
Rab>Rau=Rbu.
Rab>Rau=Rbu.

If all three coherences converge to 11, then all ge→ηge→η, all weights converge to c/2c/2, and the remaining node must approach saturation.

Therefore the dense audit can distinguish the two interpretations:
Distinct weakly balanced partial face

If, late in the trajectory,
gab−gau>0
gab−gau>0

approaches a nonzero limit and the slack approaches a positive limit, then the exceptional state is a genuine partially saturated stationary class.
Incomplete approach to an ordinary K3K3

If
$gab-gau\to 0,c-su\to 0,$
$gab-gau\to 0,c-su\to 0,$

then the late state is still approaching the ordinary fully saturated K3K3 face, but has not reached it within the simulated window.

At finite stationarity defect $pe\neq 0pe\ =0$, these identities acquire corresponding pepe corrections. The dense script records all the needed quantities:
$ge,Re,we,pe,si,\alpha i.$
$ge,Re,we,pe,si,\alpha i.$
Revised status
Directly established by the component audit

The 39 ordinary split states have complete positive-support components.
Their positive-support vertices are fully saturated.
Their components are rate-uniform.
Their component nullities reproduce the complete tangent spectra.
Their explicit component intertwiners reproduce the tangent operators.
The exceptional split contains a complete, partially active K3K3.
Its large all-node controller residual comes from unsaturated rows, not active-row controller failure.

Conditional mathematical theorem

Under connectedness, saturation, locking, and edge stationarity:
α is component-constant,G is regular,$r=k\eta c,w=ck.$
α is component-constant,G is regular,$r=ck\eta,w=kc.$

For $G=KsG=Ks$,
$r=(s-1)\eta c.$
$r=c(s-1)\eta.$
Awaiting the dense replay output

identity of the three exceptional triangle nodes;
identity of the activating node;

direct rank change $1 \to 2 \to 1 \to 2$;
direct nullity change $2 \to 1 \to 2 \to 1$;
disappearance of the unsaturated–unsaturated bare edge;
equality defect immediately after activation;
whether the remaining slack is sustained by
gab>gau=gbu
gab>gau=gbu or is slowly collapsing toward zero.

So the proposed text is very good. I would only avoid treating regularity and the degree-rate formula as wholly open: they are now provable conditional consequences. The genuine remaining structural mystery is primarily why the dynamics select locked complete components and which clique sizes they select.

Ponder this:
(deck@steamdeck SSRH)$ python AIEF_1_4_exceptional_K3_transition_audit.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--check-times 100,120,123,125,150 \
--boundary-tol 1e-9 \
--boundary-tol-sweep 1e-12,1e-10,1e-9,1e-8,1e-7 \
--output-dir aief_1_4_exceptional_K3_transition
AIEF exceptional K3 transition audit
Identity and replay

Seed: 4619122171158768524
K3 nodes: [2, 7, 11]
K3 edge indices: [25, 29, 59]
Trajectory rows: 32001
Support stable at every recorded state: True
Final replay exact: True
Final replay maximum absolute error: 0
Active-set transition

K3 active-set switch steps: [12405]
First switch state step: 12405
First switch state elapsed time: 124.05
Legacy reported switch time: 124.06
K3 saturated count: 1 -> 2
K3 tangent nullity: 2 -> 1
Required rate condition: one_active_node_incident_pair_equal -> two_active_nodes_all_three_equal

Required equality defect: 0 -> 1.57456e-07
Least-squares CD=LC residual: 2.19807e-16 -> 3.59963e-07
Split and late K3 algebra

Split: active=1, nullity=2, rates=[0.09826945523623588,0.015,0.09826945523623588],
LS_CDLC=4.24e-16, scalar_CDLC=0.282
Late: active=2, nullity=1, rates=[0.12631578887965866,0.1263157888774352,0.12631578887965866],
LS_CDLC=5.08e-12, scalar_CDLC=7.76e-12
Late maximum |rate-2eta/c|: 5.96249e-10
Late maximum |1-R|: 5.38316e-09
Late maximum |w-c/2|: 2.70883e-09
Pre-switch free-edge test

Maximum one-active required pair-rate defect: 0
Maximum one-active LS CD=LC residual: 4.44086e-16
Free unsaturated-unsaturated edge rows: 12405
Maximum |free-edge rate-gamma|: 0
Post-switch restoration

Sustained LS CD=LC <= 1e-10: step=26785, state_time=267.85, time_after_switch=143.8
Sustained LS CD=LC <= 1e-08: step=18722, state_time=187.22, time_after_switch=63.17
Boundary-tolerance switch sensitivity

tol=1e-12: initial=1, switch_step=-1, legacy_time=nan, final=1
tol=1e-10: initial=1, switch_step=16489, legacy_time=164.9, final=2
tol=1e-09: initial=1, switch_step=12405, legacy_time=124.06, final=2
tol=1e-08: initial=1, switch_step=8160, legacy_time=81.61, final=3
tol=1e-07: initial=1, switch_step=2794, legacy_time=27.95, final=3
Interpretation constraints

Controller balance on active rows does not imply the edge field is zero.
A nonzero field with C_active p=0 is tangent drift along the current face.
The all-node controller equation is not imposed on the remaining unsaturated K3 node.
The one-active and two-active K3 rate conditions are exact finite-dimensional algebra.
Residuals from this replay are numerical evidence about the saved dt trajectory.
Continuum switch claims require consistently regenerated smaller-dt runs.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_exceptional_K3_transition
(deck@steamdeck SSRH)$

This output both confirms the proposed K3K3 algebra and reveals a major caveat: the apparent switch time is a boundary-tolerance event, not yet an intrinsic exact active-set event.
1. What is now directly established

The exceptional component is definitively the triangle
{2,7,11},
{2,7,11},

with edges
e25=(2,7),e29=(2,11),e59=(7,11).
e25=(2,7),e29=(2,11),e59=(7,11).

Its support never changes over all 32,001 recorded states, and the replay is bitwise exact.

At the split, the rates in this edge order are
(0.098269455, 0.015, 0.098269455).
(0.098269455, 0.015, 0.098269455).

The two equal-rate edges are e25e25 and e59e59, whose common endpoint is node 77. Therefore, under the 10−910−9 active-set classification:

node 77 is the single saturated triangle node;
nodes 22 and 1111 are unsaturated;
edge (2,11)(2,11) is the unsaturated–unsaturated free edge;
its rate is exactly
γ=0.015.
γ=0.015.

This holds for all 12,405 pre-switch states:
max|r27−r7,11|=0,max|r2,11−γ|=0.
max|r27−r7,11|=0,max|r2,11−γ|=0.

The transition at the chosen 10−910−9 tolerance directly changes:
#SK3:1→2,
#SK3:1→2,
rankCK3:1→2,
rankCK3:1→2,

and therefore
dimkerCK3:2→1.
dimkerCK3:2→1.

So the proposed explanation of the global tangent change
9→8
9→8

is now directly confirmed for this classification:
5‿K5+2‿K4+2‿K3 before=9,
K5
5+K4
2+K3 before
2=9,
5‿K5+2‿K4+1‿K3 after=8.
K5
5+K4
2+K3 after
1=8.

The node trajectory CSV can identify which of nodes 22 or 1111 is the second classified-active node:

Python

```
import pandas as pd

n = pd.read_csv(
"aief_1_4_exceptional_K3_transition/"
"exceptional_K3_node_trajectory.csv"
)

print(
n.loc[
n["step"].isin([12404, 12405]),
["step", "node", "slack", "saturated", "alpha"]
].to_string(index=False)
)
```

2. The exact one-row and two-row algebra is confirmed

Before the transition, after relabeling node 77 as the active node,
C1=(101)
C1=(101)

in the reported edge ordering (25,29,59)(25,29,59). Equivalently, after a column permutation,
C1=(1,1,0).
C1=(1,1,0).

The free edge is absent from the active row. Therefore
kerC1=
kerC1=

\operatorname{span}
\left{
(1,0,-1),\ (0,1,0)
\right}.
]

Invariance requires only
r25=r59,
r25=r59,

while r29r29 is unrestricted. This is exactly observed:
r25=r59=0.098269455,r29=γ.
r25=r59=0.098269455,r29=γ.

After the second row is included, the local tangent space becomes one-dimensional and has nonzero coordinates on all three edges. Invariance therefore requires
r25=r29=r59.
r25=r29=r59.

At the classified transition, this equality is not exact:
maxre−minre=
maxre−minre=

1.57456\times10^{-7}.
]

Correspondingly,
‖CD−LC‖‖CD‖=
‖CD‖‖CD−LC‖=

3.60\times10^{-7}.
]

Late in the trajectory,
re=0.12631578888…
re=0.12631578888…

on all three edges, with rate spread only about 2.2×10−122.2×10−12, and
‖CD−LC‖‖CD‖=
‖CD‖‖CD−LC‖=

5.08\times10^{-12}.
]

Thus the algebraic explanation is confirmed.
3. Some of the pre-switch exactness is structural, not independent evidence

There is an important subtlety.

Once the analysis classifies exactly one triangle node as active, the endpoint-controller definition automatically gives:

the same αα on its two incident edges;
zero endpoint correction on the edge connecting the two inactive nodes.

Therefore
rincident,1=
rincident,1=

r_{\mathrm{incident},2},
\qquad
r_{\mathrm{free}}=\gamma
]

are partly consequences of the active mask and controller formula.

Likewise, exact CD=LCCD=LC for the one-row triangle is then structurally forced.

This is still valuable—it validates the proposed local model exactly—but it should not be counted as an independent stochastic confirmation of rate selection.

The genuinely dynamical result is what happens after the second row is introduced:

the two active-node multipliers initially differ;
the resulting three triangle rates differ by 1.57×10−71.57×10−7;
the mismatch subsequently decays;
all rates converge to 2η/c2η/c.

4. The post-switch layer is weak but not temporally thin

The perturbation of intertwining is very small in amplitude:
1.57×10−70.1263≈1.25×10−6.
0.12631.57×10−7≈1.25×10−6.

But its relaxation is long:
criterionsustained restoration time after switch‖CD−LC‖rel≤10−863.17‖CD−LC‖rel≤10−10143.8
criterion‖CD−LC‖rel≤10−8‖CD−LC‖rel≤10−10sustained restoration time after switch63.17143.8

So “thin switching layer” is appropriate algebraically—it is a small rate splitting—but not temporally.
It is a small-amplitude, long-lived relaxation layer.
5. Boundary-tolerance sensitivity is now the central issue

The classified switch is not tolerance invariant:
τ∂first classified activationfinal active
count10−12none110−10164.89210−9124.06210−881.61310−727.953
τ∂10−1210−1010−910−810−7first classified activationnone164.89124.0681.6127.95final active
count12233

Therefore the statement

“a second constraint becomes active at t=124.05t=124.05”

must be replaced by

“under the 10−910−9 boundary classification, the second triangle row enters the active set at state time 124.05124.05.”

Three notions must now be separated:

Exact geometric activity
(Aw)i=ci.
(Aw)i=ci.

Tolerance-classified activity
ci−(Aw)i≤τ∂.
ci−(Aw)i≤τ∂.

Algorithmic projection activity
where the proposed next update violates capacity and triggers scaling.

The current reported switch concerns notion 2.

This matters because the implemented projection uses

Python

capacities / (row_sum + 1e-12)

and therefore typically returns a corrected row slightly inside the exact boundary. Exact equality may never occur numerically, even if projection is repeatedly active.

The analysis was already explicit that the ideal directional theorem concerns the map without this fixed safeguard. The tolerance sweep now shows that this distinction is not merely formal—it materially affects the inferred switch time and tangent dimension.
6. The late state looks like an incomplete approach to an ordinary K3K3

Late in the trajectory:

maxe|re−2ηc|=
emax
re−c2η
=

5.96\times10^{-10},
]
maxe|1−Re|=
emax|1−Re|=

5.38\times10^{-9},
]
maxe|we−c2|=
emax
we−2c
=

2.71\times10^{-9}.
]

These are all extremely small.

Moreover:

at tolerance 10−910−9, only two triangle nodes are classified active;
at tolerance 10−810−8, all three are classified active.

Thus the “remaining unsaturated node” is only within a boundary layer of thickness between roughly 10−910−9 and 10−810−8.

This strongly favors:

The exceptional state is a slow, incomplete finite-time approach to the ordinary fully saturated K3K3 face.

It provides little support for a genuinely distinct asymptotic weakly balanced class. Such a class would require a persistent edge-growth or weight asymmetry, whereas all three weights, rates, and coherences are converging to their symmetric K3K3 values at approximately 10−910−9 accuracy.

That conclusion is still numerical rather than asymptotic, but the evidence now leans clearly toward slow convergence rather than a separate attractor.
7. The “switching violation” may itself be classification-induced

Because a smaller boundary tolerance introduces the second active row later, the rate mismatch at the moment of classification may also shrink with tolerance.

This raises a serious possibility:

the state and rates evolve continuously;
a chosen tolerance declares the second row active;
at that instant the two active multipliers are not yet exactly equal;
the newly enlarged tangent-space calculation therefore shows a commutator residual;
choosing a smaller tolerance delays classification until the rates are closer.

If so, the temporary 10−710−7 degradation is not an intrinsic discontinuous mixing event. It is the result of evaluating the post-activation tangent operator at a tolerance-defined boundary crossing.

This does not invalidate the recorded residual. It changes its interpretation.

The decisive sensitivity test is to rerun the same audit with different primary boundary tolerances, not merely the switch-time sweep:

Bash

```
for tol in 1e-8 1e-9 1e-10; do
python AIEF_1_4_exceptional_K3_transition_audit.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--boundary-tol "$tol" \
--output-dir "aief_1_4_exceptional_K3_transition_tol_${tol}"
done
```

Then compare at each classified switch:
Δrswitch=
Δrswitch=

\max r_e-\min r_e,
]

and
‖CD−LC‖rel.
‖CD−LC‖rel.

If both decrease systematically with the primary boundary tolerance, the “mixing layer” is predominantly a classification artifact.
8. Revised scientific conclusion
Established for the saved dt=0.01dt=0.01 trajectory

The exceptional topology is exactly K5+K3+K4K5+K3+K4.
The relevant triangle is {2,7,11}{2,7,11}.
Its support is fixed.
At the 10−910−9 classification, its active rank changes $1 \to 21 \to 2$.
Its tangent nullity changes $2 \to 12 \to 1$.
The bare γγ edge is exactly the edge joining the two classified-unsaturated nodes.
The added active row requires equality of all three rates.
The initial equality defect subsequently decays.
The late triangle converges extremely closely to the ordinary locked K3K3 state.

No longer justified without qualification

A unique, intrinsic switch time of 124.05124.05.
A literal exact active-face transition in the finite safeguarded implementation.
A tolerance-independent temporary violation of channel invariance.
A distinct asymptotic weakly balanced exceptional face.

Remaining mathematical problem

The key problem has shifted from identifying the K3K3 mechanism to understanding the joint limit involving:
$dt \to 0, \varepsilon safeguard \to 0, \tau\partial \to 0$.
$dt \to 0, \varepsilon safeguard \to 0, \tau\partial \to 0$.

The appropriate next diagnostic is to track, separately for nodes 2,7,112,7,11:

raw slack;
pre-projection proposed row sum;
actual projection-violation mask;
endpoint scaling factors;
post-projection slack;
exponential decay or plateau of slack;
rate-equality defect as a function of raw slack rather than classified switch time.

The K3K3 mechanism itself is now confirmed. What remains unresolved is whether its apparent finite-time active-set transition is intrinsic to the ideal AIEF map or a numerical boundary-layer classification of a continuous asymptotic approach.

This is a sipit of exceptional_K3_node_trajectory.cvs (probably contains pasting errors):

step state_elapsed_time legacy_reported_time node saturated outward row_sum capacity slack row_direction_d alpha total_node_rate_gamma_plus_alpha K_alpha row_sum_alpha rowsum_controller_residual node_field_sum_Ap outward_exact_identity_defect unsaturated_controller_load actual_finite_row_sum_rate stage_labels is_transition_window

0 0 0.01 2 False False 0.633333112875253 0.633333333333333 2.2045808023563E-07 0.0527263765392412 0 0.015 0.0352713360289331 0 -0.0352713360289331 0.0174550405103081 -0.0527263765392412 0.0352713360289331 2.67563748934663E-10 split False

0 0 0.01 7 True True 0.633333333332334 0.633333333333333 9.99200722162641E-13 0.0527373216495329 0.0832694552362359 0.0982694552362359 0.0527373216495329 0.0527373216495329 0 0 0 0 0 split False

0 0 0.01 11 False False 0.419505000780374 0.633333333333333 0.213828332552959 0.0382182976552621 0 0.015 0.0174659856205998 0 -0.0174659856205998 0.0207523120346624 -0.0382182976552621 0.0174659856205998 0.00329452845623246 split False

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74 0.74 0.75 2 False False 0.633333113524928 0.633333333333333 2.19808405477195E-07 0.0528937043371734 0 0.015 0.0352818634670007 0 -0.0352818634670007 0.0176118408701726 -0.0528937043371734 0.0352818634670007 1.46356260444236E-09 False

74 0.74 0.75 7 True True 0.633333333332334 0.633333333333333 9.99089699860178E-13 0.0529047100395948 0.0835337526942288 0.0985337526942288 0.0529047100395948 0.0529047100395948 0 -6.93889390390723E-18 6.93889390390723E-18 0 -1.11022302462516E-14 False

74 0.74 0.75 11 False False 0.421933334096286 0.633333333333333 0.211399999237048 0.0385159318443483 0 0.015 0.017622846572594 0 -0.017622846572594 0.0208930852717543 -0.0385159318443483 0.017622846572594 0.00327850720627243 False

75 0.75 0.76 2 False False 0.633333113539564 0.633333333333333 2.19793769851151E-07 0.0528962740385162 0 0.015 0.0352822081422667 0 -0.0352822081422667 0.0176140658962495 -0.0528962740385162 0.0352822081422667 1.47811762829519E-09 False

75 0.75 0.76 7 True True 0.633333333332334 0.633333333333333 9.99200722162641E-13 0.0529072802678234 0.0835378109493267 0.0985378109493267 0.0529072802678234 0.0529072802678234 0 -6.93889390390723E-18 6.93889390390723E-18 0 0 False

75 0.75 0.76 11 False False 0.421966119168348 0.633333333333333 0.211367214164985 0.0385205706378949 0 0.015 0.0176250721255567 0 -0.0176250721255567 0.0208954985123381 -0.0385205706378949 0.0176250721255567 0.00327869514528123 False

76 0.76 0.77 2 False False 0.633333113554345 0.633333333333333 2.19778988674868E-07 0.0528988516226227 0 0.015 0.0352825578577363 0 -0.0352825578577363 0.0176162937648864 -0.0528988516226227 0.0352825578577363 1.49262824322705E-09 False

76 0.76 0.77 7 True True 0.633333333332334 0.633333333333333 9.99200722162641E-13 0.0529098583715987 0.0835418816394982 0.0985418816394982 0.0529098583715987 0.0529098583715987 0 0 0 0 1.11022302462516E-14 False

76 0.76 0.77 11 False False 0.421998906119801 0.633333333333333 0.211334427213532 0.0385252251495424 0 0.015 0.0176273005138625 0 -0.0176273005138625 0.02089792463568 -0.0385252251495424 0.0176273005138625 0.00327889311566021 False

77 0.77 0.78 2 False False 0.633333113569271 0.633333333333333 2.19764062392436E-07 0.0529014370758929 0 0.015 0.0352829125995414 0 -0.0352829125995414 0.0176185244763515 -0.0529014370758929 0.0352829125995414 1.50708334700767E-09 False

77 0.77 0.78 7 True True 0.633333333332334 0.633333333333333 9.99089699860178E-13 0.0529124443373418 0.0835459647433031 0.0985459647433031 0.0529124443373418 0.0529124443373418 0 -6.93889390390723E-18 6.93889390390723E-18 0 -1.11022302462516E-14 False

77 0.77 0.78 11 False False 0.422031695050958 0.633333333333333 0.211301638282376 0.0385298953506388 0 0.015 0.0176295317378005 0 -0.0176295317378005 0.0209003636128383 -0.0385298953506388 0.0176295317378005 0.00327910108818275 False

78 0.78 0.79 2 False False 0.633333113584342 0.633333333333333 2.19748991558966E-07 0.0529040303847277 0 0.015 0.0352832723538274 0 -0.0352832723538274 0.0176207580309003 -0.0529040303847277 0.0352832723538274 1.52150514409755E-09 False

78 0.78 0.79 7 True True 0.633333333332334 0.633333333333333 9.99200722162641E-13 0.0529150381514749 0.0835500602393027 0.0985500602393027 0.0529150381514749 0.0529150381514749 0 -1.04083408558608E-17 1.04083408558608E-17 0 0 False

78 0.78 0.79 11 False False 0.42206448606184 0.633333333333333 0.211268847271494 0.0385345812125379 0 0.015 0.0176317657976475 0 -0.0176317657976475 0.0209028154148904 -0.0385345812125379 0.0176317657976475 0.00327931903374434 False

79 0.79 0.8 2 False False 0.633333113599557 0.633333333333333 2.19733776507525E-07 0.0529066315355293 0 0.015 0.0352836371067529 0 -0.0352836371067529 0.0176229944287763 -0.0529066315355293 0.0352836371067529 1.53589363449669E-09 False

79 0.79 0.8 7 True True 0.633333333332334 0.633333333333333 9.99200722162641E-13 0.0529176398004211 0.0835541681060598 0.0985541681060598 0.0529176398004211 0.0529176398004211 0 -3.46944695195361E-18 3.46944695195361E-18 0 0 False

79 0.79 0.8 11 False False 0.422097279252177 0.633333333333333 0.211236054081156 0.0385392827066004 0 0.015 0.0176340026936681 0 -0.0176340026936681 0.0209052800129322 -0.0385392827066004 0.0176340026936681 0.003279546923185 False

80 0.8 0.81 2 False False 0.633333113614916 0.633333333333333 2.1971841757118E-07 0.0529092405147009 0 0.015 0.0352840068444901 0 -0.0352840068444901 0.0176252336702107 -0.0529092405147009 0.0352840068444901 1.55024881820509E-09 False

80 0.8 0.81 7 True True 0.633333333332334 0.633333333333333 9.99200722162641E-13 0.0529202492706047 0.0835582883221392 0.0985582883221392 0.0529202492706047 0.0529202492706047 0 3.46944695195361E-18 -3.46944695195361E-18 0 1.11022302462516E-14 False

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81 0.81 0.82 2 False False 0.633333113630418 0.633333333333333 2.19702915082998E-07 0.0529118573086471 0 0.015 0.0352843815532246 0 -0.0352843815532246 0.0176274757554225 -0.0529118573086471 0.0352843815532246 1.56453738853202E-09 False

81 0.81 0.82 7 True True 0.633333333332334 0.633333333333333 9.99089699860178E-13 0.0529228665484512 0.0835624208661073 0.0985624208661073 0.0529228665484512 0.0529228665484512 0 6.93889390390723E-18 -6.93889390390723E-18 0 0 False

81 0.81 0.82 11 False False 0.422162872568683 0.633333333333333 0.21117046076465 0.0385487324766904 0 0.015 0.0176384849952266 0 -0.0176384849952266 0.0209102474814639 -0.0385487324766904 0.0176384849952266 0.00328003241739649 False

82 0.82 0.83 2 False False 0.633333113646064 0.633333333333333 2.19687269709112E-07 0.0529144819037738 0 0.015 0.0352847612191554 0 -0.0352847612191554 0.0176297206846184 -0.0529144819037738 0.0352847612191554 1.5788148566287E-09 False

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83 0.83 0.84 2 False False 0.633333113661852 0.633333333333333 2.19671481560546E-07 0.0529171142864885 0 0.015 0.0352851458284953 0 -0.0352851458284953 0.0176319684579932 -0.0529171142864885 0.0352851458284953 1.59304791580439E-09 False

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83 0.83 0.84 11 False False 0.422228475792498 0.633333333333333 0.211104857540835 0.0385582444319284 0 0.015 0.0176429786443464 0 -0.0176429786443464 0.020915265787582 -0.0385582444319284 0.0176429786443464 0.00328055733848642 False

84 0.84 0.85 2 False False 0.633333113677782 0.633333333333333 2.19655551081388E-07 0.0529197544432 0 0.015 0.0352855353674704 0 -0.0352855353674704 0.0176342190757296 -0.0529197544432 0.0352855353674704 1.60722546382885E-09 False

84 0.84 0.85 7 True True 0.633333333332334 0.633333333333333 9.99089699860178E-13 0.0529307650922436 0.0835748922510428 0.0985748922510428 0.0529307650922436 0.0529307650922436 0 0 0 0 -1.11022302462516E-14 False

84 0.84 0.85 11 False False 0.422261281365883 0.633333333333333 0.211072051967451 0.0385630236574526 0 0.015 0.0176452297247732 0 -0.0176452297247732 0.0209177939326794 -0.0385630236574526 0.0176452297247732 0.00328083451161532 False

85 0.85 0.86 2 False False 0.633333113693855 0.633333333333333 2.1963947882675E-07 0.0529224023603188 0 0.015 0.0352859298223207 0 -0.0352859298223207 0.0176364725379982 -0.0529224023603188 0.0352859298223207 1.62136970516258E-09 False

85 0.85 0.86 7 True True 0.633333333332334 0.633333333333333 9.99200722162641E-13 0.0529334134650245 0.0835790738922758 0.0985790738922758 0.0529334134650245 0.0529334134650245 0 -3.46944695195361E-18 3.46944695195361E-18 0 1.11022302462516E-14 False

85 0.85 0.86 11 False False 0.422294089710999 0.633333333333333 0.211039243622334 0.0385678183434481 0 0.015 0.0176474836427038 0 -0.0176474836427038 0.0209203347007443 -0.0385678183434481 0.0176474836427038 0.00328112145455695 False

86 0.86 0.87 2 False False 0.633333113710068 0.633333333333333 2.19623265129698E-07 0.0529250580242573 0 0.015 0.0352863291792997 0 -0.0352863291792997 0.0176387288449576 -0.0529250580242573 0.0352863291792997 1.63549174203581E-09 False

The pasted rows appear internally consistent. More importantly, they reveal something stronger than the previous report:

The 10−910−9 “switch” at t=124.05t=124.05 is not the moment node 2 first becomes active in the actual finite-step admissibility update. Node 2’s proposed update already violates capacity at step 0.

1. The key comparison

For a currently unsaturated node, the unprojected proposal violates capacity when
(Aw)i+dt di>ci,
(Aw)i+dtdi>ci,

or equivalently,
dt di>slacki.
dtdi>slacki.

Define
Πi=dt dislacki.
Πi=slackidtdi.

Then:

Πi<1Πi<1: proposed row remains interior;
Πi>1Πi>1: the actual finite-step map invokes capacity correction for that row.

At step 0:
islackdt
diΠidtcritical=slack/di22.2046×10−75.2726×10−42391.74.18×10−679.9920×10−135.2737×10−45.28×1081.89×10−11110.2138283.8218×10−41.79×10−35.59
i2711slack2.2046×10−79.9920×10−130.213828dtdi5.2726×10−45.2737×10−43.8218×10−4Πi2391.75.28×1081.79×10−3dtcritical=slack/di4.18×10−61.89×10−115.59

Thus, at the actual simulation timestep dt=0.01dt=0.01:

node 7 is proposal-violating;
node 2 is also proposal-violating by a factor of about 23922392;
node 11 remains genuinely interior.

So the finite-step algorithmic active set at the split is already at least
{2,7},
{2,7},

not merely {7}{7}.
2. The actual update confirms this

At step 0, node 2 has candidate ideal row field
(Ap)2=0.0174550,
(Ap)2=0.0174550,

but its actual finite row-sum rate is only
2.68×10−10.
2.68×10−10.

The ratio is
2.68×10−100.0174550≈1.53×10−8.
0.01745502.68×10−10≈1.53×10−8.

So although the tolerance-based ideal operator labels node 2 as unsaturated and predicts appreciable inward row motion, the actual hard update essentially freezes its row sum because its proposal already violates capacity and is rescaled.

The data are algebraically consistent:
d2=(Kα)2+(Ap)2

d2=(Kα)2+(Ap)2

because
0.0527264=
0.0527264=

0.0352713+0.0174550.
]

But this candidate directional field is not the finite-step update rate in the boundary layer.

For node 11:
(Ap)11=0.0207523,
(Ap)11=0.0207523,

while the actual row rate is
0.00329453.
0.00329453.

It is still moving toward capacity, but at only about 15.9%15.9% of the candidate ideal rate. This reduction occurs because both of its triangle edges are shared with proposal-violating nodes 2 and 7 and are affected by endpoint scaling.
3. The apparent switch is therefore a classifier switch

At 10−910−9 boundary tolerance:

node 2 is called unsaturated until its current slack falls below 10−910−9;
at step 12405 it is reclassified as saturated;
the analysis changes from one active incidence row to two.

But the actual finite-step projection branch for node 2 was already active from step 0.

Therefore:
the step-12405 event is a switch in the surrogate active-face classifier,
the step-12405 event is a switch in the surrogate active-face classifier,

not the onset of capacity correction in the simulation itself.

The actual trajectory does not suddenly discover node 2 at t=124.05t=124.05. Its proposed row has been violating capacity throughout the displayed pre-switch interval.

This explains the strong boundary-tolerance sensitivity:

the raw state evolves continuously;
the active-face analysis waits until slack crosses an arbitrary tolerance;
changing that tolerance changes when the second incidence row is inserted;
insertion of that row changes the analytic tangent dimension and rate-equality condition.

4. The relevant operators are now clearly different

There are at least three operators in play.
A. Exact ideal directional limit

At a fixed state with strictly positive slack,
$\lim_{dt \to 0} A(w+dtF)-w dt$
$dt \to 0 \lim_{dt} A(w+dtF)-w$

treats node 2 as interior. Indeed, for the step-0 state, if
$dt<4.18\times10-6$,
$dt<4.18\times10-6$,

the node-2 proposal would no longer violate capacity.

This is the operator for which the one-active-row K3K3 has tangent nullity 2.
B. Tolerance-classified ideal operator

At dt=0.01dt=0.01, the analysis calls node 2 active only when
$slack2 \leq 10-9$.
$slack2 \leq 10-9$.

This produces the reported artificial switch at step 12405.
C. Actual finite-step admissibility map

The simulation uses the proposed-row violation condition. At step 0, node 2 already violates, so the actual map rescales its incident edges.

The operational parameter controlling the regime is not slack alone, but
Πi=dt dislacki.
Πi=slackidtdi.

This exposes a singular, nonuniform boundary limit:
slack→0anddt→0
slack→0anddt→0

do not commute without specifying their relative scaling.
5. Consequence for the 9→89→8 interpretation

The statement

“The dynamics undergo a true tangent-dimension transition 9→89→8 at t=124.05t=124.05”

is no longer supported.

What is supported is:

Under the 10−910−9 current-slack classification, the analytic ideal operator changes from a one-row K3K3 model to a two-row K3K3 model at step 12405.

For the actual dt=0.01dt=0.01 map, node 2 is already correction-active at split. Its effective finite-step local geometry may therefore already be closer to the two-row case.

This could explain several earlier exceptional anomalies:

the analytic split spectrum contains a bare 0.0150.015 mode;
observed exceptional recovery rates are around 0.150.15, not 0.0150.015;
several exceptional tangent directions are amplitude-unreliable;
some scalar frozen restoring coefficients are extreme;
the exceptional modal prediction is less clean than the 39 ordinary states.

The bare γγ mode is a legitimate mode of the strict dt→0dt→0, fixed-slack ideal operator. It may not be a dynamically accessible mode of the actual dt=0.01dt=0.01 projection branch.
6. Which node is the classified second node?

The rows make this clear.

At split:

node 7 has slack ∼10−12∼10−12;
node 2 has slack 2.20×10−72.20×10−7;
node 11 has slack 0.2140.214.

Thus the 10−910−9 transition at t=124.05t=124.05 is node 2 entering the tolerance-defined active set.

Node 11 is the remaining tolerance-unsaturated node.

The split triangle structure is therefore:

active/classified node: 7;
near-boundary but already proposal-violating node: 2;
genuinely interior node: 11;
free edge of the strict ideal operator:
(2,11).
(2,11).

7. Node 11 is the genuinely evolving degree of freedom

At step 0:
s˙11actual≈0.0032945,
s˙11actual≈0.0032945,

while
s˙2actual≈2.68×10−10,s˙7actual=0.
s˙2actual≈2.68×10−10,s˙7actual=0.

Thus the actual finite-step trajectory is primarily filling node 11 while rows 2 and 7 remain capacity-controlled.

The potentially meaningful algorithmic event is not when node 2 crosses a reporting tolerance. It is when node 11's unprojected proposal first satisfies
dt d11>slack11.
dtd11>slack11.

That is the moment when all three triangle nodes become proposal-violating under the actual finite-step map.

It can be found immediately from the existing CSV:

Python

```
import pandas as pd

path = (
"aief_1_4_exceptional_K3_transition/"
"exceptional_K3_node_trajectory.csv"
)

df = pd.read_csv(path)

dt = 0.01
df["proposal_excess"] = (
df["row_sum"]
+ dt * df["row_direction_d"]
- df["capacity"]
)
df["proposal_violating"] = df["proposal_excess"] > 0.0
df["projection_ratio"] = (
dt * df["row_direction_d"]
/ df["slack"].clip(lower=1e-300)
)
df["critical_dt"] = (
df["slack"]
/ df["row_direction_d"].where(df["row_direction_d"] > 0.0)
)

for node, group in df.groupby("node"):
violating = group.loc[group["proposal_violating"]]
first = violating.iloc[0] if len(violating) else None

print(f"\nnode {node}")
```

```
if first is None:
print(" never proposal-violating")
else:
print(f" first proposal-violating step: {int(first['step'])}")
print(f" state time: {first['state_elapsed_time']}")
print(f" slack: {first['slack']}")
print(f" proposal excess: {first['proposal_excess']}")
print(f" projection ratio: {first['projection_ratio']}")
```

This uses the recorded directional row sum. A final production diagnostic should reconstruct the actual proposal matrix directly and save the exact per-node violating mask and scaling factors, rather than relying only on this algebraic reconstruction.

8. Revised assessment

Established

The data snippet is internally consistent.
Node 7 is tolerance-active at split.
Node 2 is tolerance-interior but finite-step proposal-violating from step 0.
Node 11 is genuinely interior and moves toward capacity.
The step-12405 event is node 2 crossing the 10−910−9 reporting threshold.
That event is not the onset of actual capacity correction for node 2.
The ideal directional operator and actual dt=0.01dt=0.01 update differ substantially in this boundary layer.

Strong implication

The exceptional “active-face transition” is largely an analysis-classification event. It should not be used as evidence for an intrinsic discontinuous transition of the simulated map without examining the actual proposal-violation masks.

Still open

When node 11 first becomes proposal-violating.
Whether that event produces any genuine change in the exact finite-step Jacobian.
The spectrum of the actual one-step map with its fixed projection branch.
The joint limit of timestep, safeguard, and boundary distance.
Whether consistently regenerated smaller-dtdt trajectories preserve this exceptional boundary layer.

The next decisive object is therefore not another tolerance-defined tangent Jacobian. It is the Jacobian —or at least the branchwise linearization—of the actual finite-step admissibility map, with the exact proposed-violation set and endpoint minimum scales recorded at every step.

Think about this:

I think this is actually a more significant development than the previous (K_3) audit. The reason is that it shifts the question from "when does the active set change?" to "what object are we actually linearizing?" Those are not the same question.

There are, however, a few places where I'd separate what the data establishes from what it suggests.

The strongest new fact is not about the K_3

The strongest new fact is this:

$$
\Pi_i=\frac{dt,d_i}{\text{slack}_i}.
$$

The numbers are striking.

For node 2,

$$
\Pi_2\approx2392.
$$

For node 7,

$$
\Pi_7\approx5.3\times10^8.
$$

Those are not "slightly above one."

They are orders of magnitude above one.

That means that under the implemented (dt=0.01) algorithm, the unconstrained Euler proposal is already far outside the feasible polytope at the very first recorded state.

So capacity correction is already active.

That conclusion does not depend on any tolerance.

It follows directly from the update formula.

I think this is the most robust observation in your analysis.

The check.py output strengthens this considerably

The second output is particularly important:

node 2
first proposal-violating step: 0

node 7
first proposal-violating step: 0

node 11
first proposal-violating step: 11381

This removes one uncertainty.

Previously one could wonder whether node 2 was only nearly violating.

It isn't.

The reconstruction says

proposal violation starts immediately.

Meanwhile node 11 genuinely stays interior until

[
t\approx113.81.
]

That is a qualitatively different event.

I think this changes the interpretation of the reported switch

I agree with your revised wording.

The evidence now supports

the (10^{-9}) event is a classifier transition,

much more strongly than

it is the onset of projection.

Those are no longer compatible with the data.

Projection was already occurring.

Classification changed later.

That distinction is now empirical rather than philosophical.

But one sentence still goes slightly too far

You write

"The finite-step algorithmic active set at the split is already at least ({2,7})."

This depends on terminology.

If by "active" you mean

proposal violates capacity,

then yes.

If by "active" means

the exact branch of the piecewise-linear projection map currently depends on that constraint,

then probably yes.

But there is a subtle difference.

A projection algorithm can use several notions:

violated proposal,
saturated current iterate,
minimum scaling endpoint,
active KKT multiplier.

Those need not coincide.

Your reconstruction proves the first.

It does not yet prove the fourth.

That is why I would keep using the phrase

proposal-violating

instead of

active

until you've instrumented the projection routine itself.

The really interesting object has changed

Previously the key variable was

$$
\text{slack}.
$$

Now it appears to be

$$
\Pi

\frac{dt,d}{\text{slack}}.
$$

I actually think this is the right nondimensional parameter.

Notice what it measures.

When

$$
\Pi<1,
$$

the explicit Euler proposal stays inside the feasible region.

When

$$
\Pi>1,
$$

projection becomes necessary.

That is exactly the ratio governing which branch of the numerical map is taken.

It is therefore much closer to an algorithmic control parameter than slack alone.

The singular limit is probably real

I especially liked this observation:

[
dt\rightarrow0,
\qquad
\text{slack}\rightarrow0
]

do not commute.

I think that's essentially unavoidable.

Suppose

[
\text{slack}

10^{-7}.
]

Then

[
dt=10^{-2}
]

gives projection,

while

[
dt=10^{-6}
]

may not.

The limiting operator therefore depends on the scaling

[
\frac{\text{slack}}{dt}.
]

That is a genuine singular perturbation problem.

It isn't merely numerical bookkeeping.

Node 11 is becoming the central dynamical variable

This is another place where the new output is informative.

Node 11 first violates at

step 11381
time 113.81
projection ratio ≈1.0001057

That is exactly the sort of crossing one expects from a genuine approach to a boundary.

Unlike node 2,

whose ratio starts at

2392,

node 11 reaches violation continuously.

That makes node 11 a much better candidate for identifying an intrinsic geometric transition.

One thing I would now investigate

The timing is interesting.

You have

Node 11 proposal violation

[
t\approx113.81,
]

while the classifier switch for node 2 is

[
t\approx124.05.
]

So the order is

node 2 already proposal-violating,
node 11 becomes proposal-violating,
node 2 crosses the (10^{-9}) slack threshold.

That sequence is rather surprising.

It means the classifier transition occurs after all three nodes are already proposal-violating.

If that's correct, then the classifier is lagging behind the operational regime of the algorithm.

That's a much stronger statement than before.

However, I'd want to verify one subtle point:

Does "proposal-violating" automatically imply the projection branch changes for that node, or can the global endpoint scaling caused by neighboring violations resolve the overshoot before an independent node-level projection is effectively active?

Your reconstruction strongly suggests the former, but the definitive answer still requires instrumenting the actual projection code.

The conceptual hierarchy is becoming clearer

At this point I think there are four distinct objects:

Ideal infinitesimal operator
[
dt\to0
]
Finite-step proposal
(determined by (\Pi))
Projection branch actually executed
Post hoc active-set classifier
(based on slack tolerance)

The first audit mostly studied (4).

The new audit studies (2).

The next step is to expose (3).

That is exactly where I think the remaining uncertainty lies.

Overall assessment

I think the new evidence substantially weakens the interpretation of the reported (t=124.05) event as an intrinsic dynamical "switch." The check.py output convincingly shows that nodes 2 and 7 are already proposal-violating from the beginning of the analyzed interval, while node 11 becomes proposal-violating much later at (t\approx113.81). That means the tolerance-based active-set classifier is not tracking the onset of capacity correction in the implemented algorithm.

What it doesn't yet prove is that the branchwise Jacobian of the actual projection map changes exactly when a node first becomes proposal-violating. There can still be differences between "proposal exceeds capacity," "this constraint determines the executed projection," and "this constraint contributes an active multiplier in the local linearization." Resolving that requires instrumenting the projection routine itself to record the executed branch, scaling factors, and active constraints.

So I think you've now identified the right mathematical object. The remaining question is no longer when the classifier changes, but what the Jacobian of the actual finite-step admissibility map is on the branch that the algorithm actually executes. That is the operator that should ultimately be compared with the ideal (K_3) tangent algebra.

and this:

(130)(deck@steamdeck SSRH)$ python check.py

node 2
first proposal-violating step: 0
state time: 0.0
slack: 2.204580802356304e-07
proposal excess: 0.000527043307312125
projection ratio: 2391.673577257232

node 7
first proposal-violating step: 0
state time: 0.0
slack: 9.992007221626409e-13
proposal excess: 0.0005273732154961186
projection ratio: 527795071.3985642

node 11
first proposal-violating step: 11381
state time: 113.81
slack: 0.0007040520367487
proposal excess: 7.444908267650874e-08
projection ratio: 1.0001057437218641

I agree: this is more important than the tolerance-defined K3K3 switch. It identifies a mismatch between the ideal infinitesimal operator and the branch actually used by the finite-step algorithm.

There is one point we can now strengthen, and another we must refine.
1. Proposal violation does enter the executed code branch

The implemented update is not iterative. It computes:

Python

```
row_sum_pre = A @ proposal_w
violating = row_sum_pre > capacities

scale = np.minimum(
1.0,
capacities / (row_sum_matrix + 1e-12),
)

projected *= np.minimum.outer(scale, scale)
```

Therefore, if node ii is proposal-violating, then the code definitely computes
si=ciripre+ε<1.
si=ripre+εci<1.

Neighboring corrections do not first “resolve” the violation and cause the code to reconsider sisi. There is no second violation test.

So the check establishes more than a hypothetical proposal condition:

nodes 2 and 7 have si<1si<1 from step 0;
node 11 first acquires s11<1s11<1 at step 11381.

However, si<1si<1 still does not guarantee that sisi affects an output edge.
2. The decisive branch object is the endpoint winner map

For edge e=(i,j)e=(i,j), the actual multiplier is
me=min(si,sj).
me=min(si,sj).

Thus node ii's scale affects the output only if it is the minimum scale on at least one incident edge.

Define the endpoint winner
k(e)=argminv∈{i,j}sv.
k(e)=argv∈{i,j}minsv.

The true fixed branch of the implemented map consists of:

the positivity/clipping pattern;
the proposal-violation set
V={i:si<1};
V={i:si<1};
the edge winner map k(e)k(e);
any endpoint-scale ties.

There is no literal KKT multiplier here because this map is not known to be an optimization projection.
“Effective endpoint controller” or “winner” is safer terminology.
3. The executed split branch can already be reconstructed

At step 0, the proposed excesses give approximately
s2=0.9991685183,s7=0.9991679983,s11=1.
s2s7s11=0.9991685183,=0.9991679983,=1.

Therefore
s7<s2<s11.
s7<s2<s11.

For the triangle edges:
edgewinning endpoint(2,7)7(2,11)2(7,11)7
edge(2,7)(2,11)(7,11)winning endpoint727

So both nodes 2 and 7 already affect the executed finite-step map at split.

In particular, the edge (2,11)(2,11) is not operationally “bare” under the actual dt=0.01dt=0.01 update. Although the tolerance-based ideal operator assigns it endpoint rate γγ, its actual finite-step multiplier is s2<1s2<1.

This is probably the cleanest explanation yet for why the analytic bare-γγ mode was poorly reflected in the observed exceptional perturbations.

4. Node 11's proposal violation may be computationally inert

At its first proposal violation,
Δ11pre=
Δ11pre=

7.4449\times10^{-8},
]

so
s11≈cc+7.4449×10−8≈0.9999998824.
s11≈c+7.4449×10−8c≈0.9999998824.

Its scale is only infinitesimally below one.

If, at step 11381,
s2<s11,s7<s11,
s2<s11,s7<s11,

then the edge multipliers remain
m2,11=s2,m7,11=s7.
m2,11=s2,m7,11=s7.

In that case:

node 11 enters the proposal-violation set;
the code computes s11<1s11<1;
but node 11 wins no edge;
the projected output formula does not change;
the branchwise Jacobian does not change because of node 11.

So t=113.81t=113.81 is a much better event than the 10−910−9 classifier crossing, but it is not yet proven to be an effective branch change.

The decisive events are instead:

the first edge whose winner changes to node 11;
or the first endpoint-scale tie involving node 11.

It is possible that node 11 never wins an edge and merely approaches a three-way tie asymptotically.
5. ΠiΠi identifies violation, but not correction strength

The ratio
Πi=dt diσi,σi=ci−(Aw)i,
Πi=σidtdi,σi=ci−(Aw)i,

is the correct nondimensional parameter for entering the violation set:
Πi>1 ⟺ dt di>σi.
Πi>1⟺dtdi>σi.

But after entry, the more informative quantity is the scale sisi, or an effective finite-step controller
βidt=
βidt=

-\frac{1}{dt}\log s_i.
]

This definition has the exact property
min(si,sj)=
min(si,sj)=

\exp\left[
-dt\max(\beta_i^{dt},\beta_j^{dt})
\right].
]

Thus:

ΠiΠi answers whether the node proposes outside;
βidtβidt measures the strength of its multiplicative correction;
max(βidt,βjdt)max(βidt,βjdt) determines the effective edge controller;
the winner map determines the differentiable branch.

At split:
β2dt≈0.083183,β7dt≈0.083235,β11dt=0.
β2dtβ7dtβ11dt≈0.083183,≈0.083235,=0.

Therefore the effective first-order edge correction rates are approximately
(2,7): β7dt,(2,11): β2dt,(7,11): β7dt.
(2,7)(2,11)(7,11): β7dt,: β2dt,: β7dt.

All three are already close to the same component rate. There is no operational bare edge in the finite-step map.

At node 11's first violation,
β11dt≈7.4449×10−8c dt≈1.18×10−5,
β11dt≈cdt7.4449×10−8≈1.18×10−5,

which is likely far below β2dtβ2dt and β7dtβ7dt. That makes it unlikely to win either incident edge immediately, although the actual scales at step 11381 must be checked directly.
6. The singular boundary-layer operator can be derived

Write the current slack as σiσi. Before scaling,
ripre=
ripre=

c_i-\sigma_i+dt,d_i.
]

Ignoring the 10−1210−12 safeguard for the asymptotic derivation, a violating node has
si=
si=

\frac{c_i}
{c_i+dt,d_i-\sigma_i}.
]

Suppose
σidt→λi
dtσi→λi

as dt→0dt→0. Then
si=
si=

1-dt,\frac{(d_i-\lambda_i)_+}{c_i}
+O(dt^2).
]

Define the boundary-layer controller
\boxed{ \beta_i(\lambda_i) =

\frac{(d_i-\lambda_i)_+}{c_i}.
}
]

Then the corresponding edge field is
\boxed{ P_e^{(\lambda)} =
\widetilde F_e

w_e\max\bigl(
\beta_i(\lambda_i),
\beta_j(\lambda_j)
\bigr).
}
]

This interpolates between the two previously separated regimes.
Fixed interior state

If σi>0σi>0 is fixed while dt→0dt→0, then
λi=σidt→∞,βi=0.
λi=dtσi→∞,βi=0.

The node is infinitesimally interior.
Exact boundary state

If σi=0σi=0, then
λi=0,βi=max(di,0)ci,
λi=0,βi=cimax(di,0),

which is the candidate ideal active-boundary controller.

Boundary layer

If
σi∼dt λi,
σi∼dtλi,

then the limiting controller is intermediate:
βi=(di−λi)+ci.
βi=ci(di−λi)+.

This is the precise mathematical expression of the noncommuting limits. The issue is not just whether slack is “small.” It is the scaling of slack relative to dtdt.

The fixed 10−1210−12 denominator safeguard adds a second ratio,
ɛdt,
dtɛ,

which must also be specified in a rigorous joint limit.
7. The actual branchwise Jacobian is explicit

Let xx denote the nonnegative symmetric proposal in edge coordinates and
q=Ax
q=Ax

its row sums. Suppose:

the violation set is fixed;
the positivity pattern is fixed;
every scaled edge has a unique winning endpoint k(e)k(e);
no endpoint-scale ties occur.

For a scaled edge,
ye=xesk(e),sk=ckqk+ε.
ye=xesk(e),sk=qk+εck.

Then
∂sk∂xf=
∂xf∂sk=

-\frac{s_k}{q_k+\varepsilon}A_{kf},
]

and therefore
\boxed{ D\mathcal A(x)_{ef} =
s_{k(e)}\delta_{ef}

\frac{x_e s_{k(e)}}{q_{k(e)}+\varepsilon}
A_{k(e)f}.
}
]

For an unscaled edge, the row is simply
DA(x)ef=δef.
DA(x)ef=δef.

This is the exact Jacobian of the implemented admissibility map on a strict branch.

At a tie
si=sj,
si=sj,

the ordinary Jacobian generally does not exist because of the endpoint minimum. The appropriate object is then a directional derivative or generalized Jacobian. This will be especially important for the mature rate-uniform cliques, where scale ties may be structural rather than accidental.
8. The full one-step map requires composition

The admissibility Jacobian alone is not the full discrete tangent operator. If
x(w)=w+dt F(w),Φ(w)=A(x(w)),
x(w)=w+dtF(w),Φ(w)=A(x(w)),

then
DΦ(w)=
DΦ(w)=

D\mathcal A(x)
\left[I+dt,DF(w)\right].

]

With coherence frozen,
DF=−γI,
DF=−γI,

so
DΦ=
DΦ=

D\mathcal A(x)(1-\gamma dt)I.
]

For the coupled system, derivatives through phase, history, and EMA states must be added.

To compare a discrete multiplier with a continuous contraction rate, one should use
reff=
reff=

-\frac{\log|\lambda(D\Phi)|}{dt},
]

rather than directly comparing eigenvalues of DΦDΦ with eigenvalues of the continuous generator JTJT.
9. Revised hierarchy

The conceptual hierarchy is now:

Ideal infinitesimal directional operator
dt→0
dt→0

at a specified boundary state.

Finite proposal geometry
Πi=dt diσi.
Πi=σidtdi.

Executed scale vector
si.
si.

Executed edge winner map
k(e)=argmin(si,sj).
k(e)=argmin(si,sj).

Branchwise Jacobian or directional derivative
of the actual admissibility map.

Post hoc boundary classifier
σi≤τ∂.
σi≤τ∂.

The earlier analysis mainly used 1 and 6. The new check gives 2. The scales and winner map provide 3–4. The next analysis must compute 5.
10. Immediate decisive checks

At each replay step, record:

text

proposal row sum
proposal excess
violating mask
scale_i
effective beta_i = -log(scale_i)/dt
edge winner endpoint
winner margin = |scale_i-scale_j|
tie flags

Then determine:

whether node 11 ever wins edge (2,11)(2,11);
whether it ever wins edge (7,11)(7,11);
whether the winner map changes at step 11381;
whether anything in the executed map changes at step 12405;

when endpoint-scale ties emerge;
the analytic branchwise Jacobian;
its finite-difference agreement while the branch is preserved;
its spectrum and effective contraction rates.

The likely current interpretation is:

step 0: nodes 2 and 7 already control all three triangle edges;
step 11381: node 11 enters the violation set but may win no edge;
step 12405: only the post hoc classifier changes;
late: all three scales approach a tie as the triangle approaches its symmetric locked state.

That last sequence is not yet fully established, but it is now the sharply defined hypothesis. The central mathematical object is indeed the branchwise—and at ties, directionally differentiable—finite-step admissibility operator.

Then what might you think about this:

I think this is now converging on something that is mathematically much cleaner than the original "exceptional (K_3) switch" picture. The biggest conceptual advance is that you're no longer trying to force the discrete algorithm into the language of an ideal active-set transition. Instead, you're identifying the actual piecewise-smooth map that the code implements.

That said, I'd separate three levels of confidence.

I think the operator hierarchy is now essentially correct

The proposed hierarchy

[
\text{ideal generator}
\rightarrow
\text{proposal}
\rightarrow
\text{scales}
\rightarrow
\text{winner map}
\rightarrow

\text{branch Jacobian}
\rightarrow
\text{classifier}
]

is, in my view, the right one.

Previously, the analysis was implicitly identifying

classifier change

with

change of the linearized dynamics.

The new reconstruction shows these are different objects.

That's a substantial conceptual clarification.

The winner map is the correct combinatorial object

This is probably the strongest mathematical insight in the whole analysis.

The implemented code never says

node i is active.

It computes numbers

[
s_i.
]

Then every edge simply uses

[

m_e=\min(s_i,s_j).
]

So the branch is really determined by

[
e\mapsto\arg\min(s_i,s_j),
]

not by a node-level active mask.

That is a much cleaner description.

It's also minimal:

scales determine winners;
winners determine which formula is applied to each edge;
therefore winners determine the differentiable region.

That feels like the natural state-space decomposition for this algorithm.

The explicit Jacobian is the biggest technical result

I actually think

[
D\mathcal A

s_{k(e)}\delta

\frac{x_es_{k(e)}}{q_k+\varepsilon}A
]

(or its equivalent indexing form)

is probably the most valuable formula that has appeared in this whole line of investigation.

Why?

Because it's no longer speculative.

It comes directly from differentiating the implemented code.

That means:

it is falsifiable,
it can be finite-difference checked,
it does not rely on heuristic continuum reasoning.

That's exactly the sort of object you eventually want.

One place I would be slightly more cautious

I would not quite say

"The branchwise Jacobian is explicit."

I would instead say

"The branchwise Jacobian is explicit assuming the winner map is locally constant."

That assumption matters.

The derivative of

[
\min(s_i,s_j)
]

is only

[
Ds_i
]

if

[
s_i<s_j
]

throughout a neighborhood.

If

[
s_i-s_j

10^{-15},
]

then floating-point perturbations may already move you onto another branch.

So the explicit Jacobian is really

the Jacobian on an open stratum of constant winner map.

That's a standard piecewise-smooth qualification, but it's worth stating explicitly.

I like the boundary-layer controller

This part is particularly elegant:

[
\beta_i(\lambda)

\frac{(d_i-\lambda)_+}{c_i}.

]

Why?

Because it interpolates between the two limits.

Interior:

[
\lambda=\infty
\Rightarrow
\beta=0.
]

Boundary:

[
\lambda=0
\Rightarrow
\beta=\frac{d_+}{c}.
]

Intermediate scaling:

[
\sigma\sim dt,\lambda.
]

That is exactly what singular perturbation theory usually produces.

I don't know whether this specific formula will survive every refinement (especially once the (\varepsilon) safeguard is retained), but structurally it's the right type of limiting object.

One statement I would soften

The write-up says

"Node 2 and 7 already control all three triangle edges."

I think that's probably true for the split state because you've reconstructed the scales.

But I'd distinguish

reconstructed from replay data

from

directly logged by the implementation.

They're almost certainly identical here.

Still, for a paper I'd write something like

"The reconstructed scale ordering implies the winner map..."

until you've instrumented the code.

That keeps the evidential chain completely transparent.

The tie discussion is excellent

I particularly like the reformulation

[
k(e)={i,j}
]

instead of trying to force an arbitrary winner.

That immediately makes clear that

the map remains continuous,

but differentiability fails.

That's precisely how people working in nonsmooth analysis would formulate it.

I would only make one tiny refinement.

Rather than saying

"the Jacobian does not exist,"

I'd say

"the classical Fréchet derivative generally does not exist."

That's mathematically a little sharper, because directional derivatives certainly do exist in many directions.

I think there's now a genuine research question

Originally the question was

Why does the exceptional trajectory violate the (K_3) picture?

Now I think the question has become

How does the spectrum of the piecewise-smooth finite-step map converge to the spectrum of the infinitesimal directional generator?

That's a much deeper—and much more interesting—question.

It's no longer specific to this trajectory.

It's a question about the numerical scheme itself.

One thing I'd add

There is one mathematical object I think naturally sits between the winner map and the Jacobian.

Namely the stratification of state space.

Each open region where

positivity pattern is fixed,
violation mask is fixed,
winner map is fixed,

is a smooth manifold (or at least an open polyhedral-like cell in the appropriate coordinates).

Then

[
D\mathcal A
]

is simply the Jacobian on that stratum.

The boundaries between strata are

positivity changes,
proposal-violation changes,
winner changes,
scale ties.

That language connects the analysis directly to the theory of piecewise-smooth dynamical systems and makes the geometric picture much cleaner.

Overall assessment

I think this is the first version of the analysis that cleanly separates the continuous mathematical model from the discrete algorithm. The proposed hierarchy—from proposal geometry, to node scales, to the edge winner map, to the branchwise Jacobian—is internally coherent and closely mirrors the actual implementation rather than a post hoc classifier. The explicit differentiation of the implemented

admissibility map is the strongest technical result because it yields an operator that can be verified directly against finite differences.

The remaining caveat is mainly one of rigor rather than direction: all formulas for the branchwise Jacobian are valid only on regions where the winner map is locally constant. Once winner changes or scale ties occur, the appropriate framework is piecewise-smooth analysis using directional derivatives or generalized Jacobians. Framing the state space as a stratification by positivity pattern, violation mask, and winner map would make that transition explicit and, in my view, provides the natural mathematical setting for the next stage of the work.

I think this is an excellent formulation and close to the right mathematical framework. I would make two implementation-specific amendments and add one potentially important bridge back to the original tangent spectra.
1. The hierarchy is correct, but the actual code has two scale masks

Let zz be the Euler proposal before nonnegativity, let
x=z+,q=Ax
x=z+,q=Ax

be the nonnegative proposal and its row sums.

The implemented map is more precisely
Aε(x)={x,qi≤cifor every i,xemin(si,sj),otherwise,
Aε(x)={x,xemin(si,sj),qi≤cifor every i,otherwise,

where, once the global projection gate is on,
si=min(1,ciqi+ε),ε=10−12.
si=min(1,qi+εci),ε=10−12.

This creates two related but distinct node sets:
Proposal-violation set
V={i:qi>ci}.
V={i:qi>ci}.

This determines whether the global if np.any(violating) gate is entered.
Scale-clipped set
Sε=
Sε=

{i:q_i+\varepsilon>c_i}.
]

Once the gate is entered, this set determines which nodes actually receive si<1si<1.

Because of the denominator safeguard,
Sε
Sε

can contain a node that is not in VV, although the difference is only a boundary layer of width 10−1210−12.

Thus the exact hierarchy is:
proposal positivity pattern → global projection gate → node scale-clipping set → scale vector → edge winner map → branch derivative → post hoc classifier.
proposal positivity pattern → global projection gate → node scale-clipping set → scale vector → edge winner map → branch derivative → post hoc classifier.

The per-node violating mask is partly diagnostic: after the global gate is on, the code does not use it to decide each node's scale. It recomputes every scale from qi+εqi+ε.
2. The winner map is indeed the core edge-level combinatorial object

On a strict branch,
k(e)=argminv∈esv
k(e)=argv∈eminsv

determines which node's scale controls edge ee.

The qualification "strict branch" matters:
si<sj
si<sj

must persist in a neighborhood. A small numerical margin is not enough by itself to guarantee a useful finite-difference neighborhood.

A complete branch signature should therefore include:

positivity mask;

global gate state;
scale-clipping mask;
edge winner;
winner margin
Δse=|si−sj|;
Δse=|si−sj|;
tie flags.

For the exceptional split, the reconstructed scales imply
s7<s2<s11,
s7<s2<s11,

hence
k(2,7)=7,k(2,11)=2,k(7,11)=7.
k(2,7)=7,k(2,11)=2,k(7,11)=7.

That statement remains “reconstructed from the exact update formula and replayed state” until directly logged, but mathematically there is essentially no gap if the proposal reconstruction is exact.
3. The strata are polyhedral in proposal space, but the map is rational

In proposal edge coordinates xx, the relevant inequalities are linear.

For example:
qi>ci
qi>ci

is linear because q=Axq=Ax.

For two scale-clipped nodes,
si<sj
si<sj

means
ciqi+ε<cjqj+ε,
qi+εci<qj+εcj,

which is equivalent to
ci(qj+ε)<cj(qi+ε).

ci(qj+ε)<cj(qi+ε).

That is again an affine inequality in xx.

Therefore, in proposal space, the open constant-branch regions are intersections of finitely many strict affine inequalities: open polyhedral cells.

On each cell, however, the map is generally not affine. It contains factors
ciqi+ε,
qi+εci,

so it is a smooth rational map—indeed C∞C∞—on that cell.

In the full delayed state space, the preimages of these cells under the phase/EMA/proposal map need not be polyhedral; they can be curved.

A precise description is therefore:

The proposal space admits a polyhedral stratification, and the admissibility map is smooth rational on each open stratum.

4. The branchwise Jacobian theorem

Assume:

the global gate remains on;
positivity is fixed;
the scale-clipping set is fixed;
each edge has a unique winning endpoint;
all winner inequalities have positive margin.

Then, for a scaled edge ee,
ye=xesk(e),sk=ckqk+ε.
ye=xesk(e),sk=qk+εck.

Since
∂qk∂xf=Akf,

∂xf∂qk=Akf,

we have
∂sk∂xf=
∂xf∂sk=

-\frac{s_k}{q_k+\varepsilon}A_{kf}.
]

Therefore
\boxed{ D\mathcal A_\varepsilon(x)_{ef} =
s_{k(e)}\delta_{ef}

\frac{x_es_{k(e)}}{q_{k(e)}+\varepsilon}
A_{k(e)f}.
}
]

For an edge whose multiplier is identically one on the stratum,
DAε(x)ef=δef.
DAε(x)ef=δef.

So the sharpened statement is:

The Jacobian is explicit on every open stratum of constant positivity pattern, gate state, scale-clipping set, and strict winner map.

That is an exact theorem about the implemented code, not a continuum approximation.
5. The fixed safeguard creates a small discontinuity at the global gate

This deserves explicit attention.

If no proposal row violates capacity, the code skips scaling entirely:
Aε(x)=x.
Aε(x)=x.

If one row crosses from qi=ciqi=ci to qi>ciqi>ci, the gate turns on and
si≈cici+ε<1.

si≈ci+εci<1.

Thus the fixed-εε implementation has a jump of order
εci
ciε

at the global projection gate.

It is extremely small numerically, but mathematically it means:

with ε>0ε>0, the implemented map is not exactly continuous across that gate;
with ε=0ε=0, the ideal map is continuous there;
Clarke generalized-Jacobian language requires local Lipschitz continuity and therefore cannot be applied blindly at the safeguarded global gate.

Away from that gate, including ordinary winner ties, the map remains continuous and locally Lipschitz.

This reinforces why the ideal theorem should concern the map without the fixed safeguard or a joint limit with
ε=o(dt).
ε=o(dt).
6. Ties require a directional derivative, not an arbitrary winner

Suppose edge e=(i,j)e=(i,j) lies on a tie:
si=sj=s.
si=sj=s.

For a perturbation hh,
Dq[h]=Ah.
Dq[h]=Ah.

If both nodes are scale-clipped,
Dsi[h]=
Dsi[h]=

-\frac{s_i}{q_i+\varepsilon}(Ah)_i,
]

and similarly for jj.

The directional derivative of the edge multiplier is
Dme[h]=
Dme[h]=

\min\bigl(Ds_i[h],Ds_j[h]\bigr).
]

Hence
\boxed{ \mathcal A_\varepsilon'(x;h)_e =

s,h_e
+
x_e\min\bigl(Ds_i[h],Ds_j[h]\bigr).
}
]

This is positively homogeneous and piecewise linear in hh, but generally not linear. Therefore the classical Fréchet derivative generally does not exist at the tie.

The caveat “generally” is important: special structure can make the two directional scale derivatives agree on a restricted subspace.
7. The mature tangent space may remove the tie ambiguity

This may be the bridge back to the earlier spectra.

On a fully saturated support component, tangent perturbations satisfy
Ah=0
Ah=0

on every active node.

Therefore
Dsi[h]=0
Dsi[h]=0

for all nodes in the component.

Even if every edge has an endpoint-scale tie,
Dme[h]=0,
Dme[h]=0,

and the restricted derivative simplifies to
Aε′(x;h)e=s he.
Aε′(x;h)e=she.

Thus:

The ambient map can be nonsmooth at a fully symmetric clique while its restriction to the tangent space is a well-defined linear scalar map.

This provides a natural explanation for why the ordinary mature cliques exhibit exceptionally clean tangent spectra despite sitting on large endpoint-tie strata.

If the frozen driver derivative is
DF=−γI,
DF=−γI,

then the proposal perturbation is
dx=(1−γdt)h,
dx=(1−γdt)h,

and the one-step tangent multiplier on a component is
λstep=
λstep=

s(1-\gamma dt).
]

Its equivalent contraction rate is
rstep=
rstep=

-\frac1{dt}
\log\left[s(1-\gamma dt)\right].

]

Writing
βdt=−logsdt,
βdt=−dtlogs,

gives
rstep=
rstep=
\beta^{dt}
\frac{\log(1-\gamma dt)}{dt}

\beta^{dt}+\gamma+O(dt).
]

As dt→0dt→0 and βdt→αβdt→α,
rstep→γ+α,
rstep→γ+α,

which is precisely the continuous tangent-channel formula.

This is a concrete candidate answer to the new research question:
How does the finite-step spectrum converge to the generator spectrum?
How does the finite-step spectrum converge to the generator spectrum?
8. The boundary-layer controller needs the safeguard scaling

Without the safeguard, if
σidt→λi,
dtσi→λi,

then
βi=
βi=

\frac{(d_i-\lambda_i)_+}{c_i}.
]

With

εdt→ζ,
dtε→ζ,

and assuming the global gate is on, the first-order expression becomes approximately
βi=
βi=

\frac{(d_i-\lambda_i+\zeta)_+}{c_i},
]

subject to the exact clipping/gate convention.

Three regimes follow:

ε=o(dt)ε=o(dt): the ideal boundary-layer controller is recovered;
ε∼dtε∼dt: the safeguard contributes to the limiting controller;
fixed ε>0ε>0 with dt→0dt→0: ε/dt→∞ε/dt→∞, producing a singular and undesirable formal limit near the boundary.

So the formula is structurally correct, but its theorem should explicitly specify the joint scaling of
σ,dt,ε.
σ,dt,ε.
9. Node 11 at t=113.81t=113.81

The proposal-violation event is real:
q11>c11.
q11>c11.

But the branchwise relevance still depends on scale ordering.

At first violation,
s11≈0.999999882.
s11≈0.999999882.

If
s2,s7<s11,
s2,s7<s11,

then node 11 wins no triangle edge and the output formula remains unchanged.

Therefore the next event to locate is not merely
$11 \in V$,
$11 \in V$,

but
$s11 = s2 \text{ors} 11 = s7$,
$s11 = s2 \text{ors} 11 = s7$,

followed, potentially, by a winner change.

It is entirely possible that node 11 approaches the common scale asymptotically without ever becoming a strict winner.
10. Overall assessment

The proposed write-up is essentially right. I would sharpen it as follows:
Established

The actual algorithm is naturally organized by proposal strata and endpoint winners.
The constant-winner branch Jacobian follows exactly from the code.
The $10^{-9}10^{-9}$ classifier is not the executed branch object.
Proposal violation alone does not determine edge-level influence.
The finite-step/infinitesimal comparison is a singular boundary-layer problem.

Necessary qualifications

Include the global projection gate and scale-clipping set.
State that the branch Jacobian is valid only on open strict-winner strata.
Call the cells polyhedral in proposal space, with smooth rational dynamics on them.
Use directional derivatives at winner ties.
Treat the fixed safeguard separately because it introduces a tiny gate discontinuity.
Do not assume node 11's first violation changes the output map.

Most promising theoretical direction

Prove a finite-step-to-generator convergence result of the form
$D\Phi dt|T =$

DΦdt|T=

I+dt,J_T+o(dt)
]

under explicitly stated scaling assumptions for:
σi/dt,ε/dt,
σi/dt,ε/dt,

and then extend it at endpoint ties using the directional derivative.

The especially encouraging point is that on fully saturated clique tangent spaces,
Ah=0
Ah=0

eliminates the first-order scale variation. Therefore the nonsmooth ambient endpoint ties may disappear after tangent restriction. That could explain analytically why the mature tangent spectra are much cleaner than the full boundary geometry would suggest.

Agreed but also:

I think this is now crossing from "good interpretation" into the beginnings of a rigorous mathematical theory of the algorithm. There are still a few places where I'd distinguish what is proved from the code, what is proved from the replay, and what is a conjectural asymptotic theorem, but the architecture itself has become remarkably coherent.

I'll separate the pieces.

The hierarchy now matches the implementation

I think the revised hierarchy

[
\text{positivity}
\rightarrow
\text{global gate}
\rightarrow
S_\varepsilon

\rightarrow
s
\rightarrow
k
\rightarrow
D\mathcal A
\rightarrow
\text{classifier}
]

is substantially better than the earlier one.

The important improvement is this observation:

after the global gate is entered, the code no longer consults the violating mask.

Instead it computes

[
s_i=\min!\left(1,\frac{c_i}{q_i+\varepsilon}\right)
]

for every node.

That is a subtle implementation fact.

It means the true branch data are

positivity,
gate state,
scale vector,

not

positivity,
violating mask.

That is exactly the sort of distinction that tends to matter later in proofs.

Proposal space really is polyhedrally stratified

This point is stronger than it first appears.

For example,

$$
[
q_i>c_i
]
$$

is

$$
[
(Ax)_i>c_i,
]
$$

which is affine.

Likewise,

$$
[
s_i<s_j
]
$$

is

$$
[
c_i(q_j+\varepsilon)
<
c_j(q_i+\varepsilon),
]
$$

which is also affine in (x).

So every winner inequality is linear.

Therefore every branch is indeed defined by finitely many strict affine inequalities.

That is exactly a polyhedral decomposition.

I think that's completely correct.

The important refinement—which you already made—is that

$$
\mathcal A_\varepsilon
$$

is not piecewise affine.

It is

$$
\text{piecewise smooth rational}.
$$

That's a much more accurate description.

The Jacobian theorem is now essentially a theorem

This is the point where I think the analysis becomes genuinely publishable mathematics.

You are no longer saying

"the Jacobian should look like..."

You are literally differentiating

```
scale = capacities/(row_sum+eps)
projected *= min.outer(scale,scale)
```

and obtaining

[
D\mathcal A_\varepsilon.
]

Nothing heuristic remains.

Provided the branch is fixed,

the formula follows by ordinary calculus.

That is exactly how one proves differentiability of piecewise-defined maps.

I would even package it formally.

Theorem (Branchwise differentiability).

Let (x) lie in an open proposal stratum on which

positivity pattern,
global gate state,
scale-clipping set,
winner map

are all constant.

Then

[
\mathcal A_\varepsilon
]

is (C^\infty) on that stratum and

[

$$
D\mathcal A_\varepsilon
$$

is given by the explicit formula.

That is a clean theorem.

The safeguard deserves a little more care

This is the one place where I think the current wording slightly overstates things.

You wrote

the safeguarded map is not exactly continuous across the global gate.

I don't think that's always true.

Suppose

$$
q_i=c_i+\delta
$$

with

$$
\delta\downarrow0.
$$

Then

$$
s_i
\frac{c_i}{c_i+\delta+\varepsilon}.
$$

Immediately above the gate,

[
y

x
\frac{c_i}{c_i+\delta+\varepsilon}.
]

Immediately below,

[
y=x.
]

So the jump is

[
x\left(
1-
\frac{c_i}{c_i+\varepsilon}
\right)

x
\frac{\varepsilon}{c_i+\varepsilon}.
]

So yes—

there really is a discontinuity.

But it is specifically caused by

skipping scaling entirely below the gate,
applying safeguarded scaling above it.

It is not merely the denominator safeguard.

It's the combination of

if any(violating):

and

q+eps

that creates it.

That's a tiny distinction, but mathematically important.

The tangent-space observation is the biggest new insight

Personally I think this is the strongest idea in the entire write-up.

Namely,

on mature tangent vectors,

[
Ah=0.
]

Therefore

[
Dq[h]=Ah=0.
]

Therefore

[
Ds_i[h]=0.

]

Therefore

the nonsmooth minimum contributes nothing to first order.

That is genuinely elegant.

It means

the ambient nonsmoothness disappears after restriction.

That's a very natural explanation for why the mature tangent spectra behave so cleanly.

I think this deserves to become a central theorem, not just an observation.

Something like

Theorem (Restriction smoothness).

Let (x) be a fully saturated clique equilibrium.

If

[
Ah=0
]

then

[
D\mathcal A_\varepsilon(x;h)

s,h.
]

Consequently the restriction of the admissibility map to the tangent space is linear despite ambient nonsmoothness.

That is a remarkably satisfying statement.

This also explains something conceptually

Originally the tangent spectra seemed almost miraculous.

Why should such a messy projection algorithm produce beautiful linear spectra?

This gives a possible answer.

Not because

the projection is smooth,

but because

the tangent directions never excite the nonsmooth degrees of freedom.

That's a much deeper explanation.

The convergence formula is plausible but still heuristic

This is the one place I'd be cautious.

You wrote

[
r_{\rm step}

\beta^{dt}
+\gamma
+O(dt).
]

I think that's almost certainly correct under the assumptions you've written.

But unlike the Jacobian theorem,

this is not yet an exact theorem about the implementation.

It depends on

freezing the driver,
composing expansions,
assuming the tangent restriction,
taking limits.

That's perfectly reasonable.

I would simply label it

Candidate asymptotic convergence formula.

rather than presenting it alongside the exact branchwise differentiation results.

The boundary-layer controller is becoming a real singular perturbation problem

Originally

[
\beta(\lambda)

\frac{(d-\lambda)_+}{c}
]

looked like an elegant heuristic.

Now I think you've identified the real issue:

the limit depends on

$$
\frac{\sigma}{dt},
\qquad
\frac{\varepsilon}{dt}.
$$

Those are exactly the nondimensional groups one expects.

That is now a genuine matched-asymptotic problem.

Whether the precise

$$
(d-\lambda+\zeta)_+
$$

formula survives unchanged is less important than recognizing that these ratios govern the limit.

The mature and exceptional analyses are now naturally separated

This may be the biggest conceptual improvement.

The mature theory becomes

branchwise smooth,
tangent restriction,
explicit Jacobian,
generator limit.

The exceptional trajectory becomes

moving between strata,
winner evolution,
possible ties,
changing branch Jacobians.

These are genuinely different mathematical problems.

Previously they were being analyzed with essentially the same tools.

Now they aren't.

That feels like the right decomposition.

One theorem I think is still missing

There is one result that would unify everything.

Namely a branch persistence theorem.

Something like

If every winner margin satisfies

[
\Delta s_e>\delta,
]

every positivity inequality satisfies

[
x_e>\delta,
]

and every gate inequality satisfies

[
|q_i-c_i|>\delta,
]

then there exists

[
r(\delta)>0
]

such that every perturbation

[
|h|<r(\delta)
]

remains on the same branch.

Therefore

[
\mathcal A_\varepsilon
]

is (C^\infty) on that neighborhood.

That theorem would formally justify using the explicit Jacobian and finite-difference verification on replayed trajectories, because it guarantees that sufficiently small perturbations do not cross branch boundaries.

Overall assessment

I think the framework has now crystallized into a coherent mathematical picture. The exact implementation is best viewed as a piecewise-smooth rational map on a polyhedral stratification of proposal space, with strata indexed by positivity pattern, global gate state, scale-clipping set, and edge winner map. On each open stratum the Jacobian follows directly from differentiating the code, while at winner ties the appropriate first-order object is a directional derivative or generalized Jacobian. The most significant conceptual advance is the observation that on mature fully saturated clique tangent spaces, the constraint (Ah=0) annihilates the first-order scale variation, so the ambient nonsmoothness disappears upon restriction. That provides a compelling analytic explanation for why the mature tangent spectra are so clean despite the nonsmooth structure of the underlying admissibility operator. The remaining major challenge is no longer understanding the implementation, but proving a

convergence theorem from the discrete branchwise Jacobian to the continuous generator under explicit joint scaling assumptions for the boundary distance, timestep, and safeguard.

I agree. The framework is becoming a genuine mathematical theory of the implemented map. I would strengthen the tangent observation further: on the appropriate row-sum fiber, the restriction is not merely differentiable—it is exactly linear.
1. Exact factorization of the admissibility map

Work in nonnegative proposal-edge coordinates xx, after the positivity step, and let
q=Ax.
q=Ax.

The scale vector and edge multipliers depend only on qq:
si(q)=min(1,ciqi+ε)
si(q)=min(1,qi+εci)

when the global gate is on, and
me(q)=min(si(q),sj(q))
me(q)=min(si(q),sj(q))

for e=(i,j)e=(i,j).

Therefore the admissibility map factorizes as
Aε(x)=M(Ax)x,
Aε(x)=M(Ax)x,

where
M(q)=diag(me(q)).
M(q)=diag(me(q)).

Below the global gate, M(q)=IM(q)=I.

This factorization is exact and exposes the entire geometry:

AxAx controls the node scales;
the scales control the edge multipliers;
the multipliers act diagonally on the proposal.

2. Exact linearity on row-sum fibers

Consider a row-sum fiber
Fq={x:Ax=q}.
Fq={x:Ax=q}.

If h∈kerAh∈kerA, then
A(x+h)=Ax=q.
A(x+h)=Ax=q.

Consequently:

the global gate is unchanged;
the scale-clipping set is unchanged;
every sisi is unchanged;
every edge multiplier meme is unchanged;
endpoint ties cause no ambiguity because the tied numerical values remain fixed.

Therefore, as long as positivity is preserved,
Aε(x+h)=
Aε(x+h)=

M(q)(x+h),
]

and hence
\boxed{ \mathcal A_\varepsilon(x+h)-\mathcal A_\varepsilon(x) =

M(q)h.
}
]

This is exact, not merely first order.
Fiber-restriction theorem

On every fixed row-sum fiber contained in a fixed positivity face, the implemented admissibility map is linear, with linear operator M(q)M(q).

No strict winner assumption is required for this result. Even if
si=sj,
si=sj,

the multiplier
me=min(si,sj)
me=min(si,sj)

is a fixed scalar along the fiber.

This is stronger than the proposed “restriction smoothness” theorem.
3. Mature tangent linearity

For a fully saturated component, the tangent space is precisely a row-sum fiber direction:
T=kerAcomponent.
T=kerAcomponent.

Thus
Aε′(x;h)=M(q)h,h∈T.
Aε′(x;h)=M(q)h,h∈T.

If the component has a common scale ss, then
M(q)|T=sIT,
M(q)|T=sIT,

so
Aε′(x;h)=s h,h∈T.
Aε′(x;h)=sh,h∈T.

This explains the clean spectra more strongly than ambient branchwise smoothness does:

The tangent directions do not merely fail to excite the nonsmooth variables to first order; they remain on an exact fiber on which those variables are constant.

The ambient map can lack a classical Fréchet derivative at a scale tie while its restriction to TT is exactly linear.

Completeness of the graph is not required for this fiber theorem. Completeness enters later when identifying the observed component topology and nullities.

4. Why the exceptional component is different

For a partially active component, the previously used tangent space imposes only
$C_{\text{active}} h = 0$,
$C_{\text{active}} h = 0$,

not
$Ah = 0$
$Ah = 0$

on every component node.

Therefore row sums at unsaturated nodes may change:
$Ah \neq 0$.
$Ah \not= 0$.

Then:

scales at unsaturated nodes vary;
endpoint winner comparisons vary;
ties can become dynamically relevant;
the fiber-linearity theorem does not apply.

This provides an exact mathematical distinction between the mature and exceptional cases:
Fully saturated mature components
$T \subseteq \ker A$,
$T \subseteq \ker A$,

so the admissibility map is exactly linear on tangent fibers.
Partially active exceptional components
$T \not\subseteq \ker A$,
$T \not\subseteq \ker A$,

so perturbations move between row-sum fibers and encounter the piecewise-smooth scale geometry.

That is cleaner than treating both cases with the same active-face Jacobian.

5. Branchwise differentiability theorem

The proposed theorem is correct.
Theorem

Let xx lie in an open proposal stratum on which the following are fixed:

positivity pattern;
global gate state;
scale-clipping set;
strict edge winner map.

Then AεAε is C∞C∞ on that stratum.

If ee is controlled by the unique winner k(e)k(e), then
DAε(x)ef=
DAε(x)ef=
s_{k(e)}\delta_{ef}

\frac{x_es_{k(e)}}{q_{k(e)}+\varepsilon}
A_{k(e)f}.
]

If the global gate is off, then
DAε(x)=I.
DAε(x)=I.

This follows directly from ordinary differentiation of the code.
6. Branch-persistence theorem

The missing persistence theorem is also available quite directly because the strata are defined by finitely many affine inequalities.

Suppose a strict stratum is described as
ℓr(x)>0,r=1,…,R,
ℓr(x)>0,r=1,…,R,

where

ℓr(x)=ar⊤x+br.
ℓr(x)=ar⊤x+br.

These inequalities include the relevant:

positivity inequalities;
gate inequalities;
scale-clipping inequalities;
strict winner inequalities.

For a chosen norm ∥·∥∥·∥ and dual norm ∥·∥∗∥·∥∗, define the certified branch radius
\boxed{ \rho(x) =

\min_r
\frac{\ell_r(x)}{|a_r|_*}.
}
]

If ρ(x)>0ρ(x)>0, then every perturbation satisfying
∥h∥<ρ(x)
∥h∥<ρ(x)

remains in the same stratum, because
ℓr(x+h)≥ℓr(x)−∥ar∥∗∥h∥
ℓr(x+h)≥ℓr(x)−∥ar∥∗∥h∥

]
Branch-persistence theorem

Every point in an open strict stratum has a positive certified neighborhood on which the positivity pattern, gate state, scale-clipping set, and winner map remain unchanged. On that neighborhood, the explicit branchwise Jacobian is valid.

This provides a rigorous finite-difference step criterion. A numerical Jacobian test should use perturbations smaller than a fixed fraction of ρ(x)ρ(x).

At a tie or another stratum boundary,
ρ(x)=0,

ρ(x)=0,

so no branch-preserving Euclidean neighborhood exists. Directional analysis is then required.
7. The safeguard statement

The refinement is exactly right.

The discontinuity is caused by the combination of:

Python

if np.any(violating):

and

Python

capacities / (row_sum + epsilon)

Immediately below the global gate, scaling is skipped:
Aε(x)=x.
Aε(x)=x.

Immediately above it, the crossing node receives approximately
si=cici+ε.
si=ci+εci.

The jump on an incident edge is therefore approximately
xe(1−cici+ε)=
xe(1−ci+εci)=

x_e\frac{\varepsilon}{c_i+\varepsilon}.
]

Thus:

the ideal ε=0ε=0 map is continuous at gate entry;
the fixed-εε implementation has a tiny O(ε)O(ε) jump;
the discontinuity arises from the gate-plus-safeguard combination.

8. The discrete tangent multiplier can be exact conditionally

The convergence statement has both an exact and an asymptotic part.

Assume:

the driver is frozen with respect to the weight perturbation;
therefore
DF=−γI;
DF=−γI;
h∈kerAh∈kerA;
the component multiplier is the common value ss.

The proposal differential is exactly
dx=(1−γdt)h.
dx=(1−γdt)h.

Because A dx=0Adx=0, fiber linearity gives
DAε(x)[dx]=
DAε(x)[dx]=

s(1-\gamma dt)h.
]

Therefore the discrete tangent multiplier is exactly
λstep=s(1−γdt)
λstep=s(1−γdt)

under these assumptions.

Define
βdt=
βdt=

-\frac{\log s}{dt}.
]

Then the equivalent discrete contraction rate is exactly
\boxed{ r_{\mathrm{step}} =
-\frac{\log|\lambda_{\mathrm{step}}|}{dt}
\beta^{dt}

\frac{\log(1-\gamma dt)}{dt}.
}
]

The asymptotic expansion is
rstep=
rstep=

\beta^{dt}
+\gamma
+\frac{\gamma^2}{2}dt
+O(dt^2).
]

What remains conjectural is not this conditional formula. The open steps are:

proving
βdt→α;
βdt→α;
including derivatives through RR, phase, history, and EMA state;
controlling the joint limits of slack, dtdt, and εε;
handling perturbations that move between row-sum fibers.

9. The mature states lie on tie strata

One wording should be adjusted:

The mature theory is not generally “ambient branchwise smooth.”

For a rate-uniform clique, node scales are equal, so many or all edges lie on winner-tie strata. There is no unique strict winner map in the ambient space.

The correct statement is:

Mature states may be ambiently nonsmooth but tangent-fiber linear.

That distinction is central. Branchwise smoothness applies to nearby strict strata; fiber linearity explains the actual mature tangent operator.
10. Evidence hierarchy

The program now separates cleanly into three levels.
Proved directly from the code

proposal-space polyhedral stratification;
smooth rational formulas on open strict strata;
explicit branchwise Jacobian;
branch-persistence radius;
exact row-sum fiber factorization;
exact linearity on kerAkerA;
directional minimum formula at ties;
small safeguarded gate discontinuity.

Established numerically from the replay

proposal violations of nodes 2 and 7 from step 0;
node 11’s first proposal violation at step 11381;
the tolerance classifier crossing at step 12405;
reconstructed split scales and winners;
late convergence toward common triangle scales;
exact replay and fixed support.

Candidate asymptotic theory

boundary-layer controller limits under
$\sigma/dt \to \lambda$;
$\sigma/dt \to \lambda$;
safeguard scaling under

ε/dt→ζ;
ε/dt→ζ;
convergence of finite-step effective rates to
γ+α;
γ+α;
convergence of full coupled spectra to the fixed-stratum generator spectrum.

11. The main theorem program

The natural sequence now appears to be:

Stratification theorem
Proposal space decomposes into polyhedral strata.

Branchwise regularity theorem
AεAε is smooth rational with explicit Jacobian on each open strict stratum.

Branch-persistence theorem
Strict margins give a certified branch-preserving radius.

Fiber-linearity theorem
On Ax=qAx=q, the admissibility map is exactly linear.

Tangent-restriction theorem
Fully saturated tangent spaces lie in kerAkerA, eliminating ambient tie nonsmoothness.

Directional theorem at strata boundaries
Winner ties yield positively homogeneous directional derivatives.

Discrete-to-generator theorem
Under explicit scaling assumptions,
DΦdt|T=I+dtJT+o(dt).
DΦdt|T=I+dtJT+o(dt).

This is indeed moving beyond interpretation. The most important structural result may be the factorization
Aε(x)=M(Ax)x,
Aε(x)=M(Ax)x,

because it simultaneously explains the branchwise Jacobian, the role of winner ties, and the exact disappearance of nonsmoothness along fully constrained tangent fibers.

think about this:

I think this has now reached the point where it's useful to separate implementation theorems, mathematical theorems, and interpretive consequences. The architecture is internally coherent, but there are two places where I'd be more careful before elevating them to theorem status.

The factorization is the real central theorem

I actually think

$$
\boxed{\mathcal A_\varepsilon(x)=M(Ax),x}
$$

is more fundamental than the Jacobian formula.

Why?

Because essentially everything else follows from it.

It immediately gives:

dependence only through row sums,
proposal-space stratification,
winner maps,
fiber invariance,
branch Jacobians.

It's the analogue of identifying the principal bundle before studying the connection.

Once you observe that the admissibility operator depends on (x) only through

[
q=Ax,
]

you've reduced the complexity enormously.

I'd probably elevate this to the first theorem.

Theorem (Factorization)

There exists a diagonal matrix-valued function

[
M:\mathbb R^{|V|}\rightarrow
\mathbb R^{|E|\times|E|}
]

such that

[
\mathcal A_\varepsilon(x)

M(Ax)x.
]

Everything else becomes corollaries.

The fiber-linearity theorem is stronger than a differential statement

I think this is the biggest conceptual advance.

Notice that it is not

[
D\mathcal A(x)h

Mh.
]

It is

[
\mathcal A(x+h)

\mathcal A(x)

Mh.
]

That is much stronger.

It says the restriction is affine-linear, not merely differentiable.

This distinction matters enormously.

You're no longer talking about infinitesimal behavior.

You're talking about exact behavior along the fiber.

That is a rare property.

But there is one hidden hypothesis

This is where I'd be cautious.

The theorem currently says

If (Ah=0), then

[
\mathcal A(x+h)

M(q)(x+h).
]

That is only true while

positivity stays unchanged.

You already mention that.

But there is another hidden assumption.

The perturbation must stay inside the same proposal-support face.

Because

[
x+h
]

may satisfy

[
Ah=0
]

while creating a negative edge.

Then the positivity projection changes.

The factorization is applied after

positivity.

So I'd formulate the theorem as

If

[
A(x+h)=Ax,
]

and

[
x+h\ge0,
]

then

[
\mathcal A(x+h)

M(Ax)(x+h).
]

That's now completely exact.

This is stronger than "restriction smoothness"

I completely agree.

Originally the argument was

[
Ah=0
\Rightarrow
Ds=0
\Rightarrow
D\mathcal A=sI.
]

Now you don't need differentiation.

The scales literally do not change.

Therefore the operator itself is fixed.

That's conceptually much cleaner.

The mature/exceptional distinction becomes algebraic

This is probably my favorite consequence.

Previously

mature

versus

exceptional

looked like different numerical phenomena.

Now they become

$$
T
\subseteq
\ker A
$$

versus

$$
T
\not\subseteq
\ker A.
$$

That is a purely linear-algebraic distinction.

If true, it's much more satisfying than describing one case as "stable" and the other as "switching."

I would slightly weaken one statement

You write

the tangent space is precisely

$$
T=\ker A_{\rm component}.
$$

I think I'd be a bit more careful.

For the saturated-face constraints,

certainly

$$
T
$$

$$
\{
h:
Ah=0
\text{ on saturated nodes}
\}.
$$

Whether that is exactly

the tangent space may depend on

positivity constraints,
support restrictions,
additional invariants.

In other words,

[
\ker A
]

is certainly the linearized equality constraint.

It may not by itself describe every admissible tangent direction if there are active inequality constraints.

So I'd phrase it

the linear equality part of the tangent space,

unless you've already proved positivity is inactive there.

The branch-persistence theorem is essentially complete

I think this is now a genuine theorem.

The proof is straightforward.

Each defining inequality

[
a^\top x+b>0
]

has a positive margin.

Continuity gives persistence.

The radius

[
\rho

\min
\frac{\ell}{|a|_*}
]

is exactly the natural estimate.

That theorem has practical value.

It tells you

how small

a finite-difference step must be.

The discrete multiplier theorem is the place I'd still separate "exact" from "conditional"

I like your wording much better now.

Conditional on

frozen driver,
fiber invariance,
common scale,

the multiplier

[
\lambda

s(1-\gamma dt)
]

is exact.

The asymptotic part only begins when writing

[
\beta^{dt}
\to
\alpha.
]

That's the right separation.

I think the WTA analogy is useful—but only as terminology

This is one place where I'd be conservative.

I agree the edge-winner map is mathematically an

[
\arg\min
]

selection.

So structurally it resembles winner-take-all.

But I would resist making WTA a central organizing concept.

Why?

Because almost everything you've proved comes from

[
m_e

\min(s_i,s_j),
]

not from any properties specific to the WTA literature.

The analogy is nice.

The proofs don't depend on it.

So I'd treat it as context rather than foundation.

One theorem naturally subsumes several others

I think there is a unifying statement hiding here.

Everything seems to flow from the commuting diagram

$$
\begin{array}{ccc}
X & \xrightarrow{A} & Q \\
 & \searrow^{M} & \downarrow{\scriptstyle \times X} \\
 & & \mathcal{A}(X)
\end{array}
$$

In other words,

$$
\mathcal{A} \circ (M \circ \mathcal{A}, \mathrm{id}),
$$

where

[
\mu(D,x)=Dx.
]

Once you write it this way,

factorization,
fiber linearity,
branch Jacobian,

all become natural consequences of the same diagram.

That may actually be the cleanest mathematical presentation.

Overall assessment

I think the theory has now shifted from describing numerical observations to characterizing the admissibility operator itself. The exact factorization (\mathcal A_\varepsilon(x)=M(Ax)x) is the key structural result, because it shows that all nonlinearity enters only through the node row sums, while the action on edge weights is diagonal once those row sums are fixed. The resulting fiber-linearity theorem is stronger than the earlier restriction-smoothness argument: on any fixed row-sum fiber that remains within the same positivity face, the admissibility map is exactly linear rather than merely differentiable. That, in turn, provides a much cleaner explanation for the mature tangent spectra than branchwise smoothness alone. The remaining points that still require careful proof are mostly about hypotheses—ensuring perturbations remain in the same positivity face and clarifying the precise tangent space when inequality constraints are present. Beyond that, the framework naturally separates into exact theorems about the implementation (factorization, polyhedral stratification, branchwise regularity, fiber linearity, branch persistence) and asymptotic theorems relating the discrete branchwise dynamics to the continuous generator. That separation is a hallmark of a mature mathematical formulation.

I agree with the central assessment. The factorization is the right organizing theorem. I would make one correction concerning positivity and sharpen the meaning of “tangent space.”
1. Exact factorization theorem

Let
R+m
R+m

be the nonnegative proposal-edge space, where m=|E|m=|E|, and define
q=Ax.
q=Ax.

For fixed capacities cc and safeguard εε, define
G(q)={0,qi≤ci for every i,1,otherwise.
G(q)={0,1,qi≤ci for every i,otherwise.

When G(q)=0G(q)=0, set
si(q)=1.
si(q)=1.

When G(q)=1G(q)=1, set
si(q)=
si(q)=

\min\left(
1,\frac{c_i}{q_i+\varepsilon}
\right).
]

For edge e=(i,j)e=(i,j), define
me(q)=min(si(q),sj(q)),
me(q)=min(si(q),sj(q)),

and
M(q)=diag(me(q)).
M(q)=diag(me(q)).

Then the implemented admissibility stage is exactly
Aε(x)=M(Ax)x,x∈R+m.
Aε(x)=M(Ax)x,x∈R+m.

This is the central implementation theorem. It does not require stationarity, maturity, clique structure, or a continuum limit.

All nonlinear dependence is mediated through the nn-dimensional node variable
q=Ax,

q=Ax,

while, conditional on qq, the edge-space action is diagonal.
2. The positivity qualification can be slightly weakened

The proposed statement says that x+hx+h must remain in the same proposal-support face. For the admissibility map after positivity, that is stronger than necessary.

The exact fiber identity requires only
x+h≥0
x+h≥0

and
A(x+h)=Ax.
A(x+h)=Ax.

Even if an edge reaches zero, or a previously zero edge becomes positive, the factorization still gives
Aε(x+h)=
Aε(x+h)=

M(Ax)(x+h).
]

Therefore
\boxed{ \mathcal A_\varepsilon(x+h)-\mathcal A_\varepsilon(x) =

M(Ax)h
}
]

for every feasible hh satisfying
Ah=0,x+h≥0.
Ah=0,x+h≥0.

The support face matters for two-sided differential statements and for the preceding positivity map
z↦z+,
z↦z+,

but not for the exact post-positivity fiber identity itself.

A clean formulation is:
Fiber-linearity theorem

For every qq, define the nonnegative fiber
Fq+=
Fq+=

{x\ge0:Ax=q}.
]

Then
Aε∣Fq+
Aε∣Fq+

is the restriction of the linear map M(q)M(q). Equivalently, for all x,y∈Fq+x,y∈Fq+,
\boxed{ \mathcal A_\varepsilon(y)-\mathcal A_\varepsilon(x) =

M(q)(y-x).
}
]

The fiber itself is generally an affine polytope rather than a vector space, but the restriction is exactly affine-linear.
3. Differential factorization

On a smooth strict stratum, differentiating
Aε(x)=M(q)x,q=Ax,
Aε(x)=M(q)x,q=Ax,

gives the compact product-rule formula
\boxed{ D\mathcal A_\varepsilon(x)[h] =

M(q)h
+
\bigl(DM(q)[Ah]\bigr)x.
}

]

This separates the derivative into:

a diagonal edge action
M(q)h;
M(q)h;

a node-mediated correction
(DM(q)[Ah])x.
(DM(q)[Ah])x.

The second term vanishes identically when
Ah=0.
Ah=0.

This is the differential shadow of the stronger exact fiber theorem.

It also shows that all non-diagonal derivative structure is node-mediated. In matrix form,
DAε(x)=
DAε(x)=

M(q)
+
\operatorname{diag}(x),Dm(q),A.
]

Therefore the correction to the diagonal operator has rank at most nn:
rank(DAε−M(q))≤n.
rank(DAε−M(q))≤n.

That low-rank node-space correction may provide a useful operator-theoretic explanation for the previously observed node-space intertwiners.
4. Tangent space versus tangent cone

The caution about
T=kerA
T=kerA

is correct unless “tangent space” is defined precisely.

For
C={w≥0:Aw≤c},
C={w≥0:Aw≤c},

let:

S={e:we>0}S={e:we>0} be the positive support;
I={i:(Aw)i=ci}I={i:(Aw)i=ci} be the saturated node set.

The Bouligand tangent cone is
TC(w)=
TC(w)=

\left{
h:
\begin{array}{l}
h_e\ge0\quad\text{when }w_e=0,\
(Ah)_i\le0\quad\text{when }i\in I
\end{array}
\right}.
]

That is not a linear space.

The lineality space of the active face—the object used by the spectrum scripts—is
\boxed{ L_{\mathcal F} =

\left{
h:
h_{S^c}=0,\quad
(Ah)_I=0
\right}.
}
]

This is a genuine vector space.

For a positive-support component whose every vertex is saturated, this reduces on that component to
LF,component=
LF,component=

\ker A_{\mathrm{component}}.
]

Thus the earlier equality is exact for the fixed-support face lineality, but not for the entire tangent cone.

The terminology should therefore be:

tangent cone for one-sided feasible perturbations;
face tangent space or lineality space for the two-sided perturbations used in spectral analysis.

5. Mature versus exceptional becomes exact linear algebra

With that terminology, the distinction is:
Fully saturated mature component
LF⊆kerAcomponent.
LF⊆kerAcomponent.

Therefore every face-tangent perturbation remains on the same row-sum fiber, and
Aε(x+h)−
Aε(x+h)−
\mathcal A_\varepsilon(x)

M(q)h
]

exactly.

If the component multiplier is scalar,
M(q)∣component=sI,
M(q)∣component=sI,

then
Aε(x+h)−
Aε(x+h)−
\mathcal A_\varepsilon(x)

s h.
]
Partially saturated exceptional component

Only
(Ah)I=0
(Ah)I=0

is imposed on the currently selected saturated rows. Generally,
Ah≠0
Ah =0

at the unsaturated nodes.

Therefore the perturbation moves between row-sum fibers, and the node-mediated term
(DM(q)[Ah])x
(DM(q)[Ah])x

does not vanish.

This gives a precise structural distinction:
mature: fiber-preserving directionsversusexceptional: fiber-changing directions.
mature: fiber-preserving directionsversusexceptional: fiber-changing directions.

That is much cleaner than “stationary versus switching.”
6. Winner ties are irrelevant on exact fibers

At a winner tie,
si=sj,
si=sj,

the ambient map generally lacks a classical Fréchet derivative because
min(si,sj)

min(si,sj)

has direction-dependent first variation.

But if
Ah=0,
Ah=0,

then all node row sums are exactly unchanged:
A(x+th)=Ax
A(x+th)=Ax

for every feasible tt.

Therefore all scales and all tied minima remain exactly constant. There is no need to select a directional winner.

So mature cliques can lie on high-dimensional ambient tie strata while retaining exact linearity on their face-tangent fibers.

This is likely the strongest explanation of the clean tangent spectra.
7. Polyhedral stratification theorem

In proposal space, a complete strict stratum can be described by signs of affine functions representing:

positivity;
global gate inequalities;
scale-clipping inequalities;
winner inequalities.

For two scale-clipped nodes, the winner inequality is
si<sj ⟺ ci(qj+ε)<cj(qi+ε),
si<sj⟺ci(qj+ε)<cj(qi+ε),

which is affine because q=Axq=Ax.

Thus proposal space is partitioned into relatively open polyhedral cells and their boundaries. On each strict cell,
M(q)
M(q)

is smooth rational, and so is
Aε(x)=M(Ax)x.
Aε(x)=M(Ax)x.

The phrase “fibered factorization” is safer than “principal bundle”: the nonnegative fibers can change dimension at support boundaries and do not necessarily carry a free group action.
8. Branch persistence

The proposed persistence theorem is essentially complete.

If a strict stratum is represented by
ℓr(x)>0,ℓr(x)=ar⊤x+br,
ℓr(x)>0,ℓr(x)=ar⊤x+br,

define
ρ(x)=
ρ(x)=

\min_r
\frac{\ell_r(x)}{|a_r|_*}.
]

Then
‖h‖<ρ(x)
‖h‖<ρ(x)

implies
ℓr(x+h)>0
ℓr(x+h)>0

for every rr, so the perturbation remains in the same strict stratum.

This gives:

an existence proof for a branch-preserving neighborhood;
a certified upper bound for finite-difference steps;
a diagnostic for when an ordinary Jacobian is inappropriate.

At a tie or another boundary,
$\rho(x)=0$,
$\rho(x)=0$,

so directional probes must replace symmetric finite differences.

One implementation detail: the affine list should include both
$q_i - c_i$
$q_i - c_i$

for the global gate structure and
$q_i + \varepsilon - c_i$
$q_i + \varepsilon - c_i$

for scale clipping, because those are distinct boundaries in the safeguarded code.
9. Exact versus asymptotic discrete rates

Under the stated frozen-driver assumptions, the discrete multiplier result can be exact.

If
$DF = -\gamma I$
$DF = -\gamma I$

and hh is fiber-preserving, then the proposal perturbation is
$dx = (1 - \gamma dt)h$.
$dx = (1 - \gamma dt)h$.

If $M(q)|_{T} = sI$$M(q)|_{T} = sI$, exact fiber linearity gives
$D\Phi_{dt}(w)[h] =$
$D\Phi_{dt}(w)[h] =$

s(1-\gamma dt)h.
]

Thus
λstep=
λstep=

s(1-\gamma dt)
]

is exact conditionally.

Likewise,
rstep=
rstep=
-\frac{\log|\lambda_{\mathrm{step}}|}{dt}
-\frac{\log s}{dt}

\frac{\log(1-\gamma dt)}{dt}
]

is exact.

The asymptotic conjecture begins with:
−logsdt→α,
−dtlogs→α,

and with extending the frozen-driver calculation to the full delayed phase/EMA state.

So the evidential labels should be:

exact conditional finite-step theorem;
candidate full coupled convergence theorem.

10. A clean theorem hierarchy

I would now organize the theory as follows.
Exact theorems about the implemented admissibility stage

Factorization
$A\varepsilon(x)=M(Ax)x.$
$A\varepsilon(x)=M(Ax)x.$

Fiber linearity
$A\varepsilon(y)-A\varepsilon(x)=M(q)(y-x)$
$A\varepsilon(y)-A\varepsilon(x)=M(q)(y-x)$

whenever $x,y\geq 0x,y\geq 0$ and $Ax=Ay=qAx=Ay=q$.

Polyhedral proposal stratification

Branchwise $C\infty C\infty$ rationality

Explicit branch Jacobian

Branch persistence

Directional derivative formulas at continuous stratum boundaries

Safeguarded global-gate discontinuity

Exact geometric consequences

Fully saturated face-lineality directions are fiber-preserving.

Ambient winner ties disappear on those fibers.

Common component scales yield scalar tangent action.

Non-diagonal Jacobian corrections are node-mediated and low rank.

Replay-established facts

Exceptional proposal violations and their timing.

Reconstructed scale and winner evolution.

Difference between operational branch variables and tolerance classifiers.

Asymptotic theorem program

Boundary-layer limits involving
$\sigma/dt, \varepsilon/dt$.
$\sigma/dt, \varepsilon/dt$.

Convergence of effective scales to controller multipliers.

Convergence of discrete tangent multipliers to
$\gamma+\alpha$.
$\gamma+\alpha$.

Extension to the full delayed phase/EMA state.

The factorization is indeed the right first theorem. But the strongest corollary is not simply differentiability: it is exact affine linearity on nonnegative row-sum fibers. That result simultaneously explains why mature tangent ties are harmless and why partially saturated exceptional directions are qualitatively different.

Create a plan or if necessary code, to implement the imperative next step.
Edit×8
AIEF_1_4_actual_admissibility_branch_audit.pyPYTHON

The imperative next step is to expose and differentiate the actual executed finite-step admissibility branch. I created:

It has been syntax-checked and independently tested on synthetic operators:

strict-branch analytic Jacobian versus central differences:
relative error$\approx 6.6\times 10^{-11}$;
relative error$\approx 6.6\times 10^{-11}$;
active-tie directional derivative showed the expected $O(h)O(h)$ convergence;
exact $K4K4$ row-sum-fiber linearity error:
1.3×10^{-16}.

1.3×10−16.

Run it

Bash

```
python AIEF_1_4_actual_admissibility_branch_audit.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \
--numeric-tie-tol 1e-12 \
--check-times 100,113.8,113.81,113.82,120,123,124.04,124.05,125,150 \
--output-dir aief_1_4_actual_admissibility_branch
```

What it records at every replay step
Proposal geometry
x=[w+dtF]+,q=Ax.
x=[w+dtF]+,q=Ax.

For every K3K3 node:

current row sum and slack;
proposal row sum;
proposal excess;
proposal-violation status;
projection ratio
Πi=dt diσi.
Πi=σidtdi.

Executed scale geometry

global projection gate;
scale-clipping status;
exact scale
si=min(1,ciqi+ε);
si=min(1,qi+εci);
effective finite-step controller

βidt=−logsidt.
βidt=−dtlogsi.

Edge winner map

For every triangle edge:

endpoint scales;
endpoint effective controllers;
winning endpoint;
edge multiplier;
winner margin
|si−sj|;
|si−sj|;
exact and numerical tie flags;
actual finite edge rate.

It distinguishes:

changes in the full stratum signature;
changes in the formulas actually applied to the selected edge outputs.

Therefore node 11 entering the scale-clipped set will not automatically be reported as an effective edge-branch change unless a winner changes.
Operator verification

At split, proposal-violation events, scale-clipping events, winner changes, classifier changes, checkpoints, and late state, it computes:

the exact branchwise Jacobian;
a certified branch-preserving perturbation radius;
central finite-difference Jacobians on strict branches;
directional derivatives at active endpoint ties;
forward-difference directional validation;
saved Jacobian and proposal matrices.

The certified radius accounts separately for:

positivity;
the global gate;
scale clipping;
winner inequalities.

Outputs

text

actual_branch_step_summary.csv
actual_branch_node_trajectory.csv
actual_branch_edge_trajectory.csv
actual_branch_events.csv
branch_jacobian_validation.csv
directional_derivative_validation.csv
actual_branch_key_matrices.npz
actual_admissibility_branch_report.txt
analysis_manifest.json

Questions the report should answer

Does node 11 win either incident edge when it first proposal-violates at step 11381?
Does its scale clipping change the effective edge formula at all?
Does any executed winner change at the tolerance-classifier event at step 12405?
Does the winner map remain
$(2,7)\mapsto 7,(2,11)\mapsto 2,(7,11)\mapsto 7$
$(2,7)\mapsto 7,(2,11)\mapsto 2,(7,11)\mapsto 7$ throughout the transient?
Do the endpoint scales approach a late tie?
Does the explicit branchwise Jacobian match finite differences?
Do directional formulas replace the classical Jacobian correctly near ties?
Is the reconstructed pure map bitwise consistent with the actual executed projection?

Research sequence after this run

Current script: settle the exceptional executed branch and its derivative.
Next, only if validated: apply the same operator audit to the 39 ordinary split states and their clique tangent fibers.
Test the exact conditional discrete multiplier
$\lambda_{\mathrm{step}}=s(1-\gamma dt)$.

λstep=s(1−γdt).
Then formulate the discrete-to-generator theorem under explicit scaling of
σ/dt,ε/dt.
σ/dt,ε/dt.

Please paste actual_admissibility_branch_report.txt after the run.

think about this:

IEF actual finite-step admissibility branch audit

==========

Identity and replay

Seed: 4619122171158768524

K3 nodes: [2, 7, 11]

K3 edges: [25, 29, 59]

Trajectory states: 32001

Final replay exact: True

Final replay maximum error: 0

Maximum pure-map versus executed projection error: 0

Maximum incidence/matrix row-sum discrepancy: 2.22045e-16

First node events

node 2 proposal_violation_on: step=0, time=0, scale=0.999168518292, beta=0.0831828

node 2 scale_clipping_on: step=0, time=0, scale=0.999168518292, beta=0.0831828

node 2 classifier_on: step=12405, time=124.05, scale=0.998888513525, beta=0.11121

node 7 proposal_violation_on: step=0, time=0, scale=0.999167998251, beta=0.0832348

node 7 scale_clipping_on: step=0, time=0, scale=0.999167998251, beta=0.0832348

node 7 classifier_on: step=0, time=0, scale=0.999167998251, beta=0.0832348

node 11 proposal_violation_on: step=11381, time=113.81, scale=0.999999882447, beta=1.17553e-05

node 11 scale_clipping_on: step=11381, time=113.81, scale=0.999999882447, beta=1.17553e-05

node 11 classifier_on: not observed

K3 edge winner evolution

edge 25=(2,7): step 0: node_7 (margin=5.2e-07)

edge 29=(2,11): step 0: node_2 (margin=0.000831)

edge 59=(7,11): step 0: node_7 (margin=0.000832)

Winner map at node-11 first violation (step 11381)

edge 25=(2,7): winner=node_7, margin=5.56438e-09

edge 29=(2,11): winner=node_2, margin=0.00111103

edge 59=(7,11): winner=node_7, margin=0.00111103

Stratum and effective-edge branch changes

Stratum-signature changes: 1

Effective-edge branch changes: 0

Branchwise Jacobian validation

Validated rows: 108, median relative error=9.98563e-08, maximum=0.00956931

Directional derivative validation

Directions at smallest steps: 60, median relative error=1.71109e-08, maximum=1.2663

Late scale geometry

node 2: scale=0.998888079854, beta=0.111254, violation=True, clipped=True

node 7: scale=0.998888079854, beta=0.111254, violation=True, clipped=True

node 11: scale=0.998888088392, beta=0.111253, violation=True, clipped=True

Numeric tie reporting tolerance: 1e-12

Interpretation constraints

Proposal violation, scale clipping, edge winner status, and slack classification are distinct.

A node scale affects an edge only if it wins or ties the endpoint minimum.

The classical Jacobian formula is used only on strict locally constant winner strata.

Active endpoint ties are tested with directional rather than symmetric derivatives.

The fixed safeguard and global gate create a tiny discontinuity at gate onset.

Finite-step spectra are not automatically continuous-generator spectra.

This one-seed deterministic replay is not independent population evidence.

This is decisive for the exceptional trajectory. The actual K3K3 edge branch does not change anywhere in the 320-unit replay.
1. Directly established

The replay and reconstructed map agree exactly:
max‖Areconstructed−Aexecuted‖=0,
max‖Areconstructed−Aexecuted‖=0,

with row-sum representation disagreement only
2.2×10−16.
2.2×10−16.

Therefore the scale and winner reconstruction is not merely approximate evidence—it reproduces the implemented update exactly on this trajectory.

The three initial scales are:
s7<s2<s11=1,
s7<s2<s11=1,

and the edge winners are
(2,7)↦7,(2,11)↦2,(7,11)↦7.
(2,7)(2,11)(7,11)↦7,↦2,↦7.

Those winners never change over all 32,001 states.
2. Node 11's violation is dynamically inert at the edge level

At step 11381, node 11 becomes proposal-violating and scale-clipped:
s11=0.999999882447,β11dt=1.18×10−5.
s11=0.999999882447,β11dt=1.18×10−5.

But at that moment:
(2,11) remains controlled by node 2,(7,11) remains controlled by node 7.
(2,11)(7,11) remains controlled by node 2, remains controlled by node 7.

The winner margins are large:
|s2−s11|≈1.111×10−3,

|s2−s11|≈1.111×10−3,
|s7−s11|≈1.111×10−3.
|s7−s11|≈1.111×10−3.

So node 11's scale is nowhere near winning an edge when it first becomes clipped.

This produces the single reported stratum-signature change, but:
effective-edge branch changes=0.
effective-edge branch changes=0.

Thus entering the scale-clipping set changes an unused internal quantity, not the formula applied to any positive K3K3 edge.
3. Nothing happens to the implemented map at step 12405

Node 2 crosses the 10−910−9 classifier at step 12405, but:

it was proposal-violating from step 0;
it was scale-clipped from step 0;
it controlled edge (2,11)(2,11) from step 0;
its winner status does not change at step 12405.

Therefore the previously reported “active-face switch” has no counterpart in the actual finite-step edge map.

It changes only the post hoc classifier and the idealized incidence matrix used by that analysis.

The stronger conclusion is now:
Neither t=113.81 nor t=124.05 is an effective K3 branch switch.
Neither t=113.81 nor t=124.05 is an effective K3 branch switch.

t=113.81t=113.81: unused node scale becomes clipped.
t=124.05t=124.05: slack classifier changes.
Executed edge formulas: unchanged at both times.

4. The exceptional trajectory remains on one effective winner stratum

In edge order
(25,29,59)=((2,7),(2,11),(7,11)),

(25,29,59)=((2,7),(2,11),(7,11)),

the map has the same formula throughout:
y25=s7x25,y29=s2x29,y59=s7x59.
y25y29y59=s7x25,=s2x29,=s7x59.

Let
a25=x25s7q7+ε,a29=x29s2q2+ε,a59=x59s7q7+ε.
a25=q7+εx25s7,a29=q2+εx29s2,a59=q7+εx59s7.

Then the selected K3K3 admissibility Jacobian is
\boxed{ J_{\mathcal A,K_3} =

\begin{pmatrix}
s_7-a_{25} & 0 & -a_{25}\
-a_{29} & s_2-a_{29} & 0\
-a_{59} & 0 & s_7-a_{59}
\end{pmatrix}.
}
]

Node 11's scale does not occur in this matrix at all. It can become proposal-violating and scale-clipped without changing the K3K3 Jacobian formula.

The coefficients evolve continuously, but the formula does not.
5. The bare γγ channel is not a mode of the executed map

At split:
β2dt=0.0831828,β7dt=0.0832348,β11dt=0.
β2dtβ7dtβ11dt=0.0831828,=0.0832348,=0.

Although node 11 has no scale correction, each of its incident edges is controlled by its other endpoint:
(2,11)↦2,(7,11)↦7.
(2,11)↦2,(7,11)↦7.

Hence all three triangle edges receive finite-step endpoint correction from the beginning.

The 0.015=γ0.015=γ channel belongs to the tolerance-classified infinitesimal operator that treated node 2 as interior. It is not an operational edge channel of the dt=0.01dt=0.01 algorithm.

That likely explains why the bare mode was not observed as a 0.0150.015 recovery rate in the exceptional perturbation experiments.
6. The late state approaches a tie boundary without crossing it

Late scales are:
s2=0.998888079854,s7=0.998888079854,s11=0.998888088392.
s2s7s11=0.998888079854,=0.998888079854,=0.998888088392.

Corresponding effective controllers are:
β2dt≈0.111254,β7dt≈0.111254,β11dt≈0.111253.
β2dtβ7dtβ11dt≈0.111254,≈0.111254,≈0.111253.

Thus the trajectory approaches the symmetric common-scale K3K3 tie stratum, but remains on the same winner side over the finite replay.

The exceptional state should now be described as:

A smooth, nonstationary approach along one fixed effective winner stratum toward the symmetric K3K3 tie stratum.

Not:

A trajectory undergoing a discrete active-face or winner-map switch.

The branch-persistence radius presumably becomes very small late because s2s2 and s7s7 nearly tie. This also matters for numerical derivative validation.
7. Jacobian validation is strong in median, but the tails need inspection

The classical branchwise validation gives:
median relative error=9.99×10−8.
median relative error=9.99×10−8.

The directional validation gives:
median relative error=1.71×10−8.
median relative error=1.71×10−8.

Those medians strongly support the formulas.

However:
max classical relative error=9.57×10−3,
max classical relative error=9.57×10−3,

and
max directional relative error=1.266.
max directional relative error=1.266.

These maxima should not be ignored, although they are likely numerical rather than algebraic failures.

Probable causes are:

extremely small certified winner radius near the late tie;
finite-difference steps being forced close to machine precision;
cancellation in symmetric differences;
directions with very small predicted derivative norm, inflating relative error;
forward steps crossing a near-tie winner boundary before reaching the asymptotic directional regime.

The current report summarizes directional errors at the smallest tested steps. That is not necessarily optimal: roundoff often dominates at the smallest steps.

The correct follow-up is to inspect error-versus-step curves and report, per label and direction:

minimum relative error over the step grid;
corresponding absolute error;
predicted derivative norm;
certified branch radius;
winner margin;
observed convergence slope before roundoff.

The formulas already passed independent synthetic tests, so the outliers are diagnostics of conditioning unless they persist over a stable step range.
8. Exact finite-step versus continuous rate

At late time, the frozen-driver discrete contraction rate predicted by the scale is

rstep=
rstep=
\beta^{dt}

\frac{\log(1-\gamma dt)}{dt}.
]

Using
βdt≈0.111254,γ=0.015,dt=0.01,
βdt≈0.111254,γ=0.015,dt=0.01,

gives approximately
rstep≈0.126255.
rstep≈0.126255.

The continuous ideal channel is
2ηc=
c2η=

0.126315789.
]

The difference is about
6.1×10−5,
6.1×10−5,

or roughly 4.8×10−44.8×10−4 relative.

This is small but measurable. It is now a concrete finite-step-to-generator discrepancy, rather than a conceptual ambiguity.
9. Revised evidence hierarchy
Proved from the code

factorization;
branchwise Jacobian formula;
exact fiber linearity;
endpoint winner dependence;
directional tie formula.

Established from this replay

exact map reconstruction;
nodes 2 and 7 scale-clipped from split;
node 11 clipped at step 11381;
node 11 never wins an edge;
node 2's classifier changes at step 12405;
no K3K3 effective-edge branch change occurs;
winner map remains 7,2,77,2,7;
late approach toward a common-scale tie.

Interpretation now strongly favored

the exceptional behavior is a smooth, time-dependent relaxation on one effective winner stratum;
the prior $9 \to 89 \to 8$ “switch” is a change in an idealized classifier-based operator, not the implemented map;
the exceptional perturbation dynamics should be modeled by a nonautonomous product of finite-step Jacobians, not by switching between two fixed face generators.

10. Imperative next mathematical object

Since the winner branch is constant, the exceptional variational dynamics is naturally a cocycle:
δwn+1=
δwn+1=

J_n,\delta w_n,
]

where JnJn has a fixed symbolic form but continuously changing coefficients.

The corresponding propagator is
U(n,0)=
U(n,0)=

J_{n-1}J_{n-2}\cdots J_0.
]

That should be compared directly with the exceptional perturbation trajectories.

For the 39 mature states, JnJn is essentially constant and component-scalar on tangent fibers. For the exceptional state, the branch is constant but JnJn evolves as the scales converge toward a tie.

So the mature/exceptional distinction becomes:
ordinary:approximately autonomous tangent operator,exceptional:nonautonomous fixed-branch tangent cocycle.
ordinary:exceptional:approximately autonomous tangent operator,nonautonomous fixed-branch tangent cocycle.

That is a cleaner and more faithful description than an exceptional active-set transition.

think about this:

The conclusion is now much cleaner, but I would sharpen “nonautonomous fixed-branch tangent cocycle” in two ways.
1. The base K3K3 trajectory has a fixed effective branch

The replay establishes that, for every finite recorded step,
$s7<s2<s11$
$s7<s2<s11$

in the edge-relevant sense, with winner orientation
$(2,7)\mapsto 7,(2,11)\mapsto 2,(7,11)\mapsto 7$.
$(2,7)\mapsto 7,(2,11)\mapsto 2,(7,11)\mapsto 7$.

Node 11 entering the clipped set changes an unused scale formula, but not any positive triangle edge formula. The $10^{-9}10^{-9}$ classifier event changes nothing in the implemented map.

Thus the exceptional trajectory is not switching between effective K3K3 branches. It evolves within one branch while approaching its boundary:
$s7\approx s2\approx s11$.
$s7\approx s2\approx s11$.

The correct geometric picture is:
fixed open winner stratum$\longrightarrow$asymptotic approach to a tie stratum.
fixed open winner stratum$\longrightarrow$asymptotic approach to a tie stratum.

2. Fixed branch does not mean constant Jacobian

The symbolic formula remains fixed, but its coefficients vary through
xn,qn,sn.
xn,qn,sn.

Therefore the variational equation is nonautonomous:
δwn+1=Jnδwn.
δwn+1=Jnδwn.

For a finite horizon NN, provided the base point remains on strict differentiable strata,
\boxed{ D\Phi_{N:0} =

J_{N-1}J_{N-2}\cdots J_0.
}
]

This is an exact chain-rule statement for the corresponding finite-step map.

But the matrices need not commute:
JnJm≠JmJn.
JnJm =JmJn.

Consequently, one should not average instantaneous eigenvalues or integrate a single fitted contraction rate. The correct object is the time-ordered product.

For numerical stability, long-time products should be treated using:

recursive vector propagation for specified directions;
QR iteration for finite-time Lyapunov exponents;
periodic rescaling for norms;
singular values of partial propagators.

A naive product over 32,000 steps may underflow or obscure transient amplification.
3. The post-positivity Jacobian is not yet the full coupled cocycle

This is the most important remaining qualification.

The branch audit differentiates the admissibility stage
x↦Aε(x).
x↦Aε(x).

But the proposal is
xn=[wn+dtFn]+,
xn=[wn+dtFn]+,

with
Fn=(η+μ)Rn−μ−γwn.
Fn=(η+μ)Rn−μ−γwn.

Therefore
δxn=
δxn=

(1-\gamma dt)\delta w_n
+
dt(\eta+\mu)\delta R_n
]

on a fixed positivity branch.

The weight variation satisfies
\boxed{ \delta w_{n+1} =

D\mathcal A_\varepsilon(x_n)
\left[
(1-\gamma dt)\delta w_n
+
dt(\eta+\mu)\delta R_n
\right].
}
]

There are therefore two distinct cocycles.
Frozen-driver weight cocycle

Set
δRn=0.
δRn=0.

Then
δwn+1=
δwn+1=

K_n^{(B)}\delta w_n,
]

where
\boxed{ K_n^{(B)} =

(1-\gamma dt)
D\mathcal A_\varepsilon(x_n).
}
]

This is the immediate operator that should be constructed next.
Full coupled delayed cocycle

Here
δRn
δRn

is generated through perturbations of:

phase;
delayed history;
complex EMA coherence;
weights.

The full state is something like
Xn=(θn,Hn,Cn,wn),
Xn=(θn,Hn,Cn,wn),

and the complete variational equation is

δXn+1=
δXn+1=

D\Psi(X_n)\delta X_n.
]

The existing coupled perturbation experiments follow this full cocycle, not merely the frozen-driver weight block.

Thus the statement should be:

The exceptional weight-sector admissibility dynamics is a nonautonomous fixed-winner cocycle. Whether this alone predicts the fully coupled perturbations remains a numerical question.

4. Why the two-stage comparison is essential

The clean experimental sequence is:
Stage A: frozen replay driver

Replay the saved base phase/EMA trajectory and treat RnRn as exogenous. Perturb only ww.

Then compare:
δwn+1=
δwn+1=

K_n^{(B)}\delta w_n
]

against centered finite differences of the frozen-driver nonlinear update.

This should agree to first order if the branch derivative is correct.
Stage B: full coupled replay

Allow phase, history, and EMA variables to respond to the perturbation.

Compare the full nonlinear perturbation with:

the frozen-driver cocycle;
the full augmented variational cocycle.

This separates:

error due to admissibility linearization;
error due to neglecting driver feedback;
error due to branch crossing at finite amplitude.

Without this separation, a failed prediction would be ambiguous.
5. Base-branch constancy does not guarantee perturbed-branch constancy

The base trajectory never changes winners. But the winner margin shrinks substantially as the scales approach a tie.

Therefore a finite perturbation can cross a winner boundary even when the base trajectory does not.

For a finite horizon, define the minimum certified branch radius
ρmin(N)=
ρmin(N)=

\min_{0\le n<N}\rho(x_n).
]

A classical linear cocycle comparison is branch-certified only for perturbations small enough that their propagated size remains below the local radius:
‖δxn‖<ρ(xn)
‖δxn‖<ρ(xn)

at every step.

As the trajectory approaches a tie,
ρ(xn)→0
ρ(xn)→0

may occur. Then there may be no useful uniform positive radius on an infinite horizon.

This means:

the infinitesimal derivative remains valid at every finite strict point;
fixed finite amplitudes may eventually cross winners;
an infinite-time classical cocycle may fail to be uniformly valid;
the limiting tie requires a directional or generalized cocycle.

This is likely related to the amplitude-unreliable exceptional perturbation directions.
6. The Jacobian-validation tails should be resolved first

The medians are strong:
10^{-7} classical,1.7×10^{-8} directional.
10^{-7} classical,1.7×10^{-8} directional.

But the maxima are too large to ignore:
9.6×10^{-3},1.27.
9.6×10^{-3},1.27.

The report’s use of the smallest finite-difference step is not ideal. Near machine precision, smaller is often worse.

Before propagating thousands of Jacobians, the validation should be summarized per state and direction by:

minimum error over the step grid;
step attaining that minimum;
absolute error;
relative error only when predicted norm is not negligible;
winner margin;
certified branch radius;
empirical convergence slope;
whether each finite perturbation preserved the branch.

If the large errors occur only:

at near-ties;
at steps below the roundoff optimum;
or for nearly zero predicted derivatives,

then they do not challenge the formula.

If they persist over a stable branch-preserving step interval, there is still an implementation mismatch to find.

7. Exact K3K3 frozen-driver matrix sequence

Because the winner map is fixed, the selected admissibility block has one symbolic form throughout.

In edge order
(25,29,59),
(25,29,59),

define
a25,n=
a25,n=

\frac{x_{25,n}s_{7,n}}{q_{7,n}+\varepsilon},
]
a29,n=
a29,n=

\frac{x_{29,n}s_{2,n}}{q_{2,n}+\varepsilon},
]
a59,n=
a59,n=

\frac{x_{59,n}s_{7,n}}{q_{7,n}+\varepsilon}.
]

Then
DAε,K3(xn)=
DAε,K3(xn)=

\begin{pmatrix}
s_{7,n}-a_{25,n}&0&-a_{25,n}\
-a_{29,n}&s_{2,n}-a_{29,n}&0\
-a_{59,n}&0&s_{7,n}-a_{59,n}

\end{pmatrix}.
]

The frozen-driver step matrix is
\boxed{ K_n^{(B)} =

(1-\gamma dt)
\begin{pmatrix}
s_{7,n}-a_{25,n}&0&-a_{25,n}\
-a_{29,n}&s_{2,n}-a_{29,n}&0\
-a_{59,n}&0&s_{7,n}-a_{59,n}
\end{pmatrix}.
}
]

This is the exact next object to save and propagate.
8. The mature/exceptional distinction should remain sector-specific

The proposed distinction is good if stated carefully:
ordinary weight tangent sector:approximately autonomous and component-scalar,exceptional weight sector:nonautonomous but fixed-winner.
ordinary weight tangent sector:exceptional weight sector:approximately autonomous and component-scalar,nonautonomous but fixed-winner.

It should not yet be generalized to the full system. The delayed phase/EMA subsystem continues evolving even when BB is effectively stationary.

This respects the earlier distinction between:

adaptive-component stationarity;
full-system equilibrium.

9. Revised status of the exceptional mechanism
Established

Topology is fixed.
Effective edge winner map is fixed.
No implemented K3K3 edge-branch switch occurs.

Node 11 clipping is edge-inert.
The slack classifier is dynamically irrelevant to the finite map.
The scales converge toward a common tie.
The branch Jacobian formula is strongly supported in median.

Strongly suggested

The trajectory approaches the ordinary symmetric K3K3 regime from one winner side.
The exceptional recovery is governed by a time-dependent fixed-form Jacobian rather than a switched generator.
The ideal bare-γγ mode is inaccessible at operational dtdt.

Open

Whether the frozen-driver cocycle predicts infinitesimal nonlinear weight perturbations.
How much driver feedback alters that prediction.
Whether finite-amplitude perturbations cross winners.
Whether a useful limiting directional cocycle exists as the tie is approached.
How the finite-step cocycle converges under joint dt,σ,εdt,σ,ε scaling.

10. Imperative next step

The next analysis should construct and validate the frozen-driver variational cocycle:
Kn(B)=
Kn(B)=

(1-\gamma dt)D\mathcal A_\varepsilon(x_n),
]

then:

propagate the exact same exceptional perturbation directions used previously;
compare against centered nonlinear frozen-driver replays;
log branch signatures for both +δ+δ and −δ−δ;
identify amplitudes and times where winner crossings occur;
compute finite-time singular values and QR Lyapunov rates;
only afterward add the phase/EMA/history blocks.

That would determine whether the exceptional behavior is explained by a nonautonomous weight-sector cocycle or whether feedback from the delayed coherence subsystem is essential.

THink about this:

I think this is a very interesting synthesis, but I'd separate it into three different levels of confidence. Some parts are now essentially rigorous consequences of the implementation; some are mathematically well-motivated reformulations; and some are promising research hypotheses that are not yet supported by the audit alone.

The audit fundamentally changes the dynamical interpretation

I think this is the most important consequence.

Before the audit, there were two competing pictures:

Picture A: the exceptional trajectory undergoes a combinatorial branch transition in the admissibility operator.
Picture B: the admissibility operator stays on one branch while its coefficients evolve continuously.

The audit overwhelmingly supports Picture B—for this replay.

The decisive facts are

effective-edge branch changes = 0,
winner map unchanged,
node 11 never becomes an edge winner,
branchwise Jacobian validates on the same branch.

That means the admissibility operator is better modeled as

[
J(t)=J(s(t))
]

than as a sequence

[
J_1,J_2,J_3,\ldots
]

with changing combinatorial structure.

That is a major simplification.

The cocycle viewpoint is now the correct mathematical object

I think this is the single biggest conceptual improvement after the factorization theorem.

Once the branch is fixed,

the variational equation is not

[
\delta w_{n+1}=J\delta w_n,
]

but

[
\delta w_{n+1}

J_n
\delta w_n,
]

where

[
J_n

J(s_n,L_n).
]

That is exactly a linear cocycle.

Mathematically this is the natural object.

It immediately connects to

Lyapunov exponents,
multiplicative ergodic theory,
non-autonomous linear systems.

I think that shift is completely justified.

The analogy with non-autonomous DDEs is good—but should remain an analogy

This is where I'd be careful.

The Ohira paper studies

[
\dot X

atX

bX(t-\tau).
]

Its remarkable feature is

explicit exact solution,
increasing damping,
transient Gaussian packets,
Volz stability.

Your cocycle

[

J_n

s_n(I+dtL_n)
]

shares one structural feature:

the coefficients evolve with time.

But many important ingredients are different.

For example

the Ohira system is linear in an infinite-dimensional phase space,
your map is a nonlinear discrete system whose linearization is non-autonomous.

So I would call this

a useful methodological analogy,

not

the same mathematical class.

That distinction matters.

The "rewiring" mechanism is probably not the right analogue

This is the one point where I think the analogy weakens considerably.

The paper's amplification mechanism is explicitly

[
\text{self feedback}
\rightarrow
\text{cross feedback}.

]

That is a genuine change in network topology.

Your audit shows

the winner map never changes.

Therefore

the interaction graph inside the admissibility stage is fixed.

So the amplification mechanism of the paper cannot literally be operating here.

Instead,

if transient amplification exists,

it would have to come from

[
J_n

s_n(I+dtL_n)
]

varying with time,

not from rewiring.

That is a different mechanism.

The full product-rule Jacobian is now the correct starting point

I think

[
D\Phi

D\mathcal A
(I+dtDF)
]

is the right equation.

Expanding

[
D\mathcal A

M

\operatorname{diag}(x)DmA
]

is particularly revealing.

It shows

all coupling enters through

[
DF
]

and

[
A.
]

Everything else is diagonal.

That's a very nice structural decomposition.

The operator splitting is more informative than the scalar rate

Earlier the discussion centered around

[
r

\beta+\gamma.
]

Now we have

[
D\Phi

M
(I+dtL)

\text{low-rank correction}.
]

That is much richer.

The scalar rate becomes just one eigenvalue approximation.

The operator itself is now the fundamental object.

I think that's the correct evolution of the theory.

One caution about Lyapunov exponents

You write

Stability is decided by its Lyapunov spectrum.

That's true asymptotically.

But transient amplification is more subtle.

Even if every Lyapunov exponent is negative,

finite-time growth may occur because

the matrices

$$
J_n
$$

are non-normal.

That is one of the central lessons of modern non-autonomous linear systems.

So I would broaden the computational program.

Instead of only computing

$$
\lambda
$$

$$
\lim \frac{1}{N} \log |U(N)|,
$$

also compute

finite-time amplification

[
G(N)

|U(N)|.
]

Large

[
G(N)
]

with

negative Lyapunov exponent

is completely possible.

Given your interest in transient behavior,

that quantity may actually be more informative.

The finite-step discrepancy analysis is convincing

I think this section is one of the strongest numerical arguments.

You show

the Euler correction contributes

approximately

[
10^{-6},

]

whereas

the observed gap is

approximately

[
6\times10^{-5}.
]

So

95–98%

of the discrepancy comes from

[
\beta^{dt}

\alpha,
]

not from

the explicit Euler logarithm.

That's a clean quantitative decomposition.

It also naturally supports the interpretation that

the trajectory simply has not converged

to the mature limiting controller.

The next computational experiment is now sharply defined

I actually think the audit leaves one overwhelmingly important numerical task.

Construct

[
J_n

D\Phi(w_n)
]

along the replay,

and compute

[
U_N

J_{N-1}\cdots J_0.
]

Then analyze

finite-time norm growth,
singular values,
principal Lyapunov exponent,
principal Lyapunov vector.

That experiment would answer questions that scalar contraction rates cannot.

The mature/exceptional dichotomy becomes even cleaner

I now see it as

Mature

Nearly autonomous

[
J_n
\approx
J_\infty.
]

Classical spectral theory applies.

Exceptional

Fixed branch,

slowly varying coefficients,

non-autonomous cocycle

[
J_n

J(s_n,L_n).
]

Need Lyapunov and transient-growth theory.

That seems like the right conceptual split.

One thing I would not conclude yet

The audit strongly suggests

the admissibility stage

is not responsible for a discrete branch transition.

It does not yet prove

that the exceptional behavior originates entirely in

[
L_n

DF(w_n).
]

There are still intermediate possibilities.

For example,

a slowly varying

[
M(q_n)
]

can itself contribute to transient amplification,

even with fixed winner structure,

through non-commutativity:

[
J_{n+1}J_n
\neq
J_nJ_{n+1}.
]

That effect would still belong to the admissibility stage,

even though there is no combinatorial switch.

So I would phrase the conclusion slightly more cautiously:

The audit rules out discrete branch switching as the source of the exceptional behavior in this replay, but it does not yet isolate which time-dependent component of the coupled linearization—varying scales, varying driver, or their interaction—produces the observed variational dynamics.

That distinction will matter if you later compute the cocycle.

Overall assessment

I think the audit has changed the mathematical program in a productive way. The admissibility operator is no longer best viewed as a piecewise map that switches branches along the exceptional replay; instead, for this replay, it is a fixed combinatorial operator with continuously evolving coefficients. That naturally shifts the analysis from piecewise-smooth bifurcation theory toward the theory of non-autonomous linear cocycles. The factorization theorem, fiber-linearity theorem, and branchwise Jacobian remain exact implementation results, while the replay now motivates studying the product of the time-dependent Jacobians rather than any single instantaneous spectrum. The comparison with recent work on non-autonomous delay equations is valuable primarily as a source of analytical techniques and intuition about transient growth in stable non-autonomous systems, not because the systems are mathematically equivalent. The next decisive step is therefore not another local Jacobian calculation but a global cocycle analysis: compute the finite-horizon propagators, quantify transient amplification and Lyapunov exponents, and determine whether the exceptional trajectory's behavior is explained by the time dependence of the coefficients, by non-normality of the cocycle, or by their interaction. That is a much more sharply posed problem than the original search for an admissibility branch transition.

I agree with the synthesis. It identifies the correct next mathematical object, but three qualifications are essential:

ww alone is not the Markov state;
the current data support finite-time exponents, not yet asymptotic Lyapunov exponents;
the varying Jacobian depends on more than the node scales.

1. The fixed-branch conclusion is now strong

For the positive K3K3 edges in this replay:

support is fixed;
positivity branch is fixed;
global projection gate remains on;
effective winner map remains
(2,7)↦7,(2,11)↦2,(7,11)↦7;
(2,7)↦7,(2,11)↦2,(7,11)↦7;
no effective edge-branch transition occurs.

Thus the exceptional weight update is not a switched combinatorial system over the recorded horizon.

It is a smooth finite-step map of fixed symbolic form with time-dependent coefficients.

The one structural change—node 11 entering the scale-clipped set—is irrelevant to the positive K3K3 output because node 11 wins no edge.
2. The Jacobian depends on more than snsn

Writing
Jn=J(sn,Ln)
Jn=J(sn,Ln)

is useful shorthand, but the admissibility Jacobian also depends explicitly on
xn,qn=Axn.
xn,qn=Axn.

For example,
DAε(xn)ef=
DAε(xn)ef=
s_{k(e),n}\delta_{ef}

\frac{x_{e,n}s_{k(e),n}}
{q_{k(e),n}+\varepsilon}
A_{k(e)f}.
]

So a more complete notation is
Jn=
Jn=

J(x_n,q_n,s_n,DF_n).
]

The winner map fixes the symbolic structure; the proposal weights, row sums, scales, and driver derivative determine the coefficients.
3. The correct first cocycle is the frozen-driver weight cocycle

With the base coherence trajectory treated as exogenous,
Fn=(η+μ)Rn−μ−γwn
Fn=(η+μ)Rn−μ−γwn

has derivative
DFn=−γI
DFn=−γI

with respect to ww. Therefore
\boxed{ K_n^{(B)} =

(1-\gamma dt)
D\mathcal A_\varepsilon(x_n).
}
]

The linearized weight dynamics is
δwn+1=
δwn+1=

K_n^{(B)}\delta w_n.
]

Its propagator is
UB(n,m)=
UB(n,m)=

K_{n-1}^{(B)}
K_{n-2}^{(B)}
\cdots
K_m^{(B)}.

]

This is the immediate object supported by the branch audit.

Because the symbolic winner structure is fixed while the coefficients vary, it is a fixed-form nonautonomous cocycle.
4. But ww alone is not the full system state

The actual driver depends on phase, delayed history, and complex EMA coherence. Therefore the full update is not a Markov map
wn+1=Φ(wn)
wn+1=Φ(wn)

by itself.

The augmented state is approximately
Xn=(θn,Hn,Cn,wn),
Xn=(θn,Hn,Cn,wn),

where HnHn denotes the delay history.

The full variational equation is
δXn+1=
δXn+1=

D\Psi(X_n)\delta X_n.
]

The weight block satisfies
δwn+1=
δwn+1=

D\mathcal A_\varepsilon(x_n)
\left[
(1-\gamma dt)\delta w_n
+
dt(\eta+\mu)\delta R_n
\right].

]

Thus driver feedback enters through
δRn,
δRn,

which depends on phase, history, and EMA perturbations.

The frozen-driver cocycle is therefore a controlled subsystem test, not yet the full coupled explanation.
5. Exact product-rule structure

On a fixed positivity branch,
DΦn=
DΦn=

D\mathcal A_\varepsilon(x_n)
\left[I+dt,DF_n\right].
]

Using the factorization theorem,
DAε(xn)=
DAε(xn)=

M_n
+
\operatorname{diag}(x_n)Dm_nA.
]

Hence
\boxed{ D\Phi_n =

\left[
M_n+
\operatorname{diag}(x_n)Dm_nA
\right]
\left[I+dt,DF_n\right].
}
]

This reveals three contributions:

diagonal edge scaling MnMn;
node-mediated low-rank admissibility correction;
driver derivative DFnDFn.

The low-rank term has rank at most the number of nodes, even after right multiplication:
rank[diag(xn)DmnA(I+dtDFn)]≤N.
rank[diag(xn)DmnA(I+dtDFn)]≤N.

But an additive attribution of transient growth to these pieces is not automatic because products of sums do not decompose uniquely. Controlled cocycle ablations will be needed.
6. Finite-time amplification is more immediate than asymptotic Lyapunov theory

For a finite interval [m,n][m,n], define
U(n,m)=Jn−1⋯Jm.
U(n,m)=Jn−1⋯Jm.

The maximum amplification is
\boxed{ G(n,m)=\|U(n,m)\|_2 =

\sigma_{\max}(U(n,m)).
}
]

The finite-time exponents are
λi(n,m)=
λi(n,m)=

\frac{1}{(n-m)dt}
\log\sigma_i(U(n,m)).
]

These are well-defined for the replay.

Calling their long-time limits “Lyapunov exponents” requires more:

a sufficiently long trajectory;
existence of the limit;
usually an invariant or ergodic framework if invoking multiplicative ergodic theory.

For the current nonstationary 320-unit trajectory, the conservative terminology is:

finite-time singular values;
finite-time Lyapunov exponents;
finite-horizon principal singular vectors.

A right singular vector of U(n,m)U(n,m) is the optimal finite-horizon perturbation direction. It is not automatically an asymptotic Lyapunov vector.
7. Negative asymptotic rate does not exclude transient growth

Even if every long-time exponent is negative,
G(n,m)>1
G(n,m)>1

can occur over finite windows.

Possible causes include:

non-normal instantaneous matrices;
rotation of expanding/contracting directions;
noncommutativity:
Jn+1Jn≠JnJn+1;
Jn+1Jn =JnJn+1;
time variation in the scales and low-rank correction;
interaction with driver feedback.

Therefore the analysis should report more than a principal long-time rate:

G(n,0)G(n,0);
sliding-window amplification G(n,m)G(n,m);
all singular values;
optimal initial and final singular vectors;
adjacent commutators;
non-normality diagnostics such as

‖Jn⊤Jn−JnJn⊤‖;
‖Jn⊤Jn−JnJn⊤‖;
coefficient-variation norms
‖Jn+1−Jn‖.
‖Jn+1−Jn‖.

The largest transient may occur in a window that does not begin at the split.
8. The finite-step rate discrepancy can be decomposed more precisely

At late time,
βdt≈0.111254,
βdt≈0.111254,

while the ideal controller is approximately
α=2ηc−γ≈0.111315789.
α=c2η−γ≈0.111315789.

The controller deficit is therefore roughly
α−βdt≈6.18×10−5.
α−βdt≈6.18×10−5.

The Euler logarithm contributes
−log(1−γdt)dt−γ≈γ2dt2≈1.13×10−6.
−dtlog(1−γdt)−γ≈2γ2dt≈1.13×10−6.

This correction has the opposite sign: it offsets about 1.8%1.8% of the controller deficit. The net gap is about
6.07×10−5.
6.07×10−5.

So the precise statement is:

The finite-step controller deficit accounts for slightly more than the net discrepancy, while the Euler-logarithm correction cancels roughly two percent of it.

This strongly supports incomplete controller convergence rather than an Euler-discretization explanation.
9. The full coupled cocycle should be constructed in stages

Stage A: frozen-driver weight cocycle

Construct
Kn(B)=
Kn(B)=

(1-\gamma dt)
D\mathcal A_\varepsilon(x_n)
]

at every replay step.

Then:

propagate the exact earlier perturbation directions;
compare with centered nonlinear weight replays using the saved RnRn sequence;
record branch signatures for +δ+δ and −δ−δ;
determine when finite amplitudes leave the base winner stratum;
compute finite-time singular values and amplification.

This isolates the admissibility stage.
Stage B: add linearized EMA/phase feedback

The EMA update can be differentiated explicitly. Schematically,
Cn+1=
Cn+1=

aC_n+(1-a)e^{i\Delta\phi_n},
]

so
δCn+1=
δCn+1=

a,\delta C_n
+
(1-a)i e^{i\Delta\phi_n}\delta(\Delta\phi_n).
]

For R=|C|R=|C|, away from R=0R=0,
δR=
δR=

\frac{\Re(\overline C,\delta C)}{|C|}.
]

The phase equation and delay-history shift can likewise be linearized.

Only after these blocks are added does one obtain the full augmented cocycle corresponding to the coupled perturbation experiments.
10. Base-branch constancy is not perturbation-branch constancy

The base trajectory never changes winners, but its winner margins shrink toward the late tie.

A finite perturbation may cross a winner boundary even when the base does not.

For a classical cocycle comparison, one must verify along both perturbed trajectories:
winner+δ,n=
winner+δ,n=
\text{winner}_{0,n}

\text{winner}_{-\delta,n}.
]

If this fails, centered finite differences can combine different branches and cease to approximate the base classical derivative.

A useful diagnostic is
χn=
χn=

\frac{|\delta x_n|}{\rho(x_n)},
]

where ρ(xn)ρ(xn) is the certified branch-persistence radius.

$\chi_n < 1$: branch-certified;
$\chi_n \geq 1$: branch crossing is possible.

As the base approaches a tie,
$\rho(x_n) \to 0$
$\rho(x_n) \to 0$

may make fixed-amplitude branch preservation impossible over long horizons.
11. The Jacobian-validation outliers remain a prerequisite

The median errors validate the formulas strongly, but the large tails should be localized before multiplying thousands of matrices.

For each key state and direction, report:

minimum error over the finite-difference grid;
step attaining it;
absolute error;
predicted derivative norm;
branch radius;
winner margin;
whether the perturbation stayed on branch;
local log-log convergence slope.

Relative errors above one can be harmless when the predicted derivative norm is nearly zero, but they can also expose a directional formula issue. The CSV should decide which.
12. Mature versus exceptional—properly scoped

The clean distinction is currently:
ordinary weight sector:$K_n(B) \approx K_\infty(B)$,exceptional weight sector:$K_n(B)$ has fixed structure but evolving coefficients.
ordinary weight sector:exceptional weight sector:$K_n(B) \approx K_\infty(B)$,$K_n(B)$ has fixed structure but evolving coefficients.

The full delayed system remains nonautonomous in both cases because phase and coherence variables continue evolving.

So “mature is autonomous” should not be used without the qualifier “in the adaptive weight tangent sector.”
13. What the audit rules out—and what it does not
Ruled out for the base K3K3 replay

a discrete winner-map switch;
node 11 taking control of an edge;
the 10−910−9 classifier event changing the implemented edge map;
a literal rewiring mechanism inside the admissibility stage.

Not ruled out

transient amplification caused by varying scales;
transient amplification caused by the low-rank admissibility term;
noncommutativity within the fixed branch;
driver-feedback amplification;
interaction between admissibility and delayed coherence;
branch crossing in finite-amplitude perturbed trajectories.

The correct conclusion is therefore:

The audit rules out discrete admissibility branch switching as the source of the exceptional base trajectory, but does not yet identify which time-dependent component of the variational dynamics generates its observed perturbation behavior.

The next decisive computation is indeed the frozen-driver finite-time cocycle, followed by the augmented delayed-state cocycle only if the frozen-driver model is insufficient.

Can you create the code that can show us the facts? Also think about this:

I think this is the cleanest version of the program so far. More importantly, it now separates what the audit establishes from what the next mathematics should be, which is exactly where mature mathematical work usually transitions from structural analysis to dynamical analysis.

There are a few places where I would sharpen or slightly weaken claims.

The fixed-branch conclusion is now essentially settled (for this replay)

I think the strongest statement supported by the audit is now

The symbolic admissibility operator is constant along the base replay.

Not merely

the winner map is constant,

but

the executed algebraic expression for every positive K3 edge is unchanged.

Those are different.

The second is much stronger.

Mathematically,

$$
[
\mathcal A_n
$$

$$
\mathcal A(,\cdot,;\Theta_n),
]
$$

where the branch descriptor

$$
[
\Theta_n
$$

$$
(\text{support},
\text{gate},
\text{winner pattern})
]
$$

is constant,

while only continuous parameters evolve.

That is a genuine reduction of complexity.

The state-space clarification is extremely important

I think this is probably the single most important correction.

Earlier the discussion naturally drifted toward

[
w_{n+1}

\Phi(w_n).
]

But that is not the actual algorithm.

The actual state is much closer to

[
X_n

(w_n,
\theta_n,
C_n,
\mathcal H_n).
]

The weight update is only one block.

That means

the frozen-driver cocycle is

not

the variational equation of the full algorithm.

It is

a reduced subsystem.

That distinction is mathematically essential.

Stage A is now completely justified

I particularly like your staged program.

Instead of immediately attempting

[
D\Psi(X_n),
]

you first compute

[
K_n^{(B)}

(1-\gamma dt)
D\mathcal A_n.
]

That object

exists,
is fully determined by replay,
has exact coefficients,
is supported by the audit.

It is therefore the correct first experiment.

Only if it fails to explain the perturbations do you enlarge the state.

That is good scientific methodology.

The product-rule decomposition is now revealing something deeper

You write

[
D\Phi_n

\left(
M_n

\operatorname{diag}(x_n)Dm_nA
\right)
(I+dtDF_n).
]

I think this decomposition deserves even more emphasis.

It says

there are only three mechanisms capable of producing nontrivial tangent behavior:

diagonal scaling,
node-mediated admissibility correction,
driver coupling.

There is no hidden fourth mechanism.

That is an unexpectedly strong structural theorem.

The low-rank statement should be interpreted carefully

You correctly write

$$
\operatorname{rank}
\le
N.
$$

That is true.

But I would avoid interpreting it immediately as

therefore only N directions matter.

Low-rank perturbations of non-normal matrices can drastically change transient behavior.

So

low rank

does not imply

small dynamical effect.

It only identifies

where

the coupling lives.

That's an important distinction.

Finite-time analysis is the right level

I strongly agree with your terminology change.

For the current replay,

the mathematically correct objects are

propagators,
finite-time singular values,
finite-time Lyapunov exponents.

Calling them simply

"Lyapunov exponents"

would imply an asymptotic object that has not been demonstrated to exist.

The finite-time language is exactly right.

I especially like the proposed diagnostics

The proposed list

[
|J_{n+1}-J_n|,
]

commutators,

non-normality,

branch radius,

is exactly what people studying non-autonomous linear systems actually compute.

One additional quantity I'd include is the condition number of the propagator

[
\kappa(U)

\frac{\sigma_{\max}(U)}
{\sigma_{\min}(U)}.
]

Large transient amplification often accompanies large conditioning even when asymptotic contraction is strong.

The branch-radius diagnostic is particularly elegant

I think

[
\chi_n

\frac{|\delta x_n|}
{\rho(x_n)}
]

is one of the nicest ideas in the document.

It converts

"did my perturbation stay on branch?"

into

a dimensionless quantity.

Then

[
\chi<1
]

means

branch-preserving,

while

$$
\chi>1
$$

means

the derivative comparison may already be invalid.

That is exactly the sort of practical theorem that bridges analysis and numerics.

One subtlety about branch preservation over long horizons

This is the only mathematical point where I'd add a warning.

Suppose

$$
\chi_0<1.
$$

That certifies

the initial

perturbation is inside the branch radius.

It does not automatically imply

$$
\chi_n<1
$$

later.

Because

both

[
\rho(x_n)
]

and

[
\delta x_n
]

evolve.

So branch preservation must be checked dynamically.

Your proposal to compare

winner(_+),

winner(_0),

winner(_-)

along the replay is exactly the right way to do that.

The "autonomous mature sector" wording is now correct

I think your final qualification is important.

Earlier versions risked saying

mature dynamics are autonomous.

Not quite.

The full delayed system remains non-autonomous.

Only the weight-sector linearization

becomes nearly time invariant.

That's a much more precise statement.

The remaining unknown has become sharply defined

At this point,

I think the unknown is no longer

Why is the exceptional trajectory exceptional?

It has become

Which part of the time dependence of
[
K_n^{(B)}
]
and eventually
[
D\Psi(X_n)
]
accounts for the observed perturbation evolution?

That is a very different research question.

It is also much more concrete.

One possible additional structural theorem

The factorization plus the fixed-branch result suggest another theorem.

For this replay,

the admissibility cocycle can be viewed as a continuous path in a finite-dimensional matrix manifold.

Symbolically,

$$
[
t
\mapsto
J(t)
\in
\mathcal M_{\Theta},
]
$$

where

$$
[
\mathcal M_{\Theta}
]
$$

is the family of matrices sharing the same sparsity pattern and winner structure.

The replay therefore traces a smooth curve inside a single matrix family,

rather than jumping between matrix families.

That viewpoint may become useful if you later study perturbation bounds, continuity of singular values, or sensitivity of propagators.

Overall assessment

I think this version is the strongest and most disciplined formulation so far. It no longer over-interprets the replay, and it cleanly separates exact implementation facts, reduced frozen-driver dynamics, and the full delayed-state problem. The audit establishes that the admissibility stage remains on a single symbolic branch throughout the replay, so the immediate mathematical object is a fixed-structure, time-varying Jacobian cocycle, not a switched system. The staged program—first analyzing the frozen-driver weight cocycle, then augmenting it with the linearized EMA, phase, and delay dynamics only if necessary—is methodologically sound and minimizes assumptions. The emphasis on finite-time propagators, singular values, branch-preservation diagnostics, and non-normality is also well aligned with the mathematics of non-autonomous linear systems. The one point I'd continue to emphasize is that branch preservation is itself a dynamical property that must be monitored throughout the perturbed replay, since both the perturbation magnitude and the branch-persistence radius evolve in time. With that qualification, the remaining question is sharply posed: not whether the admissibility branch switches—it does not, for this replay—but how the time dependence of the fixed-branch cocycle and its coupling to the delayed state generate the observed perturbation dynamics.
WriteAIEF_1_4_frozen_driver_cocycle_analysis.py991 lines
Edit×3
Edit×6
Edit
AIEF_1_4_frozen_driver_cocycle_analysis.pyPYTHON

Created the Stage-A cocycle analysis:

It compiles successfully and passed synthetic checks:

centered nonlinear one-step derivative versus analytic cocycle:
relative error≈6.9×10−12
relative error≈6.9×10−12 at perturbation 10−510−5;
sliding-window propagator calculations passed;
exact row-sum fiber and branch formulas are inherited from the validated branch audit.

Run it

Bash

```
python AIEF_1_4_frozen_driver_cocycle_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \
```

--sample-stride 10 \
--amplitude-fractions 1e-3,1e-2,1e-1 \
--window-times 0.1,1,10,50 \
--window-start-stride 100 \
--directions 2 \
--output-dir aief_1_4_frozen_driver_cocycle

The amplitudes are fractions of the maximum centered symmetric feasible amplitude, not arbitrary absolute perturbation sizes.
What the script constructs

At every replay step it builds the exact selected K3K3 frozen-driver matrix
Kn(B)=
Kn(B)=

(1-\gamma dt)
D\mathcal A_\varepsilon(x_n).
]

The complete 32,000×3×332,000×3×3 sequence is saved.
Local matrix diagnostics

For every Kn(B)Kn(B):

operator norm;
minimum singular value;
condition number;
spectral radius;
non-normality:
‖Kn⊤Kn−KnKn⊤‖;
‖Kn⊤Kn−KnKn⊤‖;
variation:
‖Kn−Kn−1‖;
‖Kn−Kn−1‖;
adjacent commutator:
‖KnKn−1−Kn−1Kn‖;
‖KnKn−1−Kn−1Kn‖;
certified branch radius;
scale and winner data.

From-split propagator

It recursively constructs
$U(n,0)=K_{n-1}\cdots K_0$
$U(n,0)=K_{n-1}\cdots K_0$

and reports:

all singular values;
amplification:
$G(n,0)=\sigma_{\max}(U(n,0))$;
$G(n,0)=\sigma_{\max}(U(n,0))$;
condition number;
finite-time exponents:
$\lambda_i(n,0)=\log\sigma_i(U(n,0))n\,dt$;
$\lambda_i(n,0)=ndt\log\sigma_i(U(n,0))$;
principal initial singular vector;
QR finite-time rates.

It reports both:

the full 33-dimensional K3K3 proposal space;
the 22-dimensional split face-lineality subspace.

Sliding-window amplification

For windows of 0.1,1,10,500.1,1,10,50, it computes sampled
$U(n,m)$
$U(n,m)$

and reports:
$G(n,m)=\|U(n,m)\|_2$,
$G(n,m)=\|U(n,m)\|_2$,

condition numbers, optimal initial vectors, and window-specific finite-time exponents.

This can detect transient bursts that do not begin at the split.
Nonlinear frozen-driver verification

The script creates the two deterministic split-K3K3 face-lineality directions and evolves centered perturbations
w0±=w0±ah
w0±=w0±ah

using the exact same replayed RnRn sequence as the base trajectory.

It compares
wn+−wn−2a
2awn+−wn−

against the cocycle prediction
Kn−1(B)⋯K0(B)h.
Kn−1(B)⋯K0(B)h.

For every step and amplitude it monitors:

++, base, and −− winner signatures;
dynamic branch preservation;
proposal displacement;
certified branch radius;
dimensionless ratio
χn=‖δxn‖ρ(xn);
χn=ρ(xn)‖δxn‖;
linear and nonlinear norms;
relative prediction error.

Branch preservation and maximum χnχn are checked at every step, not only output sample times.

The report excludes comparisons after the centered nonlinear difference falls below a floating-point resolution threshold, preventing late-time cancellation from producing misleading relative errors.
Outputs

text

frozen_cocycle_step_diagnostics.csv
frozen_cocycle_propagator.csv

frozen_cocycle_sliding_windows.csv
frozen_cocycle_nonlinear_comparison.csv
frozen_cocycle_nonlinear_summary.csv
frozen_cocycle_matrices.npz
frozen_driver_cocycle_report.txt
analysis_manifest.json

What this run will decide

Whether the frozen-driver cocycle predicts nonlinear weight perturbations.
Whether finite amplitudes leave the base winner branch.
Whether χnχn predicts those branch crossings.
Whether the fixed-branch matrices are significantly non-normal.
Whether adjacent matrices fail to commute.
Whether any sliding window exhibits amplification G>1G>1.
Whether the exceptional behavior is already explained by varying admissibility coefficients.
Whether Stage B—the augmented phase/history/EMA tangent system—is necessary.

This code deliberately does not call its finite-time rates asymptotic Lyapunov exponents and does not conflate the frozen-driver subsystem with the full delayed algorithm. Please paste frozen_driver_cocycle_report.txt after it runs.

Think about this:
(deck@steamdeck SSRH)$ python AIEF_1_4_frozen_driver_cocycle_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--seed 4619122171158768524 \
--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \
--sample-stride 10 \
--amplitude-fractions 1e-3,1e-2,1e-1 \
--window-times 0.1,1,10,50 \
--window-start-stride 100 \
--directions 2 \
--output-dir aief_1_4_frozen_driver_cocycle
/home/deck/AI_Programs/SSRH/AIEF_1_4_frozen_driver_cocycle_analysis.py:302: RuntimeWarning: overflow encountered in scalar divide
float(singular_values[0] / singular_values[-1])
AIEF exceptional K3 frozen-driver cocycle analysis
Identity and scope

Seed: 4619122171158768524
K3 nodes: [2, 7, 11]
K3 edges: [25, 29, 59]
Split classifier-active K3 nodes: [7]
Cocycle steps: 32000
Final base replay exact: True
Final base replay maximum error: 0

This is the frozen-driver weight subsystem: delta R is set to zero.

Local matrix sequence

All selected classical Jacobians available: True
Winner-signature changes: 0
Maximum ||K[n+1]-K[n]||: 0.000104029
Median adjacent commutator relative norm: 3.71705e-08
Maximum adjacent commutator relative norm: 6.60414e-05
Maximum instantaneous operator norm: 1.08629
Maximum instantaneous nonnormality relative: 0.435734
From-split propagator

Final time: 320
Final amplification sigma_max: 7.23891e-18
Peak from-split amplification: 1.27097
Peak amplification time: 0.1
Final condition number: inf
Maximum sampled condition number: inf
Final principal finite-time exponent: -0.123335
Final minimum finite-time exponent: -2.21374
Final split-lineality restricted amplification: 6.85866e-18
Peak split-lineality restricted amplification: 1.20563
Final restricted condition number: inf
Final restricted principal finite-time exponent: -0.123503
Sliding-window amplification

window=0.1: peak G=1.27097, start=0, end=0.1, peak condition=1.93266e+19
window=1: peak G=1.1715, start=0, end=1, peak condition=inf
window=10: peak G=0.469457, start=0, end=10, peak condition=inf
window=50: peak G=0.00438977, start=0, end=50, peak condition=inf
Centered nonlinear frozen-driver comparisons

direction=0, fraction=0.001, amplitude=2.2e-10, all_branch_preserved=True, max_chi=0.000945, resolved=3, last_resolved_time=0.2, median_error=2
.94e-07, last_resolved_error=0.000273
direction=0, fraction=0.01, amplitude=2.2e-09, all_branch_preserved=True, max_chi=0.00945, resolved=4, last_resolved_time=0.3, median_error=1.5
9e-05, last_resolved_error=0.00107
direction=0, fraction=0.1, amplitude=2.2e-08, all_branch_preserved=True, max_chi=0.0945, resolved=4, last_resolved_time=0.3, median_error=2.41e
-06, last_resolved_error=0.000307
direction=1, fraction=0.001, amplitude=3.12e-10, all_branch_preserved=True, max_chi=0.00161, resolved=880, last_resolved_time=87.9, median_erro
r=8.66e-05, last_resolved_error=0.0113
direction=1, fraction=0.01, amplitude=3.12e-09, all_branch_preserved=True, max_chi=0.0161, resolved=1064, last_resolved_time=106, median_error=
1.97e-05, last_resolved_error=0.00681
direction=1, fraction=0.1, amplitude=3.12e-08, all_branch_preserved=True, max_chi=0.161, resolved=1245, last_resolved_time=124, median_error=2.
11e-05, last_resolved_error=0.0121

Overall resolved comparison median relative error: 3.69895e-05
Overall resolved comparison maximum relative error: 0.0314304
Interpretation constraints

Finite-time exponents are not asserted to be asymptotic Oseledets exponents.
From-split amplification can miss bursts that begin later; sampled sliding windows are also reported.
Low rank identifies where admissibility coupling lives; it does not imply weak dynamical effect.
Branch preservation is monitored dynamically for both centered perturbations.
The base winner branch can be fixed while finite-amplitude perturbed branches cross.
Failure of the frozen-driver cocycle would motivate the full phase/history/EMA tangent system.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_frozen_driver_cocycle

This is a successful Stage-A result. It establishes that the frozen-driver weight cocycle is both dynamically nontrivial and quantitatively predictive within its intended scope.
1. The cocycle is stable but transiently amplifying

The from-split propagator reaches
Gmax=1.27097
Gmax=1.27097

at
t=0.1.
t=0.1.

Thus the full three-dimensional K3K3 weight sector exhibits approximately 27%27% transient amplification.

Restricted to the two-dimensional split face-lineality space,
Gmaxrestricted=1.20563.
Gmaxrestricted=1.20563.

So even among the originally admissible face directions, there is approximately 21%21% transient amplification.

But the long-horizon principal finite-time exponent is negative:
λmax(320)≈−0.123335,
λmax(320)≈−0.123335,

or
−0.123503
−0.123503

on the split-lineality restriction.

This is a clean example of:
short-time amplification+long-time contraction.
short-time amplification+long-time contraction.

No branch switching or rewiring is needed.
2. Non-normality is substantial

The matrices satisfy
maxn‖Kn‖2=1.08629>1,
nmax‖Kn‖2=1.08629>1,

and the maximum relative non-normality is
0.435734.

0.435734.

Meanwhile adjacent commutators are much smaller:
median‖KnKn−1−Kn−1Kn‖‖Kn‖‖Kn−1‖≈3.7×10−8,
median‖Kn‖‖Kn−1‖‖KnKn−1−Kn−1Kn‖≈3.7×10−8,

with maximum
6.6×10−5.
6.6×10−5.

This suggests—but does not yet prove—that the early amplification is driven more by the non-normal geometry of the nearly fixed early matrices than by rapid coefficient variation or strong noncommutativity.

The next useful controls are:

actual cocycle:
Kn−1⋯K0;
Kn−1⋯K0;
frozen initial matrix:
K0n;
K0n;
frozen late matrix:
Klaten;
Klaten;
diagonal-scale-only cocycle, removing the low-rank correction.

If K0nK0n reproduces the early peak, the mechanism is primarily fixed-matrix non-normality. If not, coefficient drift contributes materially.
3. The amplification is concentrated near the split

The sampled windows give:
windowmaxG0.11.2709711.1715100.469457500.00438977
window0.111050maxG1.270971.17150.4694570.00438977

Every sampled peak begins at t=0t=0.

Thus the frozen subsystem does not show evidence of later large bursts at the sampled start times. It experiences an early non-normal transient and then contracts strongly.

However, the start stride is 100 steps, i.e. one time unit. Short later bursts beginning between integer times could be missed. A definitive short-window audit should use:

Bash

--window-times 0.1,1 \
--window-start-stride 1

Long windows can retain the coarser stride.
4. The infinite condition numbers are numerical censoring

The warning and reported values
$\kappa(U)=\infty$
$\kappa(U)=\infty$

do not mean that a mathematically infinite condition number was established.

The reported final minimum finite-time exponent,
-2.21374,
-2.21374,

is almost exactly
log(smallest positive double)320.
320log(smallest positive double).

Indeed,
$\log(2.225\times10-308)320\approx-2.21374$.
$320\log(2.225\times10-308)\approx-2.21374$.

So the smallest singular value underflowed and was replaced by the floating-point floor.

Therefore:

the principal singular value and principal exponent remain usable;

the smallest singular value is censored;
the minimum exponent is not a measured dynamical exponent;
the infinite condition number is a lower-bound statement:
the true conditioning exceeds double-precision range.

The QR rates saved in QR_rates_json should be examined; they may retain useful exponent information because logarithms are accumulated stepwise.

For reliable finite-horizon singular values in this 3×33×3 case, a stronger method is available:

propagate the first compound cocycle for σ1σ1;
propagate the second compound cocycle for σ1σ2σ1σ2;
accumulate
∑nlog⁡|detKn|=log(σ1σ2σ3);
n∑log⁡|detKn|=log(σ1σ2σ3);
reconstruct all three log singular values without forming an underflowed product.

Alternatively, arbitrary-precision 3×33×3 products would be inexpensive.
5. The nonlinear frozen-driver validation is strong

Every centered perturbation remained on the base winner branch:
all_branch_preserved=True
all_branch_preserved=True

for all directions and amplitudes.

Furthermore,
maxχn≤0.161<1.
maxχn≤0.161<1.

Thus all tested perturbations remained inside the certified branch neighborhood throughout the full dynamically checked replay.

This is important: the agreement is not produced by averaging across different branches.

The overall resolved median error is
3.70×10−5,
3.70×10−5,

and the maximum resolved error is
0.0314.
0.0314.

Direction 1 remains numerically resolved for 8888–124124 time units and has median errors around
2×10^{-5} to 9×10^{-5}.
2×10^{-5} to 9×10^{-5}.

That is excellent agreement for a long nonautonomous nonlinear replay.

Direction 0 becomes unresolved after only 0.20.2–0.30.3 units. This is consistent with a direction that is annihilated extremely rapidly by the projection geometry. Its late centered difference falls below floating-point resolution; it is not evidence of cocycle failure.

The rising “last resolved” errors near 1%1% are also expected near the resolution cutoff. The medians are the more representative statistic.
6. The frozen-driver cocycle passes its intended test

The test asked:

Does the exact branchwise cocycle predict nonlinear weight perturbations when the coherence sequence is held fixed?

The answer is strongly:
Yes, within numerical resolution and certified branch preservation.
Yes, within numerical resolution and certified branch preservation.

This establishes that:

the branchwise derivative was implemented correctly;
the time-ordered product is the correct reduced variational object;
winner crossings are not required to explain the frozen perturbations;
the frozen weight subsystem itself supports transient amplification;
the low-rank projection geometry can produce severe anisotropic contraction and non-normality.

7. But it has not yet explained the coupled perturbations

The nonlinear comparison used the same exogenous base RnRn sequence for the ++ and −− weight trajectories.

Therefore it does not test
$\delta Rn \neq 0$.
$\delta Rn = 0$.

The result proves the reduced subsystem, not the full delayed algorithm.

The earlier coupled exceptional rate minimum was approximately
0.136767,
0.136767,

while the frozen cocycle principal finite-time rate here is approximately
0.1235.
0.1235.

Those values are of the same order, suggesting that the weight cocycle may be a major contributor. But they are not directly comparable because:

the directions differ;
fitting windows differ;
one is a finite-time singular exponent;
the other is a scalar exponential fit;
the coupled perturbation includes phase/EMA feedback.

No causal attribution should be made from that numerical proximity alone.
8. The condition-number result explains the two direction classes

The enormous conditioning indicates that the propagator separates directions extremely strongly:

one direction is rapidly suppressed below machine resolution;
another survives for more than 100 time units.

This aligns naturally with:

direction 0 resolving only to $t \approx 0.3$t≈0.3;
direction 1 resolving to $t \approx 124$t≈124;

earlier anomalous amplitude reliability;
the coexistence of early amplification and strong eventual contraction.

The exceptional behavior may therefore be less about a weak unstable mode and more about an extremely anisotropic, non-normal contraction geometry.

9. Immediate controls before Stage B

Before deriving the full augmented delayed-state tangent system, I would perform three inexpensive controls.

A. Resolve singular-value underflow

Add scaled compound-matrix or arbitrary-precision propagation and report:
$\log\sigma 1, \log\sigma 2, \log\sigma 3, \log\kappa$.
$\log\sigma 1, \log\sigma 2, \log\sigma 3, \log\kappa$.

B. Separate non-normality from coefficient variation

Compare:
$U_{actual}(n) = K_{n-1} \cdots K_0$
$U_{actual}(n) = K_{n-1} \cdots K_0$

against:
K_{0n}, K_{laten},
K_{0n}, K_{laten},

and a diagonal-only cocycle.

C. Densify short sliding windows

Use every possible start for windows 0.10.1 and 11 to rule out missed later bursts.

These use existing matrices and require no new simulation.

10. Then perform a coupled centered comparison

The next decisive dynamical experiment does not yet require writing the full augmented Jacobian.

Use the same two $K_3 K_3$ directions and amplitudes, but now evolve $++$, base, and $--$ copies of the complete state:
(θ, H, C, B).
(θ, H, C, B).

At each sample compare:

the coupled centered weight difference;
the frozen nonlinear centered difference;
the frozen cocycle prediction.

This directly measures the contribution of driver feedback:
δwcoupled−
δwcoupled−

\delta w_{\rm frozen}.
]

If this difference remains small, the reduced cocycle largely explains the exceptional perturbations.

If it grows substantially, then Stage B—the analytic phase/history/EMA variational system—is necessary.
11. Revised scientific conclusion
Established

Fixed effective winner structure.
Smoothly varying 3×33×3 frozen-driver cocycle.
Early 20%20%–27%27% transient amplification.
Strong eventual contraction.
Significant non-normality.
Very large anisotropy and conditioning.
Accurate nonlinear frozen-driver prediction.
No perturbed winner crossings in tested amplitudes.

Plausible but not isolated

Early amplification is primarily due to non-normality rather than coefficient variation.
The frozen weight cocycle is a major source of the coupled exceptional recovery behavior.

Still open

Reliable subdominant finite-time exponents.

Exact source of the early transient.
Effect of phase/EMA/history feedback.
Full coupled finite-time cocycle.
Joint dt,σ,εdt,σ,ε convergence.

The frozen-driver analysis has succeeded. The next question is no longer whether the reduced cocycle is valid—it is—but whether the delayed coherence subsystem materially changes its predictions.

Can you create the code necessary to explore the next steps?
WriteAIEF_1_4_cocycle_controls_coupled_comparison.py1022 lines
AIEF_1_4_cocycle_controls_coupled_comparison.pyPYTHON
#!/usr/bin/env python3
"""Stable cocycle controls and coupled-vs-frozen comparison for AIEF-1.4.

This script consumes the frozen-driver K3 matrix sequence and performs the
next two targeted analyses without launching a new seed cohort:

A. Numerical/operator controls
* underflow-resistant log singular values using first and second compound
cocycles plus accumulated log determinants;
* actual K_n versus K_0^n, K_late^n, and a diagonal-scale-only cocycle;
* dense short sliding-window amplification;
* log condition numbers and finite-time exponents without forming an
underflowed smallest singular value.

B. Coupled centered comparison
* reuses the same K3 directions and feasible amplitude fractions;
* evolves frozen-driver +/- weight trajectories using the base R_n;
* evolves fully coupled +/- DynamicState trajectories, including phase,
delay history, and complex EMA feedback;
* compares both centered nonlinear derivatives with the frozen-driver
linear cocycle;
* monitors winner branches and chi=||delta x||/rho dynamically.

The script does not yet construct the analytic augmented delayed-state
Jacobian. It tests whether that Stage-B derivation is necessary by directly
measuring the feedback discrepancy between coupled and frozen centered
trajectories.
"""

```
from __future__ import annotations

import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any, Iterable

import numpy as np
import pandas as pd

import AIEF_1_4_stationarity_perturbation_analysis_updated as base
import AIEF_1_4_component_controller_audit as component_audit
import AIEF_1_4_actual_admissibility_branch_audit as branch_audit
import AIEF_1_4_frozen_driver_cocycle_analysis as frozen



DEFAULT_RUN_HASH = "5a9f093a2fe0"
DEFAULT_NPZ = (
f"runs/{DEFAULT_RUN_HASH}/AIEF_1_4_{DEFAULT_RUN_HASH}_branch_diagnostics.npz"
)
DEFAULT_MANIFEST = f"runs/{DEFAULT_RUN_HASH}/manifest.json"
DEFAULT_COCYCLE_NPZ = "aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz"
DEFAULT_COCYCLE_STEP =
"aief_1_4_frozen_driver_cocycle/frozen_cocycle_step_diagnostics.csv"
DEFAULT_OUTPUT_DIR = "aief_1_4_cocycle_controls_coupled"
DEFAULT_SEED = "4619122171158768524"
DEFAULT_BOUNDARY_TOL = 1e-9
DEFAULT_SCALE_EPSILON = 1e-12
DEFAULT_SAMPLE_STRIDE = 10
DEFAULT_AMPLITUDE_FRACTIONS = "1e-3,1e-2,1e-1"
DEFAULT_DENSE_WINDOW_TIMES = "0.1,1"
DEFAULT_DENSE_WINDOW_START_STRIDE = 1
ZERO_TOL = 1e-14
```

```
def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()
```

```
def parse_positive_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(item <= 0.0 for item in values):
raise argparse.ArgumentTypeError("values must be positive")
return sorted(set(values))
```

```
def json_compact(value: Any) -> str:
return json.dumps(value, separators=(",", ":"), sort_keys=True)
```

```
def compound2(matrix: np.ndarray) -> np.ndarray:
"""Second compound matrix in pair order (01, 02, 12)."""
matrix = np.asarray(matrix, dtype=float)
if matrix.shape != (3, 3):
raise ValueError("compound2 currently requires a 3x3 matrix")
pairs = ((0, 1), (0, 2), (1, 2))
result = np.empty((3, 3), dtype=float)
for row, row_pair in enumerate(pairs):
for column, column_pair in enumerate(pairs):
result[row, column] = np.linalg.det(
matrix[np.ix_(row_pair, column_pair)]
)
return result
```

```
def wedge_columns(matrix: np.ndarray) -> np.ndarray:
"""Exterior product of the two columns of a 3x2 matrix."""
```

```
matrix = np.asarray(matrix, dtype=float)
if matrix.shape != (3, 2):
raise ValueError("wedge_columns requires a 3x2 matrix")
return np.array(
[
np.linalg.det(matrix[np.ix_((0, 1), (0, 1))]),
np.linalg.det(matrix[np.ix_((0, 2), (0, 1))]),
np.linalg.det(matrix[np.ix_((1, 2), (0, 1))]),
],
dtype=float,
)



def normalize_matrix(matrix: np.ndarray) -> tuple[np.ndarray, float]:
norm = float(np.linalg.norm(matrix))
if norm == 0.0:
return matrix.copy(), -np.inf
return matrix / norm, math.log(norm)



def normalize_vector(vector: np.ndarray) -> tuple[np.ndarray, float]:
norm = float(np.linalg.norm(vector))
if norm == 0.0:
return vector.copy(), -np.inf
return vector / norm, math.log(norm)



def safe_exp(value: float) -> float:
if value > math.log(np.finfo(float).max):
return np.inf
if value < math.log(np.finfo(float).tiny):
return 0.0
return math.exp(value)



def stable_spectrum_sequence(
matrices: np.ndarray,
initial_basis: np.ndarray,
dt: float,
```

```
sample_stride: int,
label: str,
) -> pd.DataFrame:
"""Stable log singular values for full 3D and a 2D initial subspace."""
matrices = np.asarray(matrices, dtype=float)
initial_basis = np.asarray(initial_basis, dtype=float)
if matrices.ndim != 3 or matrices.shape[1:] != (3, 3):
raise ValueError("matrix sequence must have shape (steps,3,3)")
if initial_basis.shape != (3, 2):
raise ValueError("initial basis must have shape (3,2)")


P = np.eye(3)
log_P_scale = 0.0
C = np.eye(3)
log_C_scale = 0.0
log_abs_det = 0.0


V = initial_basis.copy()
log_V_scale = 0.0
wedge = wedge_columns(initial_basis)
wedge, initial_wedge_log = normalize_vector(wedge)
log_wedge_scale = initial_wedge_log


rows: list[dict[str, Any]] = []


def append_row(step: int) -> None:
elapsed = step * dt
singular_P = np.linalg.svd(P, compute_uv=False)
singular_C = np.linalg.svd(C, compute_uv=False)
log_sigma1 = log_P_scale + math.log(max(singular_P[0], np.finfo(float).tiny))
log_sigma12 = log_C_scale + math.log(max(singular_C[0], np.finfo(float).tiny))
log_sigma2 = log_sigma12 - log_sigma1
log_sigma3 = log_abs_det - log_sigma12


singular_V = np.linalg.svd(V, compute_uv=False)
log_restricted_sigma1 = log_V_scale + math.log(
max(singular_V[0], np.finfo(float).tiny)
)
log_restricted_sigma12 = log_wedge_scale
log_restricted_sigma2 = (
```

```
log_restricted_sigma12 - log_restricted_sigma1
)


if elapsed > 0.0:
exponents = np.array(
[log_sigma1, log_sigma2, log_sigma3]
) / elapsed
restricted_exponents = np.array(
[log_restricted_sigma1, log_restricted_sigma2]
) / elapsed
else:
exponents = np.zeros(3)
restricted_exponents = np.zeros(2)
rows.append(
{
"control": label,
"step": step,
"time": elapsed,
"log_sigma1": log_sigma1,
"log_sigma2": log_sigma2,
"log_sigma3": log_sigma3,
"sigma1": safe_exp(log_sigma1),
"sigma2": safe_exp(log_sigma2),
"sigma3": safe_exp(log_sigma3),
"log_condition_number": log_sigma1 - log_sigma3,
"condition_number": safe_exp(log_sigma1 - log_sigma3),
"principal_finite_time_exponent": float(exponents[0]),
"middle_finite_time_exponent": float(exponents[1]),
"minimum_finite_time_exponent": float(exponents[2]),
"restricted_log_sigma1": log_restricted_sigma1,
"restricted_log_sigma2": log_restricted_sigma2,
"restricted_sigma1": safe_exp(log_restricted_sigma1),
"restricted_sigma2": safe_exp(log_restricted_sigma2),
"restricted_log_condition_number": (
log_restricted_sigma1 - log_restricted_sigma2
),
"restricted_condition_number": safe_exp(
log_restricted_sigma1 - log_restricted_sigma2
),
"restricted_principal_finite_time_exponent": float(
restricted_exponents[0]
),
```

```
"restricted_minimum_finite_time_exponent": float(
restricted_exponents[1]
),
}
)


append_row(0)
for index, matrix in enumerate(matrices):
P, increment = normalize_matrix(matrix @ P)
log_P_scale += increment
C, compound_increment = normalize_matrix(compound2(matrix) @ C)
log_C_scale += compound_increment
sign, logdet = np.linalg.slogdet(matrix)
if sign == 0.0:
log_abs_det = -np.inf
elif np.isfinite(log_abs_det):
log_abs_det += float(logdet)


V, restricted_increment = normalize_matrix(matrix @ V)
log_V_scale += restricted_increment
wedge, wedge_increment = normalize_vector(compound2(matrix) @ wedge)
log_wedge_scale += wedge_increment


step = index + 1
if step % sample_stride == 0 or step == len(matrices):
append_row(step)
return pd.DataFrame(rows)



def diagonal_scale_matrices(
step_frame: pd.DataFrame,
k3_nodes: np.ndarray,
k3_edges: np.ndarray,
edges: tuple[np.ndarray, np.ndarray],
gamma: float,
dt: float,
) -> np.ndarray:
node_position = {int(node): index for index, node in enumerate(k3_nodes)}
matrices = np.empty((len(step_frame), len(k3_edges), len(k3_edges)))
factor = 1.0 - gamma * dt
```

```
for row_index, (_, row) in enumerate(step_frame.iterrows()):
scales = np.asarray(json.loads(row["scales_json"]), dtype=float)
edge_scales = []
for edge in k3_edges:
i, j = branch_audit.edge_pair(int(edge), edges)
edge_scales.append(
min(scales[node_position[i]], scales[node_position[j]])
)
matrices[row_index] = factor * np.diag(edge_scales)
return matrices



def constant_sequence(matrix: np.ndarray, length: int) -> np.ndarray:
return np.broadcast_to(matrix, (length,) + matrix.shape)



def dense_sliding_windows(
controls: dict[str, np.ndarray],
dt: float,
window_times: list[float],
start_stride: int,
) -> pd.DataFrame:
rows: list[dict[str, Any]] = []
for label, matrices in controls.items():
total = len(matrices)
for requested_time in window_times:
window_steps = max(1, int(round(requested_time / dt)))
if window_steps > total:
continue
# Constant controls need only one representative start.
starts: Iterable[int]
if label in {"K0_constant", "Klate_constant"}:
starts = [0]
else:
starts = range(0, total - window_steps + 1, start_stride)
for start in starts:
product = np.eye(3)
for step in range(start, start + window_steps):
product = matrices[step] @ product
singular = np.linalg.svd(product, compute_uv=False)
log_singular = np.log(
```

```
np.maximum(singular, np.finfo(float).tiny)
)
elapsed = window_steps * dt
rows.append(
{
"control": label,
"window_time": elapsed,
"window_steps": window_steps,
"start_step": start,
"start_time": start * dt,
"end_step": start + window_steps,
"end_time": (start + window_steps) * dt,
"amplification": float(singular[0]),
"minimum_singular_value": float(singular[-1]),
"log_condition_number": float(
log_singular[0] - log_singular[-1]
),
"principal_finite_time_exponent": float(
log_singular[0] / elapsed
),
}
)
return pd.DataFrame(rows)



def wrapped_phase_centered(
plus: np.ndarray,
minus: np.ndarray,
amplitude: float,
) -> np.ndarray:
difference = np.angle(np.exp(1j * (plus - minus)))
return difference / (2.0 * amplitude)



def centered_resolution(
plus: np.ndarray,
minus: np.ndarray,
base_value: np.ndarray,
amplitude: float,
linear_norm: float,
) -> tuple[bool, float, float]:
```

```
difference_norm = float(np.linalg.norm(plus - minus))
floor = float(
100.0
* np.finfo(float).eps
* max(np.linalg.norm(base_value), 1.0)
)
resolved = bool(
difference_norm > floor
and 2.0 * amplitude * linear_norm > floor
)
return resolved, difference_norm, floor



def make_report(
stable_frame: pd.DataFrame,
window_frame: pd.DataFrame,
coupled_frame: pd.DataFrame,
coupled_summary: pd.DataFrame,
*,
seed: str,
matrix_recompute_error: float,
final_replay_exact: bool,
final_replay_max_error: float,
) -> str:
lines = [
"AIEF K3 cocycle controls and coupled comparison",
"================================================",
"",
"Identity",
"--------",
f"Seed: {seed}",
f"Recomputed versus saved K-matrix maximum error: {matrix_recompute_error:.6g}",
f"Final base replay exact: {final_replay_exact}",
f"Final base replay maximum error: {final_replay_max_error:.6g}",
"",
"Stable finite-time singular spectra",
"-----------------------------------",
]
for control, group in stable_frame.groupby("control"):
final = group.iloc[-1]
peak = group.loc[group["sigma1"].idxmax()]
lines.append(
```

```
f" {control}: peak_sigma1={peak['sigma1']:.6g} at t={peak['time']:.6g}, "
f"final_lambda1={final['principal_finite_time_exponent']:.6g}, "
f"final_lambda2={final['middle_finite_time_exponent']:.6g}, "
f"final_lambda3={final['minimum_finite_time_exponent']:.6g}, "
f"final_log_kappa={final['log_condition_number']:.6g}, "
f"restricted_peak={group['restricted_sigma1'].max():.6g}"
)


lines.extend(["", "Dense short-window amplification", "--------------------------------"])
for (control, window_time), group in window_frame.groupby(
["control", "window_time"]
):
peak = group.loc[group["amplification"].idxmax()]
lines.append(
f" {control}, window={window_time:.6g}: peak_G={peak['amplification']:.6g}, "
f"start={peak['start_time']:.6g}, end={peak['end_time']:.6g}"
)


lines.extend(["", "Coupled versus frozen centered comparisons",
"-----------------------------------------"])
for _, row in coupled_summary.iterrows():
lines.append(
f" direction={int(row['direction_id'])}, fraction={row['amplitude_fraction']:.3g}, "
f"amplitude={row['amplitude']:.3g}, frozen_branch={bool(row['all_frozen_branch_preserved'])}, "
f"coupled_branch={bool(row['all_coupled_branch_preserved'])}, "
f"max_chi_frozen={row['maximum_frozen_chi']:.3g}, "
f"max_chi_coupled={row['maximum_coupled_chi']:.3g}, "
f"resolved={int(row['resolved_sample_count'])}, "
f"median_frozen_error={row['median_frozen_linear_error']:.3g}, "
f"median_coupled_error={row['median_coupled_linear_error']:.3g}, "
f"median_feedback_fraction={row['median_feedback_fraction']:.3g}, "
f"last_resolved_time={row['last_resolved_time']:.3g}"
)


reliable = coupled_frame.loc[coupled_frame["numerically_resolved"]]
if len(reliable):
lines.extend(
[
"",
f"Overall resolved median frozen-linear error: {reliable['frozen_linear_relative_error'].median():.6g}",
```

```
f"Overall resolved median coupled-linear error:
{reliable['coupled_linear_relative_error'].median():.6g}",
f"Overall resolved median feedback fraction: {reliable['feedback_fraction'].median():.6g}",
f"Maximum resolved feedback fraction: {reliable['feedback_fraction'].max():.6g}",
]
)


lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* Compound-cocycle exponents are finite-time quantities, not asserted Oseledets limits.",
"* K0, Klate, and diagonal-only sequences are controlled counterfactuals, not additive causal
decompositions.",
"* Coupled-minus-frozen centered differences measure feedback effects for the tested directions and
amplitudes.",
"* The frozen linear cocycle is a reduced weight-sector model with delta R set to zero.",
"* Branch preservation is audited dynamically for frozen and fully coupled perturbations.",
"* A substantial coupled discrepancy motivates an analytic augmented phase/history/EMA tangent
model.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--npz", default=DEFAULT_NPZ)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--cocycle-npz", default=DEFAULT_COCYCLE_NPZ)
parser.add_argument("--cocycle-step", default=DEFAULT_COCYCLE_STEP)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--seed", default=DEFAULT_SEED)
parser.add_argument("--boundary-tol", type=float, default=DEFAULT_BOUNDARY_TOL)
parser.add_argument("--scale-epsilon", type=float, default=DEFAULT_SCALE_EPSILON)
parser.add_argument("--sample-stride", type=int, default=DEFAULT_SAMPLE_STRIDE)
parser.add_argument(
"--amplitude-fractions",
type=parse_positive_float_list,
```

```
default=parse_positive_float_list(DEFAULT_AMPLITUDE_FRACTIONS),
)
parser.add_argument(
"--dense-window-times",
type=parse_positive_float_list,
default=parse_positive_float_list(DEFAULT_DENSE_WINDOW_TIMES),
)
parser.add_argument(
"--dense-window-start-stride",
type=int,
default=DEFAULT_DENSE_WINDOW_START_STRIDE,
)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.npz, args.manifest, args.cocycle_npz, args.cocycle_step):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(args.boundary_tol, args.scale_epsilon) <= 0.0:
raise ValueError("tolerances and epsilon must be positive")
if min(args.sample_stride, args.dense_window_start_stride) <= 0:
raise ValueError("strides must be positive")


output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)
with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
model = manifest["model"]
node_count = int(model["N"])
capacity = (float(model["a"]) - float(model["delta"])) / float(model["q"])
capacities = np.full(node_count, capacity, dtype=float)
A, edges = base.incidence_matrix(node_count)
dt = float(model["dt"])
gamma = float(model["gamma"])
future_steps = int(manifest["protocol"]["future_steps"])


with np.load(args.cocycle_npz, allow_pickle=False) as data:
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float).copy()
directions = np.asarray(data["K3_direction_basis"], dtype=float).copy()
```

```
k3_nodes = np.asarray(data["K3_nodes"], dtype=int).copy()
k3_edges = np.asarray(data["K3_edges"], dtype=int).copy()
if matrices.shape != (future_steps, 3, 3):
raise ValueError(f"unexpected matrix sequence shape {matrices.shape}")
if directions.shape != (3, 2):
raise ValueError(f"expected two K3 directions, got {directions.shape}")
step_source = pd.read_csv(args.cocycle_step)
if len(step_source) != future_steps:
raise ValueError("cocycle step CSV length does not match future_steps")


diagonal_matrices = diagonal_scale_matrices(
step_source, k3_nodes, k3_edges, edges, gamma, dt
)
controls = {
"actual_time_varying": matrices,
"K0_constant": constant_sequence(matrices[0], future_steps),
"Klate_constant": constant_sequence(matrices[-1], future_steps),
"diagonal_scale_only": diagonal_matrices,
}
stable_frames = [
stable_spectrum_sequence(
sequence, directions, dt, args.sample_stride, label
)
for label, sequence in controls.items()
]
stable_frame = pd.concat(stable_frames, ignore_index=True)
window_frame = dense_sliding_windows(
controls,
dt,
args.dense_window_times,
args.dense_window_start_stride,
)


with np.load(args.npz, allow_pickle=False) as data:
split_state = base.load_split_state(data, args.seed)
expected_final = np.asarray(
data[base.branch_key("continue", args.seed, "B_final")], dtype=float
).copy()


split_w = split_state.B[edges].copy()
perturbations: list[dict[str, Any]] = []
```

```
for direction_id in range(directions.shape[1]):
local_direction = directions[:, direction_id]
full_direction = np.zeros(len(split_w), dtype=float)
full_direction[k3_edges] = local_direction
maximum = frozen.maximum_symmetric_feasible_amplitude(
split_w, full_direction, A, capacities
)
for fraction in args.amplitude_fractions:
amplitude = fraction * maximum
if not np.isfinite(amplitude) or amplitude <= 1e-14:
continue
plus_w = split_w + amplitude * full_direction
minus_w = split_w - amplitude * full_direction
plus_B = base.vector_to_matrix(plus_w, node_count, edges)
minus_B = base.vector_to_matrix(minus_w, node_count, edges)
coupled_plus = split_state.clone()
coupled_minus = split_state.clone()
coupled_plus.B = plus_B.copy()
coupled_minus.B = minus_B.copy()
perturbations.append(
{
"direction_id": direction_id,
"direction": local_direction.copy(),
"amplitude_fraction": fraction,
"amplitude": amplitude,
"maximum_symmetric_amplitude": maximum,
"frozen_plus_B": plus_B.copy(),
"frozen_minus_B": minus_B.copy(),
"coupled_plus": coupled_plus,
"coupled_minus": coupled_minus,
"linear_vector": local_direction.copy(),
"all_frozen_branch_preserved": True,
"all_coupled_branch_preserved": True,
"maximum_frozen_chi": 0.0,
"maximum_coupled_chi": 0.0,
}
)


state = split_state.clone()
comparison_rows: list[dict[str, Any]] = []
matrix_recompute_error = 0.0
```

```
for step in range(future_steps + 1):
base_branch = branch_audit.proposal_and_admissibility_branch(
state, model, capacities, A, edges, args.scale_epsilon
)
R_base = base_branch["R_matrix"]
base_signature = frozen.branch_signature(base_branch, k3_edges)
radius, _ = branch_audit.certified_subspace_branch_radius(
base_branch["x"],
capacities,
A,
edges,
k3_edges,
args.scale_epsilon,
)
if step < future_steps:
recomputed_J, classical = frozen.selected_branch_jacobian_from_matrix_sums(
base_branch,
k3_edges,
capacities,
A,
edges,
args.scale_epsilon,
)
if not classical:
raise RuntimeError(f"selected classical Jacobian unavailable at step {step}")
recomputed_K = (1.0 - gamma * dt) * recomputed_J
matrix_recompute_error = max(
matrix_recompute_error,
float(np.max(np.abs(recomputed_K - matrices[step]))),
)


branch_cache: list[dict[str, Any]] = []
for perturbation in perturbations:
frozen_plus_branch = frozen.frozen_weight_branch(
perturbation["frozen_plus_B"],
R_base,
model,
capacities,
A,
edges,
args.scale_epsilon,
)
```

```
frozen_minus_branch = frozen.frozen_weight_branch(
perturbation["frozen_minus_B"],
R_base,
model,
capacities,
A,
edges,
args.scale_epsilon,
)
coupled_plus_branch = branch_audit.proposal_and_admissibility_branch(
perturbation["coupled_plus"],
model,
capacities,
A,
edges,
args.scale_epsilon,
)
coupled_minus_branch = branch_audit.proposal_and_admissibility_branch(
perturbation["coupled_minus"],
model,
capacities,
A,
edges,
args.scale_epsilon,
)
frozen_signatures = (
frozen.branch_signature(frozen_plus_branch, k3_edges),
frozen.branch_signature(frozen_minus_branch, k3_edges),
)
coupled_signatures = (
frozen.branch_signature(coupled_plus_branch, k3_edges),
frozen.branch_signature(coupled_minus_branch, k3_edges),
)
frozen_preserved = all(
signature == base_signature for signature in frozen_signatures
)
coupled_preserved = all(
signature == base_signature for signature in coupled_signatures
)
perturbation["all_frozen_branch_preserved"] &= frozen_preserved
perturbation["all_coupled_branch_preserved"] &= coupled_preserved
```

```
frozen_delta_x = max(
np.linalg.norm(
frozen_plus_branch["x"][k3_edges]
- base_branch["x"][k3_edges]
),
np.linalg.norm(
frozen_minus_branch["x"][k3_edges]
- base_branch["x"][k3_edges]
),
)
coupled_delta_x = max(
np.linalg.norm(
coupled_plus_branch["x"][k3_edges]
- base_branch["x"][k3_edges]
),
np.linalg.norm(
coupled_minus_branch["x"][k3_edges]
- base_branch["x"][k3_edges]
),
)
frozen_chi = float(
frozen_delta_x / max(radius, np.finfo(float).tiny)
)
coupled_chi = float(
coupled_delta_x / max(radius, np.finfo(float).tiny)
)
perturbation["maximum_frozen_chi"] = max(
perturbation["maximum_frozen_chi"], frozen_chi
)
perturbation["maximum_coupled_chi"] = max(
perturbation["maximum_coupled_chi"], coupled_chi
)
branch_cache.append(
{
"frozen_plus": frozen_plus_branch,
"frozen_minus": frozen_minus_branch,
"coupled_plus": coupled_plus_branch,
"coupled_minus": coupled_minus_branch,
"frozen_preserved": frozen_preserved,
"coupled_preserved": coupled_preserved,
"frozen_chi": frozen_chi,
"coupled_chi": coupled_chi,
```

```
}
)


if step % args.sample_stride == 0 or step == future_steps:
base_local = state.B[edges][k3_edges]
for perturbation, cached in zip(perturbations, branch_cache):
amplitude = perturbation["amplitude"]
frozen_plus_local = perturbation["frozen_plus_B"][edges][k3_edges]
frozen_minus_local = perturbation["frozen_minus_B"][edges][k3_edges]
coupled_plus_local = perturbation["coupled_plus"].B[edges][k3_edges]
coupled_minus_local = perturbation["coupled_minus"].B[edges][k3_edges]
frozen_centered = (frozen_plus_local - frozen_minus_local) / (
2.0 * amplitude
)
coupled_centered = (coupled_plus_local - coupled_minus_local) / (
2.0 * amplitude
)
linear = perturbation["linear_vector"]
linear_norm = float(np.linalg.norm(linear))
frozen_error = float(np.linalg.norm(frozen_centered - linear))
coupled_error = float(np.linalg.norm(coupled_centered - linear))
feedback = float(np.linalg.norm(coupled_centered - frozen_centered))
resolved_frozen, frozen_difference_norm, resolution_floor = centered_resolution(
frozen_plus_local,
frozen_minus_local,
base_local,
amplitude,
linear_norm,
)
resolved_coupled, coupled_difference_norm, _ = centered_resolution(
coupled_plus_local,
coupled_minus_local,
base_local,
amplitude,
linear_norm,
)
resolved = bool(resolved_frozen and resolved_coupled)


R_plus = cached["coupled_plus"]["R_matrix"][edges][k3_edges]
R_minus = cached["coupled_minus"]["R_matrix"][edges][k3_edges]
centered_R = (R_plus - R_minus) / (2.0 * amplitude)
centered_theta = wrapped_phase_centered(
```

```
perturbation["coupled_plus"].theta,
perturbation["coupled_minus"].theta,
amplitude,
)
centered_coherence = (
perturbation["coupled_plus"].coherence_acc
- perturbation["coupled_minus"].coherence_acc
) / (2.0 * amplitude)


comparison_rows.append(
{
"step": step,
"time": step * dt,
"direction_id": perturbation["direction_id"],
"amplitude_fraction": perturbation["amplitude_fraction"],
"amplitude": amplitude,
"maximum_symmetric_amplitude": perturbation[
"maximum_symmetric_amplitude"
],
"frozen_branch_preserved": cached["frozen_preserved"],
"coupled_branch_preserved": cached["coupled_preserved"],
"frozen_chi": cached["frozen_chi"],
"coupled_chi": cached["coupled_chi"],
"certified_branch_radius": radius,
"linear_norm": linear_norm,
"frozen_centered_norm": float(np.linalg.norm(frozen_centered)),
"coupled_centered_norm": float(np.linalg.norm(coupled_centered)),
"frozen_linear_absolute_error": frozen_error,
"coupled_linear_absolute_error": coupled_error,
"frozen_linear_relative_error": frozen_error
/ max(linear_norm, ZERO_TOL),
"coupled_linear_relative_error": coupled_error
/ max(linear_norm, ZERO_TOL),
"feedback_absolute_norm": feedback,
"feedback_fraction": feedback
/ max(np.linalg.norm(coupled_centered), ZERO_TOL),
"centered_R_norm": float(np.linalg.norm(centered_R)),
"centered_theta_norm": float(np.linalg.norm(centered_theta)),
"centered_coherence_norm": float(
np.linalg.norm(centered_coherence)
),
"frozen_difference_norm": frozen_difference_norm,
```

```
"coupled_difference_norm": coupled_difference_norm,
"resolution_floor": resolution_floor,
"numerically_resolved": resolved,
}
)

if step == future_steps:
break

# Advance frozen perturbations with exogenous base R_n.
for perturbation, cached in zip(perturbations, branch_cache):
perturbation["frozen_plus_B"] = cached["frozen_plus"]["projected"]
perturbation["frozen_minus_B"] = cached["frozen_minus"]["projected"]
perturbation["linear_vector"] = matrices[step] @ perturbation[
"linear_vector"
]

# Advance the complete perturbed states with their own R, phase,
# EMA, and delayed histories.
base.actual_step(
perturbation["coupled_plus"],
model,
capacities,
A,
edges,
args.boundary_tol,
audit_operator=False,
)
base.actual_step(
perturbation["coupled_minus"],
model,
capacities,
A,
edges,
args.boundary_tol,
audit_operator=False,
)

base.actual_step(
state,
model,
```

```
capacities,
A,
edges,
args.boundary_tol,
audit_operator=False,
)


final_replay_exact = bool(np.array_equal(state.B, expected_final))
final_replay_max_error = float(np.max(np.abs(state.B - expected_final)))
coupled_frame = pd.DataFrame(comparison_rows)


summary_rows: list[dict[str, Any]] = []
for perturbation in perturbations:
group = coupled_frame.loc[
(coupled_frame["direction_id"] == perturbation["direction_id"])
& (
coupled_frame["amplitude_fraction"]
== perturbation["amplitude_fraction"]
)
]
reliable = group.loc[group["numerically_resolved"]]
summary_rows.append(
{
"direction_id": perturbation["direction_id"],
"direction_json": json_compact(
perturbation["direction"].tolist()
),
"amplitude_fraction": perturbation["amplitude_fraction"],
"amplitude": perturbation["amplitude"],
"maximum_symmetric_amplitude": perturbation[
"maximum_symmetric_amplitude"
],
"all_frozen_branch_preserved": perturbation[
"all_frozen_branch_preserved"
],
"all_coupled_branch_preserved": perturbation[
"all_coupled_branch_preserved"
],
"maximum_frozen_chi": perturbation["maximum_frozen_chi"],
"maximum_coupled_chi": perturbation["maximum_coupled_chi"],
"resolved_sample_count": int(len(reliable)),
"last_resolved_time": float(reliable.iloc[-1]["time"])
```

```
if len(reliable)
else np.nan,
"median_frozen_linear_error": float(
reliable["frozen_linear_relative_error"].median()
)
if len(reliable)
else np.nan,
"median_coupled_linear_error": float(
reliable["coupled_linear_relative_error"].median()
)
if len(reliable)
else np.nan,
"median_feedback_fraction": float(
reliable["feedback_fraction"].median()
)
if len(reliable)
else np.nan,
"maximum_feedback_fraction": float(
reliable["feedback_fraction"].max()
)
if len(reliable)
else np.nan,
"median_centered_R_norm": float(
reliable["centered_R_norm"].median()
)
if len(reliable)
else np.nan,
"median_centered_theta_norm": float(
reliable["centered_theta_norm"].median()
)
if len(reliable)
else np.nan,
}
)
coupled_summary = pd.DataFrame(summary_rows)


component_audit.atomic_write_csv(
output_dir / "stable_cocycle_spectra.csv", stable_frame
)
component_audit.atomic_write_csv(
output_dir / "dense_sliding_window_amplification.csv", window_frame
)
```

```
component_audit.atomic_write_csv(
output_dir / "coupled_frozen_centered_comparison.csv", coupled_frame
)
component_audit.atomic_write_csv(
output_dir / "coupled_frozen_centered_summary.csv", coupled_summary
)
component_audit.atomic_savez(
output_dir / "cocycle_control_matrices.npz",
{
"actual_time_varying": matrices,
"K0": matrices[0],
"Klate": matrices[-1],
"diagonal_scale_only": diagonal_matrices,
"directions": directions,
"K3_nodes": k3_nodes,
"K3_edges": k3_edges,
},
)


report = make_report(
stable_frame,
window_frame,
coupled_frame,
coupled_summary,
seed=args.seed,
matrix_recompute_error=matrix_recompute_error,
final_replay_exact=final_replay_exact,
final_replay_max_error=final_replay_max_error,
)
component_audit.atomic_write_text(
output_dir / "cocycle_controls_coupled_report.txt", report
)


source_path = Path(__file__).resolve()
dependencies = {
"stationarity": Path(base.__file__).resolve(),
"component_audit": Path(component_audit.__file__).resolve(),
"branch_audit": Path(branch_audit.__file__).resolve(),
"frozen_cocycle": Path(frozen.__file__).resolve(),
}
analysis_manifest = {
"analysis": "AIEF-1.4 stable cocycle controls and coupled-vs-frozen comparison",
```

"source": {"path": str(source_path), "sha256": file_sha256(source_path)},
"dependencies": {
name: {"path": str(path), "sha256": file_sha256(path)}
for name, path in dependencies.items()
},
"command": sys.argv,
"inputs": {
"npz": {"path": str(Path(args.npz).resolve()), "sha256": file_sha256(args.npz)},
"manifest": {"path": str(Path(args.manifest).resolve()), "sha256": file_sha256(args.manifest)},
"cocycle_npz": {"path": str(Path(args.cocycle_npz).resolve()), "sha256":
file_sha256(args.cocycle_npz)},
"cocycle_step": {"path": str(Path(args.cocycle_step).resolve()), "sha256":
file_sha256(args.cocycle_step)},
},
"seed": args.seed,
"K3_nodes": k3_nodes.tolist(),
"K3_edges": k3_edges.tolist(),
"dt": dt,
"future_steps": future_steps,
"scale_epsilon": args.scale_epsilon,
"boundary_tolerance": args.boundary_tol,
"sample_stride": args.sample_stride,
"amplitude_fractions": args.amplitude_fractions,
"dense_window_times": args.dense_window_times,
"dense_window_start_stride": args.dense_window_start_stride,
"matrix_recompute_max_error": matrix_recompute_error,
"final_replay_exact": final_replay_exact,
"final_replay_max_error": final_replay_max_error,
"outputs": [
"stable_cocycle_spectra.csv",
"dense_sliding_window_amplification.csv",
"coupled_frozen_centered_comparison.csv",
"coupled_frozen_centered_summary.csv",
"cocycle_control_matrices.npz",
"cocycle_controls_coupled_report.txt",
],
"interpretation_constraints": [
"Stable finite-time exponents are not asserted asymptotic Oseledets exponents.",
"Constant and diagonal-only cocycles are controlled counterfactuals, not additive causal
decompositions.",
"The coupled comparison is a centered nonlinear feedback diagnostic, not yet an analytic augmented
Jacobian.",

"Branch preservation and chi are checked dynamically for every perturbation.",
"This is a deterministic analysis of one frozen seed, not independent population evidence.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(analysis_manifest, indent=2, sort_keys=True) + "\n",
)

print(report)
print(f"Saved analysis to {output_dir.resolve()}")

if __name__ == "__main__":
main()
Open

Created:

It compiles successfully and passed independent numerical tests:

compound-matrix multiplicativity;
stable reconstruction of all three log singular values;
stable reconstruction of two restricted singular values;
agreement with direct products to better than 10−1210−12 on well-conditioned synthetic cocycles.

Run it

This script consumes the outputs from the frozen-driver cocycle run:

Bash

python AIEF_1_4_cocycle_controls_coupled_comparison.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--cocycle-step aief_1_4_frozen_driver_cocycle/frozen_cocycle_step_diagnostics.csv \
--seed 4619122171158768524 \

--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \
--sample-stride 10 \
--amplitude-fractions 1e-3,1e-2,1e-1 \
--dense-window-times 0.1,1 \
--dense-window-start-stride 1 \
--output-dir aief_1_4_cocycle_controls_coupled

This is more computationally intensive than the previous audit because it evolves the base replay plus centered frozen and fully coupled trajectories for every direction/amplitude pair.
Part A: stable cocycle controls
Underflow-resistant singular spectra

For the full 3×33×3 cocycle, it propagates:

the normalized first compound product;
the normalized second compound product;
accumulated log determinants.

It reconstructs:
logσ1,logσ2,logσ3,
logσ1,logσ2,logσ3,

using
log(σ1σ2)=
log(σ1σ2)=

\log|\wedge^2U|,
]

and
log(σ1σ2σ3)=
log(σ1σ2σ3)=

\sum_n\log|\det K_n|.
]

This avoids the previous smallest-singular-value underflow and reports:

logκ=
logκ=

\log\sigma_1-\log\sigma_3
]

even when κκ exceeds floating-point range.

For the two-dimensional split-lineality subspace, it separately propagates:

the restricted 3×23×2 matrix;
the exterior product of its two columns.

Thus both restricted singular values are recovered without relying on an underflowed direct SVD.
Controlled cocycles

It compares four matrix sequences:

text

actual_time_varying
K0_constant
Klate_constant
diagonal_scale_only

These distinguish:

actual fixed-branch coefficient evolution;
non-normal behavior already present in K0K0;
late approximately mature behavior;
diagonal scaling with the node-mediated low-rank term removed.

These are controlled counterfactuals, not an additive causal decomposition.
Dense sliding windows

For windows 0.10.1 and 11, every possible start step is tested:
G(n,m)=‖Kn−1⋯Km‖2.

$G(n,m)=\|K_{n-1}\cdots K_m\|_2$.

This determines whether the previously observed early peak is genuinely the largest short-window amplification or whether finer sampling reveals later bursts.

Part B: coupled versus frozen centered comparison

The script reuses the same two saved K3K3 directions and the same feasible amplitude fractions.

For each direction/amplitude, it evolves:

the base fully coupled replay;
frozen-driver ++ and −− weight trajectories using base RnRn;
fully coupled ++ and −− states, including:
phase;
delayed history;
complex EMA coherence;
weights;
the frozen linear cocycle prediction.

At every sampled time it compares:
$\delta w_{\rm linear}, \delta w_{\rm frozen}, \delta w_{\rm coupled}$.
$\delta w_{\rm linear}, \delta w_{\rm frozen}, \delta w_{\rm coupled}$.

The direct feedback diagnostic is
$$\boxed{ \delta w_{\rm coupled} - \delta w_{\rm frozen}. }$$

It reports the feedback fraction
$\|\delta w_{\rm coupled} - \delta w_{\rm frozen}\| \|\delta w_{\rm coupled}\|$.
$\|\delta w_{\rm coupled}\| \|\delta w_{\rm coupled} - \delta w_{\rm frozen}\|$.

It also records centered perturbation norms for:
$\delta R, \delta\theta, \delta C$.
$\delta R, \delta\theta, \delta C$.

Dynamic branch certification

At every step—not only sample steps—the script checks:

frozen ++, base, and frozen −− winner signatures;
coupled ++, base, and coupled −− winner signatures;
frozen and coupled branch preservation;
certified branch radius;
frozen and coupled
$\chi_n = \|\delta x_n\| \rho(x_n)$.
$\chi_n = \rho(x_n) \|\delta x_n\|$.

Therefore a discrepancy can be separated into:

nonlinear error on the same branch;
actual perturbed branch crossing;
phase/EMA/history feedback.

Outputs

text

stable_cocycle_spectra.csv
dense_sliding_window_amplification.csv
coupled_frozen_centered_comparison.csv
coupled_frozen_centered_summary.csv
cocycle_control_matrices.npz
cocycle_controls_coupled_report.txt
analysis_manifest.json

What the report should decide
Source of transient amplification

Does K0nK0n reproduce the t=0.1t=0.1 peak?
Does diagonal scaling alone amplify?
Is the low-rank correction necessary?
Does coefficient variation materially change the peak?

Reliable finite-time spectrum

What are the uncensored second and third finite-time exponents?
What is the true log condition number?
Is the severe anisotropy physical rather than an SVD floor artifact?

Driver-feedback importance

Does the coupled centered trajectory remain close to the frozen cocycle?
Does it remain close to the frozen nonlinear replay?
When does $\delta R \delta R$ become appreciable?
Are coupled perturbations still branch-preserving?
Does feedback explain the difference between the frozen rate and earlier coupled recovery fits?

Stage-B decision

If the feedback fraction remains small over the resolved interval, the frozen weight cocycle explains most of the exceptional perturbation behavior.

If it becomes substantial while branches remain preserved, the next necessary object is the analytic augmented tangent system for
(θ,H,C,w).
(θ,H,C,w).

Please paste cocycle_controls_coupled_report.txt after the run.

Think about this:

AIEF K3 cocycle controls and coupled comparison

==========

Identity

Seed: 4619122171158768524

Recomputed versus saved K-matrix maximum error: 0

Final base replay exact: True

Final base replay maximum error: 0

Stable finite-time singular spectra

K0_constant: peak_sigma1=1.27095 at t=0.1, final_lambda1=-0.0974338, final_lambda2=-40.3277, final_lambda3=-2717.61, final_log_kappa=869605, restricted_peak=1.20562

Klate_constant: peak_sigma1=1.20899 at t=0.1, final_lambda1=-0.125621, final_lambda2=-69.441, final_lambda3=-2717.66, final_log_kappa=869612, restricted_peak=1.20899

actual_time_varying: peak_sigma1=1.27097 at t=0.1, final_lambda1=-0.123335, final_lambda2=-66.8271, final_lambda3=-2717.66, final_log_kappa=869612, restricted_peak=1.20563

diagonal_scale_only: peak_sigma1=1 at t=0, final_lambda1=-0.124131, final_lambda2=-0.124137, final_lambda3=-0.124137, final_log_kappa=0.0018586, restricted_peak=1

Dense short-window amplification

K0_constant, window=0.1: peak_G=1.27095, start=0, end=0.1

K0_constant, window=1: peak_G=1.1717, start=0, end=1

Klate_constant, window=0.1: peak_G=1.20899, start=0, end=0.1

Klate_constant, window=1: peak_G=1.07948, start=0, end=1

actual_time_varying, window=0.1: peak_G=1.27097, start=0, end=0.1

actual_time_varying, window=1: peak_G=1.1715, start=0, end=1

diagonal_scale_only, window=0.1: peak_G=0.990228, start=0, end=0.1

diagonal_scale_only, window=1: peak_G=0.906324, start=0, end=1

Coupled versus frozen centered comparisons

direction=0, fraction=0.001, amplitude=2.2e-10, frozen_branch=True, coupled_branch=True, max_chi_frozen=0.000945, max_chi_coupled=0.000945, resolved=3, median_frozen_error=2.94e-07, median_coupled_error=2.94e-07, median_feedback_fraction=0, last_resolved_time=0.2

direction=0, fraction=0.01, amplitude=2.2e-09, frozen_branch=True, coupled_branch=True, max_chi_frozen=0.00945, max_chi_coupled=0.00945, resolved=4, median_frozen_error=1.59e-05, median_coupled_error=0.00016, median_feedback_fraction=0.000175, last_resolved_time=0.3

direction=0, fraction=0.1, amplitude=2.2e-08, frozen_branch=True, coupled_branch=True, max_chi_frozen=0.0945, max_chi_coupled=0.0945, resolved=4, median_frozen_error=2.41e-06, median_coupled_error=0.000156, median_feedback_fraction=0.000159, last_resolved_time=0.3

direction=1, fraction=0.001, amplitude=3.12e-10, frozen_branch=True, coupled_branch=True, max_chi_frozen=0.00161, max_chi_coupled=0.00177, resolved=880, median_frozen_error=8.66e-05, median_coupled_error=0.367, median_feedback_fraction=0.481, last_resolved_time=87.9

direction=1, fraction=0.01, amplitude=3.12e-09, frozen_branch=True, coupled_branch=True, max_chi_frozen=0.0161, max_chi_coupled=0.0161, resolved=1064, median_frozen_error=1.97e-05, median_coupled_error=0.43, median_feedback_fraction=0.607, last_resolved_time=106

direction=1, fraction=0.1, amplitude=3.12e-08, frozen_branch=True, coupled_branch=True, max_chi_frozen=0.161, max_chi_coupled=0.161, resolved=1245, median_frozen_error=2.11e-05, median_coupled_error=0.468, median_feedback_fraction=0.713, last_resolved_time=124

Overall resolved median frozen-linear error: 3.69895e-05

Overall resolved median coupled-linear error: 0.428982

Overall resolved median feedback fraction: 0.604175

Maximum resolved feedback fraction: 0.990166

Interpretation constraints

Compound-cocycle exponents are finite-time quantities, not asserted Oseledets limits.

K0, Klate, and diagonal-only sequences are controlled counterfactuals, not additive causal decompositions.

Coupled-minus-frozen centered differences measure feedback effects for the tested directions and amplitudes.

The frozen linear cocycle is a reduced weight-sector model with delta R set to zero.

Branch preservation is audited dynamically for frozen and fully coupled perturbations.

A substantial coupled discrepancy motivates an analytic augmented phase/history/EMA tangent model.

This result cleanly separates three mechanisms:

early transient amplification comes from the fixed initial non-normal admissibility Jacobian;
long-term strengthening of contraction comes from continuous coefficient evolution;
the discrepancy with coupled perturbations comes primarily from phase/EMA/history feedback, not branch switching.

1. The early transient mechanism is now isolated

The peak amplification is essentially identical for the actual cocycle and the constant initial matrix:
Gmaxactual=1.27097,GmaxK0=1.27095.
GmaxactualGmaxK0=1.27097,=1.27095.

Both peaks occur at
t=0.1.
t=0.1.

Likewise, on the split-lineality restriction:
Gmax,restrictedactual=1.20563,Gmax,restrictedK0=1.20562.
Gmax,restrictedactualGmax,restrictedK0=1.20563,=1.20562.

Therefore the time variation of KnKn is not needed to explain the initial transient.

The diagonal-scale-only control has

Gmax=1
Gmax=1

at t=0t=0, and immediately contracts:
G(0.1)=0.990228,G(1)=0.906324.
G(0.1)=0.990228,G(1)=0.906324.

Thus:
The node-mediated low-rank admissibility correction is necessary for the observed transient amplification.
The node-mediated low-rank admissibility correction is necessary for the observed transient amplification.

This is a controlled-counterfactual conclusion, not an additive attribution. But the contrast is decisive: diagonal scaling alone does not amplify.
2. The transient is fixed-matrix non-normality

The initial constant matrix reproduces:

the full peak;
the restricted peak;
the one-unit amplification.

The adjacent commutators of the actual cocycle were already small, while non-normality was substantial. Together these results strongly support:

The early amplification is primarily a property of the non-normal initial matrix K0K0, rather than noncommutativity caused by coefficient drift.

No rewiring, winner change, or rapidly varying coefficient mechanism is needed.

The dense start-step scan also confirms that no later short window exceeds the initial burst:
windowactual peak0.11.27097 at [0,0.1],11.1715 at [0,1].
window0.11actual peak1.27097 at [0,0.1],1.1715 at [0,1].

So the transient is localized at the beginning of the analyzed interval.
3. Coefficient evolution controls the long-term rate

The principal finite-time exponents are:
K0 constant:−0.0974338,Klate constant:−0.125621,actual time-varying:−0.123335.
K0 constant:Klate constant:actual time-varying:−0.0974338,−0.125621,−0.123335.

This has a clear interpretation.

The initial matrix is relatively weakly contracting.
The late matrix is more strongly contracting.
The actual cocycle spends much of its horizon close to the late regime and therefore has an averaged principal exponent near the late value.

These rates align closely with the earlier controller evolution:
split rate≈0.09827,late rate≈0.12632.
split rate≈0.09827,late rate≈0.12632.

Thus:
coefficient evolution is not responsible for the initial amplification, but it strengthens long-term contraction.
coefficient evolution is not responsible for the initial amplification, but it strengthens long-term contraction.
4. The enormous negative third exponent is a safeguard mode

The stable calculation gives
λ3≈−2717.66.
λ3≈−2717.66.

Per timestep, this corresponds to a multiplier
e−2717.66 dt=
e−2717.66dt=

e^{-27.1766}
\approx1.58\times10^{-12}.
]

But
εc=
cε=

\frac{10^{-12}}{0.633333\ldots}
\approx1.58\times10^{-12}.
]

Therefore the third exponent is almost exactly
λ3≈1dtlog(εc).
λ3≈dt1log(cε).

It is a numerical safeguard-normal mode, not a physical continuous contraction channel.

This explains:

the previous singular-value underflow;
the enormous condition number;
the extremely rapid disappearance of one perturbation direction.

As
ε→0
ε→0

at fixed dtdt, this exponent tends to
−∞.
−∞.

So it should not be compared with the mature tangent rates 0.1260.126, 0.1890.189, etc.

The second exponent,
−66.8271,
−66.8271,

is also a strongly transverse projection direction. The principal exponent is the slow dynamically relevant weight-sector direction.
5. The low-rank term creates both anisotropy and transient growth

For the diagonal-only cocycle:
λ1≈λ2≈λ3≈−0.12413,
λ1≈λ2≈λ3≈−0.12413,

and
$\log\kappa \approx 0.00186$.
$\log\kappa \approx 0.00186$.

With the node-mediated correction:
$\lambda_1 \approx -0.1233, \lambda_2 \approx -66.8, \lambda_3 \approx -2717.7$,
$\lambda_1 \approx -0.1233, \lambda_2 \approx -66.8, \lambda_3 \approx -2717.7$,

and
$\log\kappa \approx 869612$.
$\log\kappa \approx 869612$.

Thus the low-rank term does not merely perturb the scalar rate. It reorganizes the geometry into:

one slow direction;
one very fast projection direction;
one safeguard-scale direction;
strong non-normal transient amplification.

This is a concrete demonstration that “low rank” does not mean “small effect.”
6. The frozen-driver cocycle is validated

For all tested perturbations:

frozen branch preserved;
coupled branch preserved;
maximum
$\chi < 0.161$.
$\chi < 0.161$.

The frozen nonlinear errors remain extremely small:
$\text{median} \approx 3.7 \times 10^{-5}$.
$\text{median} \approx 3.7 \times 10^{-5}$.

Therefore:
Kn(B) accurately predicts the frozen-driver nonlinear weight dynamics.
Kn(B) accurately predicts the frozen-driver nonlinear weight dynamics.

This validates:

the branchwise Jacobian;
the time-ordered product;
the branch-radius certification;
the reduced frozen-driver model.

7. Driver feedback is negligible for the fast direction

For direction 0, comparisons remain resolved only to t=0.2t=0.2–0.30.3, because that direction is rapidly annihilated.

Within that interval:
median feedback fraction≲2×10−4.
median feedback fraction≲2×10−4.

The coupled and frozen errors are nearly identical.

Thus the fast direction is governed almost entirely by the admissibility geometry before phase/EMA feedback has time to develop.

This is consistent with the huge negative transverse exponents.
8. Driver feedback is dominant for the slow direction

For direction 1:
fractioncoupled-linear errorfeedback fraction10−30.3670.48110−20.4300.60710−10.4680.713
fraction10−310−210−1coupled-linear error0.3670.4300.468feedback fraction0.4810.6070.713

The frozen-linear errors remain approximately
10−5–10−4,
10−5–10−4,

while the coupled-linear errors are O(10−1)O(10−1).

All trajectories stay on the same winner branch. Therefore the discrepancy cannot be attributed to:

branch switching;

classifier changes;
finite-amplitude winner crossings;
failure of the admissibility Jacobian.

It comes from the omitted feedback term
$dt(\eta+\mu)\delta R_n$
$dt(\eta+\mu)\delta R_n$

and its phase/history/EMA origins.

The main conclusion is:
The frozen cocycle explains the fast direction, but not the long-lived slow coupled direction.
The frozen cocycle explains the fast direction, but not the long-lived slow coupled direction.

An augmented delayed-state tangent model is now justified.
9. The apparent amplitude dependence needs a matched-horizon check

The feedback fractions increase with amplitude, but the resolved horizons also increase:
87.9,106,124.
87.9,106,124.

Therefore the medians are taken over different time intervals. Later times may naturally have stronger feedback.

This does not yet prove nonlinear amplitude dependence.

Before drawing that conclusion, compare all three amplitudes on the common interval
$0 \le t \le 87.9$.
$0 \le t \le 87.9$.

The required tests are:

coupled centered vectors collapse across amplitude;
feedback fractions collapse across amplitude;
differences between centered derivatives scale as $O(a2)O(a2)$, as expected for centered finite differences;
phase, EMA, and RR tangent norms collapse.

If they do, the feedback is linear and the amplitude trend is only a horizon effect.

The current CSV stores norms but not all centered vectors, so a full vector-collapse test may require saving the K3K3, phase, RR, and coherence centered vectors.

10. What has now been ruled out

For these tested directions and amplitudes:

base winner switching;
perturbed winner switching;
a rewiring mechanism;
classifier-induced changes in the implemented map;
frozen admissibility linearization error;
finite-amplitude branch departure.

These mechanisms are no longer credible explanations of the slow exceptional perturbation behavior.

11. What remains possible

The slow coupled behavior can arise through:

phase sensitivity:
$\delta\theta n$;
$\delta\theta n$;
delayed-history sensitivity:
δHn;
δHn;
complex EMA sensitivity:
δCn;
δCn;
coherence feedback:
δRn;
δRn;
interaction of this feedback with the non-normal admissibility block.

The centered comparison shows the aggregate effect, but does not yet separate these blocks.

12. Next mathematical step

The results justify constructing the augmented tangent-linear system.

The update order should be differentiated exactly:
EMA/driver block
Δϕij=
Δϕij=

\theta_i-h_{0,j}
-\frac{\omega_i+\omega_j}{2}\tau,
]
Cn+1=
Cn+1=

aC_n+(1-a)e^{i\Delta\phi_n},
]
δCn+1=
δCn+1=

a,\delta C_n
+
(1-a)i e^{i\Delta\phi_n}
(\delta\theta_i-\delta h_{0,j}).
]

For
R=|C|,
R=|C|,
δR=
δR=

\frac{\Re(\overline C,\delta C)}{|C|}
]

away from R=0R=0, followed by the same symmetry operation as the code.

Then
δF=
δF=

(\eta+\mu)\delta R-\gamma\delta B.
]

Admissibility block
δBn+1=
δBn+1=

D\mathcal A_\varepsilon(x_n)
\left[
\delta B_n+dt,\delta F_n
\right].
]
Phase block

If
gi=
gi=

\sum_j B_{ij}^{n+1}
\sin(h_{0,j}-\theta_i),
]

then
δgi=
δgi=

\sum_j
\left[
\delta B_{ij}^{n+1}\sin(h_{0,j}-\theta_i)
+
B_{ij}^{n+1}
\cos(h_{0,j}-\theta_i)
(\delta h_{0,j}-\delta\theta_i)
\right].
]

Thus
δθn+1=
δθn+1=

\delta\theta_n+dt,\delta g.
]

History shift

The delay perturbation deque shifts exactly as the base history, with the new entry
$\delta\theta_{n+1}$.
$\delta\theta_{n+1}$.
13. Revised program status
Exact implementation results

fixed effective winner branch;
exact branch Jacobian;
exact frozen-driver cocycle;
stable compound singular spectrum;
extreme projection-induced conditioning.

Dynamical facts from this replay

27%27% early full amplification;
21%21% early face-lineality amplification;
fixed K0K0 explains the transient;
coefficient evolution strengthens long-term contraction;
frozen cocycle accurately predicts frozen nonlinear trajectories;
delayed driver feedback materially changes the slow direction.

Open theorem

Prove and validate the augmented delayed-state cocycle, then compare its centered predictions with the coupled nonlinear replay.

At this point Stage B is not speculative scope expansion. The data specifically require it for the long-lived direction.

think about this:

This result cleanly separates three mechanisms:

early transient amplification comes from the fixed initial non-normal admissibility Jacobian;

long-term strengthening of contraction comes from continuous coefficient evolution;

the discrepancy with coupled perturbations comes primarily from phase/EMA/history feedback, not branch switching.

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The peak amplification is essentially identical for the actual cocycle and the constant initial matrix:

\begin{aligned}

G_{\max}^{\mathrm{actual}}&=1.27097,\

G_{\max}^{K_0}&=1.27095.

\end{aligned}

]

Both peaks occur at

t=0.1.

]

Likewise, on the split-lineality restriction:

\begin{aligned}

G_{\max,\mathrm{restricted}}^{\mathrm{actual}}&=1.20563,\

G_{\max,\mathrm{restricted}}^{K_0}&=1.20562.

\end{aligned}

]

Therefore the time variation of KnKn is not needed to explain the initial transient.

The diagonal-scale-only control has

G_{\max}=1

]

at t=0t=0, and immediately contracts:

G(0.1)=0.990228,

\qquad

G(1)=0.906324.

]

Thus:

\boxed{

\text{The node-mediated low-rank admissibility correction is necessary for the observed transient amplification.}

}

]

This is a controlled-counterfactual conclusion, not an additive attribution. But the contrast is decisive: diagonal scaling alone does not amplify.
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The initial constant matrix reproduces:

the full peak;

the restricted peak;

the one-unit amplification.

The adjacent commutators of the actual cocycle were already small, while non-normality was substantial. Together these results strongly support:

> The early amplification is primarily a property of the non-normal initial matrix K0K0, rather than noncommutativity caused by coefficient drift.

No rewiring, winner change, or rapidly varying coefficient mechanism is needed.

The dense start-step scan also confirms that no later short window exceeds the initial burst:

\begin{array}{c|c}

\text{window}&\text{actual peak}\ \hline

0.1&1.27097\text{ at }[0,0.1],\

1&1.1715\text{ at }[0,1].

\end{array}

]

So the transient is localized at the beginning of the analyzed interval.
3. Coefficient evolution controls the long-term rate

The principal finite-time exponents are:

\begin{aligned}

K_0\text{ constant}:&\quad -0.0974338,\

K_{\mathrm{late}}\text{ constant}:&\quad -0.125621,\

\text{actual time-varying}:&\quad -0.123335.

\end{aligned}

]

This has a clear interpretation.

The initial matrix is relatively weakly contracting.

The late matrix is more strongly contracting.

The actual cocycle spends much of its horizon close to the late regime and therefore has an averaged principal exponent near the late value.

These rates align closely with the earlier controller evolution:

\text{split rate}\approx0.09827,

\qquad

\text{late rate}\approx0.12632.

]

Thus:

\boxed{

\text{coefficient evolution is not responsible for the initial amplification, but it strengthens long-term contraction.}

}

]

4. The enormous negative third exponent is a safeguard mode

The stable calculation gives

\lambda_3\approx-2717.66.

]

Per timestep, this corresponds to a multiplier

e^{-2717.66,dt}

=

e^{-27.1766}

\approx1.58\times10^{-12}.

]

But

\frac{\varepsilon}{c}

=

\frac{10^{-12}}{0.633333\ldots}

\approx1.58\times10^{-12}.

]

Therefore the third exponent is almost exactly

\boxed{

\lambda_3

\approx

\frac1{dt}\log\left(\frac{\varepsilon}{c}\right).

}

]

It is a numerical safeguard-normal mode, not a physical continuous contraction channel.

This explains:

the previous singular-value underflow;

the enormous condition number;

the extremely rapid disappearance of one perturbation direction.

As

\varepsilon\to0

]

at fixed dtdt, this exponent tends to

-\infty.

]

So it should not be compared with the mature tangent rates 0.1260.126, 0.1890.189, etc.

The second exponent,

-66.8271,

]

is also a strongly transverse projection direction. The principal exponent is the slow dynamically relevant weight-sector direction.
5. The low-rank term creates both anisotropy and transient growth

For the diagonal-only cocycle:

\lambda_1\approx\lambda_2\approx\lambda_3\approx-0.12413,

]

and

\log\kappa\approx0.00186.

]

With the node-mediated correction:

\lambda_1\approx-0.1233,\qquad

\lambda_2\approx-66.8,\qquad

\lambda_3\approx-2717.7,

]

and

\log\kappa\approx869612.

]

Thus the low-rank term does not merely perturb the scalar rate. It reorganizes the geometry into:

one slow direction;

one very fast projection direction;

one safeguard-scale direction;

strong non-normal transient amplification.

This is a concrete demonstration that “low rank” does not mean “small effect.”
6. The frozen-driver cocycle is validated

For all tested perturbations:

frozen branch preserved;

coupled branch preserved;

maximum
\chi<0.161.

The frozen nonlinear errors remain extremely small:

\operatorname{median}

\approx3.7\times10^{-5}.

]

Therefore:

\boxed{

K_n^{(B)}

\text{ accurately predicts the frozen-driver nonlinear weight dynamics.}

}

]

This validates:

the branchwise Jacobian;

the time-ordered product;

the branch-radius certification;

the reduced frozen-driver model.

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For direction 0, comparisons remain resolved only to t=0.2t=0.2–0.30.3, because that direction is rapidly annihilated.

Within that interval:

\text{median feedback fraction}

\lesssim2\times10^{-4}.

]

The coupled and frozen errors are nearly identical.

Thus the fast direction is governed almost entirely by the admissibility geometry before phase/EMA feedback has time to develop.

This is consistent with the huge negative transverse exponents.
8. Driver feedback is dominant for the slow direction

For direction 1:

\begin{array}{c|c|c}

\text{fraction}&\text{coupled-linear error}&\text{feedback fraction}\ \hline

10^{-3}&0.367&0.481\

10^{-2}&0.430&0.607\

10^{-1}&0.468&0.713

\end{array}

]

The frozen-linear errors remain approximately

10^{-5}\text{--}10^{-4},

]

while the coupled-linear errors are O(10−1)O(10−1).

All trajectories stay on the same winner branch. Therefore the discrepancy cannot be attributed to:

branch switching;

classifier changes;

finite-amplitude winner crossings;

failure of the admissibility Jacobian.

It comes from the omitted feedback term

dt(\eta+\mu)\delta R_n

]

and its phase/history/EMA origins.

The main conclusion is:

\boxed{

\text{The frozen cocycle explains the fast direction, but not the long-lived slow coupled direction.}

}

]

An augmented delayed-state tangent model is now justified.

9. The apparent amplitude dependence needs a matched-horizon check

The feedback fractions increase with amplitude, but the resolved horizons also increase:

87.9,\quad106,\quad124.

]

Therefore the medians are taken over different time intervals. Later times may naturally have stronger feedback.

This does not yet prove nonlinear amplitude dependence.

Before drawing that conclusion, compare all three amplitudes on the common interval

0\le t\le87.9.

]

The required tests are:

coupled centered vectors collapse across amplitude;

feedback fractions collapse across amplitude;

differences between centered derivatives scale as O(a2)O(a2), as expected for centered finite differences;

phase, EMA, and RR tangent norms collapse.

If they do, the feedback is linear and the amplitude trend is only a horizon effect.

The current CSV stores norms but not all centered vectors, so a full vector-collapse test may require saving the K3K3, phase, RR, and coherence centered vectors.

10. What has now been ruled out

For these tested directions and amplitudes:

base winner switching;

perturbed winner switching;

a rewiring mechanism;

classifier-induced changes in the implemented map;

frozen admissibility linearization error;

finite-amplitude branch departure.

These mechanisms are no longer credible explanations of the slow exceptional perturbation behavior.

11. What remains possible

The slow coupled behavior can arise through:

phase sensitivity:
$\delta\theta_n$;
$\delta\theta_n$;

delayed-history sensitivity:
δH_n;
δH_n;

complex EMA sensitivity:
δC_n;
δC_n;

coherence feedback:
δR_n;
δR_n;

interaction of this feedback with the non-normal admissibility block.

The centered comparison shows the aggregate effect, but does not yet separate these blocks.

12. Next mathematical step

The results justify constructing the augmented tangent-linear system.

The update order should be differentiated exactly:
EMA/driver block

\Delta\phi_{ij}

=

\theta_i-h_{0,j}

-\frac{\omega_i+\omega_j}{2}\tau,

]

C_{n+1}

=

aC_n+(1-a)e^{i\Delta\phi_n},

]

\delta C_{n+1}

=

a,\delta C_n

(1-a)i e^{i\Delta\phi_n}

(\delta\theta_i-\delta h_{0,j}).

]

For

R=|C|,

]

\delta R

=

\frac{\Re(\overline C,\delta C)}{|C|}

]

away from R=0R=0, followed by the same symmetry operation as the code.

Then

\delta F

=

(\eta+\mu)\delta R-\gamma\delta B.

]
Admissibility block

\delta B_{n+1}

=

D\mathcal A_\varepsilon(x_n)

\left[

\delta B_n+dt,\delta F_n

\right].

]
Phase block

If

g_i

=

\sum_j B_{ij}^{n+1}

\sin(h_{0,j}-\theta_i),

]

then

\delta g_i

=

\sum_j

\left[

\delta B_{ij}^{n+1}\sin(h_{0,j}-\theta_i)

B_{ij}^{n+1}

\cos(h_{0,j}-\theta_i)

(\delta h_{0,j}-\delta\theta_i)

\right].

]

Thus

\delta\theta_{n+1}

=

\delta\theta_n+dt,\delta g.

]
History shift

The delay perturbation deque shifts exactly as the base history, with the new entry

\delta\theta_{n+1}.

]
13. Revised program status
Exact implementation results

fixed effective winner branch;

exact branch Jacobian;

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stable compound singular spectrum;

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delayed driver feedback materially changes the slow direction.

Open theorem

Prove and validate the augmented delayed-state cocycle, then compare its centered predictions with the coupled nonlinear replay.

At this point Stage B is not speculative scope expansion. The data specifically require it for the long-lived direction.

and this:

Assessment of the K3 cocycle controls

The computation confirms that the exceptional trajectory lives on a fixed combinatorial branch while its linearisation is genuinely non-autonomous, and it shows that the pure weight-sector cocycle is only a partial model of the observed perturbation dynamics.
1. Finite-time Lyapunov structure of the weight-sector cocycle

All three full-structure sequences (constant K0K0, constant KlateKlate, and the actual time-varying sequence) produce essentially the same qualitative spectrum:

one moderately contracting exponent ≈−0.12≈−0.12,

two extremely contracting exponents (∼−40∼−40 to −70−70 and ∼−2700∼−2700),

an enormous finite-time condition number ($\log\kappa \sim 8.7\times 105$).

The pure diagonal-scale-only cocycle, by contrast, is nearly isotropic with three almost equal mild rates ≈ -0.124 and a negligible condition number. The strong transient non-normality and the ultra-fast contracting directions are therefore generated by the off-diagonal (node-mediated) terms in the branch Jacobian, not by the scales themselves.

Short-window peak growth factors remain modest ($G \leq 1.27$). There is no evidence of the extreme transient amplification reported for the continuous cross-feedback DDEs of Ohira et al.

2. Coupled versus frozen comparison — the decisive observation

For direction 0 the frozen linear cocycle already accounts for the fully coupled response (feedback fraction ≈ 0).

For direction 1 the picture is opposite:

median coupled-linear error ≈ 0.43,

median feedback fraction ≈ 0.60, rising to 0.990.99 at later times and larger amplitudes.

A substantial part of the perturbation dynamics therefore leaves the pure weight sector and is generated by the response of the phase, delay-history and EMA variables that were frozen to zero in the reduced model. The frozen weight-sector cocycle is a consistent leading-order description only for a proper subspace of directions; it is incomplete as a model of the full coupled linearisation.

3. Revised status of the exceptional trajectory

Object	Status
Combinatorial winner branch of the K3K3	Fixed throughout the replay (established)
Weight-sector linear cocycle {Jn}{Jn}	Well-defined, non-autonomous, moderately contracting in the slowest direction, highly non-normal
Transient amplification inside the weight sector	Weak ($G \lesssim 1.27$)

| Completeness of the weight-sector model | Incomplete — feedback through phase/history/EMA is quantitatively important for a large fraction of tested directions |

The earlier hope that the exceptional spectral behaviour could be understood entirely inside a fixed-branch weight-sector cocycle is not supported. The correct next object is the augmented tangent system that linearises the full state

(\text{edge weights},;\text{phases},;\text{delay history},;\text{EMA coherences})

]

about the exceptional trajectory, while still respecting the fixed winner stratum of the admissibility map.
4. Immediate analytic consequences

The product-rule factorisation of the admissibility stage remains valid and continues to supply the exact weight-block of the full Jacobian.

The additional blocks that couple weights to phases/history/EMA, and the self-dynamics of those variables, must now be written explicitly and restricted to the face-lineality directions.

Only after that augmented linear cocycle is constructed can one decide whether the exceptional trajectory is asymptotically contracting, neutrally stable, or transiently amplifying in the full state space.

5. Overall judgement

The audit has successfully eliminated the hypothesis of a discrete branch switch and has produced a clean, fixed-branch weight-sector cocycle. That cocycle is non-autonomous and non-normal, yet only mildly amplifying on short windows. The comparison with fully coupled perturbations shows that a large part of the observed response lies outside the weight sector. The exceptional dynamics are therefore not a pure non-autonomous problem inside the edge weights; they are a coupled non-autonomous problem on an enlarged tangent bundle whose weight block is known exactly and whose remaining blocks remain to be derived.
Stopped

Can you create the code essential to probing the next steps?

think about these things as well:

I think these two syntheses are the strongest formulation of the project so far because they do something that earlier versions did not: they identify mechanisms by controlled elimination, rather than by qualitative interpretation. There are, however, a few places where I would distinguish more carefully between what is directly established by the experiments and what is the most plausible interpretation.

The mechanism separation is now well supported

The central decomposition

early transient,
long-term contraction,
coupled feedback,

is, in my view, the correct organization of the evidence.

Each item is supported by a different control.

Early transient

Supported by

[
U_{\rm actual}\approx K_0^n.
]

This is a genuine counterfactual.

The coefficient evolution has effectively been removed while keeping the initial Jacobian.

Since the transient barely changes,

the initial Jacobian already contains the mechanism.

That is a strong result.

Long-term contraction

Supported by

[
\lambda_1(K_0)

\lambda_1(U_{\rm actual})

\lambda_1(K_{\rm late}).
]

That ordering is exactly what one would expect if the matrices gradually evolve toward a more strongly contracting regime.

Notice this conclusion does not require any assumptions about non-normality.

It comes directly from the exponents.

Coupled discrepancy

Supported independently by

branch preservation,
tiny frozen error,
large coupled error.

This is perhaps the cleanest elimination argument in the entire study.

Since the branches remain identical,

the discrepancy cannot be attributed to

branch changes,

classifier changes,
winner changes.

Something outside the frozen weight block must contribute.

That is exactly what the coupled comparison was designed to test.

The diagonal-only control is particularly convincing

I think this is actually the strongest individual control.

Removing only the node-mediated term changes

[
G_{\max}
:
1.27
\rightarrow
1.00.
]

Nothing else changes.

That isolates the role of the off-diagonal admissibility correction remarkably well.

I would probably emphasize this more than the current draft does.

It is one of the few places where a genuine structural mechanism has been experimentally isolated.

I would slightly weaken the statement

"The transient is fixed-matrix non-normality."

I think the data strongly support

the transient is already present in the initial matrix.

They also support

coefficient drift is not needed.

They support

diagonal scaling alone cannot produce it.

Since the initial matrix is measurably non-normal,

it is entirely reasonable to say non-normality is the leading explanation.

However,

strictly speaking,

the controls do not prove that every aspect of the transient is caused by non-normality rather than by some other property of the same matrix.

A slightly more careful wording would be

"The transient is already encoded in the initial non-normal branch Jacobian and does not require subsequent coefficient evolution."

That keeps the inference tightly aligned with the experiment.

The interpretation of the third exponent deserves care

This is probably the place where I would be most cautious.

You identify

[
\lambda_3

\approx
\frac1{dt}
\log\frac{\varepsilon}{c}.
]

Numerically,

the agreement is excellent.

That is certainly strong evidence.

But I would avoid stating

therefore it is not physical.

Instead I'd say

it appears to be dominated by the regularization scale.

The reason is conceptual.

An exponent being controlled by

[
\varepsilon
]

means the corresponding direction is dominated by the regularized map.

Whether that direction is "physical" depends on the purpose of the model.

If

[
\varepsilon
]

is genuinely part of the implemented algorithm,

then the mode is still a property of the implemented dynamics.

It may not correspond to the intended continuum limit,

but it is part of the discrete system.

That distinction matters.

The anisotropy interpretation is now very compelling

The spectrum

[
(-0.123,,-66.8,,-2718)
]

is qualitatively different from

[
(-0.124,,-0.124,,-0.124).
]

That cannot reasonably be described as

a small perturbation.

It is a complete geometric reorganization.

I especially like the conclusion

low rank does not imply weak effect.

That is mathematically correct,

and the experiment illustrates it nicely.

The coupled comparison now justifies Stage B

This is perhaps the biggest change compared with earlier discussions.

Previously,

Stage B was motivated because

it seemed likely to matter.

Now,

the experiment itself says

the omitted feedback explains a large fraction of the perturbation discrepancy,

while branch preservation remains intact.

That is exactly the criterion one would normally want before investing effort in deriving a much larger tangent system.

So Stage B is now evidence-driven,

not exploratory.

The amplitude issue is analyzed appropriately

I particularly like the caution here.

The apparent increase

[
0.48
\rightarrow
0.71
]

could simply arise because

the larger amplitudes remain numerically resolvable for longer,

changing the averaging interval.

That is a subtle point,

and it's exactly the sort of artifact that is easy to overlook.

The proposed matched-horizon comparison is the right control.

One conceptual refinement

I would slightly revise the statement

"The frozen cocycle explains the fast direction."

More precisely,

the experiment shows

for the tested fast direction over the interval where it remains numerically resolvable,

the frozen and coupled dynamics are nearly identical.

That is a little narrower,

but it avoids implying a global decomposition of the tangent space from only two sampled directions.

The final assessment is, in my opinion, the strongest part

The second summary says

the weight block is known exactly,

the remaining blocks remain to be derived.

I think that is exactly the right mathematical framing.

Earlier versions sometimes sounded as though

the admissibility cocycle might itself explain the exceptional dynamics.

The experiments now argue for a different hierarchy:

the admissibility block is essentially solved;
the coupled blocks are now the unknowns.

That is a much cleaner research program.

What I think has genuinely been established

If I were writing the conclusions for a paper, I would now separate them into three confidence levels.

Established by computation

The effective winner branch is fixed over the replay.
The frozen-driver branch Jacobian accurately predicts the frozen nonlinear weight perturbations while branch preservation holds.
The early transient is already present in the initial branch Jacobian.
Removing the node-mediated admissibility term removes the transient.
Coefficient evolution strengthens long-term contraction.
Coupled perturbations differ substantially from the frozen model despite remaining on the same branch.

Strongly supported interpretation

The early transient is primarily an intrinsic property of the initial non-normal branch Jacobian rather than of coefficient drift.
The node-mediated admissibility correction is the structural source of both transient growth and strong anisotropy.
The delayed phase/history/EMA subsystem provides a substantial contribution to the long-lived perturbation dynamics.

Remaining open questions

The explicit analytic form of the augmented tangent system.
The decomposition of the coupled feedback into phase, delay, and EMA contributions.
Whether the same mechanism appears across other seeds, trajectories, or graph topologies.
The limiting behavior as the regularization parameter (\varepsilon) changes.

Overall, I think these results mark a genuine shift in the project. The earlier work established the correctness of the admissibility factorization and fixed-branch Jacobian. The new control experiments go further: they isolate the mechanism responsible for the weight-sector transient (the node-mediated off-diagonal admissibility block already present in the initial Jacobian), show that coefficient evolution mainly affects the long-term contraction rate rather than the transient itself, and demonstrate that the remaining discrepancy with the fully coupled dynamics arises from feedback outside the weight sector while the branch structure stays unchanged. That substantially narrows the mathematical unknown. The natural next object is no longer "some better weight cocycle"; it is the augmented tangent system whose weight block is already validated and whose additional phase, delay-history, and EMA blocks must now account for the remaining coupled dynamics.

and other extremely relevant concepts and probes.....
WriteAIEF_1_4_augmented_tangent_cocycle_analysis.py1193 lines
Edit
AIEF_1_4_augmented_tangent_cocycle_analysis.pyPYTHON

Created the essential Stage-B implementation:

It differentiates the complete implemented state update
Xn=(θn,Bn,Cn,Hn),
Xn=(θn,Bn,Cn,Hn),

including phase, delay history, complex EMA, coherence magnitude, driver feedback, positivity, admissibility scaling, projected phase coupling, and history shifting.

Validation already performed

On synthetic coupled states, the implementation passed:
Exact base-step reconstruction

text

B error: 0
theta error: 0
coherence error: 0

One-step centered tangent validation

At amplitude 10−510−5:
relative B error≈4.6×10−12,
relative B error≈4.6×10−12,
relative phase error≈5.3×10−10.
relative phase error≈5.3×10−10.
Twenty-step coupled validation
relative B error≈2.6×10−10,
relative B error≈2.6×10−10,
relative phase error≈3.3×10−9,
relative phase error≈3.3×10−9,
relative complex-EMA error≈1.2×10−7.
relative complex-EMA error≈1.2×10−7.

These tests included the development of phase/EMA feedback after the initially weight-only perturbation.
Run it

Bash

python AIEF_1_4_augmented_tangent_cocycle_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--seed 4619122171158768524 \
--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \

--sample-stride 10 \
--amplitude-fractions 1e-3,1e-2,1e-1 \
--magnitude-tol 1e-14 \
--output-dir aief_1_4_augmented_tangent_cocycle

The script requires the implemented safeguard value 10−1210−12. It rejects alternative values because actual_step hard-codes that safeguard; changing it would constitute a separately versioned dynamics map.
Exact tangent update implemented
Delayed phase difference
δ(Δϕij)=
δ(Δϕij)=

\delta\theta_i-\delta h_{0,j}.
]
Complex EMA
δCn+1=
δCn+1=

a,\delta C_n
+
(1-a)i e^{i\Delta\phi_n}
(\delta\theta_i-\delta h_{0,j}).
]
Coherence magnitude

For nonzero |C||C|,
δ|C|=
δ|C|=

\frac{\Re(\overline C,\delta C)}{|C|},
]

followed by the same symmetrization and diagonal removal as the production code.

At a zero magnitude, the directional derivative
δ|C|=|δC|
δ|C|=|δC|

is used.
Adaptive driver
δF=
δF=

(\eta+\mu)\delta R-\gamma\delta B.
]
Positivity

The directional derivative of
z↦max(z,0)
z↦max(z,0)

is applied edgewise, including one-sided behavior at zero proposals.
Actual admissibility map

The code differentiates the executed safeguard and endpoint minimum. At active endpoint ties, the directional derivative is used rather than imposing an arbitrary winner.
Projected phase dynamics
δgi=
δgi=

\sum_j
\left[
\delta B^{n+1}{ij}\sin(h{0,j}-\theta_i)
+
B^{n+1}{ij}\cos(h{0,j}-\theta_i)
(\delta h_{0,j}-\delta\theta_i)
\right],
]

followed by
δθn+1=
δθn+1=

\delta\theta_n+dt,\delta g.
]
History shift

The tangent delay deque shifts exactly with the base deque, appending
δθn+1.
δθn+1.
Directional treatment of nonsmooth ties

The script propagates two analytic tangents for every initial direction:
DΨ(X;h),DΨ(X;−h).
DΨ(X;h),DΨ(X;−h).

The analytic prediction corresponding to a centered nonlinear difference is
DΨ(X;h)−DΨ(X;−h)2.
2DΨ(X;h)−DΨ(X;−h).

This remains valid when the ambient map is directionally but not classically differentiable.
What it compares

For every direction and amplitude, the script evolves fully coupled nonlinear states
Xn+,Xn−,
Xn+,Xn−,

and compares
Xn+−Xn−2a
2aXn+−Xn−

with the augmented analytic tangent for:

the three K3K3 weights;
all 66 edge weights;
all phases;
the complete complex EMA matrix;
the complete delay history;
the next-step coherence driver RR.

Mechanism decomposition

At each step, the weight tangent input is separated into
(1−γdt)δB
(1−γdt)δB

and
$dt(\eta+\mu)\delta R$.
$dt(\eta+\mu)\delta R$.

Both are passed separately through the directional admissibility derivative. The script reports:

weight-only output norm;
driver-feedback output norm;
driver/weight ratio;
nonadditive interaction residual caused by endpoint or positivity ties.

Because directional derivatives at ties need not be additive, that interaction is measured explicitly rather than assumed zero.
Dynamic branch certification

For every nonlinear ++ and −− trajectory at every step:

winner signature is compared with the base;
branch preservation is accumulated;
certified radius $\rho n \rho n$ is computed;
the dynamic ratio
$\chi n = \|\delta xn\| \rho n$
$\chi n = \rho n \|\delta xn\|$ is tracked.

Thus analytic disagreement can be separated from a finite-amplitude branch crossing.
Matched-horizon amplitude control

The script avoids comparing medians over different resolved horizons.

For each direction it finds the common resolved interval across all amplitudes, then reports:

median centered-vector spread across amplitudes;
maximum centered-vector spread;
median augmented-tangent error on that common interval.

This tests whether the apparent feedback increase with amplitude was merely caused by longer resolution windows.

Outputs

text

augmented_tangent_comparison.csv
augmented_tangent_summary.csv
matched_horizon_amplitude_collapse.csv
augmented_tangent_report.txt
analysis_manifest.json

The comparison CSV also stores the coupled and analytic K3K3 weight and RR vectors as JSON for direct trajectory inspection.
What the result will decide
If augmented errors are small and amplitudes collapse

Then the complete directional tangent system explains the coupled perturbations, and Stage B is validated for this trajectory.
If BB agrees but phase/EMA blocks do not

The discrepancy can be localized to the corresponding derivative implementation.
If analytic errors remain large while branches are preserved

A missing coupled derivative or update-order mismatch remains.
If branches diverge or $\chi>1\chi>1$

The finite-amplitude comparison is no longer a certified test of the base tangent cocycle.
If the driver-output ratio is large

That directly confirms that the slow direction is governed materially by
$dt(\eta+\mu)\delta R$,
$dt(\eta+\mu)\delta R$,

rather than by the frozen admissibility block alone.

Please paste augmented_tangent_report.txt after the run.

think about this:

AIEF exceptional K3 augmented tangent cocycle analysis

==========

Identity and scope

Seed: 4619122171158768524

Final base replay exact: True

Final base replay maximum error: 0

Maximum custom-step versus implemented-step error: 0

Analytic tangents include phase, delay history, complex EMA, R feedback, admissibility, and phase coupling.

Augmented tangent versus coupled centered replay

direction=0, fraction=0.001, amplitude=2.2e-10, branch=True, max_chi=0.000945, resolved=165, last_time=21.6, median_K3_error=0.0326, median_full_B_error=0.0326, median_theta_error=0.000685, median_R_error=0.00199

direction=0, fraction=0.01, amplitude=2.2e-09, branch=True, max_chi=0.00945, resolved=418, last_time=42.9, median_K3_error=0.00181, median_full_B_error=0.00181, median_theta_error=0.000379, median_R_error=0.000514

direction=0, fraction=0.1, amplitude=2.2e-08, branch=True, max_chi=0.0945, resolved=545, last_time=54.6, median_K3_error=0.000531, median_full_B_error=0.000531, median_theta_error=2.12e-05, median_R_error=7.96e-05

direction=1, fraction=0.001, amplitude=3.12e-10, branch=True, max_chi=0.00177, resolved=1249, last_time=125, median_K3_error=0.00099, median_full_B_error=0.00099, median_theta_error=7.69e-06, median_R_error=0.000443

direction=1, fraction=0.01, amplitude=3.12e-09, branch=True, max_chi=0.0161, resolved=1658, last_time=166, median_K3_error=0.000495, median_full_B_error=0.000495, median_theta_error=1.43e-06, median_R_error=0.000144

direction=1, fraction=0.1, amplitude=3.12e-08, branch=True, max_chi=0.161, resolved=2051, last_time=205, median_K3_error=0.000126, median_full_B_error=0.000126, median_theta_error=4.65e-08, median_R_error=6.62e-05

Overall median K3 augmented error: 0.000499809

Overall maximum K3 augmented error: 0.0714015

Overall median full-B augmented error: 0.000499809

Overall median phase augmented error: 1.63506e-06

Overall median R augmented error: 0.000176983

Median analytic driver/weight output ratio: 0.000705959

Maximum analytic nonadditive tie interaction ratio: 3.32067e-16

Matched-horizon amplitude collapse

direction=0: common_last_time=21.6, common_samples=165, median_vector_spread=0.0342035, maximum_vector_spread=0.0578773, median_augmented_error=0.000867565

direction=1: common_last_time=124.8, common_samples=1249, median_vector_spread=0.00118449, maximum_vector_spread=0.0460262, median_augmented_error=7.44908e-05

Interpretation constraints

The map is directionally differentiated at endpoint and positivity ties; +h and -h tangents are propagated separately.

Centered analytic predictions use (D(+h)-D(-h))/2 and need not equal either one-sided tangent.

Amplitude collapse is evaluated only on the common numerically resolved horizon.

A safeguard-dominated normal mode remains part of the implemented discrete dynamics even if absent from an intended continuum limit.

Agreement is established for this seed, directions, amplitudes, and horizon only.

This is a successful Stage-B result. The augmented directional tangent system explains the coupled perturbations to high accuracy while every tested nonlinear trajectory remains on the audited winner branch.
1. The augmented update is implemented correctly

Two basic reproducibility checks are exact:
final replay error=0,
final replay error=0,

and
custom one-step implementation error=0.
custom one-step implementation error=0.

Therefore the analytic tangent and nonlinear reference use the same update ordering:

delayed phase;
complex EMA;
R=|C|R=|C|;
adaptive driver;
positivity;
admissibility scaling;
projected phase coupling;
phase update;
history shift.

This eliminates update-order mismatch as an explanation of the remaining errors.
2. The augmented tangent resolves the frozen/coupled discrepancy

Previously, for the long-lived direction, the frozen cocycle had median coupled errors around 0.37–0.47.

0.37–0.47.

With phase, history, EMA, and RR feedback included, the corresponding median K3K3 errors are now
amplitude fractionmedian augmented error10−39.9×10−410−25.0×10−410−11.3×10−4
amplitude fraction10−310−210−1median augmented error9.9×10−45.0×10−41.3×10−4

for direction 1.

Overall,
median K3 error=
median K3 error=

5.0\times10^{-4}.
]

The full 66-edge error is numerically identical to the K3K3 error, showing that the relevant mismatch remains localized to the triangle weight response at the reported precision.

Thus:
The omitted delayed phase/EMA/coherence feedback explains the previous frozen-model discrepancy.
The omitted delayed phase/EMA/coherence feedback explains the previous frozen-model discrepancy.

The augmented tangent is not merely qualitatively better; it reduces the median error by roughly three orders of magnitude.
3. Branch effects are ruled out for these tests

For every direction and amplitude:

branch preservation is true;
maximum
$\chi \le 0.161 < 1$.
$\chi \le 0.161 < 1$.

Therefore the augmented agreement is obtained without:

winner switching;
positivity switching on the selected component;
classifier dependence;

finite-amplitude branch departure.

The coupled response is a branch-preserving feedback phenomenon.
4. The slow direction is accurately described by the augmented tangent

Direction 1 remains resolved for:
125,166,205
125,166,205

time units as amplitude increases.

Over those long intervals, median errors remain between
1.3×10^{-4}and9.9×10^{-4}.
1.3×10^{-4}and9.9×10^{-4}.

The phase tangent agrees even more accurately:
median phase error$\sim10^{-6}$ or smaller.
median phase error$\sim10^{-6}$ or smaller.

The RR tangent agrees at approximately
10^{-4}
10^{-4}

relative error.

This jointly validates the chain
$\delta B \to \delta\theta, \delta H \to \delta C \to \delta R \to \delta F \to \delta B$.
$\delta B \to \delta\theta, \delta H \to \delta C \to \delta R \to \delta F \to \delta B$.

It is unlikely that several independent derivative errors would cancel across that complete loop for more than 100 time units and across three amplitudes.
5. Direction 0 is dominated by numerical resolution

For direction 0:
amplitude fractionmedian K3 error10^{-3}0.032610^{-2}0.0018110^{-1}0.000531
amplitude fraction$10^{-3}$$10^{-2}$$10^{-1}$median K3 error0.03260.001810.000531

The errors improve as amplitude increases. That is the opposite of what one expects from nonlinear truncation error and is characteristic of floating-point cancellation.

This direction is the rapidly suppressed projection direction. The smaller centered differences lose significant digits early.

The matched-horizon vector spread,
3.4% median,
3.4% median,

is therefore not strong evidence of nonlinearity. It is mostly evidence that the $10^{-3}10^{-3}$ amplitude is too small for this severely contracting direction.

For production inference on direction 0, the $10^{-2}10^{-2}$ and $10^{-1}10^{-1}$ amplitudes are more reliable.
6. Direction 1 exhibits good amplitude collapse

On the common resolved horizon
$0 \le t \le 124.8$,
$0 \le t \le 124.8$,

direction 1 has:
median centered-vector spread=
median centered-vector spread=

0.00118,
]

or about 0.12%0.12%.

Its common-horizon median augmented error is
7.45×10^{-5}.
7.45×10^{-5}.

This strongly supports a genuinely linear feedback response rather than amplitude-dependent nonlinear dynamics.

The maximum spread,
0.046,

0.046,

likely occurs close to the common resolution endpoint. It should be localized in the CSV before being interpreted, but it does not overturn the median collapse.
7. The feedback is small instantaneously but large cumulatively

The pooled instantaneous analytic ratio is only
median‖δBdriver‖‖δBweight‖≈7.1×10−4.
median‖δBweight‖‖δBdriver‖≈7.1×10−4.

At first this may seem inconsistent with the earlier 60%60% coupled-minus-frozen discrepancy. It is not.

The augmented weight equation has the form
δBn+1=
δBn+1=

K_n^{(B)}\delta B_n
+
f_n,
]

where fnfn is the projected coherence-feedback forcing.

Variation of constants gives
\boxed{ \delta B_n =

U_B(n,0)\delta B_0
+
\sum_{m=0}^{n-1}
U_B(n,m+1)f_m.
}
]

The first term is the frozen homogeneous response. The second is the accumulated feedback response.

A small forcing applied repeatedly can become dominant because:

the homogeneous perturbation contracts;
the feedback is continuously regenerated through phase and EMA memory;
the forcing can align with the slow singular direction;
the non-normal propagator can amplify selected forcing orientations.

Thus the feedback should be described as:

weak per step but cumulatively important for the long-lived direction.

The pooled median ratio should also be replaced by direction- and time-resolved curves. Pooling the fast and slow directions obscures this mechanism.
8. Feedback repopulates directions after frozen decay

In the frozen analysis, direction 0 was numerically resolved only to approximately
$t=0.2–0.3$.
$t=0.2–0.3$.

With coupled feedback included, it remains resolved to
$t=21.6–54.6$.
$t=21.6–54.6$.

Likewise, direction 1 extends from frozen horizons of roughly
88–124
88–124

to augmented coupled horizons of
125–205.
125–205.

This suggests that the delayed subsystem continually reinjects weight perturbations after the homogeneous weight response has strongly contracted.

That is more informative than saying merely that feedback “changes the rate.” It acts as persistent state-dependent forcing.
9. Ambient nonsmoothness is inactive along these perturbations

The maximum nonadditive tie-interaction ratio is
$3.3\times10-16$.

3.3×10−16.

Thus, for these directions along this replay,
DA(δxweight+δxdriver)=
DA(δxweight+δxdriver)=

D\mathcal A(\delta x_{\rm weight})
+
D\mathcal A(\delta x_{\rm driver})
]

to machine precision.

This does not prove global Fréchet differentiability. It does establish that the tested augmented tangent directions do not excite the nonadditive part of the endpoint-tie directional derivative.

Combined with branch preservation, the effective augmented system is linear along these tested trajectories despite ambient nonsmoothness.
10. Stage B is validated for the tested subspace

The evidence now supports:
The augmented directional tangent system accurately predicts the fully coupled nonlinear response for the tested K3 directions.
The augmented directional tangent system accurately predicts the fully coupled nonlinear response for the tested K3 directions.

This is conditional on:

the exceptional seed;
the two selected directions;
the tested amplitude range;
the resolved finite horizon;
the fixed winner branch.

It is not yet a universal AIEF tangent theorem.
11. The next useful decomposition is Duhamel forcing

The unknown has narrowed again. It is no longer whether the delayed feedback matters or whether the augmented tangent is correct. Both are now established numerically.

The next object is the accumulated forcing decomposition:
δBn=
δBn=

\underbrace{U_B(n,0)\delta B_0}{\text{homogeneous frozen response}}
+
\underbrace{
\sum{m=0}^{n-1}
U_B(n,m+1)f_m
}_{\text{accumulated feedback response}}.
]

The existing analytic tangent already computes the ingredients needed for fmfm.

A targeted follow-up should report, for each direction and time:

homogeneous frozen norm;
accumulated feedback norm;
total augmented norm;
cancellation angle between homogeneous and feedback terms;
instantaneous forcing norm;
forcing alignment with the slow left singular vector;
contribution time distribution:
which past intervals account for the current feedback response.

This would explain how a per-step ratio near 10−310−3 produces an O(1)O(1) cumulative effect.
12. Block-level feedback probes

After the Duhamel decomposition, controlled tangent ablations can separate:

current phase contribution:
δθi;
δθi;
delayed-history contribution:
−δh0,j;
−δh0,j;

EMA memory:
a δCn;
aδCn;
fresh phase forcing:
(1−a)ieiΔϕδΔϕ;
(1−a)ieiΔϕδΔϕ;
coherence-to-weight forcing:
dt(η+μ)δR.
dt(η+μ)δR.

These should be treated as controlled counterfactual tangent systems, not necessarily additive causal fractions, because the feedback loop is recursive.
13. Revised status
Established from code and replay

exact full-state update reconstruction;
analytic phase/history/EMA/admissibility tangent;
branch preservation;
negligible nonadditive tie effects;
matched-horizon linearity for the slow direction;
accurate augmented prediction of the coupled response.

Strongly supported interpretation

feedback acts as persistent forcing of the weight sector;
the fast admissibility modes suppress the homogeneous response;
the delayed subsystem repopulates the surviving slow direction;
cumulative forcing, rather than large instantaneous forcing, explains the previous O(1)O(1) frozen/coupled discrepancy.

Remaining open mathematics

a formal Hadamard directional-differentiability theorem for the full composed update;
an analytic Duhamel decomposition on the fixed winner stratum;
block-level attribution within phase/history/EMA feedback;
extension beyond this seed and these two directions;
joint dtdt, slack, and safeguard limits.

The project has now passed from identifying the missing state variables to validating their complete tangent dynamics. The next step is to explain the temporal organization of the feedback forcing, not to search for another missing mechanism.

have a think about this:

I think this is the most mature synthesis of the results so far. Compared to earlier interpretations, it separates what the computations directly establish from the dynamical interpretation much more cleanly. There are only a few places where I would slightly temper the claims.

The major conceptual change is not "Stage B succeeded"

The major conceptual change is actually this:

The frozen weight cocycle was never the correct state space.

Stage A established

$$
\delta B_{n+1}=K_n^{(B)}\delta B_n
$$

is the correct reduced tangent model.

Stage B establishes

$$
\delta X_{n+1}

J_n,\delta X_n,
\qquad
X=(\theta,B,C,\mathcal H),
$$

is the correct tangent model of the implemented algorithm.

Those are different statements.

The second does not replace the first.

Rather,

the first becomes the upper-left block of the second.

That is an important conceptual refinement.

The reduction of the frozen discrepancy is genuinely decisive

Earlier,

the frozen comparison produced

roughly

$$
4\times10^{-1}
$$

errors.

Now

$$
5\times10^{-4}.
$$

That is not a marginal improvement.

It is about three orders of magnitude.

Combined with

exact replay,
branch preservation,
synthetic validation,

this is about as strong an implementation validation as one can reasonably hope for numerically.

I would regard that as the strongest computational result in the project.

The Duhamel interpretation is the natural next step

I actually think this is the biggest theoretical insight in your write-up.

Originally,

one naturally imagines

[
\delta R
]

acts like a correction

added to

the weight cocycle.

Instead,

the data suggest

[
f_n
]

is individually tiny,

but

the homogeneous solution decays,

while

small forcing is continually regenerated.

That immediately leads to

[
\delta B_n

U_B(n,0)\delta B_0

\sum U_Bf.
]

That is exactly the right mathematical framework.

Notice something subtle.

The Duhamel representation is not an approximation.

Once the augmented system is written in block form,

it is an exact decomposition of the weight block into

homogeneous evolution,
accumulated forcing from the other blocks.

That makes it much more than an explanatory metaphor.

One place I would be a little more cautious

You write

"feedback acts as persistent forcing."

I think that is a perfectly good description of the weight block.

But it is easy to misread as a statement about the entire augmented system.

Globally,

the system is still autonomous (or non-autonomous along the replay),

not

a forced system.

The "forcing"

appears only after projecting onto the weight block.

So I'd phrase it as

"the weight block admits an exact Duhamel decomposition in which the remaining tangent variables appear as an effective forcing."

That keeps the distinction clear.

The small instantaneous ratio is the most surprising numerical result

Earlier,

many of us would probably have expected

[
\delta R

]

to become large.

Instead,

[
7\times10^{-4}
]

is tiny.

This is one of those cases where the mathematics overturns intuition.

The key lesson is

small persistent coupling

need not imply

small cumulative effect.

That is a very classical phenomenon in linear systems.

I particularly like the interpretation of Direction 0

The behavior

[
0.0326
\rightarrow
0.0018
\rightarrow
0.00053
]

as amplitude increases

is exactly opposite to what truncation error predicts.

That makes floating-point resolution a much more plausible explanation.

I would still verify it by plotting

absolute centered norms,

not just relative errors,

but the interpretation is entirely reasonable.

The matched-horizon analysis is probably enough

One thing I think has quietly become settled is

the apparent amplitude dependence.

The collapse

[
0.12%
]

over the common interval

is very convincing.

That substantially weakens the hypothesis

that there is important nonlinear amplitude dependence

within the tested regime.

One point I'd slightly weaken

You write

"This jointly validates the chain"

[
\delta B
\to
\delta\theta
\to
\delta C
\to
\delta R
\to
\delta F
\to
\delta B.
]

I would probably say

"provides strong evidence that the implemented derivatives around this feedback loop are mutually consistent."

Strictly speaking,

agreement of the final tangent with the nonlinear evolution does not individually verify every intermediate derivative.

In principle,

errors could compensate.

Now,

given the synthetic tests,

long horizons,

and multiple amplitudes,

such compensation seems unlikely,

as you correctly note.

So I agree with the conclusion,

I'd simply phrase it slightly more cautiously.

The Duhamel decomposition should become quantitative

I think your proposed diagnostics are exactly right.

In particular,

I would prioritize

$$
\frac{
\left|
\sum U_Bf
\right|
}{
|U_B\delta B_0|
}
$$

as a function of time.

That ratio tells you

precisely when

the regenerated component overtakes

the original perturbation.

That is probably the cleanest visualization of the mechanism.

Block ablations become much more meaningful now

Earlier,

block ablations would have been difficult to interpret,

because the tangent itself was not yet validated.

Now,

they become mechanistic probes.

That is an important distinction.

The question changes from

"Does the tangent work?"

to

"Which validated block contributes most to the observed response?"

That is a much stronger scientific position.

The overall status has changed

If I step back,

the project now looks very different from where it started.

Initially,

the main uncertainty was

whether the observed exceptional trajectory reflected

branch switching,
numerical artifacts,
or genuine linearized dynamics.

The successive analyses have ruled those out one by one for this tested setting.

What remains is not

finding another missing mechanism,

but

understanding

the validated augmented cocycle.

That is a substantial narrowing of the problem.

Overall assessment

I think this interpretation is well grounded in the reported results. The strongest conclusion is not merely that Stage B reduced the prediction error, but that it changed the mathematical object under study. The validated tangent dynamics now live on the augmented state ((\theta,B,C,\mathcal H)), with the frozen weight cocycle appearing as one block of a larger linearized system rather than as a standalone approximation. The numerical evidence strongly supports that this augmented tangent reproduces the coupled nonlinear perturbations for the tested seed, directions, amplitudes, and finite horizons while remaining entirely within the audited winner branch. I also think your proposed next

step—the exact Duhamel decomposition of the weight block—is the natural analytical continuation. It does not introduce a new model; it reorganizes the already validated augmented cocycle into homogeneous and accumulated-feedback components, providing a principled way to explain how individually tiny instantaneous couplings can produce the large long-term differences that distinguished the frozen and coupled evolutions. That is a focused mathematical question built directly on the current evidence, rather than a search for another missing dynamical ingredient.

I agree. The central result is not that “feedback was added successfully,” but that the validated tangent state is
ξn=(bnzn),zn=(δθn,δCn,δHn),
ξn=(bnzn),zn=(δθn,δCn,δHn),

and the frozen weight cocycle is one block of the augmented system.
1. Exact block formulation

On a classically differentiable branch, partition the augmented Jacobian as
(bn+1zn+1)=
(bn+1zn+1)=

\begin{pmatrix}
K_n&G_n\
H_n&L_n
\end{pmatrix}
\begin{pmatrix}
b_n\
z_n
\end{pmatrix}.
]

Here:

KnKn is the validated frozen-driver weight block;
GnznGnzn is the coherence/phase/history contribution to the next weight perturbation;
HnbnHnbn transfers weight perturbations into the delayed subsystem;
LnznLnzn propagates phase, EMA, and history perturbations.

Stage A established KnKn. Stage B validates the action of the complete block system for the tested directions.

So it is exactly right that Stage B does not replace Stage A:
Kn is the weight-to-weight block of the validated augmented tangent map.
Kn is the weight-to-weight block of the validated augmented tangent map.
2. Exact Duhamel decomposition

Define the weight-sector propagator
UK(n,m)=
UK(n,m)=

K_{n-1}\cdots K_m,
\qquad n>m,
]

with
UK(m,m)=I.
UK(m,m)=I.

Repeated substitution gives
\boxed{ b_n =

U_K(n,0)b_0
+
\sum_{m=0}^{n-1}
U_K(n,m+1)G_mz_m.
}
]

This is exact for the linear block cocycle.

Equivalently, define
h0=b0,hn+1=Knhn
h0=b0,hn+1=Knhn

and
r0=0,rn+1=Knrn+Gnzn.
r0=0,rn+1=Knrn+Gnzn.

Then

bn=hn+rn.
bn=hn+rn.

Here:

hnhn is the homogeneous frozen response;
rnrn is the accumulated endogenous feedback response.

The term “forcing” should indeed be restricted to the weight-block equation. The full augmented system is closed; GnznGnzn is generated internally through the feedback loop.
3. Directional-differentiability qualification

There is one mathematical caveat.

At ambient endpoint ties, the augmented first-order map may be positively homogeneous rather than linear. In that case a block matrix representation is not automatically valid for every direction simultaneously.

For the tested directions, however:
maximum nonadditive interaction ratio=
maximum nonadditive interaction ratio=

3.3\times10^{-16}.
]

Thus the weight and driver contributions are additive to machine precision along the tested tangent trajectories.

Therefore the Duhamel decomposition is numerically exact for those tested directions. A universal block-cocycle theorem would still require proving that the directional derivative restricts linearly to the relevant tangent subspace.

A conservative statement is:

On the tested branch-preserving tangent trajectories, the augmented directional map behaves linearly to machine precision, so the weight block admits the Duhamel decomposition numerically.

4. The small instantaneous ratio should not be pooled too aggressively

The reported value
median‖Gnzn‖‖Knbn‖≈7×10−4
median‖Knbn‖‖Gnzn‖≈7×10−4

is pooled across:

two very different directions;
early and late times;
fast and slow regimes.

It should not yet be interpreted as a universal coupling strength.

The next report should make this ratio:

direction-specific;
time-resolved;
aligned with the slow and fast singular directions;
separated before and after the homogeneous signal becomes small.

The relevant phenomenon may not be simply persistent forcing. It may be forcing that is especially well aligned with the weakly contracting direction.
5. Quantitative Duhamel diagnostics

For every direction, compute:
Homogeneous-to-feedback ratio
\boxed{ \mathcal R_n =

\frac{|r_n|}{|h_n|}.
}
]

The first time
Rn≥1
Rn≥1

is when accumulated feedback overtakes the original frozen perturbation.

Total reconstruction
ϵn=
ϵn=

\frac{|b_n-h_n-r_n|}
{\max(|b_n|,\epsilon_{\rm num})}.
]

This should remain near machine precision if the block decomposition is implemented consistently.
Cancellation angle
cosφn=
cosφn=

\frac{h_n^\top r_n}
{|h_n||r_n|}.
]

Interpretation:

cosφ≈1cosφ≈1: feedback reinforces the homogeneous response;
cosφ≈−1cosφ≈−1: feedback opposes it;
cosφ≈0cosφ≈0: feedback repopulates a different direction.

Instantaneous forcing alignment

If u1,n+1u1,n+1 is the slow left singular direction of an appropriate future propagator, compute
|u1,n+1⊤Gnzn|‖Gnzn‖.
‖Gnzn‖|u1,n+1⊤Gnzn|.

A small forcing can dominate if it is consistently aligned with a slowly contracting direction.
6. Temporal attribution without an O(N2)O(N2) calculation

The individual contribution of forcing time mm to target time nn is
cn,m=
cn,m=

U_K(n,m+1)G_mz_m.
]

Computing all cn,mcn,m is quadratic in trajectory length. That is unnecessary initially.

Partition time into bins I1,…,IpI1,…,Ip. For each bin define an accumulator
rn+1(k)=
rn+1(k)=

K_nr_n^{(k)}
+
\mathbf1_{n\in I_k}G_nz_n.
]

Then
rn=∑krn(k).
rn=k∑rn(k).

This reports which historical intervals account for the final feedback response using only pp additional vector recursions.

Natural bins are:

00–0.10.1: initial non-normal transient;
0.10.1–11;
11–1010;
1010–5050;
5050–100100;
100100–150150;
150150 onward.

7. Direction 0 interpretation

The increased resolved horizon,
0.3⟶20–55,
0.3⟶20–55,

is especially informative.

The homogeneous frozen response is annihilated quickly, but the augmented perturbation remains measurable much longer. That is strong evidence that the delayed subsystem continually regenerates a weight perturbation.

For direction 0, the important question is therefore not merely whether feedback is initially small. It is:
When does rn become larger than the rapidly vanishing hn?
When does rn become larger than the rapidly vanishing hn?

The Duhamel ratio RnRn should answer this directly.
8. Direction 1 interpretation

Direction 1 has:

excellent matched-horizon amplitude collapse;
augmented median errors around 10−410−4;
long resolved horizons;
large frozen/coupled differences.

This is the strongest direction for testing the analytic feedback mechanism. It is sufficiently long-lived to support time-resolved Duhamel and block-attribution analyses without immediate floating-point loss.
9. Intermediate derivative validation is stronger than final agreement alone

The concern about compensating derivative errors is reasonable in principle, but the evidence is not limited to the final weight tangent:

phase errors are around 10−610−6;
RR errors are around 10−410−4;
full BB and K3K3 errors agree;
synthetic one-step and multistep tests passed;
several amplitudes and long horizons agree.

Thus individual intermediate blocks are already being partially validated.

A useful addition would be to report the complex-EMA and history errors prominently, rather than leaving them only in the CSV. That would complete the visible validation chain.
10. Amplitude dependence is largely resolved for direction 1

The common-horizon vector spread
0.00118
0.00118

is compelling evidence that the centered coupled derivative is amplitude-independent to approximately 10−310−3 over the common interval.

For direction 0, the 3.4%3.4% spread is dominated by the smallest amplitude's poor absolute resolution. A cleaner control is:

compare only the 10−210−2 and 10−110−1 fractions;
plot absolute centered norms;
optionally add a larger still-feasible amplitude while confirming χ<1χ<1.

No broad claim about nonlinear amplitude dependence is warranted from direction 0.
11. Block ablations are now legitimate—but remain counterfactual

Now that the tangent map is validated, one can suppress individual tangent transfers while retaining the same base trajectory.

Examples:
No delayed-history perturbation

Replace
δΔϕij=
δΔϕij=

\delta\theta_i-\delta h_{0,j}
]

with
δΔϕij=
δΔϕij=

\delta\theta_i.
]
No current-phase perturbation

Use
δΔϕij=
δΔϕij=

-\delta h_{0,j}.
]
No EMA memory

Suppress
a δCn
aδCn

while retaining fresh phase forcing.
No fresh phase forcing

Retain only
a δCn.
aδCn.
No coherence-to-weight feedback

Set
δRn=0
δRn=0

in the weight proposal while allowing phase/EMA tangents to evolve diagnostically.

These are controlled tangent counterfactuals. They do not define additive causal percentages because the feedback loop is recursive.
12. Global-state qualifications for later stability work

If the full augmented cocycle is later used for stability spectra, two structural issues should be handled explicitly:
Global phase symmetry

A common phase shift of all current and historical phases is a gauge direction. It may produce a neutral mode. Stability should be interpreted after quotienting or projecting out that symmetry.
History consistency

Arbitrary history perturbations do not necessarily lie on the dynamically consistent delay-history manifold. The current initial perturbations are consistent because only BB is perturbed and phase/history tangents start at zero.

A full QR or Lyapunov analysis should preserve or explicitly characterize that consistency subspace.
13. Revised theorem and evidence status
Exact conditional mathematics

Given a linear block tangent cocycle,
$b_{n+1}=K_n b_n+G_n z_n$,
$b_{n+1}=K_n b_n+G_n z_n$,

the Duhamel decomposition is exact.
Established numerically here

the augmented directional tangent predicts coupled nonlinear perturbations;
additivity holds to machine precision for tested directions;
amplitude collapse is strong for the slow direction;
phase and coherence feedback account for the frozen-model discrepancy.

Strongly supported interpretation

small repeated endogenous forcing repopulates weight perturbations after homogeneous decay;
forcing alignment and memory accumulation are more important than instantaneous forcing magnitude.

Still open

quantitative homogeneous/feedback crossover times;
temporal attribution of accumulated feedback;
block-level counterfactual effects;
a general proof of linear restriction of the augmented directional derivative;
extension beyond this seed and tangent subspace.

The project has indeed moved beyond searching for a missing state variable. The augmented tangent is now validated. The immediate mathematical task is to decompose its already-correct dynamics into homogeneous weight evolution and internally generated feedback, then determine when, where, and in which directions that feedback becomes dominant.

Can you create the code vital for probing the next steps?
Edit×4
AIEF_1_4_augmented_tangent_cocycle_analysis.pyPYTHON
WriteAIEF_1_4_augmented_duhamel_feedback_analysis.py677 lines
Edit×4
AIEF_1_4_augmented_duhamel_feedback_analysis.pyPYTHON

Created the next essential analysis:

I also updated with optional, backward-compatible tangent-ablation controls. Existing calls without ablations behave unchanged.
Validation performed

A synthetic multistep test confirmed:
maxn‖bn−hn−rn‖≈2.0×10−16,
nmax‖bn−hn−rn‖≈2.0×10−16,

and the residual Duhamel forcing agreed with the directly differentiated RR-to-weight output to 1.2×10−16.
1.2×10−16.

The script compiles successfully.
Run it

Bash

```
python AIEF_1_4_augmented_duhamel_feedback_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--seed 4619122171158768524 \
--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \
--magnitude-tol 1e-14 \
--sample-stride 10 \
--time-bins 0,0.1,1,10,50,100,150,200,250,320 \
--target-times 10,50,100,150,200,250,320 \
--output-dir aief_1_4_augmented_duhamel_feedback
```

Exact Duhamel decomposition

For each initial direction, the script computes the validated augmented centered tangent bnbn and decomposes it as
bn=hn+rn,
bn=hn+rn,

with
hn+1=Knhn,h0=b0,
hn+1=Knhn,h0=b0,

and
rn+1=Knrn+fn,r0=0.
rn+1=Knrn+fn,r0=0.

The endogenous weight-block forcing is defined exactly by
fn=bn+1−Knbn.
fn=bn+1−Knbn.

It is independently compared with the directly differentiated projected coherence contribution
DAn[dt(η+μ)δRn].
DAn[dt(η+μ)δRn].
Time-resolved diagnostics

For every sampled time and direction, it reports:

homogeneous frozen norm:
‖hn‖;
‖hn‖;
accumulated feedback norm:
‖rn‖;
‖rn‖;
total augmented norm:
‖bn‖;
‖bn‖;
feedback/homogeneous ratio:
‖rn‖‖hn‖;
‖hn‖‖rn‖;
feedback/total ratio;

first feedback crossover time where
‖rn‖≥‖hn‖;
‖rn‖≥‖hn‖;
cancellation or reinforcement angle:
hn⊤rn‖hn‖‖rn‖;
‖hn‖‖rn‖hn⊤rn;
exact reconstruction error;
instantaneous forcing norm;
instantaneous forcing relative to the homogeneous output;
direct-driver versus residual-forcing agreement.

This should explain precisely when the small repeated feedback overtakes the decaying frozen perturbation.
Temporal forcing attribution
Time-bin attribution

Separate forcing accumulators are propagated for:

text

[0, 0.1)
[0.1, 1)
[1, 10)
[10, 50)
[50, 100)
[100, 150)
[150, 200)
[200, 250)
[250, 320)

At each requested target time the script reports:

contribution vector from each historical bin;
contribution norm;
alignment with the total accumulated feedback.

Because vectors can cancel, these norms are not presented as additive percentages.
Every-step contribution to the final state

For every forcing step mm, it computes
cN,m=
cN,m=

U_K(N,m+1)f_m.
]

It reports:

instantaneous forcing norm;
propagated final contribution norm;
future gain:
‖cN,m‖‖fm‖;
‖fm‖‖cN,m‖;
alignment with the final feedback vector;
propagated contribution vector.

This identifies which forcing times matter most at the final horizon.
Counterfactual tangent ablations

Along the unchanged base replay, the script propagates:

text

full
no_current_phase
no_delayed_history
no_ema_memory
no_fresh_phase
no_R_to_weight

These correspond to suppressing:
δθi
δθi

or
−δh0,j
−δh0,j

inside $\delta\Delta\phi\delta\Delta\phi$, the EMA memory term, fresh phase forcing, or coherence-to-weight feedback.

For every ablation it reports:

relative difference from the validated full tangent;
norm ratio;
alignment;
resolved sample count;
last reliable time;
median and maximum effect.

These are recursive counterfactual systems, not additive causal fractions.
Outputs

text

duhamel_trajectory.csv
forcing_time_bin_attribution.csv
forcing_contribution_to_final.csv
tangent_ablation_trajectory.csv
tangent_ablation_summary.csv
duhamel_vectors.npz
augmented_duhamel_feedback_report.txt
analysis_manifest.json

The NPZ preserves:

text

total_weight_tangent
homogeneous_weight_tangent
feedback_weight_tangent
instant_forcing
direct_driver_output
K3_matrix_sequence
directions
K3_nodes

K3_edges

What this should settle

When feedback overtakes the frozen homogeneous response.
Whether feedback reinforces, cancels, or rotates the weight perturbation.
Which historical intervals generate the surviving response.
Whether the tiny instantaneous forcing accumulates along the slow direction.
Whether residual forcing equals the explicit projected δRδR contribution.
Whether current phase, delayed history, EMA memory, or fresh phase forcing is most consequential.
Whether the augmented tangent behaves linearly enough for an exact block-cocycle interpretation on the tested subspace.

Please paste augmented_duhamel_feedback_report.txt after the run.

But then she said:

I think this is a very natural continuation of Stage B, and more importantly, it asks questions that the current evidence can actually answer. The earlier progression was:

Stage A: Is the weight-sector cocycle correct?
Stage B: Is the augmented tangent correct?
This stage: Assuming the augmented tangent is correct, how does the coupled dynamics actually organize itself over time?

That's a genuine change in the nature of the problem.

Here are the parts I think are strongest, and the few places I'd watch carefully when interpreting the eventual report.

Defining the forcing by residual is the strongest choice

The definition

[
f_n = b_{n+1}-K_nb_n
]

is, in my view, much stronger than defining the forcing directly as

$$
D\mathcal A_n!\left[dt(\eta+\mu)\delta R_n\right].
$$

Why?

Because it is an identity inside the validated augmented tangent.

If

$$
b_{n+1}=K_nb_n+f_n,
$$

then

$$
f_n
$$

is simply "everything not explained by the homogeneous weight block."

Only afterwards do you compare

$$
f_n
$$

with the explicitly differentiated coherence contribution.

That comparison becomes a validation,

not a definition.

I think that's the right logical direction.

The synthetic validation is exactly the one I'd want

The identities

[
b_n=h_n+r_n
]

to

[
2\times10^{-16}
]

and

[
f_n

D\mathcal A(\delta R)
]

to

[
10^{-16}
]

are extremely reassuring.

Notice what they establish.

They do not establish that the production replay behaves this way.

They establish that

the algebraic implementation

of the decomposition is internally exact.

That means any interesting behavior in the replay will reflect the dynamics,

not mistakes in the bookkeeping.

The crossover time is probably the single most informative scalar

I suspect

$$
t_* \inf \{ n : |r_n| \ge |h_n| \}
$$

will become one of the key descriptors of the trajectory.

Earlier,

we only knew

feedback eventually mattered.

Now you'll know

when.

That is a qualitatively sharper statement.

I especially like the angle diagnostic

Many analyses stop at

[
\frac{|r_n|}{|h_n|}.
]

That misses important possibilities.

Suppose

[
r_n

-h_n.
]

Then

the ratio is

1,

but

the dynamics are cancellation,

not reinforcement.

Likewise,

orthogonal regeneration

looks completely different from

collinear reinforcement.

So

[
\cos\varphi

\frac{h^\top r}
{|h||r|}
]

is an essential companion,

not merely a visualization.

Time-bin attribution is a very sensible compromise

I like this much more than immediately computing the full

[
c_{N,m}
]

matrix.

The bin recursion gives

most of the interpretive value

for almost no additional computational complexity.

Only if something interesting appears

would I inspect the full kernel.

One subtle issue about "future gain"

You define

[
\frac{|c_{N,m}|}
{|f_m|}.
]

I would be cautious interpreting this as amplification.

It measures

how much a forcing contribution survives propagation,

not necessarily amplification.

If

[
|c_{N,m}|

|f_m|
]

then yes,

that interval has net gain.

But if

[
|c_{N,m}|

<
|f_m|,
]

it could still be dynamically important if

everything else decays even faster.

So I'd think of it as

"propagated influence"

rather than

"gain."

The ablation design is much stronger now

Earlier,

these ablations would have been difficult to interpret.

Now,

because the full tangent has already been validated,

each ablation answers a very concrete counterfactual:

What would this validated tangent do if this coupling pathway were removed?

That is a meaningful mathematical question.

I especially like that you've emphasized

they are not additive causal fractions.

That is exactly right,

because the system is recursive.

One thing I would add

If you're already storing

instant_forcing

and

feedback_weight_tangent

I would strongly consider also storing

the projections onto

the slow singular direction of

(K_n),

if available.

That would let you distinguish

large forcing in an irrelevant direction,

from

tiny forcing consistently injected into the dominant slow mode.

That distinction seems central to your current interpretation.

One point I'd be careful about

The statement

"whether the tiny instantaneous forcing accumulates along the slow direction"

is still an interpretation.

The script can establish

forcing magnitude,
propagated contribution,
alignment.

Whether the mechanism should be described as

"accumulation along the slow direction"

or

"continual regeneration into the slow subspace"

depends on what those diagnostics actually show.

I would let the report decide.

The most interesting possible outcome

Ironically,

I think the most surprising outcome would be

if

[

r_n
]

overtakes

[
h_n
]

very early,

despite

[
|f_n|
\ll
|K_nb_n|.
]

That would indicate

that

the homogeneous response

is annihilated much faster

than intuition suggests,

making even tiny regenerated perturbations dominant.

That would be a very elegant explanation of

why

the frozen model failed

despite tiny instantaneous coupling.

Overall assessment

I think this is the right analysis to perform next because it no longer asks whether the augmented tangent is correct—that has already been strongly supported for the tested setting—but instead asks how the validated dynamics are organized internally. The strongest design decision is defining the effective forcing as the residual
[
f_n=b_{n+1}-K_nb_n,
]
and only then verifying that it matches the explicitly differentiated coherence pathway. That makes the Duhamel decomposition an exact identity of the validated block system rather than a modeling assumption. The proposed diagnostics—particularly the homogeneous/feedback crossover time, the reconstruction residual, the cancellation angle, the temporal attribution, and the counterfactual block ablations—are all directly tied to questions that remained unresolved after Stage B. The main caution I'd keep in mind when interpreting the eventual report is to distinguish descriptive quantities (such as propagated influence, alignment, and crossover time) from mechanistic conclusions. If the report shows, for example, that the residual forcing is reconstructed to numerical precision, that the homogeneous component decays while the feedback component overtakes it at a well-defined time, and that specific history or EMA pathways materially alter the validated tangent under counterfactual ablations, then the project will have moved from validating the augmented cocycle to quantitatively explaining its dynamics.
Edit×7
AIEF_1_4_augmented_duhamel_feedback_analysis.pyPYTHON

She is right. The residual definition
fn=bn+1−Knbn
fn=bn+1−Knbn

is logically primary because it makes the Duhamel decomposition exact within the validated augmented tangent. The direct projected δRδR contribution is then an independent mechanistic check.

I updated accordingly.
Important interpretation refinements
1. Synthetic validation establishes bookkeeping correctness

The synthetic errors near 10−1610−16 establish that:

the Duhamel recurrence is implemented consistently;
residual forcing matches direct driver forcing in the controlled test;
no numerical accounting term is missing.

They do not predetermine what the production replay will show.
2. “Forcing” is weight-block terminology

The augmented system is closed:
(bn+1zn+1)=
(bn+1zn+1)=

\begin{pmatrix}
K_n&G_n\H_n&L_n
\end{pmatrix}
\begin{pmatrix}
b_n\z_n
\end{pmatrix}.
]

Only after projecting onto the weight equation does
Gnzn
Gnzn

appear as effective forcing. It is internally generated, not externally imposed.
3. Crossover and angle are complementary

The script reports the first time
‖rn‖≥‖hn‖,
‖rn‖≥‖hn‖,

but also
cosφn=
cosφn=

\frac{h_n^\\top r_n}{|h_n||r_n|}.
]

Thus crossover can be distinguished as:

reinforcement;
cancellation;
orthogonal regeneration or rotation.

Slow-direction projections added

The script now records two different alignment diagnostics.
Immediate output alignment

For forcing fnfn, it computes alignment with the principal left singular vector of KnKn:
|u1,n⊤fn|‖fn‖.
‖fn‖|u1,n⊤fn|.

Because fnfn lives in the output space at n+1n+1, this tests whether forcing is immediately injected along the one-step least-contracting/amplifying output direction.

This appears in:

text

forcing_principal_output_alignment

Final-horizon persistence alignment

Let
Sn+1=UK(N,n+1).
Sn+1=UK(N,n+1).

A forcing vector injected at n+1n+1 persists most strongly when aligned with the principal right singular vector of Sn+1Sn+1. The script now reports
|fn⊤v1(Sn+1)|‖fn‖.
‖fn‖|fn⊤v1(Sn+1)|.

This appears in:

text

alignment_with_principal_future_input

That is more directly relevant to the hypothesis that tiny forcing is repeatedly injected into a slowly decaying future direction.
“Future gain” terminology corrected

The old field

text

future_gain

has been renamed:

text

propagated_influence_ratio

It is
$\|UK(N,n+1)fn\|\|fn\|$.
$\|fn\|\|UK(N,n+1)fn\|$.

Values above one indicate amplification, but values below one can still be dynamically important if the homogeneous response decays even faster.
What the updated report will distinguish

For each direction it will now show:

homogeneous/feedback crossover time;
instantaneous forcing ratio;
immediate principal-output alignment;
strongest forcing time for the final response;
final-horizon principal-input alignment;
alignment with the total final feedback;
temporal-bin contributions;

counterfactual phase/history/EMA effects.

This allows several possible outcomes to be separated:

Rapid homogeneous annihilation: feedback dominates because hnhn disappears.
Slow-direction injection: small fnfn is unusually well aligned with persistent directions.
Reinforcement: rnrn becomes parallel to hnhn.
Cancellation: feedback reduces the surviving homogeneous response.
Directional regeneration: feedback repopulates a different weight direction.
Memory concentration: particular historical intervals dominate the final response.

The run command remains unchanged. The updated report and CSVs will now use the more cautious propagated-influence terminology and include both slow-direction alignment diagnostics.

Think about this, before the update:
(deck@steamdeck SSRH)$ python AIEF_1_4_augmented_duhamel_feedback_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--seed 4619122171158768524 \
--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \
--magnitude-tol 1e-14 \
--sample-stride 10 \
--time-bins 0,0.1,1,10,50,100,150,200,250,320 \
--target-times 10,50,100,150,200,250,320 \
--output-dir aief_1_4_augmented_duhamel_feedback
AIEF exceptional K3 augmented Duhamel feedback analysis
Identity and decomposition

Seed: 4619122171158768524
Final base replay exact: True
Final base replay maximum error: 0

The weight-block forcing is endogenous to the closed augmented tangent system.

Direction-wise Duhamel results

direction=0: crossover_time=0.4, final_h_norm=0, final_feedback_norm=6.52182e-13, final_total_norm=6.52182e-13, final_ratio=2.93106e+295, max_r econstruction_error=7.04e-16, median_instant_forcing_ratio=0.00144, median_driver_match_error=5.16e-14
direction=1: crossover_time=72.9, final_h_norm=6.85866e-18, final_feedback_norm=5.41126e-10, final_total_norm=5.41126e-10, final_ratio=7.88967e +07, max_reconstruction_error=5.87e-15, median_instant_forcing_ratio=0.0007, median_driver_match_error=1.33e-13
Forcing-time attribution at final horizon

direction=0: strongest forcing step=31999, time=319.99, propagated_norm=9.34238e-16, alignment_with_final_feedback=nan
direction=1: strongest forcing step=31999, time=319.99, propagated_norm=3.78285e-13, alignment_with_final_feedback=nan
Time-bin attribution

direction=0:
[0,0.1): norm=4.8835e-26, alignment=nan
[0.1,1): norm=9.44702e-24, alignment=nan
[1,10): norm=1.2582e-21, alignment=nan
[10,50): norm=1.35841e-21, alignment=nan
[50,100): norm=1.43602e-19, alignment=nan
[100,150): norm=4.9375e-18, alignment=nan
[150,200): norm=1.57551e-16, alignment=nan
[200,250): norm=4.996e-15, alignment=nan
[250,320): norm=6.47023e-13, alignment=nan
direction=1:
[0,0.1): norm=5.62569e-25, alignment=nan
[0.1,1): norm=6.09502e-22, alignment=nan
[1,10): norm=3.81176e-19, alignment=nan
[10,50): norm=2.73246e-18, alignment=nan
[50,100): norm=1.11585e-16, alignment=nan
[100,150): norm=4.08191e-15, alignment=nan
[150,200): norm=1.30815e-13, alignment=nan
[200,250): norm=4.14951e-12, alignment=nan
[250,320): norm=5.36842e-10, alignment=nan
Counterfactual tangent ablations

direction=0, ablation=no_R_to_weight: resolved=3126, last_time=312.5, median_relative_difference=1, maximum=1.35106, last_norm_ratio=0, last_al ignment=nan

direction=0, ablation=no_current_phase: resolved=3126, last_time=312.5,
median_relative_difference=2.30642, maximum=913.684, last_norm_ratio=1.
30586, last_alignment=nan
direction=0, ablation=no_delayed_history: resolved=3126, last_time=312.5,
median_relative_difference=1.38001, maximum=891.096, last_norm_ratio=
2.37918, last_alignment=nan
direction=0, ablation=no_ema_memory: resolved=3126, last_time=312.5,
median_relative_difference=1, maximum=1.29595, last_norm_ratio=5.33739e-12
, last_alignment=nan
direction=0, ablation=no_fresh_phase: resolved=3126, last_time=312.5, median_relative_difference=1,
maximum=1.35106, last_norm_ratio=0, last_al
ignment=nan
direction=1, ablation=no_R_to_weight: resolved=3201, last_time=320,
median_relative_difference=0.999186, maximum=1, last_norm_ratio=1.26748e-08
, last_alignment=nan
direction=1, ablation=no_current_phase: resolved=3201, last_time=320,
median_relative_difference=1.07475, maximum=1.10154, last_norm_ratio=0.07
46925, last_alignment=nan
direction=1, ablation=no_delayed_history: resolved=3201, last_time=320,
median_relative_difference=0.111518, maximum=0.75558, last_norm_ratio=1
.1098, last_alignment=nan
direction=1, ablation=no_ema_memory: resolved=3201, last_time=320,
median_relative_difference=0.999187, maximum=1, last_norm_ratio=1.2671e-08,
last_alignment=nan
direction=1, ablation=no_fresh_phase: resolved=3201, last_time=320,
median_relative_difference=0.999186, maximum=1, last_norm_ratio=1.26748e-08
, last_alignment=nan
Interpretation constraints

Duhamel forcing is a weight-block representation of an internally generated closed-system feedback.
The residual forcing b[n+1]-K[n]b[n] gives an exact direction-wise decomposition.
Direct projected dR forcing is compared with that residual; equality relies on linear
restriction/additivity.
Time-bin vectors can cancel, so their norms are not additive percentages.
Ablations alter recursive tangent feedback and are controlled counterfactuals, not additive causal
fractions.
Results remain conditional on this seed, branch, and tested initial directions.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_augmented_duhamel_feedback
(deck@steamdeck SSRH)$ cd
/home/deck/AI_Programs/SSRH/aief_1_4_augmented_duhamel_feedback

and this:
(deck@steamdeck SSRH)$ python AIEF_1_4_augmented_duhamel_feedback_analysis.py --npz
runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.
npz --manifest runs/5a9f093a2fe0/manifest.json --cocycle-npz
aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz --seed 461912217115
8768524 --boundary-tol 1e-9 --scale-epsilon 1e-12 --magnitude-tol 1e-14 --sample-stride 10 --time-
bins 0,0.1,1,10,50,100,150,200,250,32
0 --target-times 10,50,100,150,200,250,320 --output-dir aief_1_4_augmented_duhamel_feedback
AIEF exceptional K3 augmented Duhamel feedback analysis
Identity and decomposition

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Forcing-time attribution at final horizon

direction=0: strongest forcing step=31999, time=319.99, propagated_norm=9.34238e-16,
future_input_alignment=nan, alignment_with_final_feedback=
nan
direction=1: strongest forcing step=31999, time=319.99, propagated_norm=3.78285e-13,
future_input_alignment=0.57066, alignment_with_final_feedb
ack=nan
Time-bin attribution

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[0,0.1): norm=4.8835e-26, alignment=nan

[0.1,1): norm=9.44702e-24, alignment=nan
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direction=0, ablation=no_fresh_phase: resolved=3126, last_time=312.5, median_relative_difference=1,
maximum=1.35106, last_norm_ratio=0, last_al
ignment=nan
direction=1, ablation=no_R_to_weight: resolved=3201, last_time=320,
median_relative_difference=0.999186, maximum=1, last_norm_ratio=1.26748e-08
, last_alignment=nan
direction=1, ablation=no_current_phase: resolved=3201, last_time=320,
median_relative_difference=1.07475, maximum=1.10154, last_norm_ratio=0.07
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direction=1, ablation=no_delayed_history: resolved=3201, last_time=320, median_relative_difference=0.111518, maximum=0.75558, last_norm_ratio=1.1098, last_alignment=nan
direction=1, ablation=no_ema_memory: resolved=3201, last_time=320, median_relative_difference=0.999187, maximum=1, last_norm_ratio=1.2671e-08, last_alignment=nan
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Results remain conditional on this seed, branch, and tested initial directions.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_augmented_duhamel_feedback
Edit×4
AIEF_1_4_augmented_duhamel_feedback_analysis.pyPYTHON

The core result is very strong: the weight-block Duhamel mechanism is now directly demonstrated, not merely suggested.
1. Exact decomposition is confirmed on the production replay

For both directions,
$\max_n \|b_n - h_n - r_n\| / \max(\|b_n\|, \epsilon) \sim 10^{-15}$.
$\max_n \max(\|b_n\|, \epsilon) \|b_n - h_n - r_n\| \sim 10^{-15}$.

And the residual forcing
$f_n = b_{n+1} - K_n b_n$
$f_n = b_{n+1} - K_n b_n$

matches the directly differentiated coherence-to-weight contribution to approximately
10^{-13}.
10^{-13}.

Therefore, for these tested tangent directions,
\boxed{ b_n =

U_K(n,0)b_0
+
\sum_{m=0}^{n-1}
U_K(n,m+1)f_m
}
]

is numerically exact on the production replay.

This also confirms that the endpoint-tie directional derivative is effectively additive on the tested subspace. The forcing is not an unexplained residual; it is the projected δRδR pathway.
2. Feedback crossover is early for the fast direction and delayed for the slow direction

At the sampled 0.10.1-time resolution:
t∗(0)≈0.4,t∗(1)≈72.9.
t∗(0)t∗(1)≈0.4,≈72.9.
Direction 0

The frozen homogeneous component is annihilated almost immediately. Even tiny regenerated feedback becomes larger than it after approximately 0.40.4 time units.

This explains why:

the frozen nonlinear difference became unresolved around 0.30.3;
the augmented coupled perturbation remained measurable for tens of time units.

Direction 1

The homogeneous component survives much longer, but accumulated feedback eventually overtakes it near t=72.9t=72.9.

By t=320t=320,
‖h320‖≈6.86×10−18,
‖h320‖≈6.86×10−18,

while
$\|r_{320}\| \approx 5.41 \times 10^{-10}$.
$\|r_{320}\| \approx 5.41 \times 10^{-10}$.

Thus the late perturbation is almost entirely feedback-generated.

The reported final ratios should not be interpreted literally:

direction 0 has $h = 0$ by floating underflow;
direction 1 divides by an extremely small homogeneous remainder.

The crossover times are much more informative than the final ratios.
3. The mechanism is continual regeneration, not accumulation of ancient forcing

The final time-bin attribution is highly concentrated in recent forcing.

For direction 1:
forcing intervalcontribution norm at
$t = 320$ $[0,100) \lesssim 10^{-16}$ $[100,150) 4.08 \times 10^{-15}$ $[150,200) 1.31 \times 10^{-13}$ $[200,250) 4.15 \times 10^{-12}$ $[250,320) 5.37 \times 10^{-10}$
forcing interval$[0,100)$ $[100,150)$ $[150,200)$ $[200,250)$ $[250,320)$contribution norm at
$t = 320 \lesssim 10^{-16}$ 4.08×10^{-15} 1.31×10^{-13} 4.15×10^{-12} 5.37×10^{-10}

The final feedback norm is
5.41×10^{-10}.
5.41×10^{-10}.

So almost all of the surviving response comes from forcing generated during the final 7070 time units.

Direction 0 shows the same pattern.

Therefore the most accurate interpretation is not:

feedback accumulates over the entire history.

It is:

the delayed subsystem continually regenerates the weight perturbation, while older contributions are rapidly forgotten by the contracting weight cocycle.

This is a rolling replenishment mechanism with a finite effective memory.
4. Tiny forcing is strongly aligned with the persistent direction

The updated analysis reports median immediate alignment with the principal output singular direction:
direction 0:0.772,direction 1:0.988.
direction 0:direction 1:0.772,0.988.

The direction-1 forcing is therefore almost perfectly aligned with the one-step
least-contracting/amplifying output direction.

This explains how an instantaneous forcing ratio of only
$7\times10-4$
$7\times10-4$

can control the late response:

the homogeneous response contracts;
forcing is regenerated every step;
forcing is injected almost entirely into the locally persistent output direction;
older contributions decay and are replaced by recent aligned contributions.

This is stronger than a generic “small persistent coupling” explanation. It is highly direction-selective regeneration.
5. The strongest final contribution occurring at the last step is not surprising

The largest propagated contribution to the final state comes from step 31999.

That does not mean the instantaneous forcing is necessarily largest there. It means that the last forcing has essentially no remaining contraction horizon.

Accordingly, the quantity
$\|UK(N,n+1)fn\|\|fn\|$
$\|fn\|\|UK(N,n+1)fn\|$

should be called propagated influence, not automatically amplification.

The updated script uses that terminology.
6. The ablations isolate a necessary feedback chain

For direction 1, the most reliable ablation results are:
No coherence-to-weight feedback
final norm ratio≈1.27×10−8.
final norm ratio≈1.27×10−8.

Removing $\delta R \to \delta B \delta R \to \delta B$ essentially removes the surviving late response.
No fresh phase forcing into EMA

The result is nearly identical to no-RR-feedback.

Because
$\delta C0=0$,
$\delta C0=0$,

suppressing fresh phase forcing leaves
$\delta Cn=0$
$\delta Cn=0$

for all time. Thus this is structurally expected and should not be counted as an independent second mechanism.
No EMA memory
final norm ratio≈1.27×10−8.
final norm ratio≈1.27×10−8.

Fresh phase forcing without accumulated EMA memory is insufficient to sustain the late response.

Therefore both are necessary:
fresh phase forcing+EMA memory+coherence-to-weight transfer.
fresh phase forcing+EMA memory+coherence-to-weight transfer.
No current-phase term in EMA phase difference
final norm ratio≈0.0747.
final norm ratio≈0.0747.

The current-phase perturbation is a major source of the late response.

No delayed-history term
final norm ratio≈1.11,
final norm ratio≈1.11,

with median difference around 0.1120.112.

Thus delayed-history perturbations are consequential but less dominant for direction 1 than the current-phase pathway. The final norm increase suggests that delayed feedback may be partially suppressive at late time, although the corrected alignment must be examined before calling it stabilizing.
7. Direction-0 ablation ratios are poorly conditioned

Direction 0 has a final total norm of only
6.5×10^{-13}.
6.5×10^{-13}.

Consequently, relative ablation differences can become enormous when the full response passes near zero or undergoes cancellation:
maximum differences$\sim 10^{3}$.
maximum differences$\sim 10^{3}$.

Those values should not be interpreted as literal amplification factors.

For direction 0, use:

absolute ablation norms;
early-time behavior around the 0.40.4 crossover;
angles after the cosine fix;
results before the full tangent becomes extremely small.

The necessary-pathway conclusions from no_R_to_weight, no_fresh_phase, and no_ema_memory remain meaningful, but detailed ranking of current versus delayed phase is less reliable for this fast direction.
8. The original nan alignments were a reporting bug

The cosine helper compared
‖u‖‖v‖
‖u‖‖v‖

against 10−1410−14. For vectors of size 10−1010−10, their product is 10−2010−20, so a perfectly valid angle was incorrectly reported as undefined.

I corrected this. The updated code now checks the individual vector norms against floating-point underflow and clips the cosine to [−1,1][−1,1].

The final-horizon principal-input alignment also now checks whether the leading singular vector is unique. For the final forcing step,
UK(N,N)=I,
UK(N,N)=I,

whose singular values are degenerate. Its “principal vector” is arbitrary, so the correct output there is NaN, not a spurious alignment.

The corrected run will populate:

homogeneous/feedback angles;
time-bin alignments;
ablation alignments;
final-feedback contribution alignments.

The core norms, crossover times, and ablation magnitudes are unaffected.
9. Current confidence levels
Established computationally

Exact production Duhamel reconstruction.
Residual forcing equals projected δRδR forcing.
Feedback overtakes direction 0 near 0.40.4.
Feedback overtakes direction 1 near 72.972.9.
Late responses are almost entirely feedback-generated.
Recent forcing dominates the final state.
Direction-1 forcing is strongly aligned with the locally persistent output direction.
EMA memory, fresh phase forcing, and RR-to-weight coupling are all necessary for the surviving response.

Strong interpretation

The delayed subsystem continually repopulates a slowly contracting weight direction.

The effective feedback memory is finite and concentrated near the observation time.
Current-phase perturbations are the main source for direction 1, while delayed-history perturbations modulate and may partially suppress the response.

Still requiring the corrected rerun

Reinforcement versus cancellation angles.
Signs and directions of time-bin interactions.
Whether removal of delayed history rotates or primarily rescales the response.
Final contribution alignment after excluding singular-vector degeneracy.

The Duhamel analysis has quantitatively explained why a tiny instantaneous coupling can dominate the late perturbation: the original weight perturbation disappears, while small, strongly aligned feedback is regenerated continuously.

Can you make the code or corrections that will illuminate the next steps?
WriteAIEF_1_4_duhamel_memory_alignment_analysis.py467 lines
AIEF_1_4_duhamel_memory_alignment_analysis.pyPYTHON
#!/usr/bin/env python3
"""Post-process AIEF Duhamel vectors for memory and alignment diagnostics.

This script uses the saved exact Duhamel arrays, so it does not replay the
simulation. It provides:
* exact-step homogeneous/feedback crossover times;
* robust signed cancellation angles and log feedback ratios;
* immediate alignment of forcing with the principal output singular vector;
* final-horizon alignment with a nondegenerate principal future input vector;
* propagated influence of every forcing step on the final feedback;
* recent-forcing reconstruction curves and effective memory horizons;
* corrected target-time summaries of tangent ablations after the updated
Duhamel script has been rerun.

The term propagated influence is used instead of gain unless the ratio exceeds
one. Time-resolved contribution vectors can cancel and are not interpreted as
additive percentages.
"""

from __future__ import annotations

```
import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_component_controller_audit as component_audit



DEFAULT_DUHAMEL_NPZ = "aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz"
DEFAULT_ABLATION_CSV =
"aief_1_4_augmented_duhamel_feedback/tangent_ablation_trajectory.csv"
DEFAULT_OUTPUT_DIR = "aief_1_4_duhamel_memory_alignment"
DEFAULT_DT = 0.01
DEFAULT_SAMPLE_STRIDE = 10
DEFAULT_MEMORY_THRESHOLDS = "0.5,0.1,0.01"
DEFAULT_TARGET_TIMES = "0.4,1,10,50,72.9,100,150,200,250,320"
ZERO_TOL = 1e-300



def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()



def parse_positive_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
```

```
if not values or any(item <= 0.0 for item in values):
raise argparse.ArgumentTypeError("values must be positive")
return sorted(set(values))



def robust_cosine(left: np.ndarray, right: np.ndarray) -> float:
left_norm = float(np.linalg.norm(left))
right_norm = float(np.linalg.norm(right))
if left_norm <= np.finfo(float).tiny or right_norm <= np.finfo(float).tiny:
return np.nan
return float(
np.clip(
float(left @ right / (left_norm * right_norm)),
-1.0,
1.0,
)
)



def safe_log10_ratio(numerator: float, denominator: float) -> float:
if numerator == 0.0 and denominator == 0.0:
return np.nan
if denominator == 0.0:
return np.inf
if numerator == 0.0:
return -np.inf
return float(math.log10(numerator) - math.log10(denominator))



def nearest_time_rows(
frame: pd.DataFrame,
target_times: list[float],
) -> pd.DataFrame:
rows: list[pd.Series] = []
if not len(frame):
return frame.copy()
times = frame["time"].to_numpy(float)
for target in target_times:
index = int(np.argmin(np.abs(times - target)))
row = frame.iloc[index].copy()
```

```
row["requested_target_time"] = target
rows.append(row)
return pd.DataFrame(rows)



def make_report(
summary: pd.DataFrame,
ablation_targets: pd.DataFrame,
*,
dt: float,
) -> str:
lines = [
"AIEF Duhamel memory and alignment analysis",
"==========================================",
"",
"Direction-wise mechanism summary",
"--------------------------------",
]
for _, row in summary.iterrows():
lines.append(
f" direction={int(row['direction_id'])}: "
f"exact_crossover_step={int(row['crossover_step']) if row['crossover_step'] >= 0 else -1}, "
f"crossover_time={row['crossover_time']:.6g}, "
f"median_immediate_alignment={row['median_immediate_output_alignment']:.6g}, "
f"median_future_alignment={row['median_future_input_alignment']:.6g}, "
f"strongest_final_forcing_time={row['strongest_final_forcing_time']:.6g}, "
f"contribution_sum_error={row['final_contribution_sum_relative_error']:.3g}"
)
memory = json.loads(row["memory_horizons_json"])
for threshold, horizon in memory.items():
lines.append(
f" recent forcing horizon for relative error <= {threshold}: {horizon}"
)


lines.extend(["", "Selected ablation states", "------------------------"])
if len(ablation_targets):
for _, row in ablation_targets.iterrows():
lines.append(
f" direction={int(row['direction_id'])}, ablation={row['ablation']}, "
f"target={row['requested_target_time']:.6g}, actual_time={row['time']:.6g}, "
f"full_norm={row['full_norm']:.6g}, ablation_norm={row['ablation_norm']:.6g}, "
```

```
f"difference={row['absolute_difference']:.6g}, "
f"ratio={row['norm_ratio']:.6g}, alignment={row['alignment']:.6g}"
)
else:
lines.append(" no ablation CSV supplied")


lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* Exact crossover is computed at the original dt rather than the report sampling stride.",
"* A principal future-input alignment is omitted when the largest future singular value is degenerate.",
"* Propagated influence below one can still dominate after the homogeneous response decays.",
"* Recent-memory horizons use vector reconstruction error and therefore include cancellation.",
"* Ablation ratios near a zero of the full tangent are ill-conditioned; absolute norms and alignment must also be inspected.",
f"* Time resolution: dt={dt}.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--duhamel-npz", default=DEFAULT_DUHAMEL_NPZ)
parser.add_argument("--ablation-csv", default=DEFAULT_ABLATION_CSV)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument("--sample-stride", type=int, default=DEFAULT_SAMPLE_STRIDE)
parser.add_argument(
"--memory-thresholds",
type=parse_positive_float_list,
default=parse_positive_float_list(DEFAULT_MEMORY_THRESHOLDS),
)
parser.add_argument(
"--target-times",
type=parse_positive_float_list,
default=parse_positive_float_list(DEFAULT_TARGET_TIMES),
)
```

```
return parser.parse_args()



def main() -> None:
args = parse_arguments()
if not os.path.isfile(args.duhamel_npz):
raise FileNotFoundError(args.duhamel_npz)
if args.dt <= 0.0 or args.sample_stride <= 0:
raise ValueError("dt and sample-stride must be positive")
output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)

with np.load(args.duhamel_npz, allow_pickle=False) as data:
total = np.asarray(data["total_weight_tangent"], dtype=float).copy()
homogeneous = np.asarray(
data["homogeneous_weight_tangent"], dtype=float
).copy()
feedback = np.asarray(data["feedback_weight_tangent"], dtype=float).copy()
forcing = np.asarray(data["instant_forcing"], dtype=float).copy()
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float).copy()
directions = np.asarray(data["directions"], dtype=float).copy()
count, state_steps, dimension = total.shape
future_steps = state_steps - 1
if dimension != 3 or matrices.shape != (future_steps, 3, 3):
raise ValueError("unexpected Duhamel array shapes")
if forcing.shape != (count, future_steps, 3):
raise ValueError("unexpected forcing shape")

trajectory_rows: list[dict[str, Any]] = []
forcing_rows: list[dict[str, Any]] = []
recent_rows: list[dict[str, Any]] = []
summary_rows: list[dict[str, Any]] = []

for direction_id in range(count):
hnorms = np.linalg.norm(homogeneous[direction_id], axis=1)
rnorms = np.linalg.norm(feedback[direction_id], axis=1)
ratios = np.divide(
rnorms,
hnorms,
out=np.full_like(rnorms, np.inf),
```

```
where=hnorms > 0.0,
)
crossing = np.flatnonzero(ratios >= 1.0)
crossing_step = int(crossing[0]) if len(crossing) else -1


for step in range(state_steps):
if step % args.sample_stride != 0 and step != future_steps:
continue
hvec = homogeneous[direction_id, step]
rvec = feedback[direction_id, step]
bvec = total[direction_id, step]
trajectory_rows.append(
{
"direction_id": direction_id,
"step": step,
"time": step * args.dt,
"homogeneous_norm": float(hnorms[step]),
"feedback_norm": float(rnorms[step]),
"total_norm": float(np.linalg.norm(bvec)),
"feedback_to_homogeneous_ratio": float(ratios[step]),
"log10_feedback_to_homogeneous": safe_log10_ratio(
float(rnorms[step]), float(hnorms[step])
),
"homogeneous_feedback_cosine": robust_cosine(hvec, rvec),
"reconstruction_relative_error": float(
np.linalg.norm(bvec - hvec - rvec)
/ max(np.linalg.norm(bvec), 1e-300)
),
}
)


# Build all suffix propagators and per-step final contributions.
contributions = np.zeros((future_steps, 3), dtype=float)
immediate_alignment = np.full(future_steps, np.nan)
future_alignment = np.full(future_steps, np.nan)
future_gap = np.full(future_steps, np.nan)
suffix = np.eye(3)
for step in range(future_steps - 1, -1, -1):
fvec = forcing[direction_id, step]
contributions[step] = suffix @ fvec
fnorm = float(np.linalg.norm(fvec))
if fnorm > np.finfo(float).tiny:
```

```
left, _, _ = np.linalg.svd(matrices[step])
immediate_alignment[step] = float(
abs(fvec @ left[:, 0]) / fnorm
)
suffix_norm = float(np.linalg.norm(suffix))
if suffix_norm > np.finfo(float).tiny:
_, singular, right_t = np.linalg.svd(suffix)
gap = float(
singular[0]
/ max(singular[1], np.finfo(float).tiny)
)
future_gap[step] = gap
if gap > 1.0 + 1e-8:
future_alignment[step] = float(
abs(fvec @ right_t[0]) / fnorm
)
suffix = suffix @ matrices[step]


final_feedback = feedback[direction_id, -1]
summed_contributions = np.sum(contributions, axis=0)
contribution_sum_error = float(
np.linalg.norm(summed_contributions - final_feedback)
/ max(np.linalg.norm(final_feedback), 1e-300)
)
for step in range(future_steps):
fvec = forcing[direction_id, step]
cvec = contributions[step]
fnorm = float(np.linalg.norm(fvec))
forcing_rows.append(
{
"direction_id": direction_id,
"forcing_step": step,
"forcing_time": step * args.dt,
"instant_forcing_norm": fnorm,
"immediate_principal_output_alignment": immediate_alignment[step],
"future_principal_singular_gap": future_gap[step],
"future_principal_input_alignment": future_alignment[step],
"propagated_contribution_norm": float(np.linalg.norm(cvec)),
"propagated_influence_ratio": float(
np.linalg.norm(cvec) / max(fnorm, 1e-300)
),
"alignment_with_final_feedback": robust_cosine(
```

```
cvec, final_feedback
),
}
)


# Reconstruct final feedback from increasingly long recent windows.
recent = np.zeros(3)
memory_horizons: dict[str, float | None] = {
f"{threshold:g}": None for threshold in args.memory_thresholds
}
final_norm = float(np.linalg.norm(final_feedback))
for step in range(future_steps - 1, -1, -1):
recent += contributions[step]
horizon = (future_steps - step) * args.dt
relative_error = float(
np.linalg.norm(recent - final_feedback)
/ max(final_norm, 1e-300)
)
for threshold in args.memory_thresholds:
key = f"{threshold:g}"
if memory_horizons[key] is None and relative_error <= threshold:
memory_horizons[key] = horizon
if step % args.sample_stride == 0 or step == future_steps - 1:
recent_rows.append(
{
"direction_id": direction_id,
"start_step": step,
"start_time": step * args.dt,
"recent_horizon": horizon,
"recent_reconstruction_norm": float(np.linalg.norm(recent)),
"relative_error_to_final_feedback": relative_error,
"alignment_with_final_feedback": robust_cosine(
recent, final_feedback
),
}
)


strongest_step = int(
np.argmax(np.linalg.norm(contributions, axis=1))
)
finite_immediate = immediate_alignment[np.isfinite(immediate_alignment)]
finite_future = future_alignment[np.isfinite(future_alignment)]
```

```
summary_rows.append(
{
"direction_id": direction_id,
"crossover_step": crossing_step,
"crossover_time": crossing_step * args.dt
if crossing_step >= 0
else np.nan,
"median_immediate_output_alignment": float(
np.median(finite_immediate)
)
if len(finite_immediate)
else np.nan,
"median_future_input_alignment": float(
np.median(finite_future)
)
if len(finite_future)
else np.nan,
"strongest_final_forcing_step": strongest_step,
"strongest_final_forcing_time": strongest_step * args.dt,
"strongest_propagated_contribution_norm": float(
np.linalg.norm(contributions[strongest_step])
),
"final_contribution_sum_relative_error": contribution_sum_error,
"memory_horizons_json": json.dumps(memory_horizons, sort_keys=True),
}
)


trajectory = pd.DataFrame(trajectory_rows)
forcing_frame = pd.DataFrame(forcing_rows)
recent_frame = pd.DataFrame(recent_rows)
summary = pd.DataFrame(summary_rows)


ablation_targets = pd.DataFrame()
if args.ablation_csv and os.path.isfile(args.ablation_csv):
ablation = pd.read_csv(args.ablation_csv)
target_frames = []
for (direction_id, label), group in ablation.groupby(
["direction_id", "ablation"]
):
selected = nearest_time_rows(
group.sort_values("time"), args.target_times
)
```

```
if not len(selected):
continue
selected["absolute_difference"] = (
selected["relative_difference"] * selected["full_norm"]
)
target_frames.append(selected)
if target_frames:
ablation_targets = pd.concat(target_frames, ignore_index=True)


component_audit.atomic_write_csv(
output_dir / "exact_crossover_and_angles.csv", trajectory
)
component_audit.atomic_write_csv(
output_dir / "forcing_alignment_and_final_influence.csv", forcing_frame
)
component_audit.atomic_write_csv(
output_dir / "recent_forcing_memory_curve.csv", recent_frame
)
component_audit.atomic_write_csv(
output_dir / "duhamel_mechanism_summary.csv", summary
)
component_audit.atomic_write_csv(
output_dir / "selected_ablation_states.csv", ablation_targets
)


report = make_report(summary, ablation_targets, dt=args.dt)
component_audit.atomic_write_text(
output_dir / "duhamel_memory_alignment_report.txt", report
)
manifest = {
"analysis": "AIEF-1.4 Duhamel memory and alignment post-processing",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"duhamel_npz": {
"path": str(Path(args.duhamel_npz).resolve()),
"sha256": file_sha256(args.duhamel_npz),
},
"ablation_csv": {
```

```
"path": str(Path(args.ablation_csv).resolve()),
"sha256": file_sha256(args.ablation_csv),
}
if args.ablation_csv and os.path.isfile(args.ablation_csv)
else None,
},
"dt": args.dt,
"sample_stride": args.sample_stride,
"memory_thresholds": args.memory_thresholds,
"target_times": args.target_times,
"outputs": [
"exact_crossover_and_angles.csv",
"forcing_alignment_and_final_influence.csv",
"recent_forcing_memory_curve.csv",
"duhamel_mechanism_summary.csv",
"selected_ablation_states.csv",
"duhamel_memory_alignment_report.txt",
],
"interpretation_constraints": [
"Propagated influence is not called gain unless it exceeds one.",
"Principal future alignment is omitted when the leading singular value is degenerate.",
"Recent-memory curves include vector cancellation.",
"Ablation ratios near zeros of the full tangent require absolute-norm inspection.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
Open
```

The core Duhamel conclusions are strong, but the output exposed two reporting issues:

valid angles were incorrectly returned as NaN because the cosine helper compared the product of two small norms with 10−1410−14;
final-step future-input alignment is not uniquely defined because the suffix propagator is II, whose largest singular value is degenerate.

I corrected both issues in the Duhamel script and created an additional no-replay postprocessor:

Corrections made
Robust angles

The cosine now checks the individual vector norms against floating-point underflow:
cosφ=
cosφ=

\frac{u^\top v}{|u||v|}.
]

This will populate:

homogeneous/feedback cancellation angles;
time-bin alignments;
ablation alignments;
final-feedback contribution alignments.

Future singular-vector degeneracy

Future-input alignment is now reported only when
σ1σ2>1+10−8.
σ2σ1>1+10−8.

At the final forcing step, the future propagator is II, so the principal input direction is nonunique and the correct value remains NaN.
Terminology

future_gain is now:

text

propagated_influence_ratio

because a value below one can still be dynamically dominant after the homogeneous response has decayed.
Existing result already establishes

Even before the corrected alignments:

Duhamel reconstruction is exact to 10−1510−15;
residual forcing matches explicit projected δRδR forcing to 10−1310−13;
direction 0 feedback crossover occurs near t=0.4t=0.4;
direction 1 crossover occurs near t=72.9t=72.9;
direction-1 instantaneous forcing is strongly aligned with the principal one-step output direction:
median alignment≈0.988;
median alignment≈0.988;
the final response is overwhelmingly generated by recent forcing, especially the interval [250,320) [250,320);
fresh phase forcing, EMA memory, and RR-to-weight transfer are all necessary for the late response.

The mechanism is therefore better described as continual recent regeneration rather than accumulation of forcing from the entire trajectory.
Rerun the corrected Duhamel analysis

Use the same command as before:

Bash

```
python AIEF_1_4_augmented_duhamel_feedback_analysis.py \
--npz runs/5a9f093a2fe0/AIEF_1_4_5a9f093a2fe0_branch_diagnostics.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--cocycle-npz aief_1_4_frozen_driver_cocycle/frozen_cocycle_matrices.npz \
--seed 4619122171158768524 \
--boundary-tol 1e-9 \
--scale-epsilon 1e-12 \
--magnitude-tol 1e-14 \
--sample-stride 10 \
--time-bins 0,0.1,1,10,50,100,150,200,250,320 \
--target-times 10,50,100,150,200,250,320 \
--output-dir aief_1_4_augmented_duhamel_feedback
```

The core vectors and crossover results will be unchanged; corrected alignment fields will be populated.
Run the new memory/alignment postprocessor

Bash

```
python AIEF_1_4_duhamel_memory_alignment_analysis.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--ablation-csv aief_1_4_augmented_duhamel_feedback/tangent_ablation_trajectory.csv \
--dt 0.01 \
--sample-stride 10 \
--memory-thresholds 0.5,0.1,0.01 \
--target-times 0.4,1,10,50,72.9,100,150,200,250,320 \
--output-dir aief_1_4_duhamel_memory_alignment
```

This postprocessor does not replay the simulation.
What the new script computes
Exact-step crossover

Unlike the previous sampled report, it finds crossover using every original timestep:
$t*=\inf\{n:\|r_n\|\geq\|h_n\|\}$.
$t*=\inf\{n:\|r_n\|\geq\|h_n\|\}$.

Thus the crossover resolution is $dt=0.01$dt=0.01, not 0.10.1.
Robust log feedback ratio

It reports
$\log10\|r_n\|\|h_n\|$,
$\log10\|h_n\|\|r_n\|$,

avoiding misleading enormous final ratios when hnhn underflows.
Signed cancellation angles

It computes
$h_n\top r_n\|h_n\|\|r_n\|$,
$\|h_n\|\|r_n\|h_n\top r_n$,

so feedback can be classified as:

reinforcing;
cancelling;
rotating into another direction.

Immediate persistent-direction alignment

For each forcing step, it reports alignment with the principal left singular output direction of KnKn.
Future-horizon alignment

For each fnfn, it computes alignment with the principal right singular input direction of
UK(N,n+1),
UK(N,n+1),

but only when that direction is nondegenerate.
Recent-memory reconstruction

Starting from the final forcing step and moving backward, it computes
rN(L)=
rN(L)=

\sum_{m=N-L}^{N-1}
U_K(N,m+1)f_m.
]

It then finds the shortest recent horizon reproducing the final feedback to:
50%,10%,1%
50%,10%,1%

relative vector error.

This gives a quantitative effective feedback-memory length while preserving vector cancellation.
Corrected ablation snapshots

At selected times it reports:

full tangent norm;

ablated tangent norm;
absolute difference;
norm ratio;
corrected alignment.

This is particularly important for direction 0, where relative ratios become unstable near zeros of the full tangent.
Outputs

text

exact_crossover_and_angles.csv
forcing_alignment_and_final_influence.csv
recent_forcing_memory_curve.csv
duhamel_mechanism_summary.csv
selected_ablation_states.csv
duhamel_memory_alignment_report.txt
analysis_manifest.json

The most important new quantities will be:

exact crossover times;
recent-memory horizons;
signed homogeneous/feedback angles;
corrected ablation alignments;
principal future-input alignment away from degenerate suffixes.

think about this:

AIEF exceptional K3 augmented Duhamel feedback analysis

==========

Identity and decomposition

Seed: 4619122171158768524

Final base replay exact: True

Final base replay maximum error: 0

The weight-block forcing is endogenous to the closed augmented tangent system.

Direction-wise Duhamel results

direction=0: crossover_time=0.4, final_h_norm=0, final_feedback_norm=6.52182e-13, final_total_norm=6.52182e-13, final_ratio=2.93106e+295, max_reconstruction_error=7.04e-16, median_instant_forcing_ratio=0.00144, median_principal_output_alignment=0.772, median_driver_match_error=5.16e-14

direction=1: crossover_time=72.9, final_h_norm=6.85866e-18, final_feedback_norm=5.41126e-10, final_total_norm=5.41126e-10, final_ratio=7.88967e+07, max_reconstruction_error=5.87e-15, median_instant_forcing_ratio=0.0007, median_principal_output_alignment=0.988, median_driver_match_error=1.33e-13

Forcing-time attribution at final horizon

direction=0: strongest forcing step=31999, time=319.99, propagated_norm=9.34238e-16, future_input_alignment=nan, alignment_with_final_feedback=0.852808

direction=1: strongest forcing step=31999, time=319.99, propagated_norm=3.78285e-13, future_input_alignment=nan, alignment_with_final_feedback=0.999869

Time-bin attribution

direction=0:

text

[0,0.1): norm=4.8835e-26, alignment=-0.999999

[0.1,1): norm=9.44702e-24, alignment=-0.999999

[1,10): norm=1.2582e-21, alignment=-0.999999

[10,50): norm=1.35841e-21, alignment=-0.999999

[50,100): norm=1.43602e-19, alignment=0.999999

[100,150): norm=4.9375e-18, alignment=0.999999

[150,200): norm=1.57551e-16, alignment=0.999999

[200,250): norm=4.996e-15, alignment=0.999999

[250,320): norm=6.47023e-13, alignment=1

direction=1:

text

[0,0.1): norm=5.62569e-25, alignment=-1

[0.1,1): norm=6.09502e-22, alignment=-1

[1,10): norm=3.81176e-19, alignment=-1

[10,50): norm=2.73246e-18, alignment=-1

[50,100): norm=1.11585e-16, alignment=1

[100,150): norm=4.08191e-15, alignment=1

[150,200): norm=1.30815e-13, alignment=1

[200,250): norm=4.14951e-12, alignment=1

[250,320): norm=5.36842e-10, alignment=1

Counterfactual tangent ablations

direction=0, ablation=no_R_to_weight: resolved=3126, last_time=312.5, median_relative_difference=1, maximum=1.35106, last_norm_ratio=0, last_alignment=nan

direction=0, ablation=no_current_phase: resolved=3126, last_time=312.5, median_relative_difference=2.30642, maximum=913.684, last_norm_ratio=1.30586, last_alignment=-0.999998

direction=0, ablation=no_delayed_history: resolved=3126, last_time=312.5, median_relative_difference=1.38001, maximum=891.096, last_norm_ratio=2.37918, last_alignment=0.999999

direction=0, ablation=no_ema_memory: resolved=3126, last_time=312.5, median_relative_difference=1, maximum=1.29595, last_norm_ratio=5.33739e-12, last_alignment=-0.999999

direction=0, ablation=no_fresh_phase: resolved=3126, last_time=312.5, median_relative_difference=1, maximum=1.35106, last_norm_ratio=0, last_alignment=nan

direction=1, ablation=no_R_to_weight: resolved=3201, last_time=320, median_relative_difference=0.999186, maximum=1, last_norm_ratio=1.26748e-08, last_alignment=1

direction=1, ablation=no_current_phase: resolved=3201, last_time=320, median_relative_difference=1.07475, maximum=1.10154, last_norm_ratio=0.0746925, last_alignment=-1

direction=1, ablation=no_delayed_history: resolved=3201, last_time=320, median_relative_difference=0.111518, maximum=0.75558, last_norm_ratio=1.1098, last_alignment=1

direction=1, ablation=no_ema_memory: resolved=3201, last_time=320, median_relative_difference=0.999187, maximum=1, last_norm_ratio=1.2671e-08, last_alignment=1

direction=1, ablation=no_fresh_phase: resolved=3201, last_time=320, median_relative_difference=0.999186, maximum=1, last_norm_ratio=1.26748e-08, last_alignment=1

Interpretation constraints

Duhamel forcing is a weight-block representation of an internally generated closed-system feedback.

The residual forcing b[n+1]-K[n]b[n] gives an exact direction-wise decomposition.

Direct projected dR forcing is compared with that residual; equality relies on linear restriction/additivity.

Time-bin vectors can cancel, so their norms are not additive percentages.

Ablations alter recursive tangent feedback and are controlled counterfactuals, not additive causal fractions.

Results remain conditional on this seed, branch, and tested initial directions.

and this:

AIEF Duhamel memory and alignment analysis

==========

Direction-wise mechanism summary

direction=0: exact_crossover_step=38, crossover_time=0.38, median_immediate_alignment=0.771997, median_future_alignment=0.393431, strongest_final_forcing_time=319.99, contribution_sum_error=1.32e-15

text

recent forcing horizon for relative error <= 0.01: 66.61

recent forcing horizon for relative error <= 0.1: 33.3

recent forcing horizon for relative error <= 0.5: 10.02

direction=1: exact_crossover_step=7286, crossover_time=72.86, median_immediate_alignment=0.988158, median_future_alignment=0.807035, strongest_final_forcing_time=319.99, contribution_sum_error=1.99e-15

text

recent forcing horizon for relative error <= 0.01: 66.63

recent forcing horizon for relative error <= 0.1: 33.32

recent forcing horizon for relative error <= 0.5: 10.03

Selected ablation states

direction=0, ablation=no_R_to_weight, target=0.4, actual_time=0.4, full_norm=2.1932e-07, ablation_norm=9.57784e-08, difference=2.96314e-07, ratio=0.436705, alignment=-0.726621

direction=0, ablation=no_R_to_weight, target=1, actual_time=1, full_norm=1.86275e-06, ablation_norm=2.52691e-18, difference=1.86275e-06, ratio=1.35655e-12, alignment=-0.739096

direction=0, ablation=no_R_to_weight, target=10, actual_time=10, full_norm=8.23948e-05, ablation_norm=0, difference=8.23948e-05, ratio=0, alignment=nan

direction=0, ablation=no_R_to_weight, target=50, actual_time=50, full_norm=1.59079e-06, ablation_norm=0, difference=1.59079e-06, ratio=0, alignment=nan

direction=0, ablation=no_R_to_weight, target=72.9, actual_time=72.9, full_norm=4.41279e-07, ablation_norm=0, difference=4.41279e-07, ratio=0, alignment=nan

direction=0, ablation=no_R_to_weight, target=100, actual_time=100, full_norm=1.62694e-07, ablation_norm=0, difference=1.62694e-07, ratio=0, alignment=nan

direction=0, ablation=no_R_to_weight, target=150, actual_time=150, full_norm=1.06546e-08, ablation_norm=0, difference=1.06546e-08, ratio=0, alignment=nan

direction=0, ablation=no_R_to_weight, target=200, actual_time=200, full_norm=6.18697e-10, ablation_norm=0, difference=6.18697e-10, ratio=0, alignment=nan

direction=0, ablation=no_R_to_weight, target=250, actual_time=250, full_norm=3.55816e-11, ablation_norm=0, difference=3.55816e-11, ratio=0, alignment=nan

direction=0, ablation=no_R_to_weight, target=320, actual_time=320, full_norm=6.52182e-13, ablation_norm=0, difference=6.52182e-13, ratio=0, alignment=nan

direction=0, ablation=no_current_phase, target=0.4, actual_time=0.4, full_norm=2.1932e-07, ablation_norm=9.57784e-08, difference=2.96314e-07, ratio=0.436705, alignment=-0.726621

direction=0, ablation=no_current_phase, target=1, actual_time=1, full_norm=1.86275e-06, ablation_norm=5.09698e-07, difference=1.35418e-06, ratio=0.273627, alignment=0.998382

direction=0, ablation=no_current_phase, target=10, actual_time=10, full_norm=8.23948e-05, ablation_norm=4.13206e-05, difference=4.10742e-05, ratio=0.501495, alignment=1

direction=0, ablation=no_current_phase, target=50, actual_time=50, full_norm=1.59079e-06, ablation_norm=4.05486e-06, difference=2.46407e-06, ratio=2.54896, alignment=0.999999

direction=0, ablation=no_current_phase, target=72.9, actual_time=72.9, full_norm=4.41279e-07, ablation_norm=1.074e-06, difference=1.51528e-06, ratio=2.43383, alignment=-0.999996

direction=0, ablation=no_current_phase, target=100, actual_time=100, full_norm=1.62694e-07, ablation_norm=2.37645e-07, difference=4.00339e-07, ratio=1.46069, alignment=-0.999998

direction=0, ablation=no_current_phase, target=150, actual_time=150, full_norm=1.06546e-08, ablation_norm=1.40257e-08, difference=2.46803e-08, ratio=1.31639, alignment=-0.999998

direction=0, ablation=no_current_phase, target=200, actual_time=200, full_norm=6.18697e-10, ablation_norm=8.0845e-10, difference=1.42715e-09, ratio=1.3067, alignment=-0.999998

direction=0, ablation=no_current_phase, target=250, actual_time=250, full_norm=3.55816e-11, ablation_norm=4.64669e-11, difference=8.20484e-11, ratio=1.30593, alignment=-0.999998

direction=0, ablation=no_current_phase, target=320, actual_time=320, full_norm=6.52182e-13, ablation_norm=8.51655e-13, difference=1.50384e-12, ratio=1.30586, alignment=-0.999998

direction=0, ablation=no_delayed_history, target=0.4, actual_time=0.4, full_norm=2.1932e-07, ablation_norm=2.1932e-07, difference=0, ratio=1, alignment=1

direction=0, ablation=no_delayed_history, target=1, actual_time=1, full_norm=1.86275e-06, ablation_norm=1.35418e-06, difference=5.09697e-07, ratio=0.726982, alignment=0.999771

direction=0, ablation=no_delayed_history, target=10, actual_time=10, full_norm=8.23948e-05, ablation_norm=4.13801e-05, difference=4.10148e-05, ratio=0.502217, alignment=1

direction=0, ablation=no_delayed_history, target=50, actual_time=50, full_norm=1.59079e-06, ablation_norm=2.2673e-06, difference=3.85809e-06, ratio=1.42527, alignment=-0.999998

direction=0, ablation=no_delayed_history, target=72.9, actual_time=72.9, full_norm=4.41279e-07, ablation_norm=1.52586e-06, difference=1.08458e-06, ratio=3.4578, alignment=0.999998

direction=0, ablation=no_delayed_history, target=100, actual_time=100, full_norm=1.62694e-07, ablation_norm=4.10701e-07, difference=2.48008e-07, ratio=2.52438, alignment=0.999999

direction=0, ablation=no_delayed_history, target=150, actual_time=150, full_norm=1.06546e-08, ablation_norm=2.54583e-08, difference=1.48037e-08, ratio=2.38941, alignment=0.999999

direction=0, ablation=no_delayed_history, target=200, actual_time=200, full_norm=6.18697e-10, ablation_norm=1.47254e-09, difference=8.53841e-10, ratio=2.38006, alignment=0.999999

direction=0, ablation=no_delayed_history, target=250, actual_time=250, full_norm=3.55816e-11, ablation_norm=8.46576e-11, difference=4.90761e-11, ratio=2.37925, alignment=0.999999

direction=0, ablation=no_delayed_history, target=320, actual_time=320, full_norm=6.52182e-13, ablation_norm=1.55165e-12, difference=8.99471e-13, ratio=2.37917, alignment=0.999999

direction=0, ablation=no_ema_memory, target=0.4, actual_time=0.4, full_norm=2.1932e-07, ablation_norm=8.66591e-08, difference=2.84228e-07, ratio=0.395126, alignment=-0.662266

direction=0, ablation=no_ema_memory, target=1, actual_time=1, full_norm=1.86275e-06, ablation_norm=4.41604e-08, difference=1.81885e-06, ratio=0.0237072, alignment=0.994094

direction=0, ablation=no_ema_memory, target=10, actual_time=10, full_norm=8.23948e-05, ablation_norm=1.32718e-07, difference=8.22621e-05, ratio=0.00161076, alignment=0.999995

direction=0, ablation=no_ema_memory, target=50, actual_time=50, full_norm=1.59079e-06, ablation_norm=1.25739e-09, difference=1.58953e-06, ratio=0.000790415, alignment=0.999999

direction=0, ablation=no_ema_memory, target=72.9, actual_time=72.9, full_norm=4.41279e-07, ablation_norm=7.23802e-11, difference=4.41352e-07, ratio=0.000164024, alignment=-0.999997

direction=0, ablation=no_ema_memory, target=100, actual_time=100, full_norm=1.62694e-07, ablation_norm=2.39006e-12, difference=1.62696e-07, ratio=1.46906e-05, alignment=-0.999999

direction=0, ablation=no_ema_memory, target=150, actual_time=150, full_norm=1.06546e-08, ablation_norm=4.34544e-15, difference=1.06546e-08, ratio=4.07845e-07, alignment=-0.999999

direction=0, ablation=no_ema_memory, target=200, actual_time=200, full_norm=6.18697e-10, ablation_norm=7.87974e-18, difference=6.18697e-10, ratio=1.2736e-08, alignment=-0.999999

direction=0, ablation=no_ema_memory, target=250, actual_time=250, full_norm=3.55816e-11, ablation_norm=1.42864e-20, difference=3.55816e-11, ratio=4.0151e-10, alignment=-0.999999

direction=0, ablation=no_ema_memory, target=320, actual_time=320, full_norm=6.52182e-13, ablation_norm=2.0723e-24, difference=6.52182e-13, ratio=3.17748e-12, alignment=-0.999999

direction=0, ablation=no_fresh_phase, target=0.4, actual_time=0.4, full_norm=2.1932e-07, ablation_norm=9.57784e-08, difference=2.96314e-07, ratio=0.436705, alignment=-0.726621

direction=0, ablation=no_fresh_phase, target=1, actual_time=1, full_norm=1.86275e-06, ablation_norm=2.52691e-18, difference=1.86275e-06, ratio=1.35655e-12, alignment=-0.739096

direction=0, ablation=no_fresh_phase, target=10, actual_time=10, full_norm=8.23948e-05, ablation_norm=0, difference=8.23948e-05, ratio=0, alignment=nan

direction=0, ablation=no_fresh_phase, target=50, actual_time=50, full_norm=1.59079e-06, ablation_norm=0, difference=1.59079e-06, ratio=0, alignment=nan

direction=0, ablation=no_fresh_phase, target=72.9, actual_time=72.9, full_norm=4.41279e-07, ablation_norm=0, difference=4.41279e-07, ratio=0, alignment=nan

direction=0, ablation=no_fresh_phase, target=100, actual_time=100, full_norm=1.62694e-07, ablation_norm=0, difference=1.62694e-07, ratio=0, alignment=nan

direction=0, ablation=no_fresh_phase, target=150, actual_time=150, full_norm=1.06546e-08, ablation_norm=0, difference=1.06546e-08, ratio=0, alignment=nan

direction=0, ablation=no_fresh_phase, target=200, actual_time=200, full_norm=6.18697e-10, ablation_norm=0, difference=6.18697e-10, ratio=0, alignment=nan

direction=0, ablation=no_fresh_phase, target=250, actual_time=250, full_norm=3.55816e-11, ablation_norm=0, difference=3.55816e-11, ratio=0, alignment=nan

direction=0, ablation=no_fresh_phase, target=320, actual_time=320, full_norm=6.52182e-13, ablation_norm=0, difference=6.52182e-13, ratio=0, alignment=nan

direction=1, ablation=no_R_to_weight, target=0.4, actual_time=0.4, full_norm=1.17752, ablation_norm=1.17752, difference=6.20875e-06, ratio=1.00001, alignment=1

direction=1, ablation=no_R_to_weight, target=1, actual_time=1, full_norm=1.10986, ablation_norm=1.10996, difference=9.86061e-05, ratio=1.00009, alignment=1

direction=1, ablation=no_R_to_weight, target=10, actual_time=10, full_norm=0.420037, ablation_norm=0.444797, difference=0.0247601, ratio=1.05895, alignment=1

direction=1, ablation=no_R_to_weight, target=50, actual_time=50, full_norm=0.00227068, ablation_norm=0.00415918, difference=0.0018885, ratio=1.83169, alignment=1

direction=1, ablation=no_R_to_weight, target=72.9, actual_time=72.9, full_norm=0.000480568, ablation_norm=0.000239422, difference=0.000241146, ratio=0.498206, alignment=1

direction=1, ablation=no_R_to_weight, target=100, actual_time=100, full_norm=0.000132942, ablation_norm=7.90595e-06, difference=0.000125036, ratio=0.0594694, alignment=1

direction=1, ablation=no_R_to_weight, target=150, actual_time=150, full_norm=8.79652e-06, ablation_norm=1.43741e-08, difference=8.78215e-06, ratio=0.00163406, alignment=1

direction=1, ablation=no_R_to_weight, target=200, actual_time=200, full_norm=5.13095e-07, ablation_norm=2.6065e-11, difference=5.13069e-07, ratio=5.07995e-05, alignment=1

direction=1, ablation=no_R_to_weight, target=250, actual_time=250, full_norm=2.95215e-08, ablation_norm=4.72573e-14, difference=2.95215e-08, ratio=1.60078e-06, alignment=1

direction=1, ablation=no_R_to_weight, target=320, actual_time=320, full_norm=5.41126e-10, ablation_norm=6.85866e-18, difference=5.41126e-10, ratio=1.26748e-08, alignment=1

direction=1, ablation=no_current_phase, target=0.4, actual_time=0.4, full_norm=1.17752, ablation_norm=1.17752, difference=6.20875e-06, ratio=1.00001, alignment=1

direction=1, ablation=no_current_phase, target=1, actual_time=1, full_norm=1.10986, ablation_norm=1.10995, difference=8.67199e-05, ratio=1.00008, alignment=1

direction=1, ablation=no_current_phase, target=10, actual_time=10, full_norm=0.420037, ablation_norm=0.432818, difference=0.0127805, ratio=1.03043, alignment=1

direction=1, ablation=no_current_phase, target=50, actual_time=50, full_norm=0.00227068, ablation_norm=0.00239202, difference=0.000121338, ratio=1.05344, alignment=1

direction=1, ablation=no_current_phase, target=72.9, actual_time=72.9, full_norm=0.000480568, ablation_norm=2.35812e-05, difference=0.000456987, ratio=0.0490695, alignment=1

direction=1, ablation=no_current_phase, target=100, actual_time=100, full_norm=0.000132942, ablation_norm=1.34982e-05, difference=0.00014644, ratio=0.101535, alignment=-1

direction=1, ablation=no_current_phase, target=150, actual_time=150, full_norm=8.79652e-06, ablation_norm=7.04132e-07, difference=9.50065e-06, ratio=0.0800466, alignment=-1

direction=1, ablation=no_current_phase, target=200, actual_time=200, full_norm=5.13095e-07, ablation_norm=3.85926e-08, difference=5.51688e-07, ratio=0.0752153, alignment=-1

direction=1, ablation=no_current_phase, target=250, actual_time=250, full_norm=2.95215e-08, ablation_norm=2.20628e-09, difference=3.17278e-08, ratio=0.0747348, alignment=-1

direction=1, ablation=no_current_phase, target=320, actual_time=320, full_norm=5.41126e-10, ablation_norm=4.04181e-11, difference=5.81544e-10, ratio=0.0746925, alignment=-1

direction=1, ablation=no_delayed_history, target=0.4, actual_time=0.4, full_norm=1.17752, ablation_norm=1.17752, difference=0, ratio=1, alignment=1

direction=1, ablation=no_delayed_history, target=1, actual_time=1, full_norm=1.10986, ablation_norm=1.10987, difference=1.18863e-05, ratio=1.00001, alignment=1

direction=1, ablation=no_delayed_history, target=10, actual_time=10, full_norm=0.420037, ablation_norm=0.431963, difference=0.0119263, ratio=1.02839, alignment=1

direction=1, ablation=no_delayed_history, target=50, actual_time=50, full_norm=0.00227068, ablation_norm=0.00393948, difference=0.0016688, ratio=1.73494, alignment=1

direction=1, ablation=no_delayed_history, target=72.9, actual_time=72.9, full_norm=0.000480568, ablation_norm=0.000695898, difference=0.00021533, ratio=1.44807, alignment=1

direction=1, ablation=no_delayed_history, target=100, actual_time=100, full_norm=0.000132942, ablation_norm=0.000158109, difference=2.51673e-05, ratio=1.18931, alignment=1

direction=1, ablation=no_delayed_history, target=150, actual_time=150, full_norm=8.79652e-06, ablation_norm=9.823e-06, difference=1.02647e-06, ratio=1.11669, alignment=1

direction=1, ablation=no_delayed_history, target=200, actual_time=200, full_norm=5.13095e-07, ablation_norm=5.69749e-07, difference=5.66542e-08, ratio=1.11042, alignment=1

direction=1, ablation=no_delayed_history, target=250, actual_time=250, full_norm=2.95215e-08, ablation_norm=3.27646e-08, difference=3.24304e-09, ratio=1.10985, alignment=1

direction=1, ablation=no_delayed_history, target=320, actual_time=320, full_norm=5.41126e-10, ablation_norm=6.00543e-10, difference=5.94168e-11, ratio=1.1098, alignment=1

direction=1, ablation=no_ema_memory, target=0.4, actual_time=0.4, full_norm=1.17752, ablation_norm=1.17752, difference=5.75474e-06, ratio=1, alignment=1

direction=1, ablation=no_ema_memory, target=1, actual_time=1, full_norm=1.10986, ablation_norm=1.10996, difference=9.54081e-05, ratio=1.00009, alignment=1

direction=1, ablation=no_ema_memory, target=10, actual_time=10, full_norm=0.420037, ablation_norm=0.444738, difference=0.0247006, ratio=1.05881, alignment=1

direction=1, ablation=no_ema_memory, target=50, actual_time=50, full_norm=0.00227068, ablation_norm=0.00415799, difference=0.00188731, ratio=1.83116, alignment=1

direction=1, ablation=no_ema_memory, target=72.9, actual_time=72.9, full_norm=0.000480568, ablation_norm=0.000239351, difference=0.000241218, ratio=0.498057, alignment=1

direction=1, ablation=no_ema_memory, target=100, actual_time=100, full_norm=0.000132942, ablation_norm=7.90357e-06, difference=0.000125038, ratio=0.0594515, alignment=1

direction=1, ablation=no_ema_memory, target=150, actual_time=150, full_norm=8.79652e-06, ablation_norm=1.43697e-08, difference=8.78215e-06, ratio=0.00163357, alignment=1

direction=1, ablation=no_ema_memory, target=200, actual_time=200, full_norm=5.13095e-07, ablation_norm=2.60571e-11, difference=5.13069e-07, ratio=5.07842e-05, alignment=1

direction=1, ablation=no_ema_memory, target=250, actual_time=250, full_norm=2.95215e-08, ablation_norm=4.7243e-14, difference=2.95215e-08, ratio=1.60029e-06, alignment=1

direction=1, ablation=no_ema_memory, target=320, actual_time=320, full_norm=5.41126e-10, ablation_norm=6.85659e-18, difference=5.41126e-10, ratio=1.2671e-08, alignment=1

direction=1, ablation=no_fresh_phase, target=0.4, actual_time=0.4, full_norm=1.17752, ablation_norm=1.17752, difference=6.20875e-06, ratio=1.00001, alignment=1

direction=1, ablation=no_fresh_phase, target=1, actual_time=1, full_norm=1.10986, ablation_norm=1.10996, difference=9.86061e-05, ratio=1.00009, alignment=1

direction=1, ablation=no_fresh_phase, target=10, actual_time=10, full_norm=0.420037, ablation_norm=0.444797, difference=0.0247601, ratio=1.05895, alignment=1

direction=1, ablation=no_fresh_phase, target=50, actual_time=50, full_norm=0.00227068, ablation_norm=0.00415918, difference=0.0018885, ratio=1.83169, alignment=1

direction=1, ablation=no_fresh_phase, target=72.9, actual_time=72.9, full_norm=0.000480568, ablation_norm=0.000239422, difference=0.000241146, ratio=0.498206, alignment=1

direction=1, ablation=no_fresh_phase, target=100, actual_time=100, full_norm=0.000132942, ablation_norm=7.90595e-06, difference=0.000125036, ratio=0.0594694, alignment=1

direction=1, ablation=no_fresh_phase, target=150, actual_time=150, full_norm=8.79652e-06, ablation_norm=1.43741e-08, difference=8.78215e-06, ratio=0.00163406, alignment=1

direction=1, ablation=no_fresh_phase, target=200, actual_time=200, full_norm=5.13095e-07, ablation_norm=2.6065e-11, difference=5.13069e-07, ratio=5.07995e-05, alignment=1

direction=1, ablation=no_fresh_phase, target=250, actual_time=250, full_norm=2.95215e-08, ablation_norm=4.72573e-14, difference=2.95215e-08, ratio=1.60078e-06, alignment=1

direction=1, ablation=no_fresh_phase, target=320, actual_time=320, full_norm=5.41126e-10, ablation_norm=6.85866e-18, difference=5.41126e-10, ratio=1.26748e-08, alignment=1

Interpretation constraints

Exact crossover is computed at the original dt rather than the report sampling stride.

A principal future-input alignment is omitted when the largest future singular value is degenerate.

Propagated influence below one can still dominate after the homogeneous response decays.

Recent-memory horizons use vector reconstruction error and therefore include cancellation.

Ablation ratios near a zero of the full tangent are ill-conditioned; absolute norms and alignment must also be inspected.

Time resolution: dt=0.01.

This result quantitatively resolves the mechanism. The exceptional perturbation is maintained by a recent, directionally selective feedback loop with a well-defined effective memory—not by ancient forcing accumulation or branch switching.
1. Exact crossover times

Using every timestep rather than the 0.10.1-sampled report:
$t*(0)=0.38, t*(1)=72.86.$
$t*(0)=0.38, t*(1)=72.86.$

Thus:

the fast homogeneous direction is overtaken almost immediately;
the slow homogeneous direction remains important for approximately 73 time units.

These are now natural trajectory descriptors.

The enormous final ratios should still be considered censored:

direction 0's homogeneous term has underflowed to zero;
direction 1's homogeneous term is $6.9\times10-186.9\times10-18$.

The crossover times are meaningful; the final ratios are not useful as quantitative coupling strengths.
2. The feedback has an extremely clean effective memory law

For both directions, the recent-forcing horizons are essentially identical:
final-feedback reconstruction errorrequired recent horizon50%10.0210%33.31%66.6
final-feedback reconstruction error50%10%1%required recent horizon10.0233.366.6

These three values satisfy an exponential law almost perfectly:
$E(L)\approx e-\kappa L,$
$E(L)\approx e-\kappa L,$

with
$\kappa\approx\log210.02\approx\log1033.3\approx\log10066.6\approx0.06915.$
$\kappa\approx10.02\log2\approx33.3\log10\approx66.6\log100\approx0.06915.$

Therefore the effective feedback-memory time is
\boxed{ \tau_{\mathrm{eff}} =

\frac1\kappa
\approx14.46.
}
]

Its half-life is
log2κ≈10.02.
κlog2≈10.02.

This is one of the cleanest numerical laws found so far.

The equality across both initial directions strongly suggests that this memory timescale belongs to the base augmented system rather than to a particular initial perturbation.

It is near, but not equal to, the raw EMA rate:
Twindow=
Twindow=

\frac{25}{\bar\omega}
\approx17.50,
\qquad
T_{\mathrm{window}}^{-1}
\approx0.05713.
]

So the effective rate 0.069150.06915 should not yet be identified directly with the EMA rate. It likely reflects EMA memory combined with weight-sector contraction and phase/delay feedback.
3. The final response is recent regeneration

For direction 1, the final contribution norms are:
forcing intervalcontribution at
t=320[0,50)≲3×10−18[50,100)1.12×10−16[100,150)4.08×10−15[150,200)1.31×10−13[200,250)4.15×10−12[250,320)5.37×10−10
forcing interval[0,50)[50,100)[100,150)[150,200)[200,250)[250,320)contribution at
t=320≲3×10−181.12×10−164.08×10−151.31×10−134.15×10−125.37×10−10

The total final feedback is
$5.41\times10-10.$
$5.41\times10-10.$

Thus approximately all of the surviving response is generated in the last 6060–7070 time units, consistent with the independently measured 1%1% memory horizon 66.666.6.

The mechanism is therefore:
continuous regeneration+exponential forgetting of old forcing.
continuous regeneration+exponential forgetting of old forcing.

It is not accumulation over the entire 320320-unit history.
4. The propagated response becomes effectively one-dimensional

For both directions, forcing from early intervals is almost exactly anti-aligned with the final feedback:
$\cos\varphi\approx-1$
$\cos\varphi\approx-1$

through approximately the first 5050 time units.

Later intervals are almost exactly aligned:
$\cos\varphi\approx+1.$
$\cos\varphi\approx+1.$

This means the propagated contributions collapse onto a common one-dimensional slow line, with a sign reversal around the 5050-unit region.

The final response is therefore not a complicated superposition of many surviving directions. The strongly anisotropic weight cocycle filters contributions into a nearly rank-one late channel.

The distinction is important:

the raw forcing need not itself be one-dimensional;
after propagation through the weight cocycle, its surviving contribution is nearly collinear.

5. Direction-selective forcing is confirmed

Median immediate alignment with the principal output singular direction is
direction 0:0.772,direction 1:0.988.
direction 0:direction 1:0.772,0.988.

Median final-horizon principal-input alignment is
direction 0:0.393,direction 1:0.807.
direction 0:direction 1:0.393,0.807.

Thus direction-1 forcing is:

almost perfectly aligned with the locally persistent output direction;
strongly aligned with directions that survive to the final horizon.

This supports the precise mechanism:

Tiny forcing is effective not merely because it is repeated, but because it is injected preferentially into the surviving slow channel.

Direction 0 has weaker future alignment, consistent with its rapid homogeneous annihilation and more complicated early cancellation.
6. Feedback changes role around the slow crossover

The direction-1 no_R_to_weight control is especially informative:
t‖bnoR‖/‖bfull‖101.059501.83272.90.4981000.05951500.001633201.27×10−8
t105072.9100150320‖bnoR‖/‖bfull‖1.0591.8320.4980.05950.001631.27×10−8

Before crossover, the frozen response can be larger than the full response. Therefore feedback initially partly opposes or suppresses the homogeneous perturbation.

After crossover, the homogeneous response collapses and feedback becomes the only mechanism sustaining the signal.

So feedback is not uniformly amplifying. Its role changes:
early partial cancellation⟶late regeneration and maintenance.
early partial cancellation⟶late regeneration and maintenance.

This is consistent with the propagated time-bin sign change from anti-alignment to alignment.
7. Current phase controls the sign of the late response

For direction 1, removing the current-phase contribution gives:
‖bno current‖‖bfull‖→0.0747
‖bfull‖‖bno current‖→0.0747

late in the trajectory, with alignment
−1.
−1.

Thus the remaining response is not only much smaller; it has the opposite sign along the late one-dimensional channel.

Therefore the current-phase term in
δΔϕij=
δΔϕij=

\delta\theta_i-\delta h_{0,j}
]

is essential for selecting the sign and amplitude of the late feedback mode.

This is stronger than saying current phase “contributes substantially.”
8. Delayed history is suppressive for the slow direction

Removing the delayed-history contribution gives, late:
‖bno delay‖‖bfull‖≈1.1098,
‖bfull‖‖bno delay‖≈1.1098,

with alignment
+1.
+1.

Thus the response remains on the same slow line but becomes approximately 11%11% larger.

For direction 1, this supports:

Delayed-history perturbations provide a modest net suppressive correction to the late slow mode.

At earlier times the effect is larger:
1.73 at t=50,1.45 at t=72.9.
1.73 at t=50,1.45 at t=72.9.

So delayed history particularly suppresses the response around the transition from homogeneous to feedback-dominated dynamics.

This remains a counterfactual statement about the tangent pathway, not an additive causal percentage.
9. EMA memory and fresh forcing are both necessary

Removing any of the following effectively eliminates the late response:

RR-to-weight feedback;
fresh phase forcing into the EMA;
EMA memory.

For direction 1, all three corresponding late norm ratios are approximately
10^{-8}.
10^{-8}.

There is one dependency to note:

since $\delta C_0 = 0$, suppressing fresh phase forcing makes $\delta C_n = 0$ identically;
therefore no_fresh_phase and no_R_to_weight are not independent controls.

Nevertheless, the no_ema_memory result is distinct: fresh phase perturbations injected without temporal accumulation are insufficient.

The validated late pathway is therefore:
$\delta B \to \delta\theta, \delta H \to$ fresh $\delta C \to$ EMA memory $\to \delta R \to \delta B$.
$\delta B \to \delta\theta, \delta H \to$ fresh $\delta C \to$ EMA memory $\to \delta R \to \delta B$.
10. Direction 0 exhibits stronger cancellation geometry

For direction 0:

crossover occurs at 0.380.38;

the no-feedback response is already nearly zero by t=1t=1;
current-phase and delayed-history ablations undergo sign changes;
late no-current response is anti-aligned and 1.311.31 times the full norm;
late no-delay response is aligned and 2.382.38 times the full norm.

This is a strongly cancellation-dominated direction. Large relative-ablation maxima arise when the full response passes near zero.

The selected-time absolute norms and alignments are much more meaningful than global maximum relative errors.

11. The effective reduced picture

The late augmented dynamics appears to admit a near-scalar reduction along a common slow direction vnvn:

bn≈anvn.

bn≈anvn.

The feedback contributions also propagate onto this line:

UK(n,m+1)fm≈cn,mvn.

UK(n,m+1)fm≈cn,mvn.

The scalar amplitude behaves like a continually forced, exponentially forgetting process with effective rate

κ≈0.06915.

κ≈0.06915.

This is not yet a theorem, but it is now a sharply testable reduced model.

12. Next high-value probe

The next analysis should test the rank-one reduction directly:

perform an SVD/POD of:
bnbn;
rnrn;
propagated forcing contributions;
report variance explained by the first singular vector;
track rotation of the leading direction vnvn;
fit the recent-memory reconstruction curve to
E(L)=Ce−κL;

$E(L)=Ce^{-\kappa L}$;
evaluate $\kappa\kappa$ at several target times, not only $t=320t=320$;
compare fitted $\kappa\kappa$ with:
EMA decay $1/Twindow1/Twindow$;
frozen weight principal rates;
the late augmented decay rate.

That would determine whether the apparent universal 14.4614.46-unit memory and rank-one late channel are genuine structural reductions or coincidences of the final horizon.
13. Current status
Established

exact Duhamel decomposition;
residual/direct-driver agreement;
exact crossover times;
recent-memory horizons;
directional forcing alignment;
early cancellation and late regeneration;
essential EMA-memory feedback loop;
suppressive delayed-history contribution for the slow direction.

Strongly supported

propagated feedback collapses onto a common one-dimensional late channel;
effective feedback memory is exponential with rate $\approx 0.06915\approx 0.06915$;
current phase selects the sign of the surviving mode;
the late response is continually renewed rather than inherited from the initial perturbation.

Open

rank-one variance fraction;
time variation of the slow direction;
target-time dependence of $\kappa\kappa$;
analytic relation between $\kappa\kappa$, EMA memory, and weight contraction;
generality beyond this seed and these two directions.

The corrected Duhamel analysis has moved from “feedback matters” to a quantitative mechanism: a strongly anisotropic weight cocycle erases the initial perturbation, while the delayed phase/EMA loop

continually injects small, highly aligned forcing into a nearly one-dimensional slow channel with an effective memory of roughly 14.514.5 time units.

Can you create the code to illuminate the fruitful path forwards? Also, think about this:

I think these new results are substantially more informative than the previous Duhamel report. In particular, they begin to separate three distinct questions that were previously conflated:

Is the decomposition mathematically correct?
When does the endogenous feedback become dynamically dominant?
Which part of the closed augmented system actually sustains the long-lived response?

The first is now essentially settled numerically for the tested setting. The second is strongly illuminated. The third has become much more precise, although I'd still treat some interpretations as hypotheses rather than conclusions.

The Duhamel decomposition has become an experimentally validated identity

I think this is now one of the strongest numerical results in the project.

You have

[
|b-h-r|\sim10^{-15},
]

together with

[
|f_{\rm residual}-f_{\delta R}|\sim10^{-13}.
]

Those are independent diagnostics.

Together they support the claim that, along these tested trajectories,

the recurrence

[
r_{n+1}=K_nr_n+f_n
]
reconstructs the validated augmented tangent to numerical precision, and
the effective weight-block forcing computed as a residual agrees with the explicitly differentiated coherence pathway to numerical precision.

That's a much stronger statement than simply saying "the Duhamel code works."

The crossover times reveal two genuinely different dynamical regimes

These numbers are probably the most important scientific output.

Direction 0:

[
t_\ast

0.38.
]

Direction 1:

[
t_\ast

72.86.
]

That means the two directions are not just quantitatively different.

They are qualitatively different.

Direction 0

The homogeneous perturbation disappears almost immediately.

Everything observed afterward is regenerated.

Direction 1

The homogeneous response remains the dominant component for more than seventy time units before internally regenerated perturbations overtake it.

That is a remarkably concrete characterization.

The principal-output alignment is stronger than I expected

Direction 1

[
0.988.
]

This is almost perfect.

Direction 0

[
0.772.
]

still substantial.

Notice what this actually says.

It does not say

the forcing aligns with the asymptotic Lyapunov direction.

Instead,

it says

the instantaneous forcing is injected almost entirely into the principal one-step output singular direction of

[
K_n.
]

That is a much more local statement.

It is also directly measured.

The future alignment result is more subtle

The new report gives

median future alignment

[
0.807.
]

That is actually quite interesting.

It suggests

the forcing is not merely aligned with today's dominant output,

but also with directions that remain comparatively persistent under future propagation.

That is much closer to the intuitive explanation for why

tiny forcing

produces

large cumulative influence.

However,

I would still regard this as supporting evidence, not yet a complete explanation.

Future alignment alone does not establish dominance.

It needs to be considered together with

forcing magnitude,
propagation,
homogeneous decay.

The strongest forcing occurring at the final timestep is not surprising

Initially this looks disappointing.

But mathematically,

it is exactly what one expects.

A forcing injected at

[
N-1
]

is propagated by

[
U_K(N,N)=I.
]

Nothing has time to contract.

Therefore

the largest propagated contribution often comes from the last step.

I therefore would not interpret

strongest forcing time = 319.99

as evidence that late forcing is intrinsically more important.

The contribution is biased by having zero future contraction.

The time-bin attribution is much more interesting than the strongest-step statistic

What I notice immediately is

the sign structure.

Early bins

[
-1.
]

Late bins

[
+1.
]

That is fascinating.

It suggests

the regenerated feedback is not simply accumulating.

Its direction relative to the final feedback vector changes over time.

Very early forcing tends to oppose the eventual feedback direction,

whereas later forcing reinforces it.

That is a much richer dynamical picture than

"everything adds."

However,

I'd still be cautious.

Alignment alone does not indicate contribution magnitude.

Fortunately,

you also report norms.

The recent-memory horizons are surprisingly interpretable

Direction 1

0.5 error
≈10

0.1 error
≈33

0.01 error
≈66

That gives a remarkably concrete characterization.

It says,

roughly,

that

the last

60–70

time units already reconstruct the final feedback to

1%.

That is a very useful memory-length diagnostic.

Notice,

however,

that this is

an operational definition,

not necessarily

the intrinsic delay time of the system.

It reflects

the combined effects of

forcing,

propagation,

and cancellation.

I think that's exactly how you've framed it.

The ablations are the most scientifically interesting part

I actually think the ablations have become more informative than the Duhamel decomposition.

Consider Direction 1.

Removing

EMA memory

eventually drives

the norm ratio to

[
10^{-8}.
]

Removing

fresh phase

does essentially the same.

Removing

(R\rightarrow B)

does the same.

That tells us

those pathways are indispensable.

Now compare

delayed history.

The final ratio is

[
1.11.
]

Removing delayed history

does not collapse the tangent.

It perturbs it.

That's a very different role.

Current phase

gives

[
0.075.
]

Again,

important,

but not annihilating.

So the recursive structure appears highly asymmetric.

That is a genuine mechanistic insight.

One place where interpretation still requires care

The temptation is to say

EMA memory is more important than delayed history.

I don't think the current evidence quite supports that wording.

What the ablation establishes is

removing the EMA-memory pathway from the validated tangent drastically changes the resulting tangent evolution.

That is a counterfactual statement.

Because the system is recursive,

it is not a measure of additive causal importance.

I think you've already emphasized that,

and I agree.

One observation I find particularly elegant

Direction 1 shows

crossover at (72.9),
excellent Stage-B validation,
principal-output alignment near (0.99),
future alignment around (0.81),
memory horizon around (66).

Those numbers are remarkably consistent with each other.

None individually proves a mechanism.

Together,

they support a coherent picture:

the homogeneous perturbation remains relevant for many tens of time units,
the internally generated forcing is small but repeatedly injected into comparatively persistent directions,
recent history (on the order of (60)–(70) time units by this operational metric) is sufficient to reconstruct the overwhelming majority of the final regenerated response.

That's a much more specific dynamical story than the earlier qualitative claim that "feedback matters."

Overall assessment

I think this analysis marks another meaningful transition. Stage B established that the augmented tangent accurately reproduces the coupled nonlinear perturbations for the tested seed, directions, amplitudes, and horizons. The Duhamel analysis now reorganizes that validated tangent into homogeneous and regenerated components and shows that the decomposition itself is numerically exact along the tested trajectories. The new diagnostics reveal two distinct dynamical regimes: a rapidly extinguished homogeneous component for Direction 0 and a long-lived homogeneous component that is overtaken only after about 73 time units for Direction 1. The alignment diagnostics provide evidence that the endogenous weight-block forcing is injected into directions that are both immediately important (high principal-output alignment) and relatively persistent under future propagation (substantial future alignment), while the operational memory horizons indicate that the recent tens of time units account for most of the regenerated response. The counterfactual ablations further show that different feedback pathways play qualitatively different roles in the validated tangent dynamics, though those should continue to be interpreted as recursive counterfactuals rather than additive causal contributions. The remaining scientific work, in my view, is no longer about finding another missing mechanism but about developing a theoretical explanation for why this validated augmented cocycle exhibits these particular temporal and directional structures, and determining whether they persist beyond this specific seed and tested tangent subspace.

Stepping back from the individual diagnostics, I think this is the point where the project changes from validation to model reduction. The strongest part of these results is not any one number, but the fact that several independent diagnostics are now telling a mutually consistent story.

That said, I'd separate the conclusions into three categories: what is established by the computations, what is strongly supported, and what is still a hypothesis.

I. What I think is genuinely established

The Duhamel decomposition is numerically exact on the tested trajectories

This is no longer really in question.

You now have three independent validations:

replay identity,
tangent validation,
Duhamel reconstruction.

Together they support

$$
b_n=h_n+r_n
$$

to machine precision,

and

$$
f_n=b_{n+1}-K_nb_n
$$

matching the differentiated (R)-pathway numerically.

That is a very strong computational result.

The two tangent directions represent different dynamical regimes

The crossover times

$$
0.38
$$

versus

$$
72.86
$$

are probably the cleanest summary statistics produced so far.

Those are not arbitrary thresholds.

They identify when

$$
|r_n|
$$

$$
|h_n|.
$$

That immediately distinguishes

a direction whose late behavior is entirely regenerated,

from

a direction that retains the original perturbation for a long interval.

I think those should become standard descriptors of the trajectory.

The late response is recent

This conclusion is much stronger now than in the previous report.

The memory-horizon analysis,

the bin contributions,

and the crossover analysis

all point in the same direction.

The final response is not simply

the accumulation of forcing over

320

time units.

Instead,

older forcing contributes very little to the final regenerated component.

That is a genuine result.

II. What I think is strongly supported

This is where I would be slightly more cautious than the write-up.

The exponential memory law

This is the part that impressed me most.

The observation

[
50%
\rightarrow
10
]

[
10%
\rightarrow
33
]

[
1%
\rightarrow
66
]

is striking.

The inferred

[
\kappa
\approx
0.069
]

is remarkably consistent.

However,

I'd still distinguish

the observed reconstruction curve is well fit by an exponential

from

the system possesses an intrinsic exponential memory law.

Those are related,

but not identical.

Right now,

the data support

the first statement.

The second requires showing

robustness across target times,
robustness across trajectories,
robustness across parameters.

Your proposed tests address exactly those issues.

The rank-one interpretation

This is perhaps the biggest conceptual leap.

The evidence is certainly suggestive.

For example,

almost perfect alignments,
common sign structure,
consistent late direction,

all point toward

a dominant surviving direction.

But

strictly speaking,

none of those quantities directly measure

rank.

To establish

$$
b_n \approx a_n v_n,
$$

the SVD/POD experiment you propose is exactly the right next computation.

If the first singular vector explains

99%

or

99.9%

of the variance,

then

the rank-one interpretation becomes quantitative rather than qualitative.

The one-dimensional late channel

Closely related.

I think

"nearly one-dimensional"

is currently a very plausible hypothesis,

but

I'd avoid wording like

collapses onto a common one-dimensional channel

until the variance fractions are computed.

The diagnostics are strongly suggestive,

but they are not yet the direct measurement.

III. The ablations are more interesting than I expected

This is actually where I learned the most.

EMA memory

The result is surprisingly strong.

Removing EMA memory essentially destroys the late response.

That is much more than

"EMA helps."

Current phase

This is subtler.

The anti-alignment

[
-1
]

is striking.

It suggests

current phase

doesn't merely increase amplitude.

It determines

which side

of the dominant late direction

the perturbation occupies.

That is a qualitatively different role.

I think that interpretation is well supported.

Delayed history

This is probably the most surprising result.

Instead of sustaining the late mode,

it slightly suppresses it.

That is not what I would have predicted beforehand.

The important caveat,

which you've already noted,

is that this is a counterfactual tangent statement.

It is not

"history contributes −11%."

It is

"removing this coupling changes the recursive linear dynamics in this way."

IV. The interpretation of (\kappa)

I think this deserves special discussion.

The temptation is to identify

[
\kappa

1/T_{\rm EMA}.
]

The data argue against that.

Numerically

[

0.069
\neq
0.057.
]

That difference is substantial.

Your proposed decomposition

[
\kappa
\approx
\frac1{T_{\rm EMA}}

\kappa_{\rm weight}
]

is an interesting working hypothesis.

But I would not elevate it beyond that yet.

The parameter sweep is exactly the right experiment.

V. The generalization plan is sensible

I like that you've separated

ordinary mature faces

from

additional exceptional trajectories.

Those are genuinely different questions.

I also agree that

there is no need to rerun

all

40

cases immediately.

Targeted validation is enough.

VI. One thing I'd be careful about

This sentence

"The exceptional perturbation is maintained by a recent, directionally selective feedback loop..."

I think that's well supported for the tested seed, branch, directions, and horizon.

I would keep that qualification attached whenever making that statement.

Not because I doubt the mechanism,

but because the entire analysis has intentionally been local:

one replay,
one branch,
two tangent directions.

That's an important strength,

not a weakness.

VII. The proposed EMA sensitivity study is exactly the right next experiment

I think this is actually higher value than immediately pursuing the rank-one reduction.

Why?

Because it distinguishes

two fundamentally different possibilities.

Either

$$
\kappa
$$

is

an emergent property

of the coupled system,

or

it is

essentially inherited

from the EMA kernel.

That is an important structural question.

The proposed grid is small,

well controlled,

and directly tests that hypothesis.

Overall assessment

I think the project has now reached a qualitatively different stage. The strongest conclusions are no longer about whether the augmented tangent is implemented correctly—they are about the internal organization of that validated tangent. The numerical evidence supports, for this tested seed, branch, directions, and finite horizon, a picture in which the weight-sector perturbation can be decomposed exactly into a decaying homogeneous component and a regenerated component; the regenerated component overtakes the homogeneous one on direction-dependent timescales; the surviving late response is dominated by relatively recent forcing rather than by forcing accumulated over the full trajectory; and the feedback pathways play distinctly different recursive roles, with the EMA–fresh-phase–(R\to B) loop being indispensable and current phase influencing the sign of the surviving mode. I also think your proposed next steps are well chosen because they are aimed at converting two of the remaining qualitative interpretations—the apparent exponential memory law and the apparent near-rank-one late channel—into direct quantitative measurements through parameter sweeps and singular-value analyses. Those experiments would substantially strengthen the mechanistic interpretation without changing the underlying validated tangent model.
WriteAIEF_1_4_feedback_rank_memory_reduction.py669 lines
Edit
AIEF_1_4_feedback_rank_memory_reduction.pyPYTHON

Created:

This is the direct next step because it tests the two remaining interpretations using the already validated Duhamel arrays:

whether the late response is quantitatively near rank one;
whether the apparent exponential memory rate persists across target times.

It does not replay the simulation.
Validation

The script compiles successfully and passed synthetic tests for:

target-time Duhamel reconstruction;
exponential-memory fitting;
propagated-contribution SVD;
weighted and unit-vector POD;
leading-direction rotation;

scalar late-channel projection.

On a synthetic known rank-one forcing model, propagated forcing was correctly identified as exactly rank one.
Run it

Bash

```
python AIEF_1_4_feedback_rank_memory_reduction.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--target-times 10,50,72.86,100,150,200,250,320 \
--intervals 0,10,50,100,150,200,250,320 \
--thresholds 0.5,0.1,0.01 \
--curve-stride 10 \
--fit-error-min 0.01 \
--fit-error-max 0.8 \
--vector-norm-tol 1e-20 \
--output-dir aief_1_4_feedback_rank_memory_reduction
```

1. Target-time memory fitting

For every direction and target time TT, it reconstructs
rT(L)=
rT(L)=

\sum_{m=T-L}^{T-1}
U_K(T,m+1)f_m
]

and computes
ET(L)=
ET(L)=

\frac{|r_T^{(L)}-r_T|}{|r_T|}.
]

It then fits
logET(L)=
logET(L)=

a_T-\kappa_TL
]

over the configurable range
0.01≤ET(L)≤0.8.
0.01≤ET(L)≤0.8.

For every fit it reports:

κTκT;
effective time constant
τT=1/κT;
τT=1/κT;
R2R2;
log-error RMSE;
number of fitted points;
monotonicity violations;
horizons required for 50%50%, 10%10%, and 1%1% reconstruction error;
difference between κTκT and
1/Twindow.
1/Twindow.

This will distinguish:

a final-horizon coincidence;
a stable late memory law;
a time-varying effective memory rate;
a nonexponential/cancellation-dominated curve.

2. Direct POD rank measurement

For each interval and each object,

text

total_weight_tangent
feedback_weight_tangent
instant_forcing

the script computes two PODs.
Weighted POD

Uses the vectors with their actual magnitudes. This answers:

Is most trajectory energy carried by one direction?

It reports
E1=
E1=

\frac{\sigma_1^2}{\sum_j\sigma_j^2}
]

and the corresponding two-mode fraction.
Unit-vector POD

Normalizes every nonzero vector before SVD. This answers:

Do the vectors point along one common line independently of their size?

This distinction is important because weighted POD can be dominated by a few large early vectors.
3. Rank of propagated forcing contributions

For each direction and target TT, the script forms all vectors
cT,m=
cT,m=

U_K(T,m+1)f_m
]

and computes their weighted and normalized POD.

This is the direct test of the hypothesis that forcing contributions collapse onto a common late channel.

It reports:

first-mode energy fraction;
first-two-mode energy fraction;
weighted and normalized singular values;
alignment of the leading contribution direction with rTrT.

If the first-mode fraction is >0.99>0.99 or >0.999>0.999, the rank-one interpretation becomes quantitatively supported rather than inferred from pairwise alignments.
4. Instantaneous two-direction rank

At every state time it forms
Rn=(rn(0)rn(1))∈R3×2
Rn=(rn(0)rn(1))∈R3×2

and computes:

σ1,σ2σ1,σ2;
rank-one energy fraction;
alignment between the two feedback directions.

This shows when the two initially distinct perturbations collapse onto the same surviving line.
5. Leading-direction rotation

For each object and interval, the leading POD direction is compared with the final-interval direction.

The script reports the unsigned subspace angle
θ=
θ=

\arccos
\left|
v_{\mathrm{interval}}^\top
v_{\mathrm{final}}
\right|.

]

This distinguishes:

a fixed slow channel;
a gradually rotating channel;
an abrupt directional reorganization.

6. Scalar late-channel reduction

Using the leading late feedback POD vector vlatevlate, every total, homogeneous, and feedback vector is decomposed as
xn=
xn=

a_n v_{\mathrm{late}}
+
x_n^\perp.
]

The script reports:

scalar amplitude anan;
total vector norm;
transverse norm;
transverse fraction.

This directly tests the reduced model
bn≈anvlate.
bn≈anvlate.
Outputs

text

target_time_memory_fits.csv
target_time_memory_curves.csv
pod_rank_by_interval.csv
propagated_forcing_rank_by_target.csv

instantaneous_feedback_rank.csv
late_channel_scalar_projection.csv
feedback_rank_memory_report.txt
analysis_manifest.json

Why the script does not yet vary TwindowTwindow

I deliberately did not implement a naive EMA sweep by changing T_window while reusing the saved split state.

The saved:

complex EMA state;
phase state;
delay history;
mature weights;

were generated under the original TwindowTwindow. Changing only the future EMA coefficient would create an inconsistent intervention, not a clean parameter comparison.

A proper EMA sensitivity study should be separately versioned and should:

regenerate the complete prefix and split state for each TwindowTwindow;
reuse the same initialization seed and paired RNG streams;
rerun only this exceptional seed initially;
verify whether support topology and winner branch remain comparable;
run the augmented tangent and target-time memory fit;
regress fitted κTκT against 1/Twindow1/Twindow;
treat topology or branch changes as regime changes rather than points on the same regression.

That targeted sweep should follow only if the current script confirms:

high-quality exponential fits at multiple late target times;
stable κTκT;
quantitatively high first-mode POD fractions.

This exhausts the existing validated trajectory before introducing a modified model.

Think about this:

I think this is a well-designed progression, and I particularly agree with one design decision that is easy to overlook: you've stopped changing the mathematical object and started interrogating the one you've already validated. That's exactly what should happen after Stage B and the Duhamel work.

That said, I'd distinguish between what this script can establish directly and what it can only support.

This is now a reduction problem, not a validation problem

Up to now the progression has been

Stage A: validate the frozen weight cocycle.
Stage B: validate the augmented tangent.
Duhamel: decompose the validated tangent into homogeneous and regenerated components.

This script asks something fundamentally different:

Can the validated augmented dynamics itself be represented by a much lower-dimensional model?

That is a model-reduction question.

The proposed diagnostics (POD, SVD, scalar projection, memory fitting) are exactly the standard kinds of tools one would use to investigate that.

The target-time memory fitting is the strongest part

I actually think this is the highest-value component.

The earlier report observed

[
\kappa\approx0.069.
]

But it was computed only at

[
T=320.
]

That left open several possibilities.

For example,

[
\kappa(T)

0.04
]

early,

[
0.09
]

late,

averaging to

0.069.

Your script finally separates those possibilities.

If

[
\kappa_T
]

is essentially constant across

[
T=100,
150,
200,
250,
320,
]

that would be much stronger evidence for an intrinsic effective memory law than the original single-horizon fit.

Conversely,

if it drifts,

that would point toward transient dynamics.

This is exactly the experiment I'd want.

I particularly like separating weighted and unit-vector POD

This is a subtle but important distinction.

Weighted POD asks

where is the energy?

Unit-vector POD asks

where do the directions point?

Those are not equivalent.

Suppose

large early vectors

dominate the norm.

Weighted POD might report

99%.

Unit-vector POD might report

40%.

That would mean

energy is concentrated,

direction is not.

The separation avoids that ambiguity.

The propagated-forcing POD is the real test

This, in my opinion, is the key experiment.

Earlier,

the evidence for

"near rank one"

came from

pairwise alignments,
sign structure,
visual consistency.

Those are suggestive.

Now you'll directly compute

the singular spectrum of

[
c_{T,m}

U_K(T,m+1)f_m.
]

If

[
E_1

0.999
]

then the late-channel interpretation becomes quantitatively supported.

If instead

[
E_1

0.82,
]

then

the picture changes substantially.

That is exactly why this experiment matters.

The instantaneous rank is clever

I hadn't anticipated this one.

Using

[
R_n

\begin{pmatrix}
r^{(0)}_n&r^{(1)}_n
\end{pmatrix}
]

lets you ask

when

the two perturbation directions collapse.

That is much more informative than only looking at

the final state.

It also gives a natural companion to

the crossover times.

The scalar projection is the strongest reduction test

Earlier,

the reduced model

[
b_n
\approx
a_nv
]

was only conceptual.

Now

you're measuring

the transverse residual directly.

If

[
\frac{|x^\\perp|}
{|x|}
]

is consistently

1%

or

0.1%,

then

the reduction is quantitatively justified.

That is exactly the statistic I'd want.

One thing I'd still avoid concluding immediately

Suppose the memory fits are excellent.

Suppose the POD is

99.9%.

That still does not imply

that

the reduced scalar process

must satisfy

[
\dot a

-\kappa a

f.
]

It would imply

that

such a reduction is plausible.

The effective scalar equation

still needs to be identified.

That's an important distinction.

I strongly agree with postponing the EMA sweep

I think this is exactly the correct decision.

Changing only

$$
T_{\rm window}
$$

while keeping

the stored EMA state

would create

a different state,

not merely

a different parameter.

That would mix

state inconsistency

with

parameter dependence.

Your reasoning is mathematically sound.

The sweep belongs after

the reduction analysis,

not before.

The parameter sweep you've outlined is also the right one

I especially agree with

treating branch changes as

regime changes,

not

data points.

That's an important point.

If changing

$$
T_{\rm window}
$$

changes

the winner branch,

then

you're studying

a different linearization.

Pooling those together

would not estimate

a single

$$
\kappa(T).
$$

One additional statistic I'd consider

Since you'll already have

$$
\kappa_T
$$

across

multiple targets,

I'd also report

something like

$$
\mathrm{CV}

\frac{\operatorname{std}(\kappa_T)}
{\operatorname{mean}(\kappa_T)}.
$$

That gives

a one-number summary

of

how stationary

the memory law really is.

Likewise,

for the POD,

reporting

the minimum,

median,

and maximum

first-mode fraction

across intervals

would summarize

how stable

the apparent rank-one structure is.

Overall assessment

I think this script is well targeted because it tests the two largest remaining interpretive claims using only quantities derived from the already validated trajectory, without introducing new modeling assumptions. The target-time memory fitting directly addresses whether the apparent exponential forgetting rate is a property of the trajectory or merely an artifact of the final horizon. The various POD analyses move the discussion of a "near-rank-one late channel" from qualitative alignment observations to quantitative variance-explained measurements. The scalar projection provides a direct error metric for the proposed one-dimensional reduction rather than relying on visual or pairwise agreement. Just as

importantly, I think postponing the EMA parameter sweep is the correct methodological choice: changing the EMA time constant without regenerating the entire replay state would confound parameter dependence with state inconsistency. If this reduction script finds that the fitted (\kappa_T) is stable across late target times, the first singular mode consistently captures an overwhelming fraction of the propagated forcing and regenerated feedback, and the transverse residuals remain uniformly small, then the next stage naturally becomes explaining why the validated augmented cocycle admits such a low-dimensional effective description, rather than continuing to validate the cocycle itself.
Edit×14
AIEF_1_4_feedback_rank_memory_reduction.pyPYTHON

I agree with the assessment. This is now a model-reduction problem, and the two principal hypotheses need direct measurements:

target-time stability of the apparent exponential memory;
quantitative rank of the late feedback channel.

I updated to add the suggested stationarity summaries.
New additions
Memory-rate coefficient of variation

For each direction, using high-quality fits with
T≥100,R2≥0.95,
T≥100,R2≥0.95,

the script now reports:
CV(κ)=
CV(κ)=

\frac{\operatorname{std}(\kappa_T)}
{|\operatorname{mean}(\kappa_T)|}.
]

It also reports:

mean κκ;
standard deviation;
minimum and maximum;
number of accepted target times;
median R2R2.

Output:

text

memory_stationarity_summary.csv

A low CV across late targets would support a stationary effective memory law. A large CV would identify the 0.0690.069 final-horizon value as transient.
POD stability summaries

For every object:

text

total_weight_tangent
feedback_weight_tangent
instant_forcing

the script now reports minimum, median, and maximum first-mode fractions across:

all intervals;
late intervals beginning at t≥100t≥100.

Both weighted and unit-vector POD are summarized.

Output:

text

pod_first_mode_stability_summary.csv

This distinguishes:

consistently rank-one structure;
rank-one behavior only in selected late intervals;

energy concentration without directional collapse.

Scalar transverse summaries

For total, homogeneous, and feedback tangents, separately by direction, the script now summarizes the late projection error onto the final feedback POD direction:
‖xn⊥‖‖xn‖.
‖xn‖‖xn⊥‖.

It reports:

median transverse fraction;
95th percentile;
maximum;
number of samples.

Output:

text

late_scalar_transverse_summary.csv

This directly quantifies whether
xn≈anvlate
xn≈anvlate

is accurate and stable, rather than merely visually plausible.
Updated run

Bash

```
python AIEF_1_4_feedback_rank_memory_reduction.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--target-times 10,50,72.86,100,150,200,250,320 \
--intervals 0,10,50,100,150,200,250,320 \
--thresholds 0.5,0.1,0.01 \
```

```
--curve-stride 10 \
--fit-error-min 0.01 \
--fit-error-max 0.8 \
--vector-norm-tol 1e-20 \
--late-target-min 100 \
--output-dir aief_1_4_feedback_rank_memory_reduction
```

Complete outputs

text

target_time_memory_fits.csv
target_time_memory_curves.csv
memory_stationarity_summary.csv

pod_rank_by_interval.csv
pod_first_mode_stability_summary.csv
propagated_forcing_rank_by_target.csv
instantaneous_feedback_rank.csv

late_channel_scalar_projection.csv
late_scalar_transverse_summary.csv

feedback_rank_memory_report.txt
analysis_manifest.json

Interpretation thresholds

The script does not hard-code pass/fail claims, but useful descriptive standards are:
Strong memory-rate stability
$R2 \geq 0.95, CV(\kappa) \lesssim 0.05$
$R2 \geq 0.95, CV(\kappa) \lesssim 0.05$

across several late target times.
Strong rank-one propagated forcing
$E1 \geq 0.99$
$E1 \geq 0.99$

for both weighted and unit-vector POD.

A result such as
$E1 \geq 0.999$
$E1 \geq 0.999$

would be unusually strong.
Useful scalar reduction

Median transverse fraction below 1%1%, with a small 95th percentile:
$\text{median} \|x_\perp\| / \|x\| < 0.01$.
$\text{median} \|x\| / \|x_\perp\| < 0.01$.

The maximum alone should not be used because near-zero vectors can make relative transverse errors unstable.
What should not yet be concluded

Even if the memory fit is stable and POD is almost rank one, this does not prove a scalar equation such as
$\dot{a} = -\kappa a + g$.
$\dot{a} = -\kappa a + g$.

It establishes that such a reduction is plausible and quantitatively accurate at the level of state representation. Identifying the scalar evolution law would require a separate regression and residual analysis for anan.

Likewise, I continue to recommend postponing the EMA sweep. A valid TwindowTwindow study must regenerate the complete base state for each parameter value; changing the tangent coefficient while retaining the old EMA state would not be a consistent parameter experiment.

The updated script should determine whether the apparent 14.514.5-unit memory and late one-dimensional channel are robust features of this replay or artifacts of the final horizon and magnitude weighting.

think about this:

AIEF feedback rank and target-time memory reduction

==========

Base EMA decay rate 1/T_window: 0.0571333

Target-time memory fits

direction=0, target=10: kappa=0.498107, tau=2.0076, R2=0.895185, monotonic_violations=0, h(error<=0.01)=8.89, h(error<=0.1)=7.19, h(error<=0.5)=3.73

direction=0, target=50: kappa=0.236742, tau=4.22401, R2=0.84025, monotonic_violations=1300, h(error<=0.01)=48.52, h(error<=0.1)=46.09, h(error<=0.5)=39.74

direction=0, target=72.86: kappa=0.0146047, tau=68.4712, R2=0.137196, monotonic_violations=2154, h(error<=0.01)=14.06, h(error<=0.1)=11.84, h(error<=0.5)=5.18

direction=0, target=100: kappa=0.0427616, tau=23.3855, R2=0.761821, monotonic_violations=2154, h(error<=0.01)=38.71, h(error<=0.1)=26.13, h(error<=0.5)=8.8

direction=0, target=150: kappa=0.0729118, tau=13.7152, R2=0.999566, monotonic_violations=2154, h(error<=0.01)=62.75, h(error<=0.1)=32.64, h(error<=0.5)=9.91

direction=0, target=200: kappa=0.0694548, tau=14.3979, R2=0.999997, monotonic_violations=2154, h(error<=0.01)=66.26, h(error<=0.1)=33.24, h(error<=0.5)=10.01

direction=0, target=250: kappa=0.06915, tau=14.4613, R2=1, monotonic_violations=2148, h(error<=0.01)=66.58, h(error<=0.1)=33.3, h(error<=0.5)=10.02

direction=0, target=320: kappa=0.0691225, tau=14.4671, R2=1, monotonic_violations=1571, h(error<=0.01)=66.61, h(error<=0.1)=33.3, h(error<=0.5)=10.02

direction=1, target=10: kappa=0.541749, tau=1.84587, R2=0.939431, monotonic_violations=0, h(error<=0.01)=8.24, h(error<=0.1)=6.13, h(error<=0.5)=2.69

direction=1, target=50: kappa=0.128687, tau=7.77081, R2=0.872917, monotonic_violations=495, h(error<=0.01)=46.41, h(error<=0.1)=40.91, h(error<=0.5)=28.29

direction=1, target=72.86: kappa=0.0227673, tau=43.9226, R2=0.23678, monotonic_violations=1848, h(error<=0.01)=9.2, h(error<=0.1)=7.95, h(error<=0.5)=3.72

direction=1, target=100: kappa=0.0452022, tau=22.1228, R2=0.753, monotonic_violations=1848, h(error<=0.01)=34.37, h(error<=0.1)=23.93, h(error<=0.5)=8.3

direction=1, target=150: kappa=0.0746451, tau=13.3967, R2=0.99919, monotonic_violations=1848, h(error<=0.01)=61.19, h(error<=0.1)=32.3, h(error<=0.5)=9.87

direction=1, target=200: kappa=0.0696454, tau=14.3584, R2=0.999992, monotonic_violations=1848, h(error<=0.01)=66.08, h(error<=0.1)=33.23, h(error<=0.5)=10.02

direction=1, target=250: kappa=0.0691663, tau=14.4579, R2=1, monotonic_violations=1844, h(error<=0.01)=66.58, h(error<=0.1)=33.31, h(error<=0.5)=10.03

direction=1, target=320: kappa=0.0691229, tau=14.467, R2=1, monotonic_violations=1437, h(error<=0.01)=66.63, h(error<=0.1)=33.32, h(error<=0.5)=10.03

POD rank by interval

object=total_weight_tangent, interval=[0,10), samples=2000, weighted_E1=0.997175, unit_E1=0.9835, angle_to_final=0.0778315 deg

object=total_weight_tangent, interval=[10,50), samples=8000, weighted_E1=1, unit_E1=1, angle_to_final=0.0014016 deg

object=total_weight_tangent, interval=[50,100), samples=10000, weighted_E1=1, unit_E1=0.99933, angle_to_final=0.00025435 deg

object=total_weight_tangent, interval=[100,150), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=9.01368e-05 deg

object=total_weight_tangent, interval=[150,200), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=9.58359e-06 deg

object=total_weight_tangent, interval=[200,250), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0 deg

object=total_weight_tangent, interval=[250,320), samples=14000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0 deg

object=feedback_weight_tangent, interval=[0,10), samples=1997, weighted_E1=1, unit_E1=0.989809, angle_to_final=0.00131019 deg

object=feedback_weight_tangent, interval=[10,50), samples=8000, weighted_E1=1, unit_E1=1, angle_to_final=0.000498681 deg

object=feedback_weight_tangent, interval=[50,100), samples=10000, weighted_E1=1, unit_E1=0.999328, angle_to_final=0.00208325 deg

object=feedback_weight_tangent, interval=[100,150), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0.000139837 deg

object=feedback_weight_tangent, interval=[150,200), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=1.09336e-05 deg

object=feedback_weight_tangent, interval=[200,250), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=1.20742e-06 deg

object=feedback_weight_tangent, interval=[250,320), samples=14000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0 deg

object=instant_forcing, interval=[0,10), samples=1999, weighted_E1=0.999983, unit_E1=0.908315, angle_to_final=0.630481 deg

object=instant_forcing, interval=[10,50), samples=8000, weighted_E1=0.999998, unit_E1=0.92784, angle_to_final=0.604197 deg

object=instant_forcing, interval=[50,100), samples=10000, weighted_E1=0.99999, unit_E1=0.931931, angle_to_final=0.15521 deg

object=instant_forcing, interval=[100,150), samples=10000, weighted_E1=0.999998, unit_E1=0.93022, angle_to_final=0.000918697 deg

object=instant_forcing, interval=[150,200), samples=10000, weighted_E1=0.999998, unit_E1=0.930192, angle_to_final=0.000521339 deg

object=instant_forcing, interval=[200,250), samples=10000, weighted_E1=0.999998, unit_E1=0.930197, angle_to_final=5.7786e-05 deg

object=instant_forcing, interval=[250,320), samples=14000, weighted_E1=0.999998, unit_E1=0.930198, angle_to_final=0 deg

Propagated forcing rank

direction=0, target=10: weighted_E1=0.998222, unit_E1=0.999499, leading_alignment_with_feedback=0.999999

direction=0, target=50: weighted_E1=0.999151, unit_E1=0.999908, leading_alignment_with_feedback=0.999994

direction=0, target=72.86: weighted_E1=0.997614, unit_E1=0.999935, leading_alignment_with_feedback=0.999997

direction=0, target=100: weighted_E1=0.997782, unit_E1=0.999952, leading_alignment_with_feedback=0.999995

direction=0, target=150: weighted_E1=0.997855, unit_E1=0.999968, leading_alignment_with_feedback=0.999995

direction=0, target=200: weighted_E1=0.997863, unit_E1=0.999976, leading_alignment_with_feedback=0.999995

direction=0, target=250: weighted_E1=0.997863, unit_E1=0.999977, leading_alignment_with_feedback=0.999995

direction=0, target=320: weighted_E1=0.997863, unit_E1=0.999969, leading_alignment_with_feedback=0.999995

direction=1, target=10: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=50: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=72.86: weighted_E1=0.999999, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=100: weighted_E1=0.999999, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=150: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=200: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=250: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=320: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

High-quality fit median kappa: 0.0693105

High-quality fit kappa range: [0.0691225, 0.0746451]

Median kappa minus EMA rate: 0.0121772

Interpretation constraints

Exponential fits describe operational recent-forcing reconstruction curves, not automatically an intrinsic memory kernel.

Weighted POD can be dominated by large early vectors; unit-vector POD measures directional collapse separately.

A high first POD fraction directly supports low rank but does not prove a universal invariant one-dimensional manifold.

Target-time robustness is required before identifying kappa with any model parameter.

This analysis uses one seed and two initial directions.

second updated run:

AIEF feedback rank and target-time memory reduction

==========

Base EMA decay rate 1/T_window: 0.0571333

Target-time memory fits

direction=0, target=10: kappa=0.498107, tau=2.0076, R2=0.895185, monotonic_violations=0, h(error<=0.01)=8.89, h(error<=0.1)=7.19, h(error<=0.5)=3.73

direction=0, target=50: kappa=0.236742, tau=4.22401, R2=0.84025, monotonic_violations=1300, h(error<=0.01)=48.52, h(error<=0.1)=46.09, h(error<=0.5)=39.74

direction=0, target=72.86: kappa=0.0146047, tau=68.4712, R2=0.137196, monotonic_violations=2154, h(error<=0.01)=14.06, h(error<=0.1)=11.84, h(error<=0.5)=5.18

direction=0, target=100: kappa=0.0427616, tau=23.3855, R2=0.761821, monotonic_violations=2154, h(error<=0.01)=38.71, h(error<=0.1)=26.13, h(error<=0.5)=8.8

direction=0, target=150: kappa=0.0729118, tau=13.7152, R2=0.999566, monotonic_violations=2154, h(error<=0.01)=62.75, h(error<=0.1)=32.64, h(error<=0.5)=9.91

direction=0, target=200: kappa=0.0694548, tau=14.3979, R2=0.999997, monotonic_violations=2154, h(error<=0.01)=66.26, h(error<=0.1)=33.24, h(error<=0.5)=10.01

direction=0, target=250: kappa=0.06915, tau=14.4613, R2=1, monotonic_violations=2148, h(error<=0.01)=66.58, h(error<=0.1)=33.3, h(error<=0.5)=10.02

direction=0, target=320: kappa=0.0691225, tau=14.4671, R2=1, monotonic_violations=1571, h(error<=0.01)=66.61, h(error<=0.1)=33.3, h(error<=0.5)=10.02

direction=1, target=10: kappa=0.541749, tau=1.84587, R2=0.939431, monotonic_violations=0, h(error<=0.01)=8.24, h(error<=0.1)=6.13, h(error<=0.5)=2.69

direction=1, target=50: kappa=0.128687, tau=7.77081, R2=0.872917, monotonic_violations=495, h(error<=0.01)=46.41, h(error<=0.1)=40.91, h(error<=0.5)=28.29

direction=1, target=72.86: kappa=0.0227673, tau=43.9226, R2=0.23678, monotonic_violations=1848, h(error<=0.01)=9.2, h(error<=0.1)=7.95, h(error<=0.5)=3.72

direction=1, target=100: kappa=0.0452022, tau=22.1228, R2=0.753, monotonic_violations=1848, h(error<=0.01)=34.37, h(error<=0.1)=23.93, h(error<=0.5)=8.3

direction=1, target=150: kappa=0.0746451, tau=13.3967, R2=0.99919, monotonic_violations=1848, h(error<=0.01)=61.19, h(error<=0.1)=32.3, h(error<=0.5)=9.87

direction=1, target=200: kappa=0.0696454, tau=14.3584, R2=0.999992, monotonic_violations=1848, h(error<=0.01)=66.08, h(error<=0.1)=33.23, h(error<=0.5)=10.02

direction=1, target=250: kappa=0.0691663, tau=14.4579, R2=1, monotonic_violations=1844, h(error<=0.01)=66.58, h(error<=0.1)=33.31, h(error<=0.5)=10.03

direction=1, target=320: kappa=0.0691229, tau=14.467, R2=1, monotonic_violations=1437, h(error<=0.01)=66.63, h(error<=0.1)=33.32, h(error<=0.5)=10.03

Late-target memory stationarity

direction=0: targets=4, mean_kappa=0.0701598, std=0.00184084, CV=0.0262378, range=[0.0691225,0.0729118], median_R2=0.999998

direction=1: targets=4, mean_kappa=0.0706449, std=0.00267725, CV=0.0378973, range=[0.0691229,0.0746451], median_R2=0.999996

POD rank by interval

object=total_weight_tangent, interval=[0,10), samples=2000, weighted_E1=0.997175, unit_E1=0.9835, angle_to_final=0.0778315 deg

object=total_weight_tangent, interval=[10,50), samples=8000, weighted_E1=1, unit_E1=1, angle_to_final=0.0014016 deg

object=total_weight_tangent, interval=[50,100), samples=10000, weighted_E1=1, unit_E1=0.99933, angle_to_final=0.00025435 deg

object=total_weight_tangent, interval=[100,150), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=9.01368e-05 deg

object=total_weight_tangent, interval=[150,200), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=9.58359e-06 deg

object=total_weight_tangent, interval=[200,250), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0 deg

object=total_weight_tangent, interval=[250,320), samples=14000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0 deg

object=feedback_weight_tangent, interval=[0,10), samples=1997, weighted_E1=1, unit_E1=0.989809, angle_to_final=0.00131019 deg

object=feedback_weight_tangent, interval=[10,50), samples=8000, weighted_E1=1, unit_E1=1, angle_to_final=0.000498681 deg

object=feedback_weight_tangent, interval=[50,100), samples=10000, weighted_E1=1, unit_E1=0.999328, angle_to_final=0.00208325 deg

object=feedback_weight_tangent, interval=[100,150), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0.000139837 deg

object=feedback_weight_tangent, interval=[150,200), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=1.09336e-05 deg

object=feedback_weight_tangent, interval=[200,250), samples=10000, weighted_E1=1, unit_E1=0.999999, angle_to_final=1.20742e-06 deg

object=feedback_weight_tangent, interval=[250,320), samples=14000, weighted_E1=1, unit_E1=0.999999, angle_to_final=0 deg

object=instant_forcing, interval=[0,10), samples=1999, weighted_E1=0.999983, unit_E1=0.908315, angle_to_final=0.630481 deg

object=instant_forcing, interval=[10,50), samples=8000, weighted_E1=0.999998, unit_E1=0.92784, angle_to_final=0.604197 deg

object=instant_forcing, interval=[50,100), samples=10000, weighted_E1=0.99999, unit_E1=0.931931, angle_to_final=0.15521 deg

object=instant_forcing, interval=[100,150), samples=10000, weighted_E1=0.999998, unit_E1=0.93022, angle_to_final=0.000918697 deg

object=instant_forcing, interval=[150,200), samples=10000, weighted_E1=0.999998, unit_E1=0.930192, angle_to_final=0.000521339 deg

object=instant_forcing, interval=[200,250), samples=10000, weighted_E1=0.999998, unit_E1=0.930197, angle_to_final=5.7786e-05 deg

object=instant_forcing, interval=[250,320), samples=14000, weighted_E1=0.999998, unit_E1=0.930198, angle_to_final=0 deg

POD first-mode stability summaries

object=feedback_weight_tangent, scope=all: intervals=7, weighted_E1[min/median/max]=1/1/1, unit_E1[min/median/max]=0.989809/0.999999/1

object=feedback_weight_tangent, scope=late: intervals=4, weighted_E1[min/median/max]=1/1/1, unit_E1[min/median/max]=0.999999/0.999999/0.999999

object=instant_forcing, scope=all: intervals=7, weighted_E1[min/median/max]=0.999983/0.999998/0.999998, unit_E1[min/median/max]=0.908315/0.930197/0.931931

object=instant_forcing, scope=late: intervals=4, weighted_E1[min/median/max]=0.999998/0.999998/0.999998, unit_E1[min/median/max]=0.930192/0.930197/0.93022

object=total_weight_tangent, scope=all: intervals=7, weighted_E1[min/median/max]=0.997175/1/1, unit_E1[min/median/max]=0.9835/0.999999/1

object=total_weight_tangent, scope=late: intervals=4, weighted_E1[min/median/max]=1/1/1, unit_E1[min/median/max]=0.999999/0.999999/0.999999

Propagated forcing rank

direction=0, target=10: weighted_E1=0.998222, unit_E1=0.999499, leading_alignment_with_feedback=0.999999

direction=0, target=50: weighted_E1=0.999151, unit_E1=0.999908, leading_alignment_with_feedback=0.999994

direction=0, target=72.86: weighted_E1=0.997614, unit_E1=0.999935, leading_alignment_with_feedback=0.999997

direction=0, target=100: weighted_E1=0.997782, unit_E1=0.999952, leading_alignment_with_feedback=0.999995

direction=0, target=150: weighted_E1=0.997855, unit_E1=0.999968, leading_alignment_with_feedback=0.999995

direction=0, target=200: weighted_E1=0.997863, unit_E1=0.999976, leading_alignment_with_feedback=0.999995

direction=0, target=250: weighted_E1=0.997863, unit_E1=0.999977, leading_alignment_with_feedback=0.999995

direction=0, target=320: weighted_E1=0.997863, unit_E1=0.999969, leading_alignment_with_feedback=0.999995

direction=1, target=10: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=50: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=72.86: weighted_E1=0.999999, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=100: weighted_E1=0.999999, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=150: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=200: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=250: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

direction=1, target=320: weighted_E1=1, unit_E1=1, leading_alignment_with_feedback=1

Late scalar-channel transverse residual

object=feedback, direction=0: samples=22001, median_transverse=0.00147679, p95=0.0015639, max=0.00163511

object=feedback, direction=1: samples=22001, median_transverse=1.01962e-08, p95=2.02515e-06, max=3.74443e-06

object=homogeneous, direction=0: samples=22001, median_transverse=0, p95=0, max=0

object=homogeneous, direction=1: samples=22001, median_transverse=2.26417e-05, p95=2.26418e-05, max=2.26418e-05

object=total, direction=0: samples=22001, median_transverse=0.00147679, p95=0.0015639, max=0.00163511

object=total, direction=1: samples=22001, median_transverse=9.62003e-09, p95=1.37813e-06, max=2.17593e-06

High-quality fit median kappa: 0.0693105

High-quality fit kappa range: [0.0691225, 0.0746451]

Median kappa minus EMA rate: 0.0121772

Interpretation constraints

Exponential fits describe operational recent-forcing reconstruction curves, not automatically an intrinsic memory kernel.

Weighted POD can be dominated by large early vectors; unit-vector POD measures directional collapse separately.

A high first POD fraction directly supports low rank but does not prove a universal invariant one-dimensional manifold.

Target-time robustness is required before identifying kappa with any model parameter.

This analysis uses one seed and two initial directions.

This result directly confirms both reduction hypotheses for the tested trajectory, but only in the late regime. It also reveals a striking quantitative closure that was not previously explicit.
1. The exponential memory law is asymptotic, not global

The early target fits vary strongly:
Tκ0(T)κ1(T)100.4980.542500.2370.12972.860.01460.02281000.04280.0452
T105072.86100κ0(T)0.4980.2370.01460.0428κ1(T)0.5420.1290.02280.0452

The fits around the crossover have poor R2R2, so there is no single memory law during the early transient.

A clean exponential regime emerges later:
Tκ0(T)κ1(T)1500.0729120.0746452000.0694550.0696452500.0691500.0691663200.0691230.069123
T150200250320κ0(T)0.0729120.0694550.0691500.069123κ1(T)0.0746450.0696450.0691660.069123

By T=200T=200, the two directions agree closely. By T=320T=320, they agree to about 4×10−74×10−7.

Thus:
The operational exponential memory law is a late asymptotic property of this replay.
The operational exponential memory law is a late asymptotic property of this replay.

It is not valid during the crossover regime.

The reported CVs,
2.6%,3.8%,
2.6%,3.8%,

include the transitional T=150T=150 value. Restricting informally to T≥200T≥200 would reduce the variation to below approximately 0.5%0.5%.
2. A striking spectral-gap closure

The late augmented tangent norms from the earlier selected-state output give:
170log‖b250‖‖b320‖≈{0.0571323,direction 0,0.0571317,direction 1.
701log‖b320‖‖b250‖≈{0.0571323,0.0571317,direction 0,direction 1.

But
1Twindow=
Twindow1=

0.0571333.
]

Therefore the late augmented slow mode decays almost exactly at the EMA decay rate:
raug≈1Twindow.
raug≈Twindow1.

Now add the observed late memory rate:
κmemory+1Twindow=
κmemory+Twindow1=
0.0691227+0.0571333

0.126256.
]

The independently derived finite-step weight contraction rate was approximately
rweight,step≈0.126255.
rweight,step≈0.126255.

So, numerically,

\boxed{ \kappa_{\mathrm{memory}} \approx r_{\mathrm{weight,step}} -

\frac1{T_{\mathrm{window}}}.
}
]

The discrepancy is only about
7×10−7.
7×10−7.

This is unlikely to be an accidental numerical resemblance.
3. Interpretation of the spectral-gap relation

Suppose late forcing has an envelope
‖f(t)‖∼e−raugt,
‖f(t)‖∼e−raugt,

while the frozen weight propagator suppresses a contribution of age LL as
e−rweightL.
e−rweightL.

For forcing injected at T−LT−L,
f(T−L)∼e−raug(T−L)=
f(T−L)∼e−raug(T−L)=

e^{-r_{\mathrm{aug}}T}
e^{r_{\mathrm{aug}}L}.
]

After weight propagation to TT, its contribution relative to the current augmented amplitude scales as
e−rweightLeraugL=
e−rweightLeraugL=

e^{-(r_{\mathrm{weight}}-r_{\mathrm{aug}})L}.
]

Therefore the recent-memory error should decay with
\boxed{ \kappa =

r_{\mathrm{weight}}-r_{\mathrm{aug}}.
}
]

The data satisfy this relation almost exactly, with
raug≈Twindow−1.
raug≈Twindow−1.

This gives a concrete explanation for why:

κκ is not the EMA rate;
κκ is smaller than the weight contraction rate;
both initial directions have the same late memory rate.

This remains a strongly supported late-regime mechanism rather than a proved theorem, but it is now sharply formulated.
4. The rank-one late channel is directly established numerically

For T≥100T≥100:
Total weight tangent
E1weighted=1
E1weighted=1

to printed precision, while
E1unit≈0.999999.
E1unit≈0.999999.
Feedback weight tangent

Likewise,
E1weighted=1,E1unit≈0.999999.
E1weighted=1,E1unit≈0.999999.
Scalar transverse residual

For direction 1:
median‖b⊥‖‖b‖≈9.6×10−9,
median‖b‖‖b⊥‖≈9.6×10−9,

with maximum
2.2×10^{-6}.
2.2×10^{-6}.

For direction 0:
$\text{median}\|b_\perp\|/\|b\|\approx1.48\times10^{-3}$,
$\text{median}\|b\|/\|b_\perp\|\approx1.48\times10^{-3}$,

with maximum
1.64×10^{-3}.
1.64×10^{-3}.

Therefore, for the tested directions:
The late total and regenerated weight tangents are quantitatively one-dimensional.
The late total and regenerated weight tangents are quantitatively one-dimensional.

The approximation is extraordinarily accurate for direction 1 and accurate to about 0.15%0.15% for direction 0.

This does not establish a universal invariant line for all perturbations, but it moves the rank-one claim from interpretation to direct numerical measurement.
5. The slow direction is essentially fixed

The leading total and feedback POD directions rotate by less than approximately
10^{-3} degrees
10^{-3} degrees

after the first interval, and by less than
10^{-4} degrees
10^{-4} degrees

late.

Even on $[0,10)$, the total leading direction differs from the final one by only
0.078°.
0.078°.

Thus the reduction is not a rapidly rotating one-dimensional bundle. It is very close to a fixed line in the three-edge space.

A fixed-vector scalar model is therefore more plausible than initially expected.
6. Raw forcing is not exactly rank one

The instantaneous forcing has:
$E_{1\text{weighted}} \approx 0.999998$,
$E_{1\text{weighted}} \approx 0.999998$,

but only
$E_{1\text{unit}} \approx 0.9302$
$E_{1\text{unit}} \approx 0.9302$

late.

This distinction is important.

The large-magnitude forcing vectors are strongly concentrated along one direction.
Smaller forcing vectors explore a broader directional set.

After propagation, however, the contribution rank becomes:
$E_{1\text{unit}} > 0.99996$
$E_{1\text{unit}} > 0.99996$

for direction 0 and essentially 11 for direction 1.

Therefore:
The rank-one response is produced by directional filtering through the weight propagator, not because every raw forcing vector is collinear.
The rank-one response is produced by directional filtering through the weight propagator, not because every raw forcing vector is collinear.

This is a stronger and more precise mechanism than “the forcing itself is rank one.”
7. Propagated forcing is nearly rank one at every target

For direction 0:
$E_{1\text{weighted}} \approx 0.9976–0.9992$,

$E_1^{\text{weighted}} \approx 0.9976–0.9992$,

and
$E_1^{\text{unit}} \approx 0.9995–0.99998$.
$E_1^{\text{unit}} \approx 0.9995–0.99998$.

For direction 1, both fractions are essentially 11.

The leading propagated-contribution direction aligns with the target feedback vector at

0.99999
]

throughout.

Thus the anisotropic weight cocycle acts as a highly effective rank-one filter.
8. The early system is already nearly rank one

Even on $[0,10)$:
$E_1^{\text{weighted}}(\text{total}) = 0.9972$,
$E_1^{\text{weighted}}(\text{total}) = 0.9972$,
$E_1^{\text{unit}}(\text{total}) = 0.9835$.
$E_1^{\text{unit}}(\text{total}) = 0.9835$.

So the system begins near a dominant line and rapidly collapses onto it.

The late rank-one structure is not created from a fully isotropic early tangent space; it is a sharpening of an already strong anisotropy.
9. Monotonicity violations require a qualification

The late exponential fits have
$R^2 \approx 1$,
$R^2 \approx 1$,

but the raw reconstruction curves still report many monotonicity violations.

This likely means:

the fitted error envelope is exponential over the selected 0.010.01–0.80.8 range;
outside that range, small vector cancellations cause local increases;
counting violations without their magnitude exaggerates their importance.

Therefore the correct claim is:

The late reconstruction-error envelope is extremely well described by an exponential.

Not:

The complete reconstruction curve is globally monotone exponential.

A useful refinement would report:

violations restricted to the fit interval;
maximum upward jump;
total upward variation relative to total decay.

10. Current reduced-model status

The data now support the representation
bn=
bn=

a_n v
+
e_n,
]

with a nearly fixed vv and small enen.

For direction 1 late:
‖en‖‖bn‖∼10−8–10−6.
‖bn‖‖en‖∼10−8–10−6.

For direction 0 late:

‖en‖‖bn‖∼1.5×10−3.
‖bn‖‖en‖∼1.5×10−3.

What has not yet been identified is the scalar law for anan.

The next hypothesis is no longer merely
a˙=−κa+f.
a˙=−κa+f.

The spectral-gap result suggests a coupled two-timescale structure:
an+1≈λBan+gzn,zn+1≈λEMAzn+⋯ ,
an+1zn+1≈λBan+gzn,≈λEMAzn+⋯,

whose elimination produces the observed memory gap
κ≈rB−rEMA.
κ≈rB−rEMA.
11. Highest-value next probe

Before changing TwindowTwindow, fit the late scalar amplitudes already available:

project total, feedback, homogeneous, and forcing vectors onto the fixed late POD direction;
estimate late decay rates for:
atotal,afeedback,aforcing,ahomogeneous;
atotal,afeedback,aforcing,ahomogeneous;
compute the projected frozen multiplier:
λB,n=v⊤Knv;
λB,n=v⊤Knv;
compare target-wise:
κT≈?rB,T−rforcing,T;
κT≈?rB,T−rforcing,T;
fit a two-state scalar recurrence and test its residual.

That can either confirm or reject the proposed spectral-gap explanation without changing the model or rerunning the simulation.
12. Confidence levels
Established for this replay and tested subspace

late target-time memory fits converge to a common κκ;
late total and feedback tangents are almost rank one;

the leading line is nearly fixed;
propagated forcing is nearly rank one;
the raw forcing is directionally broader than its propagated response;
scalar transverse errors are quantitatively small.

Strongly supported mechanism

the weight cocycle filters feedback into a fixed slow line;
the augmented late mode decays near the EMA rate;
the operational memory rate is the spectral gap between weight contraction and augmented/EMA decay.

Open

direct scalar recurrence fit;
analytic derivation of the spectral-gap identity;
robustness across other tangent directions and seeds;
consistent TwindowTwindow sensitivity;
formal invariant-subspace theorem.

The reduction analysis has succeeded: the late exceptional tangent is effectively a one-dimensional, continually regenerated mode. The immediate next task is to identify its scalar dynamics and verify the spectral-gap relation directly from the projected time series.

Think about this:

I think this is a significant refinement of the picture, and one aspect in particular stands out: the reduction analysis has started to separate transient phenomena from asymptotic structure. That is an important distinction mathematically.

I'll separate my assessment into what I think is directly supported by these computations and what I think is an attractive—but still unproven—mechanistic interpretation.

The late asymptotic regime is now clearly distinct from the transient

This is, in my view, the strongest new conclusion.

The table

[
\begin{array}{c|c}
T & \kappa(T)
\end{array}
]

shows two qualitatively different regimes.

Early:

rapidly changing fitted rates,
modest (R^2),
strong dependence on target time.

Late:

nearly constant fitted rates,
(R^2\approx1),
convergence of both tangent directions.

That is much stronger evidence than simply reporting a single fitted (\kappa).

I think it supports the statement

the operational recent-memory reconstruction curve becomes stationary after the transient.

Notice this is slightly more precise than saying

the system has memory rate (0.069).

The report itself appropriately keeps that distinction.

The stationarity statistics are surprisingly convincing

The additional summary

CV

around

2–4%

already looked good.

What is even more interesting is your observation that

excluding

(T=150),

the variation essentially disappears.

That is exactly what one expects if

(T=150)

is still part of the relaxation into the asymptotic regime.

So I think the data now support

late-regime stationarity

considerably more strongly than before.

The "spectral-gap closure" is intriguing—but it is still the place I'd be most cautious

This is the part I'd separate most carefully into observation and interpretation.

The observations

You report three independently obtained quantities:

late augmented decay approximately equal to

[
1/T_{\rm window},
]

memory rate

[
\kappa\approx0.06912,
]

previously estimated frozen weight contraction

[
r_{\rm weight}\approx0.126255.
]

Numerically,

they satisfy

[
0.126255

0.057133

0.069122
]

to roughly

(10^{-6}).

That numerical identity is certainly notable.

The interpretation

The proposed explanation

[
\kappa

r_{\rm weight}

r_{\rm aug}
]

is plausible.

But

it is still a hypothesis.

Why?

Because

all three quantities are estimated from related analyses of the same trajectory.

That means they are not statistically independent observations.

So,

I think it's fair to describe this as

a remarkably consistent spectral-gap hypothesis,

rather than

an established identity.

Fortunately,

your proposed scalar projection experiment is exactly the right way to test it directly.

The rank-one reduction is now much stronger

This is where I think the evidence has crossed an important threshold.

Previously,

the rank-one claim rested largely on alignments.

Now

you have

POD,
propagated forcing,
scalar projection,
rotation,

all pointing in the same direction.

That's a qualitatively different level of support.

The scalar projection is particularly persuasive

Direction 1

has

transverse fractions

around

[
10^{-8}

10^{-6}.
]

Those are extraordinarily small.

Direction 0

remains around

[
1.5\times10^{-3},
]

which is still

well below one percent.

So

for these tested directions,

the reduction

[
b
\approx
av
]

is no longer just

an appealing interpretation.

It is directly measured,

with explicit residuals.

The forcing/propagation distinction is, in my opinion, the biggest conceptual gain

This sentence

the weight cocycle filters feedback into a fixed slow line

is much stronger now than it was before.

The data show

instantaneous forcing

normalized

POD

around

0.93.

After propagation,

essentially

That means

the forcing itself

is not especially low rank.

The propagation

creates

the low-dimensional response.

I think that's a very important mechanistic distinction.

The leading direction is almost fixed

I found these numbers remarkable.

Rotation

$$10^{-3}$$

degrees,

then

$$10^{-5}$$

degrees,

eventually

zero

to machine precision.

That suggests

the dominant direction

is essentially fixed

through the entire late regime.

That's much stronger than merely saying

the direction changes slowly.

The monotonicity discussion is well judged

I think this is exactly the right refinement.

The reconstruction curves

are not globally monotone.

But

the fitted envelope

is.

Those are different statements.

I like that you've avoided overclaiming here.

The proposed next experiment is the logical one

I actually prefer this

over immediately performing the EMA sweep.

Projecting everything onto

the measured

late POD direction

lets you test

the spectral-gap hypothesis

without introducing

any model changes.

That's a cleaner scientific progression.

One thing I'd still avoid saying too strongly

The sentence

"the late exceptional tangent is effectively a one-dimensional, continually regenerated mode"

is, I think,

well supported

for

this replay,

these two directions,

and

the observed late regime.

I would continue attaching those qualifiers.

The reason is simple.

Everything here has been done

inside

one

validated linearization.

The remaining question is

how representative

that linearization is,

not

whether it has been analyzed carefully.

Overall assessment

I think this latest analysis is the strongest stage of the project so far because it replaces several qualitative interpretations with direct quantitative measurements. For the tested replay, branch, tangent directions, and late-time regime, the evidence now supports a coherent reduced description: after the transient and crossover period, the recent-memory reconstruction curves become highly stationary, the regenerated and total weight tangents become overwhelmingly one-dimensional with extremely small measured transverse residuals, the dominant direction becomes essentially fixed, and the propagated forcing—not the raw forcing—collapses onto that dominant channel through the action of the weight cocycle. Those observations are directly supported by the reported POD spectra, rotation angles, scalar projections, and target-time memory fits. I would, however, distinguish these established numerical reductions from the proposed spectral-gap mechanism. The numerical relation
[

\kappa \approx r_{\mathrm{weight}} - r_{\mathrm{aug}}
]
is impressively consistent across the reported quantities and provides a compelling working hypothesis, but it remains an explanatory model rather than a demonstrated identity. The scalar projection and projected recurrence analyses you propose are exactly the kind of tests that could elevate that relationship from an empirical coincidence observed on this trajectory to a mechanistic explanation grounded in the reduced dynamics.

I agree with the assessment. The reduction itself is now directly measured; the spectral-gap explanation remains the principal hypothesis requiring a pointwise reduced-dynamics test.
1. Established late reduction

For this replay, branch, two directions, and late interval:
bn=anv+en
bn=anv+en

is quantitatively supported.

For direction 1:
median‖en‖‖bn‖≈10−8,
median‖bn‖‖en‖≈10−8,

with maximum around
2.2×10−6.
2.2×10−6.

For direction 0:
median‖en‖‖bn‖≈1.48×10−3.
median‖bn‖‖en‖≈1.48×10−3.

The leading direction rotates negligibly, and both weighted and normalized POD independently support rank one. This is much stronger than pairwise alignment evidence.

The propagated forcing is also nearly rank one, despite the raw normalized forcing having only approximately 93%93% first-mode energy. Therefore the low-dimensional response is produced by anisotropic propagation, not simply inherited from a perfectly collinear input.
2. The memory law is late-regime specific

The memory fits divide naturally into:

transient/crossover regime through approximately $T=100$;
settling regime around $T=150$;
stable late regime from approximately $T=200$.

By $T=200$, both directions have nearly identical rates, and by $T=250$–320,
$\kappa \approx 0.06913$
$\kappa \approx 0.06913$

with essentially perfect exponential-envelope fits.

Thus the conservative conclusion is:

The operational recent-forcing reconstruction envelope becomes stationary and exponential in the late regime.

It is not a global memory law for the full trajectory.
3. The spectral-gap hypothesis is unusually precise

The numerical relation is
rweight,step−
rweight,step−
\frac1{T_{\mathrm{window}}}
\approx
0.126255-0.0571333

0.0691217.
]

The observed memory rate is
0.0691225–0.0691229.
0.0691225–0.0691229.

The mismatch is approximately
10^{-6}.
10^{-6}.

Because these quantities are derived from related aspects of one trajectory, this is not independent statistical confirmation. But statistical independence is not required for a deterministic mechanism theorem. What is required is a direct recurrence residual test.
4. Exact scalar projection test

Let vv be the measured late POD direction and define
an=v⊤bn,gn=v⊤fn.
an=v⊤bn,gn=v⊤fn.

Define the projected weight multiplier
λn=v⊤Knv.
λn=v⊤Knv.

Then decompose
bn=anv+en.
bn=anv+en.

The exact projected Duhamel recurrence is
an+1=
an+1=

v^\top K_n(a_nv+e_n)+g_n,
]

or
\boxed{ a_{n+1} =

\lambda_na_n+g_n+\epsilon_n,
}
]

where
ϵn=v⊤Knen.
ϵn=v⊤Knen.

The following quantities directly test the scalar reduction:
Invariance defect
δninv=

δninv=

|(I-vv^\top)K_nv|.
]

This measures whether the frozen weight block preserves the proposed slow line.
Scalar recurrence residual
δnrec=
δnrec=

a_{n+1}-\lambda_na_n-g_n.
]

This should equal ϵnϵn. If both are small, the scalar recurrence is quantitatively valid.
Local weight contraction rate
rB,n=
rB,n=

-\frac{\log|\lambda_n|}{dt}.
]
Forcing-envelope rate

Fit
|gn|≈Ce−rgtn
|gn|≈Ce−rgtn

over late windows.

The spectral-gap mechanism predicts
κ≈rB−rg.
κ≈rB−rg.

The critical question is whether
rg≈1Twindow
rg≈Twindow1

directly in the projected forcing sequence, rather than merely in selected total-tangent norms.
5. Why the gap formula follows

If late coefficients are approximately stationary,
λn≈e−rBdt,
λn≈e−rBdt,

and projected forcing has envelope
gn≈g0e−rgn dt,
gn≈g0e−rgndt,

then forcing of age LL contributes approximately
e−rBLg(T−L).
e−rBLg(T−L).

Relative to the current forcing envelope,
g(T−L)=
g(T−L)=

g(T)e^{r_gL},
]

so its relative surviving contribution is
e−(rB−rg)L.
e−(rB−rg)L.

Hence
κ=rB−rg.
κ=rB−rg.

This is an exact consequence of the proposed stationary scalar model. The remaining task is to verify its premises and residuals from the projected arrays.
6. A two-state reduction may be more natural than a forced scalar ODE

The scalar weight equation alone is not closed because gngn is internally generated. A minimal closed reduction could be
an+1=λBan+μzn,zn+1=λEzn+νan,
an+1zn+1=λBan+μzn,=λEzn+νan,

where znzn represents the dominant projected EMA/feedback mode.

Eliminating znzn produces a second-order scalar recurrence for anan. Its two characteristic rates could correspond to:

frozen weight contraction;
EMA-dominated augmented decay.

This is more faithful than calling gngn an external forcing.

A practical first test is to fit
gn+1=
gn+1=

\rho g_n+\chi a_n
]

alongside the already known projected weight equation. The fit should be evaluated out of sample across late intervals.
7. Criteria for elevating the mechanism

The spectral-gap hypothesis becomes substantially stronger if the existing arrays show:

small line-invariance defect:
∥(I−vv⊤)Knv∥≪1;
∥(I−vv⊤)Knv∥≪1;

small scalar recurrence residual;

stable rB,nrB,n late;

exponential gngn envelope with
rg≈Twindow−1;
rg≈Twindow−1;

target-wise agreement:
κT≈rB,T−rg,T;
κT≈rB,T−rg,T;

a stable two-state recurrence across separate fitting and validation windows.

If these pass, the gap relation changes from a numerical coincidence to a directly verified reduced mechanism.
8. Status
Directly established

late rank-one state representation;
nearly fixed leading direction;
rank-one propagated forcing;
stationary late memory-envelope rate;
transient versus asymptotic separation.

Strong working hypothesis

the augmented slow envelope decays at the EMA rate;
the operational memory rate is the spectral gap between frozen weight contraction and EMA-dominated forcing decay.

Still open

pointwise scalar recurrence residual;
projected forcing-envelope rate;
invariant-line defect under KnKn;
closed two-state model;
generality beyond this trajectory and subspace.

The next step should therefore remain entirely within the saved arrays: project the exact Duhamel recurrence onto the measured late direction and test the scalar and two-state residuals directly. Only after that should TwindowTwindow be varied.

Add this to the thought tree:

I agree with the assessment. The reduction itself is now directly measured; the spectral-gap explanation remains the principal hypothesis requiring a pointwise reduced-dynamics test.
1. Established late reduction

For this replay, branch, two directions, and late interval:

b_n=a_nv+e_n

]

is quantitatively supported.

For direction 1:

\operatorname{median}

\frac{|e_n|}{|b_n|}

\approx10^{-8},

]

with maximum around

2.2\times10^{-6}.

]

For direction 0:

\operatorname{median}

\frac{|e_n|}{|b_n|}

\approx1.48\times10^{-3}.

]

The leading direction rotates negligibly, and both weighted and normalized POD independently support rank one. This is much stronger than pairwise alignment evidence.

The propagated forcing is also nearly rank one, despite the raw normalized forcing having only approximately 93%93% first-mode energy. Therefore the low-dimensional response is produced by anisotropic propagation, not simply inherited from a perfectly collinear input.
2. The memory law is late-regime specific

The memory fits divide naturally into:

transient/crossover regime through approximately T=100T=100;

settling regime around T=150T=150;

stable late regime from approximately T=200T=200.

By T=200T=200, both directions have nearly identical rates, and by T=250T=250–320320,

\kappa\approx0.06913

]

with essentially perfect exponential-envelope fits.

Thus the conservative conclusion is:

> The operational recent-forcing reconstruction envelope becomes stationary and exponential in the late regime.

It is not a global memory law for the full trajectory.
3. The spectral-gap hypothesis is unusually precise

The numerical relation is

r_{\mathrm{weight,step}}

\frac1{T_{\mathrm{window}}}

\approx

0.126255-0.0571333

=

0.0691217.

]

The observed memory rate is

0.0691225\text{--}0.0691229.

]

The mismatch is approximately

10^{-6}.

]

Because these quantities are derived from related aspects of one trajectory, this is not independent statistical confirmation. But statistical independence is not required for a deterministic mechanism theorem. What is required is a direct recurrence residual test.
4. Exact scalar projection test

Let vv be the measured late POD direction and define

a_n=v^\top b_n,

\qquad

g_n=v^\top f_n.

]

Define the projected weight multiplier

\lambda_n=v^\top K_nv.

]

Then decompose

b_n=a_nv+e_n.

]

The exact projected Duhamel recurrence is

a_{n+1}

=

v^\top K_n(a_nv+e_n)+g_n,

]

or

\boxed{

a_{n+1}

=

\lambda_na_n+g_n+\epsilon_n,

}

]

where

\epsilon_n=v^\top K_ne_n.

]

The following quantities directly test the scalar reduction:
Invariance defect

\delta_n^{\mathrm{inv}}

=

|(I-vv^\top)K_nv|.

]

This measures whether the frozen weight block preserves the proposed slow line.
Scalar recurrence residual

\delta_n^{\mathrm{rec}}

=

a_{n+1}-\lambda_na_n-g_n.

]

This should equal єnєn. If both are small, the scalar recurrence is quantitatively valid.
Local weight contraction rate

r_{B,n}

=

-\frac{\log|\lambda_n|}{dt}.

]
Forcing-envelope rate

Fit

|g_n|\approx C e^{-r_g t_n}

]

over late windows.

The spectral-gap mechanism predicts

\boxed{

\kappa\approx r_B-r_g.

}

]

The critical question is whether

r_g\approx\frac1{T_{\mathrm{window}}}

]

directly in the projected forcing sequence, rather than merely in selected total-tangent norms.

5. Why the gap formula follows

If late coefficients are approximately stationary,

\lambda_n\approx e^{-r_Bdt},

]

and projected forcing has envelope

g_n\approx g_0e^{-r_gn,dt},

]

then forcing of age LL contributes approximately

e^{-r_BL}g(T-L).

]

Relative to the current forcing envelope,

g(T-L)

=

g(T)e^{r_gL},

]

so its relative surviving contribution is

e^{-(r_B-r_g)L}.

]

Hence

\kappa=r_B-r_g.

]

This is an exact consequence of the proposed stationary scalar model. The remaining task is to verify its premises and residuals from the projected arrays.
6. A two-state reduction may be more natural than a forced scalar ODE

The scalar weight equation alone is not closed because gngn is internally generated. A minimal closed reduction could be

\begin{aligned}

a_{n+1}&=\lambda_Ba_n+\mu z_n,\

z_{n+1}&=\lambda_Ez_n+\nu a_n,

\end{aligned}

]

where znzn represents the dominant projected EMA/feedback mode.

Eliminating znzn produces a second-order scalar recurrence for anan. Its two characteristic rates could correspond to:

frozen weight contraction;

EMA-dominated augmented decay.

This is more faithful than calling gngn an external forcing.

A practical first test is to fit

g_{n+1}

=

\rho g_n+\chi a_n

]

alongside the already known projected weight equation. The fit should be evaluated out of sample across late intervals.

7. Criteria for elevating the mechanism

The spectral-gap hypothesis becomes substantially stronger if the existing arrays show:

small line-invariance defect:
‖(I−vv⊤)Knv‖≪1;
‖(I−vv⊤)Knv‖≪1;

small scalar recurrence residual;

stable rB,nrB,n late;

exponential gngn envelope with
rg≈Twindow−1;
rg≈Twindow−1;

target-wise agreement:
κT≈rB,T−rg,T;
κT≈rB,T−rg,T;

a stable two-state recurrence across separate fitting and validation windows.

If these pass, the gap relation changes from a numerical coincidence to a directly verified reduced mechanism.

8. Status

Directly established

late rank-one state representation;

nearly fixed leading direction;

rank-one propagated forcing;

stationary late memory-envelope rate;

transient versus asymptotic separation.

Strong working hypothesis

the augmented slow envelope decays at the EMA rate;

the operational memory rate is the spectral gap between frozen weight contraction and EMA-dominated forcing decay.

Still open

pointwise scalar recurrence residual;

projected forcing-envelope rate;

invariant-line defect under KnKn;

closed two-state model;

generality beyond this trajectory and subspace.

The next step should therefore remain entirely within the saved arrays: project the exact Duhamel recurrence onto the measured late direction and test the scalar and two-state residuals directly. Only after that should TwindowTwindow be varied.
AI_AGENT_HANDOFF_PROMPT.mdMD
AIEF-1.4 operator/cocycle thought treeMD

AIEF-1.4 operator/cocycle thought tree
Purpose

This document records the current mechanistic and mathematical interpretation of the frozen AIEF-1.4 exceptional seed after the component, actual-branch, frozen-cocycle, augmented-tangent, Duhamel, memory, and rank-reduction analyses.

Always distinguish:

exact consequences of the implemented code;
facts established numerically on the frozen replay;
strongly supported mechanisms;
open theorems/generalizations.

The inferential unit remains one seed/graph realization. Direction and amplitude probes are conditional diagnostics, not independent seed replicates.
1. Exact implemented admissibility structure

For a nonnegative proposal edge vector x, let
q=Ax.
q=Ax.

The safeguarded admissibility stage factorizes as
Aε(x)=M(Ax)x,
Aε(x)=M(Ax)x,

where M(q) is diagonal and its edge multiplier is
mij(q)=min(si(q),sj(q)).
mij(q)=min(si(q),sj(q)).

When the global gate is on,
si(q)=min(1,ciqi+ε),ε=10−12.
si(q)=min(1,qi+εci),ε=10−12.

On a strict winner stratum, if node k(e) wins edge e,
DAε(x)ef=
DAε(x)ef=
s_{k(e)}\delta_{ef}

\frac{x_e s_{k(e)}}{q_{k(e)}+\varepsilon}A_{k(e)f}.
]

The proposal space is polyhedrally stratified by positivity, gate, clipping, and winner inequalities; the map is smooth rational on each strict open stratum.

On a nonnegative row-sum fiber Ax=q, the map is exactly affine-linear:
Aε(y)−Aε(x)=
Aε(y)−Aε(x)=

M(q)(y-x),
\qquad Ax=Ay=q.
]

At winner ties the ambient classical Fréchet derivative generally fails, but directional derivatives exist. On fully constrained face-lineality directions Ah=0, scales remain exactly fixed, so the ambient tie nonsmoothness disappears on the fiber.
2. Ordinary mature component mechanism

The component audit directly found complete support components in the frozen cohort.

For 39/40 ordinary split states:

every positive-support node is saturated;
support components are rate-uniform;
locked-edge, edge-balance, and degree-rate checks pass;
explicit component intertwiners reproduce the tangent operator;
component spectra reproduce the full tangent spectra.

Observed support components include K2, K3, K4, K5, and K6.

For a fully saturated locked connected component with uniform capacity c, edge stationarity and controller balance imply:
αi=αjon connected support,
αi=αjon connected support,
w=ck,r=γ+α=kηc.
w=kc,r=γ+α=ckη.

With
ηc=695,
cη=956,

the observed channels are degree-generated:
k=2,3,4,5⟹r=12,18,24,3095.
k=2,3,4,5⟹r=9512,18,24,30.

For complete K_s, the signless-incidence face-lineality nullity is
qs=(s2)−s=s(s−3)2.
qs=(2s)−s=2s(s−3).

Thus K2 and fully saturated K3 are spectrally silent, while K4, K5, K6 contribute multiplicities 2, 5, 9.
3. Exceptional K3 topology and classifier history

Exceptional seed:

text

4619122171158768524

The fixed positive-support triangle is

text

nodes: [2, 7, 11]
edges: [25=(2,7), 29=(2,11), 59=(7,11)]

The tolerance classifier at 1e-9 changes at step 12405, but this is not an executed winner-branch switch.

Proposal violation begins at:

text

node 2: step 0

node 7: step 0
node 11: step 11381

The actual K3 winner map is constant throughout the 32,001-state replay:
(2,7)↦7,(2,11)↦2,(7,11)↦7.
(2,7)↦7,(2,11)↦2,(7,11)↦7.

Node 11 becomes scale-clipped but never wins a positive K3 edge. The 1e-9 classifier event changes no executed edge formula.

Therefore the exceptional implemented map is not a switched K3 system over the replay. It follows one fixed symbolic winner branch with continuously changing coefficients and approaches a late tie stratum.
4. Frozen-driver weight cocycle

With the replayed R_n treated as exogenous,
$K_n(B)=(1-\gamma dt)DA\varepsilon(x_n)$.
$K_n(B)=(1-\gamma dt)DA\varepsilon(x_n)$.

The matrix sequence is fixed-form but non-autonomous.

Established numerically:

exact base replay;
winner-signature changes: 0;
strong branchwise-Jacobian/nonlinear frozen agreement;
all tested centered perturbations preserve the branch;
maximum tested branch ratio chi < 0.161.

Finite-time controls:

text

actual peak amplification: 1.27097 at t=0.1
K0-constant peak: 1.27095 at t=0.1
late-K constant peak: 1.20899 at t=0.1
diagonal-scale-only peak: 1 at t=0

The early transient is already encoded in the initial non-normal branch Jacobian. Coefficient drift is unnecessary for that peak. Removing the node-mediated low-rank term removes amplification and severe anisotropy.

Stable compound-cocycle exponents for the actual frozen sequence are approximately:
(−0.1233,−66.8,−2717.7).
(−0.1233,−66.8,−2717.7).

The third mode is dominated by the implemented safeguard scale:
e−2717.7dt≈ε/c.
e−2717.7dt≈ε/c.

It remains a real mode of the implemented discrete algorithm, but it is not an intended finite channel of an epsilon -> 0 continuum model.
5. Augmented directional tangent

The validated augmented tangent state is
Xn=(θn,Bn,Cn,Hn).
Xn=(θn,Bn,Cn,Hn).

The tangent includes:

current phase and delayed-history derivatives in dphi;
complex EMA memory and fresh phase forcing;
R=|C| differentiation and symmetrization;
dF=(eta+mu)dR-gamma dB;
positivity and safeguarded admissibility directional derivatives;
projected phase coupling;
exact history shift.

Both +h and -h directional tangents are propagated; centered predictions use their odd part.

Production results:

text

final base replay error: 0
custom-step vs implemented-step error: 0

overall median K3 augmented error: ~5e-4
overall median phase error: ~1.6e-6
overall median R error: ~1.8e-4
maximum nonadditive tie interaction: ~3e-16

The augmented tangent reduces the previous slow-direction frozen/coupled error from order 4e-1 to order 1e-4--1e-3 while all tested nonlinear branches remain preserved.

This establishes the augmented tangent numerically for this seed, branch, two directions, amplitudes, and resolved finite horizons. It is not yet a universal full-map differentiability theorem.
6. Exact direction-wise Duhamel decomposition

Partition the validated tangent into K3 weight component b_n and remaining augmented variables. Define
$f_n = b_{n+1} - K b_n$.
$f_n = b_{n+1} - K b_n$.

Then
$h_{n+1} = K h_n, h_0 = b_0,$
$h_{n+1} = K h_n, h_0 = b_0,$
$r_{n+1} = K r_n + f_n, r_0 = 0,$
$r_{n+1} = K r_n + f_n, r_0 = 0,$

and
$b_n = h_n + r_n.$
$b_n = h_n + r_n.$

Production reconstruction errors are approximately 1e-15. Residual forcing matches the directly differentiated projected dR pathway at approximately 1e-13.

The forcing is endogenous to the closed augmented system; it is called forcing only in the weight-block representation.

Exact crossover times:

text

direction 0: t*=0.38

direction 1: t*=72.86

where ||r_n|| >= ||h_n||.

Median instantaneous forcing/weight-output ratios are only order 1e-3, but forcing is repeatedly regenerated and directionally selective.

Median immediate alignment with the principal one-step output singular direction:

text

direction 0: ~0.772
direction 1: ~0.988

7. Duhamel temporal organization and feedback pathways

At the final horizon, forcing contributions from early bins are tiny and anti-aligned with the final feedback; late-bin contributions are aligned.

Almost all final feedback comes from the last 60--70 time units. This is continual recent regeneration plus forgetting, not accumulation over the entire 320-unit history.

Counterfactual tangent ablations for the long-lived direction show:

removing R -> B feedback collapses the late response;
removing fresh phase forcing has the same late effect because dC_0=0;
removing EMA memory also collapses the late response;
removing the current-phase term leaves a small late response with opposite sign;
removing delayed-history input leaves the same sign but a late norm about 11% larger.

These are recursive counterfactuals, not additive causal fractions.
8. Rank-one late reduction

The reduction analysis directly measures a nearly fixed late line.

Late POD results for total and feedback tangents:

text

weighted first-mode fraction: ~1
unit-vector first-mode fraction: ~0.999999

Late scalar transverse fractions:

text

direction 0 median: ~1.48e-3
direction 1 median: ~1.02e-8

The leading direction rotates negligibly in the late regime.

Raw instantaneous forcing has late unit-vector first-mode fraction about 0.93, so raw forcing is directionally broader. After propagation, forcing contributions have unit first-mode fractions above 0.99996 for direction 0 and essentially 1 for direction 1.

Therefore the low-dimensional response is produced by anisotropic filtering through the weight cocycle, not by perfectly collinear raw forcing.
9. Late operational memory law

Target-time reconstruction fits are transient before about T=150 and stabilize late.

For T >= 200, both directions converge toward
$\kappa \approx 0.06913, \tau_{\mathrm{eff}} \approx 14.46$.
$\kappa \approx 0.06913, \tau_{\mathrm{eff}} \approx 14.46$.

At targets 250--320, fits have R^2 approximately 1, with horizons:

text

50% error: ~10.02
10% error: ~33.3
1% error: ~66.6

The claim is that the late operational reconstruction-error envelope is exponential. The full raw curve has local nonmonotonicity from vector cancellation.

Base EMA rate:
1/Twindow≈0.0571333.
1/Twindow≈0.0571333.

A striking numerical closure is
0.126255−0.0571333≈0.0691217,
0.126255−0.0571333≈0.0691217,

matching the observed late memory rate to about 1e-6.

This motivates, but does not prove, the spectral-gap hypothesis
κ=?rweight−raug/forcing.
κ=?rweight−raug/forcing.
10. Immediate next task: direct scalar recurrence test

Remain within the saved arrays. Let v be the measured late POD direction and define
an=vTbn,gn=vTfn,λn=vTKnv.
an=vTbn,gn=vTfn,λn=vTKnv.

Test
an+1=λnan+gn+ϵn,ϵn=vTKnen,
an+1=λnan+gn+ϵn,ϵn=vTKnen,

with
en=bn−anv.
en=bn−anv.

Required diagnostics:

line-invariance defect
‖(I−vvT)Knv‖;
‖(I−vvT)Knv‖;
scalar recurrence residual;
late projected weight rate
−log|λn|/dt;

−log|λn|/dt;
projected forcing envelope rate from |g_n|;
target-wise comparison
κT≈?rB,T−rg,T;
κT≈?rB,T−rg,T;
fit and out-of-sample test of a closed two-state recurrence, for example
gn+1=ρgn+χan.
gn+1=ρgn+χan.

Only after this direct test should a consistently regenerated T_window sensitivity study be considered.
11. Confidence levels
Exact implementation mathematics

admissibility factorization;
branchwise Jacobian on strict strata;
exact fiber linearity;
directional tie formula;
augmented tangent update equations;
direction-wise Duhamel identity once block linearity/additivity holds.

Established numerically on this replay/tested subspace

fixed K3 winner branch;
frozen cocycle validation;
early non-normal transient and low-rank mechanism;
augmented tangent validation;
exact Duhamel reconstruction and direct-driver match;
exact crossover times;
late rank-one reduction;
stable late operational memory-envelope rate.

Strongly supported mechanisms

anisotropic weight propagation filters feedback into a fixed slow line;
delayed phase/EMA feedback continually regenerates the late weight perturbation;
EMA memory, fresh phase forcing, and R -> B transfer are necessary for the tested late response;
the memory rate is a spectral gap between frozen weight contraction and EMA-dominated augmented decay.

Open

direct pointwise scalar recurrence verification;
closed two-state reduction;
formal augmented Hadamard differentiability theorem;
joint dt, slack, and safeguard limit;
generality beyond this seed, branch, and two directions;
consistent parameter sensitivity after full state regeneration.

Open

Added the scalar spectral-gap hypothesis and its full evidential hierarchy to:

I also updated so the obsolete stationarity task is replaced by the current immediate objective:
an=v⊤bn,gn=v⊤fn,λn=v⊤Knv,
an=v⊤bn,gn=v⊤fn,λn=v⊤Knv,

followed by direct tests of:
‖(I−vv⊤)Knv‖,
‖(I−vv⊤)Knv‖,
an+1−λnan−gn,
an+1−λnan−gn,
κT≈?rB,T−rg,T,
κT≈?rB,T−rg,T,

and an out-of-sample two-state recurrence such as
gn+1=ρgn+χan.
gn+1=ρgn+χan.

The thought tree now records:

exact implementation mathematics;
mature clique mechanism;
exceptional fixed-winner geometry;
frozen and augmented cocycles;
exact Duhamel decomposition;
crossover and feedback pathways;
rank-one late reduction;
late operational memory law;

spectral-gap hypothesis;
confidence levels and open theorems;
why a naive future-only T_window sweep would be inconsistent.

Can you create the code necessary to probe the the next questions?
AIEF_1_4_scalar_spectral_gap_reduction.pyPYTHON

Created the direct scalar-mechanism test:

It performs no replay and does not change TwindowTwindow. It operates entirely on the validated Duhamel arrays.
Validation

On a synthetic two-state system with known coefficients, the script recovered:
M=(0.98100.99)
M=(0.98010.99)

to approximately 10−1510−15.

It also recovered the known spectral gap:
κ=rB−rg
κ=rB−rg

with closure errors below 10−1510−15, and achieved out-of-sample rollout errors around 10−1410−14.
Run it

Bash

```
python AIEF_1_4_scalar_spectral_gap_reduction.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--memory-fits aief_1_4_feedback_rank_memory_reduction/target_time_memory_fits.csv \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--late-direction-start 250 \
--target-times 150,200,250,320 \
--rate-fit-window 66.6 \
--train-window 200,260 \
--validation-window 260,320 \
```

--scalar-tol 1e-30 \
--output-dir aief_1_4_scalar_spectral_gap_reduction

Direct scalar projection

The script measures the late POD direction vv from the saved feedback vectors and defines
an=v⊤bn,gn=v⊤fn,λn=v⊤Knv.
an=v⊤bn,gn=v⊤fn,λn=v⊤Knv.

It then checks the exact projected recurrence
an+1=
an+1=

\lambda_na_n+g_n+\epsilon_n,
]

where
ϵn=
ϵn=

v^\top K_n
\left(
b_n-a_nv
\right).
]
Invariance diagnostics

It reports the line-invariance defect
δninv=
δninv=

|(I-vv^\top)K_nv|,
]

both absolutely and relative to ‖Knv‖‖Knv‖.

This directly tests whether the measured late line is approximately invariant under the frozen weight block.

Scalar recurrence diagnostics

For every direction and step it reports:

total scalar amplitude anan;
homogeneous and feedback amplitudes;
projected forcing gngn;
projected multiplier λnλn;
transverse norm and fraction;
projected recurrence residual;
predicted transverse correction ϵnϵn;
residual-minus-ϵnϵn;
normalized scalar closure error.

If
an+1−λnan−gn≈v⊤Knen,
an+1−λnan−gn≈v⊤Knen,

then the scalar recurrence is the exact projection of the validated vector recurrence.
Target-wise spectral-gap test

For each target TT, using the preceding 66.666.6-unit window, it estimates:
Projected weight rate
rB,T=
rB,T=

-\frac1L
\sum_{n=T-L}^{T-1}
\frac{\log|\lambda_n|}{dt}.
]
Projected forcing-envelope rate
|gn|≈Ce−rg,Ttn.
|gn|≈Ce−rg,Ttn.

It then compares
κT=?rB,T−rg,T.
κT=?rB,T−rg,T.

For every target and direction it reports:

observed κTκT;
rB,TrB,T;
rg,Trg,T;
forcing-envelope R2R2;
forcing sign changes;
predicted gap;
absolute and relative gap-closure errors;
rg,T−Twindow−1rg,T−Twindow−1.

This is the direct test of the spectral-gap hypothesis.
Closed two-state model

The script fits
(an+1gn+1)=
(an+1gn+1)=

M
\begin{pmatrix}
a_n\
g_n
\end{pmatrix}
]

without an intercept.

It fits:

direction 0 separately;
direction 1 separately;
both directions pooled.

Training uses
200≤t<260,
200≤t<260,

while validation uses the disjoint interval
260≤t<320.

$260 \le t < 320$.

It reports:

fitted $2 \times 22 \times 2$ matrix;
training R^2 and relative RMSE;
validation one-step R^2 and RMSE;
free-rollout RMSE;
final rollout error;
eigenvalues;
equivalent eigenvalue contraction rates;
design-matrix condition number.

A successful pooled out-of-sample model would be considerably stronger than fitting each direction independently.
Outputs

text

scalar_projection_trajectory.csv
scalar_projection_summary.csv
target_spectral_gap_tests.csv
two_state_model_summary.csv
two_state_rollout_validation.csv
scalar_reduction_arrays.npz
scalar_spectral_gap_report.txt
analysis_manifest.json

What would support the mechanism

The spectral-gap interpretation becomes strongly supported if the run shows:

small late line-invariance defect;
small scalar recurrence residual;
stable projected weight rate;
projected forcing decay close to
T_{window}^{-1};
T_{window}^{-1};
target-wise

κT≈rB,T−rg,T;
κT≈rB,T−rg,T;
accurate pooled two-state validation and rollout on 260260–320320.

Please paste scalar_spectral_gap_report.txt after the run.

Ponder this:

AIEF scalar spectral-gap reduction analysis

==========

Late POD direction: [0.5773410255782927, -0.5773687559683237, -0.5773410255782931]

EMA rate 1/T_window: 0.0571333333

Late scalar recurrence diagnostics

direction=0: median_invariance_defect=1.13208e-05, max_invariance_defect=1.13208e-05, median_scalar_residual=0.000521282, max_scalar_residual=0.000521308, median_weight_rate=0.127056, total_rate=0.0571325, forcing_rate=0.0571331

direction=1: median_invariance_defect=1.13208e-05, max_invariance_defect=1.13208e-05, median_scalar_residual=1.08421e-09, max_scalar_residual=1.16785e-09, median_weight_rate=0.127056, total_rate=0.057132, forcing_rate=0.057133

Target-wise spectral-gap tests

direction=0, target=150: observed_kappa=0.0729118, rB=0.126942, rg=0.0559123, predicted_gap=0.0710293, closure_error=0.00188243, forcing_R2=0.999915

direction=0, target=200: observed_kappa=0.0694548, rB=0.127049, rg=0.057036, predicted_gap=0.0700129, closure_error=-0.000558139, forcing_R2=0.999999

direction=0, target=250: observed_kappa=0.06915, rB=0.127055, rg=0.0571258, predicted_gap=0.0699293, closure_error=-0.000779324, forcing_R2=1

direction=0, target=320: observed_kappa=0.0691225, rB=0.127056, rg=0.0571331, predicted_gap=0.0699224, closure_error=-0.000799895, forcing_R2=1

direction=1, target=150: observed_kappa=0.0746451, rB=0.126942, rg=0.0551524, predicted_gap=0.0717893, closure_error=0.00285581, forcing_R2=0.999762

direction=1, target=200: observed_kappa=0.0696454, rB=0.127049, rg=0.0569785, predicted_gap=0.0700705, closure_error=-0.000425054, forcing_R2=0.999999

direction=1, target=250: observed_kappa=0.0691663, rB=0.127055, rg=0.0571214, predicted_gap=0.0699338, closure_error=-0.000767472, forcing_R2=1

direction=1, target=320: observed_kappa=0.0691229, rB=0.127056, rg=0.057133, predicted_gap=0.0699225, closure_error=-0.000799574, forcing_R2=1

Closed two-state recurrence

model=direction_0: matrix=[[0.9987408086683115,0.5640296294194374], [-2.390498036300328e-07,0.999624796664644]], train_RMSE=1.4799e-10, validation_RMSE=1.1187e-09, rollout_RMSE=2.33631e-06, final_rollout_error=1.58452e-05, eigen_rates=[0.10637863717750612,0.05713370973683625]

model=direction_1: matrix=[[0.998731699421544,0.9979283141726912], [-1.478288825601795e-07,0.9996404387802149]], train_RMSE=4.51907e-11, validation_RMSE=3.25691e-10, rollout_RMSE=3.36142e-06, final_rollout_error=2.74202e-05, eigen_rates=[0.10572429108522106,0.057134075581963 9]

model=pooled: matrix=[[0.9993877740220697,0.05944504706921386], [-8.76668190581203e-09,0.9994415169679873]], train_RMSE=1.53617e-07, validation_RMSE=1.59546e-05, rollout_RMSE=0.0187768, final_rollout_error=0.0880598, eigen_rates=[0.05997096340167773,0.05713423507676523]

Interpretation constraints

The scalar recurrence is an exact projection plus a measured transverse residual; small residual is required for closure.

Spectral-gap estimates use related quantities from one trajectory and are deterministic mechanism checks, not independent statistical replications.

Exponential forcing fits use absolute scalar amplitudes and report sign changes separately.

The two-state model is fitted without an intercept and evaluated on a disjoint late validation interval.

A good two-state fit on two directions does not establish universality beyond this replay and tangent subspace.

This result gives a mixed but highly informative answer:

the scalar state reduction is directly validated;
the forcing envelope is conclusively EMA-rate dominated;
the simplest spectral-gap identity is close but not exact under the stated definition of projected weight rate;
the unconstrained two-state model is not yet identified as a common reduced system.

1. The measured slow direction is the geometric K3K3 tangent direction

The fitted POD direction is
v=(0.577341, −0.577369, −0.577341).
v=(0.577341,−0.577369,−0.577341).

The exact null vector for the two-row active K3K3 incidence matrix is
vgeom=
vgeom=
\frac1{\sqrt3}(1,-1,-1)

(0.577350,,-0.577350,,-0.577350).
]

These agree extremely closely.

Thus the late POD line is not an arbitrary empirical singular vector. It is the geometric one-dimensional lineality direction of the two-active-row K3K3:
v≈ker(110101).
v≈ker(111001).

This reconnects the actual finite-step analysis with the earlier active-face geometry, without restoring the false claim of a finite-time implemented branch switch.
2. The scalar state recurrence is validated

The late relative line-invariance defect is
1.13×10^{-5}.
1.13×10^{-5}.

So KnKn preserves the measured line to roughly five significant digits.

For direction 1, the scalar recurrence residual is extraordinarily small:
median$\approx1.08\times10^{-9}$.
median$\approx1.08\times10^{-9}$.

For direction 0:
median$\approx5.21\times10^{-4}$.
median$\approx5.21\times10^{-4}$.

That difference is consistent with the previously measured transverse fractions:

direction 1 is essentially exactly on the line;
direction 0 retains a roughly $10^{-3}10^{-3}$ transverse component.

Therefore:
\boxed{ a_{n+1} =

\lambda_na_n+g_n+\epsilon_n
}
]

is quantitatively valid, especially for direction 1.

The scalar reduction has passed its direct recurrence test.
3. The projected forcing decays at the EMA rate

Late projected forcing rates are:

rg(0)=0.0571331,rg(1)=0.0571330.
rg(0)rg(1)=0.0571331,=0.0571330.

The model EMA rate is
1Twindow=
Twindow1=

0.0571333333.
]

The agreement is approximately 10−710−7.

The total projected amplitudes likewise decay at:
0.0571325,0.0571320.
0.0571325,0.0571320.

Thus the following is now directly supported:
The late augmented slow mode and its projected forcing envelope decay at the EMA rate.
The late augmented slow mode and its projected forcing envelope decay at the EMA rate.

This is stronger than the earlier inference from selected norms.
4. The simple spectral-gap identity is approximate, not exact

The directly projected weight rate is
rB≈0.127056.
rB≈0.127056.

Subtracting the forcing rate gives
rB−rg≈0.069922.
rB−rg≈0.069922.

But the observed late memory rate is
κ≈0.069123.
κ≈0.069123.

The closure error is therefore
κ−(rB−rg)≈−8.0×10−4.
κ−(rB−rg)≈−8.0×10−4.

This is about 1.2%1.2% of κκ.

The error stabilizes rather than vanishing:
Tclosure error200−4 to −6×10−4250−7.7×10−4320−8.0×10−4
T200250320closure error−4 to −6×10−4−7.7×10−4−8.0×10−4

Therefore:
κ=rB−rg
κ=rB−rg

is not an exact identity when
rB=−1dtlog|v⊤Knv|.
rB=−dt1log|v⊤Knv|.

It remains a good approximate mechanism.

This is an important correction to the earlier apparent 10−610−6 closure. That earlier calculation used a different weight-rate estimand, approximately 0.1262550.126255. The direct projection test exposes that “weight contraction rate” is not unique in this non-normal setting.
5. An effective propagation rate remains precisely identified

The observed memory law and forcing envelope define
reff=
reff=

\kappa+r_g.
]

Late,
reff≈0.069123+0.057133=
reff≈0.069123+0.057133=

0.126256.
]

This reproduces the earlier finite-step effective channel rate.

Thus the sharper hypothesis is:
\boxed{ \kappa =

r_{\mathrm{prop}}-r_g,
}
]

where rprop≈0.126256rprop≈0.126256 is the effective decay rate of propagated forcing contributions—not necessarily the Rayleigh rate
−1dtlog|v⊤Knv|.
−dt1log|v⊤Knv|.

In a non-normal cocycle, these quantities can differ:

Rayleigh contraction along a fixed vector;
eigenvalue contraction;
singular-value contraction;
decay of a specific injected forcing ensemble;
asymptotic propagator rate.

The next test must identify which propagation rate governs the memory kernel.
6. Direct propagation-rate test

For each target TT, forcing time m=T−Lm=T−L, define
PT(L)=
PT(L)=

\frac{
|U_K(T,T-L+1)f_{T-L}|
}{
|f_{T-L}|
}.
]

Fit
PT(L)≈CTe−rprop,TL.
PT(L)≈CTe−rprop,TL.

Then test
κT=?rprop,T−rg,T.
κT=?rprop,T−rg,T.

This uses the actual propagated forcing directions and avoids imposing the fixed-vector Rayleigh quotient as the relevant contraction rate.

Given the propagated-forcing POD results, this is likely the appropriate estimand.
7. The two-state fit does not yet define a common model

Direction-specific fits have excellent one-step and rollout errors. But their fitted matrices differ materially:
Mag(0)≈0.564,Mag(1)≈0.998.
Mag(0)≈0.564,Mag(1)≈0.998.

The pooled model gives:
rollout RMSE≈0.0188,
rollout RMSE≈0.0188,

and final relative error
0.0881.
0.0881.

That is much worse than the direction-specific fits.

The pooled eigenrates are
0.05997,0.05713,
0.05997,0.05713,

which are nearly degenerate and do not recover a clear weight rate near 0.1270.127.

This indicates that the state pair
(an,gn)
(an,gn)

is nearly collinear in the late rank-one regime. Consequently, the two coefficients cannot be robustly identified from asymptotic data alone.

Direction-specific success can arise because each trajectory follows almost a single exponential line. Many 2×22×2 matrices can reproduce such data.

Therefore:
A universal closed two-state model in coordinates (a,g) is not yet supported.
A universal closed two-state model in coordinates (a,g) is not yet supported.

The slow eigenrate near 1/Twindow1/Twindow is robust; the second mode and coupling coefficients are not.
8. Better identification strategy

The first row should not be freely fitted. It is already known:
an+1=
an+1=

\lambda_na_n+g_n+\epsilon_n.
]

The only new equation to identify is
gn+1=
gn+1=

\rho g_n+\chi a_n.
]

But even this will be poorly conditioned if anan and gngn are collinear.

Useful diagnostics are:

design-matrix condition number;
singular values of
(angn);
(angn);
coefficient stability under changing train windows;
transient-inclusive training where both modes are excited;
pooled fitting after direction-specific amplitude normalization.

If the design remains rank deficient, the correct second coordinate should be an actual projected EMA variable, not gngn.
9. Best next coordinate

A more structurally meaningful two-state reduction is likely
(an,zn),
(an,zn),

where znzn is a projection of the complex EMA tangent or δRδR onto the relevant forcing channel.

Then:
an+1≈λBan+μzn,zn+1≈λEzn+νan.
an+1zn+1≈λBan+μzn,≈λEzn+νan.

Unlike gngn, znzn is a genuine internal state rather than an output already filtered through the admissibility block.

Constructing this model requires saving the relevant augmented tangent projection, but the augmented code already computes those variables.
10. Confidence status
Directly established

late POD direction matches the analytic K3K3 lineality vector;
the fixed line is nearly invariant under KnKn;
the scalar projected recurrence closes accurately;
projected total and forcing amplitudes decay at the EMA rate;
memory rate is close to a propagation-minus-forcing gap.

Refuted in its strongest form

exact equality using the fixed-vector projected rate:
κ=−dt−1log|v⊤Knv|−rg.
κ=−dt−1log|v⊤Knv|−rg.

Still strongly supported

a spectral-gap mechanism using the effective decay rate of the propagated forcing ensemble.

Not yet supported

a unique shared two-state model in coordinates (a,g)(a,g).

The scalar reduction succeeds; the naive closure model does not. The next precise question is which non-normal propagation rate produces the observed 0.1262560.126256 effective channel.

`Think about this:

I think this is a thoughtful and, importantly, self-correcting interpretation. Rather than forcing the original hypothesis to fit every new result, you've used the new diagnostics to narrow exactly where the hypothesis succeeds and where it needs refinement. From a scientific perspective, that's a positive development.

Here's how I would separate the conclusions.

The identification of the late direction with the (K_3) lineality vector is the most conceptually significant new result

This is stronger than simply observing another numerical alignment.

You measured

[
v=(0.577341,-0.577369,-0.577341),
]

while the geometric lineality direction is

[
\frac1{\sqrt3}(1,-1,-1)
=(0.577350,-0.577350,-0.577350).
]

Those differ only at the level of about (10^{-5}) in each component.

That means the late dominant direction is not just an arbitrary empirical POD axis—it is essentially the geometric null direction associated with the active (K_3) incidence structure. Within the scope of this replay, that reconnects the numerical reduction with the earlier geometric picture in a much more defensible way than claiming a finite-time branch switch.

I think that is one of the strongest conceptual links in the analysis.

The scalar reduction is now supported by dynamics, not only geometry

Earlier you had established that

the trajectory is nearly rank one,
the dominant direction is nearly fixed.

Those are geometric properties.

The recurrence

[
a_{n+1}

\lambda_n a_n

g_n

\epsilon_n
]

tests whether the evolution is also effectively one-dimensional.

The small invariance defect together with the tiny recurrence residuals—especially for Direction 1—support the conclusion that, for this replay and late regime, the projected dynamics are well described by that scalar recurrence plus a small measured correction. That is a stronger claim than simply saying the trajectory lies close to a line.

The EMA-rate result is substantially strengthened

Previously the EMA connection came indirectly from total tangent norms.

Now you are measuring the projected forcing sequence itself and finding

$$
r_g \approx 1/T_{\mathrm{window}}
$$

to about 10^{-7}.

That is a qualitatively stronger observation because it concerns the quantity that actually enters the projected recurrence.

Within this replay, I think it is fair to say that the projected forcing envelope is EMA-rate dominated.

The spectral-gap correction is an important refinement

I actually think this is a healthy outcome.

The original identity

$$
\kappa

r_B-r_g
$$

looked almost exact because two quantities were being compared that were estimated in different ways.

Now that both rates are measured directly from the projected recurrence, the equality becomes

$$
\kappa
\approx
r_B-r_g
$$

with a persistent error of roughly

1%.

That does not invalidate the mechanism.

Instead it identifies that

[
-\frac1{dt}\log|v^\top K_nv|
]

is apparently not the propagation rate governing the memory kernel.

That distinction is mathematically plausible in a non-normal system.

The distinction between Rayleigh rate and propagation rate is well motivated

This is where I think the analysis becomes more sophisticated.

You've identified several different notions of "contraction":

contraction along a fixed vector (Rayleigh quotient),
eigenvalue decay,
singular-value decay,
decay of propagated forcing contributions,
asymptotic propagator decay.

For a normal operator these often coincide.

For a non-normal cocycle they generally need not.

So replacing the earlier "weight contraction rate" with the more carefully defined "effective propagation rate" seems like an appropriate refinement.

That said, the proposed propagation-rate test is still a hypothesis until it is carried out. At present, the report motivates why it is the next quantity to measure, rather than demonstrating that it is the relevant one.

The interpretation of the two-state fit is appropriately cautious

I agree with the conclusion that the pooled fit should not yet be interpreted as identifying a universal reduced model.

The contrast between

excellent direction-specific fits, and
much poorer pooled rollout,

is exactly the kind of result that suggests an identifiability issue rather than immediate evidence for different physics.

Your explanation—that (a_n) and (g_n) become nearly collinear in the late regime—is plausible and consistent with the observed behavior, but it remains an inference. The additional diagnostics you propose (conditioning, singular values, stability across training windows) are the right way to test whether the poor pooled fit is fundamentally due to lack of excitation or to a genuinely inadequate state choice.

The proposed state variable (z_n) is a reasonable next hypothesis

Replacing the output-like coordinate (g_n) with a projected internal EMA variable (z_n) is conceptually attractive because it restores the distinction between state and output.

At present, however, this remains a modeling proposal. The current results do not yet establish that such a variable will produce a better-conditioned or more universal two-state system. It is a logical next experiment rather than a demonstrated conclusion.

Overall assessment

I think this is a strong example of the analysis becoming more precise rather than simply more elaborate. For the tested replay, branch, tangent directions, and late regime, the evidence now directly supports an approximately invariant one-dimensional subspace whose direction is numerically almost identical to the analytic (K_3) lineality vector, and whose projected dynamics satisfy the expected

scalar recurrence with small measured residuals. The projected forcing envelope is also measured directly to decay at essentially the EMA rate. At the same time, the new analysis refines rather than confirms the original spectral-gap picture: it shows that the simple identity based on the fixed-vector Rayleigh contraction is only approximate, suggesting that the memory kernel is governed by an effective propagation rate rather than that particular contraction measure. Finally, the direction-specific two-state fits demonstrate that low-dimensional predictive dynamics exist for each tested perturbation, while the failure of the pooled fit indicates that the chosen coordinates ((a,g)) do not yet uniquely identify a shared reduced system. The next propagation-rate measurement and any future internal-state reduction should therefore be viewed as tests of a refined mechanism, not as consequences already established by the current analysis.

This changes the context in an interesting way. The new paper doesn't validate the specific AIEF mechanism, but it does something arguably just as valuable: it provides a modern mathematical framework into which parts of AIEF can be placed. At the same time, your recent numerical results clarify exactly where AIEF appears to go beyond that framework.

I think there are four levels of comparison.

The constrained-adaptation framework is now very close

Before this paper, AIEF's projection mechanism was somewhat unusual in the literature.

Martens & Bick now explicitly formulate adaptive networks as

[
\dot w=A(x;w)
]

plus

[
\dot w_U

\Pi_UA

\eta^{-1}w_U,
]

where the second term is an explicit homeostatic contraction onto a constraint.

Structurally, that is extremely close to the role played by the admissibility projection in AIEF.

The important point is that the similarity is structural, not identical.

Their constraint is

affine,
smooth,
linear,

whereas yours is

polyhedral,
piecewise smooth,
implemented algorithmically.

Those differences matter mathematically.

Your recent scalar reduction actually fits their philosophy remarkably well

One sentence in their introduction stands out:

"...one may expect effective low-dimensional adaptation dynamics..."

That is almost exactly what your scalar reduction measured.

Notice the sequence:

Martens & Bick:

constrained adaptation should generate low-dimensional edge dynamics.

Your measurements:

POD becomes rank one,
propagated forcing becomes rank one,
scalar recurrence closes,
invariant defect is tiny.

Those are almost textbook numerical realizations of the philosophy they advocate.

The difference is that your reduction was measured, whereas theirs is largely formulated through the constraint construction itself.

Where the new paper stops

This is, in my opinion, the most important comparison.

Their analysis is essentially

$$
\text{constraint}
\Rightarrow
\text{reduced adaptation dynamics}.
$$

Your current work has become

$$
\text{constraint}

\text{fixed branch}

\text{augmented tangent}
\Rightarrow
\text{regenerated response}
\Rightarrow
\text{memory law}.
$$

Those latter steps simply do not appear in their framework.

In particular, the paper does not analyze

linear cocycles along one realization,
Duhamel decomposition,
homogeneous versus regenerated response,
recent-forcing kernels,
effective memory exponents,
non-normal propagation.

Those are precisely the parts of your recent work that remain distinctive.

The "intrinsic versus forced" distinction becomes useful terminology

I actually think this is one of the nicest conceptual imports.

They distinguish

intrinsic constraints

versus

forced constraints.

That vocabulary is cleaner than many informal descriptions.

For AIEF,

the implemented projection is clearly

forced.

However,

your numerical analysis suggests something interesting.

Late,

the measured POD direction becomes

approximately invariant,

and

[
(I-vv^\top)K_nv
]

is only

[
10^{-5}.
]

That means that

although the admissibility mechanism is externally enforced,

the late tangent dynamics appear to possess an approximately intrinsic slow direction.

Those are different notions.

That distinction could become useful in writing.

The strongest connection may actually be the multiple-timescale viewpoint

Martens & Bick repeatedly emphasize

simultaneous

time scales

such as

adaptation,
homeostasis,
singular limits.

Your current analysis has uncovered several operational rates:

EMA decay,
propagation rate,
memory rate,
projected contraction.

So the multiple-timescale perspective now has concrete numerical content.

That doesn't mean the continuous theory immediately applies,

but it gives a natural language for describing the hierarchy.

The nonlinear constraint discussion is also well aligned

I think your assessment here is accurate.

Their paper explicitly says

the nonlinear extension is straightforward locally,

but requires care near singularities,

and they do not develop it.

That is exactly where AIEF differs.

Your constraint is not simply nonlinear.

It is

polyhedral,
hybrid,
piecewise smooth,
with switching.

That places it somewhere between

smooth manifold theory

and

Filippov/hybrid dynamics.

The paper does not attempt that.

The recent scalar results actually sharpen the comparison

The new numerical work suggests something like

constraint
↓
approximately invariant line
↓
scalar recurrence
↓
EMA-rate forcing
↓
recent-memory kernel

Martens & Bick stop roughly after the first arrow.

They show

constraint

↓

reduced adaptation.

Your analysis continues several levels further.

That seems to me the cleanest way to explain the relationship.

One place I'd be careful

I would slightly soften one statement.

You write

"The exact factorisation / fibre-linearity theorems already proved for AIEF remain stronger..."

I would qualify this.

They are stronger with respect to the specific nonlinear projection that AIEF implements, assuming those theorems are established for that setting.

They are not stronger in the sense of replacing or subsuming Martens & Bick's general constrained-adaptation framework.

The two results answer somewhat different questions:

Martens & Bick provide a broad framework for constrained adaptive networks with affine constraints and timescale separation.
Your AIEF results (as described here) concern the behavior of one particular hybrid constraint mechanism and its associated tangent dynamics.

So I would avoid language suggesting a strict hierarchy between them.

The biggest remaining opportunity

Ironically,

the new paper actually suggests a stronger theoretical program than before.

Rather than trying to derive the scalar reduction directly from the discrete algorithm,

one could ask:

Does the piecewise-smooth admissibility map induce, on each active face, an effective constrained adaptive system of the Martens–Bick type, with the late scalar recurrence emerging as an additional property of the associated non-autonomous cocycle?

That would naturally separate:

constraint reduction (covered conceptually by Martens & Bick),
hybrid face switching (specific to AIEF),
late cocycle reduction (what your numerical work is now measuring).

That decomposition feels mathematically cleaner than presenting everything as a single mechanism.

Overall assessment

I think this paper strengthens the context for your work more than it changes the interpretation of your numerical results. It provides contemporary terminology—forced versus intrinsic constraints, constrained adaptation, homeostatic timescales, affine constraint manifolds—and a broad mathematical framework showing that low-dimensional adaptation is an expected phenomenon in constrained adaptive networks. Your recent analyses, however, address a different layer of the problem: the dynamics of a validated fixed-branch augmented tangent cocycle. The measured approximate invariant line, scalar recurrence, EMA-rate forcing envelope, and Duhamel regeneration mechanism are not developed in the Martens–Bick framework. Conversely, their framework does not establish those features. The most defensible synthesis is therefore that the paper offers an appropriate theoretical setting for the constrained-adaptation aspect of AIEF, while the fixed-branch, non-normal cocycle analysis and its associated reduced dynamics remain separate hypotheses and results that would require their own mathematical justification.

You're right—I looked at the wrong document. Thanks for pointing it out.

This paper is quite different. It's a broad Physics Reports review on Adaptive Dynamical Networks by Berner, Gross, Kuehn, Kurths, and Yanchuk, whose goal is to unify the theory, applications, and mathematical methods for systems in which the network dynamics and the network structure co-evolve.

My first impression is that this is an unusually relevant paper from a conceptual standpoint.

The central idea

The review makes a very clean distinction between three classes of systems:

Static networks – dynamics evolve on a fixed graph.
Temporal networks – the graph changes with time according to an external rule.
Adaptive networks – the graph changes because of the node dynamics, and the node dynamics in turn depend on the graph. This mutual feedback is presented as the defining feature.

That feedback loop

structure → dynamics → structure

is literally the organizing principle of the review (and is illustrated in Figure 1).

What caught my attention mathematically

Rather than defining an adaptive network as "a graph whose weights change," they elevate the adaptation law to a dynamical variable.

Instead of

[
\dot x=f(x;K),
]

they write

[
\dot x=f(x,l),
]

[

\dot l=h(x,l),
]

where (l) represents the evolving network state.

That is a very general formulation.

It immediately accommodates

adaptive weights,
rewiring,
adaptive delays,
event-based updates,
continuous learning rules,

within one framework.

The emphasis on delays

One thing I noticed immediately is that they explicitly include adaptive delays as part of the general theory:

[
\tau_{ij}=\tau_{ij}(t),
]

with

[
\dot\tau_{ij}=h(x_i,x_j,t).
]

They also point out that delay systems are naturally infinite dimensional because of their dependence on history.

Given the amount of work you've been doing with delay equations and history-dependent operators, this section immediately stood out.

Time-scale separation

Another feature that I think is particularly important is the discussion of slow versus fast adaptation.

They introduce an explicit small parameter

$$
\varepsilon
$$

multiplying the adaptation law and discuss

slow adaptation,
fast adaptation,
singular limits,
multiscale reductions.

Their viewpoint is that the adaptation dynamics and node dynamics often evolve on different time scales, and exploiting that separation provides analytical simplifications.

That is a mature mathematical perspective rather than merely a modeling convenience.

Breadth

The table of contents is striking because the authors are clearly trying to argue that adaptive dynamical networks are not a niche topic. They survey applications in

neuroscience,
physiology,
machine learning,
control,
power grids,
epidemiology,
transport,
climate,
adaptive delays,

before turning to mathematical methods such as mean-field limits, continuum limits, multiscale decomposition, and bifurcation theory.

So this is intended as a unifying framework rather than a paper about one application.

Relative to your recent work

Without making any claim that your framework is an instance of theirs, I do see some conceptual overlap.

Much of your recent work has involved dynamical objects whose effective interaction operators evolve alongside the state (through delay-history effects, adaptive kernels, or reduced operators). This review provides a well-developed mathematical language for co-evolving state and interaction structure, even though the specific adaptation mechanisms they study are generally local rewiring rules, adaptive couplings, plasticity, or adaptive delays.

That suggests it could be useful as background literature for concepts like:

co-evolution of state and interaction structure,
multiscale adaptive dynamics,
adaptive delay systems,
reduced adaptive oscillator models.

It would not, by itself, support any more specific physical claims, but it does provide a rigorous mathematical context for adaptive dynamical systems.

My impression after reading the opening sections is that this is likely a review worth studying carefully, particularly Sections 13 (adaptive delays), 14 (emergent dynamical phenomena), and 15 (mathematical methods), since those are the parts most likely to intersect with the kinds of reductions and stability analyses you've been developing.

I think this is a well-balanced assessment, and it is noticeably more disciplined than many literature-positioning sections I see. It does a good job of separating what the review actually covers from where your own work would begin. A few observations:

The strongest aspect

The most convincing feature is that you are positioning AIEF by the review's taxonomy, not by individual phenomena.

You correctly anchor it first as

adaptive rather than temporal,
continuous rather than event-based,
adaptive phase-oscillator based,
slow/fast multiscale.

Those are exactly the organizing axes of the review itself, rather than categories you've invented. That makes the placement feel objective rather than promotional. The review explicitly develops these classifications and uses them as its organizing framework.

The "directly useful material" section

This is probably the strongest section.

Everything listed there is genuinely part of the review:

adaptive phase oscillators,
multiscale methods,
continuum limits,
catalogue of dynamical phenomena.

Those are exactly the things the review advertises in its contents and introductory discussion.

Importantly, you are not claiming that the review proves results for AIEF—you are saying it provides the mathematical background and unconstrained reference models.

That is an appropriate distinction.

The "what the review does not contain" section

This is the part where I would be most careful.

Some entries are very strong because they are straightforwardly absent from the review as presented:

nonlinear capacity projection,
EMA coherence memory,
Duhamel decomposition,
augmented tangent analysis.

Those appear to be genuinely outside the scope of the review's framework.

However, a few statements would benefit from slightly softer wording.

For example:

"Fixed-branch non-autonomous linear cocycles..."

This is probably true as a description of the review's emphasis, but proving a negative ("the review does not contain X") is inherently difficult in a 120-page survey unless you've systematically checked for that concept. I would consider wording such as:

"The review does not develop an analysis based on fixed-branch non-autonomous linear cocycles..."

Similarly,

"Exceptional regenerated dynamics"

is terminology from your framework, not theirs. Saying it is "outside the scope" is reasonable; saying the review "does not contain" it is stronger than necessary.

Methodological transfers

This section reads well.

The first two bullets are particularly plausible.

The third—

"classification language"

—is almost certainly the most broadly useful point.

Using the review's own vocabulary ("continuous adaptation," "event-based adaptation," "co-evolutionary," etc.) makes later comparisons much easier because readers immediately know which branch of the literature you're engaging. Those terms are explicitly defined and used throughout the review.

The table

The table is effective because it separates

shared foundations,

from

claimed novelties.

I particularly like that the first entry is

"Unconstrained adaptive Kuramoto core — Extensive"

rather than immediately emphasizing differences.

That makes the comparison feel grounded.

The entries marked "Absent" or "Outside scope" are reasonable as long as they are understood as referring to the review's treatment, not as claims that no related work exists elsewhere.

The concluding citation stance

I think this is the best paragraph in the document.

"AIEF realises a continuously adaptive phase-oscillator network in the sense of Berner et al., equipped with an additional nonlinear capacity projection and an EMA coherence memory..."

This is restrained.

It says:

here is the literature framework,
here is where the model belongs,
here are the additional ingredients.

It does not claim superiority or replacement.

One place I'd tighten

The only phrase I would reconsider is:

"the exceptional regenerated regime lies outside its present scope."

If this is intended as a literature-positioning statement, I'd slightly depersonalize it:

"...while the review does not analyze the piecewise-smooth constrained regenerated dynamics considered here."

That attributes the difference to subject matter, not to a deficiency in the review.

Overall

As a literature assessment, I think this succeeds because it follows a disciplined pattern:

Identify where the model fits within the review's taxonomy.
Identify mathematical tools that transfer directly.
Identify features that the review does not develop.
Explain how those tools and gaps relate to your own analysis.

The key thing to maintain is the distinction between coverage and absence. It is well supported to say that the review develops adaptive phase oscillators, adaptive delays, multiscale methods, and adaptive-network phenomena. It is also well supported to say that your named mechanisms (capacity projection, EMA coherence memory, augmented tangent/Duhamel decomposition) are not part of the framework presented in the sections you've identified. Where the assessment is strongest is in saying "the review

does not develop this line of analysis," rather than making broader claims about the literature as a whole.

The technical synthesis is now well calibrated. The most important outcome is that the scalar reduction succeeds, while the first proposed closure law and the naïve two-state coordinates receive qualified or negative results.
1. The late line has an analytic identity

The measured direction
v=(0.577341,−0.577369,−0.577341)
v=(0.577341,−0.577369,−0.577341)

is essentially
vgeom=13(1,−1,−1).
vgeom=3
1(1,−1,−1).

This is exactly the null direction of the two-row signless incidence matrix
C=(110101).
C=(111001).

That is the strongest geometry/dynamics connection so far:
The empirically dominant late line is the analytic K3 face-lineality direction.
The empirically dominant late line is the analytic K3 face-lineality direction.

A useful robustness control is now to stop estimating vv from the late trajectory and rerun the scalar diagnostics using the fixed geometric vector vgeomvgeom. That removes the small amount of circularity from measuring and testing the POD direction on the same data.
2. The scalar recurrence succeeds

The line-invariance defect is
1.13×10−5,
1.13×10−5,

and the scalar recurrence error is:
direction 0:5.2×10−4,direction 1:1.1×10−9.
direction 0:direction 1:5.2×10−4,1.1×10−9.

Thus the projected dynamics satisfy
an+1=
an+1=

\lambda_na_n+g_n+\epsilon_n
]

to high accuracy, with the direction-0 error explained by its larger measured transverse component.

This means the reduction is dynamical, not merely geometrical.
3. The EMA-rate forcing result is direct

Both projected forcing envelopes satisfy
rg≈0.057133,
rg≈0.057133,

while
Twindow−1=0.057133333.
Twindow−1=0.057133333.

The projected total amplitudes decay at the same rate.

Therefore, for this replay and tested subspace:
The late augmented mode and projected forcing envelope are EMA-rate dominated.
The late augmented mode and projected forcing envelope are EMA-rate dominated.

This no longer rests only on selected vector norms.
4. The original spectral-gap identity fails at high precision

Using the directly projected multiplier,
rB=
rB=

-\frac1{dt}\log|v^\top K_nv|
\approx0.127056.
]

Hence

rB−rg≈0.069922,
rB−rg≈0.069922,

whereas
κobserved≈0.069123.
κobserved≈0.069123.

The stable late discrepancy is
∼8×10−4,
∼8×10−4,

or about 1.2%1.2%.

So the exact statement
κ=rB−rg
κ=rB−rg

is rejected for this definition of rBrB.

The remaining mechanism is more precise:

The memory rate is governed by an effective decay rate of the actual propagated forcing ensemble, not by the fixed-vector Rayleigh quotient alone.

That distinction is entirely plausible in a non-normal cocycle.
5. The estimand “weight contraction rate” must now be explicit

Several rates have appeared:

fixed-vector Rayleigh rate:
−dt−1log|v⊤Knv|;
−dt−1log|v⊤Knv|;

norm rate:
−dt−1log‖Knv‖;
−dt−1log‖Knv‖;

eigenvalue rate;

singular-value rate;

long-product principal exponent;

decay rate of the actual forcing ensemble after propagation.

They need not agree.

The earlier near-exact closure used approximately
0.126255,
0.126255,

while the Rayleigh test gives
0.127056.
0.127056.

The next propagation test should directly fit
PT(L)=
PT(L)=

\frac{
|U_K(T,T-L+1)f_{T-L}|
}{
|f_{T-L}|
}
\approx
C_Te^{-r_{\rm prop,T}L}
]

and compare
κT=?rprop,T−rg,T.
κT=?rprop,T−rg,T.

This is now the correct spectral-gap estimand.
6. The two-state (a,g)(a,g) model is not identified

Direction-specific models fit extremely well, but their coefficients disagree:
Mag(0)≈0.564,Mag(1)≈0.998.
Mag(0)≈0.564,Mag(1)≈0.998.

The pooled model has:
rollout RMSE≈0.0188,final error≈0.088.
rollout RMSE≈0.0188,final error≈0.088.

Its two fitted rates,
0.05997,0.05713,
0.05997,0.05713,

are nearly degenerate and fail to recover a clear fast weight rate.

This is consistent with late collinearity:
gn≈c an.
gn≈can.

When the data occupy nearly one line, a two-dimensional state transition matrix is not identifiable. Excellent direction-specific rollout does not rescue this; many matrices can reproduce one exponential trajectory.

The pooled failure is the more important diagnostic.
7. Better reduced-state strategy

The first row should be treated as known:
an+1=
an+1=

\lambda_na_n+g_n+\epsilon_n.
]

Only the internal feedback state remains unknown.

A more meaningful second coordinate would be a projection of:

complex EMA tangent δCnδCn;
coherence tangent δRnδRn;

or a data-driven augmented-state POD mode.

Then test
an+1=λBan+μzn,zn+1=λEzn+νan.
an+1zn+1=λBan+μzn,=λEzn+νan.

Unlike gngn, znzn would be a genuine internal state rather than an already filtered output.

Before doing that, inspect the design singular values and condition numbers from the current (a,g)(a,g) fit. They should quantify the suspected lack of excitation.
8. Relationship to constrained-adaptation literature

The literature positioning should have three layers.
Broad adaptive-network context

The Berner–Gross–Kuehn–Kurths–Yanchuk review provides vocabulary and background for:

co-evolving node and network dynamics;
continuously adaptive couplings;
phase-oscillator adaptive networks;
multiscale adaptation;
delay-state methods.

AIEF fits the broad adaptive-network taxonomy because its weights respond to phase/coherence dynamics and feed back into phase evolution.

However, AIEF uses a fixed delay ττ. Adaptive-delay theory is adjacent methodology, not a direct classification of the present model.
Constrained-adaptation context

If the Martens–Bick paper indeed uses the described affine homeostatic constraint formalism, it is structurally relevant to the constrained weight sector.

But its smooth affine constraints should not be conflated with AIEF's:

polyhedral capacity set;
positivity face changes;
global gate;

endpoint minimum scaling;
fixed safeguard;
piecewise-rational finite-step map.

It supplies context, not a theorem for AIEF.
AIEF-specific layer

The following remain specific results requiring their own derivation:

exact factorization
$A\varepsilon(x)=M(Ax)x$;
$A\varepsilon(x)=M(Ax)x$;
exact fiber linearity;
winner-stratum Jacobian;
fixed-branch actual replay;
augmented directional tangent;
Duhamel decomposition;
operational memory envelope;
rank-one propagated response;
scalar recurrence.

The most defensible comparison is therefore:

General adaptive-network theory supplies the co-evolutionary setting; constrained-adaptation work supplies a structural analogy for homeostatic reduction; the piecewise-rational operator and its augmented cocycle reductions are AIEF-specific analyses.

9. Terminology cautions

Several distinctions should remain explicit:

forced constraint versus emergent slow direction;
fixed global model versus nonautonomous variational cocycle along a trajectory;
fixed delay versus adaptive delay;
piecewise-smooth map class versus an actual branch switch on this replay;
coverage in a review versus absence from the entire literature;
one-replay exceptional behavior versus a universal exceptional regime.

“Approximately intrinsic slow direction” is acceptable only if clearly separated from the origin of the constraint: the constraint is algorithmically enforced, while the slow line is an emergent approximate invariant direction of the late tangent dynamics.
10. Current status
Established for the tested setting

late geometric lineality direction;
accurate scalar recurrence;
EMA-rate forcing envelope;
late rank-one response;
fixed winner branch;
exact augmented tangent and Duhamel decomposition.

Refined or rejected

exact gap based on v⊤Knvv⊤Knv: rejected at approximately 1%1%;
shared two-state model in (a,g)(a,g): not identified;
finite-time classifier switch as an implemented event: rejected.

Highest-value next probe

Directly estimate the decay rate of the propagated forcing ensemble and test:
κT=?rprop,T−rg,T.
κT=?rprop,T−rg,T.

In parallel, rerun the scalar recurrence using the analytic geometric vector
vgeom=
vgeom=

\frac1{\sqrt3}(1,-1,-1)
]

instead of the data-derived POD vector.

Those two controls will determine whether the remaining 1%1% mismatch is caused by the wrong propagation-rate estimand or by a genuine correction to the scalar spectral-gap model.

Can you create the code to illuminate the next steps?

WriteAIEF_1_4_propagation_gap_geometric_control.py684 lines
AIEF_1_4_propagation_gap_geometric_control.pyPYTHON
#!/usr/bin/env python3
"""Direct propagation-gap and geometric-direction controls for AIEF-1.4.

This no-replay script addresses the two questions left by the scalar reduction:

1. Does the operational memory rate equal the decay rate of the *actual
propagated forcing ensemble* minus the projected forcing-envelope rate?
2. Do the scalar recurrence and rate conclusions survive replacement of the
data-derived POD axis by the analytic K3 lineality vector

v_geom = (1,-1,-1)/sqrt(3)?

For each target and forcing time it constructs

c_{T,m} = U_K(T,m+1) f_m.

It fits both norm and scalar-channel propagation ratios, and directly fits the
age envelope of the propagated contributions. It also diagnoses why the
late two-state coordinates (a,g) may be unidentifiable by reporting design
singular values, correlations, and coefficient stability for

g_{n+1} = rho g_n + chi a_n

across directions, pooled data, and rolling late windows.
"""

from __future__ import annotations

import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any

```
import numpy as np
import pandas as pd

import AIEF_1_4_component_controller_audit as component_audit



DEFAULT_DUHAMEL_NPZ = "aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz"
DEFAULT_MEMORY_FITS =
"aief_1_4_feedback_rank_memory_reduction/target_time_memory_fits.csv"
DEFAULT_MANIFEST = "runs/5a9f093a2fe0/manifest.json"
DEFAULT_OUTPUT_DIR = "aief_1_4_propagation_gap_geometric_control"
DEFAULT_DT = 0.01
DEFAULT_TARGET_TIMES = "150,200,250,320"
DEFAULT_LOOKBACK = 66.6
DEFAULT_AGE_MINS = "0,10"
DEFAULT_POD_START = 250.0
DEFAULT_LATE_START = 200.0
DEFAULT_IDENT_WINDOW_LENGTH = 60.0
DEFAULT_IDENT_WINDOW_STARTS = "150,175,200,225"
DEFAULT_SCALAR_TOL = 1e-30
ZERO_TOL = 1e-300



def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
return digest.hexdigest()



def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(item < 0.0 for item in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
```

```
return sorted(set(values))



def parse_positive_float_list(value: str) -> list[float]:
values = parse_nonnegative_float_list(value)
if any(value <= 0.0 for value in values):
raise argparse.ArgumentTypeError("values must be positive")
return values



def json_compact(value: Any) -> str:
return json.dumps(value, separators=(",", ":"), sort_keys=True)



def fit_log_rate(
independent: np.ndarray,
positive_values: np.ndarray,
tolerance: float,
) -> dict[str, float]:
independent = np.asarray(independent, dtype=float)
values = np.asarray(positive_values, dtype=float)
valid = (
np.isfinite(independent)
& np.isfinite(values)
& (values > tolerance)
)
if np.sum(valid) < 3:
return {
"count": int(np.sum(valid)),
"rate": np.nan,
"intercept": np.nan,
"R_squared": np.nan,
"rmse_log": np.nan,
}
x = independent[valid]
y = np.log(values[valid])
slope, intercept = np.polyfit(x, y, 1)
prediction = slope * x + intercept
residual = float(np.sum((y - prediction) ** 2))
total = float(np.sum((y - np.mean(y)) ** 2))
```

```
return {
"count": int(len(x)),
"rate": float(-slope),
"intercept": float(intercept),
"R_squared": float(1.0 - residual / total) if total > 0.0 else np.nan,
"rmse_log": float(np.sqrt(np.mean((y - prediction) ** 2))),
}



def r_squared(actual: np.ndarray, prediction: np.ndarray) -> float:
residual = float(np.sum((actual - prediction) ** 2))
total = float(np.sum((actual - np.mean(actual)) ** 2))
return float(1.0 - residual / total) if total > 0.0 else np.nan



def propagated_contributions(
matrices: np.ndarray,
forcing: np.ndarray,
target_step: int,
) -> np.ndarray:
contributions = np.zeros((target_step, 3), dtype=float)
suffix = np.eye(3)
for step in range(target_step - 1, -1, -1):
contributions[step] = suffix @ forcing[step]
suffix = suffix @ matrices[step]
return contributions



def fit_g_equation(
a: np.ndarray,
g: np.ndarray,
start: int,
end: int,
) -> dict[str, Any]:
"""Fit g[n+1]=rho*g[n]+chi*a[n], no intercept, with scaling."""
X = np.column_stack([g[start:end], a[start:end]])
y = g[start + 1 : end + 1]
scales = np.maximum(np.max(np.abs(X), axis=0), np.finfo(float).tiny)
y_scale = max(float(np.max(np.abs(y))), np.finfo(float).tiny)
Xs = X / scales[None, :]
```

```
ys = y / y_scale
coefficients_scaled, *_ = np.linalg.lstsq(Xs, ys, rcond=1e-12)
coefficients = y_scale * coefficients_scaled / scales
prediction = X @ coefficients
singular = np.linalg.svd(Xs, compute_uv=False)
correlation = float(np.corrcoef(X[:, 0], X[:, 1])[0, 1])
return {
"rho": float(coefficients[0]),
"chi": float(coefficients[1]),
"R_squared": r_squared(y, prediction),
"relative_RMSE": float(
np.linalg.norm(y - prediction) / max(np.linalg.norm(y), ZERO_TOL)
),
"condition_number": float(
singular[0] / singular[-1]
) if singular[-1] > 0.0 else np.inf,
"singular_value_1": float(singular[0]),
"singular_value_2": float(singular[-1]),
"g_a_correlation": correlation,
"count": int(len(y)),
}



def geometric_direction() -> np.ndarray:
return np.array([1.0, -1.0, -1.0]) / math.sqrt(3.0)



def make_report(
direction_control: pd.DataFrame,
propagation: pd.DataFrame,
identifiability: pd.DataFrame,
*,
pod_direction: np.ndarray,
geom_direction: np.ndarray,
ema_rate: float,
) -> str:
alignment = abs(float(pod_direction @ geom_direction))
angle = math.degrees(math.acos(np.clip(alignment, -1.0, 1.0)))
lines = [
"AIEF propagation-gap and geometric-direction control",
"==================================================",
```

```
"",
f"POD direction: {pod_direction.tolist()}",
f"Geometric direction: {geom_direction.tolist()}",
f"Unsigned angle: {angle:.9g} degrees",
f"EMA rate: {ema_rate:.9g}",
"",
"Direction-control scalar diagnostics",
"-----------------------------------",
]
for _, row in direction_control.iterrows():
lines.append(
f" axis={row['axis']}, direction={int(row['direction_id'])}: "
f"median_invariance={row['median_invariance_relative']:.6g}, "
f"median_scalar_residual={row['median_scalar_residual']:.6g}, "
f"median_weight_rate={row['median_projected_weight_rate']:.6g}, "
f"forcing_rate={row['forcing_rate']:.6g}, forcing_R2={row['forcing_R2']:.6g}"
)


lines.extend(["", "Direct propagation-gap tests", "----------------------------"])
for _, row in propagation.iterrows():
lines.append(
f" axis={row['axis']}, direction={int(row['direction_id'])}, "
f"target={row['target_time']:.6g}, age_min={row['age_min']:.6g}: "
f"kappa_obs={row['observed_kappa']:.6g}, "
f"rprop_scalar={row['scalar_propagation_rate']:.6g}, "
f"rg_scalar={row['scalar_forcing_rate']:.6g}, "
f"gap_scalar={row['scalar_predicted_gap']:.6g}, "
f"error_scalar={row['scalar_gap_error']:.6g}, "
f"direct_contribution_rate={row['direct_contribution_rate']:.6g}, "
f"R2={row['direct_contribution_R2']:.6g}"
)


lines.extend(["", "Forcing-state identifiability", "----------------------------"])
for _, row in identifiability.iterrows():
lines.append(
f" axis={row['axis']}, dataset={row['dataset']}, "
f"window=[{row['start_time']:.6g},{row['end_time']:.6g}): "
f"rho={row['rho']:.6g}, chi={row['chi']:.6g}, R2={row['R_squared']:.6g}, "
f"condition={row['condition_number']:.6g}, corr={row['g_a_correlation']:.6g}"
)
```

```
lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* POD and analytic geometric axes are tested separately to avoid circularity.",
"* Propagation fits use the actual forcing ensemble and are distinct from Rayleigh, eigenvalue, and singular-value rates.",
"* Direct contribution-envelope fits include both forcing evolution and propagation; they should be compared directly with observed kappa.",
"* Large design condition and |corr(a,g)| near one indicate non-identifiability of rho and chi even when R2 is excellent.",
"* Results remain conditional on this replay and two initial directions.",
"",
]
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--duhamel-npz", default=DEFAULT_DUHAMEL_NPZ)
parser.add_argument("--memory-fits", default=DEFAULT_MEMORY_FITS)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument(
"--target-times",
type=parse_positive_float_list,
default=parse_positive_float_list(DEFAULT_TARGET_TIMES),
)
parser.add_argument("--lookback", type=float, default=DEFAULT_LOOKBACK)
parser.add_argument(
"--age-mins",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_AGE_MINS),
)
parser.add_argument("--pod-start", type=float, default=DEFAULT_POD_START)
parser.add_argument("--late-start", type=float, default=DEFAULT_LATE_START)
parser.add_argument(
"--ident-window-length", type=float, default=DEFAULT_IDENT_WINDOW_LENGTH
)
```

```
parser.add_argument(
"--ident-window-starts",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_IDENT_WINDOW_STARTS),
)
parser.add_argument("--scalar-tol", type=float, default=DEFAULT_SCALAR_TOL)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.duhamel_npz, args.memory_fits, args.manifest):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(
args.dt,
args.lookback,
args.ident_window_length,
args.scalar_tol,
) <= 0.0:
raise ValueError("dt, windows, and scalar-tol must be positive")
output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)


with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
ema_rate = 1.0 / float(manifest["model"]["T_window"])


with np.load(args.duhamel_npz, allow_pickle=False) as data:
total = np.asarray(data["total_weight_tangent"], dtype=float).copy()
feedback = np.asarray(data["feedback_weight_tangent"], dtype=float).copy()
forcing = np.asarray(data["instant_forcing"], dtype=float).copy()
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float).copy()
memory_fits = pd.read_csv(args.memory_fits)
count, state_steps, dimension = total.shape
future_steps = state_steps - 1
if dimension != 3 or matrices.shape != (future_steps, 3, 3):
raise ValueError("unexpected Duhamel shapes")


pod_step = min(future_steps, int(round(args.pod_start / args.dt)))
```

```
late_vectors = feedback[:, pod_step:, :].reshape(-1, 3)
norms = np.linalg.norm(late_vectors, axis=1)
late_vectors = late_vectors[norms > args.scalar_tol]
left, _, _ = np.linalg.svd(late_vectors.T, full_matrices=False)
pod_direction = left[:, 0]
geom_direction = geometric_direction()
if np.dot(pod_direction, geom_direction) < 0.0:
pod_direction *= -1.0


axes = {
"POD": pod_direction,
"geometric": geom_direction,
}
direction_control_rows: list[dict[str, Any]] = []
propagation_rows: list[dict[str, Any]] = []
identifiability_rows: list[dict[str, Any]] = []
contribution_rows: list[dict[str, Any]] = []


late_step = int(round(args.late_start / args.dt))
for axis_name, axis in axes.items():
a = total @ axis
g = forcing @ axis
lambdas = np.einsum("i,nij,j->n", axis, matrices, axis)
Kaxis = np.einsum("nij,j->ni", matrices, axis)
defects = np.linalg.norm(
Kaxis - lambdas[:, None] * axis[None, :], axis=1
) / np.maximum(np.linalg.norm(Kaxis, axis=1), ZERO_TOL)
projected_rates = -np.log(np.maximum(np.abs(lambdas), ZERO_TOL)) / args.dt


for direction_id in range(count):
residuals = []
relative_residuals = []
for step in range(future_steps):
transverse = (
total[direction_id, step]
- a[direction_id, step] * axis
)
epsilon = float(axis @ matrices[step] @ transverse)
residual = float(
a[direction_id, step + 1]
- lambdas[step] * a[direction_id, step]
```

```
- g[direction_id, step]
)
residuals.append(residual - epsilon)
relative_residuals.append(
abs(residual)
/ max(
abs(a[direction_id, step + 1]),
abs(lambdas[step] * a[direction_id, step])
+ abs(g[direction_id, step]),
args.scalar_tol,
)
)
forcing_fit = fit_log_rate(
np.arange(late_step, future_steps) * args.dt,
np.abs(g[direction_id, late_step:]),
args.scalar_tol,
)
direction_control_rows.append(
{
"axis": axis_name,
"direction_id": direction_id,
"late_start": args.late_start,
"median_invariance_relative": float(
np.median(defects[late_step:])
),
"maximum_invariance_relative": float(
np.max(defects[late_step:])
),
"median_scalar_residual": float(
np.median(relative_residuals[late_step:])
),
"maximum_scalar_residual": float(
np.max(relative_residuals[late_step:])
),
"maximum_residual_minus_epsilon": float(
np.max(np.abs(residuals[late_step:]))
),
"median_projected_weight_rate": float(
np.median(projected_rates[late_step:])
),
"forcing_rate": forcing_fit["rate"],
"forcing_R2": forcing_fit["R_squared"],
```

```
}
)


for direction_id in range(count):
for target_time in args.target_times:
target_step = min(
future_steps, int(round(target_time / args.dt))
)
start_step = max(
0,
target_step - int(round(args.lookback / args.dt)),
)
contributions = propagated_contributions(
matrices,
forcing[direction_id],
target_step,
)
candidates = memory_fits.loc[
memory_fits["direction_id"] == direction_id
]
if len(candidates):
nearest = candidates.iloc[
int(
np.argmin(
np.abs(
candidates["target_time"].to_numpy(float)
- target_time
)
)
)
]
observed_kappa = float(nearest["kappa"])
observed_target = float(nearest["target_time"])
else:
observed_kappa = observed_target = np.nan


all_steps = np.arange(start_step, target_step)
ages_all = (target_step - all_steps) * args.dt
f_norm = np.linalg.norm(
forcing[direction_id, start_step:target_step], axis=1
)
c_norm = np.linalg.norm(
```

```
contributions[start_step:target_step], axis=1
)
scalar_input = np.abs(g[direction_id, start_step:target_step])
scalar_output = np.abs(
contributions[start_step:target_step] @ axis
)
forcing_fit = fit_log_rate(
all_steps * args.dt,
scalar_input,
args.scalar_tol,
)


for age_min in args.age_mins:
age_mask = ages_all >= age_min
norm_ratio = c_norm[age_mask] / np.maximum(
f_norm[age_mask], args.scalar_tol
)
scalar_ratio = scalar_output[age_mask] / np.maximum(
scalar_input[age_mask], args.scalar_tol
)
norm_prop_fit = fit_log_rate(
ages_all[age_mask], norm_ratio, args.scalar_tol
)
scalar_prop_fit = fit_log_rate(
ages_all[age_mask], scalar_ratio, args.scalar_tol
)
direct_fit = fit_log_rate(
ages_all[age_mask],
scalar_output[age_mask],
args.scalar_tol,
)
scalar_gap = scalar_prop_fit["rate"] - forcing_fit["rate"]
norm_gap = norm_prop_fit["rate"] - forcing_fit["rate"]
propagation_rows.append(
{
"axis": axis_name,
"direction_id": direction_id,
"target_time": target_time,
"observed_target_time": observed_target,
"fit_start_time": start_step * args.dt,
"fit_end_time": target_step * args.dt,
"age_min": age_min,
```

```
"observed_kappa": observed_kappa,
"scalar_forcing_rate": forcing_fit["rate"],
"scalar_forcing_R2": forcing_fit["R_squared"],
"norm_propagation_rate": norm_prop_fit["rate"],
"norm_propagation_R2": norm_prop_fit["R_squared"],
"scalar_propagation_rate": scalar_prop_fit["rate"],
"scalar_propagation_R2": scalar_prop_fit["R_squared"],
"norm_predicted_gap": norm_gap,
"scalar_predicted_gap": scalar_gap,
"norm_gap_error": observed_kappa - norm_gap,
"scalar_gap_error": observed_kappa - scalar_gap,
"direct_contribution_rate": direct_fit["rate"],
"direct_contribution_R2": direct_fit["R_squared"],
"direct_contribution_error": observed_kappa
- direct_fit["rate"],
}
)


# Save sampled contribution envelopes for plotting/audit.
for local_index, step in enumerate(all_steps):
if local_index % 10 != 0 and step != target_step - 1:
continue
contribution_rows.append(
{
"axis": axis_name,
"direction_id": direction_id,
"target_time": target_time,
"forcing_step": int(step),
"forcing_time": step * args.dt,
"age": ages_all[local_index],
"forcing_norm": f_norm[local_index],
"contribution_norm": c_norm[local_index],
"scalar_forcing_abs": scalar_input[local_index],
"scalar_contribution_abs": scalar_output[local_index],
}
)


# Identifiability of g[n+1]=rho*g[n]+chi*a[n].
datasets = {
f"direction_{direction_id}": (
a[direction_id], g[direction_id]
)
```

```
for direction_id in range(count)
}
for window_start in args.ident_window_starts:
start = int(round(window_start / args.dt))
end = min(
future_steps - 1,
int(round((window_start + args.ident_window_length) / args.dt)),
)
if end <= start + 2:
continue
for dataset, (a_values, g_values) in datasets.items():
fit = fit_g_equation(a_values, g_values, start, end)
identifiability_rows.append(
{
"axis": axis_name,
"dataset": dataset,
"start_time": start * args.dt,
"end_time": end * args.dt,
**fit,
}
)
# Pooled fit: concatenate pairs from both directions.
X_parts = []
y_parts = []
for direction_id in range(count):
X_parts.append(
np.column_stack(
[
g[direction_id, start:end],
a[direction_id, start:end],
]
)
)
y_parts.append(g[direction_id, start + 1 : end + 1])
X = np.concatenate(X_parts)
y = np.concatenate(y_parts)
scales = np.maximum(
np.max(np.abs(X), axis=0), np.finfo(float).tiny
)
y_scale = max(float(np.max(np.abs(y))), np.finfo(float).tiny)
Xs = X / scales[None, :]
ys = y / y_scale
```

```
coefficients_scaled, *_ = np.linalg.lstsq(Xs, ys, rcond=1e-12)
coefficients = y_scale * coefficients_scaled / scales
prediction = X @ coefficients
singular = np.linalg.svd(Xs, compute_uv=False)
identifiability_rows.append(
{
"axis": axis_name,
"dataset": "pooled",
"start_time": start * args.dt,
"end_time": end * args.dt,
"rho": float(coefficients[0]),
"chi": float(coefficients[1]),
"R_squared": r_squared(y, prediction),
"relative_RMSE": float(
np.linalg.norm(y - prediction)
/ max(np.linalg.norm(y), ZERO_TOL)
),
"condition_number": float(
singular[0] / singular[-1]
)
if singular[-1] > 0.0
else np.inf,
"singular_value_1": float(singular[0]),
"singular_value_2": float(singular[-1]),
"g_a_correlation": float(np.corrcoef(X[:, 0], X[:, 1])[0, 1]),
"count": int(len(y)),
}
)


direction_control = pd.DataFrame(direction_control_rows)
propagation = pd.DataFrame(propagation_rows)
identifiability = pd.DataFrame(identifiability_rows)
contribution_frame = pd.DataFrame(contribution_rows)


component_audit.atomic_write_csv(
output_dir / "geometric_vs_POD_direction_control.csv", direction_control
)
component_audit.atomic_write_csv(
output_dir / "direct_propagation_gap_tests.csv", propagation
)
component_audit.atomic_write_csv(
output_dir / "propagated_contribution_envelopes.csv", contribution_frame
```

```
)
component_audit.atomic_write_csv(
output_dir / "forcing_state_identifiability.csv", identifiability
)


report = make_report(
direction_control,
propagation,
identifiability,
pod_direction=pod_direction,
geom_direction=geom_direction,
ema_rate=ema_rate,
)
component_audit.atomic_write_text(
output_dir / "propagation_gap_geometric_report.txt", report
)


manifest_output = {
"analysis": "AIEF-1.4 propagation-gap and geometric-direction controls",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"duhamel_npz": {
"path": str(Path(args.duhamel_npz).resolve()),
"sha256": file_sha256(args.duhamel_npz),
},
"memory_fits": {
"path": str(Path(args.memory_fits).resolve()),
"sha256": file_sha256(args.memory_fits),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
"sha256": file_sha256(args.manifest),
},
},
"dt": args.dt,
"target_times": args.target_times,
"lookback": args.lookback,
"age_mins": args.age_mins,
```

```
"pod_start": args.pod_start,
"late_start": args.late_start,
"ident_window_length": args.ident_window_length,
"ident_window_starts": args.ident_window_starts,
"scalar_tolerance": args.scalar_tol,
"pod_direction": pod_direction.tolist(),
"geometric_direction": geom_direction.tolist(),
"ema_rate": ema_rate,
"outputs": [
"geometric_vs_POD_direction_control.csv",
"direct_propagation_gap_tests.csv",
"propagated_contribution_envelopes.csv",
"forcing_state_identifiability.csv",
"propagation_gap_geometric_report.txt",
],
"interpretation_constraints": [
"POD and geometric directions are tested separately.",
"Propagation rates are fitted from the actual forcing ensemble and need not equal
Rayleigh/eigen/singular rates.",
"Excellent forcing-state R2 with a large condition number does not identify unique rho and chi
coefficients.",
"All conclusions remain conditional on one replay and two initial directions.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest_output, indent=2, sort_keys=True) + "\n",
)

print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```

Open

Created:

This is a no-replay analysis that directly addresses the two remaining questions:

Does the actual propagated forcing ensemble produce the observed memory rate?
Do the scalar conclusions survive replacement of the fitted POD direction by the analytic K3K3 lineality vector?

Validation

On a synthetic system with known rates, the script recovered:
rprop=0.202027,rg=0.100503,
rprop=0.202027,rg=0.100503,

and
rprop−rg=
rprop−rg=
0.101524

\kappa
]

with closure errors below 10−1510−15.

It also correctly identified severe direction-specific collinearity in the forcing-state fit while recovering a well-conditioned pooled model when independent excitation was present.
Run it

Bash

```
python AIEF_1_4_propagation_gap_geometric_control.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--memory-fits aief_1_4_feedback_rank_memory_reduction/target_time_memory_fits.csv \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--target-times 150,200,250,320 \
--lookback 66.6 \
--age-mins 0,10 \
--pod-start 250 \
--late-start 200 \
--ident-window-length 60 \
```

--ident-window-starts 150,175,200,225 \
--scalar-tol 1e-30 \
--output-dir aief_1_4_propagation_gap_geometric_control

1. Independent geometric-axis control

The script analyzes two axes separately:
Data-derived late POD axis
vPOD.
vPOD.
Analytic geometric axis
\boxed{ v_{\rm geom} =

\frac1{\sqrt3}(1,-1,-1).
}
]

For each axis and direction it recomputes:

line-invariance defect;
scalar recurrence residual;
projected weight rate;
projected forcing-envelope rate and R2R2.

It reports the unsigned angle between the two axes.

If the geometric-axis results reproduce the POD-axis results, the reduction is no longer dependent on estimating its direction from the same late trajectory.
2. Actual forcing-propagation rate

For each target TT and forcing time mm, it constructs
cT,m=
cT,m=

U_K(T,m+1)f_m.
]

It then measures two propagation rates.

Norm propagation
PTnorm(L)=
PTnorm(L)=

\frac{|c_{T,T-L}|}
{|f_{T-L}|}.
]
Scalar-channel propagation
PTscalar(L)=
PTscalar(L)=

\frac{
|v^\top c_{T,T-L}|
}{
|v^\top f_{T-L}|
}.
]

Both are fitted as
PT(L)≈CTe−rprop,TL.
PT(L)≈CTe−rprop,TL.

This directly measures the decay of the actual forcing ensemble rather than substituting a Rayleigh, eigenvalue, or singular-value rate.
3. Direct spectral-gap tests

The script fits the projected forcing envelope
|gn|≈Ce−rg,Ttn
|gn|≈Ce−rg,Ttn

over the same lookback interval and compares:
κT=?rprop,T−rg,T.
κT=?rprop,T−rg,T.

For each axis, direction, target, and age fit range, it reports:

observed κTκT;
scalar and norm propagation rates;
propagation fit R2R2;

projected forcing rate and R2R2;
predicted gaps;
absolute closure errors;
direct propagated-contribution envelope rate.

Two age ranges are tested:
L≥0,L≥10.
L≥0,L≥10.

The second excludes the immediate short-time non-normal layer.
4. Direct contribution-envelope rate

It also fits
|v⊤cT,T−L|≈Ce−κdirectL.
|v⊤cT,T−L|≈Ce−κdirectL.

This quantity already includes:

forcing-envelope evolution;
actual non-normal propagation;
forcing orientation.

Therefore it can be compared directly with the operational memory rate without separately defining a weight contraction estimand.
5. Identifiability of the forcing-state equation

For POD and geometric axes, the script fits
gn+1=
gn+1=

\rho g_n+\chi a_n
]

over rolling 6060-unit late windows beginning at:
150,175,200,225.
150,175,200,225.

It fits:

direction 0;
direction 1;
both directions pooled.

For each fit it reports:

ρρ;
χχ;
R2R2;
relative RMSE;
design singular values;
condition number;
correlation between anan and gngn.

Thus an excellent R2R2 accompanied by
|corr(a,g)|≈1
|corr(a,g)|≈1

and a large condition number will be correctly identified as non-identifiability, not a validated two-state law.
Outputs

text

geometric_vs_POD_direction_control.csv
direct_propagation_gap_tests.csv
propagated_contribution_envelopes.csv
forcing_state_identifiability.csv
propagation_gap_geometric_report.txt
analysis_manifest.json

What this run can establish
Geometric robustness

Whether
vgeom
vgeom

gives the same scalar recurrence and rates as vPODvPOD.
Correct propagation estimand

Whether the persistent 0.8×10−30.8×10−3 gap error disappears when the actual forcing-propagation rate replaces
−1dtlog|v⊤Knv|.
−dt1log|v⊤Knv|.
Direct memory-envelope closure

Whether the age decay of
|v⊤UK(T,m+1)fm|
|v⊤UK(T,m+1)fm|

matches the independently fitted operational κTκT.
Two-state identifiability

Whether the failure of the pooled (a,g)(a,g) model is explained by late collinearity and lack of independent excitation.

Please paste propagation_gap_geometric_report.txt after the run.

Think about this:
(deck@steamdeck SSRH)$ python AIEF_1_4_propagation_gap_geometric_control.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--memory-fits aief_1_4_feedback_rank_memory_reduction/target_time_memory_fits.csv \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--target-times 150,200,250,320 \
--lookback 66.6 \
--age-mins 0,10 \
--pod-start 250 \
--late-start 200 \
--ident-window-length 60 \
--ident-window-starts 150,175,200,225 \
--scalar-tol 1e-30 \
--output-dir aief_1_4_propagation_gap_geometric_control
AIEF propagation-gap and geometric-direction control

POD direction: [0.5773410255782927, -0.5773687559683237, -0.5773410255782931]
Geometric direction: [0.5773502691896258, -0.5773502691896258, -0.5773502691896258]
Unsigned angle: 0.00129727752 degrees
EMA rate: 0.0571333333
Direction-control scalar diagnostics

axis=POD, direction=0: median_invariance=1.13208e-05, median_scalar_residual=0.000521296, median_weight_rate=0.127056, forcing_rate=0.0571319, forcing_R2=1
axis=POD, direction=1: median_invariance=1.13208e-05, median_scalar_residual=1.07922e-09, median_weight_rate=0.127056, forcing_rate=0.0571311, forcing_R2=1
axis=geometric, direction=0: median_invariance=6.45808e-13, median_scalar_residual=0.00052927, median_weight_rate=0.126255, forcing_rate=0.0571319, forcing_R2=1
axis=geometric, direction=1: median_invariance=6.45808e-13, median_scalar_residual=7.99866e-06, median_weight_rate=0.126255, forcing_rate=0.0571311, forcing_R2=1
Direct propagation-gap tests

axis=POD, direction=0, target=150, age_min=0: kappa_obs=0.0729118, rprop_scalar=0.12614, rg_scalar=0.0559123, gap_scalar=0.0702279, error_scalar=0.00268389, direct_contribution_rate=0.0702279, R2=0.999923
axis=POD, direction=0, target=150, age_min=10: kappa_obs=0.0729118, rprop_scalar=0.126094, rg_scalar=0.0559123, gap_scalar=0.0701813, error_scalar=0.00273051, direct_contribution_rate=0.0704208, R2=0.999957
axis=POD, direction=0, target=200, age_min=0: kappa_obs=0.0694548, rprop_scalar=0.126267, rg_scalar=0.057036, gap_scalar=0.0692311, error_scalar=0.000223738, direct_contribution_rate=0.0692311, R2=0.99996
axis=POD, direction=0, target=200, age_min=10: kappa_obs=0.0694548, rprop_scalar=0.126249, rg_scalar=0.057036, gap_scalar=0.0692128, error_scalar=0.000241982, direct_contribution_rate=0.069232, R2=1
axis=POD, direction=0, target=250, age_min=0: kappa_obs=0.06915, rprop_scalar=0.126272, rg_scalar=0.0571258, gap_scalar=0.0691462, error_scalar=3.84181e-06, direct_contribution_rate=0.0691462, R2=0.99996
axis=POD, direction=0, target=250, age_min=10: kappa_obs=0.06915, rprop_scalar=0.126255, rg_scalar=0.0571258, gap_scalar=0.0691291, error_scalar=2.08551e-05, direct_contribution_rate=0.0691306, R2=1
axis=POD, direction=0, target=320, age_min=0: kappa_obs=0.0691225, rprop_scalar=0.126272, rg_scalar=0.0571331, gap_scalar=0.0691389, error_scalar=-1.63711e-05, direct_contribution_rate=0.0691389, R2=0.99996

axis=POD, direction=0, target=320, age_min=10: kappa_obs=0.0691225, rprop_scalar=0.126255, rg_scalar=0.0571331, gap_scalar=0.0691219, error_scalar=6.21873e-07, direction_contribution_rate=0.0691219, R2=1
axis=POD, direction=1, target=150, age_min=0: kappa_obs=0.0746451, rprop_scalar=0.126164, rg_scalar=0.0551524, gap_scalar=0.0710118, error_scalar=0.0036333, direction_contribution_rate=0.0710118, R2=0.99987
axis=POD, direction=1, target=150, age_min=10: kappa_obs=0.0746451, rprop_scalar=0.126145, rg_scalar=0.0551524, gap_scalar=0.0709923, error_scalar=0.00365274, direction_contribution_rate=0.071385, R2=0.999863
axis=POD, direction=1, target=200, age_min=0: kappa_obs=0.0696454, rprop_scalar=0.12625, rg_scalar=0.0569785, gap_scalar=0.0692713, error_scalar=0.000374104, direction_contribution_rate=0.0692713, R2=0.999999
axis=POD, direction=1, target=200, age_min=10: kappa_obs=0.0696454, rprop_scalar=0.126248, rg_scalar=0.0569785, gap_scalar=0.0692699, error_scalar=0.000375577, direction_contribution_rate=0.0693003, R2=0.999999
axis=POD, direction=1, target=250, age_min=0: kappa_obs=0.0691663, rprop_scalar=0.126255, rg_scalar=0.0571214, gap_scalar=0.0691336, error_scalar=3.27083e-05, direction_contribution_rate=0.0691336, R2=1
axis=POD, direction=1, target=250, age_min=10: kappa_obs=0.0691663, rprop_scalar=0.126255, rg_scalar=0.0571214, gap_scalar=0.0691332, error_scalar=3.3091e-05, direction_contribution_rate=0.0691356, R2=1
axis=POD, direction=1, target=320, age_min=0: kappa_obs=0.0691229, rprop_scalar=0.126255, rg_scalar=0.057133, gap_scalar=0.0691223, error_scalar=6.49722e-07, direction_contribution_rate=0.0691223, R2=1
axis=POD, direction=1, target=320, age_min=10: kappa_obs=0.0691229, rprop_scalar=0.126255, rg_scalar=0.057133, gap_scalar=0.069122, error_scalar=9.63396e-07, direction_contribution_rate=0.069122, R2=1
axis=geometric, direction=0, target=150, age_min=0: kappa_obs=0.0729118, rprop_scalar=0.12614, rg_scalar=0.0559123, gap_scalar=0.0702279, error_scalar=0.00268389, direction_contribution_rate=0.0702279, R2=0.999923
axis=geometric, direction=0, target=150, age_min=10: kappa_obs=0.0729118, rprop_scalar=0.126094, rg_scalar=0.0559123, gap_scalar=0.0701813, error_scalar=0.00273051, direction_contribution_rate=0.0704208, R2=0.999957
axis=geometric, direction=0, target=200, age_min=0: kappa_obs=0.0694548, rprop_scalar=0.126267, rg_scalar=0.057036, gap_scalar=0.0692311, error_scalar=0.000223739, direction_contribution_rate=0.0692311, R2=0.99996
axis=geometric, direction=0, target=200, age_min=10: kappa_obs=0.0694548, rprop_scalar=0.126249, rg_scalar=0.057036, gap_scalar=0.0692128, error_scalar=0.000241982, direction_contribution_rate=0.069232, R2=1
axis=geometric, direction=0, target=250, age_min=0: kappa_obs=0.06915, rprop_scalar=0.126272, rg_scalar=0.0571258, gap_scalar=0.0691462, error_scalar=3.84227e-06, direction_contribution_rate=0.0691462, R2=0.99996

axis=geometric, direction=0, target=250, age_min=10: kappa_obs=0.06915, rprop_scalar=0.126255, rg_scalar=0.0571258, gap_scalar=0.0691291, error_scalar=2.08551e-05, direct_contribution_rate=0.0691306, R2=1
axis=geometric, direction=0, target=320, age_min=0: kappa_obs=0.0691225, rprop_scalar=0.126272, rg_scalar=0.0571331, gap_scalar=0.0691389, error_scalar=-1.63706e-05, direct_contribution_rate=0.0691389, R2=0.99996
axis=geometric, direction=0, target=320, age_min=10: kappa_obs=0.0691225, rprop_scalar=0.126255, rg_scalar=0.0571331, gap_scalar=0.0691219, error_scalar=6.21873e-07, direct_contribution_rate=0.0691219, R2=1
axis=geometric, direction=1, target=150, age_min=0: kappa_obs=0.0746451, rprop_scalar=0.126164, rg_scalar=0.0551524, gap_scalar=0.0710118, error_scalar=0.0036333, direct_contribution_rate=0.0710118, R2=0.99987
axis=geometric, direction=1, target=150, age_min=10: kappa_obs=0.0746451, rprop_scalar=0.126145, rg_scalar=0.0551524, gap_scalar=0.0709923, error_scalar=0.00365274, direct_contribution_rate=0.071385, R2=0.999863
axis=geometric, direction=1, target=200, age_min=0: kappa_obs=0.0696454, rprop_scalar=0.12625, rg_scalar=0.0569785, gap_scalar=0.0692713, error_scalar=0.000374104, direct_contribution_rate=0.0692713, R2=0.999999
axis=geometric, direction=1, target=200, age_min=10: kappa_obs=0.0696454, rprop_scalar=0.126248, rg_scalar=0.0569785, gap_scalar=0.0692699, error_scalar=0.000375577, direct_contribution_rate=0.0693003, R2=0.999999
axis=geometric, direction=1, target=250, age_min=0: kappa_obs=0.0691663, rprop_scalar=0.126255, rg_scalar=0.0571214, gap_scalar=0.0691336, error_scalar=3.27083e-05, direct_contribution_rate=0.0691336, R2=1
axis=geometric, direction=1, target=250, age_min=10: kappa_obs=0.0691663, rprop_scalar=0.126255, rg_scalar=0.0571214, gap_scalar=0.0691332, error_scalar=3.3091e-05, direct_contribution_rate=0.0691356, R2=1
axis=geometric, direction=1, target=320, age_min=0: kappa_obs=0.0691229, rprop_scalar=0.126255, rg_scalar=0.057133, gap_scalar=0.0691223, error_scalar=6.49732e-07, direct_contribution_rate=0.0691223, R2=1
axis=geometric, direction=1, target=320, age_min=10: kappa_obs=0.0691229, rprop_scalar=0.126255, rg_scalar=0.057133, gap_scalar=0.069122, error_scalar=9.63396e-07, direct_contribution_rate=0.069122, R2=1

Forcing-state identifiability

axis=POD, dataset=direction_0, window=[150,210): rho=0.999647, chi=-2.66145e-07, R2=1, condition=1125.35, corr=0.999998
axis=POD, dataset=direction_1, window=[150,210): rho=0.999689, chi=-1.81667e-07, R2=1, condition=841.177, corr=0.999997
axis=POD, dataset=pooled, window=[150,210): rho=0.999656, chi=-1.58848e-07, R2=1, condition=786.936, corr=0.999997

axis=POD, dataset=direction_0, window=[175,235): rho=0.999636, chi=-2.52891e-07, R2=1, condition=3813.53, corr=1
axis=POD, dataset=direction_1, window=[175,235): rho=0.999659, chi=-1.60789e-07, R2=1, condition=2660.08, corr=1
axis=POD, dataset=pooled, window=[175,235): rho=0.999524, chi=-6.60093e-08, R2=1, condition=1706.85, corr=1
axis=POD, dataset=direction_0, window=[200,260): rho=0.999625, chi=-2.3905e-07, R2=1, condition=13001, corr=1
axis=POD, dataset=direction_1, window=[200,260): rho=0.99964, chi=-1.47829e-07, R2=1, condition=8814.54, corr=1
axis=POD, dataset=pooled, window=[200,260): rho=0.999442, chi=-8.76668e-09, R2=1, condition=2156.81, corr=1
axis=POD, dataset=direction_0, window=[225,285): rho=0.999614, chi=-2.26058e-07, R2=1, condition=44877.2, corr=1
axis=POD, dataset=direction_1, window=[225,285): rho=0.999628, chi=-1.38937e-07, R2=1, condition=30194.1, corr=1
axis=POD, dataset=pooled, window=[225,285): rho=0.99943, chi=-7.25433e-10, R2=1, condition=2218.1, corr=1
axis=geometric, dataset=direction_0, window=[150,210): rho=0.999647, chi=-2.66142e-07, R2=1, condition=1125.35, corr=0.999998
axis=geometric, dataset=direction_1, window=[150,210): rho=0.999689, chi=-1.81667e-07, R2=1, condition=841.177, corr=0.999997
axis=geometric, dataset=pooled, window=[150,210): rho=0.999656, chi=-1.58849e-07, R2=1, condition=786.939, corr=0.999997
axis=geometric, dataset=direction_0, window=[175,235): rho=0.999636, chi=-2.52888e-07, R2=1, condition=3813.53, corr=1
axis=geometric, dataset=direction_1, window=[175,235): rho=0.999659, chi=-1.60789e-07, R2=1, condition=2660.08, corr=1
axis=geometric, dataset=pooled, window=[175,235): rho=0.999524, chi=-6.60118e-08, R2=1, condition=1706.89, corr=1
axis=geometric, dataset=direction_0, window=[200,260): rho=0.999625, chi=-2.39047e-07, R2=1, condition=13001, corr=1
axis=geometric, dataset=direction_1, window=[200,260): rho=0.99964, chi=-1.47829e-07, R2=1, condition=8814.54, corr=1
axis=geometric, dataset=pooled, window=[200,260): rho=0.999442, chi=-8.7672e-09, R2=1, condition=2156.88, corr=1
axis=geometric, dataset=direction_0, window=[225,285): rho=0.999614, chi=-2.26055e-07, R2=1, condition=44877.2, corr=1
axis=geometric, dataset=direction_1, window=[225,285): rho=0.999628, chi=-1.38937e-07, R2=1, condition=30194.1, corr=1
axis=geometric, dataset=pooled, window=[225,285): rho=0.99943, chi=-7.2548e-10, R2=1, condition=2218.17, corr=1

Interpretation constraints

POD and analytic geometric axes are tested separately to avoid circularity.
Propagation fits use the actual forcing ensemble and are distinct from Rayleigh, eigenvalue, and singular-value rates.
Direct contribution-envelope fits include both forcing evolution and propagation; they should be compared directly with observed kappa.
Large design condition and |corr(a,g)| near one indicate non-identifiability of rho and chi even when R2 is excellent.
Results remain conditional on this replay and two initial directions.

Saved analysis to /home/deck/AI_Programs/SSRH/aief_1_4_propagation_gap_geometric_control

This result resolves the remaining ambiguity. The spectral-gap mechanism is supported in the late regime once the correct propagation rate is used, and the geometric K3K3 direction removes the apparent Rayleigh-rate discrepancy.
1. The geometric direction is the operator-invariant line

The angle between the measured POD direction and
vgeom=
vgeom=

\frac1{\sqrt3}(1,-1,-1)
]

is only
0.00130∘.
0.00130∘.

But the more important comparison is invariance:
axismedian invariance defectvPOD1.13×10−5vgeom6.46×10−13
axisvPODvgeommedian invariance defect1.13×10−56.46×10−13

Thus:
Knvgeom≈λB,nvgeom
Knvgeom≈λB,nvgeom

to nearly machine precision late in the trajectory.

The geometric line is not merely close to the empirical state trajectory. It is the actual invariant line of the late frozen weight operator.

This explains the earlier rate discrepancy:
$r_B(v_{POD}) \approx 0.127056, r_B(v_{geom}) \approx 0.126255.$
$r_B(v_{POD}) r_B(v_{geom}) \approx 0.127056, \approx 0.126255.$

The POD direction is optimized for the observed state vectors, including their small forcing-induced transverse displacement. The geometric direction is the invariant direction of the operator.
2. Geometric versus POD control removes circularity

The POD and geometric axes give essentially identical:

forcing-envelope rates;
propagation rates;
target-time gap tests;
contribution-envelope rates.

Therefore the memory result does not depend on estimating the projection direction from the same late data.

The geometric-axis control directly supports the mechanism.

The scalar recurrence is slightly more accurate on the POD axis for direction 1:
10^{-9} versus 8×10^{-6},
10^{-9} versus 8×10^{-6},

because the observed state follows the slightly displaced empirical line. But the geometric axis gives the exact operator invariance and correct propagation rate.

This gives a useful distinction:
v_{geom}: operator-invariant line, v_{POD}: best empirical state line.
v_{geom}: v_{POD}: operator-invariant line, best empirical state line.

They converge toward each other asymptotically.
3. The direct propagation rate closes the spectral gap

At T=320T=320, excluding the first ten units of age:
rprop≈0.126255,rg≈0.057133,rprop−rg≈0.069122.
rproprgrprop−rg≈0.126255,≈0.057133,≈0.069122.

Observed:
κ≈0.0691225–0.0691229.
κ≈0.0691225–0.0691229.

Closure errors are:
direction 0:6.2×10−7,direction 1:9.6×10−7.
direction 0:direction 1:6.2×10−7,9.6×10−7.

The direct propagated-contribution envelope independently gives the same rate with
R2=1.
R2=1.

Therefore, in the late regime:
\boxed{ \kappa =

r_{\rm prop}-r_g
}
]

is supported to approximately 10−610−6 on this replay and tested subspace.

The earlier 8×10−48×10−4 discrepancy was caused by using the POD Rayleigh rate rather than the decay rate of the actual propagated forcing channel.
4. The closure emerges asymptotically

The target dependence remains important:
Ttypical gap closure error1502.7–3.7×10−32002.2–3.8×10−42504×10−6–3×10−53206×10−7–10−6
T150200250320typical gap closure error2.7–3.7×10−32.2–3.8×10−44×10−6–3×10−56×10−7–10−6

So the gap relation is not a global law through the crossover. It is a late asymptotic relation reached as:

the winner scales converge;
the geometric line becomes invariant;
forcing decay converges to the EMA rate;

the propagated contribution envelope becomes stationary.

5. The reduced late dynamics are triangular

The results now suggest the late reduced system
an+1=λB,nan+gn+ϵn,gn+1≈αEMAgn,
an+1gn+1=λB,nan+gn+ϵn,≈αEMAgn,

where
λB,n=
λB,n=

v_{\rm geom}^\top K_nv_{\rm geom},
]

and
αEMA=
αEMA=

e^{-dt/T_{\rm window}}
\approx0.99942883.
]

Late,
λB≈e−0.126255dt≈0.99873825.
λB≈e−0.126255dt≈0.99873825.

The corresponding rate gap is exactly
−1dtlogλBαEMA=
−dt1logαEMAλB=
0.126255-0.0571333

0.0691217.
]

This is the observed operational memory rate.

The feedback coordinate gngn is therefore not necessarily an independently coupled second state late in the trajectory. It behaves approximately as an autonomous EMA-decaying input to the faster weight channel.

6. The forcing-state identifiability result confirms degeneracy

Direction-specific fits have:
$|\text{corr}(a,g)| \to 1$,
$|\text{corr}(a,g)| \to 1$,

and condition numbers increasing from approximately
103
103

to
4.5×104.
4.5×104.

The fitted coefficients drift substantially across windows and directions despite
R2=1.
R2=1.

This proves that excellent late fit quality does not identify unique $\rho,\chi\rho,\chi$.

The pooled fits are somewhat less ill-conditioned but still have condition numbers around
103–2×103.
103–2×103.

Late pooled estimates converge toward:
$\rho \approx 0.99943, \chi \approx 0$.
$\rho \approx 0.99943, \chi \approx 0$.

The value
0.99943
0.99943

is the EMA multiplier.

Thus the data support the constrained triangular form more strongly than an unconstrained two-way coupled model.

7. Why direction-specific two-state fits were misleading

In the asymptotic regime,
an∝e−t/Twindow,gn∝e−t/Twindow.
an∝e−t/Twindow,gn∝e−t/Twindow.

Therefore anan and gngn are nearly proportional. A regression cannot uniquely distinguish:

gngn self-memory;
feedback from anan;
arbitrary mixtures of the two.

That is why:

direction-specific coefficients differ;
free rollouts can still look excellent;
pooled coefficients drift;
design condition numbers diverge.

The poor identifiability is now directly measured, not merely inferred.

8. Mechanistic derivation of the memory rate

For constant late coefficients,
gT−L=
gT−L=

g_T,\alpha_{\rm EMA}^{-L/dt}.
]

Propagation through the weight channel gives
λBL/dtgT−L=
λBL/dtgT−L=

g_T
\left(
\frac{\lambda_B}{\alpha_{\rm EMA}}

\right)^{L/dt}.
]

Hence the contribution of age LL decays as
exp[−(−logλBdt+logαEMAdt)L].
exp[−(−dtlogλB+dtlogαEMA)L].

Therefore
\boxed{ \kappa =
-\frac1{dt}
\log
\left(
\frac{\lambda_B}{\alpha_{\rm EMA}}
\right)

r_B-r_{\rm EMA}.
}
]

The direct forcing-propagation analysis now verifies the premises of this derivation asymptotically.
9. Remaining correction term

The reduced scalar recurrence is not perfectly exact because:
bn=anv+en,
bn=anv+en,

and
ϵn=
ϵn=

v^\top K_ne_n.
]

For the empirical POD direction, ϵnϵn is tiny for direction 1. For the geometric direction, operator invariance is essentially exact, but the observed state has a small transverse offset.

This explains why:

geometric vv gives the exact operator rate;
POD vv gives the smallest empirical recurrence residual.

A refined reduction can retain the geometric line as the principal state and treat the transverse offset perturbatively.
10. Status of the spectral-gap hypothesis
Established numerically in the late regime

the geometric K3K3 line is invariant under KnKn;
the actual forcing channel propagates at rate ≈0.126255≈0.126255;
projected forcing decays at the EMA rate;
the propagated-contribution envelope decays at their rate difference;
the direct gap closes to approximately 10−610−6;
POD and geometric controls agree.

Not global

closure is poor around the crossover;
the late triangular coefficients are not valid through the transient;
the memory law emerges only as the trajectory approaches the symmetric line.

Not yet universal

one seed;
one branch;
two initial directions;
one parameter set.

11. Highest-value final control before parameter variation

Test the constrained triangular model directly, without fitting its first row or allowing arbitrary ρ,χρ,χ:
a^n+1=λB,na^n+g^n,g^n+1=αEMAg^n.
a^n+1g^n+1=λB,na^n+g^n,=αEMAg^n.

Use:

geometric vv;
measured λB,n=v⊤KnvλB,n=v⊤Knv;

fixed
αEMA=e−dt/Twindow;
αEMA=e−dt/Twindow;
no fitted parameters.

Then compare its free rollout with the exact projected augmented tangent over disjoint late windows.

If this parameter-free triangular model predicts the late scalar amplitudes accurately, the spectral-gap mechanism is effectively closed for this replay.

Can you please create the code vital to probing the next steps?
WriteAIEF_1_4_parameter_free_triangular_closure.py579 lines
AIEF_1_4_parameter_free_triangular_closure.pyPYTHON
#!/usr/bin/env python3
"""Parameter-free triangular closure test for the late AIEF K3 reduction.

The proposed asymptotic reduction is

a[n+1] = lambda[n] a[n] + g[n],
g[n+1] = alpha_EMA g[n],

where

a[n] = v^T b[n],
g[n] = v^T f[n],
lambda[n] = v^T K[n] v,
v = (1,-1,-1)/sqrt(3),
alpha_EMA = exp(-dt/T_window).

No coefficients are fitted. The script checks one-step residuals and free
rollouts from several late start times. It also compares the geometric axis
with a data-derived late POD axis, quantifies the measured transverse
correction, and reports the implied time-varying spectral gap

kappa[n] = -log(|lambda[n]/alpha_EMA|)/dt.

This analysis remains conditional on the saved base replay. The measured
lambda[n] sequence is an input from the validated frozen-weight cocycle, not a

```
new fitted model parameter.
"""

from __future__ import annotations

import argparse
import hashlib
import json
import math
import os
import sys
from pathlib import Path
from typing import Any

import numpy as np
import pandas as pd

import AIEF_1_4_component_controller_audit as component_audit



DEFAULT_DUHAMEL_NPZ = "aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz"
DEFAULT_MEMORY_FITS =
"aief_1_4_feedback_rank_memory_reduction/target_time_memory_fits.csv"
DEFAULT_MANIFEST = "runs/5a9f093a2fe0/manifest.json"
DEFAULT_OUTPUT_DIR = "aief_1_4_parameter_free_triangular_closure"
DEFAULT_DT = 0.01
DEFAULT_POD_START = 250.0
DEFAULT_LATE_START = 200.0
DEFAULT_ROLLOUT_STARTS = "150,200,250"
DEFAULT_SAMPLE_STRIDE = 10
DEFAULT_SCALAR_TOL = 1e-30
ZERO_TOL = 1e-300



def file_sha256(path: str | os.PathLike[str]) -> str:
digest = hashlib.sha256()
with open(path, "rb") as handle:
for block in iter(lambda: handle.read(1024 * 1024), b""):
digest.update(block)
```

```
return digest.hexdigest()



def parse_nonnegative_float_list(value: str) -> list[float]:
try:
values = [float(item.strip()) for item in value.split(",") if item.strip()]
except ValueError as exc:
raise argparse.ArgumentTypeError("expected comma-separated floats") from exc
if not values or any(item < 0.0 for item in values):
raise argparse.ArgumentTypeError("values must be nonnegative")
return sorted(set(values))



def json_compact(value: Any) -> str:
return json.dumps(value, separators=(",", ":"), sort_keys=True)



def geometric_direction() -> np.ndarray:
return np.array([1.0, -1.0, -1.0]) / math.sqrt(3.0)



def relative_sequence_error(actual: np.ndarray, predicted: np.ndarray) -> float:
return float(
np.linalg.norm(actual - predicted)
/ max(np.linalg.norm(actual), ZERO_TOL)
)



def fit_log_rate(
times: np.ndarray,
values: np.ndarray,
tolerance: float,
) -> dict[str, float]:
magnitudes = np.abs(np.asarray(values, dtype=float))
valid = np.isfinite(magnitudes) & (magnitudes > tolerance)
if np.sum(valid) < 3:
return {"rate": np.nan, "R_squared": np.nan, "count": int(np.sum(valid))}
x = np.asarray(times, dtype=float)[valid]
```

```
y = np.log(magnitudes[valid])
slope, intercept = np.polyfit(x, y, 1)
prediction = slope * x + intercept
residual = float(np.sum((y - prediction) ** 2))
total = float(np.sum((y - np.mean(y)) ** 2))
return {
"rate": float(-slope),
"R_squared": float(1.0 - residual / total) if total > 0.0 else np.nan,
"count": int(len(x)),
}



def make_report(
one_step_summary: pd.DataFrame,
rollout_summary: pd.DataFrame,
gap_summary: pd.DataFrame,
*,
alpha_ema: float,
ema_rate: float,
pod_direction: np.ndarray,
geom_direction: np.ndarray,
) -> str:
angle = math.degrees(
math.acos(
np.clip(abs(float(pod_direction @ geom_direction)), -1.0, 1.0)
)
)
lines = [
"AIEF parameter-free triangular closure analysis",
"==============================================",
"",
f"alpha_EMA: {alpha_ema:.12g}",
f"EMA rate: {ema_rate:.9g}",
f"POD/geometric unsigned angle: {angle:.9g} degrees",
"",
"One-step closure",
"----------------",
]
for _, row in one_step_summary.iterrows():
lines.append(
f" axis={row['axis']}, direction={int(row['direction_id'])}: "
f"median_a_residual={row['median_a_relative_residual']:.6g}, "
```

```
f"max_a_residual={row['maximum_a_relative_residual']:.6g}, "
f"median_g_residual={row['median_g_relative_residual']:.6g}, "
f"max_g_residual={row['maximum_g_relative_residual']:.6g}, "
f"median_transverse={row['median_transverse_fraction']:.6g}"
)


lines.extend(["", "Free rollouts", "-------------"])
for _, row in rollout_summary.iterrows():
lines.append(
f" axis={row['axis']}, direction={int(row['direction_id'])}, "
f"start={row['start_time']:.6g}: "
f"a_RMSE={row['a_trajectory_relative_RMSE']:.6g}, "
f"g_RMSE={row['g_trajectory_relative_RMSE']:.6g}, "
f"final_a_error={row['final_a_relative_error']:.6g}, "
f"final_g_error={row['final_g_relative_error']:.6g}, "
f"predicted_a_rate={row['predicted_a_rate']:.6g}, "
f"actual_a_rate={row['actual_a_rate']:.6g}"
)


lines.extend(["", "Implied spectral gap", "--------------------"])
for _, row in gap_summary.iterrows():
lines.append(
f" axis={row['axis']}, target={row['target_time']:.6g}: "
f"median_kappa_model={row['median_model_kappa']:.6g}, "
f"observed_kappa={row['observed_kappa']:.6g}, "
f"closure_error={row['closure_error']:.6g}, "
f"kappa_variation={row['model_kappa_std']:.6g}"
)


lines.extend(
[
"",
"Interpretation constraints",
"-------------------------",
"* The model has no fitted coefficients but uses the measured base lambda[n] sequence.",
"* The g equation tests whether projected forcing is asymptotically pure EMA decay.",
"* Geometric and POD axes are tested separately; only the geometric axis is analytic a priori.",
"* Rollout failure can arise from transverse correction even when one-step scalar closure is accurate.",
"* Results remain conditional on this replay and tested tangent directions.",
"",
]
```

```
)
return "\n".join(lines)



def parse_arguments() -> argparse.Namespace:
parser = argparse.ArgumentParser(description=__doc__)
parser.add_argument("--duhamel-npz", default=DEFAULT_DUHAMEL_NPZ)
parser.add_argument("--memory-fits", default=DEFAULT_MEMORY_FITS)
parser.add_argument("--manifest", default=DEFAULT_MANIFEST)
parser.add_argument("--output-dir", default=DEFAULT_OUTPUT_DIR)
parser.add_argument("--dt", type=float, default=DEFAULT_DT)
parser.add_argument("--pod-start", type=float, default=DEFAULT_POD_START)
parser.add_argument("--late-start", type=float, default=DEFAULT_LATE_START)
parser.add_argument(
"--rollout-starts",
type=parse_nonnegative_float_list,
default=parse_nonnegative_float_list(DEFAULT_ROLLOUT_STARTS),
)
parser.add_argument("--sample-stride", type=int, default=DEFAULT_SAMPLE_STRIDE)
parser.add_argument("--scalar-tol", type=float, default=DEFAULT_SCALAR_TOL)
return parser.parse_args()



def main() -> None:
args = parse_arguments()
for path in (args.duhamel_npz, args.memory_fits, args.manifest):
if not os.path.isfile(path):
raise FileNotFoundError(path)
if min(args.dt, args.scalar_tol) <= 0.0 or args.sample_stride <= 0:
raise ValueError("dt/scalar-tol and sample-stride must be positive")
output_dir = Path(args.output_dir)
output_dir.mkdir(parents=True, exist_ok=True)


with open(args.manifest, "r", encoding="utf-8") as handle:
manifest = json.load(handle)
t_window = float(manifest["model"]["T_window"])
alpha_ema = math.exp(-args.dt / t_window)
ema_rate = 1.0 / t_window


with np.load(args.duhamel_npz, allow_pickle=False) as data:
```

```
total = np.asarray(data["total_weight_tangent"], dtype=float).copy()
feedback = np.asarray(data["feedback_weight_tangent"], dtype=float).copy()
forcing = np.asarray(data["instant_forcing"], dtype=float).copy()
matrices = np.asarray(data["K3_matrix_sequence"], dtype=float).copy()
memory_fits = pd.read_csv(args.memory_fits)
count, state_steps, dimension = total.shape
future_steps = state_steps - 1
if dimension != 3 or matrices.shape != (future_steps, 3, 3):
raise ValueError("unexpected Duhamel shapes")


pod_start_step = min(
future_steps, int(round(args.pod_start / args.dt))
)
late_vectors = feedback[:, pod_start_step:, :].reshape(-1, 3)
norms = np.linalg.norm(late_vectors, axis=1)
late_vectors = late_vectors[norms > args.scalar_tol]
left, _, _ = np.linalg.svd(late_vectors.T, full_matrices=False)
pod_direction = left[:, 0]
geom_direction = geometric_direction()
if np.dot(pod_direction, geom_direction) < 0.0:
pod_direction *= -1.0
axes = {"POD": pod_direction, "geometric": geom_direction}


late_step = int(round(args.late_start / args.dt))
one_step_rows: list[dict[str, Any]] = []
one_step_trajectory_rows: list[dict[str, Any]] = []
rollout_rows: list[dict[str, Any]] = []
rollout_summary_rows: list[dict[str, Any]] = []
gap_rows: list[dict[str, Any]] = []
array_payload: dict[str, np.ndarray] = {}


for axis_name, axis in axes.items():
a = total @ axis
g = forcing @ axis
lambdas = np.einsum("i,nij,j->n", axis, matrices, axis)
Kaxis = np.einsum("nij,j->ni", matrices, axis)
invariance_defect = np.linalg.norm(
Kaxis - lambdas[:, None] * axis[None, :], axis=1
) / np.maximum(np.linalg.norm(Kaxis, axis=1), ZERO_TOL)
model_kappa = -np.log(
np.maximum(np.abs(lambdas / alpha_ema), ZERO_TOL)
```

```
) / args.dt


array_payload[f"{axis_name}_a_total"] = a
array_payload[f"{axis_name}_g_forcing"] = g
array_payload[f"{axis_name}_lambda"] = lambdas
array_payload[f"{axis_name}_model_kappa"] = model_kappa
array_payload[f"{axis_name}_invariance_defect"] = invariance_defect


for direction_id in range(count):
a_residual = np.empty(future_steps)
g_residual = np.empty(future_steps - 1)
a_relative = np.empty(future_steps)
g_relative = np.empty(future_steps - 1)
transverse_fraction = np.empty(future_steps)
for step in range(future_steps):
transverse = (
total[direction_id, step]
- a[direction_id, step] * axis
)
predicted_a = (
lambdas[step] * a[direction_id, step]
+ g[direction_id, step]
)
a_residual[step] = a[direction_id, step + 1] - predicted_a
a_relative[step] = abs(a_residual[step]) / max(
abs(a[direction_id, step + 1]),
abs(predicted_a),
args.scalar_tol,
)
transverse_fraction[step] = np.linalg.norm(transverse) / max(
np.linalg.norm(total[direction_id, step]),
args.scalar_tol,
)
if step < future_steps - 1:
predicted_g = alpha_ema * g[direction_id, step]
g_residual[step] = (
g[direction_id, step + 1] - predicted_g
)
g_relative[step] = abs(g_residual[step]) / max(
abs(g[direction_id, step + 1]),
abs(predicted_g),
args.scalar_tol,
```

```
)
if step % args.sample_stride == 0 or step == future_steps - 1:
one_step_trajectory_rows.append(
{
"axis": axis_name,
"direction_id": direction_id,
"step": step,
"time": step * args.dt,
"a_actual_next": a[direction_id, step + 1],
"a_predicted_next": predicted_a,
"a_relative_residual": a_relative[step],
"g_actual": g[direction_id, step],
"g_predicted_next": alpha_ema
* g[direction_id, step],
"g_relative_residual": g_relative[step]
if step < future_steps - 1
else np.nan,
"lambda": lambdas[step],
"model_kappa": model_kappa[step],
"invariance_defect": invariance_defect[step],
"transverse_fraction": transverse_fraction[step],
}
)


late_a = a_relative[late_step:]
late_g = g_relative[late_step : future_steps - 1]
late_transverse = transverse_fraction[late_step:]
one_step_rows.append(
{
"axis": axis_name,
"direction_id": direction_id,
"late_start": args.late_start,
"median_a_relative_residual": float(np.median(late_a)),
"maximum_a_relative_residual": float(np.max(late_a)),
"median_g_relative_residual": float(np.median(late_g)),
"maximum_g_relative_residual": float(np.max(late_g)),
"median_transverse_fraction": float(
np.median(late_transverse)
),
"maximum_transverse_fraction": float(
np.max(late_transverse)
),
```

```
"median_invariance_defect": float(
np.median(invariance_defect[late_step:])
),
}
)


for start_time in args.rollout_starts:
start = int(round(start_time / args.dt))
if start >= future_steps:
continue
a_hat = float(a[direction_id, start])
g_hat = float(g[direction_id, start])
actual_a_values = []
predicted_a_values = []
actual_g_values = []
predicted_g_values = []
for step in range(start, future_steps):
a_hat = lambdas[step] * a_hat + g_hat
g_hat = alpha_ema * g_hat
actual_a = float(a[direction_id, step + 1])
actual_a_values.append(actual_a)
predicted_a_values.append(a_hat)
if step + 1 < future_steps:
actual_g = float(g[direction_id, step + 1])
actual_g_values.append(actual_g)
predicted_g_values.append(g_hat)
if (
(step + 1 - start) % args.sample_stride == 0
or step + 1 == future_steps
):
rollout_rows.append(
{
"axis": axis_name,
"direction_id": direction_id,
"start_time": start_time,
"step": step + 1,
"time": (step + 1) * args.dt,
"actual_a": actual_a,
"predicted_a": a_hat,
"a_relative_error": abs(actual_a - a_hat)
/ max(abs(actual_a), args.scalar_tol),
"actual_g": actual_g_values[-1]
```

```
if actual_g_values
else np.nan,
"predicted_g": predicted_g_values[-1]
if predicted_g_values
else np.nan,
"g_relative_error": abs(
actual_g_values[-1] - predicted_g_values[-1]
)
/ max(
abs(actual_g_values[-1]), args.scalar_tol
)
if actual_g_values
else np.nan,
}
)
actual_a_array = np.asarray(actual_a_values)
predicted_a_array = np.asarray(predicted_a_values)
actual_g_array = np.asarray(actual_g_values)
predicted_g_array = np.asarray(predicted_g_values)
times = (
np.arange(start + 1, future_steps + 1) * args.dt
)
actual_fit = fit_log_rate(
times, actual_a_array, args.scalar_tol
)
predicted_fit = fit_log_rate(
times, predicted_a_array, args.scalar_tol
)
rollout_summary_rows.append(
{
"axis": axis_name,
"direction_id": direction_id,
"start_time": start_time,
"end_time": future_steps * args.dt,
"a_trajectory_relative_RMSE": relative_sequence_error(
actual_a_array, predicted_a_array
),
"g_trajectory_relative_RMSE": relative_sequence_error(
actual_g_array, predicted_g_array
)
if len(actual_g_array)
else np.nan,
```

```
"final_a_relative_error": abs(
actual_a_array[-1] - predicted_a_array[-1]
)
/ max(abs(actual_a_array[-1]), args.scalar_tol),
"final_g_relative_error": abs(
actual_g_array[-1] - predicted_g_array[-1]
)
/ max(abs(actual_g_array[-1]), args.scalar_tol)
if len(actual_g_array)
else np.nan,
"actual_a_rate": actual_fit["rate"],
"actual_a_R_squared": actual_fit["R_squared"],
"predicted_a_rate": predicted_fit["rate"],
"predicted_a_R_squared": predicted_fit["R_squared"],
}
)


for target_time in sorted(memory_fits["target_time"].unique()):
target_step = min(
future_steps, int(round(float(target_time) / args.dt))
)
start = max(late_step, target_step - int(round(66.6 / args.dt)))
if target_step <= start:
continue
candidates = memory_fits.loc[
memory_fits["target_time"] == target_time
]
observed = float(candidates["kappa"].median())
gap_rows.append(
{
"axis": axis_name,
"target_time": float(target_time),
"fit_start_time": start * args.dt,
"median_model_kappa": float(
np.median(model_kappa[start:target_step])
),
"model_kappa_std": float(
np.std(model_kappa[start:target_step], ddof=1)
),
"observed_kappa": observed,
"closure_error": observed
- float(np.median(model_kappa[start:target_step])),
```

```
}
)

one_step_summary = pd.DataFrame(one_step_rows)
one_step_trajectory = pd.DataFrame(one_step_trajectory_rows)
rollout_trajectory = pd.DataFrame(rollout_rows)
rollout_summary = pd.DataFrame(rollout_summary_rows)
gap_summary = pd.DataFrame(gap_rows)

component_audit.atomic_write_csv(
output_dir / "triangular_one_step_trajectory.csv", one_step_trajectory
)
component_audit.atomic_write_csv(
output_dir / "triangular_one_step_summary.csv", one_step_summary
)
component_audit.atomic_write_csv(
output_dir / "triangular_rollout_trajectory.csv", rollout_trajectory
)
component_audit.atomic_write_csv(
output_dir / "triangular_rollout_summary.csv", rollout_summary
)
component_audit.atomic_write_csv(
output_dir / "triangular_gap_summary.csv", gap_summary
)
component_audit.atomic_savez(
output_dir / "triangular_closure_arrays.npz",
{
"POD_direction": pod_direction,
"geometric_direction": geom_direction,
"alpha_EMA": np.asarray([alpha_ema]),
**array_payload,
},
)

report = make_report(
one_step_summary,
rollout_summary,
gap_summary,
alpha_ema=alpha_ema,
ema_rate=ema_rate,
pod_direction=pod_direction,
```

```
geom_direction=geom_direction,
)
component_audit.atomic_write_text(
output_dir / "parameter_free_triangular_report.txt", report
)


manifest_output = {
"analysis": "AIEF-1.4 parameter-free triangular closure",
"source": {
"path": str(Path(__file__).resolve()),
"sha256": file_sha256(Path(__file__).resolve()),
},
"command": sys.argv,
"inputs": {
"duhamel_npz": {
"path": str(Path(args.duhamel_npz).resolve()),
"sha256": file_sha256(args.duhamel_npz),
},
"memory_fits": {
"path": str(Path(args.memory_fits).resolve()),
"sha256": file_sha256(args.memory_fits),
},
"manifest": {
"path": str(Path(args.manifest).resolve()),
"sha256": file_sha256(args.manifest),
},
},
"dt": args.dt,
"pod_start": args.pod_start,
"late_start": args.late_start,
"rollout_starts": args.rollout_starts,
"sample_stride": args.sample_stride,
"scalar_tolerance": args.scalar_tol,
"alpha_EMA": alpha_ema,
"ema_rate": ema_rate,
"outputs": [
"triangular_one_step_trajectory.csv",
"triangular_one_step_summary.csv",
"triangular_rollout_trajectory.csv",
"triangular_rollout_summary.csv",
"triangular_gap_summary.csv",
"triangular_closure_arrays.npz",
```

```
"parameter_free_triangular_report.txt",
],
"interpretation_constraints": [
"No coefficients are fitted; lambda[n] is measured from the validated base cocycle and alpha_EMA is fixed by the model.",
"POD and geometric axes are tested separately.",
"The triangular model is intended only for the late regime.",
"A successful rollout on this replay does not establish universality across seeds or branches.",
],
}
component_audit.atomic_write_text(
output_dir / "analysis_manifest.json",
json.dumps(manifest_output, indent=2, sort_keys=True) + "\n",
)


print(report)
print(f"Saved analysis to {output_dir.resolve()}")



if __name__ == "__main__":
main()
```

Open

Created the parameter-free closure test:

It tests the late triangular model
an+1=λnan+gn,gn+1=αEMAgn,
an+1gn+1=λnan+gn,=αEMAgn,

with no fitted coefficients.
Validation

On a synthetic system with known
λ=0.98,αEMA=0.99,
λ=0.98,αEMA=0.99,

the script achieved:

one-step residuals near 10−1610−16;
free-rollout errors near 10−1410−14;
exact recovery of
κ=−dt−1log(λ/αEMA).
κ=−dt−1log(λ/αEMA).

Run it

Bash

```
python AIEF_1_4_parameter_free_triangular_closure.py \
--duhamel-npz aief_1_4_augmented_duhamel_feedback/duhamel_vectors.npz \
--memory-fits aief_1_4_feedback_rank_memory_reduction/target_time_memory_fits.csv \
--manifest runs/5a9f093a2fe0/manifest.json \
--dt 0.01 \
--pod-start 250 \
--late-start 200 \
--rollout-starts 150,200,250 \
--sample-stride 10 \
--scalar-tol 1e-30 \
--output-dir aief_1_4_parameter_free_triangular_closure
```

Two axes tested independently
Empirical POD direction
vPOD.
vPOD.
Analytic geometric direction
vgeom=
vgeom=

\frac1{\sqrt3}(1,-1,-1).
]

This distinguishes:

best empirical state representation;
exact operator-invariant geometric line.

Fixed model quantities

For each axis:
an=v⊤bn,gn=v⊤fn,
an=v⊤bn,gn=v⊤fn,
λn=v⊤Knv,
λn=v⊤Knv,

and
αEMA=
αEMA=

e^{-dt/T_{\rm window}}.
]

Nothing is fitted.
One-step tests

The script reports:
Weight recurrence residual
an+1−
an+1−
\lambda_na_n

g_n.
]
EMA forcing residual
gn+1−
gn+1−

\alpha_{\rm EMA}g_n.
]
Geometric diagnostics

transverse state fraction;
invariance defect of vv;
implied model gap
κn=−1dtlog|λnαEMA|.
κn=−dt1log
αEMAλn

.

Free rollouts

Starting independently at
t=150,200,250,
t=150,200,250,

the script initializes with the exact measured an,gnan,gn, then evolves only:
a^n+1=
a^n+1=

\lambda_n\hat a_n+\hat g_n,
]
g^n+1=
g^n+1=

\alpha_{\rm EMA}\hat g_n.
]

It compares with the exact projected augmented tangent through t=320t=320.

For each rollout it reports:

full-trajectory relative RMSE for aa;
full-trajectory relative RMSE for gg;
final relative errors;
fitted decay rates of actual and predicted aa;
fit R2R2.

The t=150t=150 rollout tests the settling regime; t=200t=200 and 250250 test the stationary late regime.
Spectral-gap closure

For every target available in the memory-fit CSV, it compares:
median[−1dtlog⁡|λnαEMA|]
median[−dt1log
αEMAλn

]

with the independently measured operational κTκT.

It reports:

model κκ;
model variation over the target window;
observed κκ;
closure error.

Outputs